

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 1

QP Code: 100162

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 234

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-B (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

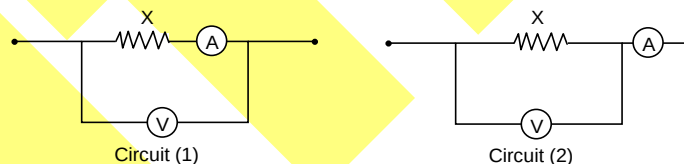
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6. A convex lens made of glass ($\mu_g = 3/2$) has focal length f in air. The image of an object placed in front of it is inverted real and magnified. Now the whole arrangement is immersed in water ($\mu_w = 4/3$) without changing the distance between object and lens. Then
 (A) the new focal length will become $4f$
 (B) the new focal length will become $f/4$
 (C) new image will be virtual and magnified
 (D) new image will be real, inverted and smaller in size
7. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).
 (A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases
 (B) the interval between $\lambda_{k\alpha}$ and λ_0 increases
 (C) the interval between $\lambda_{k\beta}$ and λ_0 increases
 (D) λ_0 does not change
 Here λ_0 is cutoff wavelength and $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ are wavelength of k_α and k_β characteristic X-rays.

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. An ammeter and a voltmeter are connected in series to a battery of emf 6 V. When a certain resistance is connected in parallel with the voltmeter the reading of the voltmeter decreases $n = 2$ times while the reading of the ammeter increases the same number of times. Then the ratio of voltmeter resistance to the ammeter resistance will be
 (A) 1 : 2
 (B) 2 : 3
 (C) 2 : 1
 (D) None of these
9. An ammeter and a voltmeter are connected as shown in figure (1) and figure (2). In order to measure an unknown resistance which connection is to be preferred,

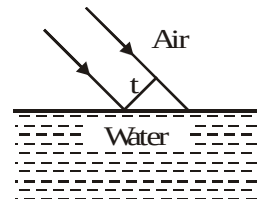


- (A) for low resistance circuit 2 and for high resistance circuit (1).
 (B) for low resistance circuit (1) and for high resistance circuit 2
 (C) For any value of resistance circuit 1.
 (D) For any value of resistance circuit 2.

Space For Rough Work

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10. There is a 90° bend in a pipe line in which water is flowing at speed v . The cross sectional area of pipe is A . The force exerted by pipe walls on water at bend is
 (A) $\sqrt{3} A v^2 \rho$ (B) $\sqrt{2} A v^2 \rho$
 (C) $A v^2 \rho$ (D) $\sqrt{5} A v \rho$
11. Calculate the capacitance of a parallel-plate capacitor of area S , separation d and dielectric constant varying linearly from ϵ_1 to ϵ_2 ($\epsilon_2 > \epsilon_1$) in the direction of the applied field.
 (A) $C = \frac{\epsilon_0 S}{d} \frac{(\epsilon_2 - \epsilon_1)}{\ln\left(\frac{\epsilon_2}{\epsilon_1}\right)}$ (B) $C = \frac{\epsilon_0 S}{d} \frac{(\epsilon_2 + \epsilon_1)}{\ln\left(\frac{\epsilon_2}{\epsilon_1}\right)}$
 (C) $C = \frac{\epsilon_0 S}{2d} \frac{(\epsilon_2 + \epsilon_1)}{\ln\left(\frac{\epsilon_2}{\epsilon_1}\right)}$ (D) none of these
12. A hoop of mass M and radius R is placed on an absolutely smooth level surface. An insect of mass m begins to slide along the hoop from the highest point with speed V relative to the hoop. Then the velocity of the centre of the hoop after time t will be
 (A) $\frac{v \cos\left(\frac{vt}{R}\right)}{1 + \frac{m}{M}}$ (B) $\frac{v \sin\left(\frac{vt}{R}\right)}{1 + \frac{M}{m}}$ (C) $\frac{v \cos\left(\frac{vt}{R}\right)}{1 - \frac{M}{m}}$ (D) $\frac{v \cos\left(\frac{vt}{R}\right)}{1 + \frac{M}{m}}$
13. A monochromatic beam of width t is incident at 45° on an air water interface as shown in the figure. The refractive index of water is μ and that of air is 1. The width of the beam in water is,
 (A) $(\mu - 1) t$ (B) μt
 (C) $\frac{\sqrt{\mu^2 - 1}}{\mu} t$ (D) $\frac{(\sqrt{2\mu^2 - 1})}{\mu} t$



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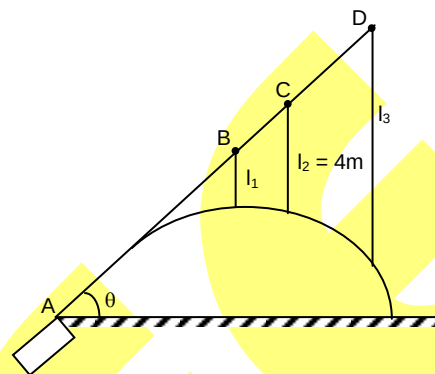
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PART – B

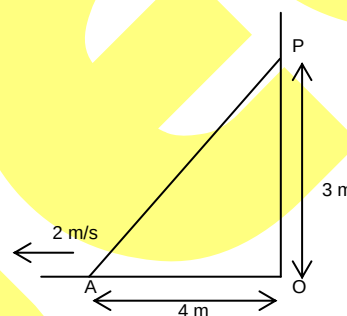
(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

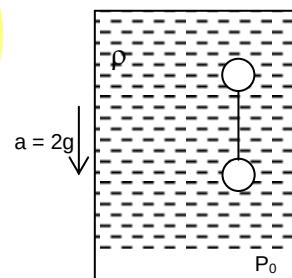
1. A stream of water coming out from a pipe at point A as shown in figure. A rod AD is along the projected velocity of stream. There are four points A, B, C, D on the rod ($AB = BC = CD$) from which string of length l_1, l_2, l_3 are hanging. These string are touching the trajectory of stream of water. If the length $l_2 = 4\text{m}$ find l_3 .



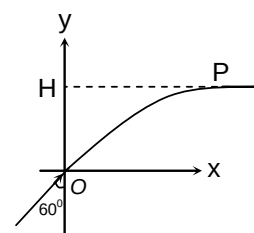
2. A ladder AP of length 5m is inclined to a vertical wall is slipping over a horizontal surface with velocity of 2 m/s when P is at a distance 3m from ground what is the velocity of centre of mass at this moment (in m/s) _____.



3. Two identical shaped spheres of specific gravity 0.8 & 1.6 are connected through string & placed in liquid container as shown in figure. if spheres are in equilibrium wrt container then tension in the string (take volume of each sphere $v_0 = 250\text{ CC}$) is xn , find the value of 'x'.



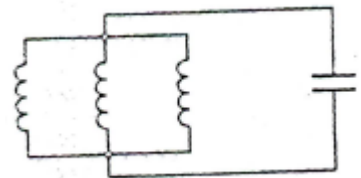
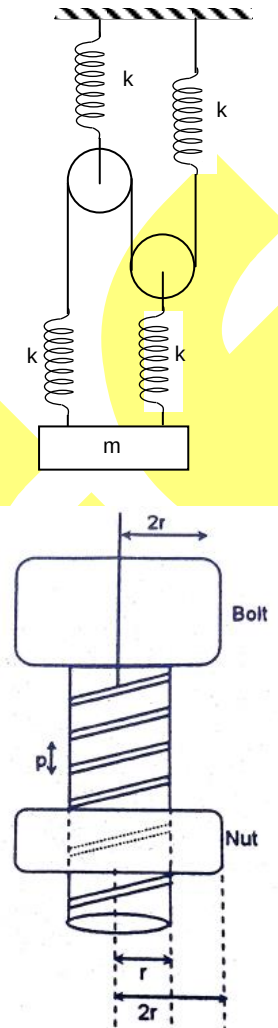
4. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \leq y \leq 1$. A ray of light is incident at origin at an angle 60° with y -axis as shown in the figure. At point P ray becomes parallel to x-axis. Find the value of 'H'.



Space For Rough Work

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5. The refractive index for water w.r.t. air for a sound wave if the velocities of sound in water and air are 1440 ms^{-1} and 340 ms^{-1} , is
6. All pulleys & ropes are ideal find out period of small oscillation of block of mass m as shown in figure. Take $\frac{M}{k} = \left(\frac{9}{10}\right)$
7. A bolt with shaft radius r has pitch of its threading equal to p . The nut has frictionless movement on threads. The bolt is rotated anti-clockwise while looking from top with an angular acceleration α such that the nut moves up by an acceleration g . Find α in terms of g and p . (take $g = 10 \text{ m/s}^2$; $p = 2 \text{ mm}$)
8. A circuit shown consists of three coils of inductances 1.0 H each and a $1.0 \mu\text{F}$ capacitance. At an instant, when charge on the capacitor is zero, currents of 1.0 A , 2.0 A and 4.0 A are flowing through the coils in the same direction. Find the maximum charge on the capacitor. (in MC)



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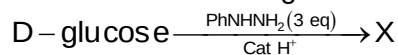
SECTION-2 : CHEMISTRY

PART – A

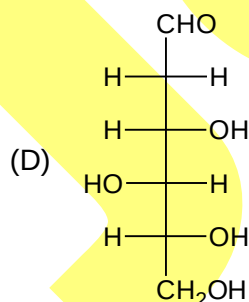
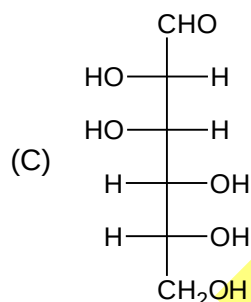
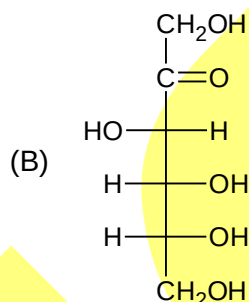
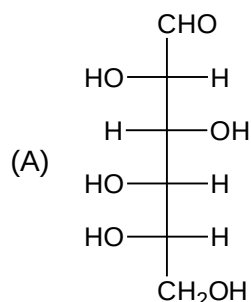
(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Consider the following reaction



Among the following the compound(s) whose derivative(s) will have the same melting point as that of X is(are)



2. The type(s) of intersection(s) that hold layers of graphite together is(are)

(A) π - π stacking (B) van der Waal's
(C) hydrogen bonding (D) coulombic

Space For Rough Work

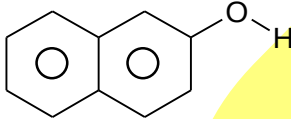
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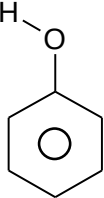
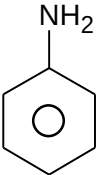
3. Which is/are correct statement about diazo-coupling reaction?

(A) Benzene diazonium ion is a strong electrophile $\left(\text{PhN}_2^+\right)$.

(B) For diazocoupling reaction of phenol pH optima should be slightly basic.

(C) pH diazo coupling reaction of Aniline pH optima should be slightly acidic.

(D) The colour of diazocoupling product with β -naphthol  is red, with

 is orange and with  is yellow.

4. Which is/are correct about allotropes of carbon?

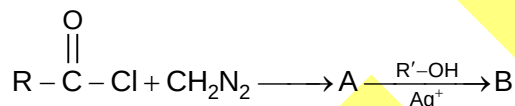
(A) Graphite is thermodynamically most stable allotrope.

(B) Molar entropy order is Fullerene > Graphite > Diamond

(C) Fullerene is aromatic

(D) Buckminster fullerene C_{60} has 12 five member and 20 six member rings

5. The intermediate formed in the reaction



Which is correct statements about above reaction?

(A) Intermediate involve is both carbene as well as carbanion.

(B) A can be formed by reaction of $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{H}$ (Carboxylic acid) with AlPO_4

(C) The nucleophile (CH_2N_2) is resonance stabilised

(D) Final product 'B' is an ester

6. When B_2H_6 is treated with excess NH_3 at low temperature. The nature of bond present in the product is/are

(A) Ionic

(B) Covalent

(C) Co-ordinate

(D) Metallic

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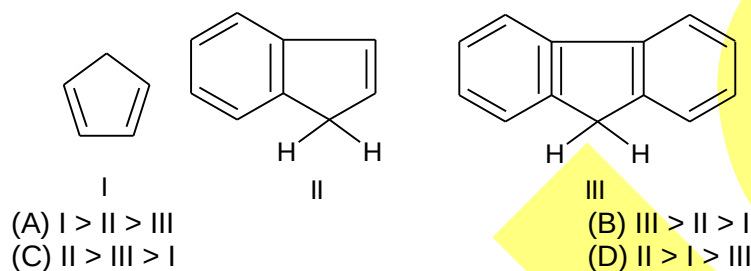
7. Spin isomerism is found in
 (A) H_2 (B) D_2
 (C) N_2 (D) F_2

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

8. The VO_4^{3-} , CrO_4^{2-} and MnO_4^- ions exhibit intense ligand to metal charge transfer transition the wavelength of this transition follows the order
 (A) $CrO_4^{2-} < VO_4^{3-} < MnO_4^-$ (B) $MnO_4^- < VO_4^{3-} < CrO_4^{2-}$
 (C) $VO_4^{3-} < CrO_4^{2-} < MnO_4^-$ (D) $CrO_4^{2-} < MnO_4^- < VO_4^{3-}$

9. Correct acidic strength of

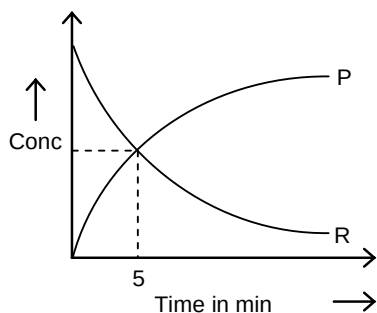


10. If EAN of metal is 36 in $[M(CO)_2(\sigma C_5H_5)(\pi C_5H_5)]$, then what is the atomic number of metal M?
 (A) 23 (B) 24
 (C) 25 (D) 26
11. A mineral consists of an equimolar mixture of the carbonates of two bivalent metals. One metal is present to the extent of 12.5% by mass. 2.8 g of the mineral on heating lost 1.32 g of CO_2 . What is the % by mass of the other metal.
 (A) 87.5 (B) 35.71
 (C) 65.11 (D) 23.21
12. The CFSE for octahedral $[CoCl_6]^{4-}$ is 18000 cm^{-1} . The CFSE for tetrahedral $[CoCl_4]^{2-}$ will be
 (A) 18000 cm^{-1} (B) 16000 cm^{-1}
 (C) 8000 cm^{-1} (D) 20000 cm^{-1}

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13.



For a given 1st order reaction $R \rightarrow 3P$

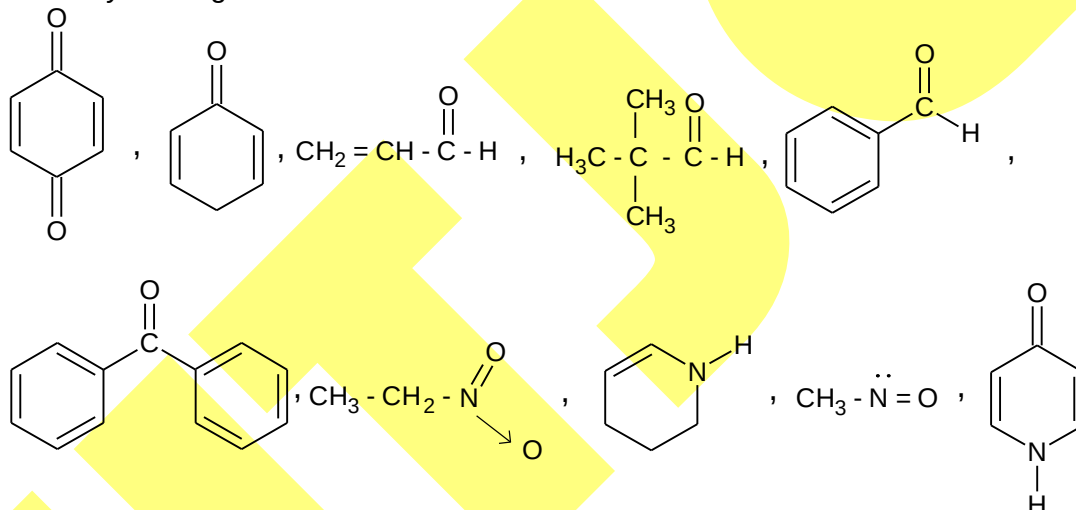
What is the time required for 20% completion of reaction in minute unit?

- (A) 4 (B) 2
(C) 3 (D) 5

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. How many of the given molecule can show tautomerism?

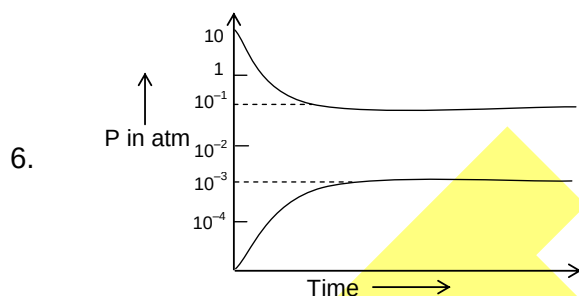


2. Number of geometrical square planar complex formed when PtCl_2 is treated with racemic mixture of alanine is
3. The molarity of 20% w/w sulphuric acid of density 1.14 g cm^{-3} is

Space For Rough Work

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4. How many of the given molecules have central atom sp^3 hybridized
 SF_4 , XeO_2F_2 , $XeOF_2$, $XeOF_4$, XeO_4 , CO_2 , SiO_2 , ClF_3 , CH_3^- , CH_3^+
5. How many of the given statement(s) is/are correct?
- Metallic bond strength in 's'-block elements decreases down the group.
 - Metallic character increases down the group in 's'-block.
 - Reactivity of 's'-block elements increases down the group.
 - Solution of alkali metal in liquid NH_3 is blue in colour due to presence of solvated electron.
 - The blue colour of alkali metals in liquid NH_3 slowly fades until it disappears owing to the formation of a metal amide and this reaction is catalysed by Fe or presence of impurities.
 - At concentrations about 3M, solution of alkali metal in liquid NH_3 are copper-bronze coloured and have a metallic lusture because metal ion clusters are formed.
 - Solutions of metals in liquid NH_3 conduct electricity better than any salt in any liquid and the conductivity is similar to that of pure metals
 - Conduction of alkali metal in liquid NH_3 is due to mainly to the presence of solvated electron.
 - Solutions of alkali metals in liquid NH_3 act as a powerful reducing agent and they will even reduce an aromatic ring.

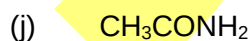
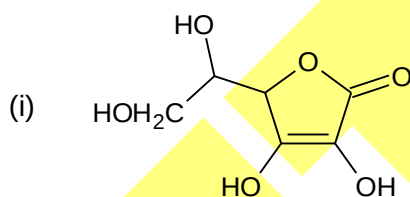
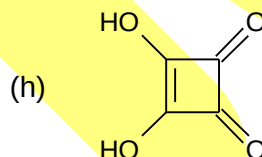
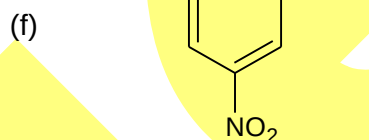
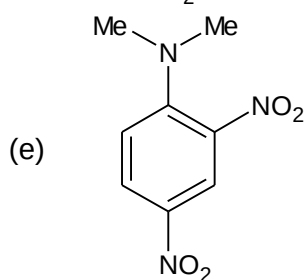
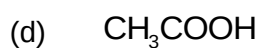
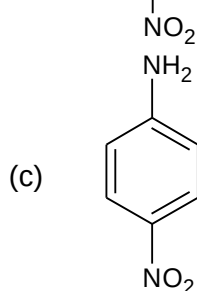
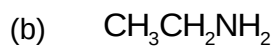
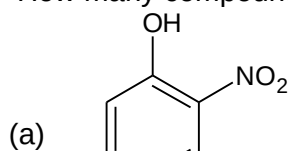


The plot of partial pressure of the product and reactant is given above. What is the value of ΔG° in L atm unit at temperature 'T'?
 [Assume $2.303 RT = 4 \text{ L atm mol}^{-1}$]

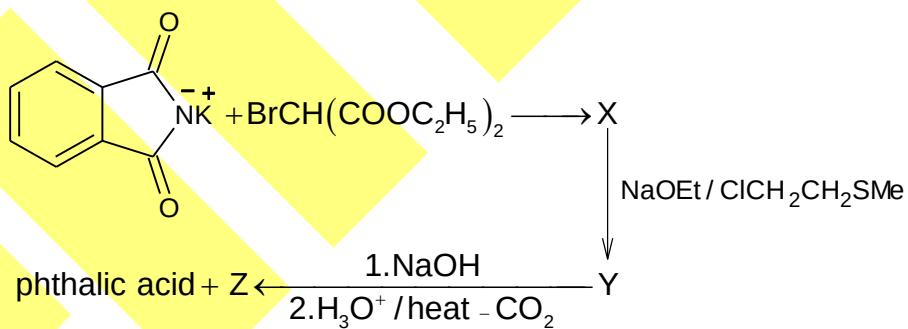
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7. How many compounds can liberate CO_2 when treated with NaHCO_3



8.



$$X = \frac{1}{4} [\text{Sum of all atoms present in Z}] + \text{no of optically active isomers of Z. Value of X is}$$

Space For Rough Work

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Which of the following statement(s) is (are) correct?

(A) If $|\vec{v} \times \vec{w}| = 0$ for all vector $\vec{w}$, then $\vec{v} = \vec{0}$

(B) If $\vec{a} + \vec{b} + \sqrt{3}\vec{c}$ is a null vector where $\vec{a}$ and $\vec{b}$ are unit vectors such that angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, then the magnitude of $\vec{c}$ is 1.

(C) If $(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = 0$, then $\vec{u} = \vec{v}$

(D) If $\vec{a}$ and $\vec{b}$ are two unit vectors then the maximum value of $|2\vec{a} + \vec{b}|$ is equal to 3.
2. Let $L_1 : \frac{x+3}{-4} = -\frac{y-6}{3} = \frac{z}{2}$ and $L_2 : \frac{x-2}{-4} = \frac{y+1}{1} = \frac{z-6}{1}$ be two lines in $\mathbb{R}^3$. Which of the following statement(s) is (are) correct?

(A) L_1, L_2 are coplanar.

(B) L_1, L_2 are skew lines

(C) Shortest distance between L_1 and L_2 is 9.

(D) $(\hat{i} - 4\hat{j} + 8\hat{k})$ is a vector perpendicular to both L_1 and L_2 .
3. Which of the following statement(s) is (are) correct?

(A) If $f(x)$ is differentiable $\forall x \in [0, 1]$, then $f'(x)$ must be bounded in $[0, 1]$

(B) There exist a bijective function $f : [0, 1] \rightarrow [0, 1]$ which is not continuous

(C) Let $f : [-1, 2] \rightarrow \mathbb{R}$ be defined as $f(x) = x^2 - \cos \pi x + 4$, then $f(c) = 2\pi$ for some $c \in (-1, 2)$

(D) If $f(x) = 3 + x + e^x$, then $(f^{-1})'(4)$ is equal to $\frac{1}{2}$.
4. Let $f(x)$ be twice differentiable function such that $f''(x) < 0$ in $[0, 2]$. Then

(A) $f(0) + f(2) = 2f(c)$, $0 < c < 2$

(B) $f(0) + f(2) = 2f(1)$

(C) $f(0) + f(2) > 2f(1)$

(D) $f(0) + f(2) < 2f(1)$

Space For Rough Work

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5. If the tangent at a point P_1 (other than $(0, 0)$) on the curve $ax^3 - y + b = 0$ meets the curve again at P_2 . The tangent at P_2 meets the curve at P_3 and so on. If the abscissae of $P_1, P_2, P_3, \dots, P_n$ form a G.P. then (a, b) may be
 (A) $(1, 0)$ (B) $(2, 7)$
 (C) $(3, 5)$ (D) $(4, 9)$
6. If three planes $P_1 = 2x + y + z - 1 = 0$, $P_2 = x - y + z - 2 = 0$ and $P_3 = \alpha x - y + 3z - 5 = 0$ intersects each other at point P on XOY plane and at point Q on YOZ plane, where O is origin then identify the correct statement(s)?
 (A) The value of α is 4
 (B) Straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$.
 (C) The length of projection of $\overline{PQ}$ on x - axis is 1
 (D) Centroid of the triangle OPQ is $\left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$
7. Which of the following statement is(are) true?
 (A) From any point on the directrix of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 (a > b)$, a pair of tangents are drawn to the auxiliary circle of the ellipse. The chord of contact will pass through the corresponding focus of the ellipse.
 (B) If two tangents to a parabola $y^2 = 4ax$ intersect on the line $x = a$ then their chord of contact always passes through the $(-a, 0)$.
 (C) If $P(x, y)$ is such that it moves on a hyperbola $\left| \sqrt{(x-3)^2 + (y-4)^2} - \sqrt{x^2 + y^2} \right| = k^2 + 1$, then number of possible integral values of k is equal to 3.
 (D) A circle passing through 3 co-normal points on the parabola $y^2 = 4x$ also passes through $(1, 0)$.

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

8. Find the locus of the centroid of an equilateral triangle inscribed in the given parabola $x^2 = 4ay$.
 (A) $9y^2 = 4a(x - 8a)$ (B) $9x^2 = 4a(y - 6a)$
 (C) $9x^2 = 4a(y - 8a)$ (D) $9x^2 = 4a(y + 8a)$

Space For Rough Work

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9. A line through the origin divides parallelogram with vertices (10, 45), (10, 114), (28, 153) and (28, 84) into two congruent pieces. The slope of the line is
(A) $\frac{19}{99}$ (B) $\frac{99}{19}$
(C) $\frac{9}{19}$ (D) $\frac{19}{9}$
10. Equation of chord AB of circle $x^2 + y^2 = 2$ passing through P (2, 2) such that $\frac{PB}{PA} = 3$, is given by
(A) $x = 3y$ (B) $x = y$
(C) $y - 2 = \sqrt{3}(x - 2)$ (D) none of these
11. The number of ways in which 3 children can distribute 10 tickets out of 15 consecutively numbered tickets among themselves such that they get consecutive blocks of 5, 3 and 2 tickets is
(A) 8C_5 (B) ${}^5C_3 3!$
(C) ${}^8C_5 (3!)^2$ (D) none of these
12. The number of ordered triplets of positive a, b, c satisfying $a + b + c = 24$, $a^2 + b^2 + c^2 = 210$ and $abc = 440$ is
(A) 3 (B) 6
(C) 9 (D) 12
13. If $\int_0^{\sin^{-1} \frac{\pi}{4}} (\pi - 4 \sin \theta) \sin 2\theta \ln(1 + \tan(\sin \theta)) d\theta$ is k, then $\frac{960k}{\pi^3 \ln 2}$ is equal to
(A) 11 (B) 10
(C) 12 (D) none of these

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. A conical vessel is to be prepared out of a circular sheet of metal of unit radius. In order that the vessel has maximum volume, the sectorial area that must be removed from the sheet is A_1 and the area of the given sheet is A_2 . Find $\frac{A_2}{A_1}$.

Space For Rough Work

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2. Let $A = \{3, 4, 5, 6\}$ and $B = \{1, 2, 3, \dots, 10\}$ be two sets. Let a function f be defined from set A to set B such that $f(i) - f(j) \geq 2 \quad \forall i > j$, where $i, j \in A$. If the number of such functions is N , then the value of N is equal to
3. If $a = 1 + \frac{x^3}{3!} + \frac{x^6}{6!} + \dots \infty$, $b = x + \frac{x^4}{4!} + \frac{x^7}{7!} + \dots \infty$, $c = \frac{x^2}{2!} + \frac{x^5}{5!} + \frac{x^8}{8!} + \dots \infty$. Find the value of $a^3 + b^3 + c^3 - 3abc =$ _____
4. Four people sit round a circular table, and each person will roll a normal six sided die once. The probability that no two people sitting next to each other will roll the same number is $\frac{N}{1296}$. Then the value of N is
5. Let two non – collinear vectors $\vec{a}$ and $\vec{b}$ inclined at an angle $\frac{2\pi}{3}$ be such that $|\vec{a}| = 3$ and $|\vec{b}| = 4$. A point P moves so that any time t the position vector $\vec{OP}$ (where O is the origin) is given as $\vec{OP} = (e^t + t^{-1})\vec{a} + (e^t - e^{-t})\vec{b}$. If the least distance of P from origin is $\sqrt{2}\sqrt{\sqrt{a} - b}$ where $a, b \in \mathbb{N}$ then find the value of $(a + b)$.
6. Lot A consists of 6 good and 4 defective articles. Lot B consists of 7 good and 3 defective articles. A new lot C is formed by taking 6 articles from lot A and 4 articles from lot B. The probability that an articles chosen at random from the lot C is defective, is $\frac{k}{25}$ then k is
7. If the dependent variable y is changed to ' z ' by the substitution $y = \tan z$ and the differential equation $\frac{d^2y}{dx^2} = 1 + \frac{2(1+y)}{1+y^2} \left(\frac{dy}{dx}\right)^2$ is changed to $\frac{d^2z}{dx^2} = \cos^2 z + k \left(\frac{dz}{dx}\right)^2$, then the value of $(305.33) \times k$ equals
8. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Let perimeter of the rectangle ABCD is P , then $103.44P$ is

Space For Rough Work

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Q.P. Code: 100162**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|---------|-------|-------|-------|
| 1. ABCD | 2. BC | 3. B | 4. D |
| 5. C | 6. AC | 7. BC | 8. C |
| 9. A | 10. B | 11. A | 12. D |
| 13. D | | | |

PART – B

- | | | | |
|---------|---------|----------|---------|
| 1. 9 | 2. 1.67 | 3. 1 | 4. 0.39 |
| 5. 0.24 | 6. 6.28 | 7. 31.43 | 8. 4.04 |

SECTION-2 : CHEMISTRY**PART – A**

- | | | | |
|---------|--------|---------|---------|
| 1. ABC | 2. AB | 3. BCD | 4. ABCD |
| 5. ABCD | 6. ABC | 7. ABCD | 8. C |
| 9. A | 10. D | 11. D | 12. C |
| 13. A | | | |

PART – B

- | | | | |
|------|------|---------|------|
| 1. 5 | 2. 6 | 3. 2.32 | 4. 3 |
| 5. 9 | 6. 8 | 7. 6 | 8. 7 |

SECTION-3 : MATHEMATICS**PART – A**

- | | | | |
|---------|---------|--------|-------|
| 1. ABD | 2. BCD | 3. BCD | 4. AD |
| 5. ABCD | 6. ABCD | 7. ABC | 8. C |
| 9. B | 10. B | 11. B | 12. B |
| 13. B | | | |

PART – B

- | | | | |
|----------|----------|-----------|------------|
| 1. 5.45 | 2. 35.00 | 3. 1.00 | 4. 630.00 |
| 5. 72.00 | 6. 9 | 7. 610.66 | 8. 8275.20 |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **ABCD**

Sol. Since the graph is like a parabola

$$\therefore \text{let } x(t) = At + Bt^2 + C$$

$$\text{From graph } x(0) = 0 \Rightarrow C = 0$$

$$x(t) = Bt^2 + At$$

$$x(4) = 0 \Rightarrow 16B + 4A = 0$$

$$\left(\frac{dx}{dt}\right)_0 = 1 \Rightarrow (A + 2Bt)_{t=0} = 1$$

Put in (i), we get $B = -\frac{1}{4}$

$$x = t - \frac{t^2}{4}$$

max x coordinate = 1 (from max and min)

→ Since motion is a straight line motion

→ total distance travelled = $2 \times 1 = 2$ m

$$\text{Average speed} = \frac{2}{4} = 0.5 \text{ m/sec}$$

2. **BC**

Sol. $|f - 256| = 2.$

3. **B**

Sol. Number of nuclei became $\frac{1}{8^{\text{th}}} \Rightarrow 3$ half lives passed

$$\Rightarrow 3t_{1/2} = 3 \text{ min} \Rightarrow t_{1/2} = 1 \text{ min}$$

$$\therefore t_{\text{mean}} = \frac{t_{1/2}}{\ln 2} = \frac{1}{0.693} \text{ min} \approx 1.44 \text{ min}$$

4. **D**

Sol. $\frac{2T}{r} = 150 \text{ Pa}$

5. **C**

Sol. For critical stability, the centre of equivalent force should be at point C. So $BC = \frac{AB}{3} = 1.5 \text{ m}$

6. **AC**

Sol. $\frac{1}{f} = (\mu_r - 1) \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$

7. **BC**

Sol. $\lambda_0 = \frac{hc}{eV}$

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8. **C**

Sol. Use Kirchhoff's loop law & Kirchhoff's junction law to make equations.

9. **A**Sol. Voltage across resistor = current $\times$ resistance.10. **B**

$$\text{Sol. } F = \frac{dp}{dt} = (\sqrt{2} v) \frac{dm}{dt} = (\sqrt{2} v \rho) \frac{d(\text{volume})}{dt} = (\sqrt{2} v \rho) (AV)$$

11. **A**

$$\text{Sol. } \frac{1}{C} = \int_0^d \frac{dx}{\epsilon_0 S \frac{\epsilon_1 + \epsilon_2 - \epsilon_{xy}}{d}}$$

12. **D**

$$\text{Sol. } V_0 = \frac{MV \cos\left(\frac{vt}{R}\right)}{M + m}$$

13. **D**

Sol. Apply Snell's law and use geometry.

PART – B1. **9**

$$\text{Sol. } \frac{\ell_2}{4} = \frac{\ell_3}{9}$$

2. **1.67**

Sol. about ICR

$$\omega \times 3 = 2 \Rightarrow \omega = \frac{2}{3}$$

$$V_{cm} = \omega \times \frac{5}{2} = \frac{2}{3} \times \frac{5}{2} = \frac{5}{3} \text{ m/s} = 1.67 \text{ m/s.}$$

3. **1**

$$\text{Sol. } F_b + m_1 g + T = 2m_1 g$$

$$F_b + T = m_1 g$$

$$F_b + m_2 g = T + 2m_2 g$$

$$F_b - T = m_2 g$$

$$\frac{F_b + T}{F_b - T} = \frac{m_1 g}{m_2 g} = \frac{\rho_1}{\rho_2} = \frac{1.6}{0.8}$$

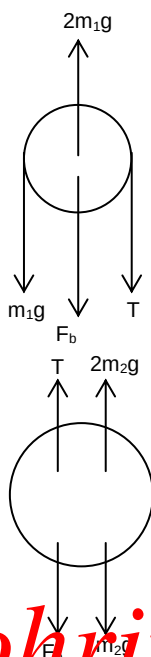
$$F_b + T = 2F_b - 2T$$

$$3T = F_b$$

$$T = \frac{F_b}{3} = \frac{V \rho g}{3}$$

$$\rho = \frac{\rho_1 + \rho_2}{2} = \frac{1.6\rho_w + 0.8\rho_w}{2} = 1.2\rho_w$$

$$T = 1 \text{ N}$$



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4. **0.39**

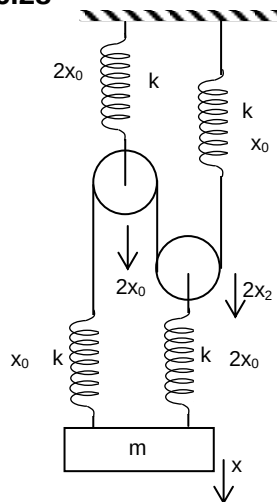
Sol. $\frac{2}{1+1+1^2} = 2 \sin(60^\circ)$

5. **0.24**

Sol. $\frac{\mu_w}{\mu_{air}} = \frac{340}{1440}$

6. **6.28**

Sol.



$$2 \times 2x_0 + x_0 + x_0 = 2x_2 + x$$

$$6x_0 = 2x_2 + x$$

$$x - x_2 = 2x_0$$

$$3x = 10x_0$$

$$x_0 = \frac{3x}{10}$$

$$kx_0 + k2x_0 = k_{eq}x$$

$$3kx_0 = k_{eq}x$$

$$3k \frac{3x}{10} = k_{eq}x$$

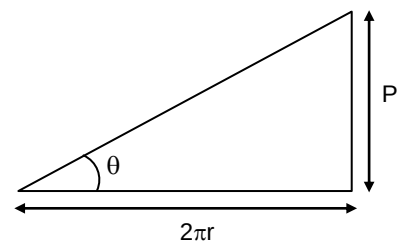
$$k_{eq} = \frac{9k}{10}$$

$$T = 2\pi \sqrt{\frac{10m}{9k}}$$

7. **31.43**

Sol. $\tan \theta = \frac{\rho}{2\pi r} = \frac{g}{r\alpha}$

$$\Rightarrow \alpha = \frac{2\pi g}{\rho}$$

8. **4.04**

Sol. $q_{max} = 7\sqrt{\frac{CL}{3}} = \frac{7}{\sqrt{3}} \text{ mA}$

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SECTION-2 : CHEMISTRY

PART – A

1. **ABC**

Sol. C2-epimer and enantiomer will give osazone having same melting point.

2. **AB**Sol. van der Waal's forces and $\pi - \pi$ stacking.3. **BCD**

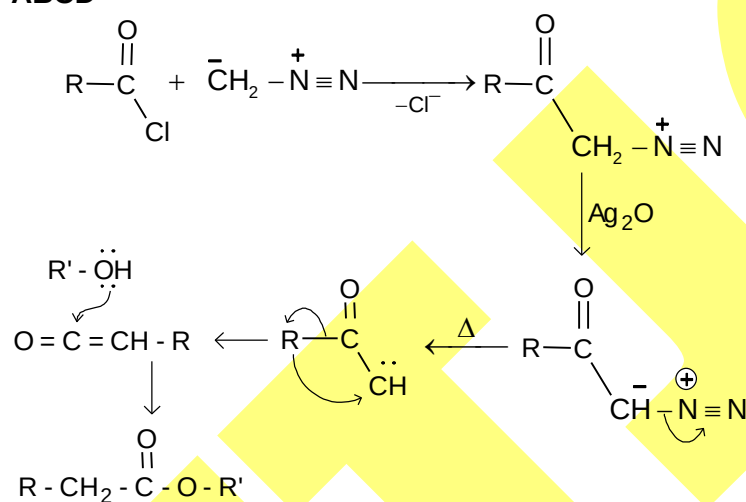
Sol. With phenol and phenol derivatives coupling occurs in weakly basic medium whereas with aniline and its derivatives it requires weakly acidic medium.

4. **ABCD**

Sol. Fact based.

5. **ABCD**

Sol.

6. **ABC**Sol. $\text{B}_2\text{H}_6 + \text{NH}_3 (\text{excess}) \xrightarrow{\text{low temp}} [\text{H}_3\text{N} \rightarrow \text{BH}_2 \leftarrow \text{NH}_3]^+ [\text{BH}_4]^-$ 7. **ABCD**

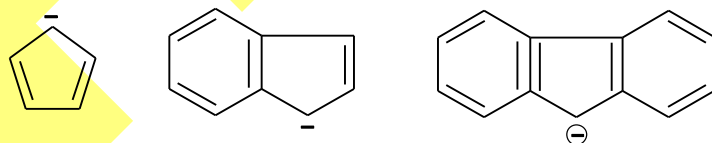
Sol.

8. **C**

Sol. Metal in high oxidation state will show efficient LMCT.

9. **A**

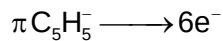
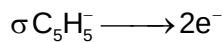
Sol.



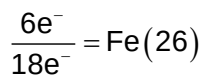
Net aromatic stabilisation decreases if an aromatic ring shares aromatic character with another ring.

10. **D**Sol. $[\text{M}(\text{CO})_2(\sigma\text{C}_2\text{H}_5)(\pi\text{C}_2\text{H}_5)]$

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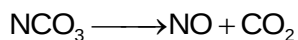
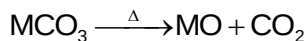


This means M is in +2 oxidation state. We need 6 more electron to have 18 electrons Fe^{2+}



11. **D**

Sol. Let metal are M and N



Carry out the required calculation

12. **C**

Sol. $\Delta_t = \frac{4}{3} \Delta_o$

$$\Rightarrow \Delta_t = 8000 \times \frac{4}{9} \text{cm}^{-1}$$

$$\Delta_t = 8000 \text{cm}^{-1}$$

13. **A**

Sol. $\text{R} \rightarrow 3\text{P}$

$$t = 0 \quad a \quad 0$$

$$t = 5 \quad a - x \quad 3x$$

$$K = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

Now do the required calculation

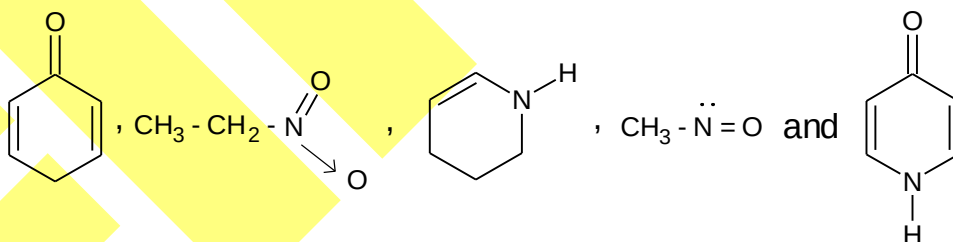
$$a - x = 3x$$

$$x = a/4$$

PART - B

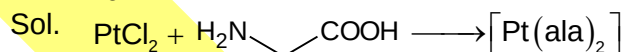
1. Sol.

5

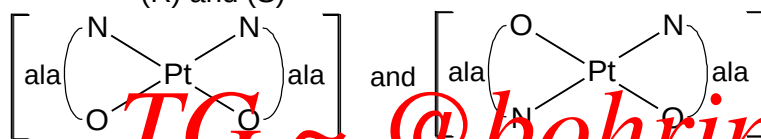


These are the molecules which can show tautomerism.

2. 6



(racemic) $\pm$
(R) and (S)



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3. **2.32**Sol. 20 g H_2SO_4 in 100 g of solution

$$\text{Volume} = \frac{100}{1.14} \text{ mL}$$

$$M = \frac{n_{\text{H}_2\text{SO}_4}}{V(l)}$$

4. 3

Sol. XeO_4 , SiO_2 , CH_3^-

5. 9

Sol. All are correct (Fact based)

6. 8

$$\text{Sol. } K_P = \frac{10^{-3}}{10^{-1}} = 10^{-2}$$

$$\Delta G^\circ = -2.303 RT \log K_P$$

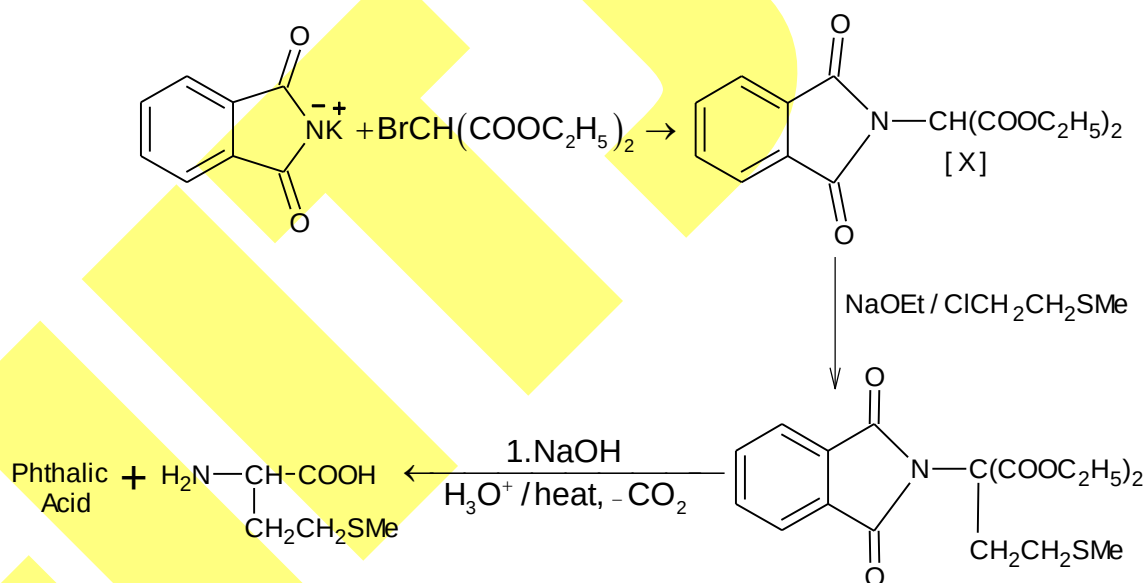
$$= -2.303 RT \log 10^{-2} = (-4)(2 \log 10) = 8$$

7. 6

Sol. Fact based

8. 7

Sol.



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SECTION-3 : MATHEMATICS

PART – A

1. ABD

Sol.

(A) If $|\vec{v} \times \vec{w}| = 0 \forall \vec{w}$ the $\vec{v} = \vec{0}$, otherwise the above relation is not possible $\Rightarrow$ True.(B) Given, $\vec{a} + \vec{b} + \sqrt{3}\vec{c} = \vec{0} \Rightarrow \vec{a} + \vec{b} = -\sqrt{3}\vec{c}$

$$\therefore |\vec{a} + \vec{b}|^2 = 3|\vec{c}|^2 \Rightarrow 2 + 2\cos\frac{\pi}{3} = 3|\vec{c}|^2$$

$$\Rightarrow |\vec{c}|^2 = 1 \Rightarrow |\vec{c}| = 1 \Rightarrow \text{True.}$$

(C) If $(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = 0 \Rightarrow |\vec{u}|^2 - |\vec{v}|^2 = 0 \Rightarrow |\vec{u}| = |\vec{v}|$ But it does not imply that $\vec{u} = \vec{v}$.e.g. Let $\vec{u} = \hat{i}$ and $\vec{v} = \hat{j}$

$$\text{Now, } (\hat{i} + \hat{j}) \cdot (\hat{i} - \hat{j}) = 0$$

But $\hat{i} \neq \hat{j} \Rightarrow \text{False.}$ (D) Given, $|\vec{a}| = 1 = |\vec{b}|$ Let $\theta = \vec{a} \wedge \vec{b}$ where $\theta \in [0, \pi]$

$$\text{Now, } |2\vec{a} + \vec{b}|^2 = 4|\vec{a}|^2 + |\vec{b}|^2 + 4\vec{a} \cdot \vec{b} = 5 + 4(1)(1)\cos\theta = 5 + 4\cos\theta$$

$$\therefore |2\vec{a} + \vec{b}|_{\max}^2 = 9, \text{ when } \cos\theta = 1 \text{ i.e., } \theta = 0^\circ$$

$$\Rightarrow \text{Maximum value of } |2\vec{a} + \vec{b}| = 3.$$

2. BCD

$$\text{Sol. Shortest distance} = \frac{|(\vec{a}_1 - \vec{a}_2) \cdot (\vec{b}_1 - \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\text{where, } \vec{a}_1 = -3\hat{i} + 6\hat{j}, \vec{a}_2 = 2\hat{i} - \hat{j} + 6\hat{k}$$

$$\vec{b}_1 = -4\hat{i} + 3\hat{j} + 2\hat{k}, \vec{b}_2 = -4\hat{i} + \hat{j} + \hat{k}$$

$$\therefore \text{S.D.} = \frac{|(-5\hat{i} + 7\hat{j} - 6\hat{k}) \cdot (\hat{i} - 4\hat{j} + 8\hat{k})|}{|\hat{i} - 4\hat{j} + 8\hat{k}|} = 9$$

Now, verify alternatives.

3. BCD

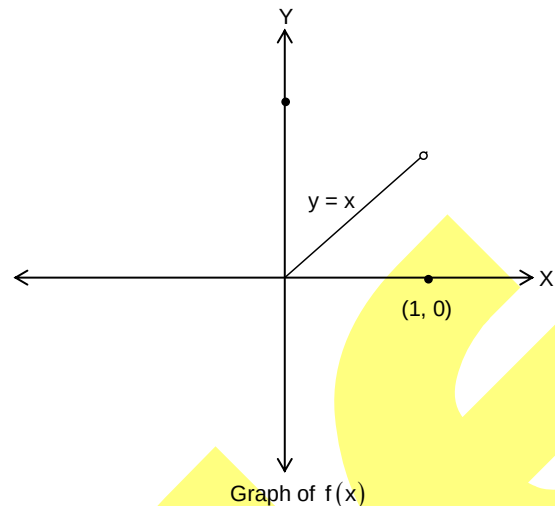
Sol.

$$(A) \text{ False, } f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

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(B) This statement is true.

$$\text{For example: } f(x) = \begin{cases} x & 0 < x < 1 \\ 1 & x = 0 \\ 0 & x = 1 \end{cases}$$



(C) $f(x) = x^2 - \cos \pi x + 4$

Now $f(-1) = 1 - (-1) + 4 = 6$

and $f(2) = 4 - 1 + 4 = 7$

As f is continuous $[-1, 2]$, so $f(c) = 2\pi$, for some $c \in (-1, 2)$

(By using intermediate value theorem)

(D) $(f^{-1})'(4) = \frac{1}{f'(0)} = \frac{1}{1+e^x} \Big|_{x=0} = \frac{1}{2}$

4. AD

Sol. By Intermediate value property

$$\frac{f(0) + f(2)}{2} = f(c), \quad 0 < c < 2$$

By mean value theorem,

$$f(1) - f(0) = f'(c_1), \quad 0 < c_1 < 1$$

$$f(2) - f(1) = f'(c_2), \quad 1 < c_2 < 2$$

By subtraction

$$f(0) + f(2) - 2f(1) = f'(c_2) - f'(c_1)$$

$$= (c_2 - c_1) f''(c), \quad c_1 < c < c_2 \Rightarrow f(0) + f(2) - 2f(1) < 0$$

$$\Rightarrow f(0) + f(2) < 2f(1)$$

5. ABCD

Sol. Given curve is $ax^3 - y + b = 0$

$$\Rightarrow y = ax^3 + b$$

Let point P_1 be $(t_1, at_1^3 + b)$

$$\text{Slope of tangent} = \frac{dy}{dx} \Big|_{P_1} = 3at_1^2$$

$$\therefore \text{Equation of tangent is } y - (at_1^3 + b) = 3at_1^2(x - t_1)$$

$$\therefore \text{Tangent meets curve at } P_2(t_2, at_2^3 + b)$$

$$\therefore (at_2^3 + b) - (at_1^3 + b) = 3at_1^2(t_2 - t_1)$$

$$\Rightarrow a(t_2^3 - t_1^3) = 3at_1^2(t_2 - t_1)$$

$$\Rightarrow t_2^2 + t_1^2 + t_2 t_1 = 3t_1^2 \quad (\because t_1 \neq t_2)$$

$$\Rightarrow t_2^2 - t_2 t_1 - 2t_1^2 = 0$$

$$\Rightarrow (t_2 + 2t_1)(t_2 - t_1) = 0$$

$$\Rightarrow t_2 = -2t_1$$

$$\text{Similarly, } t_3 = -2t_2$$

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$\therefore$ abscissae are in G.P. for all values of a and b .

6. ABCD

Sol. There planes meet at two points it means they have infinitely many solutions, so

$$\text{So, } \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ \alpha & -1 & 3 \end{vmatrix} = 0 \Rightarrow 2(-3+1) - 1(3+1) + \alpha(1+1) = 0 \Rightarrow \alpha = 4$$

$$P_1 : 2x + y + z = 1$$

$$P_2 : x - y + z = 2$$

$$P_3 : 4x - y + 3z = 5$$

P on XOY plane $= (1, -1, 0)$ (which can be obtained by putting $z = 0$ in any two of the given planes).

Q on YOZ plane $= \left(0, \frac{-1}{2}, \frac{3}{2}\right)$ (which can be obtained by putting $x = 0$ in any two of the given planes)

$\therefore$ Straight line perpendicular to plane P_3 passing through P is —

$$\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$$

$$\overrightarrow{PQ} = \hat{i} - \frac{1}{2}\hat{j} - \frac{3}{2}\hat{k}$$

$$\text{Projection of } \overrightarrow{PQ} \text{ on } x\text{-axis} \Rightarrow \left| \frac{\overrightarrow{OP} \cdot \hat{i}}{|\hat{i}|} \right| = 1$$

$$\text{Centroid of } \triangle OPQ \text{ is } \left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$$

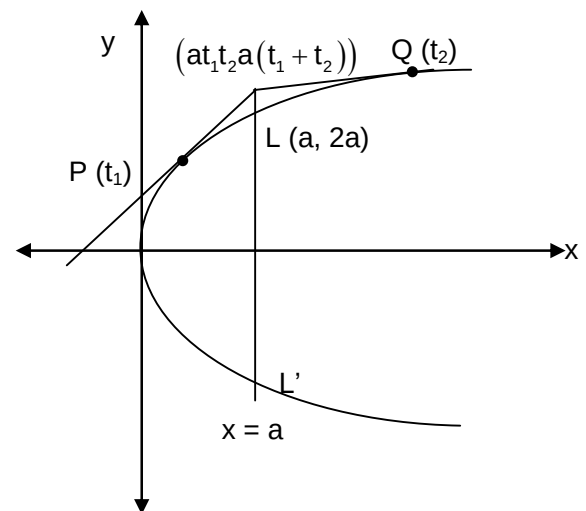
7. ABC

Sol.

(A) Let the point be $\left(\frac{a}{e}, \beta\right)$.

Chord of contact is $T = 0$

i.e., $x\left(\frac{a}{e}\right) + y\beta = a^2$, which is passing through $(ae, 0)$



(B) $T(at_1t_2, a(t_1+t_2)) \Rightarrow at_1t_2a \Rightarrow t_1t_2 = 1$

[Hence, equation of PQ is $2x - (t_1+t_2)y + 2a = 0$

$$(x+a) - \lambda y = 0]$$

$\Rightarrow$ chord PQ passes $(-a, 0)$ which is foot of directrix

$\therefore$ (B) is also true.

(C) As $|PS_1 - PS_2| = \text{constant}$

So, for hyperbola

$$0 < k^2 + 1 < 5 \Rightarrow k^2 < 4 \Rightarrow -2 < k < 2$$

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So, number of integral values of k are 3 (i.e. $k = -1, 0, 1$)

8. C

Sol. Let equation of circumcircle of triangle ABC is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots\dots\dots(1)$$

Any point on the parabola is of the form $(2at, at^2)$,

Solving with (1),

$$a^2t^4 + 2t^2a(f + 2a) + 4gat + c = 0.$$

$$\sum t_1 = 0, \sum t_1t_2 = \frac{2}{a}(f + 2a), \sum t_1t_2t_3 = \frac{-4g}{a}, t_1t_2t_3t_4 = \frac{c}{a^2}$$

Since centroid and circumcentre coincides so

$$-g = \frac{2a(t_1 + t_2 + t_3)}{3} = -\frac{2a}{3}t_4 = \frac{3g}{2a} \quad \dots\dots\dots(2)$$

$$-f = \frac{a}{3}(t_1^2 + t_2^2 + t_3^2) = \frac{a}{3}(t_1^2 + t_2^2 + t_3^2 + t_4^2 - t_4^2) = \frac{a}{3}((\sum t_1)^2 - 2\sum t_1t_2 - t_4^2)$$

$$\Rightarrow -f = \frac{a}{3}\left(0 - \frac{4}{a}(f + 2a) - t_4^2\right)$$

Putting the value of t_4 from (2),

The required locus is $9x^2 = 4a(y - 8a)$.

9. B

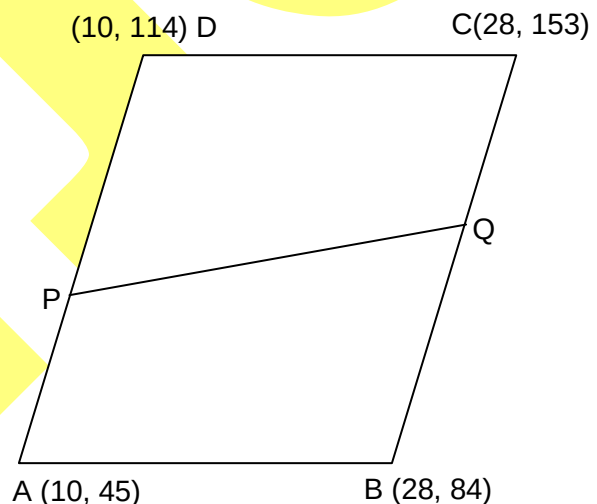
Sol. Let $AP = CQ = x$

$$\Rightarrow \frac{45 + x}{10} = \frac{153 - x}{28}$$

$$\Rightarrow 1260 + 28x = 1530 - 10x$$

$$\Rightarrow x = \frac{135}{19}$$

$$\Rightarrow \text{slope of PQ} = \frac{45 + \frac{135}{19}}{10} = \frac{99}{19}.$$



10. B

Sol. Any line passing through $(2, 2)$ will be of the form $\frac{y-2}{\sin\theta} = \frac{x-2}{\cos\theta} = r$

When this line cuts the circle $x^2 + y^2 = 2$, $(r\cos\theta + 2)^2 + (r\sin\theta + 2)^2 = 2$

$$\Rightarrow r^2 + 4(\sin\theta + \cos\theta)r + 6 = 0$$

$$\frac{PB}{PA} = \frac{r_2}{r_1}, \text{ now if } r_1 = \theta, r_2 = 3\alpha,$$

$$\text{then } 4\alpha = -4(\sin\theta + \cos\theta), 3\alpha^2 = 6 \Rightarrow \sin 2\theta = 1 \Rightarrow \theta = \frac{r}{4}.$$

So required chord will be $y - 2 = 1(x - 2) \Rightarrow y = x$.

11. B

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Sol. Problem is same as arranging 8 things out of which 5 identical i.e. $\frac{8!}{5!}$ which gives total number of ways of selecting block and distributing them away 3 children i.e. $\frac{8!}{5!} 3!$.

12. B

Sol. $ab + bc + ca = \frac{1}{2}((a+b+c)^2 - (a^2 + b^2 + c^2)) = 183$
 $\Rightarrow a, b, c$ are the roots of $x^3 - 24x^2 + 183x - 440 = 0$
 but it factorises as $(x-5)(x-8)(x-11) = 0$
 $\Rightarrow$ there are 3! solutions.

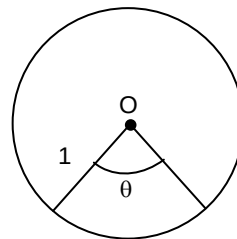
13. B

Sol. Let $I = \int_0^{\sin^{-1}\frac{\pi}{4}} (\pi - 4\sin\theta) \sin 2\theta \ln(1 + \tan(\sin\theta)) d\theta$
 put $\sin\theta = t, \cos\theta d\theta = dt$
 $I = 2 \int_0^{\pi/4} (\pi - 4t) t \ln(1 + \tan t) dt$
 $= 2 \int_0^{\pi/4} \left(\pi - 4\left(\frac{\pi}{4} - t\right) \right) \left(\frac{\pi}{4} - t \right) \ln\left(1 + \tan\left(\frac{\pi}{4} - t\right)\right) dt$
 $= 2 \int_0^{\pi/4} t(\pi - 4t) \ln 2 dt - \int_0^{\pi/4} t(\pi - 4t) \ln(1 + \tan t) dt$
 $I = \ln 2 \left[\frac{\pi}{2} \cdot \frac{\pi^2}{16} - \frac{4}{3} \times \frac{\pi^3}{64} \right] = \frac{\pi^3}{32} \ln 2 \left[1 - \frac{2}{3} \right] = \frac{\pi^3 \ln 2}{96} = k$
 $\frac{960k}{\pi^3 \ln 2} = 10.$

PART - B

1. 5.45

Sol. $\frac{A_2}{A_1} = \frac{\pi r^2}{\left(\frac{2\pi - \theta}{2\pi}\right) \cdot \pi r^2} = \frac{2\pi}{2\pi - \theta}$
 $V = \frac{\pi}{3} \left(\frac{\theta}{2\pi}\right)^2 \sqrt{1^2 - \left(\frac{\theta}{2\pi}\right)^2}$
 $\frac{dV}{d\theta} = 0 \Rightarrow \theta = \sqrt{\frac{8}{3}} \pi$
 $\frac{A_2}{A_1} = \frac{\sqrt{3}}{\sqrt{3} - \sqrt{2}} = 3 + \sqrt{6}$



2. 35.00

Sol. Let x_1 be the number of elements in B before $f(3)$, x_2 the number of elements between $f(3)$ and $f(4)$, x_3 the number of elements between $f(4)$ and $f(5)$, x_4 the number of elements between $f(5)$ and $f(6)$, and x_5 the number of elements after $f(6)$.

According to the conditions of the problem, $x_1, x_3, x_4 \geq 1$ and $x_2, x_5 \geq 0$.

$$x_1 + x_2 + x_3 + x_4 + x_5 = 6$$

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$$N = {}^7C_4$$

3. 1.00

Sol. It is obvious

$$\frac{da}{dx} = c, \frac{db}{dx} = a, \frac{dc}{dx} = b$$

$$\text{Let } f(x) = a^3 + b^3 + c^3 - 3abc$$

Differentiate with respect to x on both sides

$$\begin{aligned} f'(x) &= 3a^2 \frac{da}{dx} + 3b^2 \frac{db}{dx} + 3c \frac{dc}{dx} - 3 \left(\frac{da}{dx} bc + \frac{db}{dx} ac + \frac{dc}{dx} ab \right) \\ &= 3a^2 c + 3b^2 a + 3c^2 b - 3(bc^2 + ca^2 + ab^2) \end{aligned}$$

$$f'(x) = 0 \Rightarrow f(x) = \lambda \text{ for every } x$$

$$\text{Now, } x = 0 \Rightarrow f(0) = 1 + 0 + 0 - 0 = 1$$

$$f(x) = 1$$

4. 630.00

$$\text{Sol. } n(S) = 6.6.6.6 = 6^4 = 1296$$

Now A can throw die in 6 ways

B can throw only in 5 ways

D can also throw in 5 ways

Suppose C throws the same number as that of A

Hence 'C' can throw only in 1 ways.

$$\therefore \text{Number of ways} = 6 \cdot 5 \cdot 5 \cdot 1 = 150$$

Suppose C throws different number than A

Then A can throw in 6 ways

B can throw in 5 ways

Now C will throw different from A and B hence C can throw in 4 ways

Now D can not throw what A and C have thrown and therefore can throw in 4 ways.

$$\therefore \text{Number of ways} = 6 \cdot 5 \cdot 4 \cdot 4 = 480$$

$$\text{Total ways} = 630$$

5. 72.00

$$\text{Sol. We have } (\overline{OP})^2 = (e^t + e^{-t})^2 (\vec{a})^2 + (e^t - e^{-t})^2 (\vec{b})^2 + 2(e^t + e^{-t})(e^t - e^{-t})(\vec{a} \cdot \vec{b})$$

$$\left((\vec{a})^2 = |\vec{a}|^2 = 9, (\vec{b})^2 = |\vec{b}|^2 = 16 \text{ and } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{2\pi}{3} \right)$$

$$\Rightarrow |\overline{OP}|^2 = 9(e^t + e^{-t})^2 + 16(e^t - e^{-t})^2 + 2(e^{2t} - e^{-2t}) \cdot 3 \cdot 4 \cdot \left(-\frac{1}{2}\right) = 13e^{2t} + 5e^{-2t} - 14$$

$$\text{Now } \frac{d}{dt} |\overline{OP}|^2 = 0 \Rightarrow 26e^{2t} - 10e^{-2t} = 0 \Rightarrow e^{4t} = \frac{5}{13},$$

$$\text{So } e^{2t} = \frac{\sqrt{5}}{\sqrt{13}}$$

$$\therefore |\overline{OP}|_{\min}^2 = 13 \left(\frac{\sqrt{5}}{\sqrt{13}} \right) + 5 \left(\frac{\sqrt{13}}{\sqrt{5}} \right) - 14 \Rightarrow |\overline{OP}|_{\min}^2 = 2\sqrt{65} - 14$$

$$\Rightarrow |\overline{OP}|_{\min} = \sqrt{2\sqrt{65} - 7} = \sqrt{2}\sqrt{\sqrt{a} - b},$$

$$\text{So } a = 65, b = 7$$

$$\text{Hence } (a+b) = 72$$

6. 9

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Sol. Apply Total Probability

7. 610.66

Sol. Given $y = \tan z$

$$\frac{dy}{dx} = \sec^2 z \cdot \frac{dz}{dx} \quad (1)$$

$$\text{Now } \frac{d^2y}{dx^2} = \sec^2 z \cdot \frac{d^2z}{dx^2} + \frac{dz}{dx} \cdot \frac{d}{dx}(\sec^2 z) \quad [\text{Using product rule}]$$

$$= \sec^2 z \cdot \frac{d^2z}{dx^2} + \frac{dz}{dx} \cdot \frac{d}{dz}(\sec^2 z) \frac{dz}{dx}$$

$$\frac{d^2y}{dx^2} = \sec^2 z \cdot \frac{d^2z}{dx^2} + \left(\frac{dz}{dx}\right)^2 \cdot 2 \sec^2 z \cdot \tan z \quad (2)$$

$$\text{Now } 1 + \frac{2(1+y)}{1+y^2} \left(\frac{dy}{dx}\right)^2$$

$$= 1 + \frac{2(1+\tan z)}{\sec^2 z} \cdot \sec^4 z \cdot \left(\frac{dz}{dx}\right)^2$$

$$= 1 + 2(1+\tan z) \cdot \sec^2 z \cdot \left(\frac{dz}{dx}\right)^2$$

$$= 1 + 2\sec^2 z \left(\frac{dz}{dx}\right)^2 + 2\tan z \cdot \sec^2 z \left(\frac{dz}{dx}\right)^2 \quad (3)$$

From (2) and (3), we have RHS of (2) = RHS of (3)

$$\sec^2 z \cdot \frac{d^2z}{dx^2} = 1 + 2\sec^2 z \left(\frac{dz}{dx}\right)^2$$

$$\Rightarrow \frac{d^2z}{dx^2} = \cos^2 z + 2\left(\frac{dz}{dx}\right)^2$$

$$\Rightarrow k = 2 \text{ so } (305.33) \times k = 610.66$$

8. 8275.20

Sol. $(AB)(AD) = 200$

$$AB + AD = 2a$$

$$BD^2 = AB^2 + AD^2 = (2ae)^2$$

$$(AB + AD)^2 = AB^2 + AD^2 + 2(AB)(AD)$$

$$\Rightarrow 4a^2 = 4a^2e^2 + 400$$

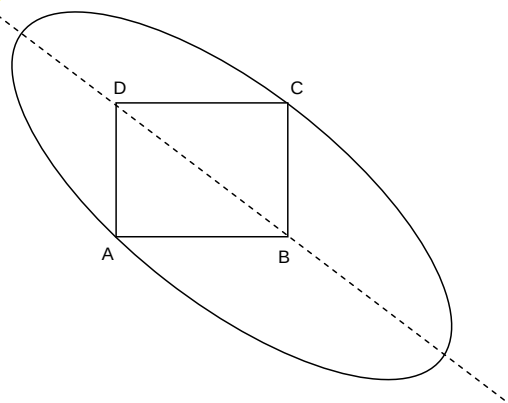
$$\Rightarrow b = 10$$

$$\Rightarrow a^2(1 - e^2) = b^2 = 100$$

$$\Rightarrow a = 20$$

$$P = 2(AB + AD) = 4a = 80$$

$$\text{Or } 103.44P = 8275.20$$



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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 2

QP Code: 100163

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 195

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+4 marks** for the correct answer and **– 2 marks** for wrong answer.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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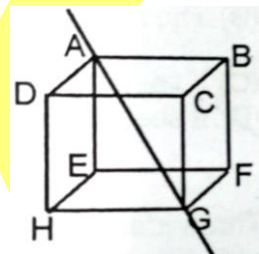
SECTION-1 : PHYSICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. An infinite line charge carrying a uniform line charge of density λ is placed along the body diagonal AG of a cube of side 'a' as shown. Then



- (A) The flux of the electric field of this charge through one of the sides

of the cube is $\frac{\lambda a}{2\sqrt{3}\epsilon_0}$

- (B) The flux of the electric field of this charge through one of the sides

of the cube is $\frac{\sqrt{3}\lambda a}{2\epsilon_0}$

- (C) The flux of the electric field of this charge through one of the sides of the cube is $\frac{\lambda a}{4\sqrt{3}\epsilon_0}$

- (D) The flux of the electric field through the face ABFE is same as that through BCGF.

2. Four waves of the equation given by

$$y_1 = 5A \sin (wt - Kx + \pi/2)$$

$$y_2 = 2A \sin (wt - Kx + 3\pi/2)$$

$$y_3 = 6A \sin (wt - Kx)$$

and $y_4 = 2A \sin (wt - Kx + \pi)$

- (A) The resulted wave will have amplitude 5A.

- (B) It will make a phase difference of 37° with y_3 .

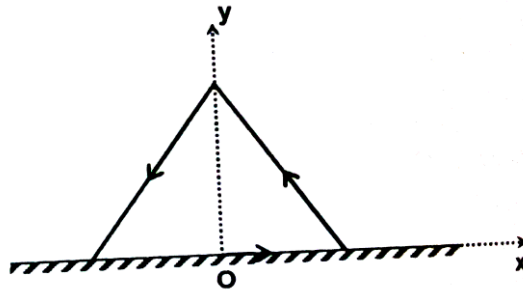
- (C) Its equation will be $y = 5A \sin (wt + 37^\circ)$.

- (D) Its equation will be $y = 5A \sin (wt - Kx + 37^\circ)$.

Space For Rough Work

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3. A conducting frame in the shape of an equilateral triangle (mass m , side a) carrying a current I is placed vertically on a horizontal rough surface (coefficient of friction is μ). A magnetic field exists such that $\vec{B} = -B_0 y \hat{i}$. Then

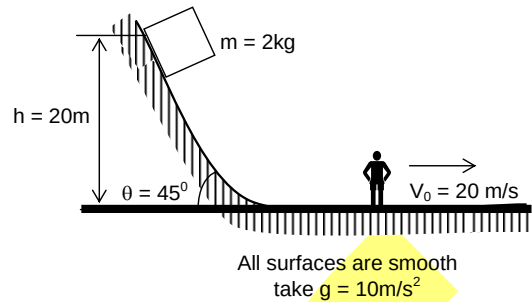


- (A) The maximum value of B_0 so that the frame does not rotate $\frac{2\mu mg}{la^2}$
- (B) The maximum value of B_0 so that the frame does not rotate $\frac{\mu mg}{la^2}$
- (C) If seen from the top, the frame will have a tendency to rotate counter clockwise.
- (D) If seen from the top, the frame will have a tendency to rotate clockwise.
4. A source is moving across a circle given by the equation $x^2 + y^2 = R^2$, with constant speed $v = \frac{330\pi}{6\sqrt{3}}$ m/s, in anti-clockwise sense. A detector is at rest at point $(2R, 0)$ w.r.t. the centre of the circle. If the frequency emitted by the source is f and the speed of sound, $C = 330$ m/s. Then
- (A) the position of the source when the detector records the maximum frequency $\left(+\frac{\sqrt{3}}{2}R, -\frac{R}{2} \right)$
- (B) the co-ordinate of the source when the detector records minimum frequency is $(0, R)$
- (C) the maximum frequency recorded by the detector is $\frac{6\sqrt{3}}{\pi + 6\sqrt{3}} f$
- (D) the minimum frequency recorded by the detector is $\frac{6\sqrt{3}}{6\sqrt{3} - \pi} f$

Space For Rough Work

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5. A block of mass m is released from top of an inclined plane of inclination $\theta = 45^\circ$ as shown in the figure. An observer is moving on the horizontal floor with constant velocity 20 m/s as shown, when the block reaches to horizontal floor.

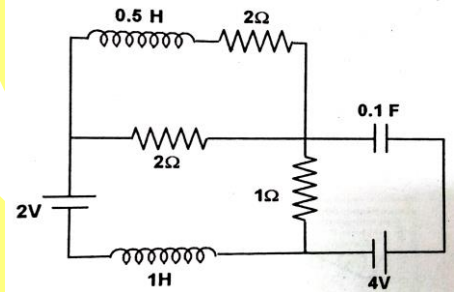


- (A) Velocity observed by observer will be 20 m/s
 (B) Change in kinetic energy of the block will be 400 J as observed by observer
 (C) Work energy theorem is not applicable in moving observer frame
 (D) Work done by normal reaction is -800 J as observed by the observer

6. The kinetic energy of particle continuously increases with time :
 (A) The resultant force on the particle must be parallel to the velocity at all the instant
 (B) The resultant force on the particle must be at an angle less than 90° all the times
 (C) Its height above the ground level must continuously decrease
 (D) The magnitude of its linear momentum increases continuously

7. The circuit shown in the figure has reached its steady state.

- (A) The charge on the capacitor is 0.1 C .
 (B) The charge on the capacitor is 0.3 C .
 (C) Power developed by the 2 Volts battery is 1 W .
 (D) Power developed by the 2 Volts battery is 2 W .

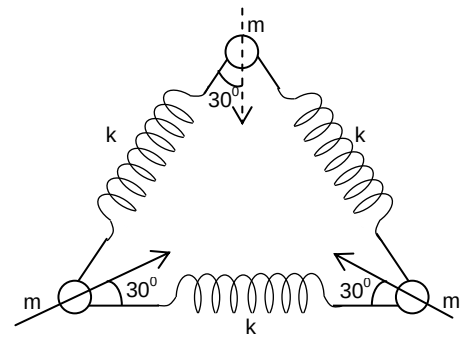


(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. Three identical point masses are placed at the vertices of an equilateral triangle and joined through identical spring of spring constant K . The system is placed on a smooth table. If the masses are displaced along line as shown in figure, then period of their oscillation will be

- (A) $2\pi\sqrt{\frac{m}{k}}$ (B) $2\pi\sqrt{\frac{m}{3k}}$
 (C) $2\pi\sqrt{\frac{m}{2k}}$ (D) $2\pi\sqrt{\frac{m}{5k}}$



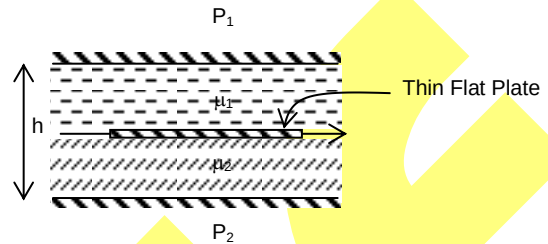
Space For Rough Work

TG ~ @bohring_bot

9. A chain of length ℓ and mass M is lumped over the hole in a horizontal plate and a short length hangs down the hole when the chain is at rest. If it is released from this position, find the velocity when the last link of the chain leaves the hole

(A) $\sqrt{\frac{3}{2}g\ell}$ (B) $\sqrt{\frac{2}{4}g\ell}$ (C) $\sqrt{\frac{2}{3}g\ell}$ (D) $\sqrt{\frac{3}{4}g\ell}$

10. A thin plate is placed between two flat surfaces, h cm apart such that the viscosity of liquids on top and bottom of the plate are μ_1 and μ_2 respectively. Find the distance of the plate from the upper surface P_1 such that the viscous resistance to uniform motion of the thin plate is minimum, [Assume h to be very small] is



(A) $y = \frac{\sqrt{\mu_2} h}{\sqrt{\mu_1} + \sqrt{\mu_2}}$ (B) $y = \frac{\sqrt{\mu_1} h}{\sqrt{\mu_1} + \sqrt{\mu_2}}$
 (C) $y = \frac{\sqrt{\mu_1} h}{\sqrt{\mu_1} - \sqrt{\mu_2}}$ (D) $y = \frac{\sqrt{\mu_2} h}{\sqrt{\mu_1} - \sqrt{\mu_2}}$

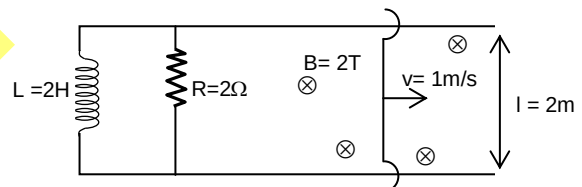
11. An a. c. source producing emf $V = V_0 [\sin \omega t + \cos 2\omega t]$ is connected in series with a capacitor and a resistor. The current found in the circuit is $i = i_1 \sin (\omega t + \phi_1) + i_2 \cos (2\omega t + \phi_2)$. Then

- (A) $i_1 = i_2$
 (B) $i_1 < i_2$
 (C) $i_1 > i_2$
 (D) i_1 may be less than, equal to or greater than i_2

12. Two long straight wires with equal cross-sectional radii a are located parallel to each other in air at a large distance b from each other. The capacitance per unit length of the wires will be

(A) $\frac{\pi\epsilon_0}{\ln\left(\frac{a}{b-a}\right)}$ (B) $\frac{\pi\epsilon_0}{\ln\left(\frac{b-a}{a}\right)}$ (C) $\frac{2\pi\epsilon_0}{\ln\left(\frac{b-a}{a}\right)}$ (D) $\frac{4\pi\epsilon_0}{\ln\left(\frac{b-a}{a}\right)}$

13. The given figure shows an inductor and resistance fixed on a conducting wire. A movable conducting wire PQ starts moving on the fixed rails from $t = 0$ with constant velocity 1 m/s . The work done by the external force on the wire PQ in 2 seconds is



- (A) 16 J (B) 32 J
 (C) 48 J (D) 64 J

Space For Rough Work

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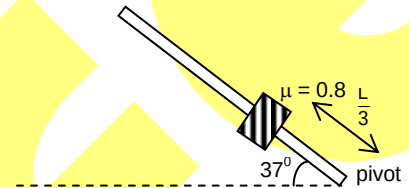
14. A hemisphere of mass $3m$ and radius R is free to slide with its base on a smooth horizontal table. A particle of mass m is placed on the top of the hemisphere. If particle is displaced with a negligible velocity, then find the angular velocity of the particle relative to the centre of the hemisphere at an angular displacement θ , when velocity of hemisphere is v .
- (A) $\frac{4v}{R \cos \theta}$ (B) $\frac{3v}{R \cos \theta}$ (C) $\frac{5v}{R \cos \theta}$ (D) $\frac{2v}{R \cos \theta}$

(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

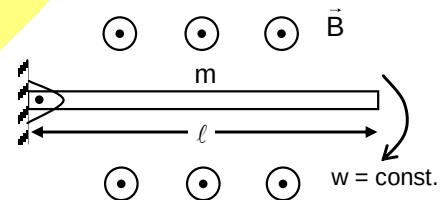
A cylindrical rod of mass m and length L is pivoted about a horizontal axis through one end and released from rest at an angle 37° with the horizontal. A small ring of same mass is kept on the rod as shown in figure



15. Angular velocity of rod about the pivot when it passes through horizontal position is
- (A) $\sqrt{\frac{9g}{4L}}$ (B) $\sqrt{\frac{4g}{3L}}$ (C) $\sqrt{\frac{12g}{7L}}$ (D) $\sqrt{\frac{2g}{3L}}$
16. Tangential acceleration of CM at the instant it passes through horizontal position is
- (A) $\frac{42g}{7}$ (B) $\frac{6g}{7}$ (C) $\frac{25g}{32}$ (D) $\frac{7g}{16}$

Paragraph for Question no. 17 to 18

A uniformly charged non-conducting rod of mass M and linear charge density λ (C/m) is rotating in horizontal plane in gravity free space with constant angular velocity w rad/sec. The rod is hinged at one end as shown in figure. A uniform magnetic field $\vec{B}$ exists in the region along vertically upward direction. Area of cross-section of rod is A .



17. Find out net force applied by hinge on the rod (when $W > \frac{B\lambda\ell}{m}$)
- (A) $\frac{mw^2\ell}{2} - \frac{B\lambda w\ell^2}{2}$ (B) 0 (C) $\frac{B\lambda w\ell^2}{2}$ (D) $\frac{mw^2\ell}{2}$
18. Tension in the rod at the mid point of rod. When $W = \frac{B\lambda\ell}{M}$
- (A) $\frac{B\lambda w\ell^2}{8A}$ (B) $\frac{W\ell}{4A}(B\lambda\ell - mw)$ (C) $\frac{mw^2\ell}{4A}$ (D) 0

Space For Rough Work

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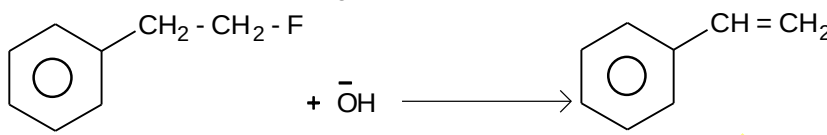
SECTION-2 : CHEMISTRY

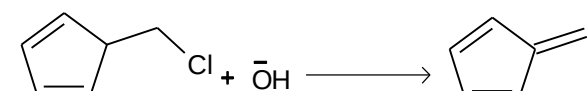
PART – A

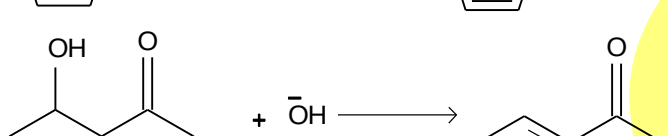
(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- Which is/are example involve E_{CB}^1 as major mechanism?

(A) 

(B) 

(C) 

(D) $CH_3CH_2Cl + ^-OH \longrightarrow CH_2 = CH_2$
- Which of the following statement(s) is/are incorrect if temperature is increased?

(A) pK_w increases (B) K_w increases
(C) pH of pure water increases (D) pOH of pure water increases
- Which of the following is/are correct option?

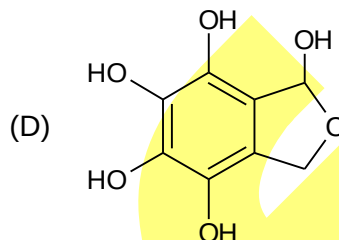
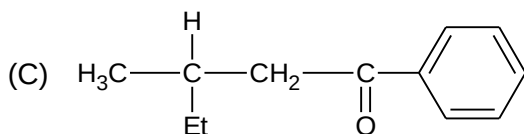
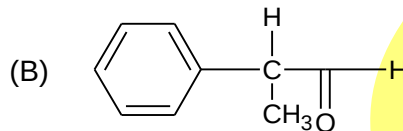
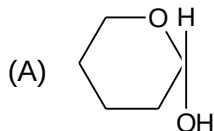
(A) Tetrapeptide represents, 4 peptide bond in a molecule of polypeptide.
(B) Glucose & Fructose can be differentiated by Tollen's reagent & Fehling's solution.
(C) One mole of glucose reacts with maximum of 3 mole of phenylhydrazine.
(D) Trehalose(also known as mycose) is the disaccharide which is non-reducing in nature.
- Which of the following statement(s) is/are false?

(A) All adiabatic processes are isentropic(or isentropic) process
(B) When $(\Delta G_{system})_{T, P} < 0$ the reaction must be exothermic
(C) $dG = VdP - SdT$ is applicable for closed system, both PV and non-PV work
(D) The heat of vaporization of water at $100^\circ C$ is 40.6 kJ.mol . When 9 g of water vapour condenses to liquid at $100^\circ C$ and 1 atm, then system $\Delta S_{system} = 54.42 \text{ J/K}$

Space For Rough Work

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5. Which is true about Scandium compound?
 (A) The Scandium carbide(ScC_2), reacts with water, liberating ethyne
 (B) Oxidation number of Scandium is +3 in Scandium carbide
 (C) Anhydrous ScCl_3 differs from AlCl_3 as ScCl_3 is monomeric while $(\text{AlCl}_3)_2$ is dimeric
 (D) $2\text{ScCl}_3 \cdot 7\text{H}_2\text{O}$ on heating gives anhydrous ScCl_3
6. In which of the following compounds, there will be either loss of optical activity or change of optical rotation, while keeping in very dilute acidic or basic aqueous solution?



7. Select the false statement(s)?
 (A) Brownian motion and Tyndall effect are shown by true solution
 (B) Sorption process is combinations of adsorption and absorption process
 (C) Micelle formation is mainly enthalpy driven process at normal temperature
 (D) Higher is the gold number greater will be the protective power of a colloid

(Single Correct Choice Type)

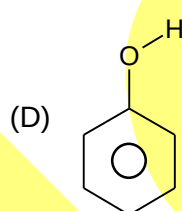
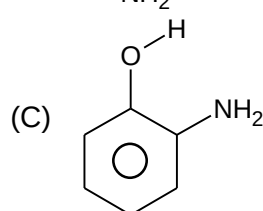
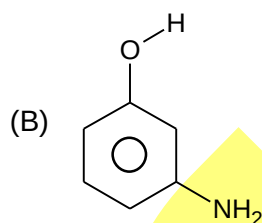
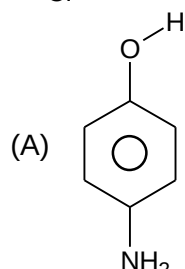
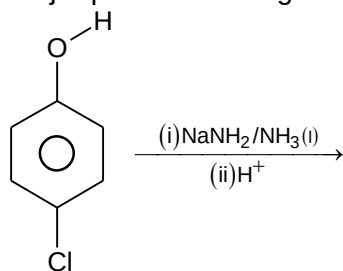
This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. Correct order for relative rate of hydrolysis
- | | | |
|-----------------------------------|---|---|
| $\text{CH}_3\text{CH}_2\text{Cl}$ | $\text{CH}_3\text{OCH}_2\text{CH}_2\text{Cl}$ | $\text{CH}_3\text{SCH}_2\text{CH}_2\text{Cl}$ |
| I | II | III |
- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$
 (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{I} > \text{III} > \text{II}$

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9. Major product in the given reaction is:



10. Cyclohexene reacts with limited amount of bromine in the presence of light to form product X ($\text{C}_6\text{H}_9\text{Br}$). The statement correct about X is

- (A) It is racemate
(B) It is a product of an addition reaction
(C) It is formed through a cationic intermediate
(D) It is optically active

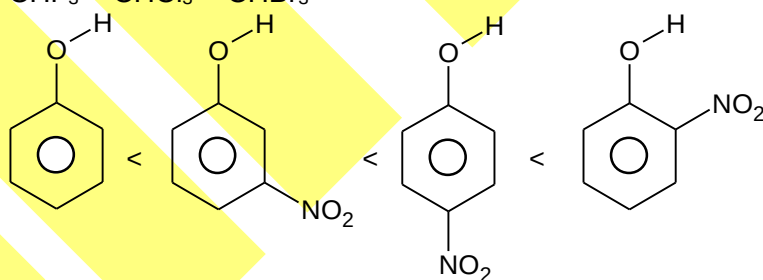
11. Which is incorrect order of acidic strength?

(A)



(B) $\text{CHF}_3 < \text{CHCl}_3 < \text{CHBr}_3$

(C)



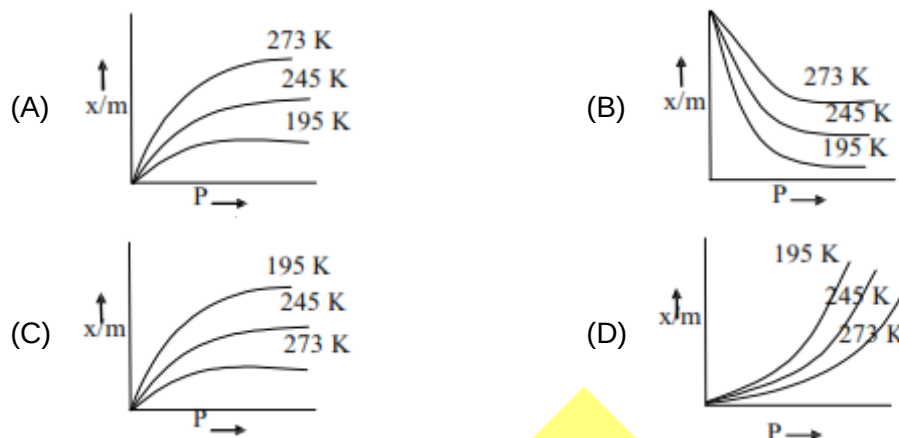
(D) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3 < \text{BI}_3$

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12. When an apple is cut, the exposed part begins to turn brown. Often the browning action can be arrested by adding a few drops of lemon juice to the exposed area. The basis for this treatment is
- (A) lemon juice is an antioxidant
(B) denaturation of proteins of the enzymes
(C) decolourization due to lemon juice
(D) activation of decolourizing enzymes under acidic medium

13. Freundlich adsorption isotherms are properly represented as in



14. A nonapeptide in rat on hydrolysis gave the following identifiable tripeptides: Gly-Ala-Phe, Ala-Leu-Val, Gly-Ala-Leu, Phe-Glu-His, and His-Gly-Ala. The sequence in the nonapeptide is
- (A) Gly-Ala-Leu-Val-Phe-Glu-His-His-Gly
(B) Ala-Phe-Leu-Val-Gly-Leu-Phe-Glu-His
(C) Gly-Ala-Phe-Glu-His-Gly-Ala-Leu-Val
(D) Phe-Ala-Leu-Val-Gly-Glu-His-Gly-Ala

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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

P_4O_{10} is one of very important phosphoric oxide having oxidation state of phosphorus as (+V). P_4O_{10} is having cyclic structure P_4O_{10} on hydrolysis consume 6 moles of H_2O in step wise manner involving different intermediate.

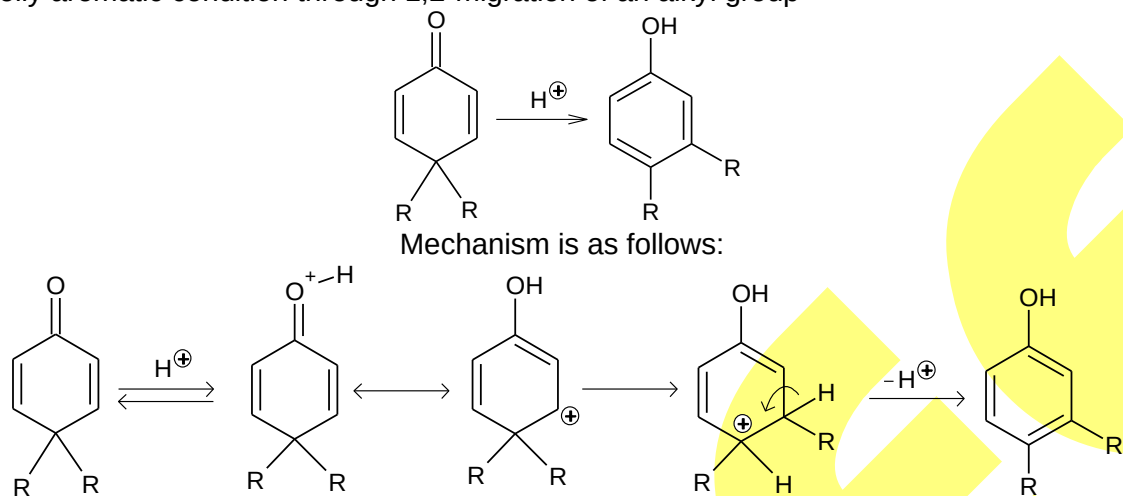
15. P_4O_{10} (1 mole) when reacts with 2 moles of H_2O gives
(A) dibasic acid (B) tribasic acid
(C) tetrabasic acid (D) monobasic acid
16. P_4O_{10} (1 mole) when reacts with 3 moles of H_2O gives
(A) dibasic acid (B) tetrabasic acid
(C) hexabasic acid (D) octabasic acid

Space For Rough Work

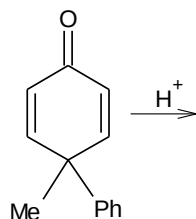
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Paragraph for Question no. 17 to 18

A rearrangement essentially, reversal of the pinacol/pinacolone change, a retro pinacol reaction, is the dienone-phenol rearrangement in which protonation of the initial dienone allows reattainment of the wholly aromatic condition through 1,2-migration of an alkyl group



17. In the reaction given below major product will be:

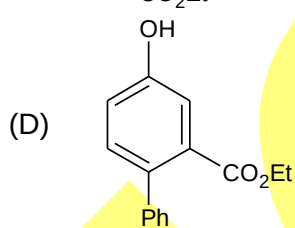
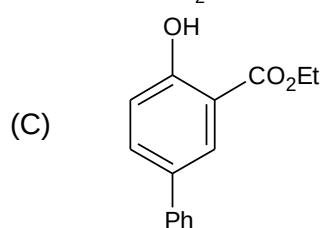
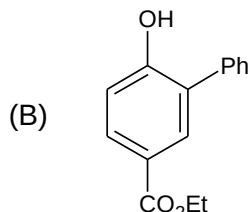
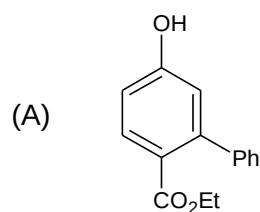
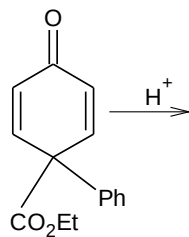


- (A)
- (B)
- (C)
- (D)

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18. In the reaction given below major product will be:



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SECTION-3 : MATHEMATICS**PART – A****(Multi Correct Choice Type)**

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- A function $y = f(x)$ satisfies the following conditions
(i) $f(1) = 1$
(ii) The tangent at any point of the graph of $f(x)$ cuts off on the y – axis an intercept equal to the square of the abscissa of the point of tangency.
Which of the following are correct?
(A) The graph of $y = f(x)$ passes through the origin
(B) $y = f(x)$ denotes a conic where coordinates of focus are $\left(1, \frac{3}{4}\right)$
(C) The area enclosed by $f(x)$, and the line $y + 3 = 0$ is $\frac{32}{3}$.
(D) $y = f(x)$ has no maxima
- The line of intersection of the plane $\vec{r} \cdot (3\hat{i} - \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} + 4\hat{j} - 2\hat{k}) = 2$ is parallel to the vector
(A) $2\hat{i} + 7\hat{j} + 13\hat{k}$ (B) $-2\hat{i} + 7\hat{j} + 13\hat{k}$
(C) $-2\hat{i} - 7\hat{j} + 13\hat{k}$ (D) $2\hat{i} - 7\hat{j} - 13\hat{k}$
- In the expansion of $(2x - 3y + z)^{10}$, then which of the following statement(s) is/are **correct**?
(A) The expansion will contain 66 terms
(B) Sum of all the coefficients is 0
(C) The coefficient of $x^2y^3z^5$ equals $\left(\frac{-10! \times 9}{5!}\right)$
(D) None of these

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4. Let $f(x) = (x^2 + 2x + 3)^2 + 2(x^2 + 2x + 3) + 3$, then which of the following statement (s) is (are) correct?
 (A) The equation $f(x) = 0$ has no real roots.
 (B) The equation $f(x) = 0$ has two real roots and two imaginary roots.
 (C) The minimum value of $f(x)$ equals 11
 (D) The minimum value of $f(x)$ equals 12
5. If ${}^{100}C_6 + 4{}^{100}C_7 + 6{}^{100}C_8 + 4{}^{100}C_9 + {}^{100}C_{10}$ has the value equal to xC_y , then the value of $(x+y)$ can be
 (A) 114 (B) 115 (C) 198 (D) 199
6. Which of the following is (are) correct?
 (A) $\lim_{n \rightarrow \infty} \left(\frac{2n^2 - 3}{2n^2 - n + 1} \right)^{\frac{n^2 - 1}{n}}$ is equal to $\sqrt{e}$.
 (B) $\lim_{x \rightarrow \infty} x \left(\frac{1}{e} - \left(\frac{x}{x+1} \right)^x \right)$ equal to $\frac{-1}{2e}$.
 (C) The coefficient of x^{50} in $(1+x^2)^{25} (1+x^{25}) (1+x^{40}) (1+x^{45}) (1+x^{47})$ is equal to $1 + {}^{25}C_5$.
 (D) If A, B are non – singular and symmetric matrices such that $AB = BA$ then $A^{-1}B^{-1}$ is symmetric matrix.
7. Let the complex number $z = x + iy$ (where $x, y \in \mathbb{R}$) satisfy $\cot^{-1}(\log_3 |2z + 1|) > \cot^{-1}(\log_3 |2z - 1|)$, then x can be
 (A) -1 (B) -2 (C) 1 (D) 2

(Single Correct Choice Type)

This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

8. Consider two concentric circle $C_1 : x^2 + y^2 - 1 = 0$ and $C_2 : x^2 + y^2 - 4 = 0$. A parabola is drawn through the points where ' C_1 ' meets the x – axis and having arbitrary tangent of ' C_2 ' as it's directrix. Find the locus of the focus of drawn parabola.
 (A) $\left(\frac{3}{4}\right)x^2 - y^2 = 3$ (B) $\left(\frac{3}{4}\right)x^2 + y^2 = 3$
 (C) $\left(\frac{3}{2}\right)x^2 + y^2 = 3$ (D) $\left(\frac{3}{4}\right)x^2 + y^2 = \frac{3}{2}$

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9. Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$

Where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers, and I is the 2×2 identity matrix. If

α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi]\}$ and

β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi]\}$

Then the value of $\alpha^* + \beta^*$ is

- (A) $-\frac{37}{16}$ (B) $-\frac{31}{16}$ (C) $-\frac{17}{16}$ (D) $-\frac{29}{16}$

10. Given $2x - y - 2z = 2$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equation has NO solution, is

- (A) 3 (B) 1
(C) 0 (D) -3

11. If $x = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$ and $y = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$, then $x^2 + y^2$ is

- (A) 1 (B) 2
(C) 3 (D) 4

12. The number of solutions of the equation $\sin^{-1}\left(\frac{x^2}{x^2+1}\right) + \cos^{-1}\left(\frac{x^2-1}{x^2+1}\right) = \pi$ is

- (A) 1 (B) 2
(C) 0 (D) 3

13. $\sum_{r=1}^{\infty} \tan^{-1}\left(\frac{1}{r^2+5r+7}\right)$ equals to

- (A) $\tan^{-1} 3$ (B) $\frac{\pi}{4}$ (C) $\sin^{-1} \frac{1}{\sqrt{10}}$ (D) $\cot^{-1} 2$

14. One of the sides of a triangle is divided into segments of 6 and 8 units by the point of tangency of the inscribed circle. If the radius of the circle is 4 units, then the length of the shortest side is

- (A) 12 units (B) 13 units
(C) 14 units (D) 15 units

Space For Rough Work

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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Let z_1 and z_2 are two complex numbers satisfying the equation $3z_1^2 - 2z_1z_2 + 2z_2^2 = 0$ and also $\operatorname{Re}\left(\frac{z_1-2}{z_1+2}\right) = 0$. Also, P_1, P_2 and O are points in complex plane corresponding to z_1, z_2 and origin respectively.

15. The value of $|z_1|$ is equal to
 (A) 1 (B) $\sqrt{5}$
 (C) 2 (D) $2\sqrt{2}$
16. Area of triangle OP_1P_2 is
 (A) $\sqrt{5}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

Paragraph for Question no. 17 to 18

Let $P(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$ ($p, q, r \in \mathbb{R}$) be divisible by x^2 and α, β and γ are the positive roots of the equation $\frac{P(x)}{x^2} = 0$.

17. The value of $(p+q+r)$ is equal to
 (A) 9 (B) 27
 (C) 81 (D) 108
18. If $\alpha - 1, \beta + 3$ and $\gamma + 7$ are the first three terms of a sequence whose sum of first n terms is given by S_n then $\sum_{n=2}^{\infty} \frac{1}{\sqrt{S_n \cdot S_{n-1}}}$ is equal to
 (A) 1 (B) $\frac{1}{4}$
 (C) $\frac{1}{2}$ (D) 2

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Q.P. Code: 100163**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|-------|--------|-------|-------|
| 1. AD | 2. ABD | 3. AD | 4. AB |
| 5. BD | 6. BD | 7. BD | 8. B |
| 9. C | 10. B | 11. B | 12. B |
| 13. B | 14. A | 15. A | 16. C |
| 17. A | 18. D | | |

SECTION-1 : CHEMISTRY**PART – A**

- | | | | |
|--------|--------|--------|---------|
| 1. ABC | 2. ACD | 3. CD | 4. ABCD |
| 5. ABC | 6. ABD | 7. ACD | 8. C |
| 9. B | 10. A | 11. C | 12. A |
| 13. C | 14. C | 15. C | 16. C |
| 17. A | 18. D | | |

SECTION-1 : MATHEMATICS**PART – A**

- | | | | |
|--------|---------|--------|-------|
| 1. ABC | 2. BD | 3. ABC | 4. AC |
| 5. AC | 6. ABCD | 7. AB | 8. B |
| 9. D | 10. D | 11. B | 12. C |
| 13. C | 14. B | 15. C | 16. A |
| 17. B | 18. C | | |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **AD**

Sol. Use Gauss law.

2. **ABD**

Sol. Amplitude added vectorily.

3. **AD**

Sol. $\vec{\tau} = \vec{M} \times \vec{B} = \frac{IB_0 a^3}{8}$
 $\tau_{\text{friction}} = \frac{\mu m g a}{4}$

4. **AB**

Sol. For maximum frequency velocity of source towards detector & for minimum frequency velocity of source away from detector.

5. **BD**

Sol. Use work energy theorem.

6. **BD**

Sol. For KE to increase power should be positive

$$\vec{f} \cdot \vec{v} > 0 \Rightarrow \theta < 90^\circ$$

$$P = \sqrt{2Km} \Rightarrow P \propto \sqrt{K}$$

7. **BD**

Sol. Voltage across capacitor = 3 V

$$q_c = 3 \times 0.1 = 0.3 \text{ C}$$

$$P = \frac{(2)^2}{2} = 2 \text{ W.}$$

8. **B**

Sol. $3 \times \frac{1}{2} k x \sqrt{3}^2 + \frac{1}{2} 3 \times mv^2 = \text{const}$

differentiating w.r.t time

$$3kx + ma = 0$$

$$a = -\left(\frac{3k}{m}\right)x$$

$$T = 2\pi\sqrt{\frac{m}{3k}}$$

9. **C**

Sol. Use concept of variable mass system

$$V = \sqrt{\frac{2gx}{3}}$$

10. **B**

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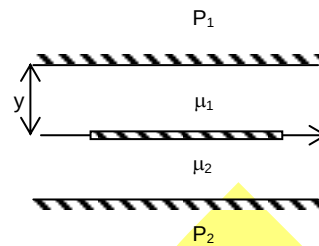
Sol. Force on both sides of the plate

$$F = v \left[\frac{\mu_1}{y} + \frac{\mu_2}{h-y} \right] A$$

Where A = area of the thin plate

For F to be minimum $\frac{dF}{dy} = 0$

$$y = \frac{\sqrt{\mu_1} h}{\sqrt{\mu_1} + \sqrt{\mu_2}}$$



11. **B**

$$\text{Sol. } i_1 = \frac{V_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} ; i_2 = \frac{V_0}{\sqrt{R^2 + \frac{1}{4\omega^2 C^2}}}$$

12. **B**

$$\text{Sol. } C = \frac{\lambda}{\Delta V}$$

$$C = \frac{\pi \epsilon_0}{\ln\left(\frac{b-a}{a}\right)}$$

13. **B**

Sol. Energy stored in inductor

$$= \frac{B^2 V^2 t^2}{2L} = 16J$$

energy dissipated in resistor

$$= \frac{V^2 B^2 t^2}{R} = 16J$$

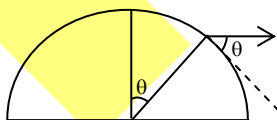
So work done by external agent = 16J + 16J = 32 J

14. **A**

Sol. By COM

$$0 = 3mv + mv_x \Rightarrow v_x = 3v$$

wrt sphere $v'_x = 4v$



15. **A**

16. **C**

Sol. By COME

$$\left[mg\left(\frac{L}{3}\right) + mg\frac{L}{2} \right] \times \frac{3}{5} = \frac{1}{2} \left[\frac{mL^2}{3} + \frac{mL^2}{9} \right] \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{9g}{4L}} = \frac{3}{2} \sqrt{\frac{g}{L}}$$

By $\tau = I\alpha$

$$2mg - N = 2ma_t$$

$$N \frac{5L}{12} = \left[\frac{mL^2}{12} + m\left(\frac{L}{12}\right)^2 + m\left(\frac{L}{12}\right)^2 \right] \alpha$$

$$\alpha \frac{5L}{12} = a_t$$

$$\Rightarrow a_t = \frac{27g}{12} \text{ and } N = \frac{7}{10} mg$$

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$$F_{ac} = (2m)\omega^2 \left(\frac{5L}{12} \right) = \frac{15mg}{8}$$

$$\text{Net reaction force} = \sqrt{N^2 + F_{ac}^2} = \frac{\sqrt{949}}{16} mg$$

17. **A**

Sol. $F_{\text{Hinge}} + F_{\text{Mag}} = m\omega^2 \frac{\ell}{2}$

18. **D**

Sol. $T + F_{\text{mag}} = \frac{M}{2} \omega^2 \left(\frac{3\ell}{4} \right)$

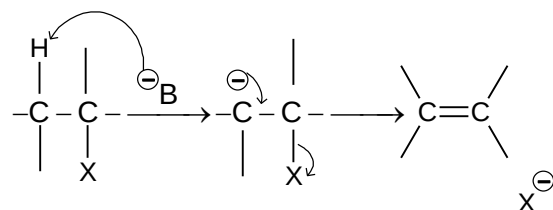
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SECTION-2 : CHEMISTRY

PART – A

1. ABC

Sol.



2. ACD

Sol. $H_2O \rightleftharpoons H^+ + OH^- \quad K_w = [H^+][OH^-]$ $T \uparrow \Rightarrow K_w \uparrow$

3. CD

Sol. (A) 4 amino acids
 (B) Both shows positive
 (C) Osazone
 (D) No free hemiacetal

4. ABCD

Sol. Theoretical
 ΔS_{system} should be negative.

5. ABC

Sol. $Sc^{3+} \cdot C_2^{2-}$ & factual

6. ABD

Sol. Enolisation & hemiacetal

7. ACD

Sol. Theoretical

8. C

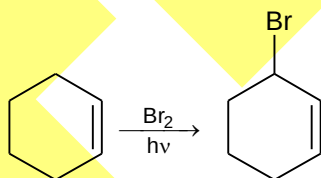
Sol. Electronegativity & nucleophilicity.

9. B

Sol. Benzyne mechanism

10. A

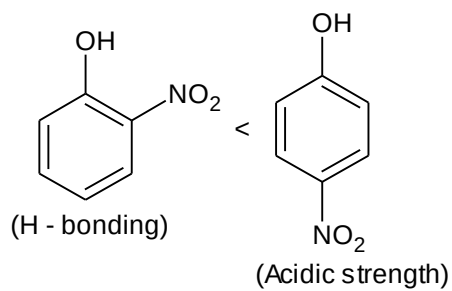
Sol.



11. C

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Sol.



12. A

Sol. Factual

13. C

Sol. Conceptual

$$\frac{x}{m} \propto P^{0-1} \text{ and } \frac{x}{m} \propto \frac{1}{T}$$

14. C

Sol. Gly-Ala-Phe-Glu-His-Gly-Ala-Leu-Val

15. C

Sol. Tetrametal Phosphoric acid $H_4P_4O_{12}$

16. C

Sol. Tetra Poly Phosphoric acid $H_6P_4O_{13}$

17. A

Sol. Migratory aptitude $-Ph > -Me$

18. D

Sol. $-COOEt$ will migrate
Stability of carbocation concerned.

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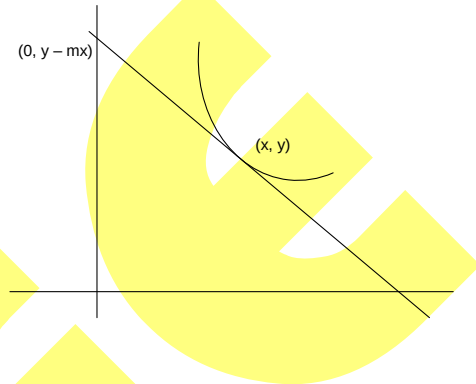
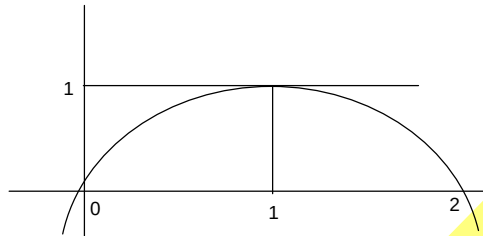
SECTION-3 : MATHEMATICS

PART – A

1. ABC

Sol. $Y - y = m(X - x)$ Put $X = 0$ $Y = y - mx$ $\therefore y - mx = x^2$ $\frac{dy}{dx} - \frac{1}{x}y = -x$ (linear

DE)

solving $y = -x^2 + cx$ $x = 1, y = 1$ $\Rightarrow c = 2$ $y = 2x - x^2$ 

2. BD

Sol. $\vec{V} - \vec{n}_1 \times \vec{n}_2 = (3\hat{i} - \hat{j} + \hat{k}) \times (\hat{i} + 4\hat{j} - 2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ 1 & 4 & -2 \end{vmatrix}$

$$= \hat{i}(2 - 4) - \hat{j}(-6 - 1) + \hat{k}(12 + 1) = -2\hat{i} + 7\hat{j} + 13\hat{k} \Rightarrow \text{B and D}$$

3. ABC

Sol. $(2x - 3y + z)^{10}$

General term = $\sum_{\alpha+\beta+\gamma=10} \frac{10!(2x)^\alpha (-3y)^\beta (z)^\gamma}{\alpha!\beta!\gamma!} \dots\dots\dots(i)$

$\therefore$ Number of terms = $\frac{12!}{10!2!} = \frac{12 \times 11}{2} = 66$.

Put $x = y = z = 1$ in (1), we get

Sum of all the coefficients 0.

Put $\alpha = 2, \beta = 3, \gamma = 5$ in (1),

We get coefficient of $x^2 y^3 z^5 = \frac{10!}{2!3!5!} (2)^2 (-3)^3 = \frac{-10! \times 4 \times 27}{2 \times 6 \times 120} = \frac{-10! \times 9}{5!}$

4. AC

Sol. $f(x) = (x^2 + 2x + 3)^2 + 2(x^2 + 2x + 3) + 3$

$$x^2 + 2x + 3 > 0 \forall x \in \mathbb{R}$$

$$\therefore f(x) > 0 \forall x \in \mathbb{R}.$$

Assume $x^2 + 2x + 3 = y$

$$f(x) = y^2 + 2y + 3 = (y + 1)^2 + 2 \Rightarrow f(x) = (x^2 + 2x + 4)^2 + 2 \Rightarrow f(x) = [(x + 1)^2 + 3]^2 + 2$$

$$f(x) \geq 9 + 2 = 11 \Rightarrow f(x)|_{\min} = 11$$

5. AC

Sol. ${}^4C_4 \cdot {}^{100}C_6 + {}^4C_3 \cdot {}^{100}C_7 + {}^4C_2 \cdot {}^{100}C_8 + {}^4C_1 \cdot {}^{100}C_9 + {}^4C_0 \cdot {}^{100}C_{10}$

Out of 104 students of which 100 are boys and 4 are girls, we have to select 10 students.

This can be done in ${}^{104}C_{10} = {}^{100}C_6 + {}^{100}C_7 + {}^{100}C_8 + {}^{100}C_9 + {}^{100}C_{10}$

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Hence, $x + y = 114$ or 198

6. ABCD

Sol.

$$(A) \quad I = e^{\lim_{n \rightarrow \infty} \frac{n^2-1}{n} \left(\frac{2n^2-3}{2n^2-n+1} - 1 \right)} = e^{\lim_{n \rightarrow \infty} \left(\frac{n-4}{2n^2-n+1} \right) \left(\frac{n^2-1}{n} \right)} = e^{\frac{1}{2}} = \sqrt{e}$$

$$\ln I = \frac{1}{2}$$

$$(B) \quad \lim_{x \rightarrow \infty} x \left(\frac{1}{e} - \left(\frac{x}{x+1} \right)^x \right)$$

$$\text{Let } x = \frac{1}{t}$$

$$\begin{aligned} \therefore \lim_{t \rightarrow 0} \frac{1}{t} \left(\frac{1}{e} - \left(\frac{1}{1+t} \right)^{1/t} \right) &= \frac{1}{e} \lim_{t \rightarrow 0} \frac{(1+t)^{1/t} - e}{t \cdot (1+t)^{1/t}} = \frac{1}{e^2} \lim_{t \rightarrow 0} \frac{e^{\frac{\ln(1+t)}{t}} - e}{t} \\ &= \frac{1}{e} \lim_{t \rightarrow 0} \frac{e^{\frac{\ln(1+t)}{t}} - 1}{\left(\frac{\ln(1+t)}{t} - 1 \right)} \cdot \left(\frac{\ln(1+t)}{t} - 1 \right) = \frac{1}{e} \lim_{t \rightarrow 0} \frac{\ln(1+t) - t}{t^2} = \frac{-1}{2e} \end{aligned}$$

(C) As we are interested in coefficient of x^{50} , we shall ignore all the terms with exponent more than 50.

$$\begin{aligned} \text{So, } (1+x^2)^{25} (1+x^{25}) (1+x^{40}) (1+x^{45}) (1+x^{47}) \\ = (1 + {}^{25}C_1 x^2 + \dots + {}^{25}C_{25} x^{50}) \times (1+x^{25} + x^{40} + x^{45} + x^{47}) = {}^{25}C_{25} + {}^{25}C_5 = 1 + {}^{25}C_5. \end{aligned}$$

$$\begin{aligned} (D) \quad (A^{-1}B^{-1})^T &= (B^{-1})^T (A^{-1})^T = (B^T)^{-1} (A^T)^{-1} = B^{-1}A^{-1} = (AB)^{-1} = (BA)^{-1} = A^{-1}B^{-1} \\ &\Rightarrow A^{-1}B^{-1} \text{ is symmetric matrix.} \end{aligned}$$

7. AB

Sol. $\log_3 |2z+1| < \log_3 |2z-1|$ (as $\cot^{-1} x$ is a decreasing function)

$$\therefore |2z+1| < |2z-1|$$

$$\text{or } (2z+1)(2\bar{z}+1) < (2z-1)(2\bar{z}-1)$$

$$2(z+\bar{z}) < -2(z+\bar{z})$$

$$z+\bar{z} < 0$$

$$2\operatorname{Re}(z) < 0$$

$$\therefore \operatorname{Re}(z) < 0$$

8. B

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Sol. Clearly the parabola should pass through (1, 0) and (-1, 0). Let directrix of this parabola be, $x \cos \theta + y \sin \theta = 2$

If $S(h, k)$ be the focus of this parabola, then distance of $(\pm 1, 0)$ from 'S' and from the directrix should be same.

$$\Rightarrow (h-1)^2 + k^2 = (\cos \theta - 2)^2$$

.....(1)

and $(h+1)^2 + k^2 = (\cos \theta + 2)^2$

.....(2)

(2) - (1) gives us; $4h = 8 \cos \theta$

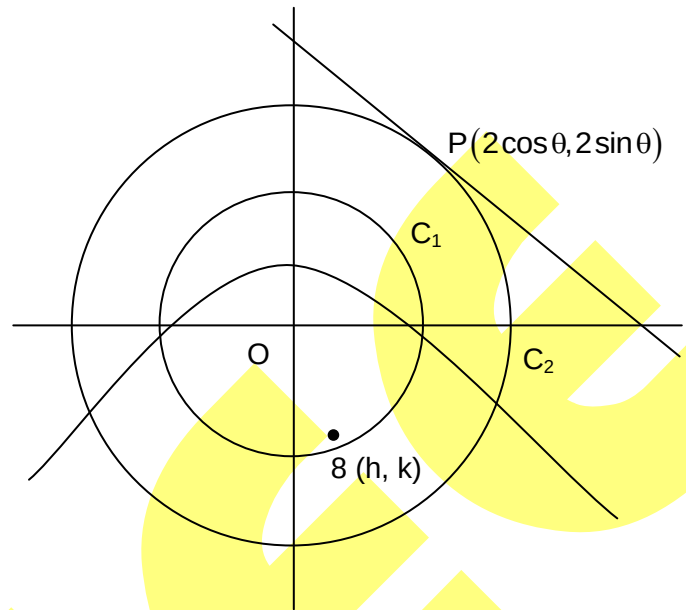
$$\Rightarrow \cos \theta = \frac{h}{2}$$

(2) + (1) gives us;

$$2(h^2 + k^2 + 1) = 2(\cos^2 \theta + 4)$$

$$\Rightarrow h^2 + k^2 + 1 = 4 + \frac{h^2}{4} \Rightarrow \frac{3}{4}h^2 + k^2 = 3$$

Hence locus of focus is: $\left(\frac{3}{4}\right)x^2 + y^2 = 3$.



9. D

Sol. $M = \alpha I + \beta M^{-1}$

Equate trace on both sides

$$\sin^4 \theta + \cos^4 \theta = 2\alpha + \beta \left(\frac{\sin^4 \theta + \cos^4 \theta}{|M|} \right)$$

Equate sum of elements of non principal diagonal

$$\cos^2 \theta - \sin^2 \theta = \beta \left(\frac{\sin^2 \theta - \cos^2 \theta}{|M|} \right)$$

$$\therefore \beta = -|M| = -[\sin^4 \theta \cos^4 \theta + \sin^2 \theta \cos^2 \theta + 2]$$

$$\alpha = \sin^4 \theta + \cos^4 \theta$$

$$\alpha^* = \alpha_{\min} = \frac{1}{2}$$

$$\beta^* = \beta_{\min} = -\frac{37}{16}$$

$$\therefore \alpha^* + \beta^* = \frac{1}{2} - \frac{37}{16} = -\frac{29}{16}$$

10. D

Sol. As $\Delta_z \neq 0$, for no solution $\Delta = 0 \Rightarrow \begin{vmatrix} 2 & -1 & -2 \\ 1 & -2 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = -3$.

11. B

Sol. $x^2 + y^2 = 3 + 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right)$

$$= 3 + \frac{2 \cos \frac{4\pi}{7} \sin \frac{3\pi}{7}}{\sin \frac{\pi}{7}} = 3 - 1 = 2$$

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12. C

$$\begin{aligned} \text{Sol. } \sin^{-1}\left(\frac{x^2}{x^2+1}\right) &= \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \\ &= \sqrt{1 - \frac{x^4}{(x^2+1)^2}} = \frac{1-x^2}{1+x^2}. \end{aligned}$$

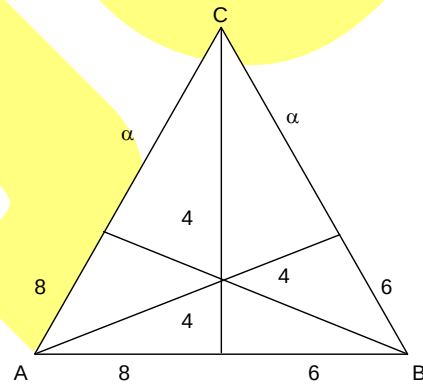
On squaring and solving, we get, $x = 0$.

13. C

$$\begin{aligned} \text{Sol. } a_n &= \tan^{-1}\left(\frac{1}{1+(n+3)(n+2)}\right) \\ &= \tan^{-1}\left(\frac{(n+3)-(n+2)}{1+(n+3)(n+2)}\right) \\ &= \tan^{-1}(n+3) - \tan^{-1}(n+2) \\ S_n &= f(n) - f(0) = \tan^{-1}(n+3) - \tan^{-1}3 \\ S_n &= \frac{\pi}{2} - \tan^{-1}3 = \cot^{-1}3 = \sin^{-1}\frac{1}{\sqrt{10}}. \end{aligned}$$

14. B

$$\begin{aligned} \text{Sol. } \tan \frac{A}{2} &= \frac{1}{2} \\ \tan \frac{B}{2} &= \frac{2}{3} \\ \Rightarrow \tan \frac{C}{2} &= \frac{4}{7} \\ \Rightarrow \alpha = 7 &\Rightarrow \text{the shortest side is 14.} \end{aligned}$$



15. C

16. A

(Sol. 15 & 16)

$$\text{Since } \operatorname{Re}\left(\frac{z_1-2}{z_1+2}\right) = 0$$

$$\frac{z_1-2}{z_1+2} \text{ is P.I.}$$

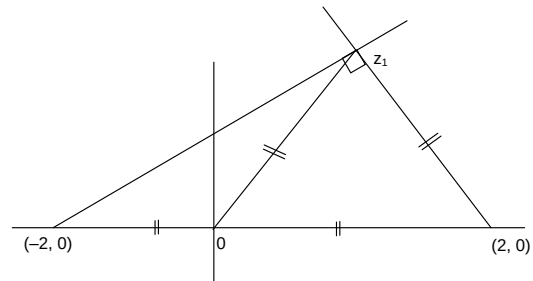
 $\Rightarrow$ line joining $(2, 0)$, z_1 and $(-2, 0)$, z_1 are perpendicular

$$\therefore |z_1| = 2$$

$$\text{Also, } 2z_2^2 - 2z_1z_2 + 3z_1^2 = 0$$

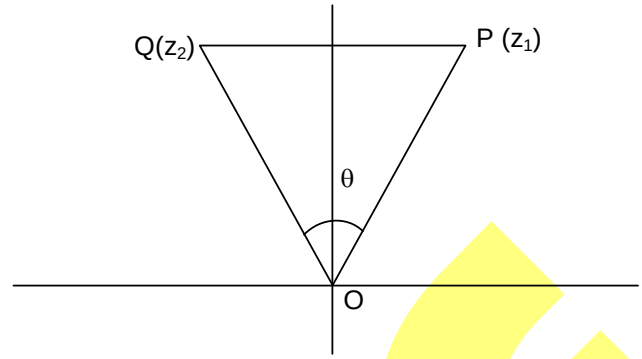
$$z_2 = \left(\frac{1+\sqrt{5}i}{2}\right)z_1$$

$$\therefore |z_2| = \frac{\sqrt{6}}{2}|z_1|.$$



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Also, $\frac{z_2}{|z_2|} = \frac{z_1}{|z_1|} e^{i\theta}$



17. B

18. C

(Sol. 17 & 18)

$$P(x) = x^5 - 9x^4 + px^3 - 27x^2 + qx + r$$

$\therefore P(x)$ is divisible by x^2

$$\therefore q = r = 0$$

$$P(x) = x^5 - 9x^4 + px^3 - 27x^2$$

$$\frac{P(x)}{x^2} = x^3 - 9x^2 + px - 27 = 0$$

$$\alpha, \beta, \gamma \in \mathbb{R}^+$$

$$\alpha + \beta + \gamma = 9, \alpha\beta\gamma = 27$$

$$\text{A.M. of } \alpha, \beta, \gamma = \frac{\alpha + \beta + \gamma}{3} = 3$$

$$\text{G.M. of } \alpha, \beta, \gamma = (\alpha\beta\gamma)^{1/3} = (27)^{1/3} = 3$$

$$\therefore \text{A.M.} = \text{G.M.} \Rightarrow \alpha = \beta = \gamma$$

$$\therefore 3\alpha = 9 \Rightarrow \alpha = 3 = \beta = \gamma$$

$$p = \alpha\beta + \beta\gamma + \gamma\alpha = 27$$

(i) $p + q + r = 27 + 0 + 0 = 27$

(ii) $\alpha - 1 = 2, \beta + 3 = 6, \gamma + 7 = 10$

$$\therefore S_n = 2 + 6 + 10 + \dots \text{to } n \text{ terms} = 2(1 + 3 + 5 + \dots \text{to } n \text{ terms}) = 2n^2$$

$$\text{Now, } \sum_{n=2}^{\infty} \frac{1}{\sqrt{S_n \cdot S_{n-1}}} = \sum_{n=2}^{\infty} \frac{1}{\sqrt{2n^2 \cdot 2(n-1)^2}} = \frac{1}{2} \cdot \sum_{n=2}^{\infty} \frac{1}{n(n-1)}$$

$$= \frac{1}{2} \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right)$$

$$= \frac{1}{2} \left(\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots \infty \right) = \frac{1}{2}$$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 3

QP CODE: 100174

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 204

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & B** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are **not allowed**.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-04)** – Contains **Four (04)** multiple choice questions which have ONLY ONE CORRECT answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **PART-A (05–12)** contains Eight (8) Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (iii) **PART – B (1 – 3)**: This section contains **Three (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iv) **Part-B (4 – 8)** contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

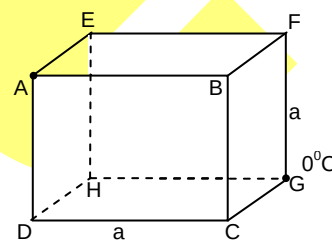
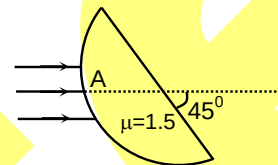
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SECTION – I : PHYSICS**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- The diameter of a wire is measured with a screw gauge having 50 divisions on circular scale and by one complete rotation of circular scale, main scale moves 0.5 mm. If reading of screw gauge is 0.250 cm. The minimum percentage error in the reading will be
 (A) 0.4 (B) 0.8 (C) 4 (D) 5
- A parallel narrow beam of light is incident on the surface of a transparent hemisphere of radius R and refractive index $\mu = 1.5$ as shown. The distance (from A) of the image formed by refraction at the spherical surface only is
 (A) $\frac{R}{2}$ (B) $3R$ (C) $\frac{R}{3}$ (D) $2R$
- A cubical frame is made by connecting 12 identical uniform conducting rods as shown in the figure. In the steady state the temperature of junction A is 100°C while that of the G is 0°C . Then,
 (A) B will be Hotter than H (B) Temperature of F is 40°C
 (C) Temperature of D is 66.67°C (D) Temperature of E is 50°C
- The figure shows a charge q placed inside a cavity in an uncharged conductor. Now if an external electric field is switched on:
 (A) only induced charges on outer surface will redistribute.
 (B) only induced charges on inner surface will redistribute.
 (C) both induced charges on outer and inner surfaces will redistribute.
 (D) force on charge q placed inside the cavity will change.



Space For Rough Work

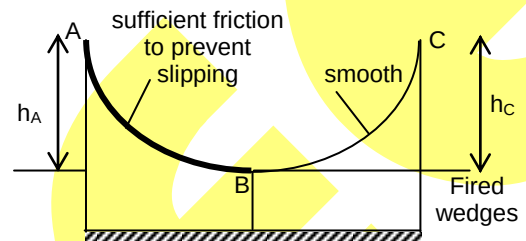
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PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A sphere A moving with a speed u and rotating with an angular velocity ω , makes a head on elastic collision with an identical stationary sphere B. There is no friction between the surfaces of A and B. Disregard gravity. Then
 (A) A will stop translating but continue to rotate with an angular speed ω .
 (B) A will come to rest and stop rotating.
 (C) B will move with a speed u without rotating.
 (D) B will move with a speed u and rotate with an angular velocity ω .

6. The spherical ball is released from point A. If C be the point upto which the ball goes on smooth surface then (assuming h_A , h_C represent the height of point A and C from horizontal level B and k_A , k_C are the kinetic energies of spheres at points A and C respectively)

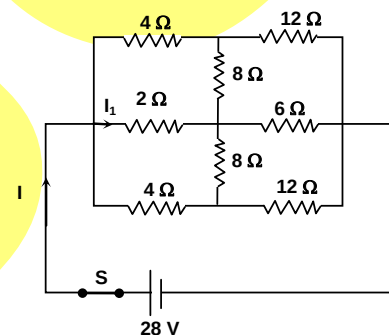


- (A) $h_A > h_C$ and $k_A < k_C$
 (C) $h_A = h_C$

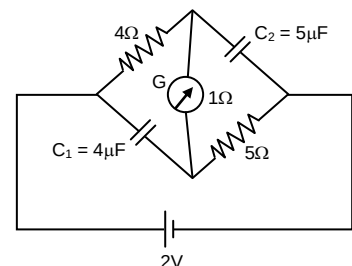
- (B) $h_A > h_C$ and $k_B > k_C$
 (D) $h_A > h_C$ & $k_A > k_C$

7. In the circuit shown, the current

- (A) $I = 8$ A
 (B) $I = 7$ A
 (C) $I_1 = 3.5$ A
 (D) $I_1 = 4$ A



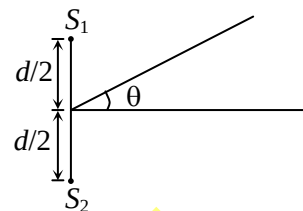
8. In the circuit shown, the cell is ideal with emf = 2V. The resistance of the coil of the galvanometer $G = 1\Omega$. Then in steady state
 (A) No current flows in G .
 (B) 0.2 A current flow in G .
 (C) Potential difference across $C_1 = 1.2$
 (D) Potential difference across $C_2 = 2$ V



Space For Rough Work

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9. In an interference arrangement, similar to Young's double slit experiment, the slits S_1 and S_2 are illuminated with coherent microwave sources, each of frequency 10^6 Hz. The sources are synchronized to have zero phase difference. The slits are separated by distance $d = 150.0$ m. The intensity $I_{(\theta)}$ is measured as a function of θ , where θ is defined as shown in the figure. If I_0 is maximum intensity, then $I_{(\theta)}$ for $0 \leq \theta \leq 90$ is given by

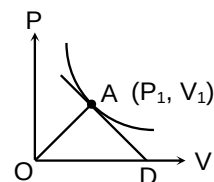


- (A) $I_{(\theta)} = I_0$ for $\theta = 0^\circ$ (B) $I_{(\theta)} = (I_0 / 2)$ for $\theta = 30^\circ$
 (C) $I_{(\theta)} = (I_0 / 4)$ for $\theta = 90^\circ$ (D) $I_{(\theta)}$ is constant for all values of θ
10. The electron in a hydrogen atom makes a transition $n_1 \rightarrow n_2$, where n_1 and n_2 are the principal quantum numbers of two states. Assume the Bohr model to be valid. If the time period of the electron in the initial state is eight times that in the final state then the possible values of n_1 and n_2 are
- (A) $n_1 = 4, n_2 = 2$ (B) $n_1 = 8, n_2 = 2$ (C) $n_1 = 8, n_2 = 1$ (D) $n_1 = 6, n_2 = 3$

11. A string of length l is stretched along the x-axis and is rigidly clamped at $x = 0$ and $x = l$. Transverse vibrations are produced in the string. For n th harmonic which of the following equations may represent the shape of the string at any time t . (Assume amplitude at antinodes to be a)

- (A) $y = a \cos(\omega t) \cos\left(\frac{n\pi x}{l}\right)$ (B) $y = a \sin(\omega t) \cos\left(\frac{n\pi x}{l}\right)$
 (C) $y = a \cos(\omega t) \sin\left(\frac{n\pi x}{l}\right)$ (D) $y = a \sin(\omega t) \sin\left(\frac{n\pi x}{l}\right)$

12. n moles of an ideal gas undergo an isothermal process at temperature T . P-V graph of the process is as shown in the figure. A point A (P_1, V_1) is located on the P-V curve. Tangent at point A, cuts the V-axis at point D. AO is the line joining the point A to the origin O of PV diagram. Then,



- (A) coordinates of points D is $\left(\frac{3V_1}{2}, 0\right)$
 (B) coordinates of points D is $(2V_1, 0)$
 (C) area of the triangle AOD is nRT
 (D) area of the triangle AOD is $\frac{3}{4}nRT$

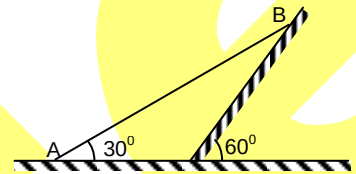
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PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

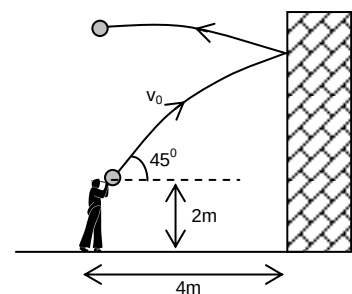
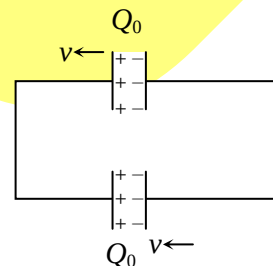
1. A bus is moving towards a huge wall with a velocity of 5 m/s. The driver sounds a horn of frequency 200Hz. The frequency of the beats heard by a passenger of the bus will be (In Hz) nearly (velocity of sound in air = 338 m / s)
2. An ideal gas is expanding such that $PT^2 = \text{constant}$. The coefficient of volume expansion of the gas is A/T where T is temperature in kelvin. Find the value of A .
3. In the figure shown, the instantaneous speed of end A of the rod is v to the left. The angular velocity of the rod of length L is equal to $n\frac{v}{L}$ then find the value of 'n'.

**PART – B (Maximum Marks: 15)**

(Numerical Type)

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

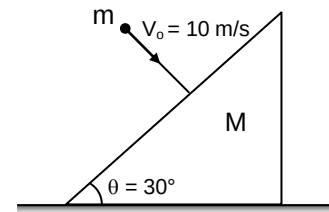
4. Two identical capacitor connected as shown and having initial charge Q_0 . Separation between plates of capacitor is d_0 . Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed v towards left while other plate of capacitor remains fixed. (given $\frac{Q_0 v}{2d} = 1$ amp). Find the value of current (in amp) in the circuit.
5. A boy stand $l = 4$ m away from a vertical wall and throws a ball. The ball leaves the boy's hand at $h = 2$ m above the ground with initial velocity $v_0 = 10\sqrt{2}$ m/s at an angle 45° from the horizontal. After striking the wall elastically the ball rebounds and hit the ground at $4K$ (in m). Find K (take $g = 9.8$ m/s²)



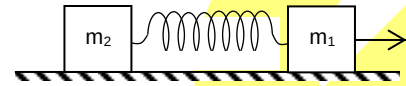
Space For Rough Work

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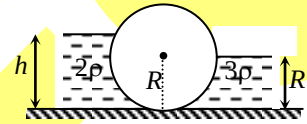
6. A ball of mass 1 kg moving with velocity 10 m/s collides perpendicularly on a smooth stationary wedge of mass 2 kg placed on smooth horizontal surface. If the coefficient of restitution is $e = 7/20$ then find the velocity of ball after the collision. [in ms^{-1}]



7. Two blocks of masses $m_1 = 1\text{kg}$ and $m_2 = 2\text{kg}$ are connected by a non deformed light spring. They are lying on a rough horizontal surface. The coefficient of friction between the blocks and the surface is 0.4, what minimum constant force F (in N) has to be applied in horizontal direction to the block of mass m_1 in order to shift the other block? ($g = 9.8 \text{ m/s}^2$)



8. In the figure shown, the heavy cylinder (radius R) resting on a smooth surface separates two liquids of densities 2ρ and 3ρ . If the height h for the equilibrium of cylinder is equal to be $R\sqrt{x}$. Then find the value of 'x'.



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SECTION – II : CHEMISTRY

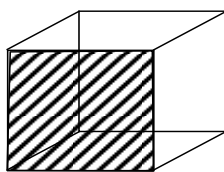
PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

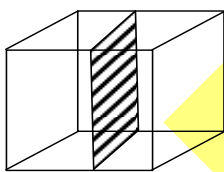
1. Copper reduces NO_3^- into NO and NO_2 depending upon concentration of HNO_3 in solution. Assuming $[\text{Cu}^{2+}] = 0.1 \text{ M}$, and $P_{\text{NO}} = P_{\text{NO}_2} = 10^{-3} \text{ bar}$, at which concentration of HNO_3 , thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same?

[Given : $E_{\text{Cu}^{2+}|\text{Cu}}^0 = +0.34 \text{ volt}$, $E_{\text{NO}_3^-|\text{NO}}^0 = +0.96 \text{ volt}$, $E_{\text{NO}_3^-|\text{NO}_2}^0 = +0.79 \text{ volt}$]

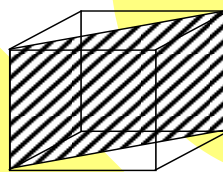
- (A) $10^{1.23} \text{ M}$ (B) $10^{0.56} \text{ M}$
(C) $10^{0.66} \text{ M}$ (D) $10^{0.12} \text{ M}$
2. Following four planes (M_1, M_2, M_3, M_4) in an ideal Zinc Blend type unit cell are shown in the figure below. Consider the following statements and choose which of the following statement is incorrect



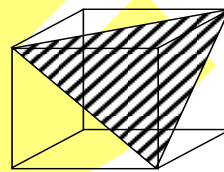
M_1



M_2

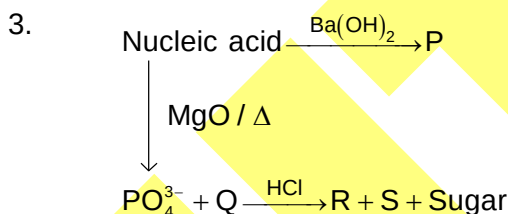


M_3



M_4

- (A) Effective number of anions present on lattice sites on M_1 is 1.
(B) If ions along M_2 are removed, the formula of solid changes.
(C) All the cations are either located on lattice sites on M_3 or touches M_3 .
(D) All the ions present at lattice sites on M_4 are closely packed.



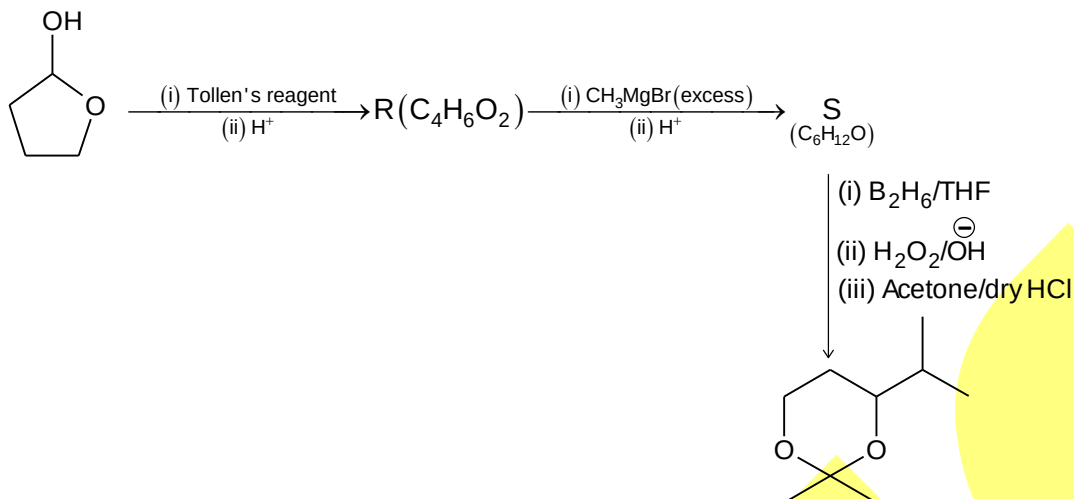
In the scheme shown above P, Q, R and S are

- (A) P = purine base, (Q) = pyrimidine base, (R) = nucleotides, (S) = nucleosides
(B) P = (Q) = nucleotides, (R) = pyrimidine base, (S) = purine base
(C) P = nucleosides, (Q) = nucleotides, (R) = (S) = purine base
(D) P = nucleotides, (Q) = nucleosides, (R) = pyrimidine base, (S) = purine base

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4.



Choose the incorrect statement(s)

- (A) R cannot be prepared by reacting cyclopentanone with m-CPBA.
 (B) S contains 11 H-atoms bonded to sp^3 -hybridized atoms
 (C) Both R & S are cyclic compounds.
 (D) Both R & S contains atleast one sp^2 -hybridized C-atom.

PART – A (Maximum Marks: 32)

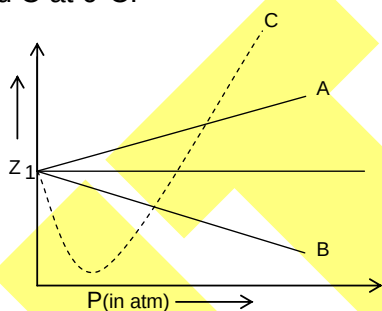
This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The correct statement(s) among the following is(are)
- (A) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure p_1 .
 (B) If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic
 (C) The work done by the gas is more when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions.
 (D) The change in internal energy of the gas is (i) zero, if it is expanded reversibly with $T_1 = T_2$ and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$.
6. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The TRUE statement(s) regarding this adsorption is(are)
- (A) O_2 is physisorbed
 (B) heat is released
 (C) occupancy of π_{2p}^* of O_2 is increased
 (D) bond length of O_2 is decreased

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7. Which is/are not correctly stated about graphite?
 (A) Electrical conductivity perpendicular to the planes is high.
 (B) Bond order of C – C bond is $3/2$.
 (C) Electrical conductivity parallel to the planes increases as temperature is raised.
 (D) It can act as either electron donor or an electron acceptor.
8. Which is/are correct statement?
 (A) There cannot be two probabilities for an electron at any position in atom
 (B) Probability cannot change abruptly from one point to the other
 (C) For large distance from the nucleus probability should be smaller
 (D) The total probability of electron some where in space around the nucleus approaches to 1
9. When sodium is dissolved in liquor ammonia it will produce
 (A) Ammoniate electron (B) Sodium ion
 (C) Amide ion (D) Hydroxyl ion
10. When sulphur is heated with $\text{NaOH}_{(\text{aq})}$ the compounds formed is/are
 (A) Na_2SO_4 (B) Na_2S
 (C) $\text{Na}_2\text{S}_2\text{O}_3$ (D) NaHSO_3
11. What volume of 0.1 M KMnO_4 in acidic medium is required for complete oxidation of 100 ml of 0.1 M FeC_2O_4 and 100 ml of 0.1 M Ferric oxalate separately?
 (A) 60 ml of KMnO_4 with FeC_2O_4 (B) 40 ml of KMnO_4 with FeC_2O_4
 (C) 40 ml of KMnO_4 with ferric oxalate (D) 120 ml of KMnO_4 with ferric oxalate
12. The given graph represents the variation of compressibility factor, Z Vs P for three real gases A, B and C at 0°C .



Identify the correct statement.

- (A) For gas A, $a = 0$ and its dependence on P is linear at all pressures
 (B) For gas B, $b = 0$ and its dependence on P is linear at all pressures
 (C) For gas C, neither a nor $b = 0$. By knowing the minima and point of intersection with $Z = 1$, a and b can be calculate
 (D) At high pressure, the slope is positive for all real gases

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PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. A decapeptide (Mol. Wt. 796) on complete hydrolysis gives glycine (Mol. Wt. 75), alanine and phenylalanine. Glycine contributes 47.0 % to the total weight of the hydrolyzed products. The number of glycine units present in the decapeptide is
2. The number of unpaired electrons in the complex $[\text{Mn}(\text{acac})_3]$ is (Atomic number of Mn = 25)
3. The number of all possible isomers for the molecular formula C_6H_{14} is

PART – B (Maximum Marks: 15)

(Numerical Type)

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

4. Hydrolysis of 15.45 g of benzonitrile produced 10.98 g of benzoic acid. The percentage yield of acid formed is _____
5. The total number of isomers possible for the molecular $[\text{Co}(\text{NH}_3)_4\text{Cl}(\text{NO}_2)]^+$ is _____
6. Specific rotation of the (R)-enantiomer of a chiral compound is 48. The specific rotation of a sample of this compound which contains 25% of (s)-enantiomer is
7. If the percentage of the number of coordinate-covalent bonds present in the dimer of aluminium chloride is x, what is the value of $\frac{x}{10}$?
8. Consider the complexes of manganese
 $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (I)
 $[\text{Mn}(\text{CN})_6]^{4-}$ (II)
 If a = Number of electrons present in the t_{2g} orbitals in complex(II)
 b = Number of unpaired electrons of metal ion in complex(I)
 & c = Number of electrons present in the e_g orbital of complex(I)
 What is the value of $\frac{a+b+c}{10}$?

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SECTION – III : MATHEMATICS**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- Let function $g(x)$ be differentiable and $g'(x)$ is continuous in $(-\infty, \infty)$ with $g'(2)=14$, then $\lim_{x \rightarrow 0} \frac{g(2 + \sin x) - g(2 + x \cos x)}{x - \sin x}$ is equal to:
 (A) 7 (B) 14
 (C) 28 (D) 56
- $\int \frac{(2x+1)}{(x^2+4x+1)^{3/2}} dx$
 (A) $\frac{x^3}{(x^2+4x+1)^{1/2}} + C$ (B) $\frac{x}{(x^2+4x+1)^{1/2}} + C$
 (C) $\frac{x^2}{(x^2+4x+1)^{1/2}} + C$ (D) $\frac{1}{(x^2+4x+1)^{1/2}} + C$
- Given a function g continuous on $\mathbb{R}$ such that $\int_0^1 g(t) dt = 2$ and $g(1) = 5$. If $f(x) = \frac{1}{2} \int_0^x (x-t)^2 g(t) dt$, then the value of $(f'''(1) - f''(1))$ is equal to:
 (A) 0 (B) 3
 (C) 5 (D) 7
- If $r, k, p \in \mathbb{W}$, then $\sum_{r+k+p=10} {}^{30}C_r \cdot {}^{20}C_k \cdot {}^{10}C_p$ is equal to
 (A) $\binom{60}{50}$ (B) $\binom{60}{30}$
 (C) $\binom{60}{20}$ (D) $\binom{30}{10} \binom{30}{20}$

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PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. If sides AB, BC and CA of a triangle ABC are represented by $x + 2 = 0$, $3x + y = 0$ and $x + 3y + 2 = 0$ respectively, then identify the correct statement.
- (A) $\sum \tan A = \frac{4}{3}$ (B) $\prod \tan A = -\frac{4}{3}$
 (C) $\sum \tan A \tan B = -\frac{41}{9}$ (D) $\sin^2(A + B) + \cos^2 C = \frac{5}{4}$
6. In triangle ABC, $a=4$ and $b=c=2\sqrt{2}$. A point P moves within the triangle such that the square of its distance from BC is half the area of rectangle contained by its distances from the other two sides. If D be the centre of locus of P, then
- (A) locus of P is an ellipse with eccentricity $\sqrt{\frac{2}{3}}$
 (B) locus of P is a hyperbola with eccentricity $\sqrt{\frac{3}{2}}$
 (C) area of the quadrilateral ABDC = $\frac{16}{3}$ sq. units
 (D) area of the quadrilateral ABDC = $\frac{32}{3}$ sq. units
7. $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c} = 3$ and $|\vec{a} + \vec{b} - \vec{c}|^2 + |\vec{b} + \vec{c} - \vec{a}|^2 + |\vec{c} + \vec{a} - \vec{b}|^2 = 36$ then
- (A) $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{9}{2}$ (B) $\vec{a}, \vec{b}, \vec{c}$ are co-planar
 (C) $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a} \neq 0$ (D) $\vec{a}, \vec{b}, \vec{c}$ represents sides of triangle
8. If $|\sin x| + |\cos x| \geq 1$ then x can lie in the interval
- (A) $\left[n\pi - \frac{\pi}{4}, n\pi\right]$ (B) $\left[n\pi, n\pi + \frac{\pi}{4}\right]$
 (C) $\left[2n\pi, 2n\pi + \frac{\pi}{4}\right]$ (D) $\left[(4n+1)\frac{\pi}{4}, (4n+3)\frac{\pi}{4}\right]$ where $n \in \text{integer}$.

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9. If the sum of n terms of series $\frac{5}{1.2.3^1} + \frac{7}{2.3.3^2} + \frac{9}{3.4.3^3} + \dots$ is S_n then
- (A) $S_3 = \frac{107}{108}$ (B) $S_n = 1 - \frac{1}{3^n \cdot n}$
 (C) $\lim_{n \rightarrow \infty} S_n = 1$ (D) $\lim_{n \rightarrow \infty} S_n = 3$
10. If $f: \mathbb{R} \rightarrow \left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ is a function defined by $f(x) = \tan^{-1}\left(x^4 - x^2 - \frac{7}{4} + \tan^{-1} x\right)$ and f is surjective then:
- (A) $\cos^{-1}\left(\frac{1-\alpha^2}{1+\alpha^2}\right) = 2$ (B) $\alpha + \frac{1}{\alpha} = 2 \operatorname{cosec} 2$
 (C) $\sin^{-1}\left(\frac{2\alpha}{\alpha^2+1}\right) = \pi - 2$ (D) $\tan^{-1}\left(\frac{2\alpha}{\alpha^2-1}\right) = 2 - \pi$
11. The possible value(s) of k for which $\lim_{x \rightarrow \infty} \frac{2x^3 - (\tan^{-1} x)^3}{\frac{8}{\pi} x^3 \cot^{-1}|kx| + k^2 x^6 \sin \frac{1}{x^3} - 3kx^3} = \frac{1}{2}$, is:
- (A) 0 (B) -1
 (C) 2 (D) 4
12. Let function $y = f(x)$ satisfies the differential equation $x^2 \frac{dy}{dx} = y^2 e^{\frac{1}{x}}$ ($x \neq 0$) and $\lim_{x \rightarrow 0^-} f(x) = 1$. Identify the correct statement(s):
- (A) Range of $f(x)$ is $(0, 1) - \left\{\frac{1}{2}\right\}$ (B) $f(x)$ is bounded
 (C) $\lim_{x \rightarrow 0^+} f(x) = 1$ (D) $\int_0^e f(x) dx > \int_0^1 f(x) dx$

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PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. If A, B and C are angles of a triangle, then the minimum value of

$$E = \frac{\cos\left(\frac{B-C}{2}\right)}{\cos\left(\frac{B+C}{2}\right)} + \frac{\cos\left(\frac{C-A}{2}\right)}{\cos\left(\frac{C+A}{2}\right)} + \frac{\cos\left(\frac{A-B}{2}\right)}{\cos\left(\frac{A+B}{2}\right)}$$

2. Let $f(x)$ is a polynomial function and $(f(\alpha))^2 + (f'(\alpha))^2 = 0$, then find $\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left[\frac{f'(x)}{f(x)} \right]$, where $[.]$ denotes greatest integer function, is _____
3. α, β are roots of equation $x^2 - p(x+1) - c = 0$ then find the value of $\frac{\alpha^2 + 2\alpha + 1}{\alpha^2 + 2\alpha + c} + \frac{\beta^2 + 2\beta + 1}{\beta^2 + 2\beta + c}$

PART – B (Maximum Marks: 15)

(Numerical Type)

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

4. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$ are two sets. Let m is the number of one – one functions $f : A \rightarrow B$ such that $f(i) \neq i \forall i \in A$ and n is the number of one – one functions $f : A \rightarrow B$ such that $|f(i) - i| \leq 3 \forall i \in A$, then find the value of $(m+n)$
5. Let $A(1, -1), B(4, -2)$ and $C(9, 3)$ be the vertices of the triangle ABC. A parallelogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.

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6. If the maximum area of the region enclosed by the curves $y = |x|e^{|x|}$ and the line $y = a$ ($0 \leq a \leq e$) in $x \in [-1, 1]$ is A , then find the value of $[A]$.
[Note : $[k]$ denotes greatest integer function less than or equal to k .]
7. Let $a_r = r({}^7C_r)$, $b_r = (7-r)({}^7C_r)$ and $A_r = \begin{bmatrix} a_r & 0 \\ 0 & b_r \end{bmatrix}$. If $A = \sum_{r=0}^7 A_r = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, then the value of $a+b$ must be $(k+895)$ then k is
8. The least possible value of a for which $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ does not have a real solution is
-

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Answers & Solutions

SECTION – I : PHYSICS

PART – A

1. **A**

Sol. $LC = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$

$$\text{Percentage error} = \frac{0.01 \times 10^{-3}}{0.25 \times 10^{-2}} \times 100$$

$$= 0.4$$

2. **B**

Sol. $\frac{1.5}{V} - \frac{1}{-\infty} = \frac{1.5 - 1}{R}$

$$V = 3R.$$

3. **B**

Sol. $100^\circ\text{C} - lr - \frac{l}{2}r - lr = 0^\circ\text{C}$

$$\Rightarrow lr = 40^\circ\text{C} ; \therefore t_F = 0^\circ + lr = 40^\circ\text{C}$$

4. **A**

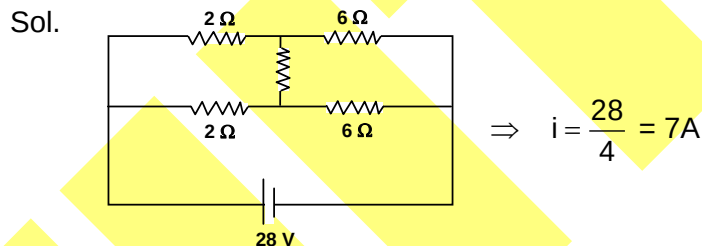
Sol. Due to electrostatic shielding.

5. **AC**

Sol. After elastic collision, translational velocities get interchanged.

6. **AB**

Sol. Apply energy conservation.

7. **BC**8. **B**

Sol. $i = \frac{V}{R_{\text{eq}}} = \frac{2}{10} = 0.2 \text{ A}$

9. **AB**

Sol. $\cos^2\left(\frac{\phi}{2}\right), \phi = \frac{d \sin \theta}{\lambda} \times 2\pi$

For $\theta = 0^\circ, \phi = 0, I_{(\theta)} = I_0$

For $\theta = 30^\circ, \phi = \frac{\pi}{2}, I_{(\theta)} = \frac{I_0}{2}$

10. **AD**

Sol. $T \propto n^3$

11. **CD**

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Sol. at $x = 0$, $\sin\left(\frac{n\pi x}{l}\right) = 0$

12. **BC**

Sol. $\tan \alpha = -[\text{slope at } (P_1, V_1)] = \frac{P_1}{V_1}$

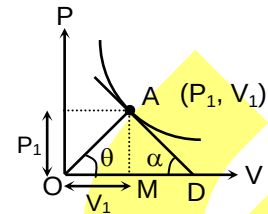
$$\tan \theta = \frac{P_1}{V_1}$$

$$\therefore \alpha = \theta$$

$$OM = MD = V_1$$

$$\therefore OD = 2V_1$$

$$\text{Area of triangle } AOD = \frac{1}{2}(AM \times OD) = \frac{1}{2}(P_1 \times 2V_1) = P_1 V_1 = nRT$$



PART - B

1. **6**

Sol. _____

2. **3**

Sol. $PT^2 = k$

$$\gamma = \frac{1}{V} \left(\frac{dV}{dT} \right) \quad \dots(i)$$

$$\left(\frac{nRT}{V} \right) T^2 = k$$

$$\Rightarrow \frac{T^3}{V} = k'$$

$$\Rightarrow V = \frac{T^3}{k'} \quad \dots(ii)$$

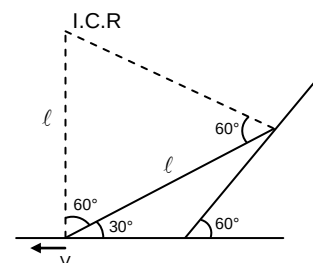
$$\frac{dV}{dT} = \frac{3T^2}{k'} \quad \dots(iii)$$

From (i), (ii) and (iii)

$$\gamma = \left(\frac{k'}{T^3} \right) \times \left(\frac{3T^2}{k'} \right) = \frac{3}{T}$$

3. **1**

Sol. $\omega = \frac{v}{L}$



4. **2**

Sol. $\frac{q_1}{C_1} = \frac{q_2}{C_2}$; $q_1 + q_2 = 2Q_0$

$$C_1 = \frac{\epsilon_0 A}{d_0 + vt} ; C_2 = \frac{\epsilon_0 A}{d_0 - vt}$$

$$\frac{q_1}{q_2} = \frac{d_0 - vt}{d_0 + vt}$$

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$$q_2 \left(\frac{d_0 - vt}{d_0 + vt} \right) + q_2 = 2Q_0$$

$$q_2 \left[\frac{2d_0}{d_0 + vt} \right] = 2Q_0$$

$$q_2 = \frac{2Q_0}{2d_0} (d_0 + vt)$$

$$I = \frac{dq_2}{dt} = \frac{Q_0 v}{d_0} = 2 \text{ amp}$$

5. **4.50**

Sol. The velocity component at the point of projection are

$$u_x = 10 \text{ m/s}, u_y = 10 \text{ m/s}$$

$$t = 0.4 \text{ s}$$

$$v_x = 10 \text{ m/s}, v_y = 6.1 \text{ m/s}$$

$$h = 3.2 \text{ m}$$

$$H = 5.2 \text{ m}$$

The time taken by the ball to hit the ground

$$5.2 = -6.1T + \frac{1}{2}gT^2$$

$$T = 1.8 \text{ s}$$

$$x = v_x T = 10 \times 1.8 = 18 \text{ m.}$$

$$4K = 18 \quad ; \quad \therefore K = 4.5$$

6. **2**

Sol. Equation of Newtons collision law

$$e = \frac{v_1 + \frac{v_2}{2}}{v_0}$$

$$2v_1 + v_2 = 7$$

... (i)

From momentum conservation

$$mv \sin 30 = -mv_1 \sin 30 + mv_2$$

$$5 = -\frac{v_1}{2} + 2v_2$$

... (ii)

$$\text{Solving } v_1 = 2 \text{ m/s}$$

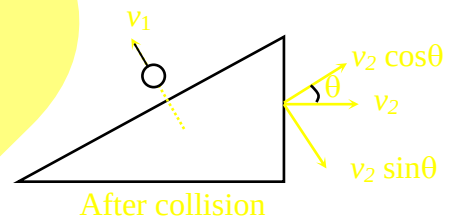
7. **7.64**

$$\text{Sol. } Fx - \mu m_1 gx - \frac{1}{2} kx^2 = 0$$

$$kx = \mu m_2 g \text{ for just shifting } m_2$$

$$Fx - \mu m_1 gx - \frac{1}{2} \mu m_2 gx = 0$$

$$F = \mu \left(m_1 + \frac{m_2}{2} \right) g = 0.4 \left(1 + \frac{2}{2} \right) (9.8) = 7.64 \text{ N}$$

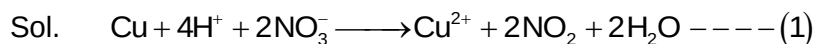
8. **1.50**Sol. $\rho h^2 = \text{constant}$ 

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SECTION – II : CHEMISTRY

PART – A

1. C



According to question

ΔG of equation (1) = ΔG of equation (2)

$\therefore E_{\text{Cell}}(\text{equation -1}) = E_{\text{Cell}}(\text{equation -2})$

or, $(E_{\text{cathode}} - E_{\text{anode}})(\text{equ}^n.1) = (E_{\text{cathode}} - E_{\text{anode}})(\text{equ}^n.2)$

$\therefore E_{\text{NO}_3^-/\text{NO}_2} - E_{\text{Cu}^{2+}/\text{Cu}} = E_{\text{NO}_3^-/\text{NO}} - E_{\text{Cu}^{2+}/\text{Cu}}$

$\therefore E_{\text{NO}_3^-/\text{NO}_2} = E_{\text{NO}_3^-/\text{NO}}$

$$\text{or, } E_{\text{NO}_3^-/\text{NO}_2}^0 - \frac{0.0591}{2} \log \frac{(p_{\text{NO}_2})^2}{[\text{H}^+]^4 [\text{NO}_3^-]^2} = E_{\text{NO}_3^-/\text{NO}}^0 - \frac{0.0591}{6} \log \frac{(p_{\text{NO}})^2}{[\text{NO}_3^-]^2 [\text{H}^+]^8}$$

$$\text{or, } 0.79 - \frac{0.0591}{2} \log \frac{(10^{-3})^2}{(x^4)(x^2)} = 0.96 - \frac{0.0591}{6} \log \frac{(10^{-3})^2}{(x^2)(x)^8}$$

On solving $x = 10^{0.66}$

2. C

Sol. (A) Number of corner atoms at four corners = 4. Effective number of atoms = $4 \times \frac{1}{4} = 1$

Each corner in the plane touches four unit cells.

(B) M_2 contains the face centre ions. $\therefore$ Formula will be changed.

(C) Incorrect as cations are present in tetrahedral voids.

(D) Close packing takes place along face diagonals.

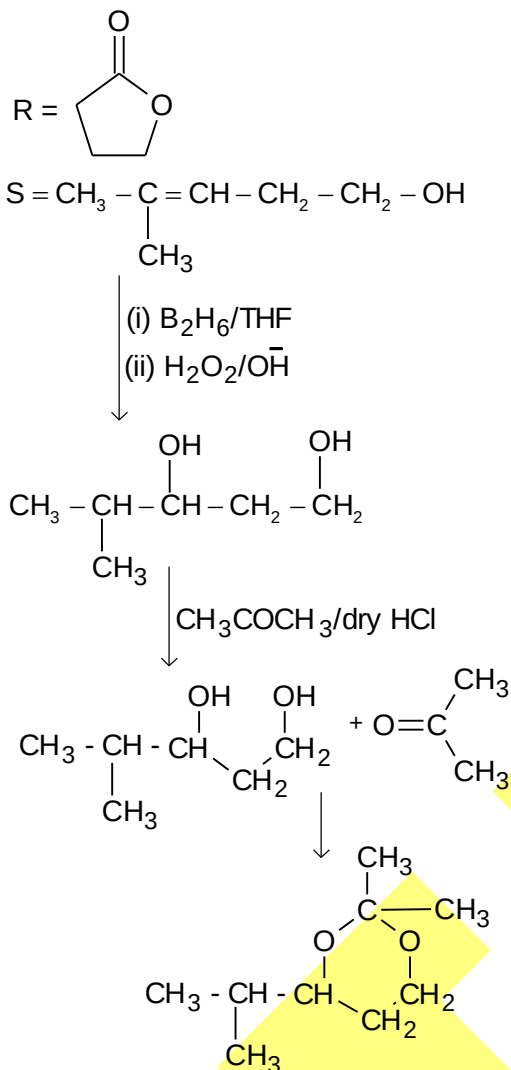
3. D

Sol. P is nucleotides and Q is nucleoside, R and S are bases.

4. C

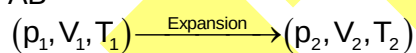
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Sol.

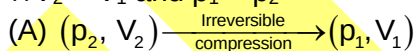


5. AB

Sol.



$$\therefore V_2 > V_1 \text{ and } p_1 > p_2$$



$$\text{Work done} = -p_1(V_1 - V_2)$$

$$\text{Since } V_2 > V_1, W_{\text{irr}} = +ve \text{ and } p_1 > p_2$$

This is called maximum work because p_{ext} is p_1 which is larger than p_2 .

$\therefore$ This is correct.

(B) For free expansion, $p_{\text{ext}} = 0$ and $W = 0$

In isothermal process

$$\Delta T = 0, \Delta U = 0$$

$$\Delta U = q - W$$

$$\text{or, } q - W = 0$$

Since q is zero for free expansion

$$W = 0, \text{ or } p_{\text{ext}}\Delta V = 0$$

$$\therefore p_{\text{ext}} = 0$$

In adiabatic process

$$\Delta U = q - W$$

$$\text{Since } q = 0, \Delta U = -W$$

$$\text{For free expansion, } \Delta U = 0$$

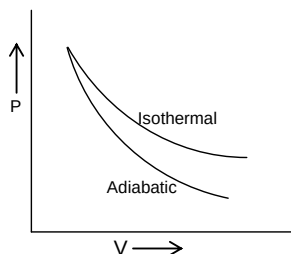
$$\therefore W = 0$$

$$p_{\text{ext}}\Delta V = 0, p_{\text{ext}} = 0$$

(C) is incorrect

Work done in isothermal process is higher than that in adiabatic process

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Work is area under the curve. So area under isothermal curve is higher than that in adiabatic curve.

(D) is incorrect

For isothermal process, the change in internal energy = 0 as $\Delta T = 0$

For adiabatic process, expansion of gas decreases internal energy, which is used to do work.

$\therefore \Delta U < 0$ in adiabatic process.

6. BC

Sol. Adsorption of O_2 on metal surface is called chemisorption. Since electrons are transferred from metal to oxygen, heat will be released as electron affinity of oxygen. The electrons are absorbed in the outermost molecular orbital (π_y^*) of O_2 .

7. ABC

Sol. (A) is incorrect, conduction of electricity taking place along the plane of graphite

(B) is incorrect, bond order of graphite = $\frac{4}{3}$

(C) is correct

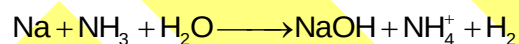
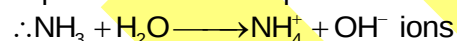
(D) is correct, carbon can lose and accept electrons.

8. ABCD

Sol. Probability depends on distance from the nucleus. The probability is maximum ($\psi^2 = 1$) around the nucleus and decreases away from it.

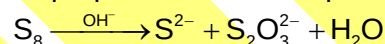
9. BD

Sol. Liquor ammonia is aqueous solution of NH_3 .



10. BC

Sol. Disproportionation of sulphur takes place



11. AD

Sol. For FeC_2O_4 ($n = 3$)

$$M_{eq} \text{ of } FeC_2O_4 = M_{eq} \text{ of } MnO_4^-$$

$$\text{or, } 100 \times 0.1 \times 3 = V \times 0.1 \times 5$$

$$\therefore V = 60 \text{ mL}$$

For $Fe_2(C_2O_4)_3$, $n = 6$

$$M_{eq} \text{ of } Fe_2(C_2O_4)_3 = M_{eq} \text{ of } MnO_4^-$$

$$\text{or, } 100 \times 0.1 \times 6 = V \times 0.1 \times 5$$

$$\therefore V = 120 \text{ mL}$$

12. ACD

Sol. Gas B is incorrectly represented. When $a = 0$, only positive deviation takes place.

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PART – B

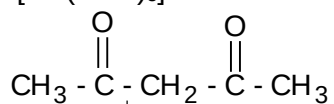
1. 6

Sol. Molar mass of decapeptide = 796 g mol^{-1} Decapeptide contains nine peptide linkages. So one mole of decapeptide absorbs 9 moles of H_2O for hydrolysis. $\therefore$ Mass of all amino acids obtained after hydrolysis = $796 + 18(9) = 958 \text{ g mol}^{-1}$

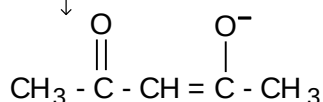
Glycine contribute = 47%

 $\therefore$ Mass of glycine = $\frac{47}{100} \times 958 = 450.26 \text{ g}$ Molar mass of glycine = 75 g mol^{-1} $\therefore$ Number of glycine molecules = $\frac{450.26}{75} = 6$

2. 4

Sol. In $[\text{Mn}(\text{acac})_3]$ the metal ion is Mn^{3+} and acac is a monodentate ligand.

Tautomerisation

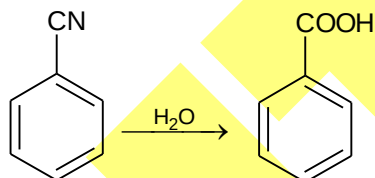
 Mn^{3+} contains $7 - 3 = 4$ unpaired electrons.

3. 5

Sol. C_6H_{14} has five chain isomers.

4. 60

Sol.

Molar mass of $\text{PhCN} = 103$ Mol. Mass of $\text{PhCOOH} = 122$ Moles of $\text{PhCN} = \frac{15.45}{103} = 0.15$ $\therefore$ 0.15 mole PhCN can produce 0.15 mole of PhCOOH or, 15.45 g PhCN can produce $(0.15 \times 122) = 18.3 \text{ g PhCOOH}$ $\therefore$ Product yield = $\frac{\text{Expt.yield}}{\text{Max.yield}} \times 100 = \frac{10.98}{18.3} \times 100 = 60\%$

5. 4

Sol. It produces two geometrical and two linkage isomers.

6. 24

Sol. Sp rotation of R-enantiomer = 48° Sp rotation of S-enantiomer = -48°

25% S isomer contains 75% R isomer

 $\therefore$ 100% R $\rightarrow 48^\circ$ $\therefore$ 75% R $\rightarrow \frac{48}{100} \times 75 = 36^\circ$ 25% S $\rightarrow \frac{-48}{100} \times 25 = -12$ $\therefore$ Sp rotation of sample = $36 - 12 = 24^\circ$ *Telegram ~ @bohring_bot*

7. 2.5

Sol. Total number of bonds = 8
Number of coordinate bonds = 2
 $\therefore \% \text{ of coordinate bonds} = \frac{2}{8} \times 100 = 25\%$

$$\frac{x}{10} = \frac{25}{10} = 2.5$$

8. 1.2

Sol. $a = 5, b = 5, c = 2$
 $\frac{a+b+c}{10} = \frac{12}{10} = 1.2$

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SECTION – III : MATHEMATICS

PART – A

1. C

$$\begin{aligned} \text{Sol. } \lim_{x \rightarrow 0} \left(\frac{g(2 + \sin x) - g(2 + x \cos x)}{\sin x - x \cos x} \right) & \cdot \left(\frac{\sin x - x \cos x}{x - \sin x} \right) \\ &= g'(2) \lim_{x \rightarrow 0} \left(\frac{\sin x - x \cos x}{x - \sin x} \right) \left(\frac{0}{0} \right) \\ &= g'(2) \lim_{x \rightarrow 0} \left(\frac{x \sin x}{1 - \cos x} \right) = 2g'(2) \end{aligned}$$

2. B

$$\begin{aligned} \text{Sol. } \int \frac{2x+1}{(x^2+4x+1)^{3/2}} dx &= \int \frac{2x+1}{x^3 \left(1 + \frac{4}{x} + \frac{1}{x^2} \right)^{3/2}} dx = \int \frac{2x^{-2} + x^{-3}}{\left(1 + \frac{4}{x} + \frac{1}{x^2} \right)^{3/2}} dx \\ \text{Now put } \frac{1}{x^2} + \frac{4}{x} + 1 &= t^2. \end{aligned}$$

3. B

$$\begin{aligned} \text{Sol. } f'(x) &= \frac{1}{2} \int_0^x 2(x-t)g(t)dt + \frac{1}{2}(x-x)^2 g(x) - 0 \\ &= \int_0^x (x-t)g(t)dt \\ f''(x) &= \int_0^x g(t)dt + (x-x)g(x) - 0 \\ f'''(x) &= g(x) \\ f'''(1) - f''(1) &= g(1) - \int_0^1 g(t)dt \\ &= 5 - 2 = 3 \end{aligned}$$

4. A

$$\text{Sol. Required sum of coefficient of } x^{10} \text{ in the expansion of } (1+x)^{30} (1+x)^{20} (1+x)^{10} \text{ i.e. } {}^{60}C_{10}.$$

5. BC

$$\text{Sol. } m_{AC} = \frac{-1}{3} = \tan \theta$$

$$\begin{aligned} \tan A &= \tan(90 + 180 - \theta) \\ &= \tan(270 - \theta) = \cot \theta = -3 \\ &\dots\dots(1) \end{aligned}$$

$$\tan C = \left| \frac{m_{BC} - m_{AC}}{1 + m_{BC} m_{AC}} \right| = \left| \frac{-3 + \frac{1}{3}}{1 + 1} \right| = \frac{4}{3}$$

.....(2)

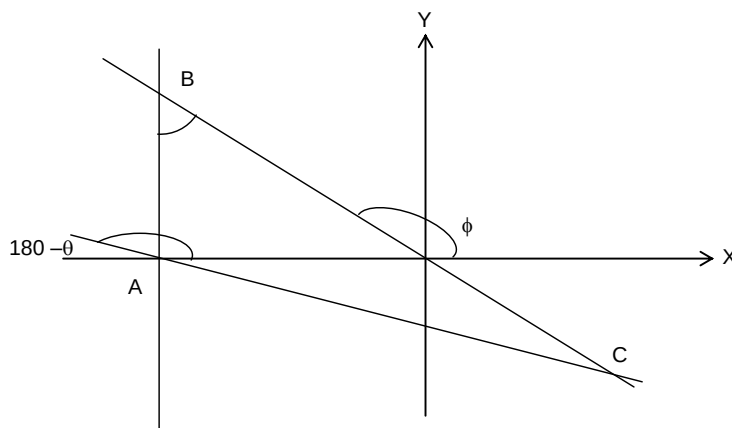
$$\tan \phi = -3$$

$$B = \phi - 90^\circ$$

$$\tan B = \tan(\phi - 90) = -\cot \phi = \frac{1}{3}$$

$$\text{So } \tan A + \tan B + \tan C = -3 + \frac{1}{3} + \frac{4}{3} = -\frac{4}{3}$$

Now check Others.



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6. AC

Sol. $\triangle ABC$ is isosceles right angled at A
 Let P be (h, k)
 Let ABC be triangle as shown in figure
 Equation BC : $y = 0$

$$AB : y - x = 0$$

$$AC : x + y - 4 = 0$$

Given condition

$$k^2 = \frac{1}{2} \frac{|h+k-4|}{\sqrt{1+1}} \cdot \frac{|k-h|}{\sqrt{1+1}}$$

Replace (h, k) by (x, y)

$$\Rightarrow y^2 = |(x+y-4)(y-x)|$$

$$= -\frac{1}{4}(x+y-4)(x-y)$$

$$\Rightarrow 4y^2 = -(x^2 - y^2 - 4(x-y))$$

$$x^2 + 3y^2 - 4x + 4y = 0$$

$$(x-2)^2 + 3\left(y^2 + \frac{4}{3}y + \frac{4}{9}\right) = 4 + \frac{4}{3}$$

$$(x-2)^2 + 3\left(y + \frac{2}{3}\right)^2 = \frac{16}{3}$$

$$\frac{(x-2)^2}{\frac{16}{3}} + \frac{\left(y + \frac{2}{3}\right)^2}{\frac{16}{9}} = 1$$

$$\frac{16}{9} = \frac{16}{3}(1-c^2)$$

$$\frac{1}{3} = 1-c^2 \Rightarrow c = \sqrt{\frac{2}{3}}$$

$$D = \left(2, -\frac{2}{3}\right)$$

$$\text{Area} = \text{Ar}(\triangle ABC) + \text{Ar}BCD$$

$$= \frac{1}{2} \cdot 4 \cdot 2\sqrt{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times 4 \times \frac{2}{3}$$

$$= 4 + \frac{4}{3} = \frac{16}{3}$$

7. ABCD

Sol. $|\vec{a} + \vec{b} - \vec{c}|^2 + |\vec{b} + \vec{c} - \vec{a}|^2 + |\vec{c} + \vec{a} - \vec{b}|^2 = 36$

$$\Rightarrow 3(\vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c}) - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{a} \cdot \vec{c} = 36$$

$$\Rightarrow 3(3+3+3) - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 36$$

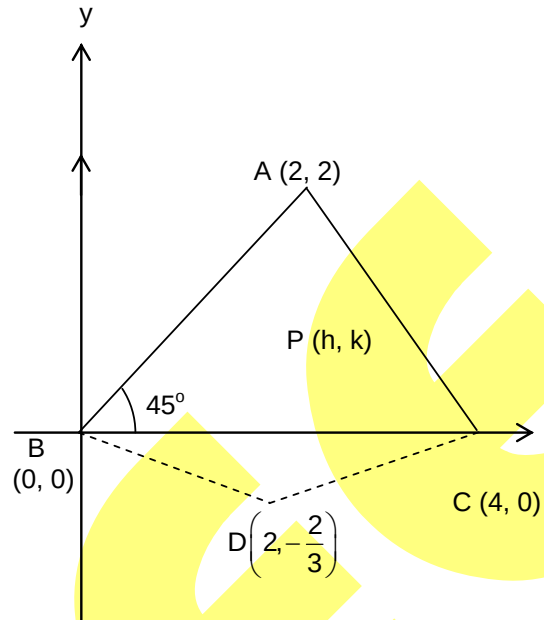
$$\Rightarrow \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-9}{2}$$

$$2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} + 9 = 0$$

$$2\vec{a} \cdot \vec{b} + 2\vec{b} \cdot \vec{c} + 2\vec{c} \cdot \vec{a} + \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{c} \cdot \vec{c} = 0$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}|^2 = 0$$

$$\Rightarrow \vec{a} + \vec{b} + \vec{c} = 0$$



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Now verify (B), (C), (D)

8. ABCD

Sol. $|\sin x| + |\cos x| \geq 1 \forall x \in \mathbb{R}$

Note $1 \leq |\sin x| + |\cos x| < \sqrt{2}$ so $x \in \mathbb{R}$

$\Rightarrow$ a, b, c, d all are correct.

9. AC

$$\begin{aligned} \text{Sol. } S_n &= \sum_{r=1}^n \frac{2r+3}{r(r+1)3^r} = \sum_{r=1}^n \frac{3(r+1)-r}{r(r+1)3^r} = \sum_{r=1}^n \frac{1}{r3^{r-1}} - \frac{1}{(r+1)3^r} \\ &= 1 - \frac{1}{(n+1)3^n} \end{aligned}$$

10. ABC

$$\text{Sol. } f(x) = \tan^{-1}\left(x^4 - x^2 + \frac{1}{4} - 2 + \tan^{-1} \alpha\right) = \tan^{-1}\left(\left(x^2 - \frac{1}{2}\right)^2 + \tan^{-1} \alpha - 2\right)$$

For 'f' to be surjective, $\tan^{-1} \alpha - 2 = -1 \Rightarrow \alpha = \tan 1$

Now, verify the options.

11. ABD

Sol. When $k = 0$

$$\begin{aligned} \text{Lt}_{x \rightarrow \infty} \frac{2x^3 - (\tan^{-1} x)^3}{\frac{8}{\pi} x^3 \cdot \frac{\pi}{2} + 0 + 0} &= \text{Lt}_{x \rightarrow \infty} \frac{2 - \left(\frac{\tan^{-1} x}{x}\right)^3}{4} = 2 - 1 \\ &= \frac{2-0}{4} = \frac{1}{2} \end{aligned}$$

So k can be 0.

$$\begin{aligned} \text{If } k \neq 0 \text{ then } \text{Lt}_{x \rightarrow \infty} \frac{2x^3 - (\tan^{-1} x)^3}{x^3 \left[\frac{8}{\pi} \cot^{-1} |kx| + k^2 \frac{\sin \frac{1}{x^3}}{\frac{1}{x^3}} - 3k \right]} \\ \text{Lt}_{x \rightarrow \infty} \frac{2 - \left(\frac{\tan^{-1} x}{x}\right)^3}{\frac{8}{\pi} \cot^{-1}(kx) + \frac{k^2 \sin^{-1} \frac{1}{x^3}}{\frac{1}{x^3}} - 3k} = \frac{2-0}{\frac{8}{\pi} \times 0 + k^2 - 3k} = \frac{1}{2} \end{aligned}$$

$$\Rightarrow k^2 - 3k = 4$$

$$\Rightarrow k = 4, -1$$

12. ABD

$$\text{Sol. } x^2 \frac{dy}{dx} = y^2 e^{\frac{1}{x}} \Rightarrow \frac{dy}{y^2} = \frac{e^{\frac{1}{x}}}{x^2} dx$$

Integrating both sides,

$$-\frac{1}{y} = \int \frac{e^{\frac{1}{x}}}{x^2} dx + C \Rightarrow \text{Putting } \frac{1}{x} = t \Rightarrow -\frac{1}{x^2} dx = dt \Rightarrow -\frac{1}{y} = \int e^t dt + C \Rightarrow$$

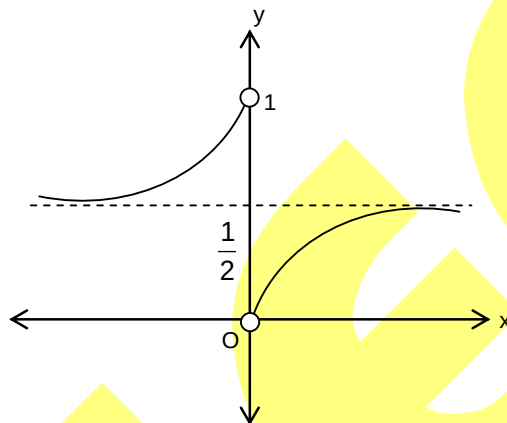
$$\frac{-1}{y} = e^{-t} + C \Rightarrow \frac{-1}{y} = -e^{\frac{1}{x}} + C$$

$$\therefore \lim_{x \rightarrow 0^-} f(x) = 1 \Rightarrow -1 = 0 + C \Rightarrow C = -1$$

$$\frac{-1}{y} = -e^{\frac{1}{x}} - 1 \Rightarrow y = \frac{1}{1 + e^{\frac{1}{x}}} \Rightarrow \frac{dy}{dx} = \frac{1}{\left(1 + e^{\frac{1}{x}}\right)^2} e^{\frac{1}{x}} \left(\frac{-1}{x^2}\right) = -\frac{e^{\frac{1}{x}}}{x^2 \left(1 + e^{\frac{1}{x}}\right)^2}$$

$$\therefore \frac{dy}{dx} > 0 \forall x \in \mathbb{R} - \{0\}$$

$$\lim_{x \rightarrow \pm\infty} \frac{1}{1 + e^{\frac{1}{x}}} = \frac{1}{2} \text{ and } \lim_{x \rightarrow 0^+} \frac{1}{1 + e^{\frac{1}{x}}} = 0$$



$\therefore$ Graph of the function is
From the graph options (A) (B) and (D) are correct.

PART - B

1. 6

Sol.

$$\frac{\cos \frac{B-C}{2} \cdot \cos \frac{A}{2}}{\cos \left(90 - \frac{A}{2}\right) \cos \frac{A}{2}} + \frac{\cos \left(\frac{C-A}{2}\right) \cos \frac{B}{2}}{\cos \left(90 - \frac{B}{2}\right) \cos \frac{B}{2}} + \frac{\cos \left(\frac{A-B}{2}\right) \cos \frac{C}{2}}{\cos \left(90 - \frac{C}{2}\right) \cos \frac{C}{2}}$$

$$\frac{2 \cos \frac{B-C}{2} \sin \frac{B+C}{2}}{2 \sin \frac{A}{2} \cos \frac{A}{2}} + \frac{2 \cos \left(\frac{C-A}{2}\right) \sin \left(\frac{C+A}{2}\right)}{2 \sin \frac{B}{2} \cos \frac{B}{2}} + \frac{2 \cos \left(\frac{A-B}{2}\right) \sin \left(\frac{A+B}{2}\right)}{2 \sin \frac{C}{2} \cos \frac{C}{2}}$$

$$\frac{\sin B + \sin C}{\sin A} + \frac{\sin A + \sin C}{\sin B} + \frac{\sin A + \sin B}{\sin C}$$

$$\frac{\sin B}{\sin A} + \frac{\sin C}{\sin A} + \frac{\sin A}{\sin B} + \frac{\sin C}{\sin B} + \frac{\sin A}{\sin C} + \frac{\sin B}{\sin C} \geq 6$$

2. 1

Sol. $[f(\alpha)]^2 + [f'(\alpha)]^2 = 0$ both $f(\alpha)$ and $f'(\alpha)$ are real so $f(\alpha) = 0 = f'(\alpha)$

$$\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left[\frac{f'(x)}{f(x)} \right]$$

$$\lim_{x \rightarrow \alpha} \frac{f(x)}{f'(x)} \left(\frac{f'(x)}{f(x)} - \left\{ \frac{f'(x)}{f(x)} \right\} \right) \quad \{.\} \text{ represents fraction part function}$$

$$= \lim_{x \rightarrow \alpha} 1 - \frac{f(x)}{f'(x)} \left\{ \frac{f'(x)}{f(x)} \right\}$$

$$1 - 0 \times \text{any number is } [0, 1)$$

3. 1

Sol. $\alpha + \beta = p$

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$$\alpha\beta = -p - c$$

$$\alpha\beta = -(\alpha + \beta) - c$$

$$\Rightarrow \alpha + \beta + \alpha\beta = -c$$

$$\alpha + \beta + \alpha\beta + 1 = -c + 1$$

$$\Rightarrow (\alpha + 1)(\beta + 1) = 1 - c$$

$$\frac{(\alpha + 1)^2}{(\alpha + 1)^2 + c - 1} + \frac{(\beta + 1)^2}{(\beta + 1)^2 + c - 1}$$

$$\Rightarrow \frac{(\alpha + 1)^2}{(\alpha + 1)^2 - (\alpha + 1)(\beta + 1)} + \frac{(\beta + 1)^2}{(\beta + 1)^2 - (\alpha + 1)(\beta + 1)}$$

$$= \frac{\alpha + 1}{\alpha - \beta} + \frac{\beta + 1}{\beta - \alpha} = 1$$

4. 94

Sol. $m = {}^5C_4 \cdot 4! = 5! = \text{Total}$

When exactly 2 elements of A maps to itself i.e. $f(3) = 3, f(4) = 4$

$\therefore$ From 5, 6, 7 select any 2 in ${}^3C_2 \times 2! = 6$

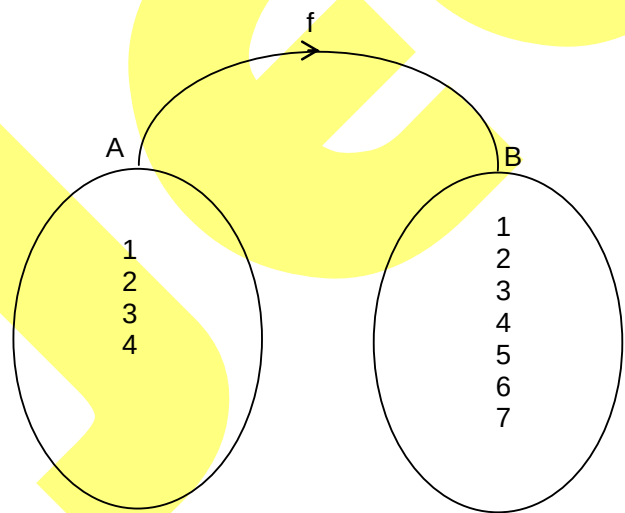
When exactly one element of A maps to itself say $f(3) = 3$

Now 4 can be map in 3 ways and remaining elements $3 \times 2 = 6$

$$\therefore {}^2C_1 \times 3 \times 6 = 36$$

$$\text{Total} = 36 + 6 = 42 \Rightarrow 5! - 42 = 78 = m$$

$$n = {}^2C_1 \cdot {}^2C_1 \cdot {}^2C_1 \cdot {}^2C_1 = 16$$



5. 5

Sol. Let $AF = x = DE$ and $AE = y = DF$

As $\triangle CAB$ is $\triangle CED$

$$\text{So, } \frac{CE}{CA} = \frac{DE}{AB} \Rightarrow \frac{b-y}{b} = \frac{x}{c} \Rightarrow y = b \left(1 - \frac{x}{c} \right)$$

(Here $BC = a$, $AC = b$ and $AB = c$)

Now, area of parallelogram AFDE

$$= S = (AF)(EM) = xy \sin A \Rightarrow S = x \cdot b \left(1 - \frac{x}{c} \right) \sin A$$

Note: $\sin A$ is fixed ... (i)

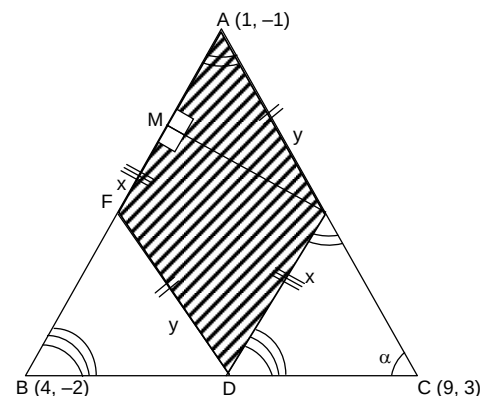
Now, differentiating both sides of equation (i) with respect to x , we get

$$\frac{dS}{dx} = \frac{b}{c} (c - 2x) \sin A = 0 \Rightarrow x = \frac{c}{2}$$

$$\text{Also, } \left. \frac{d^2S}{dx^2} \right|_{x=\frac{c}{2}} = \frac{-2b}{c} < 0$$

So, S is maximum when $x = \frac{c}{2}$.

$$\text{Now, } S_{\max} = \frac{1}{4} bc \sin A$$



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$$= \frac{1}{2} \left(\frac{1}{2} bc \sin A \right) = \frac{1}{2} [\text{area}(\triangle ABC)] = \frac{1}{4} \begin{vmatrix} 1 & -1 & 1 \\ 4 & -2 & 1 \\ 9 & 3 & 1 \end{vmatrix} = \frac{20}{4} = 5$$

(square units.)

6. 3

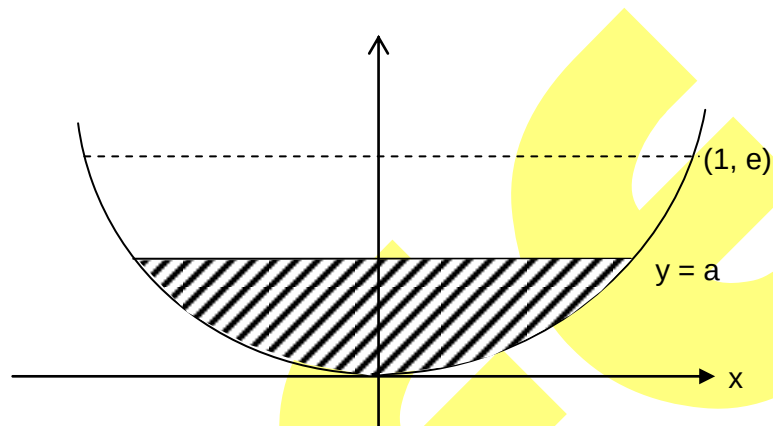
Sol. $f(x) = \begin{cases} xe^x, & x \geq 0 \\ -xe^{-x}, & x \leq 0 \end{cases}$

For maximum area $a = e$

$$A_{\max} = 2 \int_0^1 (e - xe^x) dx$$

$$= 2 \left[ex - (xe^x - e^x) \right]_0^1$$

$$= 2(e - 1)$$

Hence, $[A] = 3$ 

7. 1

Sol. $\sum_{r=0}^7 A_r = \begin{bmatrix} \sum_{r=0}^7 r({}^7C_r) & 0 \\ 0 & \sum_{r=0}^7 (7-r){}^7C_r \end{bmatrix}$

$$= \begin{bmatrix} 7 \cdot 2^6 & 0 \\ 0 & 7 \cdot 2^7 - 7 \cdot 2^6 \end{bmatrix}$$

$$= \begin{bmatrix} 7 \cdot 2^6 & 0 \\ 0 & 7 \cdot 2^6 \end{bmatrix}$$

$$a + b = 2 \cdot 7 \cdot 2^6 = 14 \times 64 = 896$$

8. 5

Sol. $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$
 $\frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)} + \frac{a}{30} = 0$
 $\Rightarrow \frac{x-3}{x+4} + \frac{9}{30} = 0, x \neq 1, 2$

Above equation do not have solution then $-\frac{a}{30}$ should not belong to range of

$$y = \frac{x-3}{x+4}, x \neq 1, 2$$

$$\text{Range of above is } R - \left\{ 1, \frac{-2}{5}, -\frac{1}{6} \right\}$$

$$\text{So, } \frac{-a}{30} = 1, \frac{-2}{5}, -\frac{1}{6}$$

$$\Rightarrow a = -30, 12, 5$$

So least value of $a = 5$

$$-\frac{a}{30} = \frac{x-3}{x+4}$$

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FIITJEE RBT-2 for (JEE-Advanced)

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 4

QP CODE: 100175

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

- Attempt ALL the questions. Answers have to be marked on the OMR sheets.
- This question paper contains **Three Sections**.
- Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
- Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
- Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
- Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

- Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
- On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
- OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Part.

- (i) **PART-A (01-08)** contains (8) Multiple Choice Questions which have **One or More Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (09-12)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **-1 marks** for wrong options chosen.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

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SECTION – I : PHYSICS

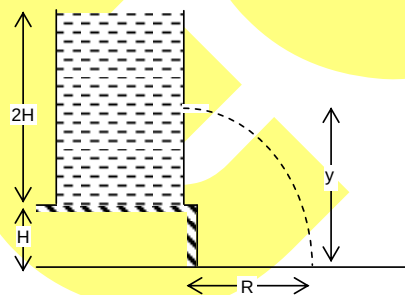
(PART – A)

(One or More Than One Options Correct Type)

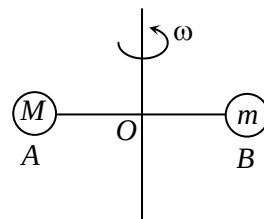
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. An artificial satellite is in a circular orbit around the earth. The universal gravitational constant starts decreasing at time $t = 0$ at a constant rate with respect to time t . Then the satellite has its
 (A) path gradually spiralling out away from the centre of the earth.
 (B) path gradually spiralling in towards the centre of the earth.
 (C) angular momentum about the centre of the earth remains constant.
 (D) potential energy increases.

2. A tank is filled upto a height $2H$ with a liquid and is placed on a platform of height H from the ground. The distance y from the ground where a small hole is made in the tank, to get the maximum horizontal range R is
 (A) $2H$
 (B) $3H/2$
 (C) $5H/4$
 (D) H



3. Two balls of mass $M = 9\text{ g}$ and $m = 3\text{ g}$ are attached by massless threads AO and OB . The length AB is 1 m . They are set in rotational motion in a horizontal plane about a vertical axis at O with constant angular velocity ω . The ratio of length AO and OB $\left(\frac{AO}{OB}\right)$ for which the tension in threads are same will be



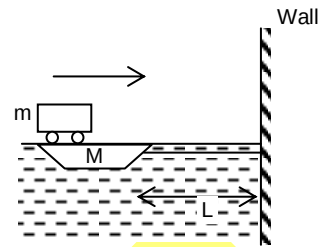
- (A) $\frac{1}{3}$ (B) 3 (C) $\frac{2}{3}$ (D) $\frac{3}{2}$
4. A particle moves along positive branch of the curve $y = \frac{x}{2}$ where $x = \frac{t^3}{3}$, x and y are measured in metres and t in seconds, then:
 (A) The velocity of particle at $t = 1\text{ s}$ is $\hat{i} + \frac{1}{2}\hat{j}$
 (B) The velocity of particle at $t = 1\text{ s}$ is $\frac{1}{2}\hat{i} + \hat{j}$
 (C) The acceleration of particle at $t = 1\text{ s}$ is $2\hat{i} + \hat{j}$
 (D) The acceleration of particle at $t = 2\text{ s}$ is $\hat{i} + 2\hat{j}$

Space For Rough Work

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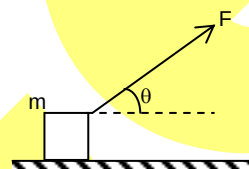
5. A car of mass m is initially at rest on the boat of mass M tied to the wall of dock through a massless inextensible string. The car accelerates from rest to velocity v_0 in times t_0 . At $t = t_0$ the car applies brake and comes to rest relative to the boat in negligible time. Neglect friction between the boat and water; the time ' t ' at which boat will strike the wall is

- (A) $t_0 + (L/v_0)$ (B) $t_0 + \frac{L(M+m)}{mv_0}$
 (C) $\frac{LM}{mv_0}$ (D) none of these



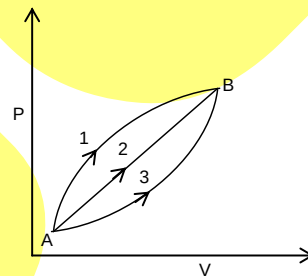
6. A force F acts on a smooth block of mass m placed on a horizontal floor at an angle θ with horizontal. On the block

- (A) the net force = $F \cos \theta$ if $F \sin \theta < mg$
 (B) acceleration = $\frac{F \cos \theta}{m}$ when $F > mg \operatorname{cosec} \theta$
 (C) acceleration = $\frac{F}{m}$ if $F \sin \theta > mg$
 (D) $N = mg - F \sin \theta$ if $F \sin \theta < mg$



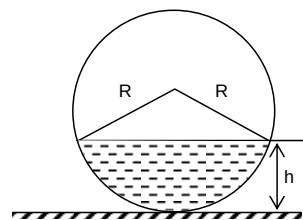
7. In figure, a certain mass of gas traces three paths 1,2,3 from state A to state B. If work done by the gas along three paths are W_1, W_2, W_3 respectively, then

- (A) $W_1 < W_2 < W_3$
 (B) $W_1 = W_2 = W_3$
 (C) $W_1 > W_2 > W_3$
 (D) cannot say



8. A liquid is filled in a spherical container of radius R till a height h . At this position the liquid surface at the edges is also horizontal. The contact angle is

- (A) 0 (B) $\cos^{-1}\left(\frac{R-h}{R}\right)$
 (C) $\cos^{-1}\left(\frac{h-R}{R}\right)$ (D) $\sin^{-1}\left(\frac{R-h}{R}\right)$



Space For Rough Work

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

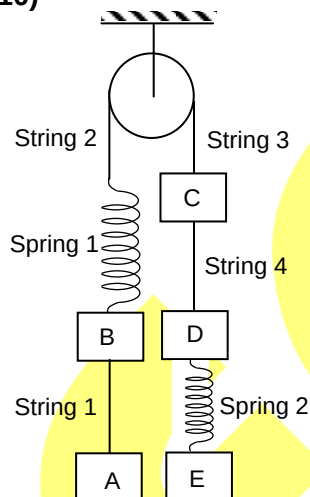
List Match Set (09-10)

The system shown below is initially in equilibrium

$$m_A = m_B = 3\text{kg}$$

$$m_C = m_D = m_E = 2\text{kg}$$

$$\text{Take } g = 10 \text{ m/s}^2$$



List-I gives the four strings while List-II the value of the tension in the strings.

List – I		List – II	
(I)	String 1	(P)	10 N
(II)	String 2	(Q)	20 N
(III)	String 3	(R)	30 N
(IV)	String 4	(S)	40 N
		(T)	60 N
		(U)	0 N

9. If the spring 1 is cut, match the correct option for the tension in the strings just after the cutting.
- (A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ P (B) I $\rightarrow$ U, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ P
 (C) I $\rightarrow$ U, II $\rightarrow$ U, III $\rightarrow$ U, IV $\rightarrow$ P (D) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ P
10. If the spring 2 is cut, match the correct option for the tension in the strings just after the cutting.
- (A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ R (B) I $\rightarrow$ R, II $\rightarrow$ U, III $\rightarrow$ U, IV $\rightarrow$ R
 (C) I $\rightarrow$ U, II $\rightarrow$ R, III $\rightarrow$ R, IV $\rightarrow$ P (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ R

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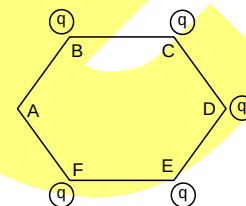
List Match Set (11-12)

If some charges are placed at the corners of a regular hexagon and E is the electric field due to one charge at the centre.

List – I		List – II	
(I)	If charge at B is removed, then electric field at centre become	(P)	Zero
(II)	If charge at C is removed, then electric field at centre become.	(Q)	$2E$
(III)	If charge at D is removed, then electric field at centre become.	(R)	$\frac{2E}{\sqrt{3}}$
(IV)	If charge at B and C both are removed, then electric field at centre become	(S)	$\sqrt{3} E$
		(T)	$\frac{E}{2}$
		(U)	E

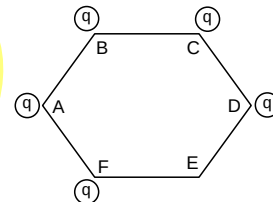
11. If five charges are kept at five vertices of regular hexagon as shown, then

- (A) I $\rightarrow$ S, II $\rightarrow$ U, III $\rightarrow$ Q, IV $\rightarrow$ P
 (B) I $\rightarrow$ S, II $\rightarrow$ U, III $\rightarrow$ P, IV $\rightarrow$ Q
 (C) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ U
 (D) I $\rightarrow$ Q, II $\rightarrow$ U, III $\rightarrow$ P, IV $\rightarrow$ S



12. If five charges are kept at five vertices of regular hexagon as shown, then

- (A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ U
 (B) I $\rightarrow$ P, II $\rightarrow$ U, III $\rightarrow$ R, IV $\rightarrow$ S
 (C) I $\rightarrow$ P, II $\rightarrow$ U, III $\rightarrow$ S, IV $\rightarrow$ U
 (D) I $\rightarrow$ Q, II $\rightarrow$ U, III $\rightarrow$ S, IV $\rightarrow$ R



(PART – B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

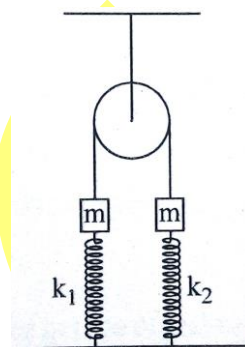
1. At a distance of 20 m from a point source, the loudness is 30 dB. After a minimum distance of $k(300\sqrt{10})$ metre the sound is inaudible. Value of 'k' is

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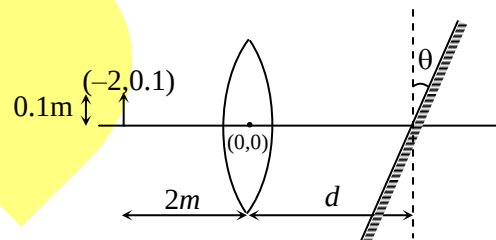
2. A proton is fired from very far away towards a nucleus with charge $= 120 e$, where e is the electronic charge. It makes a closest approach of 10 fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is: (take the proton mass, $m_p = (5/3) \times 10^{-27} \text{ kg}$; $h/e = 4.2 \times 10^{-15} \text{ J.s./C}$; $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ m/F}$; $1 \text{ fm} = 10^{-15} \text{ m}$)

3. On the diagram, two blocks of equal mass are connected by an ideal string. The values of $m = 1 \text{ kg}$ and $k_2 = 100 \text{ N/m}$ are given. Initially, both springs are relaxed. Then the left block is slowly pulled down a distance 0.1 m and released. The maximum possible value of k_1 (in N/m) for which both blocks will have same magnitude of acceleration just after releasing, is k . Both spring have natural length of 2m . Then find the value of $\frac{k}{100}$.



4. A cube of wood supporting 200 gm mass just floats in water. When the mass is removed, the cube is raised by 2 cm . The size of the cube is 20 k cm , then the value of k is
5. Two vibrating strings of the same material but of lengths L and $2L$ have radii $2r$ and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the one of length L with frequency f_1 and other with frequency f_2 . Find the ratio $\frac{f_1}{f_2}$?

6. A convex lens of focal length 1.5m is placed in a system of coordinate axis such that its optical centre is at origin and principal axis coinciding with the x -axis. An object and a plane mirror are arranged on the principal axis as shown in figure. Find the value of d (in m) so that y -coordinate of final image (after refraction and reflection) is 0.3m . (Take $\tan \theta = 0.3$)



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SECTION – II : CHEMISTRY

(PART – A)

(One or More Than One Options Correct Type)

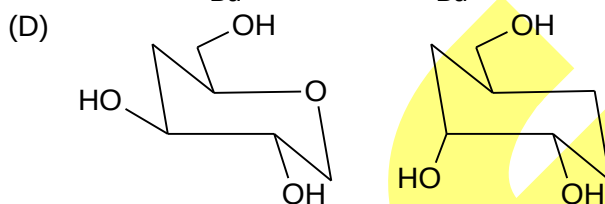
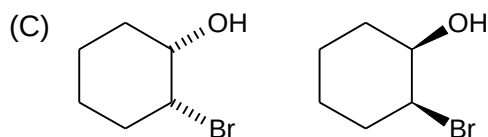
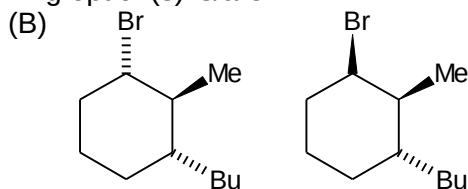
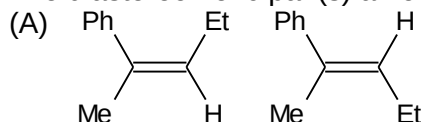
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

- Compound/s which contain $3c - 2e^-$ bond.
 (A) $(BeH_2)_n$ (B) $Al_2(CH_3)_6$
 (C) B_2H_6 (D) Al_2Cl_6
- $\psi(x, y, z)$ describe the wave function of a particle the probability of finding the particle between x and $x + dx$, y and $y + dy$, z and $z + dz$ can be expressed as
 (A) $\psi^*(x, y, z) \psi(x, y, z)$ (B) $|\psi(x, y, z)|^2 dx dy dz$
 (C) $\psi^*(x, y, z) \psi(x, y, z) dx dy dz$ (D) $\int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz \psi^*(x, y, z) \psi(x, y, z)$
- In water, the enthalpy of a protein in its folded state (H_f) is lower than that in its unfolded state (H_{uf}). The entropies of the folded and unfolded states are S_f and S_{uf} respectively. The condition(s) under which this protein spontaneously folds in water at a temperature T is(are)
 (A) $S_{uf} < S_f$ (B) $S_{uf} = 0$
 (C) $S_{uf} = S_f$ (D) $(S_f - S_{uf}) > H_f - H_{uf}/T$
- The transition metal complex(es) with zero magnetic moment, zero dipole moment and CFSE of $-2.4\Delta_0$ is(are)
 (A) $[Mn(CO)_5(CH_3)]$ (B) $[Trans-Ni(ethylenediamine)_2Cl_2]$
 (C) $[trans-Co(CN)_4(H_2O)_2]^-$ (D) $[Trans-Fe(CN)_4Cl_2]^{4-}$
- The organometallic reagent(s) among the following is/are
 (A) lithium divinylcuprate (B) lithium diisopropylamide
 (C) potassium tert-butoxide (D) isopropyl magnesiumiodide
- The correct statement(s) among the following is/are
 (A) secondary structure of a polypeptide describes the number and type of amino acid residues
 (B) uracil is a pyrimidine nucleobase
 (C) natural fatty acids have odd number of carbon atoms
 (D) reaction of D-glucose with $Ca(OH)_2$ gives a product mixture containing (D)-fructose, D-mannose and (D)-glucose

Space For Rough Work

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7. The diastereomeric pair(s) among the following option(s) is/are



8. With reference to the variation of molar conductivity (Λ_m) with concentration for a strong electrolyte in aq. solution, the incorrect statement is

- (A) the asymmetric effect contributes to decrease Λ_m whereas the electrophoretic effect contributes to increase Λ_m .
 (B) the asymmetric effect contributes to increase Λ_m whereas the electrophoretic effect contributes to decrease Λ_m .
 (C) both asymmetry effect and electrophoretic effect contribute to decrease Λ_m .
 (D) both asymmetry effect and electrophoretic effect contribute to increase Λ_m .

This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Match the lists and answer the question.

List- I		List - II	
(I)	One mole of nitrogen gas contained in a cylinder at 300 K is allowed to expand isothermally against an external pressure of 5 atm	(P)	$w > 0$
(II)	3 moles of helium gas at 1 atm are compressed reversibly and isothermally at 400 K to 5 atm pressure	(Q)	$\Delta U = 0$
(III)	An ideal gas is compressed reversibly and adiabatically from a volume of 1 dm^3 to a volume of 0.25 dm^3	(R)	$\Delta H = 0$
(IV)	One mole of liquid benzene is converted reversibly into vapour at its boiling point by heating it	(S)	$\Delta H > \Delta U$

9. Which combination is correct according to the above list?

- (A) I $\rightarrow$ P
 (B) II $\rightarrow$ S
 (C) III $\rightarrow$ PS
 (D) IV $\rightarrow$ PR

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10. Which combination is correct according to the above list?
 (A) I $\rightarrow$ QR (B) II $\rightarrow$ QS (C) III $\rightarrow$ Q (D) IV $\rightarrow$ PR

List Match Set (11-12)

Match the lists and answer the question.

When we titrate sodium carbonate solution(in beaker) with hydrochloric acid

List- I		List - II	
(I)	At the start of titration	(P)	Buffer solution of HCO_3^- & CO_3^{2-}
(II)	Before the first equivalent point	(Q)	Buffer solution of H_2CO_3 & HCO_3^-
(III)	At the first equivalent point	(R)	Amphiprotic anion $\text{pH} = \frac{1}{2}(\text{pK}_{a_1} + \text{pK}_{a_2})$
(IV)	Between the first and second equivalent points	(S)	Hydrolysis of CO_3^{2-}

11. Which combination is correct according to the above list?
 (A) I $\rightarrow$ S (B) II $\rightarrow$ R (C) III $\rightarrow$ Q (D) IV $\rightarrow$ P
12. Which combination is correct according to the above list?
 (A) I $\rightarrow$ P (B) II $\rightarrow$ R (C) III $\rightarrow$ S (D) IV $\rightarrow$ Q

(PART - B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

- The enthalpy of formation for $\text{CH}_4(\text{g})$, $\text{C}(\text{g})$ and $\text{H}(\text{g})$ are -75, 717 and 218 kJ mol^{-1} respectively the enthalpy of the C - H bond in kJ mol^{-1} is _____
- The number of P - H bonds in Hypophosphoric acid is _____
- Number of vertices in an icosahedral dodeca borane is _____
- The number of gauche butane-interaction(s) in cis-1, 2-dimethyl cyclohexane the following compound is _____
- 5.3 g of benzaldehyde was reacted with excess of acetophenone to produce 5.2 g of enone product as per the reaction shown below the yield of the reaction is _____% (round off to the nearest integer)
- Assume that the reaction of MeMgBr with ethylacetate proceeds with 100% conversion to give tert butanol. The volume of 0.2 M solution of MeMgBr required to convert 10 mL of a 0.025 M solution of ethylacetate to tert butanol is _____mL (Round off to one decimal place)

Space For Rough Work

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SECTION – III : MATHEMATICS**(PART – A)****(One or More Than One Options Correct Type)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

- Consider the function $f(x) = xe^x + (xe^x)^{-1}$ where $x \in \mathbb{R} - \{0\}$ then which of following statement(s) is CORRECT?
 (A) $f(x)$ attains it's local maxima at $x = x_0$ where $x_0 \in (0,1)$
 (B) $f(x)$ attains local minima at $x = -1$
 (C) $f(x)$ attains local minima at $x = x_0$ where $x_0 \in (1,\infty)$
 (D) $f(x)$ attains it's local minima at $x = x_0$ where $x_0 \in (0,1)$
- $f : \mathbb{R} \rightarrow \mathbb{R}$ is one – one, onto and differentiable function and graph of $y = f(x)$ is symmetrical about the point $(4, 0)$, then
 (A) $f^{-1}(2010) + f^{-1}(-2010) = 8$
 (B) $\int_{-2010}^{2018} f(x) dx = 0$
 (C) If $f'(-100) > 0$, then roots of $x^2 - f'(10)x - f'(10) = 0$ are non real
 (D) If $f'(10) = 20$, then $f'(-2) = 20$
- The value of x in $\left(0, \frac{\pi}{2}\right)$ satisfying equation $\frac{\sqrt{5}-1}{\sin x} + \frac{\sqrt{10+2\sqrt{5}}}{\cos x} = 8$ is
 (A) $\frac{\pi}{10}$ (B) $\frac{3\pi}{10}$ (C) $\frac{9\pi}{10}$ (D) $\frac{7\pi}{10}$
- Point M moved on the circle $(x-4)^2 + (y-8)^2 = 20$. Then it broke away from it and moving along a tangent to the circle cut the x – axis at point $(-2, 0)$. The coordinates of the point on the circle at which the moving point broke away is
 (A) $\left(\frac{42}{5}, \frac{36}{5}\right)$ (B) $\left(-\frac{2}{5}, \frac{44}{5}\right)$ (C) $(6, 4)$ (D) $(2, 4)$

Space For Rough Work

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5. If f is an odd continuous function in $[-1, 1]$ and differentiable in $(-1, 1)$ then which of the following statement(s) is (are) correct?
 (A) $f'(a) = f(1)$ for some $a \in (-1, 0)$
 (B) $f'(b) = f(1)$ for some $b \in (0, 1)$
 (C) $n(f(\alpha))^{n-1} f'(\alpha) = (f(1))^n$ for some $\alpha \in (-1, 0)$ and $\forall n \in \mathbb{N}$
 (D) $n(f(\beta))^{n-1} f'(\beta) = (f(1))^n$ for some $\beta \in (0, 1)$ and $n \in \mathbb{N}$
6. Two soldiers X and Y shoot each other. If one or both are hit then shooting is over. If both shoot miss then they repeat the process. Suppose the results of shots are independent and that each shot of X will hit Y with probability $\frac{2}{5}$ and each shot of Y will hit X with probability $\frac{1}{5}$.
 (A) The probability that X is not hit is $\frac{8}{13}$ (B) The probability that X is not hit is $\frac{8}{25}$
 (C) The probability that both are hit is $\frac{2}{25}$ (D) The probability that both are hit is $\frac{2}{13}$
7. Let P be the plane consisting of all points that are equidistant from A $(-4, 2, 1)$ and B $(2, -4, 3)$ and Q denotes the plane $x - y + Cz = 1$, where $C \in \mathbb{R}$.
 (A) If $C = \frac{1}{3}$ then plane P will be parallel to Q
 (B) If $C = -1$ then plane P will be perpendicular to Q
 (C) Volume of tetrahedron formed by plane P with coordinate planes will be $\frac{2}{81}$
 (D) If line $L: \frac{x-1}{1} = \frac{y+2}{3} = \frac{z-7}{-1}$ intersect plane P at $R(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma = 12$
8. Let $A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ where $x, y, z \in \mathbb{N}$ if $|\text{adj}(\text{adj}(\text{adj}(\text{adj} A)))| = 4^8 \cdot 5^{16}$ then the number of such matrix is p and value of $\int_{-\pi/2}^{\pi/2} 2^{\sin x} dx + \int_{5/2}^4 \sin^{-1}(\log_2(x-2)) dx = \frac{q\pi}{4}$, then $p+q$ is
 (A) 40 (B) 41 (C) 42 (D) 44

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Consider two circles C_1 of radius a and C_2 of radius b ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of conditions listed in list- I, the ratio of $\frac{b}{a}$ is given in list II

	List – I		List – II
(I)	C_1 and C_2 touch each other.	(P)	$2 + \sqrt{2}$
(II)	C_1 and C_2 are orthogonal	(Q)	3
(III)	C_1 and C_2 intersect so that common chord is longest	(R)	$2 + \sqrt{3}$
(IV)	C_1 passes through center of C_2	(S)	$3 + 2\sqrt{2}$
		(T)	$3 - 2\sqrt{2}$

9. Which is correct option?

- (A) I $\rightarrow$ R
(C) III $\rightarrow$ Q

- (B) II $\rightarrow$ T
(D) IV $\rightarrow$ S

Consider two circles C_1 of radius a and C_2 of radius b ($b > a$) both lying in the first quadrant and touching the coordinate axes. In each of conditions listed in list- I, the ratio of $\frac{b}{a}$ is given in list II

	List – I		List – II
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(III)	C_1 and C_2 intersect so that common chord is longest	(R)	$2 + \sqrt{3}$
(IV)	C_1 passes through center of C_2	(S)	$3 + 2\sqrt{2}$
		(T)	$3 - 2\sqrt{2}$

10. Which is correct option?

- (A) I $\rightarrow$ Q
(C) III $\rightarrow$ T

- (B) II $\rightarrow$ S
(D) IV $\rightarrow$ P

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List Match Set (11-12)

Consider square matrices of order 2 which has it is elements 0, 1, 2 and 4. N denotes the number of such matrices all elements of which are different

	List – I		List – II
(I)	Possible non negative values of $\det(A)$ is	(P)	2
(II)	Sum of values of determinants corresponding to N matrices is	(Q)	4
(III)	If absolute value of $(\det A)$ is least then possible value of $ \text{adj}(\text{adj}(\text{adj})) $ is	(R)	-2
(IV)	If $(\det A)$ is algebraically least, then possible values of $\det(4A^{-1})$ is	(S)	0
		(T)	8

11. Which is correct option?

(A) I $\rightarrow$ Q, R

(B) II $\rightarrow$ S

(C) III $\rightarrow$ Q, P

(D) IV $\rightarrow$ T

Consider square matrices of order 2 which has it is elements 0, 1, 2 and 4. N denotes the number of such matrices all elements of which are different

	List – I		List – II
(I)	Possible non negative values of $\det(A)$ is	(P)	2
(II)	Sum of values of determinants corresponding to N matrices is	(Q)	4
(III)	If absolute value of $(\det A)$ is least then possible value of $ \text{adj}(\text{adj}(\text{adj})) $ is	(R)	-2
(IV)	If $(\det A)$ is algebraically least, then possible values of $\det(4A^{-1})$ is	(S)	0
		(T)	8

12. Which is correct option?

(A) I $\rightarrow$ P, Q, T

(B) II $\rightarrow$ R

(C) III $\rightarrow$ P, S

(D) IV $\rightarrow$ Q

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(PART – B)**(Numerical Type)**

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30).

- A 3×3 determinant has entries either 1 or -1 . Let S be the set of all determinants such that the product of elements of any row or column is -1 . For example $\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{vmatrix}$ is an element of S and number of elements in S is m . Let $P = \begin{bmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{bmatrix}$ and trace of the matrix $\text{adj}(\text{adj}P)$ is n then find the value of $\frac{m}{n}$.
- Let z_1 and z_2 be two complex numbers such that $|z_2| = 1$ and $\frac{z_1 - 1}{z_1 + 1} = \left(\frac{z_2 - 1}{z_2 + 1} \right)^2$, then find the value of $|z_1 - 1| + |z_1 + 1|$.
- Let $\langle a_n \rangle$ is an infinite geometric sequence with first term $2 \cot x$ and common ratio $\sin^2 x$ and $\langle b_n \rangle$ is an infinite geometric sequence with first term $\sin 2x$ and common ratio $\sin^2 x$. If $x \in \left(0, \frac{\pi}{4} \right]$, then the minimum value of $\sum_{i=1}^{\infty} a_i - \sum_{j=1}^{\infty} b_j$ is
- Let $\vec{a} = (2 + \sin \theta) \hat{i} + \cos(\theta) \hat{j} + \sin(2\theta) \hat{k}$
 $\vec{b} = \sin\left(\theta + \frac{2\pi}{3}\right) \hat{i} + \cos\left(\theta + \frac{2\pi}{3}\right) \hat{j} + \sin\left(2\theta + \frac{4\pi}{3}\right) \hat{k}$
 $\vec{c} = \sin\left(\theta - \frac{2\pi}{3}\right) \hat{i} + \cos\left(\theta - \frac{2\pi}{3}\right) \hat{j} + \sin\left(2\theta - \frac{4\pi}{3}\right) \hat{k}$
 be three vectors where $\theta \in \left(0, \frac{\pi}{2} \right)$. Let V is the maximum volume of tetrahedron whose coterminal edges are represented by the vector $2\vec{b} \times \vec{c}, 3\vec{c} \times \vec{a}$ and $\vec{a} \times 4\vec{b}$ then $\frac{V}{6} =$

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5. Considering the curve $xy = 15!$. Let n be the number of points (α, β) lying on it, where $\alpha, \beta \in \mathbb{I}^+$ and $\text{HCF}(\alpha, \beta) = 1$ then $\left[\frac{n}{13} \right]$ equals (where $[.]$ represents G.I.F.)
6. Let $g: \mathbb{R} - \left\{ \frac{3}{2} \right\} \rightarrow \mathbb{R} - \left\{ \frac{3}{2} \right\}$ be defined as $g(x) = \frac{3x+5}{2x-3}$. If $g_{2014}(x) = \frac{px-q}{-rx+s}$, where $g_{2014}(x) = \underbrace{g(g(g(\dots)))}_{2014 \text{ times}}(x)$, then find the value of $(p+q+r+s)$.
-

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Answers & Solutions

SECTION – I : PHYSICS

(PART – A)

1. **AC**

Sol. $\frac{GMm}{r^2} = \frac{mv^2}{r}$

If G keeps decreasing, earth will attract satellite with smaller force, hence leading to spiral outward path.

2. **B**Sol. R to be maximum.

$$Y = \frac{3H}{2} = \frac{\text{Total height}}{2}.$$

3. **A**

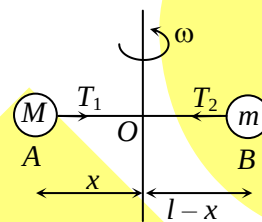
Sol. $T_1 = T_2$

$$\Rightarrow M\omega^2 x = m\omega^2 (l - x)$$

$$x = \frac{ml}{M + m}$$

$$l - x = \frac{Ml}{M + m}$$

$$\frac{AO}{OB} = \frac{x}{l - x} = \frac{m}{M} = \frac{3}{9} = \frac{1}{3}$$

4. **AC**

Sol. $\frac{dx}{dt} = t^2 \quad \dots(i)$

$$y = \frac{1}{2} t^3$$

$$\frac{dy}{dt} = \frac{t^2}{2} \quad \dots(ii)$$

$$t = 1, \quad v_x = 1, \quad v_y = \frac{1}{2}; \quad \vec{v} = \hat{i} + \frac{1}{2}\hat{j}$$

$$\frac{d^2x}{dt^2} = 2t \quad \dots(iii)$$

$$\frac{d^2y}{dt^2} = t \quad \dots(iv)$$

$$\text{at } t = 1 \text{ s } a_x = 2 \text{ and } a_y = 1$$

$$\therefore \vec{a} = 2\hat{i} + \hat{j}.$$

5. **B**

Sol. After brakes applied,

$$\text{Common velocity of car and boat} = \left(\frac{mv_0}{m + M} \right)$$

$$\text{Time taken to hit wall} = \frac{L(M + m)}{mv_0}$$

6. **AD**

Sol. Draw F.B.D. and apply Newton's law of motion.

7. **C**

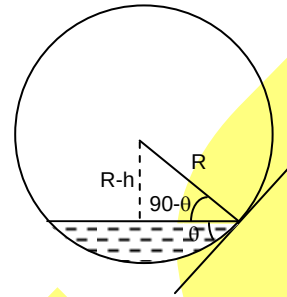
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Sol. Work done by gas = $\int P dv$ = Area under P-V graph.

$$\therefore W_1 > W_2 > W_3$$

8. **B**

Sol. $\sin(90 - \theta) = \frac{R - h}{R}$



9. **C**

Sol. Just after cutting force of spring will be zero whereas the force of other spring will be unchanged.

10. **A**

Sol. Just after cutting force of spring will be zero whereas the force of other spring will be unchanged.

11. **B**

Sol. Apply principle of superposition of electric fields.

12. **C**

Sol. Apply principle of superposition of electric fields.

(PART - B)

1. **0.67**

Sol. Loudness (dB) = $10 \log_{10} \left(\frac{I}{I_0} \right)$ ($I_0 \rightarrow$ Minimum audible intensity)

$$30 = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

$$\frac{I}{I_0} = 1000$$

$$\Rightarrow I = (1000)I_0$$

For point source, intensity $\propto \frac{1}{(\text{Distance})^2}$

$$(1000)I_0 \propto \frac{1}{(20)^2}$$

$$\Rightarrow I_0 \propto \frac{1}{(20)^2 (1000)}$$

$$\Rightarrow \text{Required distance} = 20\sqrt{1000} = 200\sqrt{10}$$

2. **7**

Sol. $\frac{P^2}{2m} = KE = \frac{9 \times 10^9 \times 120e^2}{10 \times 10^{-15}}$

$$\lambda = \frac{h}{P} = \frac{h}{e\sqrt{360 \times 10^{-3}}} = \frac{4.2 \times 10^{-15}}{0.6} = 7.$$

3. **3**

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Sol. Let T be the tension in the ideal string and 'a' be the acceleration of the blocks at the instant of release. For the block on the left, the upward acceleration may be found from

$$T + k_1x - mg = ma$$

For the block on the right, the downward acceleration may be found from

$$k_2x + mg - T = ma$$

Adding the equations gives the acceleration of the blocks as

$$a = (k_1 + k_2)x/(2m)$$

However, subtracting the equations gives

$$T = mg - (k_1 - k_2)x/2$$

for maximum value of k_1T will be zero.

$$mg = \left(\frac{k_1 - k_2}{2} \right) x \quad ; \quad k_1 = 300.$$

4. **0.50**

Sol. Let the length of each side of the cube is x , $x^2 \times 2 \times 1 \times g = 200 \times g$

$$\Rightarrow x = 10 \text{ cm.} \quad \therefore k = \frac{1}{2}.$$

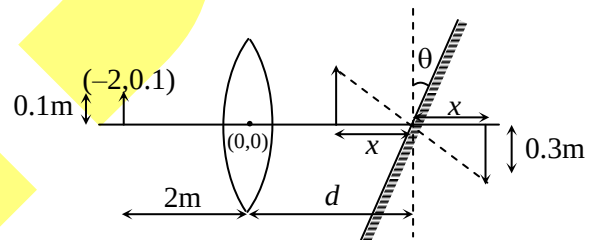
5. **1**

$$\text{Sol. } v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{T}{\pi \rho r^2}}$$

$$\therefore \frac{v_1}{v_2} = \frac{1}{2} \quad ; \quad \frac{f_1}{f_2} = \frac{v_1 / 2L_1}{v_2 / 2L_2} = \frac{v_1}{v_2} \times \frac{L_2}{L_1} = \frac{1}{2} \times 2 = 1.$$

6. **5**

$$\begin{aligned} \text{Sol. } \frac{1}{v} - \frac{1}{u} &= \frac{1}{f} \\ \frac{1}{v} - \frac{1}{-2} &= \frac{1}{1.5} \quad ; \quad v = 6\text{m} \\ m &= \frac{v}{u} = -3 \\ x &= 6 - d; \quad \tan \theta = \frac{0.3}{x} \\ \Rightarrow d &= 5\text{m}. \end{aligned}$$



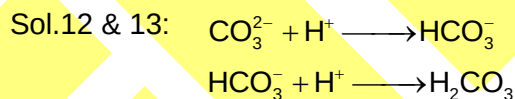
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SECTION – II : CHEMISTRY**(PART – A)**

1. ABC
Sol. Bridge bonding
 $\text{Al}_2\text{Cl}_6 \Rightarrow 3\text{c} - 4\text{e}^-$ Bond
2. BC
Sol. $|\psi|^2 dv \Rightarrow 3\text{c} - 4\text{e}^-$ bond
3. ABCD
Sol. $uf \rightarrow f$
 $\Delta G = \Delta H - T\Delta S$
For spontaneous $\Delta G = -ve$
4. CD
Sol. 'VBT' with SFL pairing occur.
5. AD
Sol. Ogranometallic $\Rightarrow$ Carbon – Metal bond.
6. BD
Sol. Theoretical & Conceptual
7. ABD
Sol. Non-mirror images, stereoisomer.
8. ABD
Sol. "Counter ion effect + solution effect" to opposite charges electrodes.
9. C
10. A
Sol.9 & 10: $W_{\text{Expansion}} < 0$ $\Delta U = \Delta H = 0$ (for isothermal process)
 $W_{\text{Compression}} > 0$

11. A

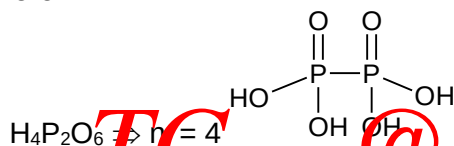
12. D

Before first equivalent point $\Rightarrow \text{CO}_3^{2-} + \text{HCO}_3^-$ At first equivalent point $\Rightarrow \text{HCO}_3^-$ After first equivalent point $\Rightarrow \text{HCO}_3^- + \text{H}_2\text{CO}_3$ **(PART – B)**

1. 416
Sol. $\text{CH}_4(\text{g}) \longrightarrow \text{C}(\text{g}) + 4\text{H}(\text{g})$
 $\Delta H_{\text{Rxn}} = 1664, E_{\text{C-H}} = \frac{1664}{4} = 416$

2. Zero

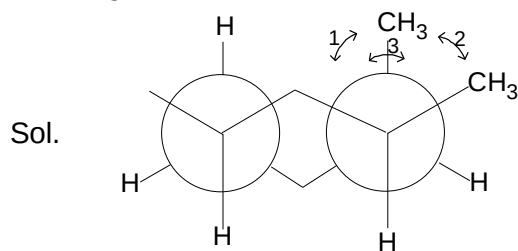
Sol.

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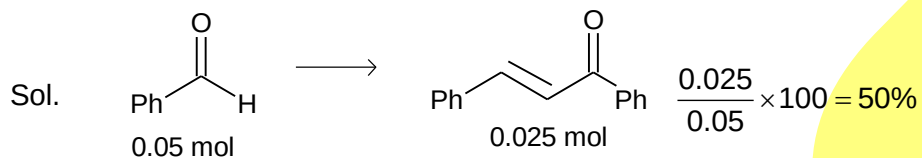
3. 12

Sol. $B_{12}H_{12}^{2-}$

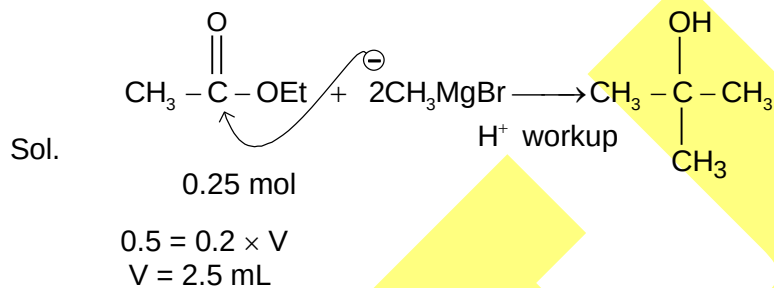
4. 3



5. 50



6. 2.5



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SECTION – III : MATHEMATICS

(PART – A)

1. D

Sol. $f(x) = xe^x + (xe^x)^{-1}$

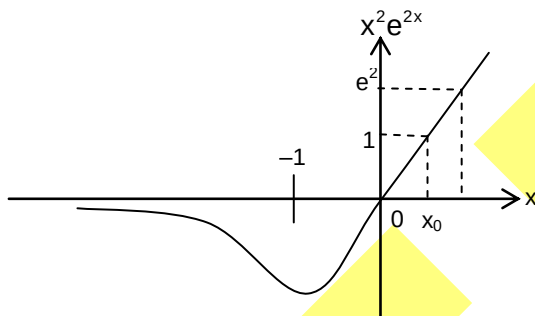
$$= xe^x + \frac{1}{x}e^{-x}$$

$$f'(x) = (x+1)e^x - \frac{1}{x^2}e^{-x} - \frac{1}{x}e^{-x}$$

$$= (x+1)e^x - \frac{e^{-x}}{x^2}(1+x)$$

$$= \frac{(1+x)}{x^2}e^{-x} [x^2e^{2x} - 1]$$

$$f'(x) = 0 \Rightarrow 1+x=0 \text{ or } x^2e^{2x} - 1 = 0$$

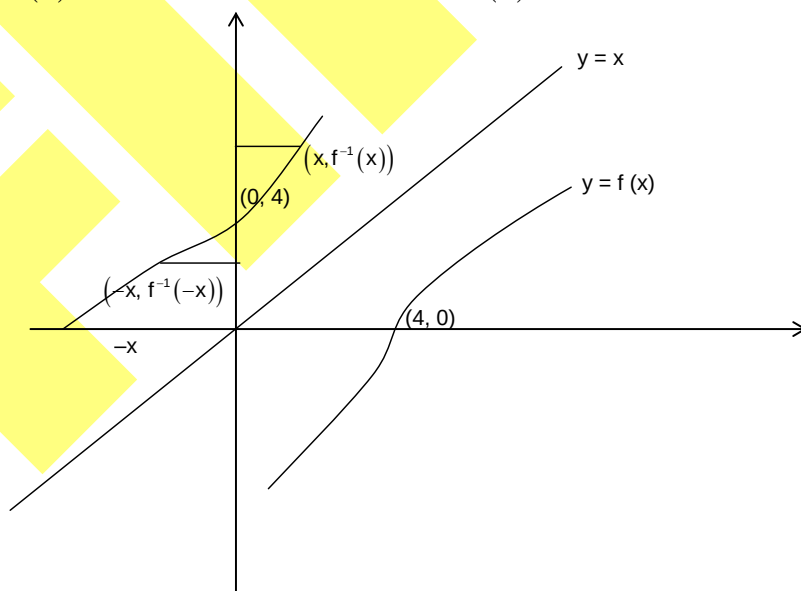


From figure $x^2e^{2x} - 1 = 0$
at $x = x_0 \in (0, 1)$

So $x = -1$ and x_0 are two points of extrema. At $x = x_0$, $x^2e^{2x} - 1$ changes sign from –ive to +ive $\Rightarrow x_0$ is point of minima. At $x = -1$ sign of $f(x)$ changes from +ive to –ive $\Rightarrow x = -1$ is point of maxima.

2. ABD

Sol. $f(x)$ is symmetric about $(4, 0) \Rightarrow f^{-1}(x)$ is symmetric about $(0, 4)$



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- (a) Mid point of $(x, f^{-1}(x))$ and $(-x, f^{-1}(-x))$ is $(0, 4)$

$$\Rightarrow \left(\frac{x - x}{2}, \frac{f^{-1}(x) + f^{-1}(-x)}{2} \right) = (0, 4)$$

$$\Rightarrow \frac{f^{-1}(x) + f^{-1}(-x)}{2} = 4 \Rightarrow f^{-1}(x) + f^{-1}(-x) = 8$$

- (b) $f(x)$ is symmetric about $(4, 0)$

$$\Rightarrow f(4+x) = -f(4-x)$$

$$x \rightarrow 4-x$$

$$\Rightarrow f(8-x) = -f(x)$$

$$\Rightarrow f(x) + f(8-x) = 0 \quad \dots\dots\dots(1)$$

$$I = \int_{-2010}^{2018} f(x) dx \quad \dots\dots\dots(2)$$

$$I = \int_{-2010}^{2018} f(2018 - 2010 - x) dx = \int_{-2010}^{2018} f(8-x) dx \quad \dots\dots\dots(3)$$

$$(2) + (3)$$

$$2I = \int_{-2010}^{2018} [f(x) + f(8-x)] dx = 0$$

$$\Rightarrow I = 0$$

- (c) Since $f(x)$ is one to one and $f'(-100) > 0 \Rightarrow f(x)$ is increasing function

$$\text{So } f'(10) > 0$$

$$D = (f'(10)^2 - 4x_1[-f'(10)]) = [f'(10)]^2 + 4f'(10) > 0$$

$$\Rightarrow \text{roots are always real.}$$

- (d) $f(x) + f(8-x) = 0$

$$\text{Diff. w.r.t. } \Rightarrow f'(x) - f'(8-x) = 0 \quad \dots\dots\dots(4)$$

$$\text{Put } x=10$$

3. AB

$$\text{Sol. } \frac{(\sqrt{5}-1)\cos x}{\sqrt{(\sqrt{5}-1)^2 + 10 + 2\sqrt{5}}} + \frac{\sqrt{10+2\sqrt{5}}\sin x}{\sqrt{(\sqrt{5}-1)^2 + 10 + 2\sqrt{5}}} = \frac{8\sin x \cos x}{\sqrt{(5-1)^2 + 10 + 2\sqrt{5}}}$$

$$\left(\frac{\sqrt{5}-1}{4} \right) \cos x + \frac{\sqrt{10+2\sqrt{5}}}{4} \sin x = \frac{8}{4} \sin x \cos x$$

$$\sin \frac{\pi}{10} \cos x + \cos \frac{\pi}{10} \sin x = \sin 2x$$

$$\sin \left(x + \frac{\pi}{10} \right) = \sin 2x$$

$$\sin 2x = \sin \left(x + \frac{\pi}{10} \right)$$

$$\Rightarrow 2x = x + \frac{\pi}{10} \text{ or } \pi - \left(x + \frac{\pi}{10} \right)$$

$$\Rightarrow x = \frac{\pi}{10} \text{ or } \frac{3\pi}{10}$$

4. BC

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Sol. A and B are intersection of chord of contact with circle.

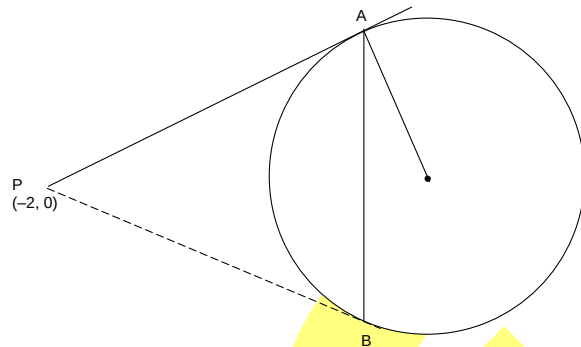
$$x^2 + y^2 - 8x - 16y + 60 = 0 \quad \dots\dots(1)$$

Equation of AB.

$$-2x + 0y + 4(x - 2) - 8(y + 0) + 60 = 0$$

$$-6x - 8y + 68 = 0 \Rightarrow 3x + 4y - 34 = 0 \quad \dots(2)$$

Now solve (1) and (2)



5. ABD

Sol. Clearly, $f(-1) = -f(1)$, $f(0) = 0$

For option (A) and (B), apply LMVT for the function $y = f(x)$ in $[-1, 0]$ and $[0, 1]$ respectively

For option (D), apply LMVT to the function $y = (f(x))^n$ in $[0, 1]$

6. AD

Sol. Probability that X is not hit is
$$= \frac{2}{5} \cdot \frac{4}{5} + \frac{3}{5} \cdot \frac{4}{5} \cdot \frac{2}{5} \cdot \frac{4}{5} + \dots\dots = \frac{\frac{2}{5} \cdot \frac{4}{5}}{1 - \frac{2}{5} \cdot \frac{4}{5}} = \frac{8}{13}$$

Probability that both are hit is
$$= \frac{2}{5} \cdot \frac{1}{5} + \frac{3}{5} \cdot \frac{4}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} + \dots\dots = \frac{\frac{2}{5} \cdot \frac{1}{5}}{1 - \frac{2}{5} \cdot \frac{4}{5}} = \frac{2}{13}$$

7. AD

Sol. Clearly plane P will be perpendicular bisector of line joining A and B.

$\therefore$ It passes through $(-1, -1, 2)$ and dir of its normal are $(3, -3, 1)$

$\therefore$ Equation of plane P is

$$3(x+1) - 3(y+1) + 1(z-2) = 0$$

$$\Rightarrow 3x - 3y + z = 2 \quad \dots\dots\dots(1)$$

(a) $\therefore$ P is parallel to Q

$$\therefore \frac{1}{3} = \frac{1}{3} = \frac{C}{1} \quad \Rightarrow \quad C = \frac{1}{3}$$

(b) If P is perpendicular to Q then $1 \cdot 3 - 1 \cdot (-3) + 1 \cdot C = 0 \Rightarrow C = -6$

(c) $\therefore$ Plane P intersects coordinate axes at points with position vectors $\frac{2}{3}\hat{i}$, $-\frac{2}{3}\hat{j}$ and $2\hat{k}$

$$\therefore \text{volume of tetrahedron} = \frac{1}{6} \left\| \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} \quad 2\hat{k} \right\| = \frac{4}{27}$$

(d) Point on line L can be taken as $(\lambda + 1, 3\lambda - 2, -\lambda + 7)$

$\therefore$ It is on plane P.

$$\therefore 3(\lambda + 1) - 3(3\lambda - 2) + (-\lambda + 7) = 2$$

$$\Rightarrow 7\lambda = 14 \Rightarrow \lambda = 2$$

$\therefore$ point of int. = $(3, 4, 5)$

$$\therefore \alpha + \beta + \gamma = 12$$

8. B

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Sol. $|A| = x + y + z$

$$|\text{adj}(\text{adj}(\text{adj}(\text{adj})))| = |A|^{(n-1)^4} = |A|^{16}$$

$$|A|^{16} = 4^8 \cdot 5^{16} = 2^{16} \cdot 5^{16}$$

$$|A| = 2 \times 5 = 10 = x + y + z$$

$$x + y + z = 10$$

$$\text{Number of natural numbers solutions} = {}^{10-1}C_{3-1} = {}^9C_2 = 36$$

$$\Rightarrow p = 36$$

$$\int_{-\pi/2}^{\pi/2} 2^{\sin x} dx + \int_{5/2}^4 \sin^{-1}(\log_2(x-2)) dx$$

$$\text{sub } x-2 = t$$

$$+ \int_{1/2}^2 \sin^{-1}(\log_2 t) dt$$

$$\int_{-\pi/2}^{\pi/2} 2^{\sin x} dx + \int_{1/2}^2 \sin^{-1}(\log_2 x) dx = \frac{\pi}{2} 2^{\sin \pi/2} - \left(\frac{-\pi}{2}\right) 2^{\sin(-\pi/2)}$$

$$= \frac{\pi}{2} \cdot 2 + \frac{\pi}{2} \cdot \frac{1}{2}$$

$$= \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

$$\text{So, } p = 36, q = 5$$

9. C

10. D

(Sol. 9 & 10)

For C_1 : Center (a, a), radius = a

C_2 : Center (b, b), radius = b

(i) $C_1 C_2 = r_1 r_2 \Rightarrow \sqrt{(a-b)^2 + (a-b)^2} = (a+b)$
 $\Rightarrow \sqrt{2}(b-a) = a+b$
 $\Rightarrow \frac{b}{a} = 3 + 2\sqrt{2}$

(ii) $C_1 C_2^2 = r_1^2 + r_2^2$
 $(a-b)^2 + (a-b)^2 = a^2 + b^2 \Rightarrow a^2 - 4ab + b^2 = 0$
 $\Rightarrow \frac{b^2}{a^2} - \frac{4b}{a} + 1 = 0$
 $\frac{b}{a} = 2 \pm \sqrt{3}$ but $\frac{b}{a} \geq 1$
 $\Rightarrow \frac{b}{a} = 2 + \sqrt{3}$

(iii) Common chord is diameter of smaller circle

$$x^2 + y^2 - 2bx - 2by + b^2 = 0 \quad \dots\dots\dots(i)$$

$$x^2 + y^2 - 2ax - 2ay + a^2 = 0 \quad \dots\dots\dots(ii)$$

$$(2) - (1)$$

$$2x + 2y - a + b$$

passes through (a, a)

$$2a + 2a - a + b$$

$$\Rightarrow \frac{b}{a} = 3$$

(iv) $(x-b)^2 + (y-b)^2 = b^2$ passes through (a_1, a)

11. B

12. A

(Sol. 11 & 12)

Total matrices = $4! = 24$

For non negative $\det(A)$ possibilities

$$\begin{bmatrix} 1 & 0 \\ 4 & 2 \end{bmatrix} = 2, \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix} = 4, \begin{bmatrix} 2 & 0 \\ 1 & 4 \end{bmatrix} = 8$$

So, (i) P, Q, T

(iii) least absolute value of $\det A = 2$

$$|\text{adj adj adj}| = |A|^{(n-1)^3} = |A|^{(2-1)^3} = |A| = 2 \Rightarrow (P)$$

(ii) Possible value of $\det A$ are $\pm 2, \pm 4, \pm 8$

So sum = 0 $\Rightarrow (S)$

(iv) $(\det A)_{\text{least}} = -8$ so $\det(4A^{-1}) = 4^2 \frac{1}{(\det A)}$

$$= \frac{16}{-8} = -2 \Rightarrow (R)$$

(PART - B)

1. 8

Sol. $\therefore$ Elements of determinant can be 1 or -1

$\therefore$ First two places of first row can be filled in $2 \times 2 = 4$ ways and last place can be filled in only one way to product be 1.

$\therefore$ Second row can be made in 4 ways and third row can be made in 1 way only.

$\therefore$ Number of elements in $S = 16 = m$

$$\therefore |P| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1(-2+3) = 1$$

$$\therefore \text{adj}(\text{adj}P) = |P|^{n-2} \cdot P = P$$

$$\Rightarrow \text{tr}(\text{adj}(\text{adj}P)) = \text{tr}(P) = 2 = n$$

$$\therefore \frac{m}{n} = 8$$

2. 2

Sol. $\therefore |z_2| = 1 \Rightarrow z_2 = e^{i\theta} = \cos\theta + i\sin\theta$

$$\therefore \frac{z_2 - 1}{z_2 + 1} = \frac{\cos\theta - 1 + i\sin\theta}{\cos\theta + 1 + i\sin\theta}$$

$$= \frac{-2\sin^2 \frac{\theta}{2} + i \cdot 2\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{2\cos^2 \frac{\theta}{2} + i \cdot 2\sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

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$$= \frac{i \sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} = i \tan \frac{\theta}{2}$$

$$\therefore \frac{z_1 - 1}{z_1 + 1} = -\tan^2 \frac{\theta}{2} \Rightarrow \left(\frac{z_1 - 1}{z_1 + 1} \right) = -\tan^2 \frac{\theta}{2}$$

$$\Rightarrow z_1 = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} = \cos \theta \text{ which is purely real and lies in } [-1, 1]$$

$$\therefore |z_1 - 1| + |z_1 + 1| = |\cos \theta - 1| + |\cos \theta + 1| = 2$$

3. 2

$$\text{Sol. } a_n = \frac{2 \cot x}{1 - \sin^2 x} = \frac{2 \cot x}{\cos^2 x} = \frac{2}{\sin x \cos x}$$

$$b_n = \frac{\sin 2x}{1 - \sin^2 x} = \frac{2 \sin x \cos x}{\cos^2 x} = 2 \tan x$$

$$a_n - b_n = \frac{2}{\sin x \cos x} - 2 \tan x$$

$$= \frac{2(1 - \sin^2 x)}{\sin x \cos x} = \frac{2 \cos^2 x}{\sin x \cos x}$$

$$= 2 \cot x$$

$$\text{Since } \cot x \downarrow \text{ in } \left(0, \frac{\pi}{4} \right]$$

$$\text{So, minimum} = 2 \cot \frac{\pi}{4} = 2$$

4. 2

$$\text{Sol. } \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{vmatrix} 2 + \sin \theta & \cos \theta & \sin 2\theta \\ \sin \left(\theta + \frac{2\pi}{3} \right) & \cos \left(\theta + \frac{2\pi}{3} \right) & \sin \left(2\theta + \frac{4\pi}{3} \right) \\ \sin \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta - \frac{2\pi}{3} \right) & \sin \left(2\theta - \frac{4\pi}{3} \right) \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2$$

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{vmatrix} 2 + \sin \theta & \cos \theta & \sin 2\theta \\ 2 \sin \theta \cos \frac{2\pi}{3} & 2 \cos \theta \cos \frac{2\pi}{3} & 2 \sin 2\theta \cos \frac{4\pi}{3} \\ \sin \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta - \frac{2\pi}{3} \right) & \sin \left(2\theta - \frac{4\pi}{3} \right) \end{vmatrix}$$

$$= \begin{vmatrix} 27 \sin \theta & \cos \theta & \sin 2\theta \\ -\sin \theta & -\cos \theta & -\sin 2\theta \\ \sin \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta - \frac{2\pi}{3} \right) & \sin \left(\theta - \frac{4\pi}{3} \right) \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2$$

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$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & 0 & 0 \\ -\sin\theta & -\cos\theta & -\sin 2\theta \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$$

$$= 2 \left(\sin 2\theta \cos\left(\theta - \frac{2\pi}{3}\right) - \cos\theta \sin\left(2\theta - \frac{4\pi}{3}\right) \right)$$

$$= 2 \left(\sin 2\theta \cos\left(\theta - \frac{2\pi}{3}\right) - \cos\theta \sin\left(2\theta - \frac{4\pi}{3}\right) \right)$$

$$= 2 \left\{ \sin 2\theta \left[\cos\theta \left(\frac{1}{-2} \right) + (\sin\theta) \frac{\sqrt{3}}{2} \right] - \cos\theta \left[\sin 2\theta \left(-\frac{1}{2} \right) - (\cos 2\theta) \left(-\frac{\sqrt{3}}{2} \right) \right] \right\}$$

$$= \sin 2\theta [-\cos\theta + \sqrt{3} \sin\theta] - \cos\theta [\sqrt{3} \cos^2\theta - \sin^2\theta]$$

$$= \sqrt{3} (\sin 2\theta \sin\theta - \cos\theta \cos 2\theta) = -\sqrt{3} \cos 3\theta$$

$$V = \frac{1}{6} [2\vec{b} \times \vec{c} \quad 3\vec{c} \times \vec{a} \quad \vec{a} \times 4\vec{b}] = \frac{24}{6} [a, b, c]^2$$

$$= \frac{24 \times 3}{6} \cos^2 3\theta$$

$$\text{So, } V = 12 \cos^2 3\theta$$

$$\text{So } V \leq 12$$

$$\text{Hence } V_{\max} = 12$$

5. 4

Sol. $15! = 2^{11} \cdot 3^6 \cdot 7^3 \cdot 11 \cdot 13$

All the powers of single prime number has two possibilities either is α or β so total pair of

$$(\alpha, \beta) = 2^5 = 32$$

6. 2

Sol.
$$\text{gog} = \frac{3 \left(\frac{3x+5}{2x-3} \right) + 5}{2 \left(\frac{3x+5}{2x-3} \right) - 3} = \frac{19x}{19} = x$$

$$\text{gogog} = g(x), \text{gogogog} = x$$

$$\Rightarrow \text{gogogog} \text{ up to } n \text{ times} = \begin{cases} x, & \text{if } n \text{ is even} \\ \frac{3x+5}{2x-3} & \text{if } n \text{ is odd} \end{cases}$$

$$\text{So, } g_{2014}(x) = x \text{ In comparing } p=1, q=0, r=0, s=1$$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 1

QP Code:

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 234

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-B (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

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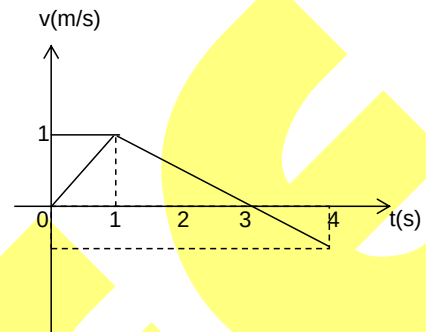
SECTION-1 : PHYSICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The velocity time graph of a particle is shown for time interval $t = 0$ to $t = 4$ sec. Choose correct options.

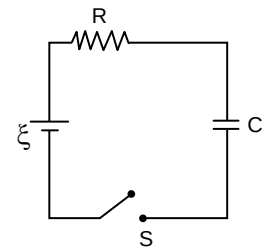


- (A) the displacement of the particle is equal to $\frac{5}{4}$ m
 (B) the distance covered by the particle is equal to $\frac{7}{4}$ m
 (C) the average velocity of the particle is $\frac{1}{2}$ m/s over 4 s
 (D) the motion is uniformly accelerated for 4s

2. Using screw gauge the radius of the wire is found to be 2.50 mm. The length of wire is measured by using a scale and is found to be 50.0 cm. If mass of wire is measured as 25 g, then mark the correct statement(s) [Take $\pi = 3.14$]

- (A) The density has to be computed upto 2 significant digit.
 (B) The least count of scale used to measure length of wire is 1 mm.
 (C) The density of wire is 2.5 g/cm^3 .
 (D) The least count of screw gauge is 0.01 mm.

3. As situation shown in figure the maximum value of rate of energy stored in the capacitor after the switch is closed

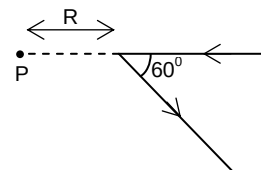


- (A) $\frac{\xi^2}{2R}$
 (B) $\frac{\xi^2}{4R}$
 (C) $\frac{\xi^2}{8R}$
 (D) none of these

4. A metal cylinder of mass 0.5 kg is heated electrically by a 12 W heater in a room at 15°C . The cylinder temperature rises uniformly to 25°C in 5 min and finally becomes constant at 45°C . Assuming that the rate of heat loss is proportional to the excess temperature over the surroundings,

- (A) the rate of loss of heat of the cylinder to surrounding at 20°C is 2 W
 (B) the rate of loss of heat of the cylinder to surrounding at 45°C is 12 W
 (C) specific heat capacity of metal is $\frac{240}{\ln(3/2)} \text{ J/kg}^\circ\text{C}$
 (D) none of the above

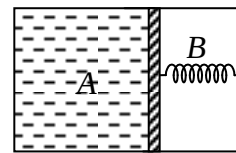
5. A long straight wire, carrying a current I is bent at its mid point to form an angle of 60° . At a point P, distance R from the point of bending the magnetic field is



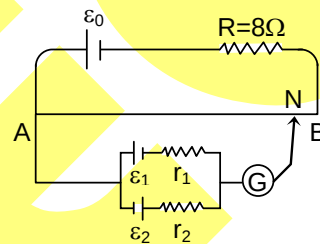
- (A) $\frac{(\sqrt{2}-1)\mu_0 I}{4\pi R}$
 (B) $\frac{(\sqrt{2}+1)\mu_0 I}{4\pi R}$
 (C) $\frac{\mu_0 I}{4\sqrt{3}\pi R}$
 (D) $\frac{\mu_0 I}{8R}$

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6. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant k is connected with piston and the wall of the container as shown. Initially spring is unstretched. Gas in chamber A is allowed to expand. Let in equilibrium spring is compressed by x_0 . Then



- (A) final pressure of the gas is $\frac{kx_0}{S}$
- (B) work done by the gas is $\frac{1}{2}kx_0^2$
- (C) change in internal energy of the gas is $\frac{1}{2}kx_0^2$
- (D) temperature of the gas is decreased.
7. A battery of emf $\varepsilon_0 = 12V$ is connected across a 4m long uniform wire having resistance $4\Omega/m$. The cells of small emfs $\varepsilon_1 = 2V$ and $\varepsilon_2 = 4V$ having internal resistance 2Ω and 6Ω respectively are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to :



- (A) $\frac{1}{6}m$
- (B) $\frac{1}{3}m$
- (C) 25 cm
- (D) 50 cm

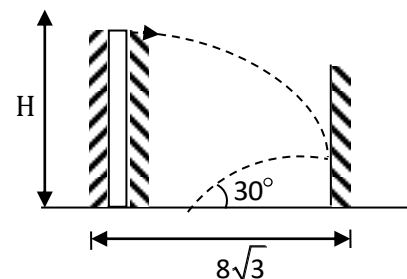
(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. An object of mass m is thrown vertically upwards in air with an initial speed v_0 . The air applies a drag force which is proportional to the instantaneous velocity of the particle in the opposite direction of the motion of ball. Then the time it takes to reach maximum height above ground is $[V_T = mg/c]$ $[\vec{F} = -c(\vec{v})]$

- (A) $\frac{V_T}{g} \ln \left[1 - \frac{v_0}{g} \right]$
- (B) $\frac{V_T}{g} \ln \left[1 + \frac{v_0}{V_T} \right]$
- (C) $\frac{V_T}{2g} \ln \left[1 + \frac{2v_0}{V_T} \right]$
- (D) $\frac{V_T}{g} \ln \left[1 + \frac{2v_0}{V_T} \right]$

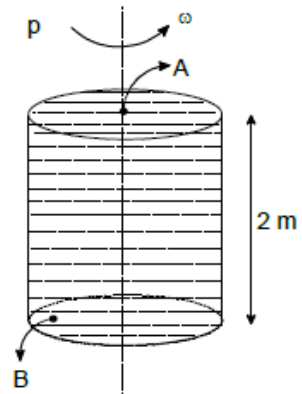
9. A ball is thrown horizontally from the top of a tower of unknown height. Ball strikes a vertical wall whose plane is normal to the plane of motion of ball. Collision is elastic and ball falls on ground at mid-point between tower and wall. Ball strikes the ground at angle of 30° with horizontal. The height of tower is



- (A) 2 m
- (B) 4 m
- (C) 6 m
- (D) 8 m
10. A 50 Hz 20V source is connected to a resistance of 100Ω an inductance of 1 H and a capacitance of $10\mu F$ all in series. Calculate approximate time (in sec) which the resistance (thermal capacity $2J/^\circ C$) will get heated by $10^\circ C$.
- (A) 4
- (B) 5
- (C) 6
- (D) 7

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11. A closed tank of water is rotating about a vertical axis as shown in figure and at same time the entire tank is accelerated upward at 4 m/s^2 . If the rate of rotation is 10 rad/sec . The difference in pressure between points A and B is equal to $(10 \times a.01) \text{ kPa}$. There is a point B at the bottom of the tank at a radial distance of 0.5 m from the axis of rotation and point A is at the top on the axis of rotation. Find the value of a . (take $g = 9.8 \text{ m/s}^2$)
- (A) 1 (B) 2
(C) 3 (D) 4

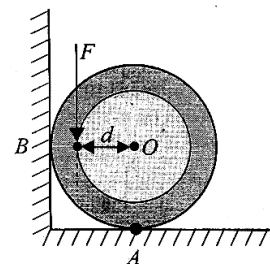


12. If $\vec{A} \times \vec{B} = \vec{C} + \vec{D}$, then the correct alternative is
- (A) $\vec{B}$ is parallel to $(\vec{C} + \vec{D})$
 (B) $\vec{A}$ is perpendicular to $\vec{C}$
 (C) component of $\vec{C}$ along $\vec{A}$ = component of $\vec{D}$ along $\vec{A}$
 (D) component of $\vec{C}$ along $\vec{A}$ = - component of $\vec{D}$ along $\vec{A}$
13. If θ is the angle with horizontal with which a projectile must be fired to escape from earth's gravitational pull then
- (A) $0^\circ \leq \theta < 45^\circ$ (B) $0^\circ \leq \theta \leq 180^\circ$ (C) $\theta = 90^\circ$ (D) $\theta = 45^\circ$

PART – B (Numerical based)

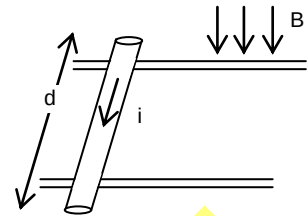
This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

- In a thermally isolated system, two boxes filled with an ideal gas are connected by a valve. When the valve is in closed position, states of the box 1 and 2, respectively, are $(1 \text{ atm}, V, T)$ and $(0.5 \text{ atm}, 4V, T)$. When the valve is opened, what is approximately the final pressure (in atm) of the system?
- Three simple harmonic motions of equal amplitude 2.25 m and equal time periods in the same direction combine. The phase of the second motion is 60° ahead of the first and the phase of the third motion is 60° ahead of the second. Find the amplitude of the resultant motion (in m).
- On a horizontal frictionless frozen lake, a girl (36 kg) and a box (9 kg) are connected to each other by means of a rope. Initially they are 18 m apart. The girl exerts a horizontal force on the box, pulling it towards her. How far has the girl travelled (in m) when she meets the box?
- A solid cylinder with $r = 0.1 \text{ m}$ and mass $M = 2 \text{ kg}$ is placed such that it is in contact with the vertical and a horizontal surface as shown in figure. The coefficient of static friction is $\mu = (1/3)$ for both the surfaces. Find the distance d (in cm) from the centre of the cylinder at which a force $F = 40 \text{ N}$ should be applied vertically so that the cylinder just starts rotating in anticlockwise direction.



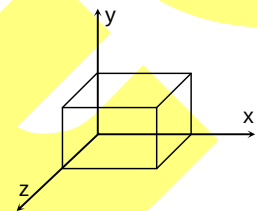
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5. A cylindrical uniform rod of mass 0.72 kg and radius 6 cm rests on two parallel rails, that are $d = 50$ cm apart. The rod carries a current $I = 48$ A (In the direction shown) and rolls along the rails without slipping. If it starts from rest, uniform magnetic field of magnitudes 0.25 T is directed perpendicular to the rod and the rail, then the friction force (In N) between rod and rails is



6. A wire of length L and 3 identical cells of negligible internal resistance are connected in series, when the temperature of the wire is raised by (ΔT) in time t due to the current. The same temperature rise is observed in the same time when N similar cells are connected in series with a wire of length $2L$ but of same material and cross-section. Find the value of N .
7. A thin wire of area of cross-section $A = 10^{-2} \text{ m}^2$ is used to make a ring of radius $r = 10^{-1} \text{ m}$. This ring is placed on a smooth horizontal floor & is given angular velocity $\omega = 20 \text{ rad/s}$ about its centre. Find out stress in the ring (mass per unit length of wire $\lambda = 1 \text{ kg/m}$)

8. Electric field $\vec{E} = x\hat{i} + y\hat{j}$ exist in the region. The flux linked with the surface of cube of side 'a' as shown in the figure is ka^3 then 'k' is



Space For Rough Work

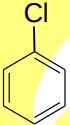
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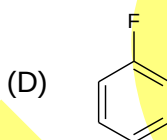
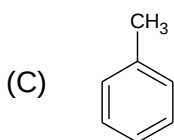
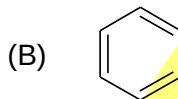
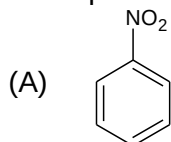
SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Which of the following compound(s) is/are more reactive than chlorobenzene  towards electrophilic attack?



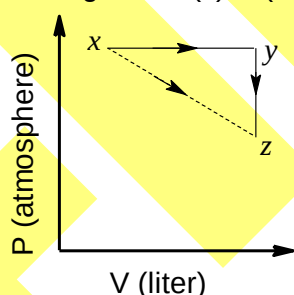
2. For a reaction $A \rightarrow B$, E_a for the forward reaction ($E_{a,f}$) and backward reaction ($E_{a,b}$) is 19 kJ/mole and 9 kJ/mole respectively. Potential energy of A is 12 kJ/mole. Find out which of the following options are correct?

- (A) Heat of reaction is 10 kJ/mol
 (B) It is an exothermic reaction
 (C) The threshold energy of reaction is 31 kJ/mol
 (D) The potential energy of product is 22 kJ/mol

3. Which of the following hydrogen halides react(s) with $\text{AgNO}_3(\text{aq})$ to give a precipitate that dissolves in $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$?

- (A) HCl (B) HF
 (C) HBr (D) HI

4. For an ideal gas, consider only $P-V$ work in going from an initial state X to the final state Z. The final state Z can be reached by either of the two paths shown in the figure. Which of the following choice(s) is (are) correct? [Take ΔS as change in entropy and w as work done]



- (A) $\Delta S_{x \rightarrow z} = \Delta S_{x \rightarrow y} + \Delta S_{y \rightarrow z}$
 (C) $w_{x \rightarrow y \rightarrow z} = w_{x \rightarrow z}$

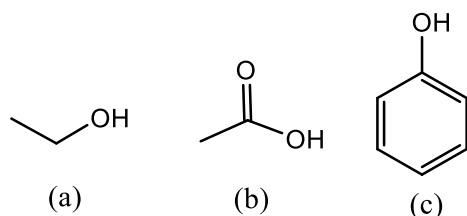
- (B) $w_{x \rightarrow z} = w_{x \rightarrow y} + w_{y \rightarrow z}$
 (D) $\Delta S_{x \rightarrow y \rightarrow z} = \Delta S_{x \rightarrow z}$

5. A salt heated with dil. H_2SO_4 and subsequently treated with few drops of dil. KMnO_4 . The solution loses the colour of KMnO_4 indicates the presence of:

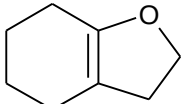
- (a) Nitrite ion (b) nitrate ion
 (c) Oxalate ion (d) sulfite ion

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6. 1 mole benzene ($p_{\text{benzene}}^0 = 42 \text{ mm of Hg}$) and 2 mole toluene ($p_{\text{toluene}}^0 = 36 \text{ mm of Hg}$) will have
 (A) total vapour pressure 38 mm of Hg
 (B) mole fraction of vapours of benzene above liquid mixture is 7/19
 (C) positive deviation from Raoult's law
 (D) negative deviation from Raoult's law
7. Correct acidity order of the following substrate in H_2O solvent.

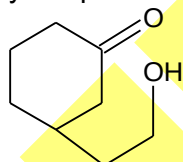
(A) $b < c < a$ (B) $b > a$ (C) $b > c > a$ (D) $c > b$ **(Single Correct Choice Type)**

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

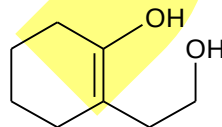
8.  $\xrightarrow[\text{H}_2\text{O}]{\text{H}^+}$ 'P' (major),

Identify the product 'P'

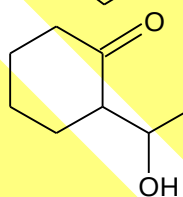
(A)



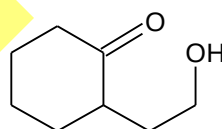
(B)



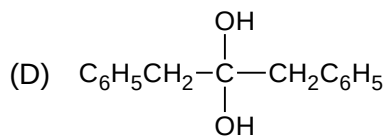
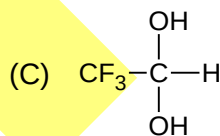
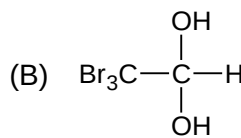
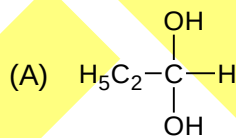
(C)

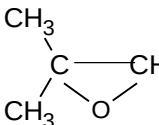


(D)



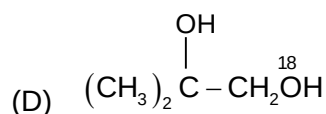
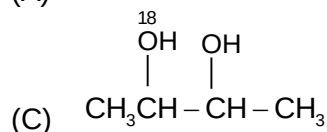
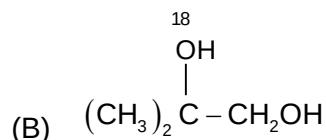
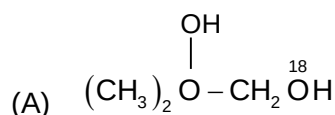
9. Which of the following 'gem' diols is stable?



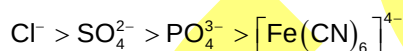
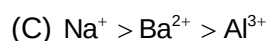
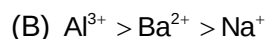
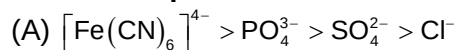
10.  $\xrightarrow{\text{H}_3\text{O}^+}$ A (major product)

The product (A) has the structure

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11. The flocculating power of the given ions for the specified colloidal sols will be such that:
Arsenic sulphide sol. **Ferric hydroxide sol**



12. In an amino acid the carboxylic acid group has $K_a = 10^{-4}$ and amino group has $K_b = 10^{-5}$ at 298 K. The isoelectric point of that amino acid is

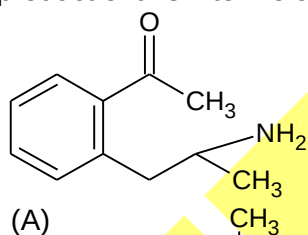
(A) 4

(B) 5

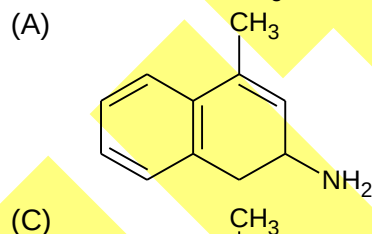
(C) 6.5

(D) 4.5

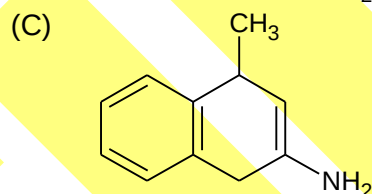
13. The compound shown below undergoes a spontaneous cyclization on long standing, the product of this intermolecular reaction is:



(B)



(D)



PART – B

(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

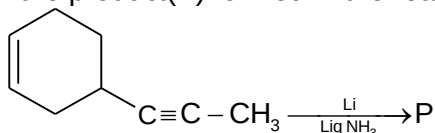
- The successive ionization energies of an s-block element are 8.6, 12.9, 1630.2, 2406.2 eV, etc. If atomic mass of the elements is 39.8 g mol⁻¹, what will be the molar mass of its normal oxide?
- 5 mole of a mixture of Na₂C₂O₄ and Na₂CO₃ required 400 mL of 0.5 M acidified KMnO₄ for complete oxidation. What is the mole fraction of Na₂C₂O₄ in the mixture?

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3. If the wavelength(λ) of the electronic transition $n = 4 \rightarrow n = 1$ in a hydrogen atom is expressed as $\left(\frac{x}{y} \times 10^{-7}\right) \text{ m}$. What is the value of $\left(\frac{x+y}{10}\right)$?
[Rydberg constant = 10^7 m^{-1}]

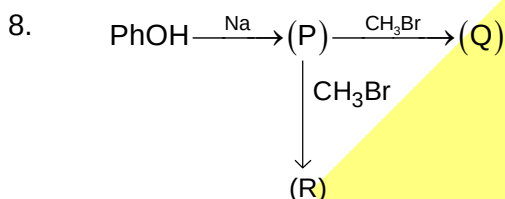
4. $\text{CH}_4 \xrightarrow[h\nu]{\text{Cl}_2(\text{excess})} \text{A} + \text{B} + \text{C} + \text{D}$
If x is the molar mass of the most acidic product and y is the molar mass of the heaviest product, then what is the value of $(x + y)$ in g mol^{-1} unit?

5. If the product(P) formed in the reaction



Has molar mass M_0 then $\frac{M_0}{4}$ is

6. pH of solution obtained by mixing 20 mL of 0.2 M NaOH with 50 mL of 0.2 M acetic acid ($K_P = 1.8 \times 10^{-5}$) is
7. 100 mL of hard water is passed through a column of the ion exchange resin RH_2 . The water coming off the column requires 15.17 mL of 0.0265 M NaOH for its titration. The hardness of water as ppm of Ca^{2+} is



If (R) is more reactive than (Q) and both of them contain different functional groups, if the molar mass of (R) is X , what is $\frac{X}{10}$?

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The equation $\sin^4 x + \cos^4 x = a$ has a real solution for
 (A) all values of a (B) $a = \frac{1}{2}$
 (C) $a = \frac{7}{10}$ (D) $a = 1$
2. Let PQR be a triangle. Let $\vec{a} = \overrightarrow{QR}$, $\vec{b} = \overrightarrow{RP}$ and $\vec{c} = \overrightarrow{PQ}$. If $|\vec{a}| = 12$, $|\vec{b}| = 4\sqrt{3}$ and $\vec{b} \cdot \vec{c} = 24$, then which of the following is (are) true?
 (A) $\frac{|\vec{c}|^2}{2} - |\vec{a}| = 12$ (B) $\frac{|\vec{c}|^2}{2} + |\vec{a}| = 30$
 (C) $|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = 48\sqrt{3}$ (D) $\vec{a} \cdot \vec{b} = -72$
3. If $\alpha = x_1 x_2 x_3$ and $\beta = y_1 y_2 y_3$ are two 3 – digit numbers, then the number of pairs of α and β can be formed so that α can be subtracted from β without borrowing, is
 (A) $2! \times 10! \times 10!$ (B) $(45)(55)^2$
 (C) $3^2 \cdot 5^3 \cdot 11^2$ (D) 136125
4. If all the roots of the equation $z^4 + az^3 + bz^2 + cz + d = 0$ ($a, b, c, d \in \mathbb{R}$) are of unit modulus, then
 (A) $|a| \leq 4$ (B) $|b| \leq 4$
 (C) $|c| \leq 4$ (D) $|b| \leq 6$
5. Which of the following statement(s) is (are) correct?
 (A) The coefficient of x^2 in the expansion of $\sum_{r=0}^{100} {}^{100}C_r (x-4)^{100-r} (5)^r$ is equal to 4950.
 (B) If the sixth term in the expansion of $\left(x^{\frac{-8}{3}} + x^2 \log_{10} x \right)$ is 5600, then x is equal to 1000.
 (C) Let $A_n = {}^nC_0 {}^nC_1 + {}^nC_1 {}^nC_2 + \dots + {}^nC_{n-1} {}^nC_n$ and $\frac{A_{n+1}}{A_n} = \frac{15}{4}$, then the sum of possible values of n is equal to 6.
 (D) If $A_k = \frac{{}^nC_k}{{}^nC_k + {}^nC_{k+1}}$ and $\sqrt[3]{\sum_{k=0}^{n-1} A_k} = 4$, then n is equal to 128.
6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Defined $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0 \\ \min\{f(x), g(x)\} & \text{if } x > 0 \end{cases}$$
 The number of points at which $h(x)$ is not differentiable is

- (A) more than 2
(C) 3

- (B) more than 2 but less than 4
(D) less than 5

7. Identify correct statement(s) about conic $\sqrt{(x-5)^2 + (y-7)^2} + \sqrt{(x+1)^2 + (y+1)^2} = 12$.
 (A) centre is (2, 3) (B) hyperbola with foci (5, 7) and (-1, -1)
 (C) ellipse with major axis $4x - 3y + 1 = 0$ (D) eccentricity is $\frac{5}{7}$

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. If $\tan\theta_1, \tan\theta_2, \tan\theta_3$ and $\tan\theta_4$ are the roots of the equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$ then $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4)$ is equal to
 (A) $\sin \beta$ (B) $\cos \beta$
 (C) $\tan \beta$ (D) $\cot \beta$
9. If a vertex of triangle is (3, 3) and the mid points of two sides through this vertex are $\left(2, \frac{2}{3}\right)$ and $\left(4, \frac{3}{2}\right)$ then the centroid of the triangle is given by
 (A) (1, 3) (B) (3, 0)
 (C) (3, 4/9) (D) (0, 3)
10. Three normals are drawn from the point (14, 7) to the parabola $y^2 - 16x - 8y = 0$. The coordinates of the feet of the normals are
 (A) (0, 0), (8, -16), (3, -4) (B) (0, 0), (8, 16), (3, -4)
 (C) (0, 0), (-8, 16), (3, -4) (D) None of these
11. For two 3×3 matrices A and B, let $A + B = 2B'$ and $3A + 2B = I_3$, where B's is the transpose of B and I_3 is 3×3 identity matrix. Then
 (A) $10A + 5B = 3I_3$ (B) $5A + 10B = 2I_3$
 (C) $3A + 6B = 2I_3$ (D) $B + 2A + I_3$
12. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A+Bx)(x-A)^2$, then the ordered pair (A, B) is equal to:
 (A) (4, 5) (B) (-4, -5)
 (C) (-4, 3) (D) (-4, 5)
13. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous onto function satisfying $f(x) + f(-x) = 0, \forall x \in \mathbb{R}$. If $f(-3) = 2$ and $f(5) = 4$ in $[-5, 5]$, then the equation $f(x) = 0$ has:
 (A) exactly three real roots (B) exactly two real roots
 (C) at least five real roots (D) at least three real roots

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PART – B

(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

- Through a given point P (5, 2), m secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1), A_2(B_2), A_3(B_3), A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5, PA_2 + PB_2 = 6, PA_3 + PB_3 = 7, PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$.
Find the value of $\frac{\left(\sum_{i=1}^5 PA_i^2 + \sum_{i=1}^5 PB_i^2\right)}{2}$.
[Note : $A_r(B_r)$ denotes that the line passing through P (5, 2) meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r]
- The quadratic polynomial $P(x) = ax^2 + bx + c$ has two different zeroes including -2. The quadratic polynomial $Q(x) = ax^2 + cx + b$ has two different zeroes including 3. If α and β be the other zeroes of P(x) and Q(x) respectively, then find the value of $\frac{3\alpha}{5\beta}$.
- Let L denotes the number of surjective functions $f : A \rightarrow B$, where set A contains 4 elements and set B contains 3 elements.
M denotes number of elements in the range of the function.
 $f(x) = \sec^{-1}(\operatorname{sgn} x) + \operatorname{cosec}^{-1}(\operatorname{sgn} x)$ where $\operatorname{sgn} x$ denotes signum function of x.
And N denotes coefficient of t^5 in $(1+t^2)^5(1+t^3)^8$. Find the value of $1.285(LM+N)$.
- Let $0 \leq \beta_r \leq 1$ and $\sum_{r=1}^k \cos^{-1} \beta_r = \frac{k\pi}{2}$ for any $k \geq 1$ and $A = \sum_{r=1}^k (\beta_r)^r$, then $\lim_{x \rightarrow A} \frac{(1+x^2)^{1/3} - (1-2x)^{1/4}}{x+x^2}$ is equal to λ , where $6.258\lambda =$
- Given a function g continuous on R such that $\int_0^1 g(t) dt = 2$ and $g(1) = 5$. If $f(x) = \frac{1}{2} \int_0^x (x-t)^2 g(t) dt$, then the value of $1.85(f'''(1) - f''(1))$ is equal to:
- If $F(x) = \int \frac{(1+x)[(1-x+x^2)(1+x+x^2)+x^2]}{1+2x+3x^2+4x^3+3x^4+2x^5+x^6} dx$ then find the value of $6.76[F(99) - F(3)]$.
[Note: [k] denotes greatest integer less than or equal to k.]
- If $6 + f''(x) + f''(x) = x^2 + f^2(x)$ is the differential equation of a curve and let P be the point of minima of this curve then the number of tangents which can be drawn from P to the circle $x^2 + y^2 = 4$ is k where $8.71k$ equals to
- If a, b, c, d are distinct integers form an increasing AP such that $d = a^2 + b^2 + c^2$, then find the value of $9.69(a+b+c+d)$.

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Space For Rough Work

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Q.P. Code:**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|-------|---------|-------|--------|
| 1. AB | 2. ABCD | 3. B | 4. ABC |
| 5. C | 6. ABCD | 7. C | 8. B |
| 9. C | 10. B | 11. D | 12. D |
| 13. B | | | |

PART – B

- | | | | |
|---------|---------|---------|---------|
| 1. 0.60 | 2. 4.50 | 3. 3.60 | 4. 6 |
| 5. 2 | 6. 6.00 | 7. 400 | 8. 2.00 |

SECTION-2 : CHEMISTRY**PART – A**

- | | | | |
|--------|--------|--------|-------|
| 1. BCD | 2. ACD | 3. ACD | 4. AC |
| 5. ACD | 6. AB | 7. B,C | 8. D |
| 9. C | 10. B | 11. B | 12. C |
| 13. D | | | |

PART – B

- | | | | |
|---------|---------|----------|----------|
| 1. 55.8 | 2. 0.1 | 3. 3.1 | 4. 273.5 |
| 5. 30.5 | 6. 3.82 | 7. 80.40 | 8. 10.8 |

SECTION-3 : MATHEMATICS**PART – A**

- | | | | |
|--------|---------|--------|--------|
| 1. BCD | 2. ACD | 3. BCD | 4. ACD |
| 5. ACD | 6. ABCD | 7. AC | 8. D |
| 9. C | 10. B | 11. A | 12. D |
| 13. D | | | |

PART – B

- | | | | |
|-----------|----------|----------|----------|
| 1. 107.50 | 2. 6.60 | 3. 97.66 | 4. 3.13 |
| 5. 5.55 | 6. 20.28 | 7. 17.42 | 8. 17.38 |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **AB**

Sol. Displacement = Area under v-t graph.
For distance, consider magnitude of area only.

2. **ABCD**

Sol. Based on fact of errors and significant figures.

3. **B**

Sol. $q = q_0(1 - e^{-t/RC})$
 $U = \frac{q^2}{2C} = \frac{q_0^2(1 - e^{-t/RC})^2}{2C}$
 Rate = $\frac{dU}{dt}$

Rate will be maximum when $\frac{d^2U}{dt^2} = 0$.

4. **ABC**

Sol. For cylinder:

$$\left(\frac{dq}{dt}\right)_{\text{net}} = ms \frac{dT}{dt}$$

= Rate of heat inflow – Rate of heat loss to surroundings.

5. **C**

Sol. $B = \frac{\mu_0 I}{4\pi d} (\sin \theta_1 + \sin \theta_2)$
Use this formula for both straight sections.

6. **ABCD**

Sol. Final gas pressure $P = \frac{F}{A} = \frac{kx}{S}$

Work done by gas = Spring energy = $\frac{1}{2}kx_0^2$.

Temperature of gas will decrease.

7. **C**

Sol. Due to auxiliary battery E_0 , find current in potentiometer wire. Then find $V_A - V_N$. This must be equal to potential difference produced by E_1 and E_2 combinely.

8. **B**

Sol. $m \frac{dV}{dt} = -mg - cv$
 $m \frac{dv}{mg + cv} = -dt$
 $m \int_{v_0}^0 \frac{dV}{mg + cv} = - \int_0^t dt$

9. **C**

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Sol. Assuming wall to be absent, it can be treated as standard projectile case having range
 $= 8\sqrt{3} + 4\sqrt{3}$
 $= 12\sqrt{3} \text{ m.}$

10. **B**

Sol. $I_{\text{rms}} = \frac{V_{\text{rms}}}{Z}$
 $= \frac{20}{\sqrt{(X_L - X_C)^2 + R^2}}$
 $\Delta q = C\Delta T = (I_{\text{rms}})^2 R t$

11. **D**

Sol. Pressure difference along vertical $= \rho g_{\text{eff}} h$
 $= \rho(g + a)h$
 Pressure difference along horizontal $= \rho \left(\frac{r}{2} \omega^2 \right) (r)$

12. **D**

Sol. $\vec{A} \times \vec{B} = \vec{C} + \vec{D}$
 $\vec{A} \cdot (\vec{A} \times \vec{B}) = \vec{A} \cdot \vec{C} + \vec{A} \cdot \vec{D}$
 $0 = \vec{A} \cdot \vec{C} + \vec{A} \cdot \vec{D}$
 $\vec{A} \cdot \vec{C} = -\vec{A} \cdot \vec{D}$
 $\frac{\vec{A} \cdot \vec{C}}{A} = \frac{-\vec{A} \cdot \vec{D}}{A}$
 $\Rightarrow$ option (D).

13. **B**

Sol. Escape speed is independent of angle of projection.

PART - B

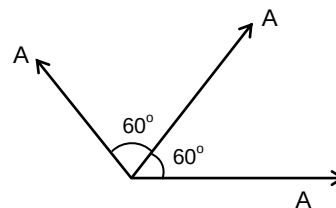
1. **0.60**

Sol. Number of moles of system remains conserved.

$$\frac{(1)V}{RT} + \frac{(0.5)(4V)}{RT} = \frac{PV}{RT} + \frac{P(4V)}{RT}$$

2. **4.50**

Sol. Use vector resultant method to find resultant amplitude.



3. **3.60**

Sol. As $F_{\text{ext}} = 0$, girl and box will meet at their centre of mass.

4. **6**

Sol. At the verge of rotation:
 For translation:

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$$F_{\text{net}} = 0 \quad (\text{Along horizontal and vertical})$$

$$\tau_{\text{net}} = 0 \quad (\text{About centre of cylinder})$$

$$\text{Use: } f = \mu \times N \quad (N \rightarrow \text{Normal reaction from adjoining surface})$$

5. **2**

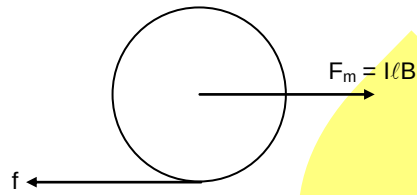
Sol. For translation:

$$lB - f = ma \quad \dots(1)$$

For rotation about centre:

$$f(R) = \left(\frac{mR^2}{2} \right) (\alpha) \quad \dots(2)$$

$$\text{Use } \alpha = \frac{a}{R}$$



6. **6.00**

Sol. Only wire will contribute to heating as cells are ideal.

$$I = \frac{\text{Net E.M.F.}}{\text{Resistance}}$$

$$I^2 R t = m \Delta T$$

7. **400**

Sol. $T d\theta = \lambda r d\theta \omega^2 r$

$$T = \lambda r^2 \omega^2$$

$$\text{Stress} = \frac{T}{A} = \frac{\lambda r^2 \omega^2}{A} = 4$$

8. **2.00**

Sol. Flux = $2a^3$.

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SECTION-2 : CHEMISTRY

PART – A

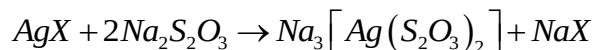
1. BCD

Sol. Fluorine is more activating than chlorine due to greater +R effect.

2. ACD

Sol. Heat of reaction = $19 - 9 = 10$ (It is not an exothermic reaction)

3. ACD

Sol. $HX + AgNO_3 \rightarrow AgX \downarrow + HNO_3$ ($X = Cl, Br, I$)

4. AC

Sol. $\Delta S_{x \rightarrow z} = \Delta S_{x \rightarrow y} + \Delta S_{y \rightarrow z}$ [entropy (S) is a state function, hence additive]
 $W_{x \rightarrow y \rightarrow z} = W_{x \rightarrow y}$ [work done in $Y \rightarrow Z$ is zero as it is an isochoric process]

5. ACD

Sol. Nitrate ion is not oxidized by $KMnO_4$.

6. AB

Sol. $p = p_A^0 x_A + p_B^0 x_B$ A $\rightarrow$ Benzene, B $\rightarrow$ Toluene

$$= 42 \times \frac{1}{3} + 36 \times \frac{2}{3}$$

$$= \frac{114}{3} = 38 \text{ mm of Hg}$$

$$\text{Mole fraction of benzene in vapour} = \frac{p_{\text{benzene}}}{p_{\text{total}}} = \frac{42/3}{38} = 7/19$$

7. B, C

Sol. b $\rightarrow pK_a = 4.76$ c $\rightarrow pK_a = 9.95$ a $\rightarrow pK_a \approx 16 - 17$

8. D

Sol. Cleavage of ether takes place.

9. C

Sol. gem diols are stabilised by electron withdrawing groups.

10. B

Sol. Cleavage of ether takes place in acidic medium.

11. B

Sol. 1. Negatively charged arsenic sulphide solution is coagulated by higher valence of active cation.

2. Positive sol. $[Fe(OH)_3]$ is coagulated by higher valency of active anion.

12. C

Sol. Isoelectronic point = $\frac{4+9}{2} = 6.5$

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13. D

Sol. Cyclisation takes place by reaction between NH_2 group with the ketone part of the molecule.**PART – B**

1. 55.8

Sol. There is a large jump from I.E₃ which means the atom contains two valence electron. $\therefore$ It's valency is 2

Let the atom is M

So formula of it's oxide = Mo

Molar mass = $39.8 + 16 = 55.8 \text{ g mol}^{-1}$

2. 0.1

Sol. Na_2CO_3 does not react with KMnO_4/H^+ M_{eq} of $\text{KMnO}_4/\text{H}^+ = 400 \times 0.5 \times 5 = 1000$ $\therefore M_{\text{eq}}$ of $\text{Na}_2\text{C}_2\text{O}_4 = 1000$ or $\frac{W}{E} \times 1000 = 1000$ $\therefore \frac{W}{E} = 1$ n-factor of $\text{Na}_2\text{C}_2\text{O}_4$ is 2 as it is converted to CO_2 $\therefore \text{Moles of Na}_2\text{C}_2\text{O}_4 = \frac{\text{Equivalent}}{\text{n-factor}} = \frac{W/E}{2} = \frac{1}{2}$ $\therefore \text{Mole fraction} = \frac{1/2}{5} = 0.1$

3. 3.1

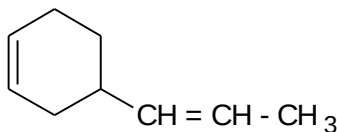
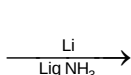
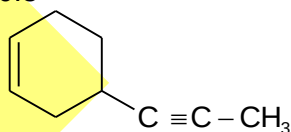
Sol. $\frac{1}{\lambda} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ or, $\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{4^2} \right] = R \left[1 - \frac{1}{16} \right] = \frac{15R}{16}$ or, $\lambda = \frac{16}{15R} = \frac{16}{15} \times 10^{-7} = \left(\frac{x}{y} \times 10^{-7} \right)$ $\therefore \frac{x+y}{10} = \frac{16+15}{10} = 3.1$

4. 273.5

Sol. $x = \text{CHCl}_3$, y is CCl_4 Molar mass of $\text{CHCl}_3 = 119.5$ Molar mass of $y = 154$ $x + y = 119.5 + 154 = 273.5$

5. 30.5

Sol.

 $M_0 = 122, \frac{M_0}{4} = 30.5$

6. 3.82

Sol. $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

Initial

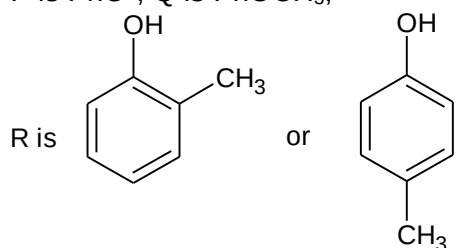
 $50 \times 0.2 \quad 20 \times 0.2$ *IG @bohring_bot*

10 m moles 4 m moles
After Neutralization 6 –

$$\text{pH} = \text{p}K_a + \log \frac{\text{Salt}}{\text{Acid}} = 4.76 + \log \frac{4}{6} = 3.82$$

7. 80.40
Sol. M_{eq} of $\text{Ca}^{2+} = 15.17 \times 0.0265 \times 1 = 0.402$
Hardness of water = $\frac{0.402 \times 20}{100 \times 1000} \times 10 = 80.4$

8. 10.8
Sol. P is PhO^- , Q is PhOCH_3 ,



Molar mass of R is 10.8

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SECTION-3 : MATHEMATICS

PART – A

1. BCD

Sol. We have $\sin^4 x + \cos^4 x \leq \sin^2 x + \cos^2 x$, as $|\sin x| \leq 1$ and $|\cos x| \leq 1$
 $\Rightarrow a \leq 1$ (1)

Next, $\sin^4 x + \cos^4 x = a$

$$\Rightarrow (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = a$$

$$\Rightarrow \frac{1}{2} \sin^2 2x = 1 - a$$

$$\Rightarrow 1 - a \leq \frac{1}{2} \quad [\because \sin^2 2x \leq 1]$$

$$\Rightarrow \frac{1}{2} \leq a \quad (2)$$

From (1) and (2) we get $\frac{1}{2} \leq a \leq 1$. Note that $a = \frac{1}{2}$ for $x = \frac{\pi}{4}$ and $a = 1$ for $x = \frac{\pi}{2}$.

2. ACD

Sol. As $\vec{a} + \vec{b} + \vec{c} = 0$

$$\Rightarrow a^2 = b^2 + c^2 + 2\vec{b} \cdot \vec{c}$$

$$\Rightarrow 144 = 48 + c^2 + 48 \Rightarrow c^2 = 48 \Rightarrow c = 4\sqrt{3}$$

Again, $c^2 = a^2 + b^2 + 2\vec{a} \cdot \vec{b}$

$$\Rightarrow \vec{a} \cdot \vec{b} = \frac{48 - 144 - 48}{2} = -72$$

$$|\vec{a} \times \vec{b} + \vec{c} \times \vec{a}| = |\vec{a} \times \vec{b} + \vec{a} \times \vec{b}| = 2|\vec{a} \times \vec{b}|$$

$$= 2\sqrt{a^2 b^2 - (\vec{a} \cdot \vec{b})^2} = 2\sqrt{12^2 \cdot 48 - (-72)^2}$$

$$= 2 \cdot 12 \sqrt{48 - 36} = 2 \cdot 12 \cdot 2\sqrt{3} = 48\sqrt{3}$$

3. BCD

Sol. Since, α can be subtracted from β without borrowing, if $y_i \geq x_i$, for $i = 1, 2, 3$

Let $x_i = \lambda$

If $i = 1$, then $\lambda = 1, 2, 3, \dots, 9$ and if $i = 2$ and 3 , then $\lambda = 0, 1, 2, 3, \dots, 9$

Hence, total number of ways of choosing the pairs α, β

$$= \left(\sum_{\lambda=1}^9 (10 - \lambda) \right) \left(\sum_{\lambda=0}^9 (10 - \lambda) \right)^2 = (45)(55)^2$$

4. ACD

Sol. $z^4 + az^3 + bz^2 + cz + d = 0$ ($a, b, c, d \in \mathbb{R}$)

Let roots are z_1, z_2, z_3, z_4

$$|z_1| = |z_2| = |z_3| = |z_4| = 1$$

$$z_1 + z_2 + z_3 + z_4 = -4$$

$$z_1 z_2 + z_1 z_3 + z_1 z_4 + z_2 z_3 + z_2 z_4 + z_3 z_4 = b$$

$$z_1 z_2 z_3 + z_1 z_2 z_4 + z_1 z_3 z_4 + z_2 z_3 z_4 = -c$$

$$z_1 z_2 z_3 z_4 = d$$

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$$|z_1 + z_2 + z_3 + z_4| \leq |z_1| + |z_2| + |z_3| + |z_4|$$

$$|a| \leq 4; |b| \leq 6; |c| \leq 4; |d| = 1$$

5. ACD

Sol.

(a) Clearly, $E = ((x-4) + 5)^{100} = (1+x)^{100}$
 $\therefore$ Coefficient of x^2 in $(1+x)^{100} = {}^{100}C_2 = 4950$

(b) Clearly, ${}^8C_5 (x^{-8/3})^3 (x^2 \log_{10} x)^5 = 5600$
 $\Rightarrow x^2 (\log_{10} x)^5 = 100$
 $\Rightarrow x = 10$

(c) Clearly

$$A_n = {}^nC_0 {}^nC_{n-1} + {}^nC_1 {}^nC_{n-2} + \dots + {}^nC_{n-1} {}^nC_0$$

$$= {}^{2n}C_{n-1}$$

(We can also explain by using theory of permutation).

Now, $\frac{A_{n+1}}{A_n} = \frac{(2n+2)!}{(n)!(n+2)!} \times \frac{(n-1)!(n+1)!}{(2n)!}$
 $= \frac{(2n+2)(2n+1)}{n(n+2)} = \frac{15}{4}$

$$\Rightarrow n = 2, 4$$

Hence, Sum = $2 + 4 = 6$

(d) As $A_k = \frac{{}^nC_k}{{}^{n+1}C_{k+1}} = \frac{k+1}{n+1}$

$$\Rightarrow \sum_{k=0}^{n-1} A_k = \frac{n}{2}$$

Now, $\sum_{k=0}^{n-1} A_k = 64$

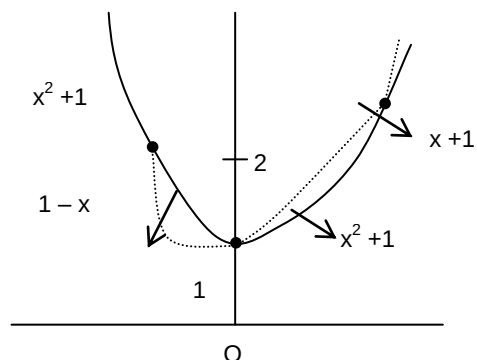
$$\Rightarrow \frac{n}{2} = 64 \Rightarrow n = 128$$

6. ABCD

Sol.

$$h(x) = \begin{cases} x^2 + 1 & x \leq -1 \\ |x| + 1 & -1 < x < 0 \\ x^2 + 1 & 0 \leq x < 1 \\ |x| + 1 & x > 1 \end{cases}$$

Clearly 3 points of non-differentiability.



7. AC

Sol. $\sqrt{(x-5)^2 + (y-7)^2} + \sqrt{(x+1)^2 + (y+1)^2} = 12$

SP + S'P = ca $\Rightarrow$ conic is ellipse

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$$2ae = \sqrt{36 + 64}$$

$$2a = 12 \Rightarrow e = \frac{5}{6}$$

$$\text{Major axis } y - 7 = \frac{7+1}{5+1}(x - 5)$$

$$\Rightarrow 4x - 3y + 1 = 0$$

8. D

Sol. From the given equation we get

$$S_1 = \tan \theta_1 + \tan \theta_2 + \tan \theta_3 + \tan \theta_4 = \sin 2\beta.$$

$$S_2 = \sum \tan \theta_1 \tan \theta_2 = \cos 2\beta$$

$$S_3 = \sum \tan \theta_1 \tan \theta_2 \tan \theta_3 = \cos \beta$$

$$\text{and } S_4 = \tan \theta_1 \tan \theta_2 \tan \theta_3 \tan \theta_4 = -\sin \beta$$

$$\text{Now } \tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) = \frac{S_1 - S_3}{1 - S_2 + S_4}.$$

$$= \frac{\sin 2\beta - \cos \beta}{1 - \cos 2\beta - \sin \beta} = \frac{\cos \beta (2 \sin \beta - 1)}{\sin \beta (2 \sin \beta - 1)} = \cot \beta$$

9. C

Sol. Let $A \equiv (3, 3)$

Given that $F \equiv \left(2, \frac{2}{3}\right)$ and $E \equiv \left(4, \frac{3}{2}\right)$ are mid points of AB and AC.

Let $B \equiv (x_1, y_1)$ and $C \equiv (x_2, y_2)$

Using mid point formula.

$$\text{Co-ordinate of } F \equiv \left(2, \frac{2}{3}\right) = \left(\frac{3+x_1}{2}, \frac{3+y_1}{2}\right)$$

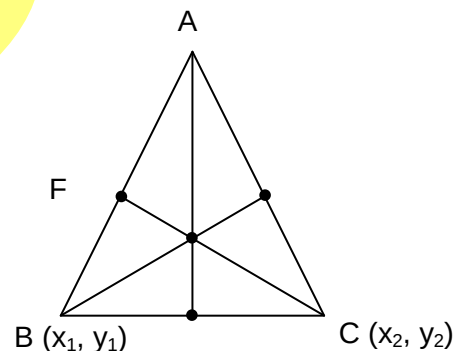
$$\Rightarrow (x_1, y_1) = \left(1 - \frac{5}{3}\right)$$

$$\text{Co-ordinate of } E = \left(4, \frac{3}{2}\right) = \left(\frac{3+x_2}{2}, \frac{3+y_2}{2}\right)$$

$$\Rightarrow (x_2, y_2) = (5, 0)$$

$\therefore$ Centroid of triangle is

$$G(x, y) = \left(\frac{3+1+5}{3}, \frac{3-\frac{5}{3}+0}{3}\right) = \left(3, \frac{4}{9}\right)$$



10. B

Sol. Given parabola is $y^2 - 16x - 8y = 0$ (1)

Let the coordinates of the feet of the normal from $(14, 7)$ be $P(\alpha, \beta)$

Now, equation of the tangent at $P(\alpha, \beta)$ to parabola (1) is $y\beta - 8(x + \alpha) - 4(y + \beta) = 0$

$$\text{Or, } (\beta - 4)y = 8x + 8\alpha + 4\beta \quad \text{.....(2)}$$

$$\text{Its slope} = \frac{8}{\beta - 4}$$

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Equation of normal to parabola (1) at (a, b) is $\gamma - \beta = \frac{4-\beta}{8}(x - \alpha)$.

It passes through (14, 7),

$$\therefore 7 - \beta = \frac{4-\beta}{8}(14 - \alpha) \text{ or } \alpha = \frac{6\beta}{\beta - 4} \quad \dots\dots\dots(3)$$

Also, (α, β) lies on parabola (1), $\therefore \beta^2 - 16\alpha - 8\beta = 0$

Putting the value of α from (3) in (4), we get

$$\beta^2 - \frac{96\beta}{\beta - 4} - 8\beta = 0 \quad \Rightarrow \quad \beta(\beta^2 - 12\beta - 64) = 0$$

$$\Rightarrow \beta(\beta - 16)(\beta + 4) = 0$$

$$\therefore \beta = 0, 16, -4$$

From (3), when $\beta = 0$, $\alpha = 0$

When $\beta = 16$, $\alpha = 8$ and when $\beta = -4$, $\alpha = 3$

Hence, the feet of the normals are (0, 0), (8, 16) and (3, -4).

11. A

Sol. $A + B = 2B'$

$$\Rightarrow (A + B)' = (2B')'$$

$$\Rightarrow A' + B' = 2B \quad (\because (A + B)' = A' + B')$$

$$\Rightarrow B = \frac{A' + B'}{2}$$

$$\text{Now, } A + \left(\frac{B' + A'}{2}\right) = 2B' \quad [\because A + B = 2B']$$

$$\Rightarrow 2A + B' + A' = 4B' \Rightarrow 2A + A' = 3B'$$

$$\Rightarrow A = \frac{3B' - A'}{2} \quad \dots\dots\dots(1)$$

$$\text{Also, } 3A + 2B = I_3$$

$$\Rightarrow 3\left(\frac{3B' - A'}{2}\right) + 2\left(\frac{A' + B'}{2}\right) = I_3$$

$$\Rightarrow \left(\frac{9B' + 2B'}{2}\right) + \left(\frac{2A' - 3A'}{2}\right) = I_3$$

$$\Rightarrow (11B' - A')' = (2I_3)'$$

$$\Rightarrow 11B - A = 2I_3 \quad \dots\dots\dots(2)$$

Multiplying (2) by 3 and then adding (1) and (2), we get

$$35B = 7I_3 \Rightarrow B = \frac{I_3}{5}$$

$$\text{From (2), } 11\frac{I_3}{5} - A = 2I_3 \Rightarrow 11\frac{I_3}{5} - 2I_3 = A$$

$$\Rightarrow A = \frac{I_3}{5}$$

$$\therefore 5A = 5B = I_3 \Rightarrow 10A + 5B = 3I_3$$

12. D

$$\text{Sol. } (5x - 4) \begin{vmatrix} 1 & 2x & 2x \\ 1 & x - 4 & 2x \\ 1 & 2x & x - 4 \end{vmatrix}$$

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$$= (5x-4) \begin{vmatrix} 1 & 2x & 2x \\ 0 & -(x+4) & 0 \\ 0 & 0 & -(x+4) \end{vmatrix}$$

$$= (5x-4)(x+4)^2 = (A+Bx)(x-A)^2$$

$$\Rightarrow A = -4, B = 5$$

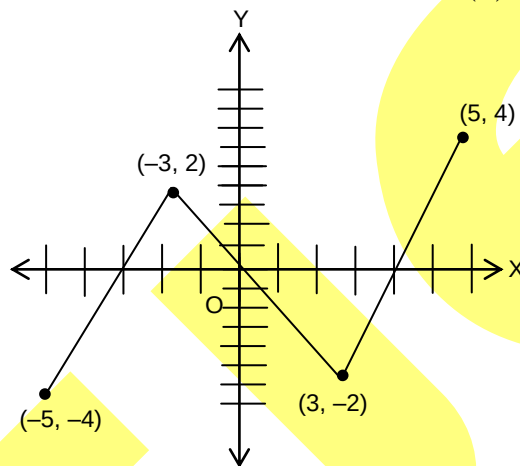
13. D

Sol. $f(x) + f(-x) = 0$

$\Rightarrow f(x)$ is an odd function. Since points $(-3, 2)$ and $(5, 4)$ lie on the curve.

$\therefore (3, -2)$ and $(-5, -4)$ will also lie on the curve

For minimum number of roots, graph of continuous function $f(x)$ is as follows.

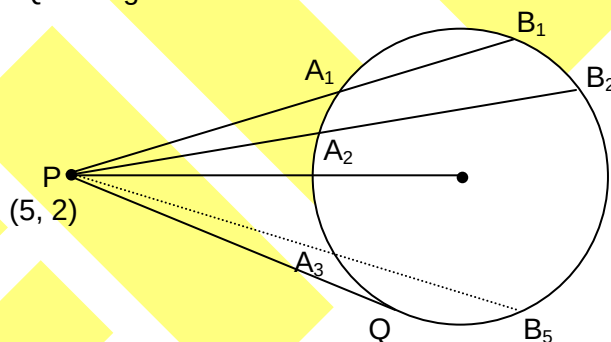


From the above graphs of $f(x)$ it is clear that equation $f(x) = 0$ has at least three real roots.

PART - B

1. 107.50

Sol. PQ is tangent to circle



$$PA_1 \cdot PB_1 = PQ^2 = 4$$

$$PA_2 \cdot PB_2 = 4$$

:

$$PA_5 \cdot PB_5 = 4$$

Given $PA_1 + PB_1 = 5$

:

$$PA_5 + PB_5 = 9$$

Square and Add

$$(PA_1^2 + PA_5^2) + (PB_1^2 + PB_2^2 + PB_5^2)$$

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$$= [5^2 + 6^2 + \dots + 9^2] - 2[4 + 4 + 4 + 4 + 4]$$

$$= 255 - 40 = 215$$

2. 00006.60

Sol. $P(x) = ax^2 + bx + c = 0 \quad (-2, \alpha)$

$$Q(x) = ax^2 + cx + b = 0 \quad (3, \beta)$$

$$S_p + P_q = 0$$

$$[S_p \rightarrow \text{sum of } P(x) \text{ and } P_q \rightarrow \text{Product of } P(x)]$$

$$\text{and } P_p + S_Q = 0$$

$$\therefore \alpha - 2 + 3\beta = 0$$

$$\Rightarrow \alpha + 3\beta = 2 \quad \dots(i)$$

$$\text{and } -2\alpha + 3 + \beta = 0$$

$$\Rightarrow -2\alpha + \beta = -3 \quad \dots(ii)$$

From equation (i) and (ii), we get

$$\alpha = \frac{11}{7} \text{ and } \beta = \frac{1}{7}$$

$$\frac{\alpha}{\beta} = \frac{11}{7} \cdot \frac{7}{1} = 11$$

3. 97.66

Sol. $L: 3^4 - [{}^3C_1(2^4 - 2) + {}^3C_2] = 36$

M: If $x > 0$, $\text{sgn}(x) = 1$

$$f(x) = 0 + \frac{\pi}{2} = \frac{\pi}{2}$$

For $x = 0$ $f(x)$ is not defined

$$\therefore \text{For } x < 0, f(x) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

$$\therefore M = 1$$

$$N: \text{Coefficient of } t^5 = \text{coefficient of } t^2 \text{ in } (1+t^2)^5 \times \text{coefficient of } t^3 \text{ in } (1+t^3)^8$$

$$= 5 \times 8 = 40$$

Hence, $L = 36$; $M = 1$ and $N = 40$

$$\Rightarrow LM + N = 36 + 40 = 76$$

4. 3.13

Sol. Since maximum value of $\cos^{-1} x$ in $[0, 1] = \frac{\pi}{2}$, $\sum_{r=1}^k \cos^{-1} \beta_r = \frac{k\pi}{2}$ is possible if and only if

$$\text{each } \cos^{-1} \beta_r = \frac{\pi}{2} \Leftrightarrow \beta_r = 0.$$

$$A = \sum_{r=1}^k (\beta_r)^r = 0$$

$$\text{Thus, } \lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - (1-2x)^{1/4}}{x + x^2}$$

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$$= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{x^2}{3} + 0(x^4)\right) - \left(1 - \frac{x}{2} + 0(x^2)\right)}{x(1+x)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} + \frac{x}{3} + 0(x^2)}{1+x} = \frac{1}{2}.$$

Thus the required limit is 249.

5. 5.55

Sol. $f(x) = \frac{1}{2} \left(x^2 \cdot \int_0^x g(t) dt - 2x \cdot \int_0^x t^2 + \int_0^x t^2 \cdot g(t) dt \right)$

$\therefore$ Differentiate both sides with respect to x , we get

$$f'(x) = x \cdot \int_0^x g(t) dt - \int_0^x t \cdot g(t) dt \quad \dots\dots(i)$$

Again differentiate both sides with respect to x , we get

$$f''(x) = \int_0^x g(t) dt \quad \dots\dots(ii)$$

$$\therefore f''(1) = 2$$

Also, $f'''(x) = g(x)$

$$\Rightarrow f'''(1) = g(1) = 5$$

6. 20.28

Sol. $F(x) = \int \frac{(1+x)(1+x^2)^2}{(1+x)^2(1+x^2)^2} dx = \ln(1+x) + c$

$$\therefore [F(99) - F(3)] = [\ln 25] = 3$$

7. 17.42

Sol. At $P(x, y)$ for minima, $f'(x) = 0$, $f''(x) > 0$

$$\Rightarrow x^2 + f^2(x) - 6 > 0 \Rightarrow x^2 + y^2 > 6$$

(i.e., P lies outside $x^2 + y^2 = 6$)

$\therefore$ Number of tangents to $x^2 + y^2 = 4$ is 2

8. 17.38

Sol. Here, sum of numbers i.e. $a + b + c + d$ is not given.

Let, $b = a + D$, $c = a + 2D$, $d = a + 3D$, $\forall D \in \mathbb{N}$

According to hypothesis,

$$a + 3D = a^2 + (a + D)^2 + (a + 2D)^2$$

$$\Rightarrow 5D^2 + 3(2a - 1)D + 3a^2 - a = 0 \quad \dots\dots(i)$$

$$\therefore D = \frac{-3(2a - 1) \pm \sqrt{9(2a - 1)^2 - 20(3a^2 - a)}}{10}$$

$$= \frac{-3(2a - 1) \pm \sqrt{(-24a^2 - 16a + 9)}}{10}$$

$$\text{Now, } -24a^2 - 16a + 9 \geq 0$$

$$\Rightarrow 24a^2 + 16a - 9 \leq 0$$

$$\Rightarrow -\frac{1}{2} - \frac{\sqrt{70}}{3} \leq a \leq -\frac{1}{3} + \frac{\sqrt{70}}{12}$$

$$\Rightarrow a = -1, 0 \quad \therefore a \in \mathbb{N}$$

TC ~ @bohiring_bot

When $a = 0$ from Equation (i), $D = 0, \frac{3}{5}$ (not possible $\because D \in \mathbb{N}$) and for $a = -1$

From equation (i), $D = 1, \frac{4}{5}$

$\therefore D = 1$ $[\because D \in \mathbb{N}]$

$\therefore a = -1, b = 0, c = 1, d = 2$

Then $a + b + c + d = -1 + 0 + 1 + 2 = 2$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 2

QP Code:

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 195

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+4 marks** for the correct answer and **– 2 marks** for wrong answer.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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SECTION-1 : PHYSICS

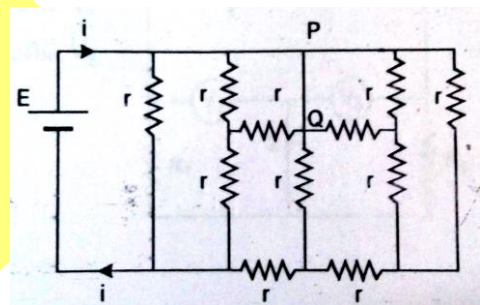
PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

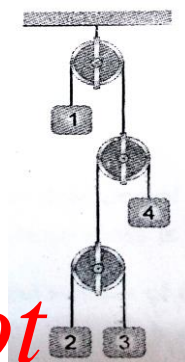
- The electric field strength in a region is given as $\vec{E} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$. Then select the correct option(s).
 (A) The net charge inside a sphere of radius 'a' with its centre at origin will be $2\pi\epsilon_0 a$
 (B) The net charge inside a sphere of radius 'a' with its centre at origin will be $4\pi\epsilon_0 a$
 (C) The net charge inside a sphere of radius '2a' with its centre at origin will be $4\pi\epsilon_0 a$
 (D) The net charge inside a sphere of radius '2a' with its centre at origin will be $8\pi\epsilon_0 a$
- A solution contains a mixture of two isotopes A (half life = 10 days) and B (half life = 5 days). Total activity of the mixture is 10^{10} disintegrations per second at time $t = 0$, the activity reduces to 20% in 20 days. Then select correct option(s).
 (A) Initial activity of A is 0.73×10^{10} dps.
 (B) Initial activity of B is 0.27×10^{10} dps.
 (C) The ratio of initial number of their nuclei is $N_A : N_B$ is 5.4
 (D) None of these

- If $r = 1 \Omega$ and $E = 10 \text{ V}$ in the network shown in figure. Then answer the following:
 (A) The value of current I is 22.85 A.
 (B) The value of current I is 42.85 A.
 (C) The current in branch PQ is 7.62 A.
 (D) The current in branch PQ is 3.8 A.



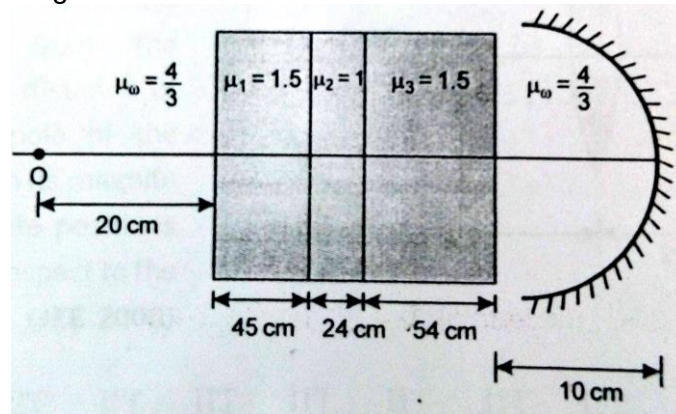
- An aluminium wire of cross-sectional area 10^{-6} m^2 is joined to a steel wire of the same cross-sectional area. This compound wire is stretched on a sonometer pulled by a weight of 10 kg. The total length of the compound wire between the bridges is 1.5 m of which the aluminium wire is 0.6 m and the rest is steel wire. Transverse vibrations are set up in the wire by using an external source of variable frequency. The density of aluminium is $2.6 \times 10^3 \text{ kg/m}^3$ and that of steel is $1.04 \times 10^4 \text{ kg/m}^3$. ($g = 10 \text{ m/s}^2$)
 (A) The lowest frequency of excitation for which standing waves are formed such that joint of wires is a node is nearly 82 Hz.
 (B) The lowest frequency of excitation for which standing waves are formed such that joint of wires is a node is nearly 164 Hz.
 (C) A total of 4 nodes are formed on composite wire.
 (D) A total of 5 nodes are formed on composite wire.

- In the arrangement shown in figure, all pulleys are smooth and massless. When the system is released from rest, accelerations of blocks 2 and 3 relative to 1 are 1 m/s^2 downwards and 5 m/s^2 downwards. Acceleration of block 3 relative to 4 is zero. Then
 (A) The absolute acceleration of block 1 is 2 m/s^2 upward.
 (B) The absolute acceleration of block 2 is 1 m/s^2 downward.
 (C) The absolute acceleration of block 3 is 3 m/s^2 downward.
 (D) The absolute acceleration of block 4 is 3 m/s^2 downward.



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6. A composite slab consisting of different media is placed in front of a concave mirror of radius of curvature 150 cm. The whole arrangement is placed in water ($\mu_w = 4/3$). An object O is placed at a distance 20 cm from the slab. The refractive indices are $\mu_1 = 1.5$, $\mu_2 = 1$, $\mu_3 = 1.5$. The position of final image formed is

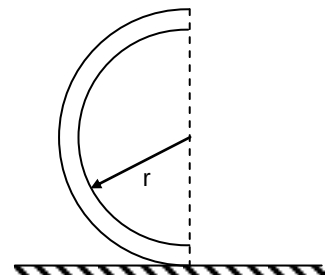


- (A) 130 cm on left of mirror
(B) 153 cm on left of mirror
(C) Final image is real
(D) Final image is virtual
7. Two identical soap bubbles each of radius 1 cm come close to each other and join together maintaining a common surface between them.
- (A) Their common surface will be curved.
(B) Their common surface will be flat.
(C) Area of their common surface will be $\frac{3\pi}{2} \text{ cm}^2$
(D) Area of their common surface will be $\frac{3\pi}{4} \text{ cm}^2$

(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

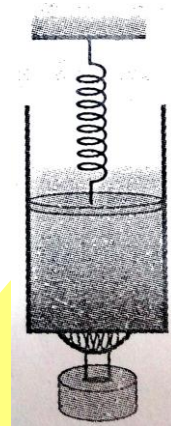
8. A ball of mass "m" moving with kinetic energy 3 J collides with a stationary ball of mass 2m in a head-on elastic collision. During collision, what will be maximum deformation potential energy stored in system?
- (A) 1 J
(B) 2 J
(C) 2.5 J
(D) 1.5 J
9. A semicircular ring of mass m and radius r is released from rest in the position shown with its lower edge resting on a horizontal surface. Find the minimum coefficient of static friction μ_s which is necessary to prevent any initial slipping of the ring.
- (A) 0.2
(B) 0.3
(C) 0.4
(D) 0.5



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10. A gas is inside a cylinder closed by a piston. The piston is held from above by a spring whose elastic properties obey Hooke's law. If the gas is heated slowly then determine the work done by the gas in the process if volume of the gas varies from V_1 to V_2 and the pressure varies from P_1 to P_2 .

- (A) $(P_1 + P_2)(V_2 - V_1)$ (B) $\frac{(P_1 + P_2)(V_2 - V_1)}{2}$
 (C) $\frac{(P_1 + P_2)(V_2 - V_1)}{4}$ (D) Data insufficient

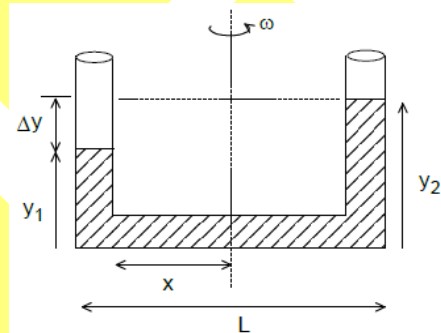


11. An air sealed spherical tank of 1.2 m radius is half filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of 10 m/s^2 . The maximum pressure at any point on the tank is [Assume near vacuum condition]

- (A) $4800\sqrt{2} \text{ N/m}^2$ (B) 4800 N/m^2
 (C) $3036\sqrt{2} \text{ N/m}^2$ (D) 9600 N/m^2

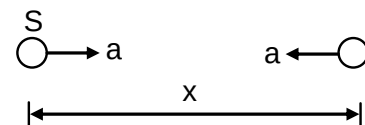
12. An U-shaped tube contains a liquid of density ρ and it rotates about an axis as shown in the figure. Given $L = 2 \text{ m}$, $\Delta y = 1.6 \text{ m}$ and $\omega = 4 \text{ rad/s}$. Then the value of 'x' is

- (A) 0.25 m
 (B) 0.5 m
 (C) 0.75 m
 (D) 1 m



13. A source S and a detector O are initially at a distance of $x = 1 \text{ km}$. Both start moving towards one another with same acceleration $a = 10 \text{ m/s}^2$. Frequency of source is $f = 2000 \text{ Hz}$. Find the frequency observed by the detector at time $t = 4 \text{ second}$. Speed of sound in air is $v = 300 \text{ m/s}$.

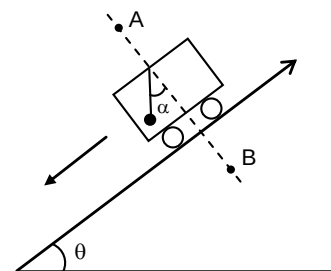
- (A) 2241 Hz (B) 2341 Hz (C) 2441 Hz



- (D) 2541 Hz

14. A pendulum hangs from roof of a cart which slides down the smooth incline as shown. Line AB is perpendicular to incline. What is angle " α " made by string of pendulum with line AB in equilibrium position?

- (A) 0°
 (B) θ
 (C) $\frac{\theta}{2}$
 (D) greater than θ



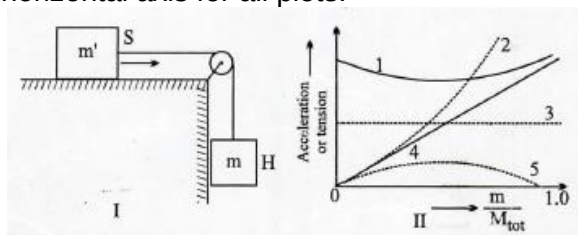
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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Two containers of sand S and H are arranged like the blocks in figure I. The containers alone have negligible mass; the sand in them has a total mass M_{tot} ; the sand in the hanging container H has mass m . You are to measure the magnitude of the acceleration of the system in a series of experiments where m varies from experiment to experiment but M_{tot} does not; that is, you will shift sand between the containers before each trial. $\frac{m}{M_{\text{tot}}}$ is taken on the horizontal axis for all plots.



15. The plot in figure II which gives the acceleration magnitude of the containers (taken on y-axis) against ratio $\left(\frac{m}{M_{\text{tot}}}\right)$ is:
 (A) 1 (B) 3 (C) 4 (D) 5
16. The curve which gives tension in the connecting string (taken on y-axis) against ratio $\left(\frac{m}{M_{\text{tot}}}\right)$ is:
 (A) 1 (B) 2 (C) 4 (D) 5

Paragraph for Question no. 17 to 18

The main scale of a vernier callipers reads in millimetre and its vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws of the instrument touch each other the seventh division of the vernier scale coincide with a main scale division and the zero of the vernier lies to the right of the zero of main scale. Furthermore, when a cylinder is tightly placed along its length between the two jaws, the zero of the vernier scale lies slightly to the left of 3.2 cm; and the fourth vernier division coincides with a scale division. Choose from following the correct option.

17. The zero error in the apparatus is
 (A) -0.07 cm (B) 0.07 cm
 (C) 0.03 cm (D) -0.03 cm
18. The measured value of the length of the cylinder is
 (A) 3.14 cm (B) 3.24 cm
 (C) 3.07 cm (D) 3.17 cm

Space For Rough Work

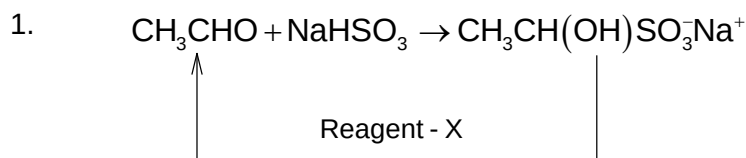
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SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.



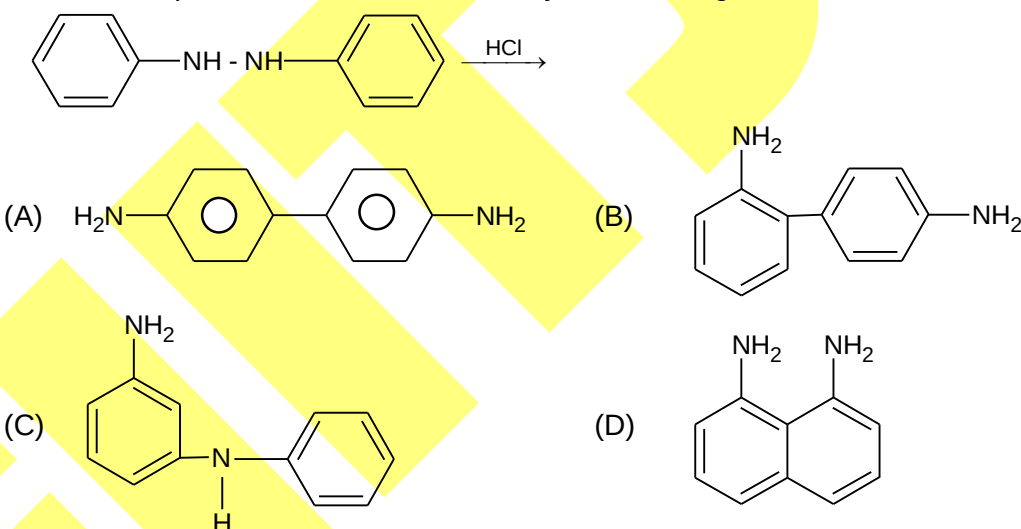
In the above change, the reagent (X) can be

- (A) CCl_4 (B) HCl
(C) NaOH (D) DMF

2. Which of the following reaction(s) takes place in presence of NaOH ?

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}=\text{CH}_2 + \text{H}_2\text{O}$
(B) $\text{CH}_3\text{CHO} + \text{CH}_3\text{CHO} \longrightarrow \text{CH}_3\text{CH}=\text{CHCHO} + \text{H}_2\text{O}$
(C) $\text{CH}_3\text{C}(=\text{O})-\text{O}-\text{C}(=\text{O})-\text{CH}_3 \longrightarrow \text{CH}_3\text{COO}^-$
(D) $\text{CH}_3\text{CONH}_2 \longrightarrow \text{CH}_3\text{CN} + \text{H}_2\text{O}$

3. Find out the products which are formed by the following reaction:



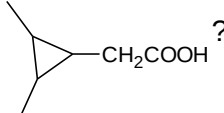
4. Assuming ideal behaviour, when a solution of 1 mol of liquid A in 9 mol of liquid B is mixed at 300 K than for the solution which of the following is/are correct.

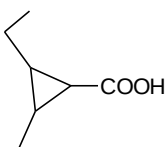
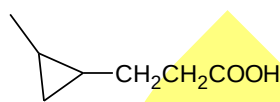
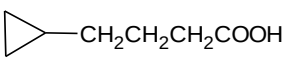
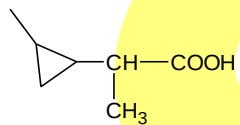
- (A) $\Delta A_{\text{mix}} = 0$ (B) $\Delta S_{\text{mix}} = 27.04 \text{ JK}^{-1}$
(C) $\Delta H_{\text{mix}} = 0$ (D) $\Delta G_{\text{mix}} = -8112 \text{ J}$

5. Choose the correct statement(s):

- (A) Of the two isomeric liquids A and B with their normal b.p's 127°C and 27°C , respectively if the K_b for B is 6 K Kg mol^{-1} then that for A K_b almost equal to 8 k Kg mol^{-1} (K_b = molal elevation constant)
(B) At the b.p. the molar free energy of each phase must be equal
(C) For an ideal solution of two liquids, $\Delta S_{\text{system}} = +\text{Ve}$ and $\Delta S_{\text{surr}} = 0$
(D) Minimum boiling azeotrope results when mixture of two volatile liquids show negative deviation from Raoult's law

6. Which of the following statements is/are correct?
- (A) Boiling point of liquid racemic mixture is same as that of pure enantiomer.
 (B) Melting point of solid racemic mixture is same as that of pure enantiomer.
 (C) In osmosis there is movement of solvent molecules from higher concentration (i.e. solvent) to lower concentration.
 (D) In osmosis there is movement of solvent molecules from lower concentration (i.e. solute) to higher concentration.

7. Which of the following acid(s) is are more acidic than  ?

- (A)  (B) 
 (C)  (D) 

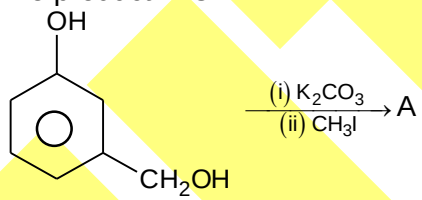
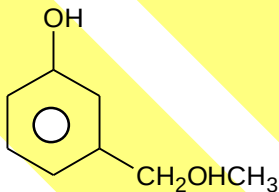
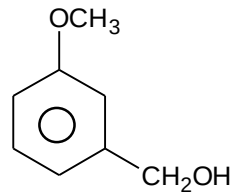
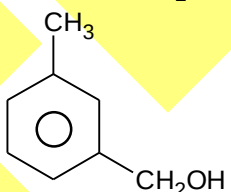
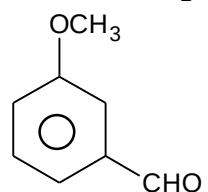
(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

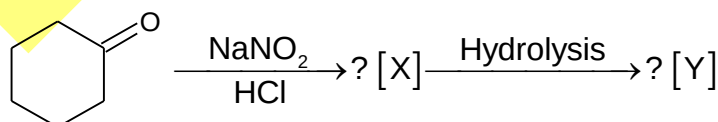
8. The compound that will react most readily with NaOH to form methanol is:-

- (A) $(\text{CH}_3)_4\text{NI}^+$ (B) $\text{CH}_3 - \text{O} - \text{CH}_3$
 (C) $(\text{CH}_3)_3\text{Si}^+$ (D) $(\text{CH}_3)_3\text{C} - \text{Cl}$

9. The product A is

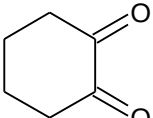
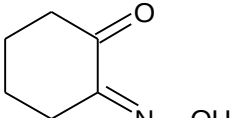
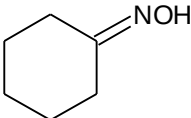
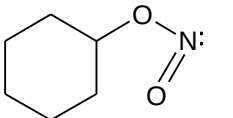
- 
- (A)  (B) 
 (C)  (D) 

- 10.

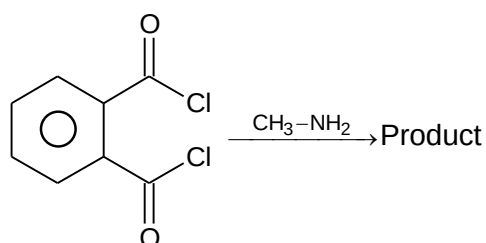


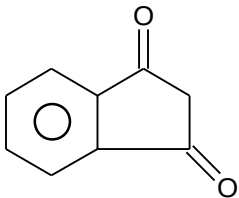
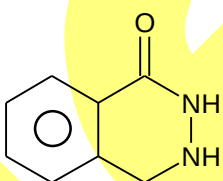
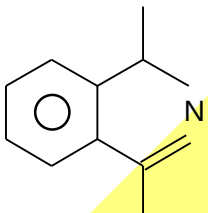
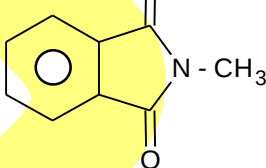
The end product [Y] is

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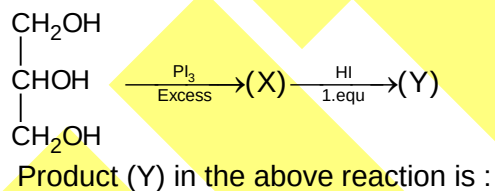
- (A) 
- (B) 
- (C) 
- (D) 

11.

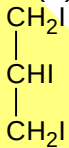
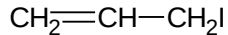
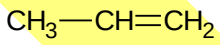
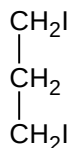


- (A) 
- (B) 
- (C) 
- (D) 

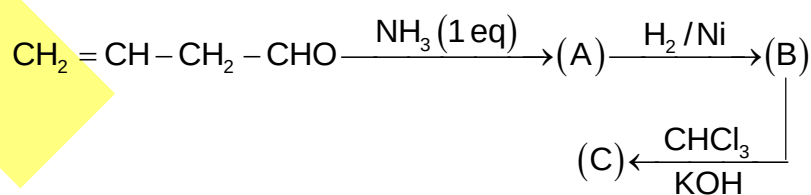
12.



Product (Y) in the above reaction is :

- (A) 
- (B) 
- (C) 
- (D) 

13.

How many pi(π) bond(s) is/are present in one molecule of (C)?

- (A) 1 (B) 2
(C) 4 (D) 3

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14. Which of the following resonating structure is most stable?

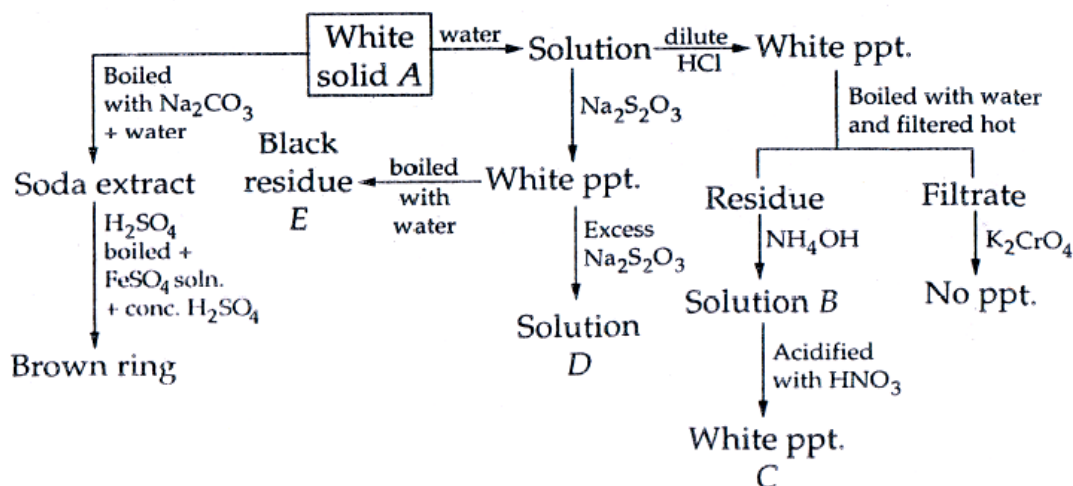
- (A) $\overset{+}{\text{CH}_2} - \text{CH} = \text{CH} - \overset{-}{\text{CH}_2}$ (B) $\text{CH}_2 = \text{CH} - \overset{+}{\text{CH}} - \overset{-}{\text{CH}_2}$
 (C) $\overset{+}{\text{CH}_2} - \overset{-}{\text{CH}} - \text{CH} = \text{CH}_2$ (D) $\overset{-}{\text{CH}_2} - \overset{+}{\text{CH}} - \overset{-}{\text{CH}} - \overset{+}{\text{CH}_2}$

(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Consider the following sequence of reactions.



15. The change from B to C involves the reaction

- (A) $[\text{Hg}(\text{NH}_3)_4]^{2+} + 2\text{H}^+ + \text{Cl}^- \longrightarrow \text{Hg}(\text{NH}_2)\text{Cl} \downarrow + 3\text{NH}_4^+$
 (B) $[\text{Pb}(\text{NH}_3)_4]^{2+} + 4\text{H}^+ + 2\text{Cl}^- \longrightarrow \text{PbCl}_2 \downarrow + 4\text{NH}_4^+$
 (C) $[\text{Pb}(\text{OH})_4]^{2-} + 4\text{H}^+ + 2\text{Cl}^- \longrightarrow \text{PbCl}_2 \downarrow + 4\text{H}_2\text{O}$
 (D) $[\text{Ag}(\text{NH}_3)_2]^{+} + 2\text{H}^+ + \text{Cl}^- \longrightarrow \text{AgCl} \downarrow + 2\text{NH}_4^+$

16. The white ppt. 'C' is

- (A) PbCl_2 (B) PbSO_4 (C) AgCl (D) BaSO_4

Paragraph for Question no. 17 to 18

An aromatic tertiary alcohol upon acid catalysed dehydration gives a product (I). Reductive ozonolysis of (I) forms compounds (J) and (K). Compound (J) upon reaction with NaOH gives benzyl alcohol and compound (L), whereas (K) on heating with NaOH gives only (M)

Answer the following questions on the basis of the above write up.

17. The structure of the compound (I) is:

- (A) $\text{CH}_3\text{CH} = \underset{\text{Ph}}{\text{C}} - \text{Ph}$ (B) $\text{PhCH} = \underset{\text{Ph}}{\text{C}} - \text{CH}_3$
 (C) $\text{PhCH} = \text{CH} - \text{CH}_3$ (D) $\text{PhCH}_2\text{CH} = \text{CH}(\text{Ph})$

18. The structure of compounds J, K and L respectively are:

- (A) PhCHO , PhCH_2CHO and $\text{PhCH}_2\text{COONa}$
 (B) PhCHO , PhCOCH_3 and PhCOONa
 (C) PhCHO , PhCOCH_3 and $\text{PhCH}_2\text{COONa}$
 (D) PhCHO , PhCOPh and PhCOONa

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SECTION-3 : MATHEMATICS**PART – A****(Multi Correct Choice Type)**

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. A line L passing through the point P (1, 4, 3), is perpendicular to both the lines $\frac{x-1}{2} = \frac{y+3}{1} = -\frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$.

If the position vector of point Q on L is (a_1, a_2, a_3) such that $(PQ)^2 = 357$, then $(a + a_2 + a_3)$ can be:

- (A) 16 (B) 15
(C) 2 (D) 1

2. If $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \log(1+x) - 2x^3 + x^4}$ exists and is finite, then

- (A) $a = b$ (B) $c = 0$
(C) $a = 6$ (D) $c = 2$

3. Which of the following definite integral(s) vanishes?

- (A) $\int_0^{\pi/2} \ln(\cot x) dx$ (B) $\int_0^{2\pi} \sin^3 x dx$
(C) $\int_{1/e}^e \frac{dx}{x(\ln x)^{1/3}}$ (D) $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}} dx$

4. Let f be a real valued function satisfying $f\left(\frac{x}{y}\right) = f(x) - f(y)$ and $\lim_{x \rightarrow 0} \frac{f(1+x)}{x} = 3$. The

area bounded by the curve $y = f(x)$, the y – axis and the line $y = 3$ is

- (A) e (B) 2e
(C) 3e (D) 4e

5. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 5, & x \leq 1 \\ -2x + \log_2(b^2 - 2), & x > 1 \end{cases}$. If $f(x)$ has greatest value at $x = 1$, then

$[b^2 \in (2, \lambda)]$. Then, λ is

- (A) multiple of 13 (B) even number
(C) divisible by 5 (D) odd number

6. The area of the region bounded by the curve $y = \frac{16-x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ where $[.]$ denotes the greatest integer function) is

- (A) $\frac{1}{3}(4-\pi)^{3/2}$ (B) $8(4-\pi)^{3/2}$
(C) $\frac{8}{3}(4-\pi)^{3/2}$ (D) $\left(\frac{3}{8(4-\pi)^{3/2}}\right)^{-1}$

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7. If $x = 2 + 5i$ (where $i^2 = -1$) and $2\left(\frac{1}{1! - 9!} + \frac{1}{3! 7!}\right) + \frac{1}{5! 5!} = \frac{2^a}{b!}$, then the value of $(x^3 - 5x^2 + 33x - 19)$ is equal to:
- (A) $a + 1$ (B) b
 (C) $a - b + 11$ (D) $a + b - 9$

(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. A square, of each side 2, lies above the x - axis and has one vertex at the origin. If one of the sides passing through the origin makes an angle 30° with the positive direction of the x - axis, then the sum of the x - coordinates of the vertices of the square is
- (A) $\sqrt{3} - 2$ (B) $2\sqrt{3} - 1$
 (C) $\sqrt{3} - 1$ (D) $2\sqrt{3} - 2$
9. All the chords of the hyperbola $3x^2 - y^2 - 2x + 4y = 0$ subtending a right angle at the origin pass through the fixed points
- (A) $(1, -2)$ (B) $(-1, 2)$
 (C) $(1, 2)$ (D) none of these
10. The sum of two natural numbers n_1 and n_2 is known to be equal to 100. The probability that their product being greater than 1600, is equal to:
- (A) $\frac{20}{33}$ (B) $\frac{58}{99}$
 (C) $\frac{13}{33}$ (D) $\frac{59}{99}$
11. The reflection of the complex number $\frac{4+3i}{1+2i}$, in the straight line $iz = \bar{z}$ is
- (A) $2+i$ (B) $2-i$
 (C) $1+2i$ (D) $1-2i$
12. Let a, b, c be distinct positive numbers such that each of the quadratics $ax^2 + bx + c, bx^2 + cx + a$ and $cx^2 + ax + b$ is non - negative for all $x \in \mathbb{R}$. If $R = \frac{a^2 + b^2 + c^2}{ab + bc + ca}$, then
- (A) $1 \leq R < 4$ (B) $1 < R \leq 4$
 (C) $1 \leq R \leq 4$ (D) $1 < R < 4$
13. Let $P = \begin{bmatrix} (-z)^r & z^{2s} \\ z^{2s} & z^r \end{bmatrix}$, where $r, s \in \{1, 2, 3\}$ and $z = \frac{-1+i\sqrt{3}}{2}$. If $P^2 = -I_2$, then total number of ordered pairs (r, s) is
- (A) 0 (B) 1
 (C) 2 (D) 3
14. Let $S_n = \cot^{-1}\left(3x + \frac{2}{x}\right) + \cot^{-1}\left(6x + \frac{2}{x}\right) + \cot^{-1}\left(10x + \frac{2}{x}\right) + \dots + n \text{ terms}$, where $x > 0$. If $\lim_{n \rightarrow \infty} S_n = 1$, then x equals
- (A) $\frac{\pi}{4}$ (B) 1
 (C) $\tan 1$ (D) $\cot 1$

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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Let f be an even function satisfying $f(x-2) = f\left(x + \left[\frac{6x^2+13}{x^2+2}\right]\right) \forall x \in \mathbb{R}$ and

$$f(x) = \begin{cases} 3x, & 0 \leq x < 1 \\ 4-x, & 1 \leq x \leq 4 \end{cases}$$

[Note : $[y]$ denotes greatest integer function of y .]

15. The area bounded by the graph of $f(x)$ and the x -axis from $x = -1$ to $x = 9$ is:
- (A) $\frac{31}{2}$ (B) 15
(C) 12 (D) $\frac{15}{2}$
16. The value of $f(-89) - f(-67) + f(46)$ is equal to :
- (A) 4 (B) 5
(C) 6 (D) 7

Paragraph for Question no. 17 to 18

Let a differentiable function ' f ' satisfies the functional rule $f(xy) = f(x) + f(y) + xy - x - y \forall x, y > 0$ and $f'(1) = 4$.

17. If $f(x_0) = 0$, then x_0 lies in the interval
- (A) $(0, 1)$ (B) $(1, e)$
(C) (e, e^2) (D) (e^2, e^3)
18. If $\int e^{f(x)} dx = e^x (ax^2 + cx + d) + \lambda$, then the value of $(a + b + c + d)$ is equal to:
- (A) -1 (B) -2
(C) 3 (D) 6

Space For Rough Work

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Q.P. Code:**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|--------|--------|-------|-------|
| 1. BD | 2. ABC | 3. AC | 4. BD |
| 5. ACD | 6. BC | 7. BD | 8. B |
| 9. C | 10. B | 11. C | 12. B |
| 13. B | 14. A | 15. C | 16. D |
| 17. B | 18. C | | |

SECTION-1 : CHEMISTRY**PART – A**

- | | | | |
|--------|--------|-------|--------|
| 1. BC | 2. BC | 3. AB | 4. BCD |
| 5. ABC | 6. ACD | 7. BC | 8. A |
| 9. B | 10. A | 11. D | 12. C |
| 13. B | 14. B | 15. B | 16. A |
| 17. B | 18. B | | |

SECTION-1 : MATHEMATICS**PART – A**

- | | | | |
|--------|--------|---------|-------|
| 1. BD | 2. ABC | 3. ABC | 4. C |
| 5. ABC | 6. CD | 7. ABCD | 8. D |
| 9. B | 10. D | 11. D | 12. D |
| 13. B | 14. D | 15. B | 16. A |
| 17. A | 18. B | | |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **BD**

Sol. At a point P(x, y, z) on sphere of radius 'a' unit vector $\perp$ to surface:

$$\hat{n} = \frac{x}{a}\hat{i} + \frac{y}{a}\hat{j} + \frac{z}{a}\hat{k}$$

$$\vec{E} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$$

$$\vec{E} \cdot \hat{n} = \frac{1}{a} = \text{Independent of point P.}$$

$$\Rightarrow \phi = \frac{1}{a} \times 4\pi a^2 = \frac{q}{\epsilon_0}$$

2. **ABC**

Sol. Initial activities:

$$A_{O1} + A_{O2} = 10^{10}$$

After 20 days:

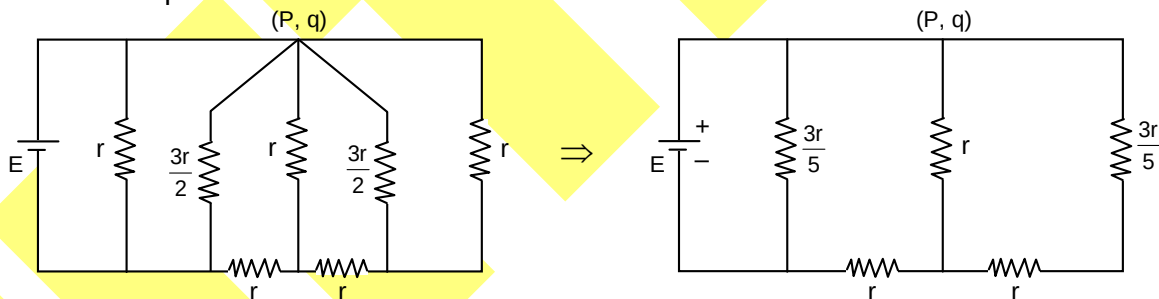
$$A_1 = \left(\frac{1}{2}\right)^2 A_{O1},$$

$$A_2 = \left(\frac{1}{2}\right)^4 A_{O2}$$

$$A_1 + A_2 = 0.2 \times 10^{10}.$$

3. **AC**

Sol. Circuit simplification:



4. **BD**

Sol. Let $n_a \rightarrow$ Number of loops on Al wire.

$N_s \rightarrow$ Number of loops on steel wire.

$$f_a = f_s$$

$$n_a \left(\frac{V_a}{2\ell_a} \right) = n_s \left(\frac{V_s}{2\ell_s} \right)$$

$$\Rightarrow \frac{n_a}{n_s} = \left(\frac{V_s}{V_a} \right) \left(\frac{\ell_a}{\ell_s} \right) ; \quad \frac{V_s}{V_a} = \sqrt{\frac{\rho_a}{\rho_s}} \quad \left(\text{Using } V = \sqrt{\frac{T}{\mu}} \right)$$

5. **ACD**

Sol. String constrains and $\sum_{i=1}^n \vec{T}_i \cdot \vec{a}_i = 0$

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6. **BC**

Sol. Shift produced by three slabs:

$$\Delta x_1 = \left(1 - \frac{\mu_w}{\mu_1}\right)(45) = 5 \text{ cm}$$

$$\Delta x_2 = \left(\frac{\mu_w}{\mu_2} - 1\right)(24) = 8 \text{ cm}$$

$$\Delta x_3 = \left(1 - \frac{\mu_w}{\mu_3}\right)(54) = 6 \text{ cm}$$

 $(\Delta x_1, \Delta x_3) \longrightarrow$ Towards mirror

 $\Delta x_2 \longrightarrow$ Away from mirror.

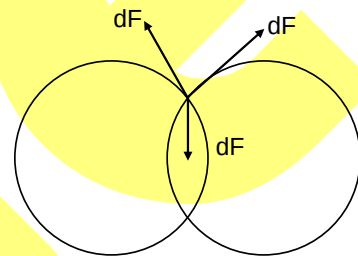
Net shift = 5 + 6 - 8 = 3 cm

Now, solution can be proceeded with.

7. **BD**

Sol. As pressure on both sides of common surface is equal, hence it will be flat.

By symmetry of surface tension forces (dF) on common interface, angle between any two surfaces will be 120° .

8. **B**

Sol. Conservation of momentum:

$$mu = (3m)V$$

$$V = \frac{u}{3} = \text{Common speed at maximum deformation.}$$

Maximum deformation energy = Maximum loss of kinetic energy during impact.

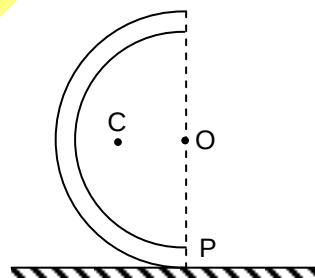
$$= \frac{1}{2}mu^2 - \frac{1}{2}(3m)V^2$$

9. **C**

Sol. As there is no slipping I.C.R is at point P.

C $\longrightarrow$ Centre of mass

$$OC = \frac{2r}{\pi}$$

10. **B**Sol. When piston moves up by distance Δx :

$$\text{Gas pressure} = P = P_0 + \frac{mg}{A} + \frac{k\Delta x}{A}$$

$$P = P_0 + \frac{mg}{A} + \frac{k}{A^2}(V - V_i)$$

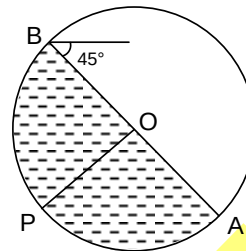
Draw P-V graph which will be straight line.

Work done = Area under P-V graph.

11. **C**

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- Sol. In frame of reference of tank, g_{eff} will be along OP where $OP \perp AB$. Maximum pressure will be at point P.



12. **B**

Sol. Centrifugal force on horizontal liquid makes the liquid rise in vertical section.

13. **B**

Sol. Let t be the time when sound is emitted which is received at $t = 4$ sec.

$$\text{Then: } \frac{1}{2}at^2 + \frac{1}{2}a(4)^2 + v(4-t) = 1000$$

$$\frac{1}{2}(10)t^2 + \frac{1}{2}(10)(16) + (300)(4-t) = 1000$$

$$t = 0.95 \text{ sec.}$$

Hence, pulse emitted at 0.95 sec will be received at $t = 4$ sec.

14. **A**

Sol. Due to effect of pseudo force, g_{eff} in frame of cart will be along line AB. Hence, $\alpha = 0^\circ$.

15. **C**

Sol. Let acceleration of system = a

$$mg - T = ma \quad \dots(1)$$

$$(M - m)a = T \quad \dots(2)$$

16. **D**

Sol. As in problem 15, tension (T) can also be calculated.

17. **B**

Sol. Zero error = M.S.R + (V.S.R) (L.C) (with closed scale)

$$= 0 + (7) \left(\frac{1}{10} \text{ mm} \right)$$

$$= 0.07 \text{ cm.}$$

18. **C**

Sol. Length = M.S.R + (V.S.R) (L.C) - (Z.E.)

$$= 3.1 \text{ cm} + 4 \times (0.01 \text{ cm}) - (0.07) \text{ cm}$$

$$\text{Length} = 3.07 \text{ cm.}$$

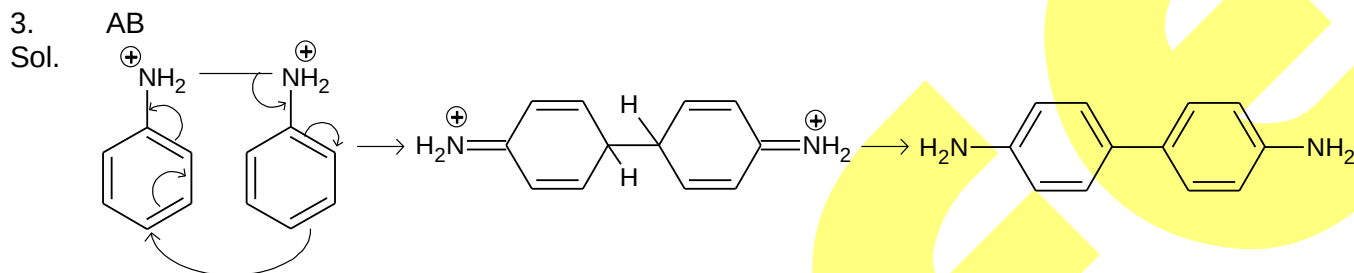
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SECTION-2 : CHEMISTRY

PART – A

1. BC
Sol. Hydrolysis in acidic or basic medium.

2. BC
Sol. Aldol & I hydrolysis



4. BCD
Sol. $\Delta S_{\text{mixing}} = -n_{\text{total}} \ell \sum \chi_i \ln \chi_i$
 $= -10 \times 8.314 \times (-0.23 - 0.0948) = -0.32 \times 10 \times 9.314 = 27 \text{ JK}^{-1}$

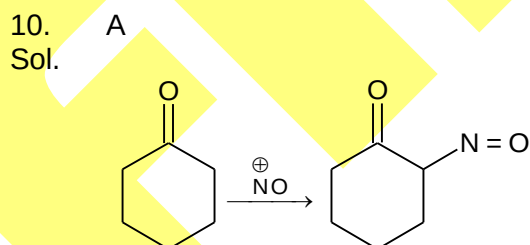
5. ABC
Sol. $K_f = \frac{MR(T_b^0)^2}{\Delta H_{\text{vap}}}$, at equilibrium $\Delta G = 0$

6. ACD
Sol. Theoretical

7. BC
Sol. Acidic strength $\propto$ Stability of conjugate base.

8. A
Sol. Good leaving group & electronegativity.

9. B
Sol. Acidity Phenol > Alcohol



11. D
Sol. $S_{N_{AE}}$ of acid halide.

12. C
Sol. $\text{CH}_2 = \underset{(X)}{\text{CH}} - \text{CH} - \text{I} \xrightarrow{\text{HI}} \text{CH}_3 - \underset{(Y)}{\text{CH}} = \text{CH}_2$

13. B

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Sol. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{N} \equiv \text{C}$ (imine followed by reduction followed by carbylamine)

14. B

Sol. Theoretical, opposite charge close

15. D

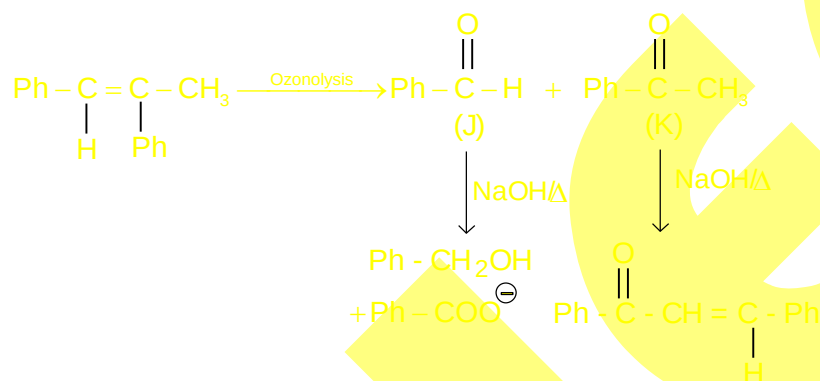
16. C

Sol for 15 & 16. A = AgNO_3 , B = $[\text{Ag}(\text{NH}_3)_2]^+$, C = $\text{AgCl} \downarrow$, D = $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, E = Ag_2S

17. B

18. B

Sol for 17 & 18.



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SECTION-3 : MATHEMATICS

PART – A

1. BD

Sol. Equation of the line passing through P(1, 4, 3) is: $\frac{x-1}{a} = \frac{y-4}{b} = \frac{z-3}{c}$ (i)

Since equation (i) is perpendicular to $\frac{x-1}{2} = \frac{y+3}{1} = \frac{z-2}{4}$ and $\frac{x+2}{3} = \frac{y-4}{2} = \frac{z+1}{-2}$

Hence $2a+b+4c=0$ and $3a+2b-2c=0$

$$\therefore \frac{a}{-2-8} = \frac{b}{12+4} = \frac{c}{4-3}$$

$$\Rightarrow \frac{a}{-10} = \frac{b}{16} = \frac{c}{1}$$

Hence the equation of the lines is $\frac{x-1}{-10} = \frac{y-4}{16} = \frac{z-3}{1}$ (ii)

Now any point Q on (2) can be taken as $(1-10\lambda, 16\lambda+4, \lambda+3)$

$$\therefore \text{Distance of Q from P (1, 4, 3)} = (10\lambda)^2 + (16\lambda)^2 + \lambda^2 = 357$$

$$\Rightarrow (100+256+1)\lambda^2 = 357$$

$$\Rightarrow \lambda = 1 \text{ or } -1$$

$\therefore$ Q is $(-9, 20, 4)$ or $(11, -12, 2)$

Hence $a_1 + a_2 + a_3 = 15$ or 1

2. ABC

Sol.
$$\lim_{x \rightarrow 0} \frac{a \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots \right) - bx + cx^2 + x^3}{2x^2 \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right) - 2x^3 + x^4}$$

$$= \lim_{x \rightarrow 0} \frac{(a-b)x + cx^2 + (1-a/6)x^3 + ax^5/120 + \dots}{2x^2/3 - x^6/2 + \dots}$$

For this limit to exist we must have $a-b=0$, $c=0$, and $1-a/6=0$, that is, $a=b=6$ and $c=0$

3. ABC

Sol.

(A) $I = \int_0^{\pi/2} \ln(\cot x) dx \Rightarrow I = \int_0^{\pi/2} \ln(\tan x) dx$

$$I = - \int_0^{\pi/2} \ln(\cot x) dx \Rightarrow I = -I \Rightarrow I = 0$$

(B) $I = \int_0^{2\pi} \sin^3 x dx = - \int_0^{2\pi} \sin^3 x dx \Rightarrow I = 0$

(C) $I = \int_{1/e}^e \frac{-\left(\frac{1}{t^2}\right) dt}{\frac{-1}{t} (\ln t)^{1/3}} = \int_{1/e}^e \frac{dt}{t (\ln t)^{1/3}}$

$$I = -I$$

$$I = 0$$

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(D) $\sqrt{\frac{1+\cos 2x}{2}} > 0$
 $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}} dx > 0$

4. C

Sol. Given $f\left(\frac{x}{y}\right) = f(x) - f(y)$... (1)

Putting $x = y = 1$, we get

$$\text{Now, } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right)}{h} \quad (\text{From (1)})$$

$$= \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right)}{\left(\frac{h}{x}\right)x}$$

$$\Rightarrow f'(x) = \frac{3}{x}$$

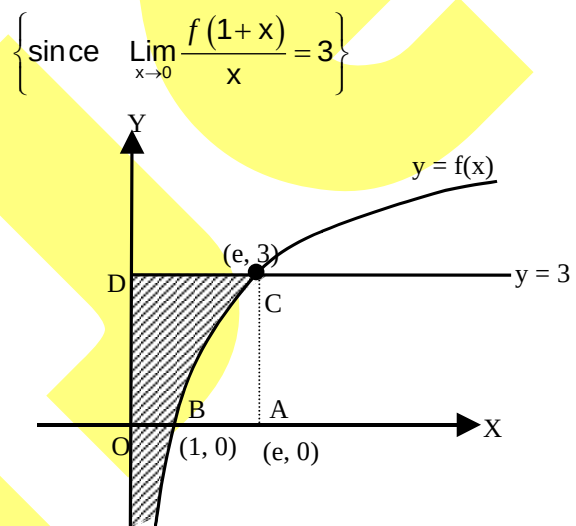
$$\Rightarrow f(x) = 3 \ln x + c$$

Putting $x = 1$

$$\Rightarrow c = 0$$

$$\Rightarrow f(x) = 3 \ln x = y \quad (\text{say})$$

$$\begin{aligned} \therefore \text{Required area} &= \int_{-\infty}^3 x \, dy \\ &= \int_{-\infty}^3 e^{y/3} dy \\ &= 3 \left[e^{y/3} \right]_{-\infty}^3 \\ &= 3(e - 0) \end{aligned}$$



5. ABC

Sol. For $x \leq 1$ $f'(x) = 3x^2 - 2x + 10$

$$= 3 \left\{ \left(x - \frac{1}{3} \right)^2 + \frac{29}{9} \right\} > 0$$

$\therefore f(x)$ is increasing function for $x \leq 1$ Now, for $x > 1$,

$$f'(x) = -2 < 0$$

So, $f(x)$ is decreasing function, for $x > 1$

Now, $f(x)$ will have greatest value at $x = 1$ if

$$\Rightarrow \lim_{x \rightarrow 1^+} f(x) \leq f(1)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(1+h) \leq 5$$

$$\Rightarrow \lim_{h \rightarrow 0} -2(1+h) + \log_2(b^2 - 2) \leq 5$$

$$\Rightarrow -2 + \log_2(b^2 - 2) \leq 5$$

$$\Rightarrow \log_2(b^2 - 2) \leq 7$$

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$$\Rightarrow b^2 - 2 \leq 2^7$$

$$\Rightarrow b^2 \leq 130$$

but $b^2 - 2 > 0$

$$\therefore 2 < b^2 \leq 130$$

Hence, $\lambda = 130$

6. CD

Sol. $0 \leq \sin^2 x \leq 1$

$$\Rightarrow -1 \leq -\sin^2 x \leq 0$$

$$\therefore [\sin^2 x] = 0 \text{ or } -1$$

but $\sec^{-1}(0)$ is not defined

hence $y = \sec^{-1}[-\sin^2 x] = \sec^{-1}(-1) = \pi$

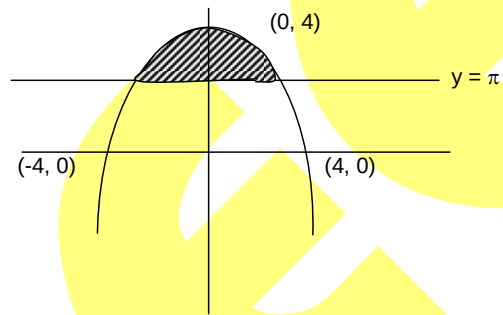
now $\pi = \frac{16 - x^2}{4}$

$$\Rightarrow x^2 = 16 - 4\pi = 4(4 - \pi)$$

$$x = \pm 2\sqrt{4 - \pi}$$

The required area =

$$\int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}} \left(\frac{16 - x^2}{4} - \pi \right) dx = \frac{8}{3} (4 - \pi)^{3/2}$$



7. ABCD

Sol. $2\left(\frac{1}{1!9!} + \frac{1}{3!7!}\right) + \frac{1}{5!5!} = \frac{2^a}{b!}$

$$\Rightarrow \frac{2}{1!9!} + \frac{2}{3!7!} + \frac{1}{5!5!} = \frac{2^a}{b!}$$

$$\Rightarrow \frac{1}{10!} \left(\frac{2!10!}{1!9!} + \frac{2!10!}{3!7!} + \frac{10!}{5!5!} \right) = \frac{2^a}{b!}$$

$$\Rightarrow \frac{1}{10!} (2 \cdot {}^{10}C_1 + 2 \cdot {}^{10}C_3 + {}^{10}C_5) = \frac{2^a}{b!}$$

$$\Rightarrow \frac{1}{10!} ({}^{10}C_1 + {}^{10}C_1 + {}^{10}C_3 + {}^{10}C_3 + {}^{10}C_5) = \frac{2^a}{b!}$$

$$\Rightarrow \frac{1}{10!} ({}^{10}C_1 + {}^{10}C_9 + {}^{10}C_3 + {}^{10}C_7 + {}^{10}C_5) = \frac{2^a}{b!}$$

$$\Rightarrow \frac{1}{10!} ({}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + {}^{10}C_7 + {}^{10}C_9) = \frac{2^a}{b!}$$

$$\Rightarrow \frac{2^{10-1}}{10!} = \frac{2^a}{b!}$$

$$\therefore a = 9, b = 10$$

Also, $x = 2 + 5i$

$$\therefore (x - 2)^2 = (5i)^2$$

$$\Rightarrow x^2 - 4x + 29 = 0$$

$$\therefore x^3 - 5x^2 + 33x - 19 = (x - 1)(x^2 - 4x + 29) + 10$$

$$= 0 + 10$$

$$(\because x^2 - 4x + 29 = 0)$$

$$= 10 = b$$

8. D

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Sol. Side of square = 2

$$\frac{x}{\cos 30^\circ} = \frac{y}{\sin 30^\circ} = 2$$

$$\Rightarrow x = \frac{2\sqrt{3}}{2} = \sqrt{3} \text{ and } y = 1$$

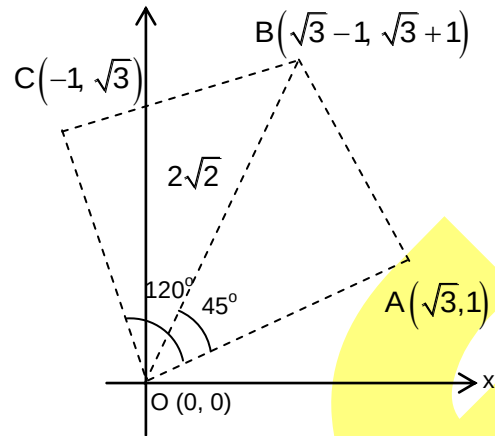
$$\frac{x}{\cos 120^\circ} = \frac{y}{\sin 120^\circ} = 2$$

$$\Rightarrow x = -1, y = \sqrt{3}$$

$$\frac{x}{\cos 75^\circ} = \frac{y}{\sin 75^\circ} = 2\sqrt{2}$$

$$\Rightarrow x = \sqrt{3} - 1 \text{ and } y = \sqrt{3} + 1$$

$$\therefore \text{Required sum} = 0 + \sqrt{3} + \sqrt{3} - 1 + (-1) \\ = 2\sqrt{3} - 2$$



9. B

Sol. Let $ax + by = 1$ be the chord (1)

Making the equation of hyperbola homogeneous using (1),

We get

$$3x^2 - y^2 + (-2x + 4y)(ax + by) = 0$$

$$\text{or, } (3 - 2a)x^2 + (-1 + 4b)y^2 + (-2b + 4a)xy = 0$$

Since the angle subtended at the origin is a right angle, so, coefficient of x^2 + coefficient of $y^2 = 0$

$$\Rightarrow (3 - 2a) + (-1 + 4b) = 0 \Rightarrow a = 2b + 1$$

$$\therefore \text{The chords are } (2b + 1)x + by - 1 = 0$$

$$\text{Or, } b(2 + y) + (x - 1) = 0,$$

which, clearly, pass through the fixed point (1, -2)

10. D

Sol. Total number of ways in which $n_1 + n_2 = 100$ is equal to 99.

Now, $n_1, n_2 > 1600$

$$\Rightarrow n_1(100 - n_1) > 1600$$

$$\Rightarrow n_1^2 - 100n_1 + 1600 < 0$$

$$\Rightarrow (n_1 - 80)(n_1 - 20) < 0$$

$$\Rightarrow 20 < n_1 < 80$$

$$\Rightarrow 21 \leq n_1 \leq 79$$

Thus, number of favourable ways

$$= 79 - 21 + 1 = 59$$

$$\text{Hence, required probability} = \frac{59}{99},$$

11. D

Sol. We have,

$$z_1 = \frac{4 + 3i}{1 + 2i} = \frac{(4 + 3i)(1 - 2i)}{(1 + 2i)(1 - 2i)}$$

$$= \frac{10 - 5i}{5} = 2 - i \text{ which represents the point whose coordinates are } (2, -1)$$

Also, we have,

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$$iz = \bar{z}$$

$$\Rightarrow i(x + iy) - (x - iy) = 0 \quad [\text{Putting } z = x + iy]$$

$$\Rightarrow i(x + y) - (x + y) = 0$$

$$\Rightarrow (i - 1)(x + y) = 0 \text{ which represents the line } y = -x$$

Hence, reflection of the point $(2, -1)$ in the line $y = -x$ gives the point $(1, -2)$ which is equivalent to $1 - 2i$ in the argand plane.

The correct option is (D)

12. D

Sol. Given, $b^2 \leq 4ac$, $c^2 \leq 4ab$ and $a^2 \leq 4ac$

Equality cannot hold simultaneously

$[\because a, b, c \text{ are different}]$

$$\therefore a^2 + b^2 + c^2 < 4(ab + bc + ca) \Rightarrow R < 4$$

$$\text{Also, } a^2 + b^2 + c^2 - ab - bc - ca$$

$$\Rightarrow \frac{1}{2}[(b - c)^2 + (c - a)^2 + (a - b)^2] > 0 \Rightarrow R > 1$$

The correct option is (D)

13. B

Sol. $z = \omega$

$$P^2 = \begin{bmatrix} (-\omega)^r & \omega^{2s} \\ \omega^{2s} & \omega^r \end{bmatrix} \begin{bmatrix} (-\omega)^r & \omega^{2s} \\ \omega^{2s} & \omega^r \end{bmatrix} = \begin{bmatrix} \omega^{2r} + \omega^{4s} & \omega^{r+2s}(1 + (-1)^r) \\ \omega^{r+2s}(1 + (-1)^r) & \omega^{2r} + \omega^{4s} \end{bmatrix} = -I_2$$

$$\Rightarrow 1 + (-1)^r = 0 \Rightarrow r = 1 \text{ or } 3 \text{ and } \omega^{2r} + \omega^{4s} = -1 \quad \dots\dots\dots(i)$$

Possible options are $(r, s) = (1, 1), (1, 2), (1, 3), (3, 1), (3, 2), (3, 3)$ of which one $(1, 1)$ satisfies (i).

14. D

$$\text{Sol. As, } T_n = \cot^{-1} \left[\frac{(n+1)(n+2)}{2} x + \frac{2}{x} \right]$$

$$\Rightarrow T_n = \tan^{-1} \left(\frac{2x}{(n+2)(n+1)x^2 + 4} \right)$$

$$\therefore T_n = \tan^{-1} \left(\left(\frac{n+2}{2} \right) x \right) - \tan^{-1} \left(\left(\frac{n+1}{2} \right) x \right)$$

$$\text{So, } S_n = \tan^{-1} \left(\left(\frac{n+2}{2} \right) x \right) - \tan^{-1} x$$

$$\Rightarrow \lim_{n \rightarrow \infty} S_n = \frac{\pi}{2} - \tan^{-1} x$$

$$= \cot^{-1} x = 1 \quad (\text{given})$$

$$\Rightarrow x = \cot 1$$

15. B

$$\text{Sol. Required area} = 2 \times \frac{1}{2} \times 4 \times 3 + 2 + 2 \times \frac{1}{2} \times 1 \times 3 = 15$$

16. A

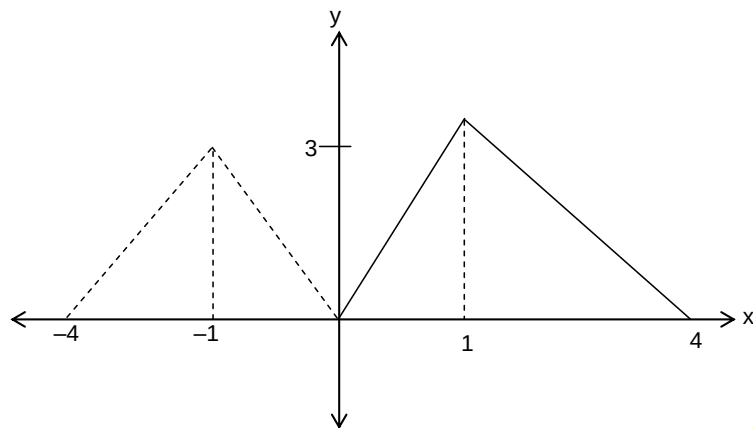
$$\text{Sol. } f(-89) - f(-67) + f(46) = f(-1) - f(-3) + f(-2)$$

$$= 3 - 1 + 2 = 4$$

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$$f(x-2) = f(x+6)$$

$$\Rightarrow f(x) = f(x+8)$$



17. A

18. B

(Sol. 17 to 18)

$$f(xy) = f(x) + f(y) + xy - x - y$$

Partially differentiating w.r.t. y (taking x as a constant)

$$f'(xy) \cdot x = f'(y) + x - 1$$

Putting $y = 1$

$$f'(x) = x = f'(1) + x - 1$$

$$f'(x) \cdot x = 3 + x$$

$$f'(x) = \frac{3}{x} + 1$$

Integrating both sides.

$$f(x) = 3 \ln x + x + c$$

From the given functional rule

$$x = y = 1, f(1) = 1$$

$$f(x) = 3 \ln x + x + c$$

$$\therefore f(x) = 3 \ln x + x$$

$$(i) f(x) = 0 \Rightarrow 3 \ln x + x = 0$$

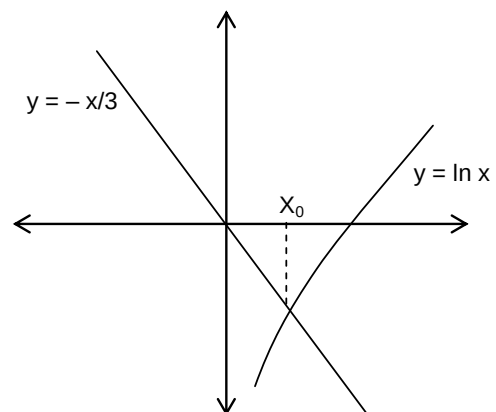
$$\Rightarrow \ln x = \frac{-x}{3}$$

Clearly, x_0 lies in the interval $(0, 1)$.

$$(ii) \int e^{f(x)} dx = e^x (ax^3 + bx^2 + cx + d) + \lambda$$

$\Rightarrow$

$$\int e^{3 \ln x + x} dx = e^x (ax^3 + bx^2 + cx + d) + \lambda$$



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PHYSICS, CHEMISTRY & MATHEMATICS**QP Code: 100182****Paper-1****Time Allotted: 3 Hours****Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

- (i) **PART – A (01–04):** This section contains **Four(04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.

PART – A (05–10): This section contains **Six (06)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.

Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.

Zero Marks: 0 If none of the bubbles is darkened.

Negative Marks: –2 In all other cases.

For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **–2 marks**, as a wrong option is also darkened.

- (ii) **PART – B (1–3):** This section contains **Three (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. Each question carries **+4 marks** for correct answer and **there will be no negative marking**.

- (iii) **PART – B(4–9):** This section contains **Six (06)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places. Each question carries **+2 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)**IG ~ @bohring_bot**

SECTION – I : PHYSICS

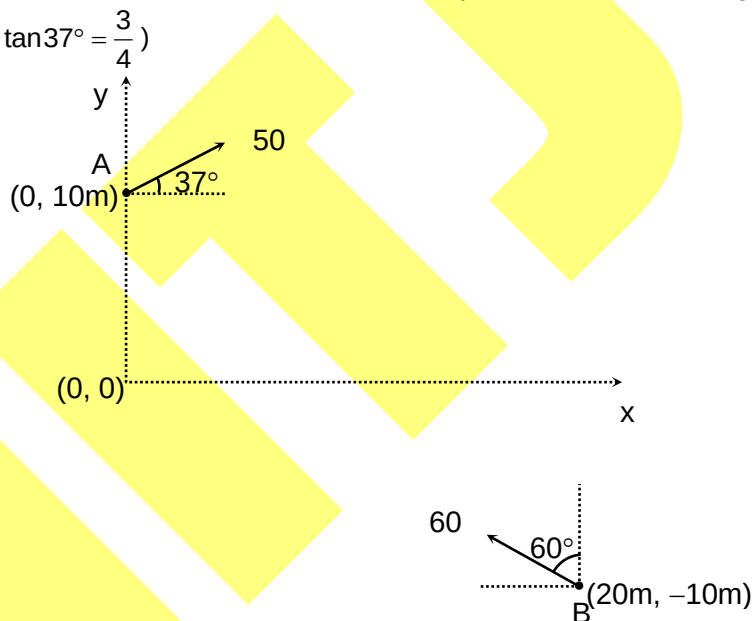
PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A circular disc of radius R carries surface charge density $\sigma(r) = \sigma_0 \left(1 - \frac{r^2}{R^2}\right)$, where σ_0 is a constant and r is the distance from the centre of the disc. Assume that disc is in xz -plane and its axis is y -axis. We have an imaginary hollow sphere of radius R and centre at $\left(0, \frac{3}{5}R, 0\right)$. What is the flux of electric field through this imaginary sphere?

- (A) $\frac{408}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$ (B) $\frac{272}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$
 (C) $\frac{544}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$ (D) $\frac{136}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$

2. Two particles A and B are projected in the vertical plane with velocity 50 m/s at an angle of 37° with horizontal and with velocity 60 m/s at an angle of 60° with vertical respectively as shown in the figure. Find the minimum distance between the particles A and B during the motion? (Take $g = 10 \text{ m/s}^2$ and $\tan 37^\circ = \frac{3}{4}$)



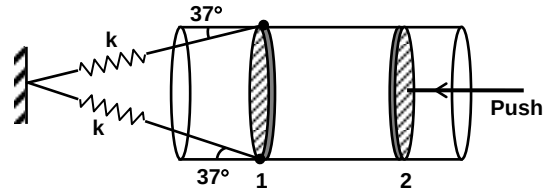
- (A) 10 m
(C) 20 m

- (B) 15 m
(D) 8 m

Space For Rough Work

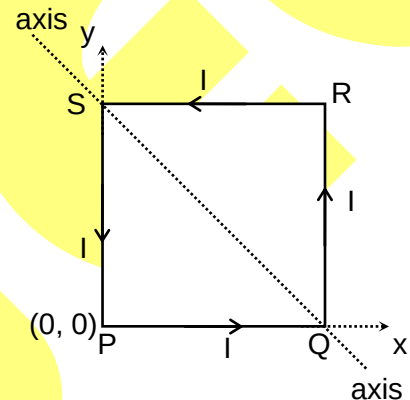
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3. A long container has air enclosed inside at room temperature and atmospheric pressure 10^5 N/m^2 between two movable pistons. It has a volume $20,000 \text{ cc}$. The area of cross section is 100 cm^2 and force constant of springs is $k = 781.25 \text{ N/m}^2$. Now push the right piston isothermally and slowly till it reaches the original position of the left piston-1 which is movable. What is the final length of air column. Assume that spring is initially relaxed and length of spring is very large as compare to separation between two piston.



- (A) 100 cm
(B) 75 cm
(C) 50 cm
(D) 200 cm

4. A uniform rigid, square loop of mass ' m ' and side ' ℓ ' is free to rotate about an axis in xy plane passing through two corner as shown in figure. A constant current I is flowing in the loop. In space magnetic field $\vec{B}(x) = B_0 \left(1 + \frac{x}{\ell} \right) (\hat{i} + \hat{k})$ is present. Initially loop is held at rest in xy -plane. On releasing its instantaneous angular acceleration will be (space is gravity free space)



- (A) $\frac{9IB}{\sqrt{2}m}$
(B) $\frac{5IB}{2\sqrt{2}m}$
(C) $\frac{9IB}{2\sqrt{2}m}$
(D) $\frac{5\sqrt{2}IB}{2m}$

Space For Rough Work

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PART – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

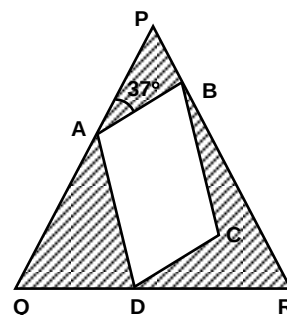
5. A gaseous mixture at 300 K and $2 \times 10^5 \text{ N/m}^2$ pressure contains 6g of H_2 and 8g of He. The mixture is expanded four times its initial volume, through an isobaric heating process. Then it is **isochorically** cooled until its temperature again becomes 300 K. After that the gas mixture is isothermally compressed to its original volume. Choose the correct option(s). (take $\ln 2 = 0.695$)

- (A) Ratio of molar specific heat (γ) of the mixture is $\frac{31}{21}$
 (B) Ratio of heat absorb in isobaric expansion to that of heat rejected in isochoric cooling is $\frac{31}{21}$
 (C) Heat rejected in isothermal compression is more than the heat rejected in isochoric cooling.
 (D) Efficiency of the complete cycle is $\frac{161}{930}$.

6. An electron, initially at rest, is released far away from a proton (fixed in space). The de-broglie wavelength of the electron when it is at distance r from proton is λ . The ratio of its Kinetic energy at this distance with the kinetic energy of the electron in ground state of hydrogen atom (Bohr model) with the radius of its orbit being r , is μ . Pick the correct option(s).

- (A) $\lambda \propto \sqrt{r}$ (B) $\lambda \propto \frac{1}{r}$
 (C) $\mu = \frac{1}{2}$ (D) $\mu = 2$

7. A rectangular cavity ABCD is carved inside an equilateral prism PQR of refractive index $\left(\frac{8}{5}\right)$ as shown in figure. If $\delta_{\min}$ is the deviation in a ray entering at face PQ and emerging at face PR, without passing through AB or CD, then which of the following is/are correct.

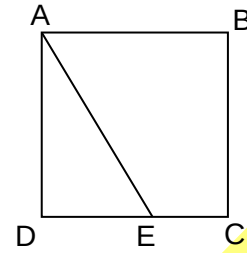


- (A) There is no effect of cavity of the value of $\delta_{\min}$.
 (B) The minimum deviation $\delta_{\min} = 46^\circ$
 (C) If the cavity is filled with water ($\mu_w = 4/3$), the $\delta_{\min} = 46^\circ$
 (D) If the cavity is filled with a liquid of refractive index $\mu = 2$, then $\delta_{\min} = 23^\circ$

Space For Rough Work

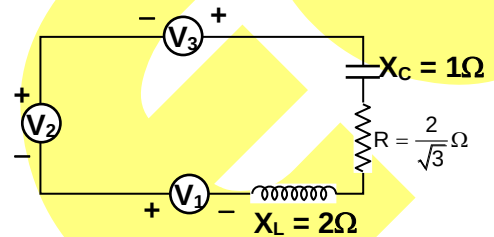
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8. ABCD is square as shown in figure, resistance of side AB, BC, CD and DA are 8Ω , 2Ω , 14Ω and 22Ω respectively. A point E lies on the CD such that a uniform wire of resistance 32Ω is connected across AE and constant potential difference is applied across A and C then B and E are equipotential. Choose the correct option(s).



- (A) Resistance of DE is 10Ω
 (B) Resistance of EC is 10Ω
 (C) Equivalent resistance across AC is $\frac{20}{3}\Omega$
 (D) Equivalent resistance across AC is 10Ω

9. Three alternating voltage sources $V_1 = 3\sin\omega t$, $V_2 = 5\sin(\omega t + 30^\circ)$ and $V_3 = 5\sin(\omega t - 127^\circ)$ volt connected across a resistor, inductor and capacitor as shown in the figure, then choose the correct option(s)



- (A) The maximum current in the resistors is 3 ampere.
 (B) The maximum current in the resistor is 2 ampere.
 (C) Effective impedance of the circuit is $\sqrt{\frac{7}{3}}\Omega$
 (D) Maximum potential of the three AC source is $\sqrt{21}$ volts.
10. Two vernier calipers A and B both have M.S.D. of 1 mm. 10 vernier scale division have same length as 7 main scale division of vernier A and 8 vernier scale division have same length as 3 main scale division for vernier B.
- (A) The least count of vernier A is more than the vernier B.
 (B) The least count of vernier A is less than the vernier B.
 (C) The least count of vernier B is 0.125 mm.
 (D) The magnitude of difference in least count of vernier A and vernier B is 0.025 mm.

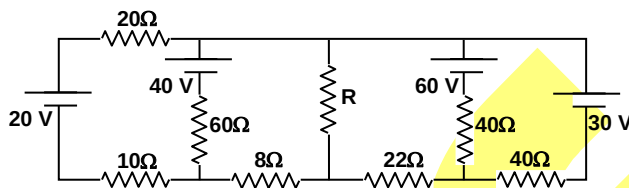
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PART – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. In the given circuit, the value of R so that thermal power generated in R will be maximum is $4.2\text{m}\Omega$. Find the value of m .



2. Velocity of an object in rectilinear motion is given as function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. Its average speed over the time interval from $t = 0$ second to $t = 2$ seconds, is $\frac{8K}{27}$ m/s. Find the value of K .
3. An open pipe 68.48 cm long and a closed pipe 58.4 cm long, both having same diameter, are producing their second overtone, and these are in unison. Determine the end correction in centimeter of these pipes.

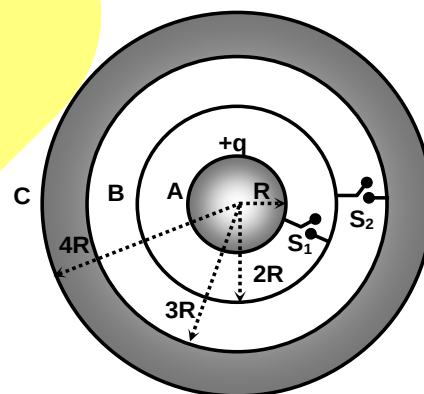
PART – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 4 and 5**Question Stem**

Three conducting bodies A (solid sphere), B (spherical shell) and C (hollow sphere with thickness R) are arranged as shown in the figure. A charge q is given to the inner sphere. When S_1 , S_2 both are open capacity of the system is C_1 and when S_1 is closed but S_2 is open capacity is C_2 , then $(C_2 - C_1) = \frac{X\pi\epsilon_0 R}{55}$.

When S_2 is closed but S_1 is opened capacity is C_3 and When both S_1 and S_2 are closed capacity is C_4 then $(C_4 - C_3) = \frac{Y\pi\epsilon_0 R}{3}$.



4. The value of X is
5. The value of Y is

Space For Rough Work

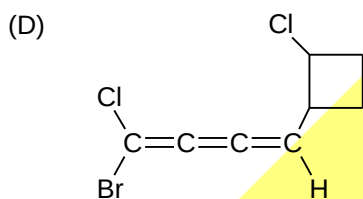
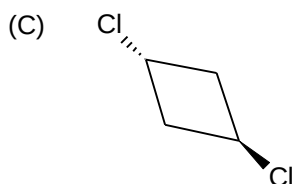
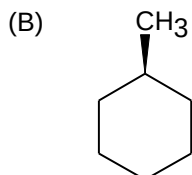
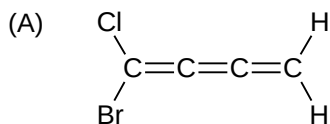
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SECTION – II : CHEMISTRY

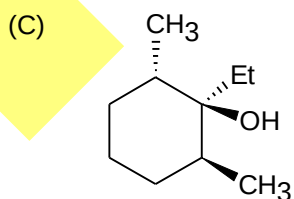
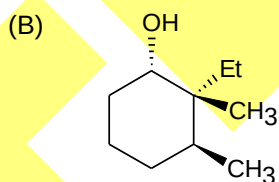
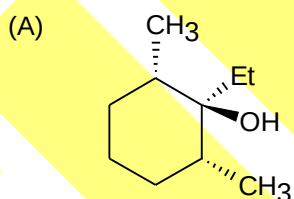
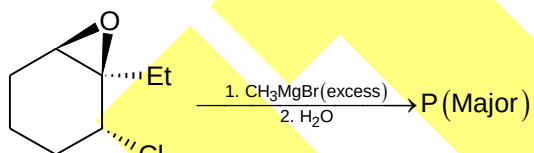
PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

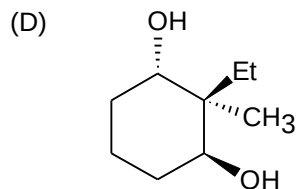
1. Optically active is



2.



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3. 4 mole of a mixture O_2 and O_3 is reacted with excess of acidified solution of KI. The liberated iodine require 1 L of 2 M hypo solution for complete reaction. The mole percent of O_3 in the initial sample is

(A) 25

(B) 30

(C) 75

(D) 50

4. Square pyramidal shape is

(A) XeF_2

(B) ICl_4^-

(C) XeF_5^+

(D) XeF_5^-

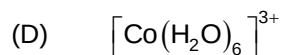
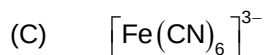
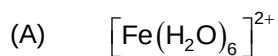
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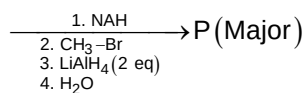
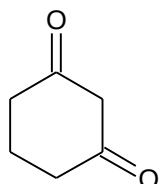
PART – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. High spin complex is/are



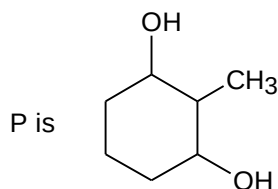
6.



→ P (Major)

Choose correct option is/are

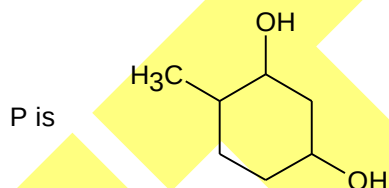
(A)



(B) Number stereo isomers in P is 4

(C) Number of stereo isomer in P is 3

(D)



7. The correct statement for $2p_z$ orbital is/are

(A) $\Psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2} \left(\frac{Zr}{a}\right) e^{-2r/2a} \cos(\theta)$

(B) $\Psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2} \left(2 - \frac{Zr}{a}\right) e^{-2r/2a}$

(C) xy plane is nodal plane

(D) $2p_z$ ungerade orbital

Space For Rough Work

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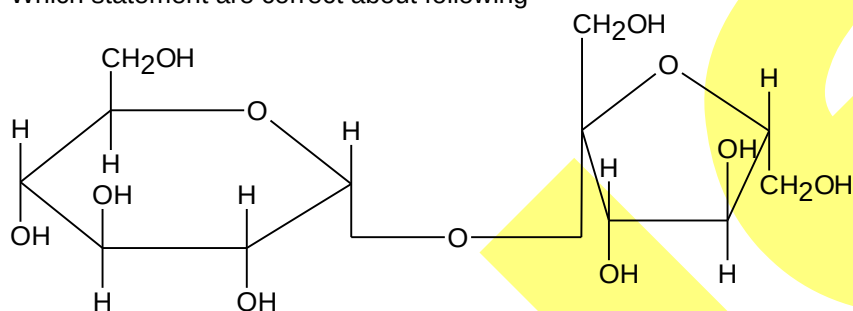
8. Which of the following complexes is/are shows the co-ordination isomerism?

- (A) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ (B) $[\text{Co}(\text{en})_3][\text{Cr}(\text{CN})_6]$
 (C) $[\text{Ag}(\text{NH}_3)_2][\text{AgCl}_2]$ (D) $[\text{Co}(\text{NH}_3)_6][\text{Co}(\text{NO}_2)_6]$

9. Choose homopolymer:

- (A) Butadiene-styrene (Buna-S)
 (B) Polythene
 (C) Polypropene
 (D) Nylon 6, 6

10. Which statement are correct about following



- (A) Due to presence of Hemiacetal linkage it is non reducing sugar
 (B) It is sucrose
 (C) Due to absence of Hemiacetal group it can't give Tollen's test
 (D) Left side is a α -D-Glucose and right side unit is β -D-fructose

PART – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. Number of monohalogen derivative (excluding stereo isomers) for the following is

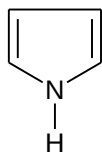


Space For Rough Work

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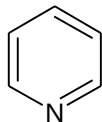
2. Total number of correct statement among the following is

(A)

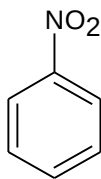


is aromatic

(B)



is more reactive than



for EAS reaction

(C)

In $\text{H}_3\text{C}-\left(\text{C}(=\text{O})\right)-\dot{\text{C}}\text{H}_2$ the presence of $-\text{C}(=\text{O})-$ group stabilise the radical

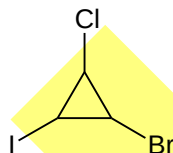
(D)

Adiabatic irreversible expansion for ideal gas, $[\Delta S]_{\text{system}} = 0$

(E)

In $\text{F}-\dot{\text{C}}(\text{F})-\text{F}$ the state of hybridization of radical is sp^3

3. Total number of diastereomeric pair for



is x then value of $\frac{x}{6}$ is

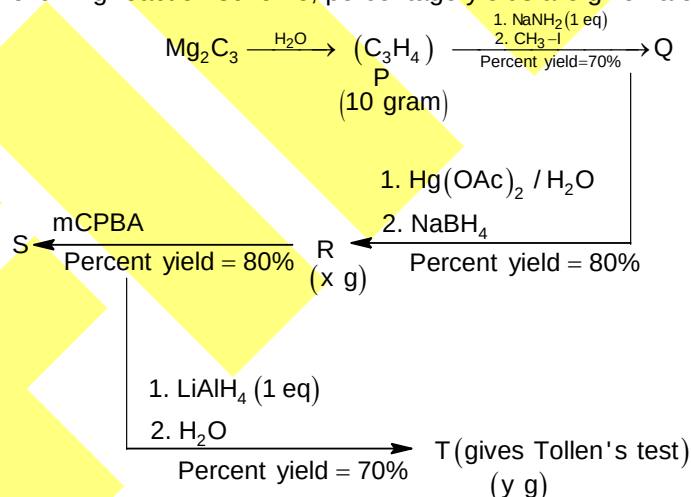
PART – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 4 and 5

Question Stem

For the following reaction scheme, percentage yields are given along the arrow



If mass of R and T are x and y g respectively then

4. The value of x is _____
5. The value of y is _____

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SECTION – III : MATHEMATICS**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. $\int_0^2 (\sqrt{1+x^3} + \sqrt[3]{x^2+2x})$ is equal to
(A) 4
(B) 5
(C) 6
(D) 7
2. Assume that $P = \{(x, y) : x^2 + 2y^2 = 10\}$ and $Q = \{(x, y) : y = mx + c\}$. If $P \cap Q = \phi$ for all $m \in [-1, 1]$, then minimum positive integral value of c is
(A) 3
(B) 4
(C) 5
(D) 10
3. The locus of the centre of the circle which touches the circle $(x - 1)^2 + y^2 = 9$ and the line $x = 6$ is a curve whose directrix is
(A) $x = 1$
(B) $y = 9$
(C) $y = -4$
(D) $x = 9$
4. If $f(x)$ is a 4 degree polynomial satisfying $f(x) = \frac{x}{x+1}$ for $x = 0, 1, 2, 3, 4$, then $f(6)$ is
(A) $\frac{12}{7}$
(B) 0
(C) 1
(D) -6

Space For Rough Work

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PART – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. The value of definite integral $\int_{-\infty}^k \frac{(\sin^{-1} e^x + \sec^{-1} e^{-x})}{(\tan^{-1} e^k + \tan^{-1} e^x)(e^x + e^{-x})} dx$ (where $k \in \mathbb{R}$) is
- (A) $\frac{\pi}{2} \ln 2 (2 \tan^{-1} e^k)$
- (B) independent of k
- (C) dependent on k
- (D) $\frac{\pi}{2} \ln 2$
6. Let $g(x)$ be a cubic polynomial with leading coefficient unity such that the roots of $g(x) = 0$ are the squares of the roots of $x^3 + x + 1 = 0$, then
- (A) $g(3)$ is a prime number
- (B) last two digit of $(g(1))^{50}$ are 49
- (C) when $(g(2))^{2022}$ is divided by 5, remainder is 2
- (D) $g(x) = 0$ has all 3 roots real
7. Let P and Q be two square matrix of order 3, then which of the following statement is/are always CORRECT?
- (A) PQP^T is symmetric matrix
- (B) $PQ - QP$ is skew symmetric matrix
- (C) If $Q = |P|P^{-1}$, $|P| \neq 0$, then $\text{adj}(P^T) - Q$ is skew symmetric matrix
- (D) If $Q + P^T = 0$ and P is skew symmetric matrix and Q^{15} is also skew symmetric matrix

Space For Rough Work

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8. A tangent L_1 is drawn to the curve $x^2 - 4y^2 = 16$ at point A in first quadrant whose abscissa is 5. Another tangent L_2 parallel to L_1 meets the curve at B. L_3 and L_4 are normal to the curve at A and B lines L_1, L_2, L_3, L_4 forms a rectangle, then
- (A) equation of normal at B is $12x + 10y + 75 = 0$
- (B) area of rectangle $L_1 L_2 L_3 L_4$ is $\frac{2400}{61}$
- (C) radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$
- (D) radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$
9. The integral point satisfying $|z - 1| + |z + 1| \leq 6$ in the argand plane for which $||z + \omega^2| - |z + \omega||$ is minimum (where ω is complex root of unity) is/are
- (A) (2, 0)
- (B) (1, 2)
- (C) (-3, 0)
- (D) (0, 2)
10. Let for $x \in [1, \infty)$, $\int_1^x f(x) dx = \left(\frac{x^2+1}{2x}\right) \left(f(x) - x + \frac{2x}{x^2+1}\right) = \ln 2$, then for $x \in [1, \infty)$ which of the following is are TRUE?
- (A) $f(1) = \ln 2$
- (B) Range of $g(x) = \frac{xf'(x) - f(x)}{x}$ is $[0, 2]$
- (C) $\lim_{x \rightarrow 3} \left[\frac{f(x)}{x} \right] = 2$
- (D) $h(x) = \frac{f(x)}{x}$ is an odd function
(where $[.]$ denotes greatest integer function)

Space For Rough Work

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PART – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. If P is the number of ways in which a person can walk up a stairway which has 11 steps if he can take 1 or 2 steps up the stairs at a time, then $\frac{P}{16}$ is equal to
2. In a triangle ABC, side AC = 4 units and $\sin A \sin B + \sin B \sin C + \sin C \sin A = \frac{9}{4}$. If A is the area of the triangle ABC, then $\frac{A}{\sqrt{3}}$ is equal to
3. If p is the least positive integer which satisfies the equation $|x - 2| + |x^2 - 9x + 20| = |x^2 - 8x + 18|$ and q is the minimum value of $\sin^2 x + \sin x + 1$ and α is the only positive root of $((2021)^{2022} - 1)x^2 + (p - (2021)^{2022})x = 1$, then the value of $(\alpha^{2022} - 1)p^2 + (p - 1 - \alpha^{2017})pq + 8q$ is

PART – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 4 and 5**Question Stem**

Let $A_n = [A_{ij}]$ be a $n \times n$ matrix such that $A_{ij} = \begin{cases} k & ; \quad i = j \\ 1 & ; \quad |i - j| = 1 \\ 0 & ; \quad \text{otherwise} \end{cases}$. Let B_n denote the determinant of matrix A_n

4. If $k = 2$, then B_{2022} is equal to
5. If $k = 1$, then $\sum_{p=1}^{2022} |B_p|$ is equal to (where $|\cdot|$ denotes absolute value function)

Space For Rough Work

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Question Stem for Question Nos. 6 and 7**Question Stem**

Let $A_1, A_2, A_3, \dots, A_{12}$ be a regular polygon of 12 sides whose centre is origin. Let the complex numbers representing vertices $A_1, A_2, A_3, \dots, A_n$ be $z_1, z_2, z_3, \dots, z_{12}$ respectively. Let $OA_1 = OA_2 = OA_3 = \dots, OA_{12} = 1$ (where O is origin)

6. The value of $|A_1 A_2| \times |A_1 A_3| \times |A_1 A_4| \times \dots |A_1 A_{12}|$ is equal to
7. The value of $|A_1 A_2|^2 + |A_1 A_3|^2 + \dots |A_1 A_n|^2$ is

Question Stem for Question Nos. 8 and 9**Question Stem**

Suppose 12 students of a class are asked a question successively in a random order and exactly 4 of them known the answer. Assume that none of them are aware of the answers given by other students, answer the following

8. If the probability that 6th student to be asked the question is the first one to know the answer is k , then $99k$ is equal to
9. If the probability that 1st and 12th student knows the answer is k , then $99k$ is equal to

Space For Rough Work

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Q. P. Code: 100182

ANSWERS**SECTION-1 : PHYSICS****PART – A**

- | | | | |
|------------|-------------|------------|---------|
| 1. B | 2. C | 3. A | 4. A |
| 5. A, B, D | 6. A, D | 7. A, B, C | 8. A, C |
| 9. A, C, D | 10. B, C, D | | |

PART – B

- | | | | |
|----------|----------|----------|-----------|
| 1. 4 | 2. 4 | 3. 2 | 4. 288.00 |
| 5. 32.00 | 6. 25.00 | 7. 25.00 | 8. 02.00 |
| 9. 03.00 | | | |

SECTION – 2 : CHEMISTRY**PART – A**

- | | | | |
|---------|-------------|------------|------------|
| 1. D | 2. A | 3. A | 4. C |
| 5. A, B | 6. A, B | 7. A, C, D | 8. A, B, D |
| 9. B, C | 10. B, C, D | | |

PART – B

- | | | | |
|----------|----------|----------|----------|
| 1. 4 | 2. 3 | 3. 4 | 4. 10.08 |
| 5. 03.44 | 6. 01.00 | 7. 04.00 | 8. 01.08 |
| 9. 00.36 | | | |

SECTION – 3 : MATHEMATICS**PART – A**

- | | | | |
|---------|-------------|---------|---------------|
| 1. C | 2. B | 3. D | 4. B |
| 5. B, D | 6. A, B | 7. C, D | 8. A, B, C, D |
| 9. A, C | 10. A, B, C | | |

PART – B

- | | | | |
|------------|----------|----------|------------|
| 1. 9 | 2. 4 | 3. 6 | 4. 2023.00 |
| 5. 1348.00 | 6. 12.00 | 7. 24.00 | 8. 04.00 |
| 9. 09.00 | | | |

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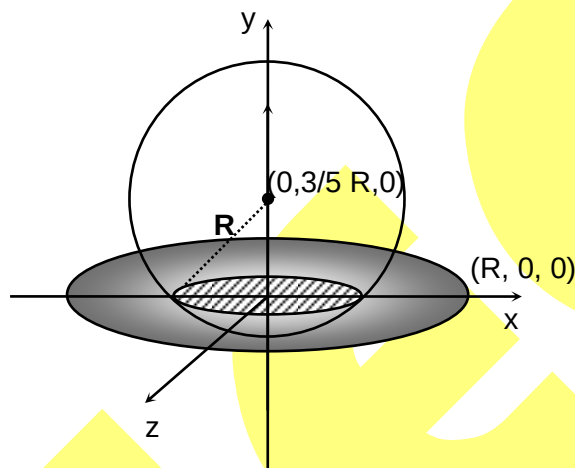
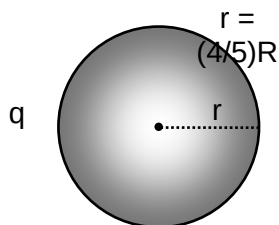
Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1.
Sol.

B



Charge on circular part

$$q = \int \sigma(r) \cdot 2\pi r \cdot dr$$

$$= \int_0^{(4/5)R} 2\pi\sigma_0 \left(r - \frac{r^3}{R^2} \right) dr = \frac{272}{625} (\sigma_0 \pi R^2)$$

$$\phi = \frac{q}{\epsilon_0} = \frac{272}{625} \left(\frac{\sigma_0 \pi R^2}{\epsilon_0} \right)$$

2.
Sol.

C

The path of projectile motion as observed by the other projectile motion is a straight line. The vertical component 30 m/s will get cancelled. Hence B will travel horizontal with respect to A. Hence minimum separation = 20 m.

3.
Sol.

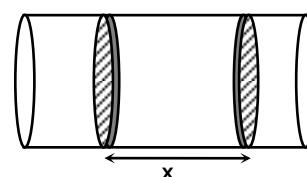
A

$$P_1 V_1 = P_2 V_2$$

$$(10^5) (20,000 \times 10^{-6}) = \left\{ 10^5 + \frac{(2k \cos^2 37^\circ)x}{100 \times 10^{-4}} \right\} (100 \times 10^{-4})x$$

$$x^2 + x - 2 = 0$$

$$x = 1 \text{ m} = 100 \text{ cm}$$



4.

A

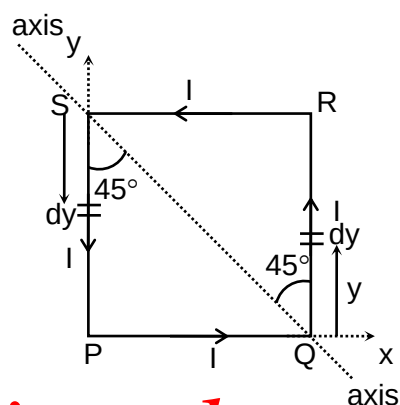
Sol.

Torque on the loop due to magnetic field

$$\tau = \int_0^\ell (IB_0 dy) \frac{y}{\sqrt{2}} + \int_0^\ell (IB_0 dy) \frac{y}{\sqrt{2}}$$

$$I\alpha = \frac{3IB_0 \ell^2}{2\sqrt{2}}$$

$$\frac{1}{6} m \ell^2 \alpha = \frac{3IB_0 \ell^2}{2\sqrt{2}} \Rightarrow \alpha = \left(\frac{9IB_0}{\sqrt{2}m} \right)$$



5.

A, B, D

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Sol. $f = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{3 \times 5 + 2 \times 3}{3 + 2} = \frac{21}{5}$

$$\gamma = 1 + \frac{2}{f} = \frac{31}{21}$$

Let initial volume and pressure are V_0 and P_0 respectively
Heat absorb in isobaric expansion

$$Q_1 = \frac{\gamma}{\gamma - 1} (3P_0 V_0) = 9.3P_0 V_0$$

Heat rejected in isochoric cooling

$$Q_2 = \frac{1}{\gamma - 1} (3P_0 V_0) = 6.3P_0 V_0$$

Heat rejected in isothermal compression $Q_3 = nRT_0 \ln 4 = P_0 V_0 \ln 4$

$$\text{Efficiency} = \frac{Q_1 - Q_2 - Q_3}{Q_1} = \left(\frac{161}{930} \right)$$

6. A, D

Sol. $\frac{1}{2} \mu u^2 - \frac{ke^2}{r} = 0$

$$\Rightarrow u = \sqrt{\frac{2k}{mr}} e$$

$$\lambda = \frac{h}{\mu u} \propto \frac{1}{u} \propto \sqrt{r}$$

$$KE_1 = \frac{1}{2} \mu u^2 = \frac{ke^2}{r}$$

For Bohr model, $\frac{1}{2} m v^2 = \frac{ke^2}{2r} = KE_2$

$$\Rightarrow \mu = \frac{KE_1}{KE_2} = 2$$

7. A, B, C

Sol. The deviation produced by ABCD is zero. Hence the cavity will not have any effect on the deviation.

$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{8}{5}$$

$$\sin\left(\frac{\delta_{\min} + A}{2}\right) = \left(\frac{8}{5}\right) \sin\left(\frac{60^\circ}{2}\right)$$

$$\delta_{\min} = 46^\circ$$

8. A, C

Sol. Equivalent resistance between A and E

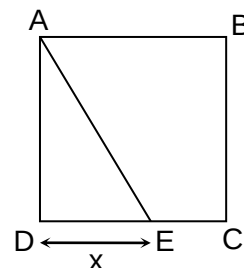
$$Y = \frac{x + 22}{x + 22 + 32}$$

For B and E to be equipotential

$$\frac{R_{AE}}{R_{AR}} = \frac{R_{EC}}{R_{RC}}$$

$$\Rightarrow \frac{(22 + x) \times 32}{(54 + x) \times (14 - x)} = \frac{8}{2} = 4$$

Solve to get : $x = 10 \Omega$



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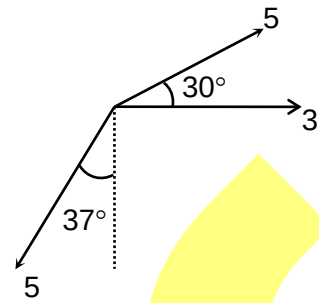
9. A, C, D

$$\text{Sol. } z = \sqrt{(2-1)^2 + \left(\frac{2}{\sqrt{3}}\right)^2} = \sqrt{\frac{7}{3}} \Omega$$

$$V_1 = 3 \sin \omega t, V_2 = 5 \sin(\omega t + 30^\circ), V_3 = 5 \sin(\omega t - 127^\circ)$$

$$V_{\max} = \sqrt{21} \text{ volts}$$

$$I_{\max} = \frac{V_{\max}}{z} = \frac{\sqrt{21}}{\sqrt{\frac{7}{3}}} = 3 \text{ amp}$$



10. B, C, D

$$\text{Sol. For A, } 1V = \frac{7S}{10} = 0.7 \text{ mm}$$

$$\text{Least count of A} = 5 - 0.7 \times 7 = 0.1 \text{ mm}$$

$$\text{For B, } 1V = \frac{3S}{8} = 0.375 \text{ mm}$$

$$\text{Least count of B} = 2 - 0.375 \times 5 = 0.125$$

$$\text{Difference} = 0.125 - 0.1 = 0.025 \text{ mm}$$

PART - B

1. 4

$$\text{Sol. } R = \frac{28 \times 42}{70} = 16.8 \Omega$$

2. 4

$$\text{Sol. } v = 4t - 3t^2$$

$$S = 2t^2 - t^3 + C$$

$$\text{When } v = 0, \Rightarrow t = 0 \text{ and } t = 4/3$$

$$S(t=0) = C, S\left(t = \frac{4}{3}\right) = \frac{32}{27} + C, S(t=2) = 0$$

$$\text{Distance traveled} = \frac{32}{27} + \frac{32}{27} = \frac{64}{27}$$

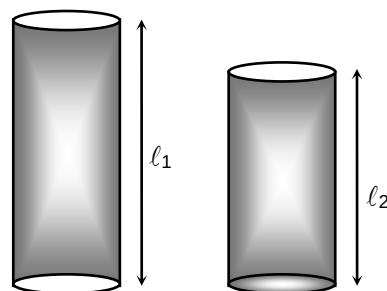
$$\text{Average speed} = \frac{64}{27 \times 2} = \frac{32}{27} \text{ m/s}$$

3. 2

$$\text{Sol. } \frac{3v}{2\ell_1} = \frac{5v}{4\ell_2}$$

$$\frac{3v}{2(\ell_1 + 2e)} = \frac{5v}{4(\ell_2 + e)}$$

$$e = \frac{12\ell_2 - 10\ell_1}{8} = 2 \text{ cm}$$



4. 288.00

5. 32.00

Sol. (for Q.45):

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When S_1, S_2 both are open capacity of the system is C_1 . Then C_1 .

$$C_1 = \frac{\frac{4\pi\epsilon_0(3R)(R)}{(3R-R)}(4\pi\epsilon_0 4R)}{\frac{4\pi\epsilon_0(3R)(R)}{(3R-R)} + 4\pi\epsilon_0 4R}$$

$$C_1 = \frac{(4\pi\epsilon_0 R)\left(\frac{3}{2}\right)(4)}{\left(\frac{3}{2} + 4\right)} = \frac{(4\pi\epsilon_0 R)(3 \cdot 4)}{11}$$

$$C_1 = \frac{48\pi\epsilon_0 R}{11}$$

When S_1 is closed but S_2 is open capacity is C_2 . Find C_2 .

$$C_2 = \frac{\frac{4\pi\epsilon_0(3R)(2R)}{(3R-2R)}(4\pi\epsilon_0 \cdot 4R)}{\frac{4\pi\epsilon_0(3R)(2R)}{(3R-2R)} + 4\pi\epsilon_0 4R} = \frac{(4\pi\epsilon_0 R)(6)(4)}{(6+4)} = \frac{48\pi\epsilon_0 R}{5}$$

When S_2 is closed but S_1 is opened capacity is C_3 . Find C_3

$$C_3 = \frac{\frac{4\pi\epsilon_0(2R)(R)}{(2R-R)}(4\pi\epsilon_0 4R)}{\frac{4\pi\epsilon_0(2R)(R)}{(2R-R)} + 4\pi\epsilon_0 4R} = \frac{4\pi\epsilon_0 R \cdot 2 \cdot 4}{6} = \frac{16\pi\epsilon_0 R}{3}$$

When both S_1 and S_2 are closed. Capacity is C_4 is . Then C_4 is equal to

$$C_4 = 4\pi\epsilon_0 4R$$

$$C_4 = 16\pi\epsilon_0 R$$

6. 25.00

7. 25.00

Sol. (for Q.6-7):

When B and D is cut

$$mg - T = ma$$

taking torque about mid point of rod

$$T \cdot \frac{\ell}{2} = \frac{m\ell^2}{12} \times \frac{a}{\ell/2}$$

$$T = mg/4$$

8. 02.00

9. 03.00

Sol. (for Q.8-9):

There is a dark fringe at O if path difference = $\lambda/2$

$$\Delta x = SO_2O - SO_1O = \frac{\lambda}{2}$$

$$2\sqrt{D^2 + d^2} - 2D = \frac{\lambda}{2}$$

$$2D\left(1 + \frac{d^2}{2D^2}\right) - 2D = \left(\frac{\lambda}{2}\right)$$

$$d_{\min} = \sqrt{\frac{D\lambda}{2}}$$

The bright fringe is formed at P if the path difference = λ

$$\Delta x = SO_1P - SO_2P = \lambda$$

$$\left\{ \left(D + \sqrt{D^2 + X^2} \right) - \left(\sqrt{D^2 + d^2} + \sqrt{D^2 + (x-d)^2} \right) \right\} = \lambda$$

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$$D + D \left(1 + \frac{x^2}{2D^2} \right) - D \left(1 + \frac{d^2}{2D^2} \right) - D \left(1 + \frac{(x-d)^2}{2D^2} \right) = \lambda$$

$$\frac{x^2}{2D} - \frac{d^2}{2D} - \frac{(x-d)^2}{2D} = \lambda$$

$$x = 2\sqrt{\frac{\lambda D}{2}} = 3 d_{\min}$$

SECTION - 2 : CHEMISTRY

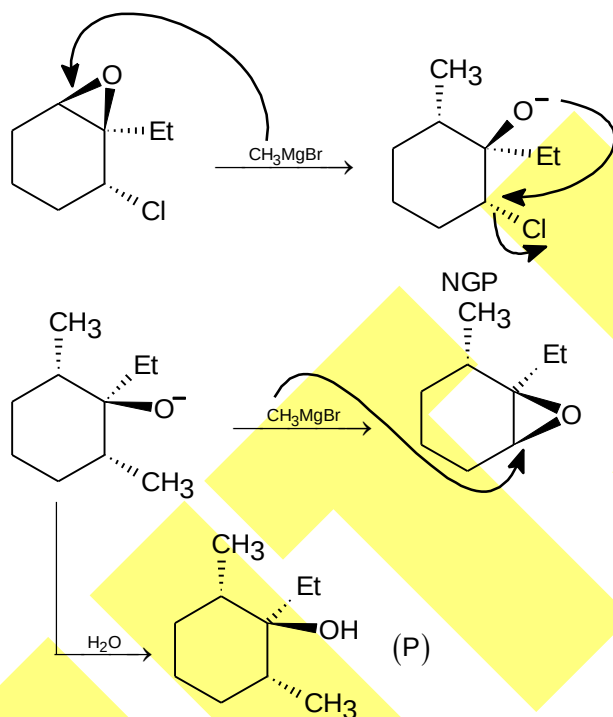
PART - A

1. D

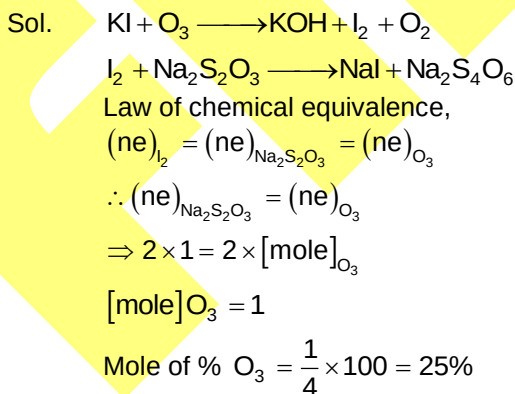
Sol. Due to absence of plane of symmetry. Option (D) is optically active.

2. A

Sol.



3. A



TG ~ @bohring_bot

4. C

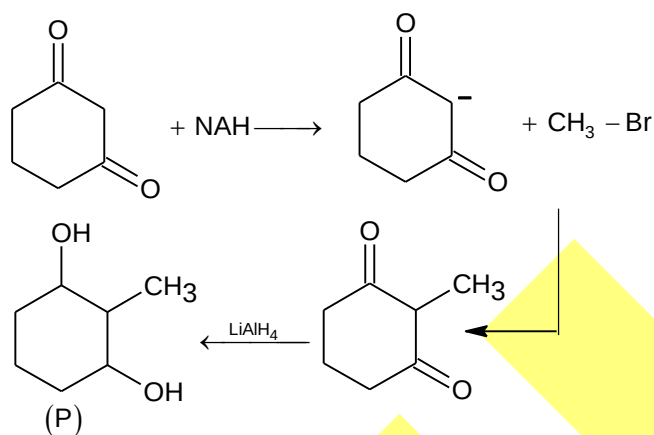
Sol. $\text{XeF}_2 \longrightarrow \text{linear}$
 $\text{ICl}_4^- \longrightarrow \text{square planar}$
 $\text{XeF}_5^+ \longrightarrow \text{square pyramidal}$
 $\text{XeF}_5^- \longrightarrow \text{pentagonal planar}$

5. A, B

Sol. A and B are high spin complex [NO pairing] C and D are low spin complex.

6. A, B

Sol.



Number of stereo isomers in P is 4.

7. A, C, D

Sol. $\psi_{n,\ell,m} \propto \left(\frac{Z}{a}\right)^{3/2} \left(\frac{Zr}{a}\right) e^{-2r/2a} \cos(\theta)$ and xy plane is nodal plane
 Number of nodal plane = 1
 $2p_z$ is ungerade orbital.

8. A, B, D

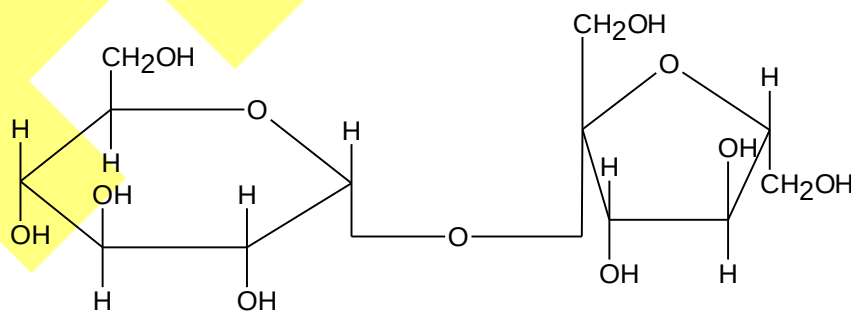
Sol. $[\text{Ag}(\text{NH}_3)_2][\text{AgCl}_2]$ will not show the co-ordination isomerism.

9. B, C

Sol. Polythene and polypropene are homopolymer.

10. B, C, D

Sol.



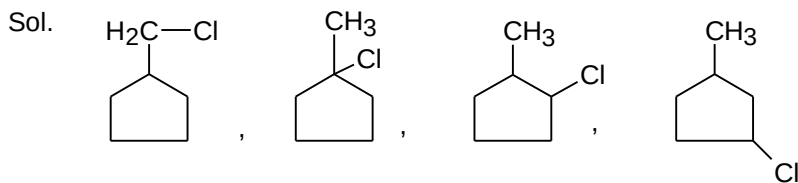
Structure is of sucrose which is non reducing sugar due to absence of Hemiacetal group.

PART - B

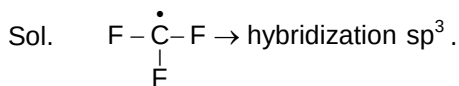
1.

4

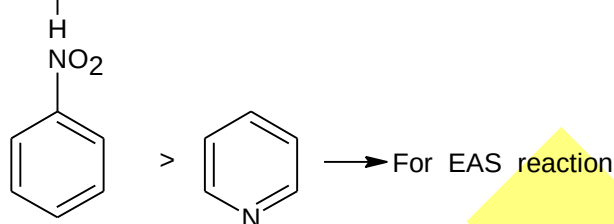
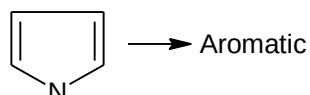
TG ~ @bohring_bot



2. 3



$-\text{C}(=\text{O})-$ group stabilize the radical.



$[\Delta S]_{\text{system}} > 0$ for adiabatic irreversible process.

3. 4

Sol. Number of diastereomeric pair 8_{C_2-4}

$$= \frac{|8|}{|2| |6|} - 4$$

$$= \frac{7 \times 8}{2} - 4$$

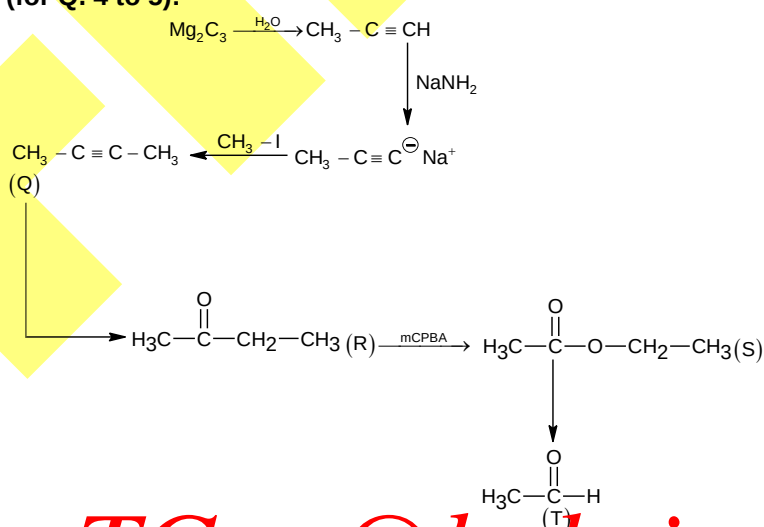
$$x = 24$$

$$\therefore \frac{x}{6} = 4$$

4. 10.08

5. 03.44

Sol. (for Q. 4 to 5):



$$\therefore [\text{Mass}]_{\text{CH}_3-\text{C}\equiv\text{C}-\text{H}} = 10 \times [\text{M}]_{\text{CH}_3-\text{C}\equiv\text{C}-\text{H}} = 40 \text{ gram/mol}$$

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$$\therefore [\text{Mole}]_{\text{CH}_3-\text{C}\equiv\text{CH}} = \frac{10}{40} = \frac{1}{4}$$

$$\therefore \text{Mole of Q} = \frac{1}{4} \times \frac{70}{100} = \frac{7}{40}$$

$$\therefore \text{Mole of R} = \frac{7}{40} \times \frac{80}{100}$$

$$\therefore \text{Mass of R} = \frac{7}{40} \times \frac{80}{100} \times [72]$$

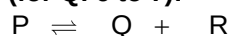
$$x = 10.08 \text{ g}$$

$$\text{Mass of T} = \frac{7}{40} \times \frac{80}{100} \times \frac{80}{100} \times \frac{70}{100} \times [44]$$

6. 01.00

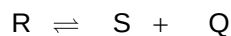
7. 04.00

Sol. (for Q. 6 to 7):



$$2 \quad 0 \quad 0$$

$$2-x \quad x+y \quad x-y$$



$$x \quad 0 \quad 0$$

$$x-y \quad y \quad y+x$$

$$2-x+x+y+x-y+y=4$$

$$x+y=2$$

$$\frac{x-y}{x+y} = \frac{1}{5}; y = \frac{2}{3}x$$

$$\text{Hence, } x=1.2$$

$$y=0.8$$

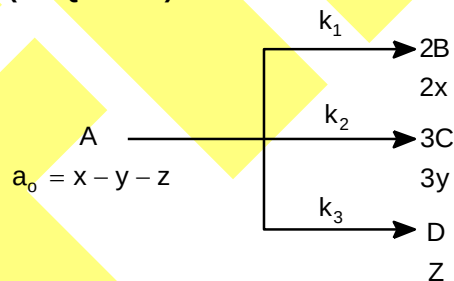
$$K_{C_1} = W = \frac{(x-y)(x+y)}{2-x} = \frac{2 \times 0.4}{0.8} = 1$$

$$K_{C_2} = Z = \frac{(x+y)(y)}{x-y} = \frac{2 \times 0.8}{0.4} = 4$$

8. 01.08

9. 00.36

Sol. (for Q. 8 to 9):



$$A_t = a_0 \times e^{-(k_1+k_2+k_3)t}$$

$$= 1 \times e^{-[0.5]4.606}$$

$$\therefore A_t = [10]^{-1} = \frac{1}{10} = 0.1 \text{ M}$$

$$\frac{k_1}{k_2} = \frac{x}{y} \Rightarrow \frac{1}{2} = \frac{x}{y} \quad \dots (1)$$

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$$\frac{k_2}{k_3} = \frac{y}{z} \Rightarrow y = z \quad \dots (2)$$

$$A_t = a_0 - [x + y + z]$$

$$0.1 = 1 - [x + y + z]$$

$$x + y + z = 1 - 0.1 = 0.9 \quad \dots (3)$$

$$A_t = 0.1 \text{ M}, B_t = 2x = 0.36 \text{ M}$$

$$C_t = 3y = 1.08 \text{ M}, D_t = 0.36 \text{ M}$$

SECTION - 3 : MATHEMATICS

PART - A

1. C

Sol. Let $(x^2 + 2x)^{\frac{1}{3}} = y \Rightarrow x^2 + 2x + 1 = y^3 + 1 \Rightarrow x = -1 + (y^3 + 1)^{\frac{1}{3}} \Rightarrow f^{-1}(x) = -1 + (x^3 + 1)^{\frac{1}{3}}$

$$\int_0^2 (f^{-1}(x) + 1 + f(x)) dx \Rightarrow \int_0^2 f^{-1}(x) dx$$

Let $f^{-1}(x) = t \Rightarrow x = f(t) \Rightarrow \frac{dx}{dt} = f'(t) ; \int t f'(t) dt = t f(t) - \int f(t) dt = 2f(2) - \int_0^2 f(x) dx$

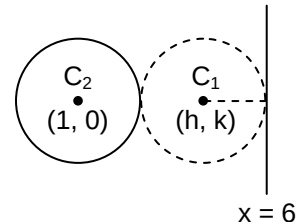
$$\Rightarrow \int_0^2 (f^{-1}(x) + 1 + f(x)) dx = 2f(2) - \int_0^2 f(x) dx + \int_0^2 1 dx + \int_0^2 f(x) dx = 2f(2) + 2 = 2 \times 2 + 2 = 6$$

2. B

Sol. $10m^2 + 5 - c^2 < 0$
Thus $c^2 > 15$

3. D

Sol. $C_1 C_2 = r_1 + r_2 \Rightarrow \sqrt{(h-1)^2 + k^2} = r + 3$ and $h = 6 - r$
 $\Rightarrow r = 6 - h ; (h-1)^2 + k^2 + r^2 + 9 + 6r$
 $\Rightarrow h^2 + 1 - 2h + k^2 = (6-h)^2 + 9 + 6(6-h)$
 $\Rightarrow h^2 + 1 - 2h + k^2 = 36 + h^2 - 12h + 9 + 36 - 6h$
 $\Rightarrow k^2 = 16h + 80 \Rightarrow y^2 = -16(x-5)$
 Equation of directrix is $x - 5 = 4 \Rightarrow x = 9$



4. B

Sol. Let $g(x) = f(x)(x+1) - x \Rightarrow g(0) = g(1) = g(2) = g(3) = g(4) = 0$
 $\therefore g(x) = f(x)(x+1) - x = kx(x-1)(x-2)(x-3)(x-4)$
 Put $x = -1 \Rightarrow 1 = kx - 1x - 2x - 3x - 4x - 5 \Rightarrow k = -\frac{1}{120}$
 $\therefore f(x)(x+1) - x = -\frac{x(x-1)(x-2)(x-3)(x-4)}{120}$
 Put $x = 6 \Rightarrow f(6) \times (7-6) = -\frac{6 \times 5 \times 4 \times 3 \times 2}{120} \Rightarrow f(6) = 0$

5. B, D

Sol. $\int_{-\infty}^k \frac{(\sin^{-1} e^x + \cos^{-1} e^x) e^x}{(\tan^{-1} e^k + \tan^{-1} e^x) e^{2x} + 1} \cdot \text{Let } \tan^{-1} e^x = t$

$$\frac{\pi}{2} \int_0^{\tan^{-1}(e^k)} \frac{1}{t + (\tan^{-1} e^k)} dt$$

TG ~ @bohring_bot

6. A, B

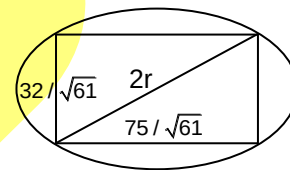
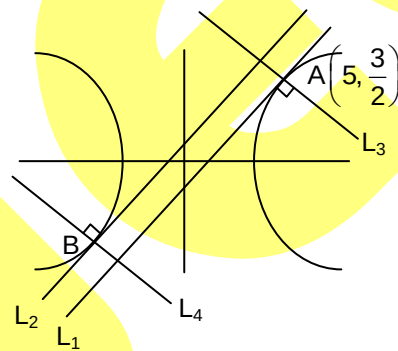
Sol. $\alpha^2 = x \Rightarrow \alpha = \sqrt{x}$ replace x by $\sqrt{x}$ in $x^3 + x + 1 = 0$. We get $g(x) = x^3 + 2x^2 + x - 1$
 $\Rightarrow g(3) = 47 \Rightarrow g(1) = 3 \Rightarrow 3^{50} = (10 - 1)^{25} = {}^{25}C_0 10^{25} + \dots + {}^{25}C_{23} 10^2 + {}^{25}C_{24} 10 - 1$
 $\Rightarrow g'(x) > 0 \forall x \in \mathbb{R} \therefore g(x)$ has only 1 real root

7. C, D

Sol. $(PQP^T)^T = PQ^TP^T \Rightarrow PQP^T$ is not symmetric
 $(PQ - QP)^T = Q^TP^T - P^TQ^T$
 $Q = |P| \frac{\text{adj } P}{|P|} \Rightarrow Q = \text{adj } P$
 $\Rightarrow (\text{adj } (P^T) - Q)^T = (\text{adj } (P^T) - \text{adj } P)^T = \text{adj } P - \text{adj } P^T$
and $Q = -P^T \Rightarrow Q = P \Rightarrow Q$ is also skew symmetric

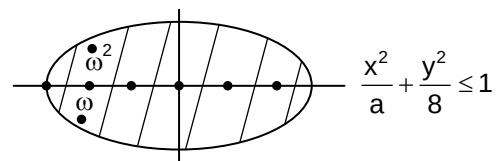
8. A, B, C, D

Sol. Equation of L_1
 $5x - 4y \times \frac{3}{2} = 16$
 $5x - 6y = 16$
 $\therefore$ Equation of tangent of B
 $5x - 6y = -16$
 $\therefore$ Equation of normal at A
 $6x + 5y = \frac{75}{2}$
 $\therefore$ Equation of normal at B is
 $6x + 5y = -\frac{75}{2}$
Distance between tangents $\frac{32}{\sqrt{61}}$
Distance between normals $\frac{75}{\sqrt{61}}$



9. A, C

Sol. $||z + \omega^2| - |z + \omega||$ is minimum when z lies on the major axis of the ellipse



10. A, B, C

Sol. $\int_1^x f(x) dx = \frac{f(x)}{2} \left(x + \frac{1}{x} \right) - \frac{x^2 + 1}{2} + 1$ using Leibnitz Rule
 $f(x) = \frac{f'(x)}{2} \left(x + \frac{1}{x} \right) + \frac{f(x)}{2} \left(1 - \frac{1}{x^2} \right) - x \Rightarrow \frac{xf'(x) - f(x)}{x^2} = \frac{2x}{x^2 + 1}$
Integrating both sides, we get $\frac{f(x)}{x} = \ln(x^2 + 1)$

PART – B

1. 9

Sol. Let x_1 be the number of times he took 1 step up the staircase
Let x_2 be the number of times he took 2 step up the staircase
 $x_1 + 2x_2 = 1$

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			Number of ways
$x_1 = 1$	$x_2 = 6$	1, 2, 2, 2, 2, 2	$\frac{6!}{1! 5!} = 6$
$x_1 = 3$	$x_2 = 4$	1, 1, 1, 2, 2, 2, 2	$\frac{7!}{3! 4!} = 35$
$x_1 = 5$	$x_2 = 3$	1, 1, 1, 1, 1, 2, 2, 2	$\frac{8!}{3! 4!} = 56$
$x_1 = 7$	$x_2 = 2$	1, 1, 1, 1, 1, 1, 1, 2, 2	$\frac{9!}{2! 7!} = 36$
$x_1 = 9$	$x_2 = 1$		$\frac{10!}{9! 1!} = 10$
$x_1 = 11$	$x_2 = 0$		1
			144

$$\therefore P = 144$$

2. 4

Sol. $2 \sin A \sin B + 2 \sin B \sin C + 2 \sin C \sin A = \frac{9}{2}$

$$\cos(A - B) - \cos(A + B) + \cos(B - C) - \cos(B + C) + \cos(C - A) - \cos(C + A) = \frac{9}{2}$$

$$\Rightarrow \cos(A - B) + \cos(B - C) + \cos(C - A) + \cos A + \cos B + \cos C = \frac{9}{2}$$

$$\Rightarrow \cos(A + B) + \cos(B - C) + \cos(C - A) = \frac{9}{2} - (\cos A + \cos B + \cos C) \geq \frac{9}{2} - \frac{3}{2} \geq 3$$

$$\Rightarrow \cos(A - B) + \cos(B - C) + \cos(C - A) = 3$$

$$\Rightarrow A = B = C \therefore A = \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3}$$

3. 6

Sol. $(x - 2)(x^2 - 9x + 20) \geq 0 \Rightarrow x \in [2, 4] \cup [5, \infty)$

$$\therefore p = 2$$

$$\Rightarrow \left(\sin x + \frac{1}{2}\right)^2 + \frac{3}{4}, \text{ least value is } \frac{3}{4} \therefore q = \frac{3}{4}$$

$$\Rightarrow x = 1 \text{ satisfies it } \therefore \alpha = 1 \text{ and other root is } \frac{-1}{(2021)^{2022} - 1}$$

$$\therefore \alpha = 1, q = \frac{3}{4}, p = 2$$

4. 2023.00

5. 1348.00

Sol. (for Q.4.-5):

$$A_1 = [k] \Rightarrow B_1 = k; A_2 = \begin{bmatrix} k & 1 \\ k & 1 \end{bmatrix} \Rightarrow B_2 = k^2 - 1; A_3 = \begin{bmatrix} k & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & k \end{bmatrix} \Rightarrow B_3 = kB_2 - 1B_1$$

$$\text{Similarly, } A_4 = \begin{bmatrix} k & 1 & 0 & 0 \\ 1 & k & 1 & 0 \\ 0 & 1 & k & 1 \\ 0 & 0 & 1 & k \end{bmatrix} \Rightarrow B_4 = kB_3 - B_2$$

$$\text{Thus, } B_n = kB_{n-1} - B_{n-2}. \text{ If } k = 2, B_1 = 2, B_2 = 1, B_3 = 4, B_4 = 5$$

TCG @bohring_bot

Thus, $B_{2022} = 2023$

If $k = 1$, $B_n = B_{n-1} - B_{n-2}$; $B_1 = 1$, $B_2 = 0$, $B_3 = B_2 - B_1 = -1$, $B_4 = B_3 - B_2 = -1$

$B_5 = B_4 - B_3 = 0$; $B_6 = 1$ and it goes on

$B_{6n+1} = 1$; $B_{6n+2} = 0$; $B_{6n+3} = -1 \Rightarrow B_{6n-4} = -1$; $B_{6n+5} = 0$; $B_{6n} = 1$

Thus, $\sum_{k=1}^{2022} |B_p| = \frac{4}{6} \times 2022 = 1348$

6. 12.00

7. 24.00

Sol. (for Q. 6.-7):

$z_1, z_2, z_3, \dots, z$ are also the roots of $z^{12} = 1$ i.e. 12th roots of unity

$z^{12} - 1 = (z - z_1)(z - z_2)(z - z_3) \dots (z - z_{12})$

$$\frac{z^{12} - 1}{z - 1} = (z - z_2)(z - z_3)(z - z_4) \dots (z - z_{12})$$

$$1 + z + z^2 + \dots + z^{11} = (1 - z_1)(1 - z_3)(1 - z_4) \dots (1 - z_{12})$$

$$\text{Put } z = 1, 12 = |1 - z_2| |1 - z_3| |1 - z_4| \dots |1 - z_{12}|$$

$$\Rightarrow |z_2 - z_1|^2 = |z_1|^2 |(e^{i2\pi/12} - 1)|^2 = 2^2 \sin^2 \frac{\pi}{12} \Rightarrow |A_1 A_2|^2 + |A_1 A_3|^2 + \dots + |A_1 A_{12}|^2$$

$$= 4 \left[\sin^2 \frac{\pi}{12} + \sin^2 \frac{2\pi}{12} + \sin^2 \frac{3\pi}{12} + \dots + \sin^2 \frac{11\pi}{12} \right] = 2 \left[11 - \left(\cos \frac{2\pi}{12} + \cos \frac{4\pi}{12} + \dots + \cos^2 \frac{2\pi}{12} \right) \right]$$

$$= 2(11 - (-1)) = 24$$

8. 04.00

Sol. $\frac{4}{7} \times \frac{{}^8C_5}{{}^{12}C_5}$

9. 09.00

Sol. $\frac{4}{12} \times \frac{3}{11} = \frac{1}{11}$

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PHYSICS, CHEMISTRY & MATHEMATICS**QP Code: 100183****Paper-2****Time Allotted: 3 Hours****Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are **not** allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

- (i) **PART – A (01 –06):** This section contains SIX (06) questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.

Zero Marks: 0 If none of the bubbles is darkened.

Negative Marks: –2 In all other cases.

For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.

PART – A (07 –10): This section contains TWO (02) paragraphs. Based on each paragraph, there are TWO (02) questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.

- (ii) **PART – B (1 – 3):** This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. Each question carries **+4 marks** for correct answer and **there will be no negative marking**.

- (iii) **PART – B (4 – 9):** This section contains **SIX (06)** questions stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round -off the value to **TWO** decimal places. Each question carries **+2 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

IG ~ @bohring_bot

BATCHES – Two Yr CRP2123(AII)

SECTION – I : PHYSICS

PART – A (Maximum Marks: 24)

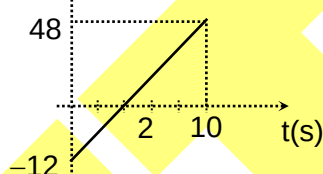
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two stars S_1 and S_2 of mass M each are moving in circular orbits about their common centre of mass 'O' due to mutual gravitational force of attraction. The separation between them is $4R$. An asteroid of mass m ($m \ll M$) is in a coplanar circular orbit about same centre 'O'. It is observed that the asteroid is also moving in a circular orbit with same angular speed. Neglect size of stars and asteroid compared with R . Then choose the correct option(s).

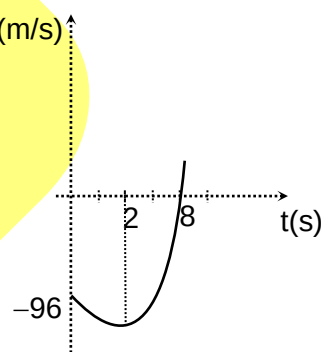
- (A) Time period of revolution of stars and asteroid is $8\pi\sqrt{\frac{2R^3}{GM}}$
- (B) Radius of the circle traced by asteroid is $2\sqrt{3}R$
- (C) The total energy of the asteroid is $\frac{-5GMm}{16R}$
- (D) Radius of the circle traced by the star S_1 is $2R$.

2. The position of a particle which moves along a straight line is defined by the relation $x = t^3 - 6t^2 - 96t + 37$, where x is expressed in meters and t is in seconds. Which of the graph(s) represent(s) the motion of the particle?

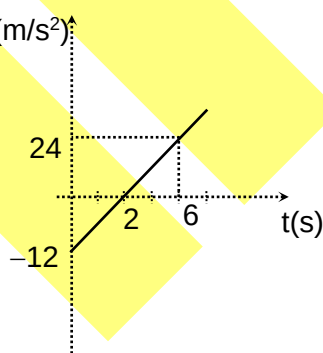
(A) $a(m/s^2)$



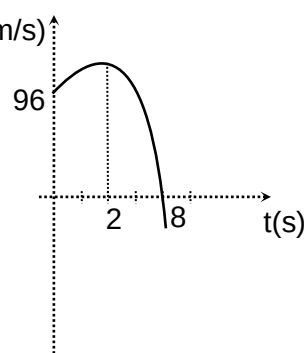
(B) $v(m/s)$



(C) $a(m/s^2)$



(D) $v(m/s)$



Space For Rough Work

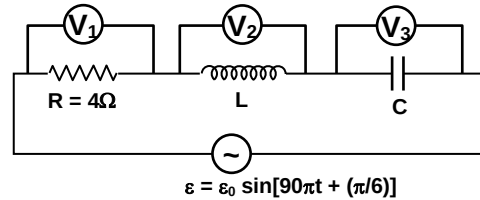
TG ~ @bohring_bot

3. A flat circular annular disc of mass M , outer radius a and inner radius b . The measured value of M , a and b are, $M = (0.30 \text{ kg} \pm 0.0003 \text{ kg})$, $a = (4 \text{ cm} \pm 0.001 \text{ cm})$ and $b = (2 \text{ cm} \pm 0.001 \text{ cm})$. Then,
- (A) Percentage error in the calculation of a and b are in ratio 1 : 2.
 - (B) Percentage error in the calculation M is 0.1%
 - (C) Percentage error in the calculation of moment of inertia of the disc about an axis passing through the centre and perpendicular to the plane of disc is 0.16%
 - (D) Percentage error in the calculation of moment of inertia of the disc about an axis passing through the centre and perpendicular to the plane of disc is 1.6%
4. When a yellow light of wavelength $6000 \text{ \AA}$ from a sodium lamp falls on a certain photocell, a negative potential of 0.30 volt is needed to stop all the electrons from reaching the collector. Then choose correct option(s).
- (A) Work function of the given photocell is 2.5 eV (approximately).
 - (B) Work function of the given photocell is 1.78 eV (approximately).
 - (C) A negative potential of 0.6 Volt will be needed to stop all the electrons if light of wavelength $\lambda = 4000 \text{ \AA}$ is used
 - (D) A negative potential of 1.32 Volt (approximately) will be needed to stop all the electrons if light of wavelength $\lambda = 4000 \text{ \AA}$ is used
5. 10g of ice at 0°C is mixed with 5g of steam at 100°C . If latent heat of fusion of ice is 80 cal-g^{-1} and latent heat of vaporization of 540 cal-g^{-1} . When thermal equilibrium is reached then which of the following is/are correct?
- (A) Temperature of the mixture is 0°C
 - (B) Temperature of mixture is 100°C
 - (C) Mixture contains 13.3 g of water and 1.67 g of steam
 - (D) Mixture contains 5.3 g of ice and 9.7 g of water

Space For Rough Work

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6. In the figure shown the reading of voltmeter are $V_1 = 80$ V, $V_2 = 90$ V and $V_3 = 30$ V. Then choose the correct option(s).



- (A) The peak value of current is $20\sqrt{2}$ ampere.
 (B) The peak value of emf is $100\sqrt{2}$ volts.
 (C) The value of L, is $\frac{1}{20\pi}$ H
 (D) The value of C for which the circuit will produce maximum heat is $\frac{1}{405\pi}$ F

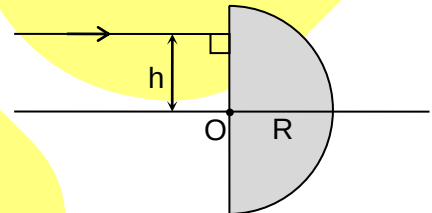
PART – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

A semi-cylinder made of a transparent plastic has a refractive index of $\mu = 5/3$ and radius of $R = 30$ cm lies in vacuum. There is a narrow incident light ray perpendicular to the flat side of the semi-cylinder at height 'h' from the axis of symmetry. (take $\tan 37^\circ = 3/4$)

Now answers the following questions.



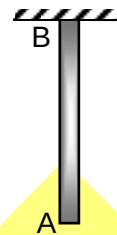
7. What is the maximum value of 'h' so that the light ray can still leave the other side of the semi-cylinder?
- (A) 12 cm (B) 8 cm
 (C) 18 cm (D) 24 cm
8. When the refractive index of transparent plastic is $\mu = \sqrt{2}$, the value of 'h' is chosen such that total internal reflection just takes place, then time for which light remains inside the cylinder, is
- (A) 10^{-9} s (B) 2×10^{-9} s
 (C) 4×10^{-9} s (D) 8×10^{-9} s

Space For Rough Work

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Paragraph for Question Nos. 09 and 10

Consider a rope AB of non-uniform density as shown in the figure. If we consider direction A to B as the direction of increasing y , with $y = 0$, at A. The linear mass density of rope is given by $\mu = \left(\frac{y}{\ell}\right)^3 \mu_0$, where μ_0 is a positive constant and ℓ is length of the rope. Now one pulse is produced at the lower end A. Then answer the following questions



9. The time to move the pulse from A to B.

(A) $\sqrt{\frac{\ell}{g}}$

(B) $2\sqrt{\frac{\ell}{g}}$

(C) $3\sqrt{\frac{\ell}{g}}$

(D) $4\sqrt{\frac{\ell}{g}}$

10. Acceleration of the pulse when $y = \ell/4$.

(A) $g/4$

(B) $g/2$

(C) g

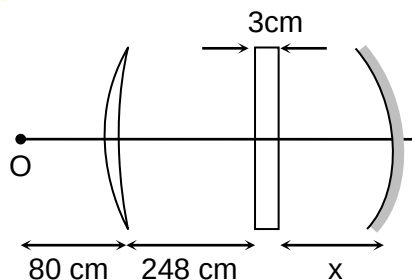
(D) $g/8$

PART – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. Two rods each of length L_1 and one rod of length L_2 form isosceles triangle having base of length L_2 . Co-efficient of linear expansion of base is $4 \times 10^{-6} / ^\circ\text{C}$ while for rod of length L_1 is $25 \times 10^{-6} / ^\circ\text{C}$. Find the ratio $\left(\frac{L_2}{L_1}\right)$, if the length of latitude remains same when the temperature of the system is increased by 10°C .

2. A concavo-convex lens made of glass of refractive index 1.5 has surfaces of radii 20 cm and 60 cm. A concave mirror of radius of curvature 20 cm is placed co-axially to the lens. A glass slab of thickness 3 cm and refractive index 1.5 is placed close to the mirror in the space between the mirror and lens as shown in the figure. The distance of the nearest surface of the slab from lens is 248 cm. An object is placed 80 cm to the left of the lens. If the final position of the image formed after the refraction from lens, refraction from slab, reflection from mirror, refraction from the slab and lens is at the object O, find the distance x (in cm) of the mirror from nearer surface of slab.



Space For Rough Work

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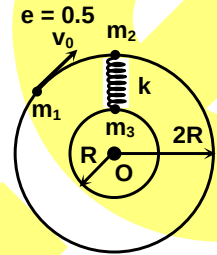
3. A train moving towards a hill at a speed 144 km/hr, sounds whistle of frequency 300 Hz. Wind is blowing from the hill with a velocity 360 km/hr at an angle of 53° with the railway track. If the speed of sound in air is $\frac{400}{3}$ m/s, then find the frequency (in Hz) heard by a man on the hill.

PART – B (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 4 and 5

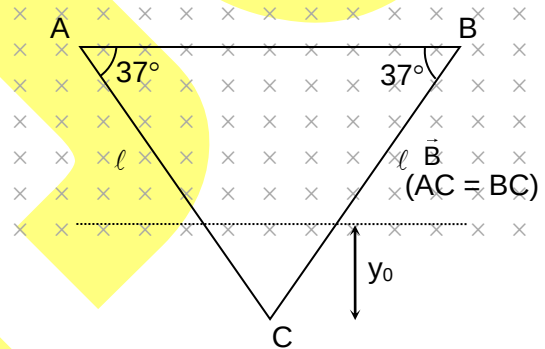
Three particles each of mass 'm' can slide on fixed frictionless circular track in the same horizontal plane as shown in the figure. Particle m_1 moves with velocity v_0 and hits particle m_2 , the coefficient of restitution being $e = 1/2$. Assume m_2 and m_3 are initially in rest and lie along a radial line before impact and the spring is initially unstretched. The velocity of m_3 when extension in the spring is maximum, is X m/s and the maximum extension in the spring is Y meter. (Take $v_0 = 20$ m/s and $\frac{m}{k} = 5$ kg/(N/m))



4. The value of X is
5. The value of Y is

Question Stem for Question Nos. 6 and 7

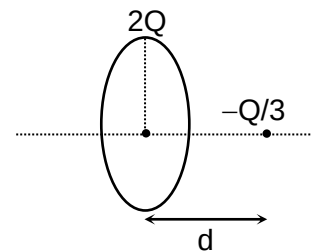
A current carrying loop is in the shape of isosceles triangle of mass m lies in the vertical plane. There exists uniform horizontal magnetic field $\vec{B}$ in the region as shown in the figure. The current in the loop is i. The value of y_0 is X meter when loop is in equilibrium. If the loop is displaced slightly in its plane perpendicular to its side AB from its equilibrium position and released, then time period of its oscillation is Y sec. (neglect the induced emf in the loop, take $\frac{mg}{iB} = \frac{80}{3}$, $\pi = 3.14$ and $g = 10$ m/s²)



6. The value of X is.....
7. The value of Y is.....

Question Stem for Question Nos. 8 and 9

A ring of mass $2M$, radius R and charge $2Q$ is placed as shown in the figure. A point charge $-\frac{Q}{3}$ and mass M is placed on the axis of ring at a distance d from the centre of the ring as shown in the figure. ($d \ll R$). When system is released it performs oscillations. The energy of oscillation of the system is $\frac{Q^2}{m\pi\epsilon_0 R^n}$ and the time period of oscillation is $T = \sqrt{\frac{X\pi^3\epsilon_0 MR^3}{YQ^2}}$.



8. The value of $|m - n|$ is
9. The value of $|X - Y|$ is

Space For Rough Work

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SECTION – II : CHEMISTRY

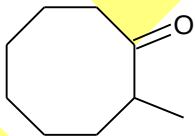
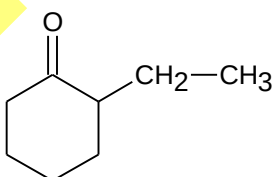
PART – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. Among the following intensive properties are
 - (A) E_{cell}°
 - (B) E_{cell}
 - (C) Normality
 - (D) Entropy

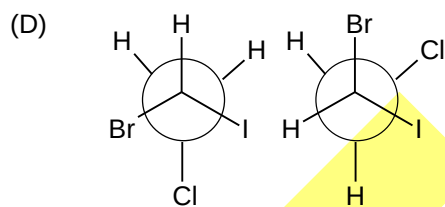
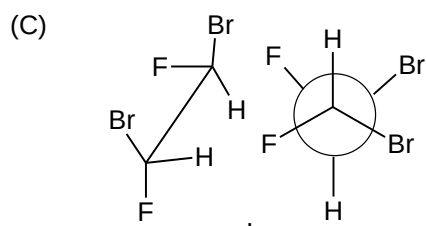
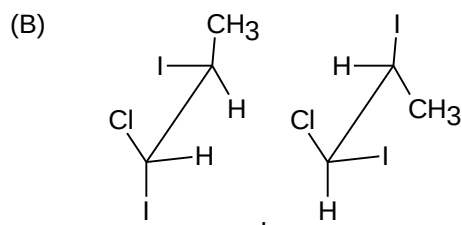
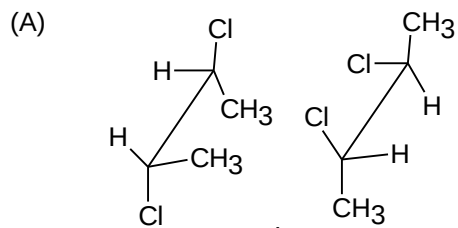
2. Which of the following solution is/are hypertonic at constant T with 0.4 M NaCl showing 80% dissociation?
 - (A) 0.7 M glucose
 - (B) 0.3 M Na_3PO_4 showing 90% dissociation
 - (C) 1 M $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ showing 30% association
 - (D) 0.5 M MgCl_2 showing 20% dissociation

3. In isothermal compression of ideal gas
 - (A) W is +ve
 - (B) ΔH is zero
 - (C) ΔS_{gas} is – ve
 - (D) ΔE is zero

4. Tautomer of which of the following can show geometrical isomerism
 - (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
 - (B) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
 - (C) 
 - (D) 

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5. Which of the following pairs of compounds is/are identical?



6. Which set of quantum numbers is/are not consistent with the theory?

(A) $n = 2, \ell = 0, m = -1, s = +\frac{1}{2}$

(B) $n = 2, \ell = 0, m = 0, s = -\frac{1}{2}$

(C) $n = 1, \ell = -1, m = 0, s = -\frac{1}{2}$

(D) $n = 3, \ell = 3, m = -3, s = +\frac{1}{2}$

Space For Rough Work

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PART – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

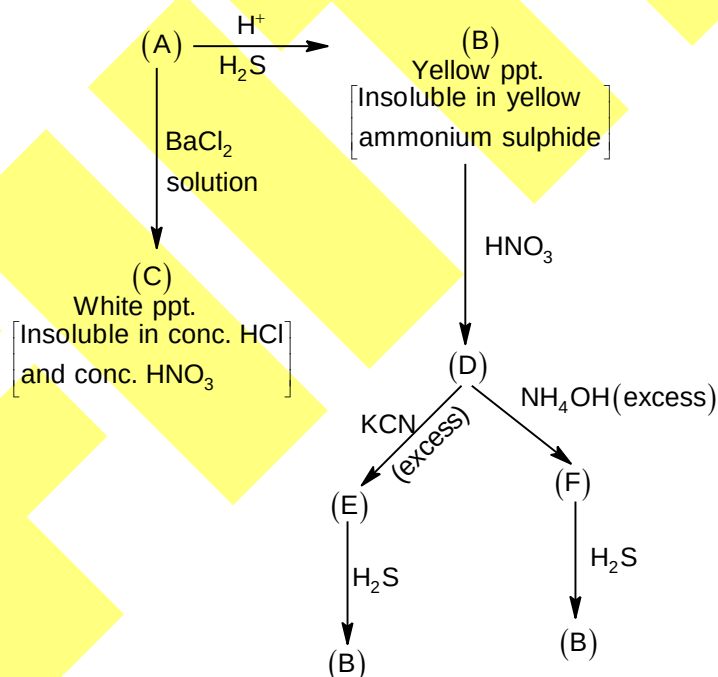
An orange solid (A) on heating gives a green residue (B), a colourless gas (C) and water vapours. The dry gas (C) on passing over heated Mg gave a white solid (D). (D) on reaction with water gave a gas (E) which formed black precipitate with mercurous nitrate solution.

7. Select the incorrect statement

- (A) The central atom (s) of the anion of solid (A) has sp^3 hybridization.
- (B) The orange solid (A) is diamagnetic in nature.
- (C) The anion of orange solid (A) is oxidizing in nature.
- (D) All metal oxygen bond lengths are equal in anion of solid (A).

8. Which of the following is false for the gas (E)?

- (A) It gives a deep blue colouration with CuSO_4 solution.
- (B) It is oxidized to a colourless gas (neutral oxide) at 1200 K in presence of a catalyst Pt/Rh in air.
- (C) It gives the same gas (C) with potassium permanganate solution.
- (D) It gives black precipitate with HgCl_2 .

Paragraph for Question Nos. 09 and 10

9. What is "C"?

- (A) PbCl_2
- (B) BaSO_4
- (C) BaSO_3
- (D) PbSO_4

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10. What is "B"?

(A) CdS

(B) PbCrO_4

(C) HgO

(D) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$

PART – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

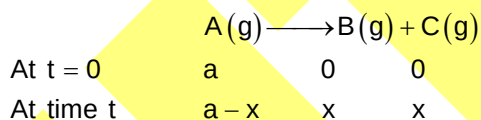
- Number of co-ordination isomerism for the complex $\left[\text{Pt}(\text{NH}_3)_4\right]\left[\text{PtCl}_4\right]$ is (including itself)
- Number of stereo isomers in the complex $[\text{Ma}_2\text{b}_2\text{c}_2]$ [where a, b, c are monodentate ligand]
- The following equilibrium are established in mixing of two gases x_2 and y in sealed container at constant temperature T.
 $3x_2(\text{g}) \rightleftharpoons x_6(\text{g}), K_p = 1.6 \text{ atm}^{-2}$
 $x_2(\text{g}) + y(\text{g}) \rightleftharpoons x_2y(\text{g}), K_p = z \text{ atm}^{-1}$
 When $x_2(\text{g})$ and $y(\text{g})$ are mixed in 2 : 1 molar ratio, the total pressure of gases at equilibrium is found to be 1.4 atm and partial pressure of $x_6(\text{g})$ is 0.2 atm. The value of $6W$ is.

PART – B (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 4 and 5

A reaction is said to be first order if its rate is proportional to the concentration of reactant. Let us consider a reaction



The rate of reaction is given by the expression $\frac{dx}{dt} = k(a - x)$ and integrated rate equation for given reaction

is represented as $k = \frac{1}{t} \ln\left(\frac{a}{a-x}\right)$ where a = initial concentration and $(a - x)$ = concentration of A after time t .

- Thermal decomposition of compound A is a first order reaction. If 75% of A is decomposed in 100 min. How long will it take for 90% of the compound to decompose?
[Given: $\log 2 = 0.30$]
- Consider a reaction $\text{A}(\text{g}) \longrightarrow 3\text{B}(\text{g}) + 2\text{C}(\text{g})$ with rate constant $1.386 \times 10^{-2} \text{ min}^{-1}$. Starting with 2 moles of A in 12.5 litre vessel initially, if reaction is allowed to take place at constant pressure and at 298 K then find the concentration of B after 100 min? [$e^{-1.386} = 0.25$]

Space For Rough Work

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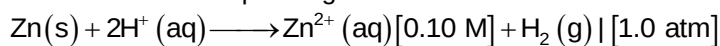
Question Stem for Question Nos. 6 and 7

0.257 g of an organic compound is analysed for nitrogen by Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.1 M HCl which required 23.2 ml of $\frac{N}{10}$ NaOH for the complete neutralization.

6. What is percentage of 'N' in the compound?
7. In Carius method of estimation of halogen, 0.15 g of an organic compound gave 0.12 g of AgBr. What is % of Bromine in the compound?
[Take : $M_{\text{AgBr}} = 188 \text{ g/mol}$; $M_{\text{Br}} = 80 \text{ g/mol}$]

Question Stem for Question Nos. 8 and 9

EMF of a cell corresponding to the reaction:



is 0.61 V at 25°C. Also, at 25°C, $E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76 \text{ V}$. $\left[\frac{2.303RT}{F} = 0.06 \right]$

8. The pH of solution in cathode chamber is.....
9. ΔG of the above cell reaction is y, then the value of |y| in kJ? (Given $1F = 96500$)

Space For Rough Work

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SECTION – III : MATHEMATICS

PART – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. If the value of limit ; $\lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \frac{k-1}{k!} \right)$ is L, then (where $[.]$ denotes greatest integer function)

(A) $\left[\frac{[L] + 2L}{[L]} \right] = 3$

(C) $\left[\frac{[L] + [L-1]}{2[L]} \right] = 0$

(B) $\left[\frac{[L] + 2L}{[L]} \right] = 4$

(D) $\left[\frac{[L] + [L-1]}{2[L]} \right] = 1$

2. Suppose that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the relation $f(x + g(y)) = 2x + y + 5$, then if function $g(x + f(y)) = k_1(x) + k_2(y) + k_3$, then

(A) $2k_1 + k_2 = 1$

(C) $k_3 - k_1 = 0$

(B) $2k_1 + k_2 = 2$

(D) $k_3 - k_1 = 2$

3. ${}^m C_1 \cdot {}^n C_m - {}^m C_2 \cdot {}^{2n} C_m + {}^m C_3 \cdot {}^{3n} C_m \dots$ is equal to

(A) coefficient of x^n in $(1 - (1 + (1 - x)^n)^m)$

(C) coefficient of x^m in $(1 - (1 - (1 + x)^n)^m)$

(B) coefficient of x^n in $(1 - (1 - (1 + x)^n)^m)$

(D) coefficient of x^m in $(1 - (1 + (1 - x)^n)^m)$

4. Given an alphabet with three letters a, b, c, then the number of words of n letter which contain even number of a's?

(A) $2^n \left(\left(1 + \frac{1}{2} \right)^n + \left(1 - \frac{1}{2} \right)^n \right)$

(C) $\frac{1}{4}(3^n + 1)$

(B) $2^{n-1} \left(\left(1 + \frac{1}{2} \right)^n + \left(1 - \frac{1}{2} \right)^n \right)$

(D) $\frac{1}{2}(3^n + 1)$

Space For Rough Work

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5. For each integer $n \geq 1$. Let $a_n = 1 + \frac{1}{2} + \frac{1}{3} \dots \frac{1}{n} - \log n$ and $b_n = a_n - \frac{1}{n}$, then
- (A) $a_{n+1} > a_n$ (B) $a_{n+1} < a_n$
- (C) $b_{n+1} > b_n$ (D) $b_{n+1} < b_n$
6. Let point A is taken on line $x + y - 9 = 0$ and B and C are taken on circle $2x^2 + 2y^2 - 8x - 8y - 1 = 0$ such that line AB passes through centre of circle and $\angle BAC = 45^\circ$, then range of x coordinate of point A is superset of which of the following?
- (A) $[3, 4]$ (B) $[2, 4]$
- (C) $\left[\pi, \frac{3\pi}{2}\right]$ (D) $[\pi, 5]$

PART – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 7 and 8

N dice are rolled once, then the probability of getting sum t is expressed as $P_d(t, N)$. Where dice is a d sided dice with face marked 1, 2, 3, 4, d.

7. Which of the following is an **incorrect** combination?
- (A) $P_6(14, 3) = P_6(14, 5)$ (B) $P_3(5, 2) = P_3(5, 3)$
- (C) $P_3(10, 4) = P_3(10, 5)$ (D) $P_4(9, 3) = P_4(9, 4)$
8. The value of $P_9(15, 2)$ is equal to
- (A) $\frac{4}{71}$ (B) $\frac{4}{81}$
- (C) $\frac{8}{81}$ (D) $\frac{8}{71}$

Space For Rough Work

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Paragraph for Question Nos. 09 and 10

Let consider lines $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $2x + 5y - z + 47 = 0 = 2x + y + z + 7$

9. The shortest distance between both given line is
- (A) $2\sqrt{30}$
- (B) $3\sqrt{30}$
- (C) $\sqrt{30}$
- (D) $\sqrt{75}$
10. Equation of plane parallel to both of lines and which lies midway between the lines
- (A) $2x + 5y - z + 2 = 0$
- (B) $2x - 5y + z - 2 = 0$
- (C) $2x + 5y - z - 2 = 0$
- (D) $2x - 5y - z - 2 = 0$

PART – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

- Let R be the circum-radius of triangle ABC and let G and H be its centroid and orthocentre respectively. Let F be the midpoint of GH, then if $AF^2 + BF^2 + CF^2 = kR^2$, then k is equal to
- In the coordinate plane, the area of the region bounded by the line passing through (1, 2) with the slope α and the parabola $y = x^2$ is denoted by $S(\alpha)$ where α varies in the range of $0 \leq \alpha \leq 6$, then find the value of α such that $S(\alpha)$ is minimized
- Given a triangle ABC, let a, b, c denotes the length of its sides and m, n, p the length of its medians. For every positive real α , let $\lambda(\alpha)$ be the real number satisfying $a^\alpha + b^\alpha + c^\alpha = (\lambda(\alpha))^\alpha \{m^\alpha + n^\alpha + p^\alpha\}$, then of value of $\lambda(2)$ is k, then $[k]$ is (where $[.]$ denotes the greatest integer function)

Space For Rough Work

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PART – B (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

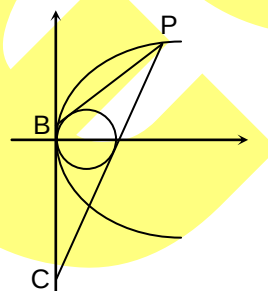
Question Stem for Question Nos. 4 and 5

If the abscissa and the ordinate of an equilateral triangle are the roots of $a_1x^3 + b_1x + c_1 = 0$ and $a_2x^3 + b_2x + c_2 = 0$, then if $k_1 \left[\frac{b_1}{a_1} + \frac{b_2}{a_2} \right] + k_2 R^3 = 0$ (where R is the circumradius of triangle ABC); k_1 and k_2 are coprime to each other, then

4. Value of k_1 is
5. Value of k_2 is

Question Stem for Question Nos. 6 and 7

As is shown in figure, $P(x_0, y_0)$ is a moving point on the parabola $y^2 = 2x$, point B and C are on y -axis, and the sides of triangle PBC touches the circle $(x - 1)^2 + y^2 = 1$.



6. The minimum value of area of triangle PBC is _____ (in sq. unit)
7. The value of x_0 when the area of $\triangle PBC$ is minimum

Question Stem for Question Nos. 8 and 9

The largest area of rectangle whose

8. Base is on x -axis and upper two vertices lie on parabola $y = 12 - x^2$
9. Base is along with x -axis and inscribed in equilateral triangle having base as x -axis and side length 20 unit

Space For Rough Work

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QP Code: 100183

ANSWERS**SECTION-1 : PHYSICS****PART – A**

- | | | | |
|---------------|---------------|------------|---------|
| 1. A, B, C, D | 2. A, B, C | 3. A, B, C | 4. B, D |
| 5. B, C | 6. A, B, C, D | 7. C | 8. C |
| 9. D | 10. D | | |

PART – B

- | | | | |
|----------|----------|----------|----------|
| 1. 5 | 2. 10 | 3. 2100 | 4. 06.00 |
| 5. 15.00 | 6. 10.00 | 7. 06.28 | 8. 09.00 |
| 9. 15.00 | | | |

SECTION – 2 : CHEMISTRY**PART – A**

- | | | | |
|------------|------------|---------------|---------|
| 1. A, B, C | 2. B, C | 3. A, B, C, D | 4. B, C |
| 5. A | 6. A, C, D | 7. D | 8. D |
| 9. B | 10. A | | |

PART – B

- | | | | |
|-----------|----------|----------|-----------|
| 1. 2 | 2. 6 | 3. 9 | 4. 166.67 |
| 5. 00.07 | 6. 17.72 | 7. 34.04 | 8. 03.00 |
| 9. 117.73 | | | |

SECTION – 3 : MATHEMATICS**PART – A**

- | | | | |
|---------|------------|------|---------|
| 1. A, C | 2. B, D | 3. C | 4. B, D |
| 5. B, C | 6. A, C, D | 7. C | 8. B |
| 9. B | 10. A | | |

PART – B

- | | | | |
|---------------------------------|----------|----------|----------|
| 1. 3 | 2. 2 | 3. 1 | 4. 02.00 |
| 5. 03.00 | 6. 08.00 | 7. 04.00 | 8. 32.00 |
| 9. 86.50 (Range 86.50 to 86.60) | | | |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. A, B, C, D

Sol. For the stars

$$\frac{GM^2}{(4R)^2} = M(2R)\omega^2$$

$$\omega = \sqrt{\frac{GM}{32R^3}} \quad \dots(i)$$

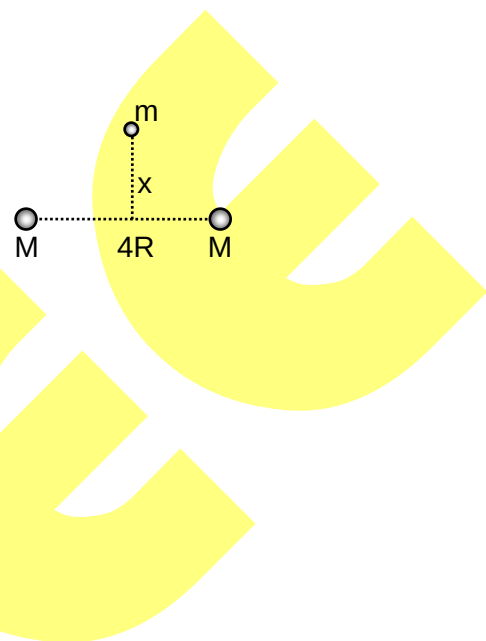
The asteroid should be equidistant from both the stars.

$$\frac{2GMmX}{[(X^2 + (2R)^2)^{3/2}]} = mX\omega^2 \quad \dots(ii)$$

From (i) and (ii)

$$X = 2\sqrt{3}R$$

$$\text{Total energy} = -\frac{2GMm}{4R} + \frac{1}{2}m\left(2\sqrt{3}R\sqrt{\frac{GM}{32R^3}}\right)^2 = -\frac{5GMm}{16R}$$



2. A, B, C

Sol. $x = t^3 - 6t^2 - 96t + 37 \quad \dots(i)$

$$v = 3t^2 - 12t - 96 \quad \dots(ii)$$

$$a = 6t - 12 \quad \dots(iii)$$

at $t = 0$, $v = -96$ m/s

when $v = 0$, $t = 8$ sec

and when v is maximum $t = 2$ sec.

3. A, B, C

Sol. Percentage error in a , b and M are $\frac{\Delta a}{a} \times 100 = \frac{0.001}{4} \times 100 = 0.025\%$

$$\frac{\Delta b}{b} \times 100 = \frac{0.001}{2} \times 100 = 0.05\%$$

$$\frac{\Delta M}{M} \times 100 = \frac{0.0003}{0.3} \times 100 = 0.1\%$$

Moment of Inertia

$$I = \frac{M}{2}(a^2 + b^2)$$

$$\frac{\Delta I}{I} = \frac{\Delta M}{M} + \frac{2a \cdot \Delta a + 2b \cdot \Delta b}{a^2 + b^2}$$

$$\frac{\Delta I}{I} = 0.001 + \frac{2(4)(0.001) + 2(2)(0.001)}{20}$$

$$\frac{\Delta I}{I} = 0.001 + \frac{0.008 + 0.004}{20} = 0.0016$$

$$\frac{\Delta I}{I} \times 100 = 0.16\%$$

4. B, D

Sol. As we know that,

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$$\frac{hc}{\lambda} = \phi + K \Rightarrow \frac{12400}{\lambda} = \phi + v_s, \text{ where } \phi \text{ is in electron Volt and } \lambda \text{ is in } \text{\AA}$$

When light of wavelength $\lambda = 6000 \text{ \AA}$ is used

$$\frac{12400}{6000} = \phi + 0.3 \Rightarrow \phi = 1.77 \approx 1.78 \text{ eV (approximately)}$$

When light with wavelength $\lambda = 4000 \text{ \AA}$ is used

$$\frac{12400}{4000} = \phi + eV$$

$$\Rightarrow eV = (3.1 - 1.78) \text{ e volt} = 1.32 \text{ eV}$$

$$\Rightarrow V = 1.32 \text{ Volt}$$

5. B, C

Sol. Heat gain by ice to melt = $mL_f = 10 \times 80 = 800 \text{ cal}$

Heat gain by 10 g water to raise its temperature from 0°C to $100^\circ\text{C} = 10 \times 1 \times 100 = 1000 \text{ cal}$

Total heat gain = 1800 cal

Mass of steam converted into water

$$\therefore 1800 = m \times 540 \Rightarrow m = 3.33 \text{ g}$$

$\therefore$ equilibrium temperature 100°C

Amount of water = $10 + 3.33 = 13.33 \text{ g}$

Amount of steam = $5 - 3.33 = 1.67 \text{ g}$

6. A, B, C, D

Sol. (A) $i_{\text{rms}} = \frac{80}{4} = 20 \text{ amp}$, $i_{\text{peak}} = 20\sqrt{2} \text{ amp}$

(B) $\varepsilon_{\text{rms}} = \sqrt{(40)^2 + (40 - 10)^2} = 100\text{V}$, $\varepsilon_0 = 100\sqrt{2}\text{V}$

(C) $\omega L i_{\text{rms}} = 90 \Rightarrow L = \frac{90}{10 \times 90\pi} = \frac{1}{20\pi} \text{ H}$

(D) For maximum heat, current through the circuit will be maximum $\omega L = \frac{1}{\omega C}$

$$C = \frac{1}{405\pi} \text{ F}$$

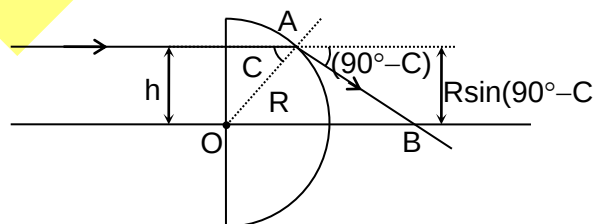
7. C

Sol. $\frac{5}{3} \sin C = 1 \sin 90^\circ$

$$\sin C = 3/5 \quad (C \rightarrow \text{critical angle})$$

$$C = 37^\circ$$

$$h = R \sin 37^\circ = \frac{3}{5} R = \frac{3}{5} (30) = 18 \text{ cm}$$



8. C

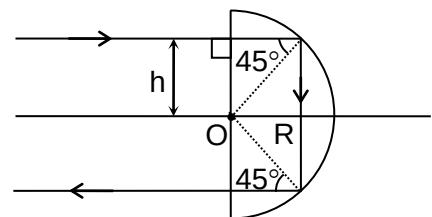
Sol. when $\mu = \sqrt{2}$

$$\sqrt{2} \sin C = 1 \quad (C \rightarrow \text{critical angle})$$

$$C = 45^\circ$$

Total distance travel in semi-cylinder

$$= \left(\frac{R}{\sqrt{2}} \right) 4 = 2\sqrt{2}R = (2\sqrt{2})(30) = 60\sqrt{2} \text{ cm}$$



$$\text{Time} = \frac{60\sqrt{2}}{C_m} \times 10^{-2} = \frac{60\sqrt{2} \times 10^{-2}}{\left(\frac{3 \times 10^8}{\sqrt{2}} \right)} = \frac{120 \times 10^{-2}}{3 \times 10^8} = 4 \times 10^{-9} \text{ s} \quad (C_m \rightarrow \text{speed of light in the medium})$$

9. D

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10. D

Sol. (for Q. 9-10):

$$dm = \mu dy$$

$$m = \int_0^y \mu_0 \left(\frac{y}{\ell} \right)^3 dy = \left(\frac{\mu_0}{\ell^3} \right) \left(\frac{y^4}{4} \right)$$

Velocity of wave

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{gy}{4}}$$

$$\frac{dy}{dt} = \frac{\sqrt{gy}}{2}$$

$$\int_0^{\ell} \frac{dy}{\sqrt{y}} = \frac{\sqrt{g}}{2} \int_0^t dt$$

$$t = \left(4 \sqrt{\frac{\ell}{g}} \right)$$

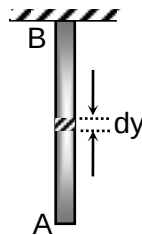
$$v = \frac{\sqrt{gy}}{2}$$

$$v^2 = \frac{gy}{4}$$

$$2v \frac{dv}{dy} = \frac{g}{4}$$

$$v \frac{dv}{dy} = \left(\frac{g}{8} \right)$$

$$a = g/8$$



PART - B

1. 5

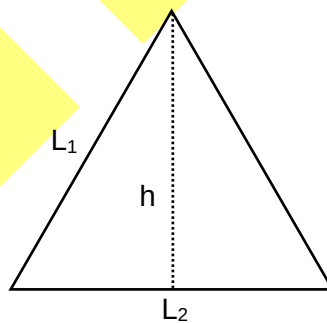
$$\text{Sol. } h^2 = L_1^2 - \frac{L_2^2}{4}$$

Differentiating and using $dL = L \alpha \Delta t$

$$0 = 2L_1\alpha_1 L_1 \Delta t - \frac{1}{4} 2L_2\alpha_2 L_2 \Delta t$$

$$L_1 L_1 \alpha_1 \Delta t = \frac{1}{4} L_2 L_2 \alpha_2 \Delta t$$

$$\Rightarrow \frac{L_2}{L_1} = 2 \sqrt{\frac{\alpha_1}{\alpha_2}} = 2 \sqrt{\frac{25 \times 10^{-6}}{4 \times 10^{-6}}} = 5$$



2. 10

$$\text{Sol. For lens, } \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Here, $R_2 = +60$ cm and $R_1 = 20$ cm

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{20} - \frac{1}{60} \right)$$

$$\Rightarrow \frac{1}{60} = \frac{1}{v} + \frac{1}{80}$$

$$v = 240$$

$$\text{Shift due to slab, } s = 2 \left[1 - \frac{1}{1.5} \right] = 1 \text{ cm}$$

For final image to form at O, ray should retrace its path after reflection from mirror. So, I_2 should be at centre of curvature of mirror i.e.

$$7 + 3 + x = 20$$

$$x = 10 \text{ cm}$$

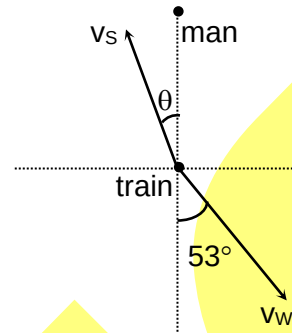
3. 2100

Sol. $v_w \rightarrow$ velocity of wind
 $v_s \rightarrow$ velocity of sound
 $v_s \sin \theta = v_w \sin 53^\circ$

$$\sin \theta = \frac{v_w}{v_s} \left(\frac{4}{5} \right) = \frac{100}{(400/3)} \times \frac{4}{5} = \frac{3}{5}$$

$$\theta = 37^\circ$$

$$f' = f \left(\frac{v_s \cos \theta - v_w \cos 53^\circ}{v_s \cos \theta - v_w \cos 53^\circ - v_T} \right) = 300 \times 7 = 2100 \text{ Hz}$$



4. 06.00

5. 15.00

Sol. (for Q. 4-5):

By the conservation of linear momentum between m_1 and m_2 and by using coefficient of restitution velocity m_2 just after collision $\frac{3v_0}{4}$.

After the collision, the maximum extension in the spring occurs when angular velocity of m_2 and m_3 about O become same. Now by conservation of angular momentum about 'O' and energy conservation

$$\text{Velocity of } m_2 = 2R\omega = \frac{3}{5}v_0$$

$$\text{Velocity of } m_3 = R\omega = \frac{3}{10}v_0 \text{ and } \Delta x \text{ maximum} = \frac{3}{4}v_0 \sqrt{\frac{m}{5k}}$$

6. 10.00

7. 06.28

Sol. (for Q. 6-7):

$$i \left(\frac{8y_0}{3} \right) B = mg$$

$$y_0 = \frac{3mg}{8iB}$$

$$y_0 = 10$$

If loop is displaced by small y downward.

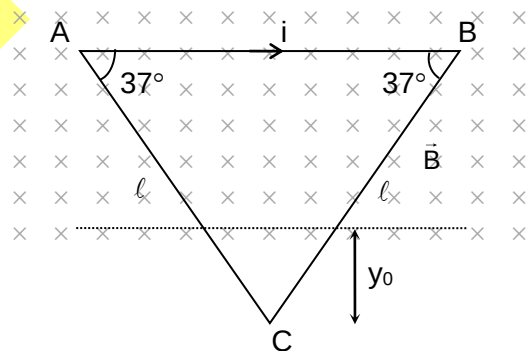
Then,

$$ma = mg - iB \left[\frac{8}{3}(y + y_0) \right]$$

$$a = - \left(\frac{8iB}{3m} \right) y$$

$$\omega = \sqrt{\frac{8iB}{3m}}$$

$$T = \frac{2\pi}{\omega} = 2\pi$$



8. 09.00

9. 15.00

Sol. (for Q. 8-9):

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Difference of potential energy of two configuration when point charge is at rest at a distance d to when it passes through centre.

The energy of oscillation system

$$E = \frac{2KQ^2}{3\sqrt{R^2 + d^2}} - \frac{2KQ^2}{3R} = \frac{Q^2 d^2}{12\pi\epsilon_0 R^3}$$

$$|m - n| = 12 - 3 = 9$$

Force between the ring and the charge particle

$$F = -\left(\frac{2KQ^2}{3R^3}\right)d$$

$$M_{eq} = \frac{2}{3}M$$

$$\text{Hence time period of oscillation is } T = \sqrt{\frac{16\pi^3\epsilon_0 MR^3}{Q^2}}$$

SECTION – 2 : CHEMISTRY

PART – A

1. A, B, C

Sol. E_{cell}^0 , E_{cell} and normality are intensive property, where as entropy is extensive property.

2. B, C

Sol. $[\pi_{obs}]_{0.4 \text{ M NaCl}} = i \times C \times R \times T$

$$= i \times 0.4 \times R \times T$$

$$= [1 + (n-1)\alpha] \times 0.4 \times R \times T$$

$$= [1 + 0.8] \times 0.4 \times R \times T$$

$$= 1.8 \times 0.4 \times R \times T$$

$$= 0.72 \text{ RT}$$

$$[\pi_{obs}]_{0.3 \text{ M Na}_3\text{PO}_4} = [1 + (4-1) \times 0.9] \times 0.3 \times R \times T$$

$$= 1.11 \text{ RT}$$

$$[\pi_{obs}]_{0.7 \text{ M Glucose}} = 0.7 \times R \times T$$

$$= 0.7 \text{ RT}$$

$$[\pi_{obs}]_{1\text{MH}-\overset{\text{O}}{\parallel}\text{C}-\text{OH}} = [1 + (2-1)0.3] \times R \times T$$

$$= 1.3 \times R \times T$$

$$[\pi_{obs}]_{0.5 \text{ M MgCl}_2} = 0.5 \times R \times T [1 + (2) \times 0.2]$$

$$= R \times T \times 0.5 [1.4]$$

$$= 0.7 \text{ RT}$$

Here only option (B) and (C) have greater osmotic pressure then given solution.

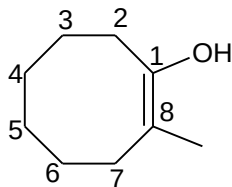
3. A, B, C, D

Sol. In isothermal ideal gas compression as volume decrease. Hence, W is +ve, ΔH is zero, ΔS_{gas} is -ve and ΔE is zero

4. B, C

Sol. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel}\text{C} - \text{H}$ will show geometrical isomerism.

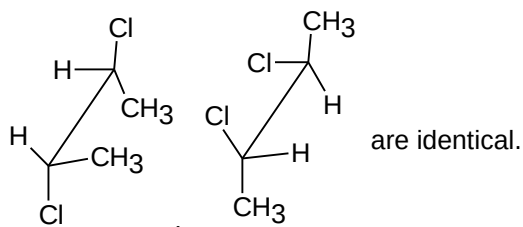
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will show geometrical isomerism

5. A

Sol.

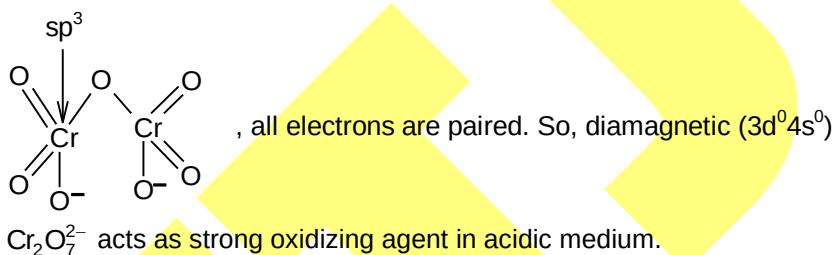


6. A, C, D

Sol. (A) $\ell = 0$, then $m \neq 1$. Hence, (A) is incorrect.
 (B) It is the correct set.
 (C) If $n = 1$, then $\ell = -1$. Hence, (C) is not correct.
 (D) If $n = 3$, then $m \neq 3$. Hence, (D) is also incorrect.

7. D

Sol.



8. D

Sol.

(E) = NH_3
 (A) $\text{Cu}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$ (deep blue colouration)
 (D) $\text{HgCl}_2 + \text{NH}_3 + \text{H}_2\text{O} \longrightarrow \text{HgO} \cdot \text{Hg}(\text{NH}_2)\text{Cl} \downarrow$ (white)

9. B

10. A

Sol.

(for Q. 9 to 10):

$A \xrightarrow{\text{H}^+ / \text{H}_2\text{S}} \text{B}$ (Yellow ppt.)

Only CdS and As_2S_3 are yellow

CdS insoluble in Yellow Ammonium Sulphide

As_2S_3 soluble in Yellow Ammonium Sulphide

So, B is CdS

$A \xrightarrow{\text{BaCl}_2} \text{C}$ (White ppt.)

BaSO_4 are white ppt.

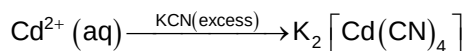
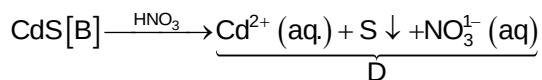
BaSO_4 Insoluble in mineral acids

BaSO_3 soluble in mineral acids

So, C is BaSO_4

$\therefore$ A is CdSO_4

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[D]

PART – B

1. 2

Sol. $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{PtCl}_3\text{NH}_3]$

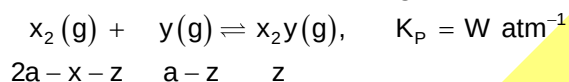
2. 6

Sol. Number of stereo isomers for the complex is 6.

3. 9

Sol. $3x_2(\text{g}) \rightleftharpoons x_6(\text{g}), K_p = 1.6 \text{ atm}^{-2}$

$$2a - x - z \quad \frac{x}{3}$$



$$2a - x - z \quad a - z \quad z$$

$$\frac{x}{3} = 0.2 \Rightarrow x = 0.6$$

$$1.6 = \frac{P_{x_6}}{[P_{x_2}]^3} \Rightarrow [P_{x_2}]^3 = \frac{0.2}{1.6}$$

$$P_{x_2} = \frac{1}{2} = 0.5 \text{ atm}$$

$$\text{Now, } 0.5 + 0.2 + a - z + z = 1.4$$

$$a = 0.7$$

$$2a - x - z = 0.5$$

$$\Rightarrow 2 \times 0.7 - 0.6 - 0.5 = z$$

$$z = 0.3$$

$$K_p = W = \frac{0.3}{0.5 \times [0.7 - 0.3]} = 1.5$$

$$\text{Hence, } 6W = 6 \times 1.5 = 9$$

4. 166.67

Sol. 1st order kinetic

$$t = \frac{2.303}{k} \log \frac{a_0}{a_t}$$

$$\frac{0.6932}{50} = \frac{2.303}{t} \log \frac{100}{10}$$

5. 00.07

Sol. $\text{A}(\text{g}) \longrightarrow 3\text{B}(\text{g}) + 2\text{C}(\text{g})$

$$t = 0 \quad \left[\frac{2}{12.5} \right] \quad 0 \quad 0$$

$$t = t \quad \frac{2-x}{V} \quad \frac{3x}{V} \quad \frac{2x}{V}$$

$$\text{Now, } t = \frac{2.303}{K} \log \frac{a_0}{a_t}$$

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$$\Rightarrow 100 = \frac{2.303}{1.386 \times 10^{-2}} \log \frac{2/12.5}{a_t}$$

$$\Rightarrow at = 0.04$$

$$\text{Now, } \frac{2-x}{V} = 0.04 \quad \dots (1)$$

And as the P, T constant

$$\frac{2}{12.5} = \frac{2+4x}{V} \quad \dots (2)$$

On solving Eq. (1) and (2), we get

$$[B_t] = 0.07 \text{ M}$$

6. 17.72

Sol. Meq. of NH_3 = Meq. of HCl – Meq. of NaOH

$$\frac{W \times 1000}{17} = 5 - 2.32 = 2.68$$

$$W = \frac{17 \times 2.68}{1000} = 0.04556$$

$$\%N = \frac{0.04556}{0.257} \times 100$$

$$= 17.72\%$$

7. 34.04

$$\text{Sol. } \%Br = \frac{80}{188} \times \frac{0.12}{0.15} \times 100 = 34.04\%$$

8. 03.00

$$\text{Sol. } E = 0.61 = 0.76 - \frac{0.06}{2} \log \frac{[Zn^{2+}]}{[H^+]^2}$$

$$\Rightarrow \log \frac{(0.1)}{[H^+]^2} = \frac{0.76 - 0.61}{0.03}$$

$$\Rightarrow \text{pH} = 3$$

9. 117.73

$$\text{Sol. } \Delta G = -nFE_{\text{cell}}$$

$$|\Delta G| = \frac{2 \times 0.61 \times 96500}{1000} \text{ kJ}$$

$$= 117.73 \text{ kJ}$$

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SECTION – 3 : MATHEMATICS

PART – A

1. A, C

$$\text{Sol. } \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \frac{k-1}{k!} \right) = \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \sum_{k=2}^i \left\{ \frac{1}{(k-1)!} - \frac{1}{k!} \right\} \right)$$

$$L = \lim_{n \rightarrow \infty} \left(n+1 - \sum_{i=2}^n \left(1 - \frac{1}{i!} \right) \right); L = \lim_{n \rightarrow \infty} \left(2 + \sum_{i=2}^n \frac{1}{i!} \right)$$

$$\therefore L = e$$

2. B, D

Sol. Let $g(0) = c$ then put $x = x$ and $y = 0$ in functional equation $f(x + g(y)) = 2x + y + 5$

$$\Rightarrow f(x + c) = 2x + 5 \Rightarrow f(x) = 2x + 5 - 2c$$

$$\Rightarrow f(x + g(y)) = 2x + 2(g(y)) - 2c + 5 \text{ compare with give equation}$$

$$\Rightarrow g(y) = \frac{y}{2} + c \Rightarrow g(x + f(y)) = \frac{x + f(y)}{2} + c \Rightarrow g(x + f(y)) = \frac{x + 2y - 2c + 5}{2} + c = \frac{x + 2y + 5}{2}$$

3. C

$$\text{Sol. } {}^m C_1 \cdot {}^n C_m - {}^m C_2 \cdot {}^{2n} C_m + {}^m C_3 \cdot {}^{3n} C_m \dots (-1)^{m-1} \cdot {}^m C_m \cdot {}^{mn} C_n$$

$\Rightarrow$ Coefficient of x^m in expression

$${}^m C_1 \cdot (1+x)^n - {}^m C_2 \cdot (1+x)^{2n} + {}^m C_3 \cdot (1+x)^{3n} \dots {}^m C_m \cdot (1+x)^{mn} \cdot (-1)^{m-1}$$

$$\Rightarrow {}^m C_0 - [{}^m C_0 - {}^m C_1 \cdot (1+x)^n + {}^m C_2 \cdot (1+x)^{2n} - {}^m C_3 \cdot (1+x)^{3n} \dots {}^m C_m \cdot (-1)^{mn} \cdot (1+x)^{mn}]$$

$$\Rightarrow \text{Coefficient of } x^m \text{ in } {}^m C_0 - (1 - (1+x)^n)^m \Rightarrow \text{Coefficient of } x^m \text{ in } (1 - (1 - (1+x)^n))^m$$

4. B, D

Sol. If there is $2k$ occurrence of a then this can be done by ${}^n C_{2k}$ ways and remaining positions can be

$$\text{filled in } 2^{n-2k} \text{ ways. So total number of ways are } \sum_{k=0}^n {}^n C_{2k} 2^{n-2k} = \frac{1}{2}(3^n + 1)$$

5. B, C

$$\text{Sol. Let } f(x) = \frac{1}{1+x}, \text{ then } \frac{1}{n} \left(\frac{1}{1+\frac{1}{n}} \right) < \int_0^{\frac{1}{n}} f(x) dx < \frac{1}{n}, \frac{1}{1+n} < \ln \left(1 + \frac{1}{n} \right) < \frac{1}{n}$$

$$\text{Now, } a_n - a_{n+1} = -\log n - \frac{1}{n+1} + \log(n+1) = \log \left(1 + \frac{1}{n} \right) - \frac{1}{n+1} > 0$$

$$b_{n+1} - b_n = a_{n+1} - a_n - \frac{1}{n+1} + \frac{1}{n} = \frac{1}{n+1} - \log \left(1 + \frac{1}{n} \right) - \frac{1}{n+1} + \frac{1}{n} = -\log \left(1 + \frac{1}{n} \right) + \frac{1}{n} > 0$$

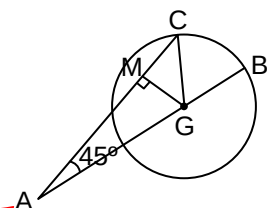
6. A, C, D

Sol. Let coordinate of $A(a, 9-a)$, then the distance of the centre of circle

$$x^2 + y^2 - 4x - 4y - \frac{1}{2} = 0 \text{ from AC will be}$$

$$= |AG| \sin 45^\circ$$

$$= \sqrt{(a-2)^2 + (7-a)^2} \cdot \frac{1}{\sqrt{2}}$$



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$$\text{Now, } d \leq \sqrt{\frac{17}{2}} \Rightarrow \sqrt{(a-2)^2 + (7-a)^2} \cdot \frac{1}{\sqrt{2}} \leq \sqrt{\frac{17}{2}}$$

$$\Rightarrow a^2 - 9a + 18 \leq 0 \Rightarrow a \in [3, 6]$$

7. C

8. B

Sol. (for Q. 7 & 8):

We can determine the value of $P_d(t, N)$ as = $\frac{\text{coefficient of } x^t \text{ in } (x^1 + x^2 + x^3 + \dots + x^d)^N}{d^N}$

$$\text{The value of } P_3(5, 2) = P_3(5, 3) = \frac{2}{9}; P_9(15, 2) = \frac{4}{81}$$

$$P_6(14, 3) = P_6(14, 5) = \frac{5}{72}; P_3(10, 4) = P(10, 6) = \frac{10}{81}$$

$$P_4(9, 3) = P_4(9, 4) = \frac{5}{32}$$

9. B

10. A

Sol. (for Q. 9 & 10):

$$\text{Equation of line-1} \equiv \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$$

$$\text{Equation of line-2} \equiv 2x + 5y - z + 47 = 2x + y + z + 7 = 0 \text{ in symmetric form } \frac{x-\frac{3}{2}}{-3} = \frac{y+10}{2} = \frac{z}{4}$$

Shortest distance between both line

$$= \frac{|\vec{AB} \cdot (\vec{n}_1 \times \vec{n}_2)|}{|\vec{n}_1 \times \vec{n}_2|} = 3\sqrt{30}$$

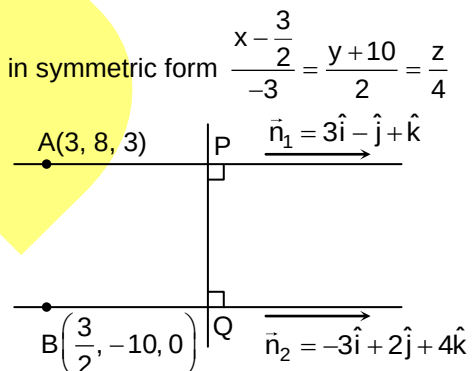
$$\text{Now, P is point on AP} \Rightarrow P = (3 + 3r, 8 - r, 3 + r)$$

$$Q \text{ is point of BQ} = Q = \left(\frac{3}{2} - 3r', -10 + 2r', 4r'\right)$$

$$\text{Now, } \vec{PQ} \parallel (\vec{n}_1 \times \vec{n}_2)$$

$$\Rightarrow \frac{\frac{3}{2} + 3r + 3r'}{-2} = \frac{19 - r - 2r'}{-5} = \frac{3 + r - 4r'}{1} \Rightarrow r = 0 \text{ and } r' = \frac{3}{2}$$

$$\therefore P = (3, 8, 3) \text{ and } Q = (-3, -7, 6) \text{ midpoint of PQ} = \left(0, \frac{1}{2}, \frac{9}{2}\right),$$

so required plane $2x + 5y - z + 2 = 0$ **PART – B**

1. 3

Sol. Let vertices $\vec{A}, \vec{B}, \vec{C}$ and circumcentre as origin, then

$$\vec{H} \text{ (orthocentre)} = \vec{A} + \vec{B} + \vec{C}$$

$$\vec{G} \text{ (centroid)} = \frac{\vec{H}}{3}$$

$$\vec{F} = \frac{\vec{G} + \vec{H}}{2} = \frac{2\vec{H}}{3}$$

$$\text{Now, } AF^2 + BF^2 + CF^2 = (\vec{A} - \vec{F})(\vec{A} - \vec{F}) + (\vec{B} - \vec{F})(\vec{B} - \vec{F}) + (\vec{C} - \vec{F})(\vec{C} - \vec{F})$$

$$= \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} + \vec{C} \cdot \vec{C} - 2(\vec{A} + \vec{B} + \vec{C}) \cdot \vec{F} + 3\vec{F} \cdot \vec{F} = 3R^2 - \vec{F} \cdot (2(\vec{A} + \vec{B} + \vec{C})) - 3\vec{F} \cdot \vec{F} = 3R^2$$

TG ~ @bohring_bot

2. 2

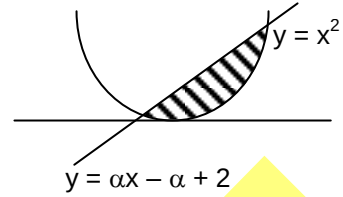
Sol. Let point of intersection of line
 $y = \alpha x - \alpha + 2$ with $y = x^2$ are $x = x_1$ and $x = x_2$
 Then, $x_1 + x_2 = \alpha$; $x_1 x_2 = \alpha - 2$

$$S(\alpha) = \int_{x_1}^{x_2} ((\alpha x - \alpha + 2) - x^2) dx$$

For area to be minimized $\frac{d(S(\alpha))}{d\alpha} = 0$

$$\Rightarrow \frac{d(S(\alpha))}{d(\alpha)} = \frac{d}{d\alpha} \int_{x_1}^{x_2} ((\alpha x - \alpha + 2) - x^2) dx = \int_{x_1}^{x_2} \frac{d}{d\alpha} (\alpha x - \alpha + 2 - x^2) dx$$

$$= \int_{x_1}^{x_2} (x - 1) dx = (x_2 - x_1) \left(\frac{x_1 + x_2}{2} - 1 \right) = 0 \Rightarrow x_1 + x_2 = 2 \Rightarrow \alpha = 2$$



3. 1

Sol. $m = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2}$; $n = \frac{1}{2} \sqrt{2a^2 + 2c^2 - a^2}$; $p = \frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2}$

$$\Rightarrow a^2 + b^2 + c^2 = (\lambda(2))^2 \{m^2 + n^2 + p^2\}$$

$$\Rightarrow a^2 + b^2 + c^2 = (\lambda(2))^2 \left\{ \frac{1}{4} \left\{ (2b^2 + 2c^2 - a^2) + (2a^2 + 2c^2 - b^2) + (2a^2 + 2b^2 - c^2) \right\} \right\}$$

$$\Rightarrow \lambda(2) = \frac{2}{\sqrt{3}}$$

4. 02.00

5. 03.00

Sol. (for Q. 4-5):

Let the vertices of triangle ABC are (x_1, y_1) , (x_2, y_2) , (x_3, y_3)
 Then x_1, x_2, x_3 are root of $a_1 x^3 + b_1 x + c_1 = 0$

$$\Rightarrow x_1 + x_2 + x_3 = 0 \text{ and } \sum x_1 x_2 = \frac{b_1}{a_1}$$

$$\text{Similarly } y_1 + y_2 + y_3 = 0 \text{ and } \sum y_1 y_2 = \frac{b_2}{a_2}$$

$\Rightarrow$ Circumcentre is at origin

$$\Rightarrow x_1^2 + y_1^2 = x_2^2 + y_2^2 = x_3^2 + y_3^2 = R^2 \Rightarrow x_1^2 + x_2^2 + x_3^2 + y_1^2 + y_2^2 + y_3^2 = 3R^2$$

$$\Rightarrow (x_1 + x_2 + x_3)^2 - 2 \sum x_1 x_2 + (y_1 + y_2 + y_3)^2 - 2 \sum y_1 y_2 = 3R^2$$

$$\Rightarrow -2 \left[\sum x_1 x_2 + \sum y_1 y_2 \right] = 3R^2 \Rightarrow -2 \left[\frac{b_1}{a_1} + \frac{b_2}{a_2} \right] = 3R^2$$

6. 08.00

7. 04.00

Sol. Let point B(0, b) and C(0, c) and assume $b > c$

The equation of PB is $y - b = \frac{y_0 - b}{x_0} (x)$

$$\Rightarrow (y_0 - b)x - x_0 y + x_0 b = 0 \text{ as its tangent of circle } (x - 1)^2 + y^2 = 1$$

$$\Rightarrow 1 = \left| \frac{y_0 - b + x_0 b}{\sqrt{(y_0 - b)^2 + x_0^2}} \right| \Rightarrow (x_0 - 2)b^2 + 2y_0 b - x_0 = 0 \text{ similarly } (x_0 - 2)c^2 + 2y_0 c - x_0 = 0$$

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$$\Rightarrow b+c = \frac{-2y_0}{x_0-2} \text{ and } bc = \frac{-x_0}{x_0-2} \Rightarrow (b-c)^2 = \frac{4x_0^2 + 4y_0^2 - 8x_0}{(x_0-2)^2} \Rightarrow b-c = \frac{2x_0}{x_0-2} \text{ (as } y_0^2 = 2x_0)$$

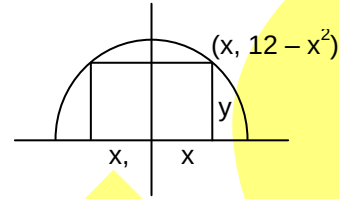
$$\text{Now, area of } \Delta PBC = \frac{1}{2}(b-c)x_0 = \frac{x_0^2}{x_0-2} = (x_0-2) + \frac{4}{(x_0-2)} + 4 \geq 8$$

8. 32.00

Sol. $A = 2x(12 - x^2)$

$$\frac{dA}{dx} = 2(12 - x^2 + (-2x(2x))) = 24 - 6x^2$$

For maximum $x = 2$, $A = 32$ sq units



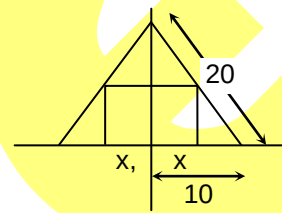
9. 86.50
(Range 86.50 to 86.60)

Sol. $A = 2x\sqrt{3}(10 - x)$

$$\frac{dA}{dx} = 2\sqrt{3}(10 - 2x)$$

Maximize when $x = 5$

$$A = 50\sqrt{3}$$



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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 3

QP CODE: 100187

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 204

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & B** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are **not allowed**.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-04)** – Contains **Four (04)** multiple choice questions which have ONLY ONE CORRECT answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **PART-A (05–12)** contains **Eight (8)** Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if **(A), (C) and (D)** are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only **(A) and (D)** will result in **+2 marks**; and darkening **(A) and (B)** will result in **-2 marks**, as a wrong option is also darkened.
- (iii) **PART – B (1 – 3)**: This section contains **Three (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iv) **Part-B (4 – 8)** contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

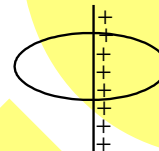
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SECTION – I : PHYSICS**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A paper ball (assumed to be carried easily by wind) is to be thrown in a trash can located $\frac{6\sqrt{3}}{5}$ m north of the point of projection. Wind is blowing towards east with speed 2 m/s. The paper ball is thrown at an angle 37° with the horizontal and it falls on the trash can. What is the speed with which it was thrown?
 (A) 4 m/s (B) 5 m/s (C) 6 m/s (D) 7 m/s

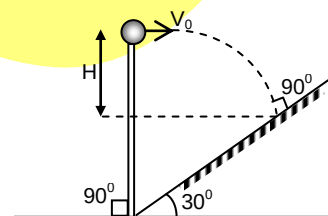
2. A very long uniformly charged rod falls with a constant velocity V through the center of a circular loop making an angle 37° with axis of loop. Then the magnitude of induced emf in loop is (charge per unit length of rod = λ)



- (A) $\frac{\mu_0}{2\pi} \lambda V^2$ (B) $\frac{\mu_0}{2} \lambda V^2$ (C) $\frac{\mu_0}{2\lambda} V$ (D) zero

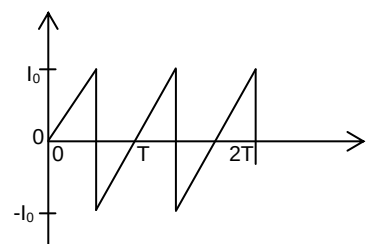
3. In the given figure, the angle of inclination of the inclined plane is 30° . The horizontal velocity V_0 so that the particle hits the inclined plane perpendicularly is

- (A) $v_0 = \sqrt{\frac{2gH}{3}}$ (B) $V_0 = \sqrt{\frac{2gH}{7}}$
 (C) $V_0 = \sqrt{\frac{gH}{5}}$ (D) $V_0 = \sqrt{\frac{gH}{7}}$



4. For a circuit, current variation with time is shown in figure, corresponding rms current is

- (A) I_0 (B) $I_0 / \sqrt{2}$
 (C) $I_0 / \sqrt{3}$ (D) $I_0 / 2$



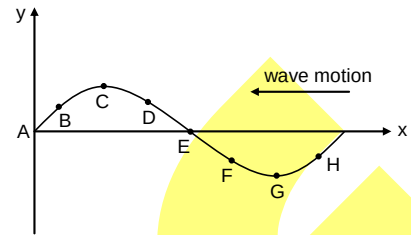
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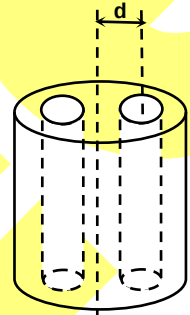
PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A transverse wave is travelling along a stretched string from right to left. The figure shown represents the shape of the string (snap shot) at a given instant. At this instant:
- (A) the particles at A, B and H have upward velocity.
 (B) the particles at D, E and F have downward velocity.
 (C) the particles at C, E and G have zero velocity.
 (D) the particles at A and E have maximum velocity.

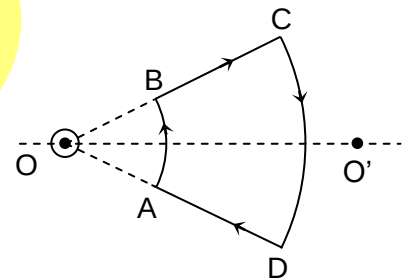


6. An infinitely long cylindrical conductor of radius R contains current of uniform density J along axis of the cylinder. Two infinitely long cylindrical holes of radius r are drilled symmetrically throughout the length of the cylinder. The axes of the holes are parallel to the axis of cylinder and at distance d from it in the same plane.



- (A) Magnetic field on the axis of one of the hole will be $\frac{\mu_0 J}{2} \left(d - \frac{r^2}{2d} \right)$
 (B) Magnetic field on the axis of one of the hole will be $\frac{\mu_0 J}{2} \left(d - \frac{r^2}{d} \right)$
 (C) Magnetic field on the axis of cylinder will be $\frac{\mu_0 J r^2}{2 d}$
 (D) Magnetic field on the axis of cylinder will be zero

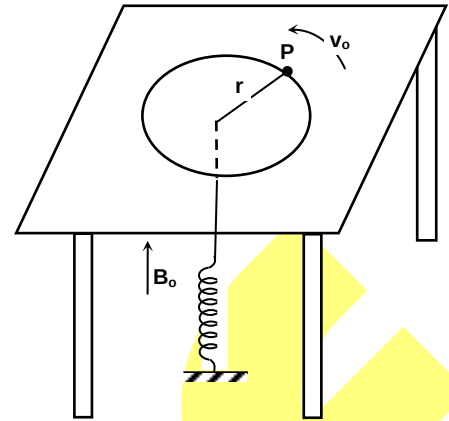
7. An infinite current carrying wire coming outward passes through point O and is perpendicular to the plane containing a current carrying loop ABCD as shown in the figure. Choose the correct option(s).
- (A) net force on the loop is zero
 (B) Net torque on the loop is zero
 (C) As seen from O, the loop rotates clockwise
 (D) As seen from O, the loop rotates anticlockwise



Space For Rough Work

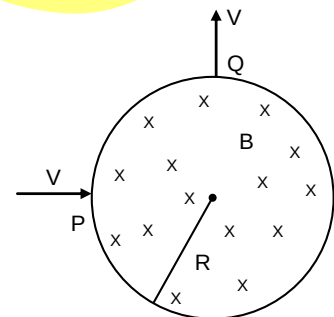
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8. A particle mass m charge q is moving on a circular path on the surface of a frictionless table with speed v_0 where a magnetic field B_0 exists uniformly over the whole region. It is attached by a string which passes through a hole in the table to a spring as shown. The spring is stretched by x_0 . If now the magnetic field is increased slowly to $2B_0$,
- (A) The extension in the spring will increase
 (B) The speed of the particle will increase
 (C) The speed of the particle will decrease
 (D) The kinetic energy of the particle will decrease



9. Two projectile are thrown at the same time from two different points. The projectile thrown from the origin has initial velocity $3\hat{i} + 3\hat{j}$ with respect to earth. The projectile has initial velocity $a\hat{i} + b\hat{j}$ with respect to earth thrown from the point $(10, 5)$. ($\hat{i}$ is a unit vector along horizontal, $\hat{j}$ along vertical). If the projectile collides after two second, then the
- (A) value of a is -2 (B) value of a is $\frac{1}{2}$
 (C) value of b is $\frac{1}{2}$ (D) value of b is -2

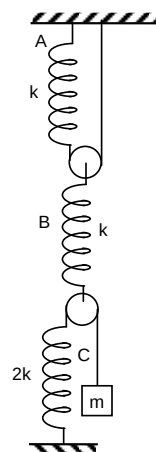
10. A particle of charge q mass m enters normally (at point P) in a region of magnetic field with speed v . It comes out normally from Q after time T as shown in figure. The magnetic field B is present only in region of radius R and is constant and uniform. Initial and final velocities are along radial direction and they are perpendicular to each other. For this to happen, which of the following expression(s) is/are correct?



11. In the given figure, the block is attached with a system of three ideal springs A, B, C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block. Then (pulleys are ideal)

(A) $T = 2\pi\sqrt{\frac{11m}{2k}}$

- (B) the deformation of the spring A is $(2/11)$ times the displacement of the block.
 (C) the deformation of the spring C is $(1/11)$ times the displacement of the block.
 (D) the deformation of the spring B is $(4/11)$ times the displacement of the block.



Space For Rough Work

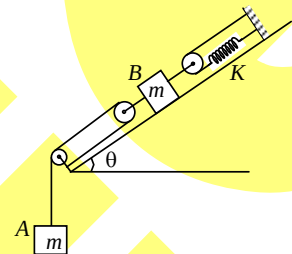
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12. The distance between a screen and an object is 120 cm. A convex lens is placed closed to the object and is moved along the line joining object and screen, towards the screen. Two sharp images of the object are found on the screen. The ratio of magnification of two real images is 1 : 9. Then,
- (A) focal length of the lens is 22.5 cm.
 (B) smaller image is brighter than the larger one.
 (C) larger image is brighter than the smaller one.
 (D) brightness of both the images is same.

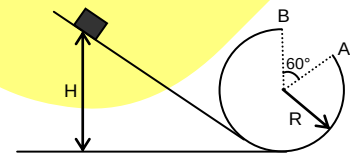
PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

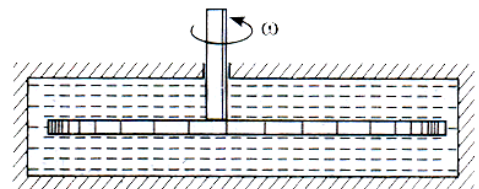
1. Two blocks A and B, each of mass m are connected by means of a pulley-spring system on a smooth inclined plane of inclination θ as shown in the figure. All the pulleys and spring are ideal. Now, B is slightly displaced from its equilibrium position. It starts to oscillate. Time period of oscillation of B is $n\pi$, then find n . (Take $m = 4$ kg, $K = 5$ N/m, $\pi = 3.14$)



2. A small particle slides from height $H = 45$ cm as shown and then loops inside the vertical loop of radius R from where a section of angle $\theta = 60^\circ$ has been removed. If $R = (1/N)$ meter, such that after losing contact at A and flying through the air, the particle will reach at the point B. Find N . Neglect friction everywhere.



3. A thin circular disc of radius R is made to rotate with a constant angular speed ω within a oil filled (coeff. of viscosity η) cylindrical box as shown in the figure. The clearance between the disc & the horizontal planes of the cylindrical box is very small & is equal to h . Considering that the vertical side of the cylindrical box is almost in contact with the disc, the power to be supplied to the system to maintain the constant angular speed is $\frac{\pi\eta\omega^x R^y}{h}$. Then $x + y$ is

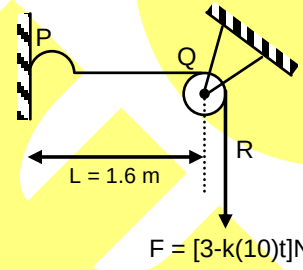


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PART – B (Maximum Marks: 15)**(Numerical Type)**

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

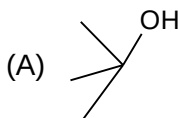
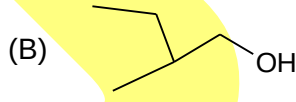
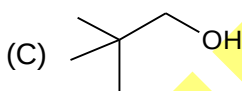
4. A ring of mass 3.15 kg is rolling without slipping with linear velocity 1 m/sec on a smooth horizontal surface. A rod of same mass is fitted along its one diameter. Find total kinetic energy of the system (in J).
5. Potential energy of a particle moving along x-axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position of stable equilibrium state.
6. As shown in figure light string PQR is stretched by force $F = (3 - 10kt)$ N, where k is a constant and t is time in second. At time $t = 0$, a pulse is generated at the end P of the string. For the value of k (in N/s) if the value of force becomes zero as the pulse reaches point Q. Given mass per unit length of string is 0.03 kg/m.
- 
7. In a moving coil galvanometer, torque on the coil can be expressed as $\tau \propto i$, where i is current through the wire. The rectangular coil of the galvanometer having number of turns N , area A and moment of inertia I is placed in magnetic field B . The torsion constant of spring in the galvanometer is C and charge Q is passed through the coil almost instantaneously. If the maximum angle through which coil deflected is $\frac{KNABQ}{\sqrt{3CI}}$. (Ignore the damping in mechanical oscillations). Then the value of K is
8. 2 kg of ice at -22.5°C is mixed with 2.5 kg of water at 25°C in an insulating container. If the specific heat of ice and water are $0.5 \text{ cal/gm}^\circ\text{C}$ and $1 \text{ cal/gm}^\circ\text{C}$ respectively. Latent heat of fusion of ice = 80 cal/gm . Find the amount of water present in the container (In kg).

Space For Rough Work

TG ~ @bohring_bot

SECTION – II : CHEMISTRY**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- Choose correct statement from the following
 (A) Bond angle: $\text{OCl}_2 > \text{BCl}_3 > \text{NCl}_3$
 (B) Strength of backbonding: $\text{BCl}_3 > \text{OCl}_2 > \text{NCl}_3$
 (C) Bond order: $\text{NCl}_3 > \text{OCl}_2 > \text{BCl}_3$
 (D) Dipole moment: $\text{NCl}_3 > \text{BCl}_3 > \text{OCl}_2$
- $\text{R}-\text{C}\equiv\text{C}-\text{R}' \xrightarrow{\text{LiAlH}_4} \text{RCH}=\text{CHR}'$
 In the above reaction, R and R' should be
 (A) $\text{R} = \text{CHO}$ and $\text{R}' = \text{CH}_3$
 (B) $\text{R} = \text{CH}_3$ and $\text{R}' = \text{CH}_2\text{OH}$
 (C) $\text{R} = \text{C}_2\text{H}_5$ and $\text{R}' = \text{CO}$
 (D) $\text{R} = \text{CH}_3$ and $\text{R}' = \text{COOH}$
- The S – O bond order of SOF_2 is greater than that of SOCl_2 . This is due to
 (A) polarity of O – F bond is greater than that of O – Cl bond
 (B) the $\text{O} \rightarrow \text{S}$ back bond in SOF_2 is stronger than that in SOCl_2
 (C) the $\text{S} \rightarrow \text{O}$ back bond in SOF_2 is stronger than that in SOCl_2
 (D) the $\angle \text{FSF}$ is greater than $\angle \text{ClSCl}$
- Which alcohol can't be formed by acidic hydration of any alkene?
 (A) 
 (B) 
 (C) 
 (D) 

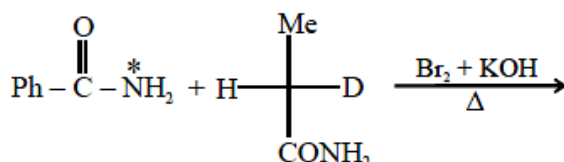
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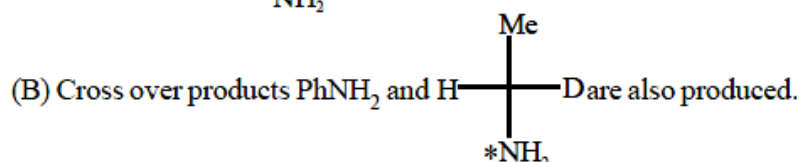
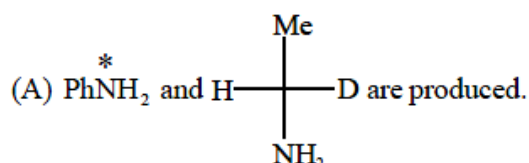
PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

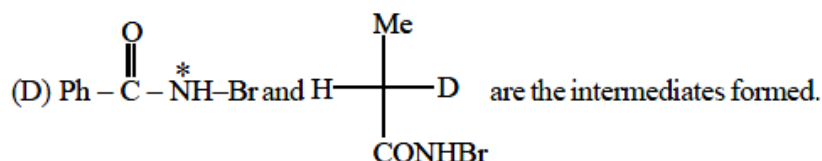
5.



Which of the following statement(s) is/are correct regarding the given reaction?

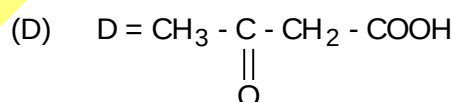
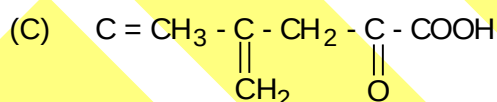
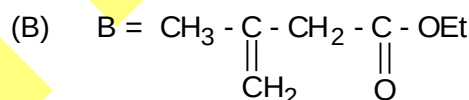
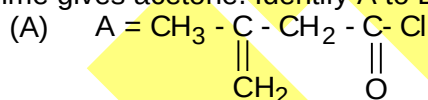


(C) Migration of $-\text{R}$ (alkyl group or aryl) of $\text{R}-\text{CONH}_2$ takes place with retention of configuration.



6.

Compound A ($\text{C}_5\text{H}_7\text{OCl}$) reacts rapidly with ethanol and catalytic amount of acid to form a pleasant smelling substance B ($\text{C}_7\text{H}_{12}\text{O}_2$). A also reacts with H_2O to form C with neutralization equivalent of 100. A, B and C all react with Br_2 water. Acid C, which can also be obtained by acidic hydrolysis of B is oxidized to new acid D ($\text{C}_4\text{H}_6\text{O}_3$) and CO_2 . D on heating with soda lime gives acetone. Identify A to D.



7.

Choose the correct statement

(A) d_{yz} orbital lies in the xz plane

(B) p_z orbital lies along the x axis

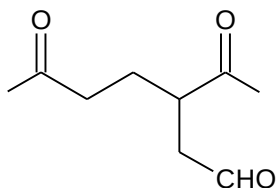
(C) lobes of $d_{x^2-y^2}$ orbital are at 90° with the z axis

(D) lobes of d_{xy} orbital are at 90° with the z axis

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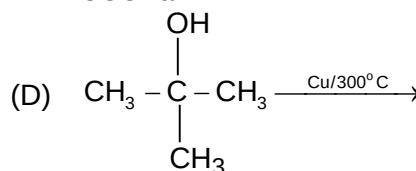
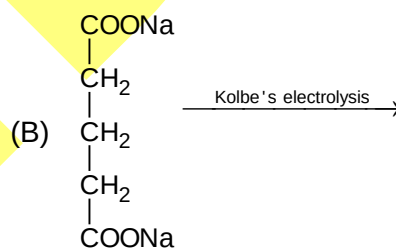
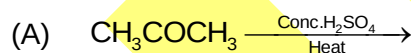
8. The spectra of which complex(s) is/are spin forbidden as well as Laporte forbidden?
 (A) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
 (C) $[\text{Co}(\text{CN})_6]^{3-}$ (D) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
9. A hydrocarbon X ($\text{C}_{10}\text{H}_{16}$) upon catalytic hydrogenation gives 4-methyl-1-isopropyl cyclohexane. Also (X) upon ozonolysis followed by hydrolysis in the presence of Zn gives CH_2O and



The correct statement(s) concerning (X) is/are

- (A) Structure of (X) is (B) (X) has two chiral carbon.
 (C) (X) has one chiral carbon. (D) With excess of HCl, (X) gives racemic dichloride.

10. Which of the following reactant(s) produce(s) racemic mixtures when treated with HBr?
 (A) Cis-2-butene (B) Trans-2-butene
 (C) 1-butene (D) Isobutene
11. Which of the following increases by increasing the temperature of a chemical reaction?
 (A) Rate constant (B) Half-time
 (C) Rate of reaction (D) Temperature coefficient
12. Which reaction produces cyclic products?

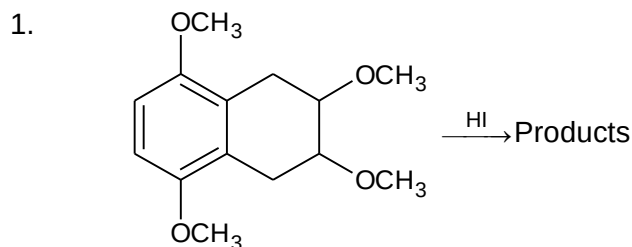


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PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.



If x moles of HI can be completely consumed by one mole of the above compound, what is the value of x ?

2. $\text{FeCl}_3 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 If ' x ' moles of KSCN will be completely consumed by one mole of product in above reaction, to produce blood red colouration, what is the value of $x/3$?
3. If the pH of 0.1 M aqueous solution of NH_4NO_3 is x , what is the value of x ? [K_b of $\text{NH}_4\text{OH} = 10^{-5}$]

PART – B (Maximum Marks: 15)

(Numerical Type)

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

4. 2.5 mole of KCl is added to one Kg of water at 1 atm. At what temperature in kelvin unit will the water boil if the degree of dissociation of KCl is 0.8? [K_b of $\text{H}_2\text{O} = 0.52 \text{ K kg mol}^{-1}$]
5. 14 g of a metal oxide of formula MO is added to a container containing 500 mL of 1.25 M HCl solution. After complete reaction, the reaction mixture is treated with 250 mL of 0.5 M NaOH to neutralize the excess acid. If the atomic mass of M is expressed as $(2x + 15)$. What is x ?
6. The successive ionization energies of a s-block element are 8.6, 12.9, 1630.2, 2406.2 eV, etc
 If atomic mass of the elements is 39.8 g mol^{-1} , what will be the molar mass of its normal oxide?

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7. The molecular mass of a polyhydric aliphatic alcohol, increases by 252 unit if it reacts with $\text{CH}_3\text{COCl}/(\text{CH}_3\text{CO})_2\text{O}$. if the number of OH groups present in the alcohol is x , what is the value of $\frac{x}{4}$?

8. $\text{A}_2(\text{g}) \longrightarrow \text{A}(\text{g}) + \text{A}(\text{g})$

The heat of reaction of above decomposition of a diatomic molecule $\text{A}_2(\text{g})$, into monoatomic gases $\text{A}(\text{g})$ is 40.5 kJ mol^{-1} at 320 K. What will be the enthalpy change of the reaction at 330 K in kJ mol^{-1} unit?

$$[C_P \text{ of monoatomic gases} = \frac{5R}{2}]$$

$$[C_P \text{ of diatomic gases} = \frac{7R}{2}]$$

$$[R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}]$$

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SECTION – III : MATHEMATICS

PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. Let $a, b, p, q \in \mathbb{Q}$ and suppose that $f(x) = x^2 + ax + b = 0$ and $g(x) = x^3 + px + q = 0$ have a common irrational root, then
 (A) $f(x)$ divides $g(x)$ (B) $g(x) \equiv xf(x)$
 (C) $g(x) \equiv (x - b - q)f(x)$ (D) none of these
2. If $x = (7 + 4\sqrt{3})^{2n} = [x] + f$, then $x(1-f)$ is equal to
 (A) 1 (B) 2
 (C) 3 (D) 4
3. Consider the points $P = (-\sin(\beta - \alpha), -\cos\beta)$, $Q = (\cos(\beta - \alpha), \sin\beta)$ and $R = (\cos(\beta - \alpha + \theta), \sin(\beta - \theta))$, where $0 < \alpha, \beta < \frac{\pi}{4}$ then
 (A) P lies on the line segment RQ (B) Q lies on the line segment PR
 (C) R lies on the line segment QP (D) P, Q, R are non – collinear
4. For $a > b > c > 0$, the distance between $(1, 1)$ and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then
 (A) $a + b - c > 0$ (B) $a - b + c < 0$
 (C) $a - b + c > 0$ (D) $a + b - c < 0$

PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. The number of triangles that can be formed with the sides of lengths a, b and c where a, b, c are integers such that $a \leq b \leq c$ is
 (A) $\frac{1}{4}(c+1)^2$ when c is odd (B) $\frac{1}{2}c(c+1)$ when c is odd
 (C) $\frac{1}{4}c(c+2)$ when c is even (D) $\frac{1}{4}c^2$ when c is even

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6. If $x \in \mathbb{R}$, and $S = 1 - C_1 \frac{1+x}{1+nx} + C_2 \frac{1+2x}{(1+nx)^2} - C_3 \frac{1+3x}{(1+nx)^3} + \dots$ upto $(n+1)$ terms then S
- (A) equals x^2 (B) equals 1
(C) equals 0 (D) is independent of x
7. A and B are two points on the x – axis and y – axis respectively. Two circles are drawn passing through the origin and having centre at A and B.
- (A) Equation of the common chord is $ax - by = 0$
(B) mid – point of the common chord is $\left(\frac{ab^2}{a^2 + b^2}, \frac{a^2b}{a^2 + b^2} \right)$
(C) AB bisect the common chord.
(D) AB is perpendicular to the common chord.
8. The equation $\frac{b \cos x}{2 \cos 2x - 1} = \frac{b + \sin x}{(\cos^2 x - 3 \sin^2 x) \tan x}$ posses a solution if
- (A) $b < \frac{1}{2}$ (B) $0 < b < \frac{1}{2}$ (C) $b < 0$ (D) $b = \frac{1}{2}$
9. Normals are drawn from the point P (15, 12) to the parabola $y^2 = 4x$. If the feet of the normal form a ΔABC , then which of the following is/are true.
- (A) Centroid of ΔABC is $\left(\frac{26}{3}, 0 \right)$
(B) Centroid of ΔABC is $\left(\frac{22}{3}, 0 \right)$
(C) Area of triangle formed by tangents to the parabola at A, B and C is 35 sq. units
(D) Area of triangle formed by tangents to the parabola at A, B and C is 140 sq. units
10. Let z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$, then which of the following is/are true?
- (A) maximum value of $|z_1 + z_2|$ is 3 (B) minimum value of $|2z_1 - z_2|$ is 0
(C) maximum value of $|2z_1 + z_2|$ is 4 (D) minimum value of $|2z_1 - 3z_2|$ is 5
11. If pair of tangents are drawn from centre C of the circle $x^2 + y^2 - 5x = 0$ to the nearer branch of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ to touch it at point A and B, then
- (A) product of the slopes of lines CA and CB is $-\frac{64}{75}$
(B) product of the slopes of lines CA and CB is $-\frac{32}{25}$
(C) circumcentre of ΔABC lies inside the ΔABC .
(D) circumcentre of ΔABC lies outside the ΔABC .

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12. If $a+b+c=4$ and $a^2+b^2+c^2+3(ab+bc+ac)=21$ where $a,b,c \in \mathbb{R}$, then which of the following is/are true
- (A) minimum value of $(a+b)(b+c)(a+c)$ is 18
- (B) minimum value of $(a+b)(b+c)(a+c)$ is 24
- (C) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{490}{27}$
- (D) maximum value of $(a+b)(b+c)(a+c)$ is $\frac{471}{27}$

PART – B (Maximum Marks: 9)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

1. Consider the number $N=10!$. If the number of positive divisors of N is λ , then $\frac{\lambda}{30}$ equals
2. Let L_1, L_2 and L_3 be the length of tangents drawn from a point P to the circles $x^2+y^2=4$, $x^2+y^2-4x=0$ and $x^2+y^2-4y=0$ respectively. If $L_1^4=L_2^2L_3^2+16$, then locus of P are the curves C_1 (a straight line) and C_2 (a circle). Let the circumcentre of the triangle formed by C_1 and other two lines which are inclined at an angle of 45° with C_1 and is tangent to C_2 is (α, β) , then $(\alpha+2\beta)$ equals
3. Let $S_n = \sum_{r=1}^n r \cdot 2^{\frac{r}{2}(1+(-1)^r)} \cdot 3^{\frac{r}{2}(1-(-1)^r)}$. If $S_{20} = \alpha \cdot 3^{21} + \beta \cdot 2^{22} + \frac{391}{288}$, then the value of $\left\lceil \frac{32\alpha - 9\beta}{4} \right\rceil$ is (where $[.]$ denotes the greatest integer function)

PART – B (Maximum Marks: 15)
(Numerical Type)

This section contains **Five (05)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

4. If $[16 \cos x + 12 \sin x] = 4k + 10$, then the maximum value of k is _____

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5. The length of the semi – major axis of an ellipse is 8 and the eccentricity is $\frac{1}{2}$. If Δ denotes the area of the rectangle formed by joining the vertices of the latera recta of the ellipse then the value of $\frac{\Delta}{120}$ is equal to
6. The centres of two circles C_1 and C_2 each of unit radius are at a distance of 6 units from each other. Let P be the mid point of the line segment joining the centres of C_1 and C_2 and C be a circle touching C_1 and C_2 externally. If a common tangent to C_1 and C passing through P is also a common tangents to C_2 and C_1 , then the one fifth of the radius of the circle C is
7. Value of x if x satisfies the equation $\log_{1-x}(3) - \log_{1-x}(2) = \frac{1}{2}$ is
8. The sum of all the integral roots of $(\log_5 x)^2 + \log_{5x}\left(\frac{5}{x}\right) = 1$ is _____
-

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QP Code: 100187
Answers
SECTION – I : PHYSICS
PART – A

- | | | | |
|---------------|----------------|-----------------|---------------|
| 1. B | 2. D | 3. A | 4. C |
| 5. ABD | 6. AD | 7. AC | 8. ACD |
| 9. AC | 10. ABC | 11. ABCD | 12. AB |

PART – B

- | | | | |
|-------------|----------------|----------------|----------------|
| 1. 2 | 2. 5 | 3. 6 | 4. 5.25 |
| 5. 5 | 6. 1.25 | 7. 1.73 | 8. 3 |

SECTION – II : CHEMISTRY

PART – A

- | | | | |
|---------------|----------------|----------------|----------------|
| 1. B | 2. B | 3. B | 4. C |
| 5. ACD | 6. ABD | 7. CD | 8. AB |
| 9. AC | 10. ABC | 11. ACD | 12. ABC |

PART – B

- | | | |
|---|-------------------------------------|---------------|
| 1. 7 | 2. 4 | 3. 5 |
| 4. 375.34 (Range 375.2 to 375.4) | 5. 12.5 (Range 12.4 to 12.6) | 7. 1.5 |
| 6. 55.8 (Range 55.7 to 55.9) | | |
| 8. 40.62 (Range 40.6 – 40.7) | | |

SECTION – III : MATHEMATICS

PART – A

- | | | | |
|--------------|----------------|----------------|---------------|
| 1. A | 2. A | 3. D | 4. A |
| 5. AC | 6. CD | 7. ABCD | 8. ABC |
| 9. AC | 10. ABC | 11. AC | 12. AC |

PART – B

- | | | | |
|---------------|---------------|-----------------|----------------|
| 1. 9 | 2. 0 | 3. 4 | 4. 2.50 |
| 5. 0.8 | 6. 1.6 | 7. -1.25 | 8. 6 |

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Answers & Solutions

SECTION – I : PHYSICS

PART – A

1. **B**

Sol. From range:

$$\frac{2u_x u_y}{g} = \frac{6\sqrt{3}}{5} \quad u_x u_y = 6\sqrt{3} \quad \dots(1)$$

$$\text{Resultant of 2 and } u_x \Rightarrow \sqrt{4 + u_x^2}$$

Whose resultant with u_y makes an angle 37° .

$$\text{So, } \frac{u_y^2}{4 + u_x^2} = \frac{9}{16} \quad \dots(2)$$

Solving (1) + (2), we will get $u_x = 2\sqrt{3}$, $u_y = 3$ & $u_z = 2 \text{ m/s}$

$$\text{So net } \Rightarrow \sqrt{u_x^2 + u_y^2 + u_z^2} = 5 \text{ m/s.}$$

2. **D**Sol. $\phi_B = 0$

$$\because \vec{B} \perp \vec{S} \quad ; \quad \therefore \frac{d\phi_B}{dt} = 0$$

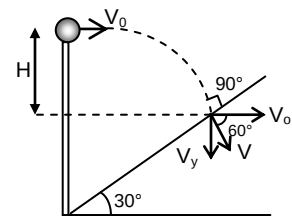
3. **A**

Sol.

$$v_y = \sqrt{2gH}$$

$$\frac{v_y}{v_x} = \frac{\sqrt{2gH}}{v_0} = \tan 60^\circ = \sqrt{3}$$

$$v_0 = \sqrt{\frac{2gH}{3}}$$

4. **C**

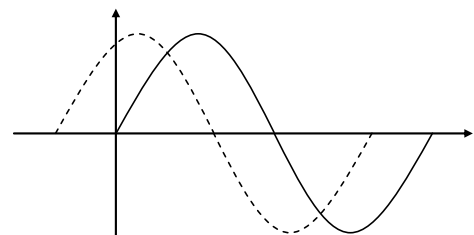
$$\text{Sol. } i = \frac{2I_0}{T} t \quad [0 < t < T/2]$$

$$i_{\text{rms}} = \frac{\frac{2I_0}{T} \sqrt{\int_0^{T/2} t^2 dt}}{\sqrt{T/2}} = \frac{I_0}{\sqrt{3}}$$

5. **ABD**

Sol.

Results can be obtained by looking at shape of string after a short time (shown dotted). Also, each particle of string executes SHM about mean position which is on x-axis.

6. **AD**

Sol.

Use ampere's law to find magnetic field due to solid cylinder current $= \mu_0 Jd/2$

Due to cylinder on its own axis is 0

Applying superposition principle we can get the required answers.

7. **AC**

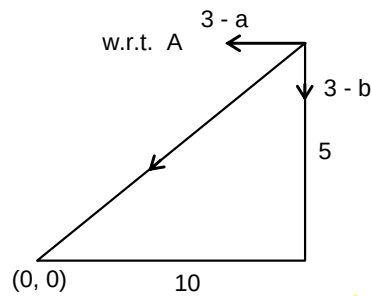
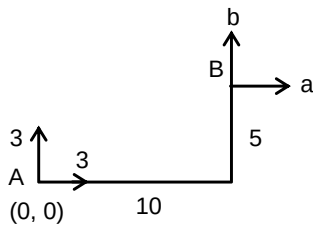
Sol.

Force on branch BC is coming out of the plane of paper and force on AD is going into the plane of paper and net force on the loop is zero.

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8. **ACD**Sol. $\vec{F}_m = q(\vec{v} \times \vec{B})$ and apply Newton's second law of motion.9. **AC**

Sol.

In vertical direction $\left(s = \frac{d}{t}\right)$

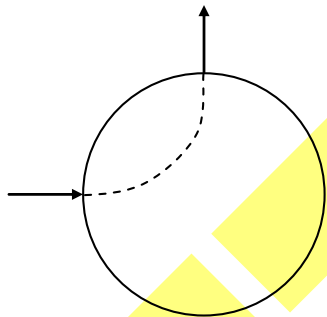
$$3 - b = \frac{5}{2} \rightarrow b = \frac{1}{2}$$

In horizontal direction

$$3 - a = \frac{10}{2} \rightarrow a = -2.$$

10. **ABC**

Sol.



So, its radius is also R.

$$\text{So, } R = \frac{mv}{qB} \rightarrow B = \frac{mv}{qR}$$

$$T = \frac{2\pi M}{4qB} = \frac{\pi m}{2qB}$$

So, A, B & C are correct.

11. **ABCD**

Sol.

Let elongation in spring A, B and C be x_1 , x_2 and x_3 respectively.
Considering spring forces and constraint relations

$$x_2 = 4x_3 \quad \dots(i)$$

$$x_2 = 2x_1 \quad \dots(ii)$$

$$\text{and } x_1 + 2x_2 + x_3 = x \quad \dots(iii)$$

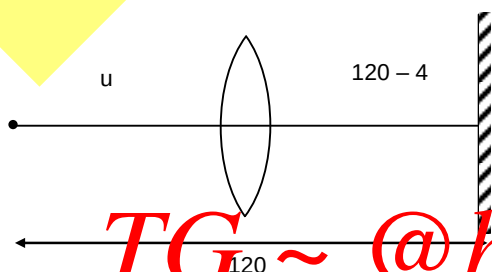
$$\Rightarrow x_1 = \left(\frac{2}{11}\right)x ; x_2 = \left(\frac{4}{11}\right)x ; x_3 = \left(\frac{1}{11}\right)x$$

$$\text{Also, } F = 2K\left(\frac{x}{11}\right)$$

$$\Rightarrow T = 2\pi\sqrt{\frac{11m}{2k}}$$

12. **AB**

Sol.



For two different positions of lens

$$1 = \frac{u}{120-u} ; 9 = \frac{120-u}{u}$$

$$\frac{1}{9} = \frac{u^2}{(120-u)^2} \rightarrow \frac{1}{3} = \frac{u}{120-u}$$

$$u = 30 ; v = 90$$

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$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{90} + \frac{1}{30}$$

$$P = 22.5$$

PART – B

1. 2

Sol. Let elongation of spring be x_0 in equilibrium. Then,

$$2T + mg \sin \theta = 2kx_0 \quad \dots(i)$$

$$\text{and } T = mg \quad \dots(ii)$$

Let Block B is displaced by x down the inclination

F.B.D. of B

$$-ma_B = 2k(x_0 + 2x) - 2T' - mg \sin \theta \quad \dots(iii)$$

F.B.D. of A

$$mg - T' = ma_A$$

$$\text{Also, } a_A = 2a_B$$

$$T' = mg - 2ma_B$$

$$-ma_B = 2kx_0 + 4kx - 2mg + 4ma_B - mg \sin \theta$$

$$-ma_B = 4kx + 4ma_B$$

$$a_B = -\frac{4k}{5m}x$$

$$\therefore T = 2\pi\sqrt{\frac{5m}{4k}}$$

$$T = 6.28 \text{ s.}$$

2. 5

Sol. Conservation of mechanical energy

$$mgH = mgh + \frac{1}{2}mv^2$$

$$0.45 mg = 3mg \frac{R}{2} + \frac{1}{2}mv^2$$

$$9 - 30R = v^2 \quad \dots(i)$$

$$AC = \frac{R\sqrt{3}}{2}$$

For reaching at B let particle takes t sec.

$$\frac{v}{2} \times t = \frac{R\sqrt{3}}{2} \quad \dots(ii)$$

$$\frac{v\sqrt{3}}{2}t - \frac{1}{2}g \times t^2 = \frac{R}{2} \quad \dots(iii)$$

On solving, $v^2 = 15R$ by eq. (i)

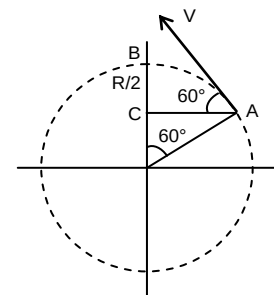
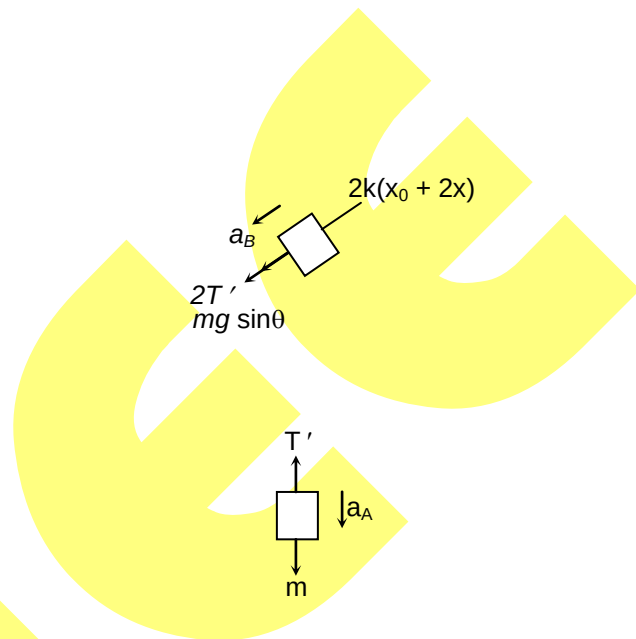
$$9 - 30R = 15R \Rightarrow R = \frac{9}{45} \times 100 = 20 \text{ cm}$$

 $= 1/5 \text{ meter.}$

3. 6

$$\text{Sol. } dF = \left[\eta (2\pi r dr) \frac{\omega r}{h} \right] 2$$

$$dP = dF.$$

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$$= \frac{4\eta\pi}{h}(\omega r)^2 r dr$$

$$\int_0^P dP = \frac{4\eta\pi\omega^2}{h} \int_0^R r^3 dr = \frac{\eta\pi\omega^2 R^4}{h}$$

$$x + y = 6$$

4. **5.25**

Sol. K.E. = $\frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2} \frac{m(2r)^2}{12} \omega^2 + \frac{1}{2}mr^2\omega^2 = 5.25.$

5. **5**

Sol. At stable equilibrium, U is minimum.

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} > 0$$

$$= \frac{1}{dx} \left(\frac{x^3}{3} - \frac{ax^2}{2} + 20x \right) = 0.$$

$$\Rightarrow x^2 - 9x + 20 = 0. \Rightarrow (x-5)(x-4) = 0.$$

$x = 5$ and $x = 4$ are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2} > 0$. i.e. at $x = 5$.

6. **1.25**

Sol. For pulse $\frac{dx}{dt} = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{3-10kt}{\mu}}$

$$\Rightarrow \int_0^L dx = \int_0^{t_0} \sqrt{\frac{3-10kt}{\mu}} dt$$

And $3 - 10kt_0 = 0$

7. **1.73**

Sol. $\tau = NIAB = C\theta$

$$\theta = \frac{NABQ}{C}$$

So, $K = \sqrt{3}$

$K = 1.73$

8. **3**

Sol. $2000 \times 15 \times 0.5 + m \times 80 = 2500 \times 25 \times 1$

$m = 500 \text{ gm}$

Water = $2.5 \text{ kg} + 0.5 \text{ kg} = 3 \text{ kg}.$

TG ~ @bohring_bot

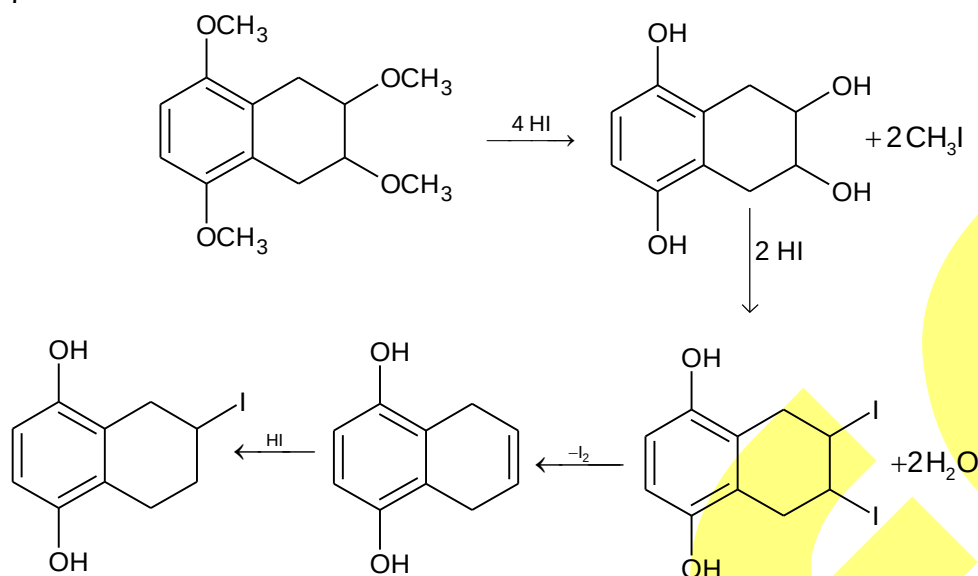
SECTION – II : CHEMISTRY

PART – A

1. B
Sol. Back bonding in BCl_3 is easier as it is electron deficient compound. Between OCl_2 and NCl_3 the electronegative difference between atoms in OCl_2 is greater than between atoms in NCl_3 .
 $\therefore \text{OCl}_2$ forms stronger back bond than NCl_3 .
2. B
Sol. R or R' can't be CHO, CO, COOH as they are reduced by LiAlH_4 .
3. B
Sol. The $\text{O} \rightarrow \text{S}$ back bond in SOF_2 is stronger than $\text{O} \rightarrow \text{S}$ in SOCl_2 . Since F is more electronegative than Cl, the S atom in SOF_2 strongly attract the lone pair of oxygen to form $p\pi - d\pi$ back bond.
4. C
Sol. No alkene can produce the alcohol in (C) because loss of H_2O can't form any alkene.
5. **ACD**
Sol. This is Hoffmann's bromamide reaction.
6. **ABD**
Sol. The alkene part of the carboxylic acid derivatives reacts with $\text{Br}_2/\text{H}_2\text{O}$.
7. **CD**
Sol. $d_{x^2-y^2}$ lie along X- and Y- axis, which are perpendicular to Z-axis.
8. AB
Sol. Due to $t_{2g}^3 e_g^2$ crystal field electronic configuration of Mn^{2+} and Fe^{3+} ions
9. AC
Sol. Cleavage of $\text{C} = \text{C}$ bond takes place during ozonolysis.
10. ABC
Sol. Racemic mixture is formed from compounds containing chiral carbon atom.
11. ACD
Sol. $t_{1/2}$ only decreases by increasing temperature.
12. ABC
Sol. In A, B and C the products are mesitylene, cyclohexane and benzene respectively.

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PART – B

1. 7
Sol.

2. 4



One mole Fe^{3+} consumes three moles of SCN^- ion. Four moles of Fe^{3+} will consume $4 \times 3 = 12$ moles $\text{Fe}(\text{SCN})_3$.

$$\therefore x = 4$$

3. 5

Sol. $\text{pH} = \frac{1}{2}[\text{p}K_w - \text{p}K_b - \log C] = \frac{1}{2}[14 - 5 - \log 10^{-1}] = \frac{1}{2} \times 10 = 5$

4. 375.34 (Range 375.2 to 375.4)

Sol. $\Delta T_b = iK_b m = 1.8 \times 0.52 \times 2.5 = 2.34$

$$\therefore T_b - T_b^0 = 2.34$$

$$\text{or } T_b - 100^\circ\text{C} = 2.34$$

$$\therefore T_b = 102.34^\circ\text{C} = 375.34 \text{ K}$$

5. 12.5 (Range 12.4 to 12.6)

Sol. M_{eq} of HCl added = $500 \times 1.25 = 625$

$$M_{\text{eq}}$$
 of NaOH used = $250 \times 0.5 = 125$

$$M_{\text{eq}}$$
 of HCl reacted with MO = $625 - 125 = 500$

$$500 = M_{\text{eq}}$$
 of MO = $\frac{W}{E} \times 1000 = \frac{14}{E} \times 1000 \Rightarrow E = 28, \text{molar mass} = 28 \times 2 = 56$

$$\therefore \text{At weight of MO} = 56 - 16 = 40$$

$$\therefore 2x + 15 = 40$$

$$\therefore x = 12.5$$

6. 55.8 (Range 55.7 to 55.9)

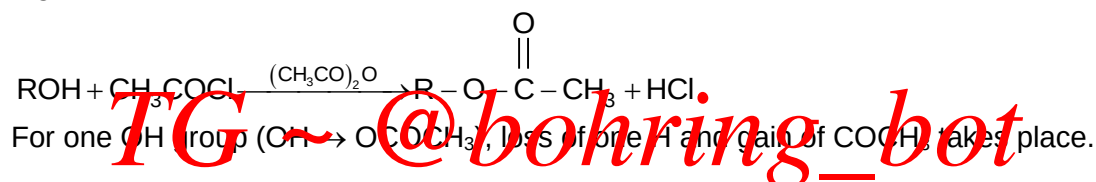
Sol. The atom contains two unpaired electrons.

$$\therefore \text{Formula of oxide} = \text{MO}$$

$$\text{Molar mass} = 39.8 + 16 = 55.8$$

7. 1.5

Sol.



$\therefore$ For one OH group, the mass increases by 42 units.

$$\therefore \text{Number of OH group} = \frac{252}{42} = 6 = x$$

$$\therefore \frac{x}{4} = \frac{6}{4} = 1.5$$

8. 40.62(Range 40.6 – 40.7)

Sol. $\Delta H_2 = \Delta H_1 + n\Delta C_p dT$

$$= 40.5 + n\Delta C_p \int_{320}^{330} dT$$

$$= 40.5 + 0.1(1.5 R) (330-320)$$

$$= 40.5 + 1.5R \times 10 = 40.5 + 15R$$

$$= 40.5 + 15 \times 8.3 = 40.62$$

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SECTION – III : MATHEMATICS

PART – A

1. A

Sol. Let $\alpha \in \mathbb{R} - \mathbb{Q}$ be a common root of $f(x) = 0$ and $g(x) = 0$. Then $\alpha^2 = -a\alpha - b$. Substituting this in $\alpha^3 + p\alpha + q = 0$, we get $(a^2 - b + p)\alpha + ab + q = 0$

As α is irrational and $a, b, p, q \in \mathbb{Q}$, $p = b - a^2$, $q = -ab$. This gives, $g(x) = (x - a)f(x)$.

2. A

Sol. We have $7 - 4\sqrt{3} = \frac{1}{7 + 4\sqrt{3}}$

$$\therefore 0 < 7 - 4\sqrt{3} < 1 \Rightarrow 0 < (7 - 4\sqrt{3})^{2n} < 1$$

$$\text{Let } F = (7 - 4\sqrt{3})^{2n}.$$

$$\text{Then } x + F = (7 + 4\sqrt{3})^{2n} + (7 - 4\sqrt{3})^{2n}$$

$$= 2 \left[{}^{2n}C_0 7^{2n} + {}^{2n}C_2 7^{2n-2} (4\sqrt{3})^2 + {}^{2n}C_4 7^{2n-4} (4\sqrt{3})^4 + \dots + {}^{2n}C_{2n} (4\sqrt{3})^{2n} \right]$$

$= 2m$, where m is some positive integer.

$$\Rightarrow [x] + f + F = 2m \Rightarrow f + F = 2m - [x]$$

Since $0 \leq f < 1$ and $0 < F < 1$, we get $0 < f + F < 2$. Also, since $f + F$ is an integer, we

must have $f + F = 1$. Thus, $x(1 - f) = xF = (7 + 4\sqrt{3})^{2n} (7 - 4\sqrt{3})^{2n}$

$$= (49 - 48)^{2n} = 1^{2n} = 1.$$

3. D

Sol. Put $\beta - \alpha = \phi$ and consider the determinant $\Delta = \begin{vmatrix} -\sin \phi & -\cos \beta & 1 \\ \cos \phi & \sin \beta & 1 \\ \cos(\phi + \theta) & \sin(\beta - \theta) & 1 \end{vmatrix}$

Using $R_3 \rightarrow R_3 - \cos \theta R_2 - \sin \theta R_1$

$$\Delta = \begin{vmatrix} -\sin \phi & -\cos \beta & 1 \\ \cos \phi & \sin \beta & 1 \\ 0 & 0 & 1 - \cos \theta - \sin \theta \end{vmatrix}$$

$$= (1 - \cos \theta - \sin \theta) \cos(\phi + \beta)$$

$$= (1 - \cos \theta - \sin \theta) \cos(2\beta - \alpha)$$

$$= \left[1 - \sqrt{2} \sin \left(\theta + \frac{\pi}{4} \right) \right] \cos(2\beta - \alpha)$$

$$\text{As } 0 < \theta < \frac{\pi}{4} \Rightarrow \frac{\pi}{4} < \theta + \frac{\pi}{4} < \frac{\pi}{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}} < \sin \left(\theta + \frac{\pi}{4} \right) < 1 \text{ and } 0 < \alpha, \beta < \frac{\pi}{4} \Rightarrow \frac{-\pi}{4} < 2\beta - \alpha < \frac{\pi}{2}$$

$$\Rightarrow \cos(2\beta - \alpha) \neq 0$$

Thus $\Delta \neq 0$ and the points P, Q, R are non-collinear.

4. A

Sol. The lines $ax + by + c = 0$ and $bx + ay + c = 0$ intersect in $\left(\frac{-c}{a+b}, \frac{-c}{a+b} \right)$.

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Distance between $(1, 1)$ and $\left(\frac{-c}{a+b}, \frac{-c}{a+b}\right)$

$$= \sqrt{2} \left| 1 + \frac{c}{a+b} \right| = \sqrt{2} \frac{a+b+c}{a+b}.$$

As $\sqrt{2} \frac{a+b+c}{a+b} < 2\sqrt{2}$, we get $a+b+c < 2(a+b)$

$$\Rightarrow a+b-c > 0.$$

5. AC

Sol. Let $c = 2m+1$ where m is a positive integer. Since $c < a+b$ and $a \leq b$, we get $c < 2b \Rightarrow b > \frac{1}{2}c = m + \frac{1}{2}$. Thus, b can take values from $m+1$ to $2m+1$.

If $b = 2m+1$, a can be $1, 2, \dots, (2m+1)$. Thus, there are $(2m+1)$ values.

If $b = 2m$, a can be $2, 3, \dots, 2m$. In this case a can take $(2m-1)$ values. And so on. When $b = m+1$, a can take just one value viz. $(m+1)$. Thus, there are $(2m+1) + (2m-1) + \dots + 1 = (m+1)^2$.

6. CD

Sol. Putting $\frac{1}{1+nx} = y$, we can write $S = [C_0 - C_1 y + C_2 y^2 - \dots + (-1)^n C_n y^n]$

$$-xy [C_1 - 2C_2 y + 3C_3 y^2 - \dots + (-1)^{n-1} n C_n y^{n-1}]$$

$$= (1-y)^n + xy \frac{d}{dt} \{(1-t)^n\}_{t=y}$$

$$= (1-y)^n - nxy(1-y)^{n-1}$$

$$= (1-y)^{n-1} [1-y-nxy]$$

$$= (1-y)^{n-1} [1-(1+nx)y] = 0.$$

7. ABCD

Sol. Let $OA = a$ and $OB = b$ (figure), so that the equation of the line AB is $\frac{x}{a} + \frac{y}{b} = 1$. Equations of the circles passing through the origin and having centres at $A(a, 0)$ and $B(0, b)$ are, respectively.

$$x^2 - 2ax + y^2 = 0 \quad (1)$$

$$\text{and } x^2 - 2by + y^2 = 0 \quad (2)$$

so that the equation of the common chord OP is $ax - by = 0$ (3)

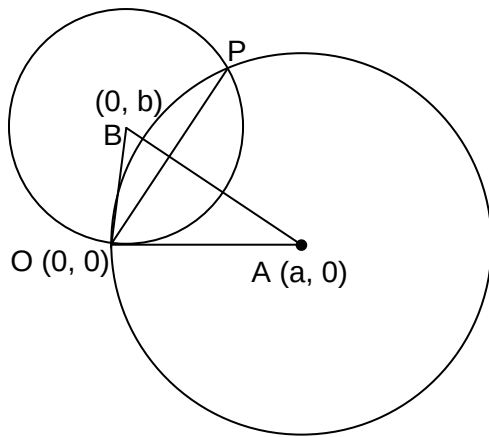
which is perpendicular to AB

This has one end at the origin O and the other end P is given by solving (2) and (3).

$$\left(\frac{b}{a}y\right)^2 - 2by + y^2 = 0$$

$$\Rightarrow y = \frac{2a^2b}{a^2 + b^2} \Rightarrow x = \frac{2ab^2}{a^2 + b^2}$$

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Therefore, the mid – point of the common chord OP is $\left(\frac{ab^2}{a^2+b^2}, \frac{a^2b}{a^2+b^2} \right)$ which lies on AB

8. ABC

Sol. The conditions for the existence of a solution are

$$1. 2\cos 2x - 1 \neq 0 \Rightarrow \cos 2x \neq \frac{1}{2} \Rightarrow 2x \neq \frac{\pi}{6}$$

$$2. \tan x \neq 0 \Rightarrow x \neq \pm \frac{\pi}{2}$$

$$3. \cos^2 x - 3\sin^2 x \neq 0 \Rightarrow \tan^2 x \neq \frac{1}{3} \Rightarrow x \neq \pm \frac{\pi}{6}$$

$$\begin{aligned} \text{As } 2\cos 2x - 1 &= 2(\cos^2 x - \sin^2 x) - 1 \\ &= 2(\cos^2 x - \sin^2 x) - (\cos^2 x + \sin^2 x) \\ &= \cos^2 x - 3\sin^2 x \end{aligned}$$

So the given equation can be written as $b\sin x = b + \sin x$

$$\Rightarrow \sin x = \frac{b}{b-1}$$

Which is possible if $-1 \leq \frac{b}{b-1} \leq 1$

$$\Rightarrow \frac{2b-1}{b-1} \geq 0 \text{ and } \frac{1}{b-1} \leq 0$$

$$\Rightarrow 2b-1 \leq 0 \Rightarrow b \leq \frac{1}{2}$$

But for $b = \frac{1}{2}$, $\sin x = -1$ which is not possible as $x \neq \frac{-\pi}{2}$.

Hence equation has a solution if $b < \frac{1}{2}$.

9. AC

Sol. $tx + y = 2t + t^3$ passes through $(13, 12)$ $t^3 - 13t - 12 = 0$

$$\Rightarrow t^2(t+1) - t(t+1) - 12(t+1) = 0$$

$$(t+1)(t-4)(t+3) = 0$$

$$A, B, C = (t^2, 2t)$$

$$A(1, -2); B(16, 8); C(9, -6)$$

$$\text{Centroid} = \left(\frac{26}{3}, 0 \right), \text{Area} = \frac{1}{2} \begin{vmatrix} 1 & -2 & 1 \\ 16 & 8 & 1 \\ 9 & -6 & 1 \end{vmatrix}$$

$$A = 70$$

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Area of triangle formed by tangents $= \frac{70}{2} = 35$.

10. ABC

Sol. $|z_1| = 1$ and $|z_2| = 2$

(A) $|z_1 + z_2| \leq |z_1| + |z_2| \leq 3$

(B) $|2z_1 - z_2| \geq |2|z_1| - |z_2|| \geq 0$

(C) $|2z_1 + z_2| \leq 2|z_1| + |z_2| \leq 4$

(D) $|2z_1 - 3z_2| \geq |2|z_1| - 3|z_2|| \geq 4$

11. AC

Sol. centre: $\left(\frac{5}{2}, 0\right)$; $y = mx \pm \sqrt{25m^2 - 16}$

$$\frac{5}{2}m = \pm \sqrt{25m^2 - 16} \Rightarrow \frac{25m^2}{4} = 25m^2 - 16$$

$\frac{75m^2}{4} = 16 \Rightarrow m^2 = \frac{64}{75} = 0$, product of slopes of CA . CB is $-\frac{64}{75}$ circumcentre of $\triangle ABC$ lies inside the $\triangle ABC$

12. AC

Sol. $(a+b) + (b+c) + (a+c) = 8$

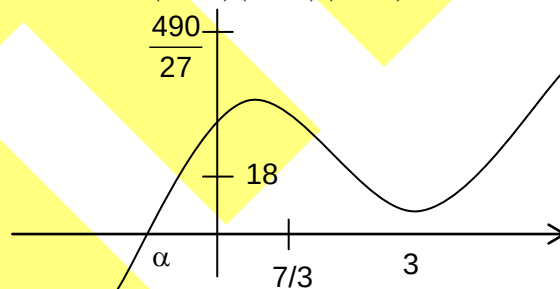
Let $a+b = \alpha, b+c = \beta, a+c = \gamma$

$$\Rightarrow \alpha + \beta + \gamma = 8$$

$$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = a^2 + b^2 + c^2 + 3(ab + bc + ac)$$

$$= 21$$

$\Rightarrow x^3 - 8x^2 + 21x - \lambda = 0$ where $\lambda = (a+b)(b+c)(a+c)$ these equations have 3 real roots.

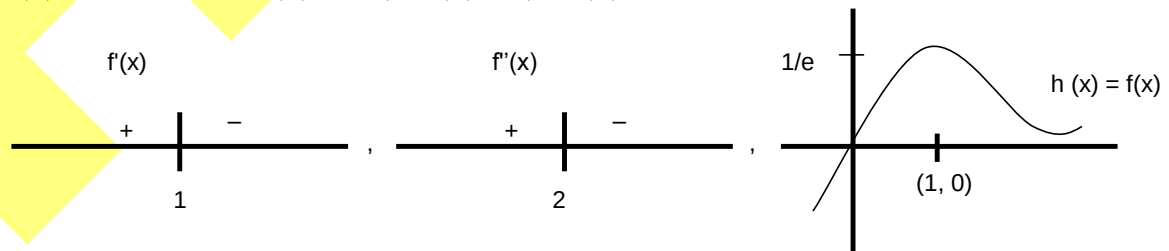


$$f(x) = xe^{-x}, f'(x) = -(x-1)e^{-x}, f''(x) = e^{-x}(x-2)$$

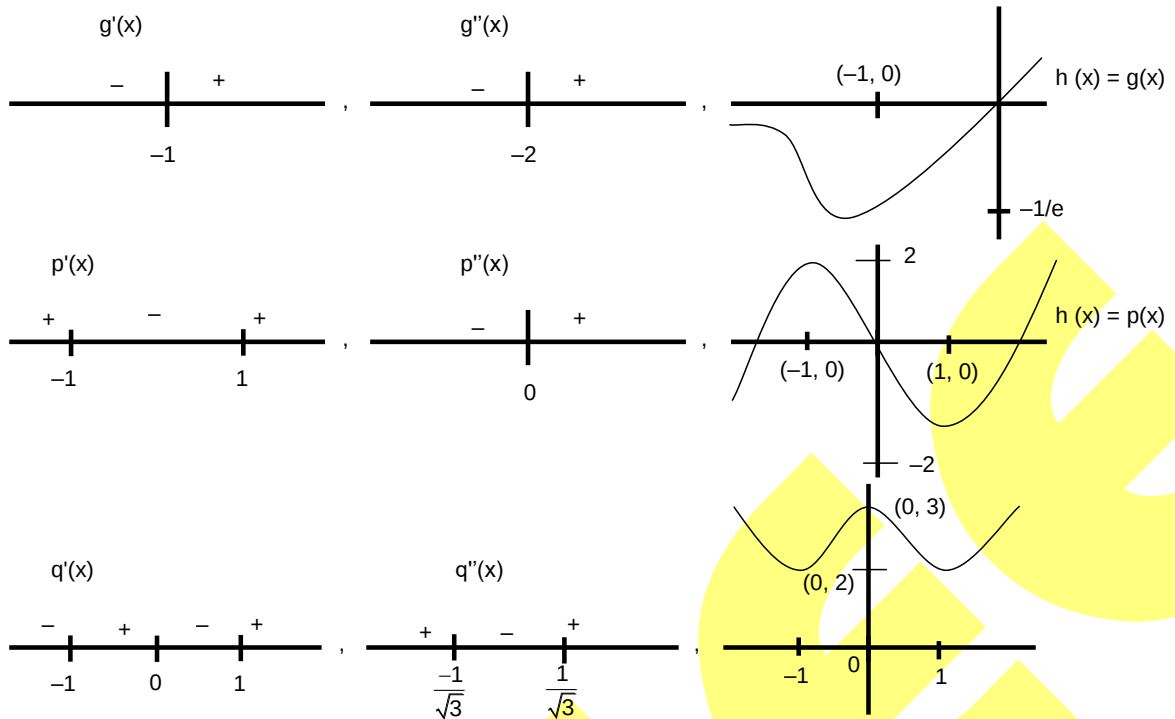
$$g(x) = xe^x, g'(x) = (x+1)e^x, g''(x) = (x+2)e^x$$

$$p(x) = x^3 - 3x, p'(x) = 3x^2 - 3, p''(x) = 6x$$

$$q(x) = x^4 - 2x^2 + 3, q'(x) = 4x(x-1)(x+1), q''(x) = 12x^2 - 4$$



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- (I), (i) (iv), (R)
 (II), (i) (iv), (R, S)
 (III), (ii) (iv), (P, Q, R)
 (IV), (ii) (iii), (P, S)

PART - B

1. 9

Sol. $10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$

Number of divisors $= 9 \times 5 \times 3 \times 2 = 270 = \lambda$

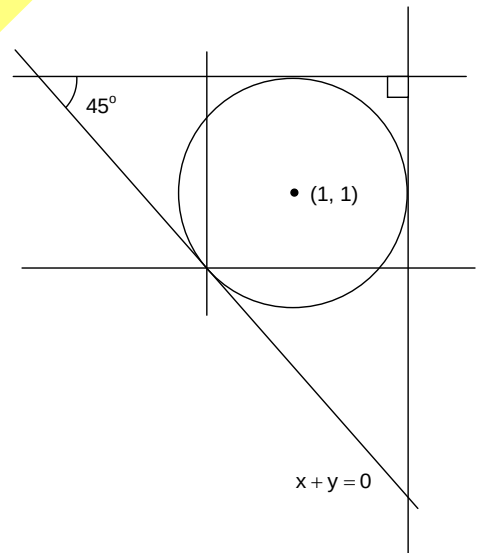
2. 0

Sol. $(h^2 + k^2 - 4)^2 = (h^2 + k^2 - 4h)(h^2 + k^2 - 4k) + 16$

$\Rightarrow (h + k)(h^2 + k^2 - 2h - 2k) = 0$ circle

$C_1: x^2 + y^2 - 2x - 2y = 0$ and

$C_2: x + y = 0$ circumcentre will be mid point of hypotenuse in right angle triangle $(\alpha, \beta) = (0, 0)$



3. 4

Sol. $S_n = 1.3 + 2.2^2 + 3.3^3 + 4.2^4 + \dots$ upto n - terms

$S_n = (1.3 + 3.3^3 + 5.3^5 + \dots) + (2.2^2 + 4.2^4 + 6.2^6 + \dots)$

$S_1 = 1.3 + 3.3^3 + 5.3^5 + \dots$ upto 10 - terms, $S_1 = \frac{15}{32}(5.3^{21} + 1)$

$S_2 = 2.2^2 + 4.2^4 + 6.2^6 + \dots$ upto 10 - terms, $S_2 = \frac{58.2^{22} + 8}{9}$

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$$S_{20} = S_1 + S_2 = \frac{15}{32}(5 \cdot 3^{21} + 1) + \left(\frac{58 \cdot 2^{22} + 8}{9} \right)$$

$$= \frac{75}{32} \cdot 3^{21} + \frac{58}{9} \cdot 2^{22} + \frac{391}{288}$$

4. 2.50

Sol. $16 \cos x + 12 \sin x = \sqrt{16^2 + 12^2} \cos(x - \alpha), \alpha = \tan^{-1}\left(\frac{3}{4}\right).$

5. 0.8

Sol. Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a^2 = 8^2$ and $b^2 = 8^2 \left(1 - \frac{1}{4}\right)$

Vertices of the latus recta are $\left(\pm ae, \pm \frac{b^2}{a}\right)$

So $\Delta = 2ae \times \frac{2b^2}{a} = 4b^2e$

$$= 4 \times 8^2 \times \frac{3}{4} \times \frac{1}{2} = 12 \times 8$$

$$\Rightarrow \frac{\Delta}{120} = 0.8$$

6. 1.6

Sol. Let A_1, A_2 and M be the centres of the circles C_1, C_2 and C respectively. Let the common tangent through P to C_1 and C touch C_1 at B_1 , C at B_2 and C_2 also at B_2 .

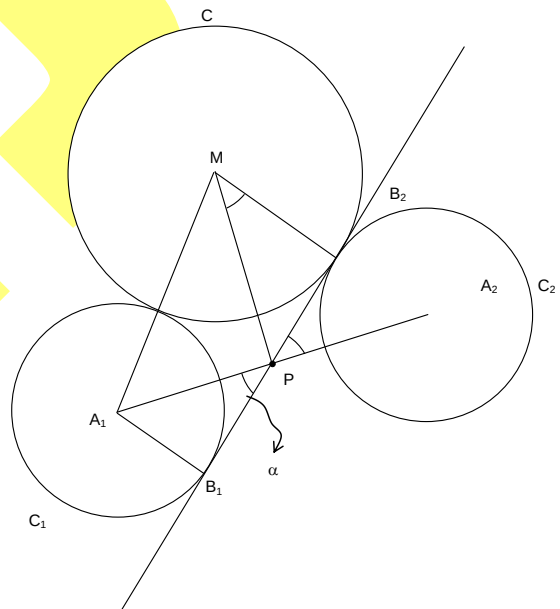
From right angled triangle A_1B_1P if

$$\angle A_1PB_1 = \alpha, \sin \alpha = \frac{A_1B_1}{A_1P} = \frac{1}{3}$$

$$\Rightarrow \cos \alpha = \frac{2\sqrt{2}}{3} \Rightarrow PB_1 = 2\sqrt{2} = PB_2$$

From triangle MPB_2 , $\tan \alpha = \frac{PB_2}{MB_2} = \frac{2\sqrt{2}}{r}$

$$\Rightarrow \frac{1}{2\sqrt{2}} = \frac{2\sqrt{2}}{r} \Rightarrow r = 8.$$



7. -1.25

Sol. $1-x > 0, 1-x \neq 1 \Rightarrow x < 1, x \neq 0$

Also, $\log_{1-x}\left(\frac{3}{2}\right) = \frac{1}{2} \Rightarrow \frac{3}{2} = (1-x)^{1/2}$

$$\Rightarrow \frac{9}{4} = 1-x \Rightarrow x = 1 - \frac{9}{4} = -\frac{5}{4}.$$

$$\Rightarrow -4x = 5$$

$$\Rightarrow x = -1.25$$

8. 6

Sol. Clearly $x > 0$ and $x \neq \frac{1}{5}$ **TC @bohring_bot**

$$\log_{5x} \left(\frac{5}{x} \right) = \frac{\log_5 5 - \log_5 x}{\log_5 5 + \log_5 x}$$

Putting $\log_5 x = t$, then equation (1) becomes

$$t^2 + \frac{1-t}{1+t} = 1 \Leftrightarrow t^3 + t^2 - 2t = 0$$

$$\Leftrightarrow t(t-1)(t+2) = 0 \Leftrightarrow t = 0, 1, -2$$

So integral roots of (1) are 1 and 5.

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FIITJEE RBT-5 for (JEE-Advanced)

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 4

QP CODE: 100188

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Part.

- (i) **PART-A (01-08)** contains (8) Multiple Choice Questions which have **One or More Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (09-12)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **-1 marks** for wrong options chosen.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

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SECTION – I : PHYSICS

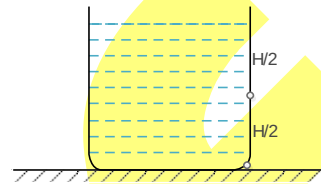
(PART – A)

(One or More Than One Options Correct Type)

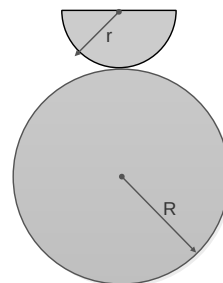
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. There is a cylindrical container of height H and base area A . It is filled with a fluid and two small holes of area a each are made at mid-height and at bottom of the container.

$$\frac{A}{a} \sqrt{\frac{H}{g}} = 600 \text{ s. Then}$$



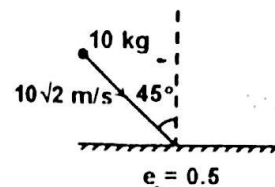
- (A) Time taken to half-empty the tank lies between 150 s and 160 s
 (B) Time taken to half-empty the tank lies between 155 s and 170 s
 (C) Time taken to empty the tank lies between 760 s and 775 s
 (D) Time taken to empty the tank lies between 755 s and 770 s
2. Choose the correct statement about X-rays
 (A) Probability of K_α radiation is more than K_β radiation.
 (B) A K_α photon is more energetic than a K_β photon.
 (C) A photon of cutoff wavelength is least energetic.
 (D) As we move down the periodic table, the energy of K_α X-ray photon goes on decreasing.
3. A solid hemisphere of radius $r = 20$ cm is placed on top of a fixed, very rough sphere of radius R in equilibrium such that curved surface of hemisphere is in contact with fixed sphere. There may not be any slipping between the surfaces. A very slight angular displacement is given to the hemisphere. Then
- (A) Hemisphere cannot oscillate if $\frac{r}{R} < 0.2$
 (B) Hemisphere cannot oscillate if $\frac{r}{R} > 0.6$
 (C) If $\frac{r}{R} = 0.2$ then time period of oscillation is 0.24π s
 (D) If $\frac{r}{R} = 0.5$ then time period of oscillation is π s



Space For Rough Work

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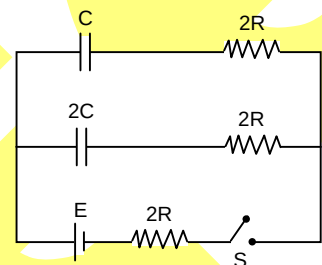
4. A ball of mass 10 Kg hits a smooth horizontal surface with a speed of $10\sqrt{2}$ m/s at an angle 45° as shown in figure. The co-efficient of restitution between the ball and the surface is 0.5 and the ball remains in contact with the surface for 0.1 s. $\vec{F}$ is the instantaneous force exerted by the surface on the ball at any time t during the collision.



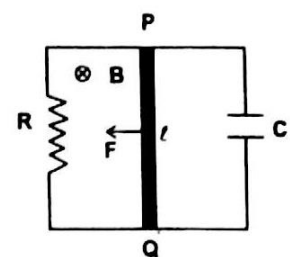
- (A) The speed with which ball rebounds is $\sqrt{125}$ m/s
 (B) During collision $|\vec{F}|$ may be less than or equal to 1800 N
 (C) During collision $|\vec{F}|$ may be less than or equal to 100 N
 (D) The average force exerted by ground during collision is 1600 N

5. In the circuit shown in figure, switch S is closed at time $t = 0$. Select the correct statement(s)

- (A) rate of increase of charge is same in both capacitors at $t = 0$
 (B) ratio of charge stored in C to that in 2C at any instant is 1:2
 (C) time constant of both the capacitors are equal
 (D) steady state charge in capacitors C and 2C are in ratio 1:2



6. A conducting rod PQ of length l and mass m is dragged with a constant force F along two parallel smooth rails separated by a distance l . There is a magnetic field B in the region as shown in the adjacent figure. Then choose the correct statement(s). [Also, $m = B^2 L^2 C$ and at $t = 0$, velocity = 0].



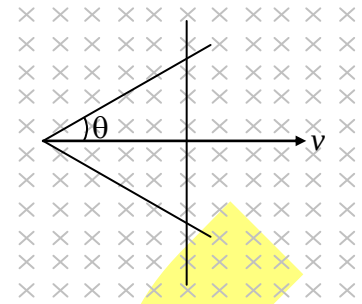
- (A) terminal velocity of rod, $V_T = \frac{2FR}{B^2 l^2}$
 (B) terminal velocity of rod, $V_T = \frac{FR}{B^2 l^2}$
 (C) Maximum charge on the capacitor, $q_{\max} = \frac{FCR}{BI}$
 (D) At $t = 2RC$, current through rod is $\frac{F}{Bl} \left(1 - \frac{1}{2e}\right)$

Space For Rough Work

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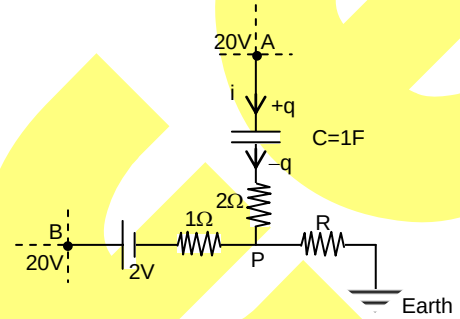
7. Two straight conducting rails form a right angle where their ends are joined. A conducting bar in contact with the rails start at the vertex at $t=0$ and moves with a constant velocity v along them as shown. A magnetic field B is directed into the page. The induced emf in the circuit at any time t is proportional to

- (A) t^0 (B) t
(C) v (D) v^2



8. The network shown in the figure is part of some bigger circuit. Charge on capacitor ($C = 1 \text{ F}$) at any time t is $q = 3(1 - e^{-t})$ in coulombs. If at $t = 0$ the potentials of point A and B is 20 V, then at $t = 0$ match the following.

- (A) current in branch AP is 3A
(B) current through R is 7A
(C) value of R is 2Ω
(D) value of R is 1Ω



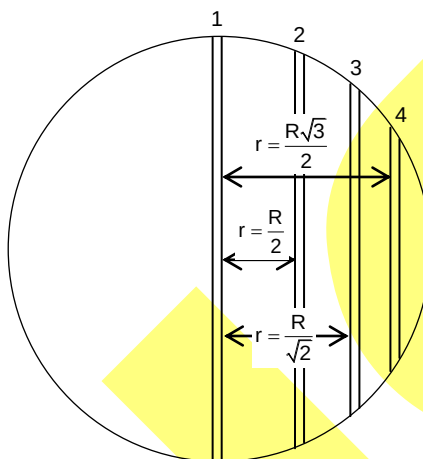
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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Consider a planet of radius R having density ρ . The four tunnels (1, 2, 3 and 4) are dug side at the distances from its centre are $r = 0$, $r = \frac{R}{2}$, $r = \frac{R}{\sqrt{2}}$ and $r = \frac{R\sqrt{3}}{2}$ respectively as shown. Four objects each of mass 1 kg are dropped inside the tunnel from the surface of planet at $t = 0$.



List-I gives the above four tunnels while List-II the magnitude of some quantity.

List – I		List – II	
(I)	Tunnel '1'	(P)	0
(II)	Tunnel '2'	(Q)	$\frac{2\pi G\rho R}{3}$
(III)	Tunnel '3'	(R)	$\frac{2\sqrt{2}\pi G\rho R}{3}$
(IV)	Tunnel '4'	(S)	$\frac{2\pi G\rho R}{\sqrt{3}}$
		(T)	$\frac{4\pi G\rho}{3}R$

9. The normal reaction between objects and mid of tunnels, the correct match is
 (A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
 (B) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P
 (C) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ S
 (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ R
10. The magnitude of maximum acceleration of the objects in the tunnels
 (A) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ Q
 (B) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ Q
 (C) I $\rightarrow$ T, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ R
 (D) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T

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List Match Set (11-12)

11. A ball is attached to a string and moves in vertical circle. There is no resistive force. Match the column-I to column-II.

List – I		List – II	
(I)	Minimum	(P)	Tension in the string when ball is at the lowest point.
(II)	Maximum	(Q)	Tension in the string when ball is at the highest point.
(III)	Towards the centre	(R)	The angle between tension and weight of ball when string becomes horizontal.
(IV)	Perpendicular	(S)	The direction of tension in the string
		(T)	Magnitude of tangential acceleration when string becomes horizontal.
		(U)	Magnitude of centripetal acceleration when string becomes horizontal.

(A) I → Q, II → P, III → U, IV → RS
(C) I → Q, II → PT, III → SU, IV → R

(B) I → T, II → R, III → SU, IV → Q
(D) I → Q, II → PT, III → S, IV → U

12. A small bob of mass $m = 0.01$ kg is hanging from the roof of a moving car in equilibrium making angle θ with vertical. If initial velocity of car was zero and acceleration of car is $a = 2$ m/s².

List – I		List – II	
(I)	Work done by tension in ground frame in first 10 second	(P)	20 m/s
(II)	Work done by tension in the frame of car in first 10 second	(Q)	2J
(III)	Velocity of bob in ground frame after 10sec	(R)	0J
(IV)	Work done by gravity	(S)	–ve
		(T)	+ve
		(U)	10 m/s

(A) I → QT, II → R, III → PT, IV → R
(C) I → Q, II → P, III → S, IV → T

(B) I → QR, II → S, III → PT, IV → Q
(D) I → U, II → P, III → S, IV → R

Space For Rough Work

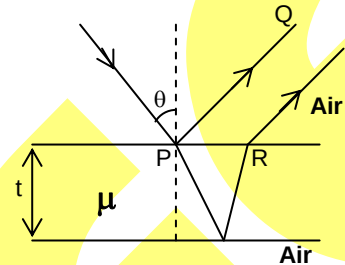
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(PART – B)**(Numerical Type)**

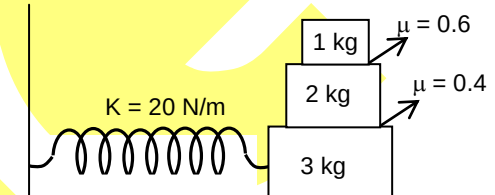
Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. Two spherical bodies A (of radius 6 cm) and B (of radius 36 cm) are at temperatures T_1 and T_2 respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500nm. Considering them to be black bodies, the ratio of the rate of total energy radiated by A and B will be

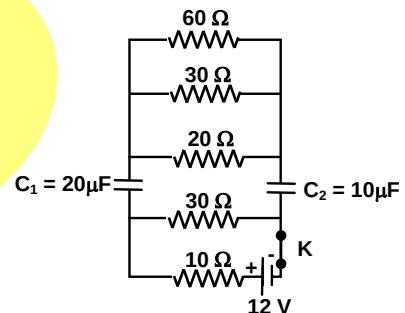
2. Rays PQ and RS (Wavelength λ) interfere constructively for small θ and corresponding minimum thickness of the slab is $\frac{n\lambda}{12\mu}$, then $n =$



3. What can be the maximum amplitude (in m) of the system so that there is no slipping between any of the blocks



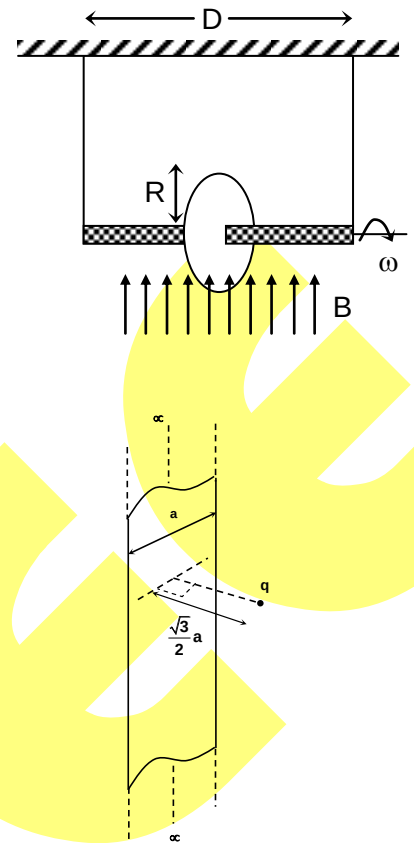
4. For the circuit arrangement shown in the figure, find the potential difference (in V) across C_2 in the steady state condition.



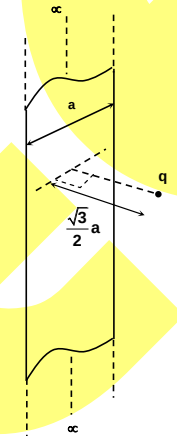
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5. The axle of a circular wheel of radius R is held horizontally by two identical strings of equal lengths separated by a distance D . The tension in each string is T_0 . The rim of the wheel carries a total charge $+Q$ distributed uniformly on it. The wheel is vertical and is kept in a uniform vertical magnetic field $\vec{B}$. It is now rotated at an angular speed ω . The string break at a tension of $3T_0/2$. If the maximum possible value of ω at which the wheel can be rotated without breaking a string is $\frac{4KT_0D}{5QR^2B}$. Then find the value of K .



6. There is an infinite long straight surface having width ' a '. A point charge q is placed at a perpendicular distance $\frac{\sqrt{3}}{2}a$ from the surface symmetrically as shown in the figure. If the flux linked with this infinitely long surface due to charge q is $\frac{q}{k\epsilon_0}$. Then find the value of k .



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SECTION – II : CHEMISTRY

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

- Which statements about corrosive sublimate(HgCl_2) are correct?
 (A) It is prepared by heating mercury in chlorine
 (B) It oxidizes stannous chloride
 (C) It is highly poisonous
 (D) It sublimes on heating
- $$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{O} - \text{CH} \begin{matrix} \text{Ph} \\ \text{Ph} \end{matrix} \xrightarrow{\text{CH}_3\text{OH}/\text{H}^+} \text{Product(s)}$$

(A) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{OH}$

(C) $\begin{matrix} \text{Ph} \\ \text{CH} - \text{OH} \\ \text{Ph} \end{matrix}$

(B) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{OCH}_3$

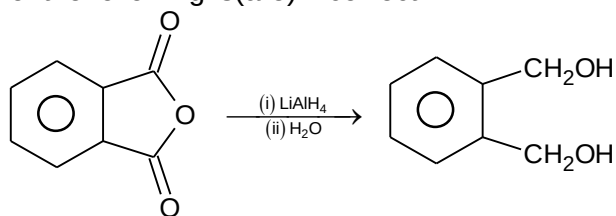
(D) $\begin{matrix} \text{Ph} \\ \text{CH} - \text{OCH}_3 \\ \text{Ph} \end{matrix}$
- Select the correct match
 (A) $\text{Fe}^{3+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow$ Blue colour ppt. (B) $\text{Fe}^{3+} + [\text{Fe}(\text{CN})_6]^{3-} \rightarrow$ Red Brown colouration
 (C) $\text{Fe}^{2+} + [\text{Fe}(\text{CN})_6]^{3-} \rightarrow$ Blue colour ppt. (D) $\text{Fe}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \rightarrow$ Red Brown colouration
- Let the height of hcp unit cell is 'h'. The height of octahedral voids from the base is
 (A) $\frac{h}{2}$ (B) $\frac{h}{3}$
 (C) $\frac{h}{4}$ (D) $\frac{3h}{4}$
- The significance of Heisenberg uncertainty principle in the atom is/are
 (A) it rules out the concept of orbit
 (B) it rules out the existence of nucleus
 (C) it introduces the probability of position and momentum of an electron in an atom
 (D) it introduces the probability of finding e^- in a given orbit

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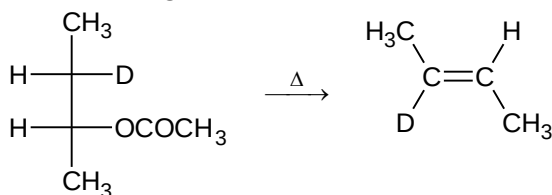
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6. Which of the following is(are) incorrect?

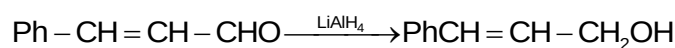
(A)



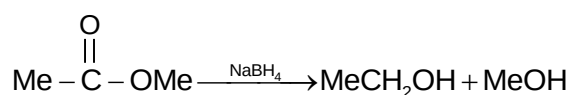
(B)



(C)



(D)



7. Select correct statement(s)

(A) liquefaction of gases in a continuous transition

(B) Boyle temperature $T_b = \frac{2a}{Rb}$

(C) van der Waal's equation is not very accurate near the critical states

(D) for an ideal gas $T_c = 0$

8. Which of the following factors is/are responsible for the extent of positive deviation?

(A) Difference in polarity of the molecules

(B) Difference in the length of hydrocarbon chain in case of their solution

(C) Difference in intermolecular forces of attraction

(D) Association of either of the constituent in the liquid state

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Match the Lists and answer the following question.

List – I		List – II	
(I)	Cubic	(P)	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$
(II)	Tetragonal	(Q)	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$
(III)	Orthorhombic	(R)	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$
(IV)	Monoclinic	(S)	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$

9. Which combination is correct according to the above list?
 (A) I $\rightarrow$ Q (B) II $\rightarrow$ S
 (C) III $\rightarrow$ R (D) IV $\rightarrow$ P
10. Which combination is correct according to the above list?
 (A) I $\rightarrow$ S (B) II $\rightarrow$ Q
 (C) III $\rightarrow$ P (D) IV $\rightarrow$ R

List Match Set (11-12)

Match the Lists and answer the following question.

List – I		List – II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	

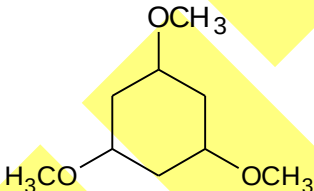
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11. Which combination is correct according to the above list?
 (A) I $\rightarrow$ ST (B) II $\rightarrow$ RS (C) III $\rightarrow$ PQ (D) IV $\rightarrow$ RT
12. Which combination is correct according to the above list?
 (A) I $\rightarrow$ RS (B) II $\rightarrow$ Q (C) III $\rightarrow$ PQ (D) IV $\rightarrow$ P

(PART – B)**(Numerical Type)**

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. pH of 0.1 M NH_4HCO_3 solution, given $K_b = 1.8 \times 10^{-5}$ K_1 of $\text{H}_2\text{CO}_3 = 4.2 \times 10^{-7}$ M and $K_2(\text{HCO}_3^{-1}) = 4.8 \times 10^{-11}$
2. Which of the following cations can form soluble complex when treated with conc. NaOH(excess)
 Cu^{2+} , Pb^{2+} , Zn^{2+} , Ni^{2+} , Al^{3+} , Fe^{3+} , Sn^{2+} , Cd^{2+} , Co^{2+}
3. 3.24 g of mercuric nitrate is dissolved in 1 kg of water, the freezing point of the solution is found to be -0.0558°C . Sum of degree of dissociation and van't Hoff factor for mercuric nitrate is
 (Given: $K_{f_{\text{H}_2\text{O}}} = 1.86 \text{ mol}^{-1} \text{ K kg}$; molecular mass of mercuric nitrate = 324.6)
4. In the following reactions,
 $\text{Na}_2\text{SO}_3 + \text{S} \xrightarrow{\text{boiling water}} [\text{X}]$ (Colorless Solid)
 $\text{AgBr} \xrightarrow{\text{excess}[\text{X}]} [\text{Y}]$ (soluble Complex)
 $\text{X} + \text{Cl}_2 + \text{H}_2\text{O} \longrightarrow [\text{Z}] + \text{HCl}$
 Sum of sulphur atoms in compound [X], [Y] and [Z]

5. 
 How many maximum moles of HI is needed for complete reaction of one mole of the above compound?
6. 9 gm oleum completely neutralizes 100 mL 2 M NaOH. Percentage oleum is

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SECTION – III : MATHEMATICS

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. If $y = f(x)$ is a solution of $\frac{dy}{d(x^2)} + \frac{y}{x^2} = \frac{\sin^{-1}x}{2x^2}$ and $f(0) = 0$, then

(A) $f(1) = \frac{\pi}{8}$

(C) $f\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{4} - \frac{\pi}{6}$

(B) $f(-1) = -\frac{\pi}{8}$

(D) $x^2 f(x) = \frac{1}{2}(2x^2 - 1)\sin^{-1}x + \frac{1}{4}x\sqrt{1-x^2} + \frac{1}{4}$

2. If $f(x) = \begin{vmatrix} 5 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 5 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}$, then which of the following is/are correct

(A) $f(x)$ is a non – monotonic function

(B) $f(x) = 0$ has no real roots

(C) $f'(x)$ has infinitely many maxima and minima

(D) Number of integers in the range of $f(x)$ is 201

3. If $\cos \frac{2\pi}{2019} \cdot \cos \frac{4\pi}{2019} \cdot \cos \frac{6\pi}{2019} \dots \cos \frac{2018\pi}{2019}$ is 2^{-k} where $k = \frac{p}{q}$, $p, q \in \mathbb{N}$ and p and q are relatively prime, then

(A) $(p+q)$ is a 4 – digit number

(C) $(p+q)$ is a 3 – digit number

(B) sum of the digits in $(p+q)$ is 2

(D) number of divisors of p is 8

4. For the curve $\sin x + \sin y = 1$, $\lim_{x \rightarrow 0} (x^\alpha) \frac{d^2 y}{dx^2}$ exists then α can be

(A) $\frac{3}{2}$

(C) $\frac{5}{2}$

(B) 2

(D) 1

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5. A point P moves inside a square with vertices A (1, 1), B (-1, 1), C(-1, -1) and D (1, -1) such that minimum $\{PA, PB, PC, PD\} \leq 1$. If the area bounded by the curve traced out by moving point P is λ , then which of the following is/are correct
 (A) λ is a rational number
 (B) $[\lambda] = 3$ (where $[.]$ represents greatest integral function)
 (C) λ is an irrational number
 (D) $[\lambda] = 0$ (where $[.]$ represents greatest integral function)
6. $\lim_{x \rightarrow 0} \frac{x^2 \sin \beta x}{\tan \alpha x - \sin x} = 4$ where $\alpha, \beta \in \mathbb{R}$, then which of the following is/are correct
 (A) $\alpha = 1$
 (B) $\beta = 2$
 (C) $\alpha = 2$
 (D) $\beta = 1$
7. Let A and B be two matrices different from I such that $AB = BA$ and $A^n - B^n$ is invertible for some positive integer n . If $A^n - B^n = A^{n+1} - B^{n+1} = A^{n+2} - B^{n+2}$, then
 (A) $I - A$ is singular
 (B) $I - B$ is singular
 (C) $A + B = AB + I$
 (D) $(I - A)(I - B)$ is non-singular
8. $a, b \in \mathbb{R}$, $|a| \leq 1$, $|b| \leq 1$, let $I(a, b) = \int_0^\pi (a \sin x + b \cos x)^3 dx$ then
 (A) $I(a, b)$ is independent of a
 (B) $I(a, b)$ is independent of b
 (C) Maximum value of $I(a, 1)$ is $\frac{10}{3}$
 (D) minimum value of $I(-1, b)$ is $-\frac{10}{3}$

Space For Rough Work

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

	List – I		List – II
(I)	If $(n-1)(2n-1)$ is the total number of odd numbers less than 1000 formed by using 0, 3, 5 and 7 (repetition of digits not allowed), then n is	(P)	8
(II)	Number of triangles formed by the vertices of the regular polygon of 10 sides is n and no side of the polygon is the sides of the triangle then $\frac{n}{10}$ is	(Q)	7
(III)	If five – digit numbers divisible by 4 can be formed by digit 1, 2, 3, 4 and 5 (the digits cannot be repeated in the same number) is $(n+2)(n-3)$, then n is	(R)	6
(IV)	A student writes 4 quizzes of 4 marks each. If the number of ways of getting total 8 marks is n , then $\left[\frac{n}{16} \right]$ is ([.] denotes greatest integer function)	(S)	5
		(T)	4

9. Which of the following is the correct combination?
 (A) I $\rightarrow$ T (B) I $\rightarrow$ Q (C) II $\rightarrow$ Q (D) II $\rightarrow$ U
10. Which of the following is the correct combination?
 (A) I $\rightarrow$ Q (B) II $\rightarrow$ S (C) III $\rightarrow$ R (D) IV $\rightarrow$ T

List Match Set (11-12)

	List – I		List – II
(I)	A vector perpendicular to the line $x = 2t + 1, y = t + 2$ and $z = -t - 3$	(P)	$7\hat{i} + 3\hat{j} + 5\hat{k}$
(II)	A vector parallel to the planes $x + y + z - 3 = 0$ and $2x - y + 3z = 0$	(Q)	$4\hat{i} - \hat{j} - 3\hat{k}$
(III)	A vector along which the distance between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{-1}$ and $\vec{r} = (3\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + \hat{j} - 2\hat{k})$ is the shortest	(R)	$-11\hat{i} + 7\hat{j} + 5\hat{k}$
(IV)	A vector normal to the plane $\vec{r} = -\hat{i} + 4\hat{j} - 6\hat{k} + \lambda(\hat{i} + 3\hat{j} - 2\hat{k}) + \mu(-\hat{i} + 2\hat{j} - 5\hat{k})$	(S)	$\hat{i} + 3\hat{j} + \hat{k}$
		(T)	

11. Which of the following is not the correct combination?
 (A) II $\rightarrow$ Q (B) I $\rightarrow$ P (C) III $\rightarrow$ P (D) IV $\rightarrow$ R
12. Which of the following is the correct combination?
 (A) I $\rightarrow$ R (B) II $\rightarrow$ R (C) III $\rightarrow$ P (D) IV $\rightarrow$ S

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(PART – B)**(Numerical Type)**

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. If the differential equation of all straight lines which are at a fixed distance of 10 units from origin is $(y - xy_1)^2 = 400A(1 + y_1^2)$ then A is equal to _____ units.
2. There are n different objects, 1, 2, 3, ..., n, distributed at random in n places marked 1, 2, 3, ..., n. If p is the probability that at least three of the objects occupy places corresponding to their number, then 6p is equal to
3. If $\int_0^{\pi/2} \sin \theta \log \sin \theta d\theta = \log\left(\frac{5A}{e}\right)$ then A is equal to
4. If p_1, p_2 are respectively the perpendicular distances of the points with position vectors $\mathbf{a} = 3\mathbf{i} - 5\mathbf{j} + 8\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$ from the plane $\mathbf{r} \cdot (2\mathbf{i} + 2\mathbf{j} - \mathbf{k}) = 12$, then $\frac{p_2}{37} + \frac{p_1}{32}$ is equal to
5. A variable plane at a unit distance from the origin cuts the coordinate axes at points A, B, C. If the centroid (x, y, z) of the triangle ABC satisfies the equation $x^{-2} + y^{-2} + z^{-2} = k$, then the value of $\frac{k}{4}$ is
6. Let $p(x)$ be a polynomial of degree 4 having extremum at $x = 1, 2$ and $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2}\right) = 2$. Then the value of $p(2)$ is

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QP Code: 100188
Answers
SECTION – I : PHYSICS

(PART – A)

- | | | | |
|--------|--------|-------|--------|
| 1. BCD | 2. A | 3. BC | 4. ABD |
| 5. AD | 6. BCD | 7. BD | 8. ABC |
| 9. A | 10. A | 11. C | 12. A |

(PART – B)

- | | | | |
|---------|------|---------|------|
| 1. 2.25 | 2. 3 | 3. 1.20 | 4. 6 |
| 5. 1.25 | 6. 6 | | |

SECTION – II : CHEMISTRY

(PART – A)

- | | | | |
|---------|--------|--------|---------|
| 1. ABCD | 2. AD | 3. ABC | 4. CD |
| 5. AC | 6. BCD | 7. CD | 8. ABCD |
| 9. A | 10. D | 11. B | 12. D |

(PART – B)

- | | | | |
|---------|----------|------|------|
| 1. 7.83 | 2. 4 | 3. 4 | 4. 7 |
| 5. 6 | 6. 108.8 | | |

SECTION – III : MATHEMATICS

(PART – A)

- | | | | |
|-------|---------|--------|--------|
| 1. AB | 2. ABCD | 3. AB | 4. ABC |
| 5. BC | 6. AB | 7. ABC | 8. CD |
| 9. A | 10. B | 11. B | 12. C |

(PART – B)

- | | | | |
|---------|---------|--------|---------|
| 1. 0.25 | 2. 0.17 | 3. 0.4 | 4. 1.25 |
| 5. 2.25 | 6. 0 | | |

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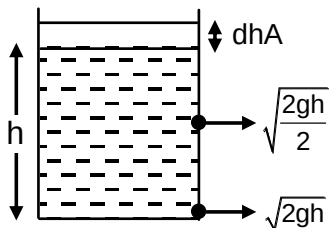
Answers & Solutions

SECTION – I : PHYSICS

(PART – A)

1. **BCD**

Sol.



$$\text{Liquid coming out} = \sqrt{2g \frac{h}{2}} a \cdot dt + \sqrt{2gh} \text{ adt}$$

$$A(-dh) = \sqrt{gh} \text{ adt} + \sqrt{2gh} \text{ adt}$$

2. **A**Sol. Probability of K_α radiation is more than K_β radiation.3. **BC**Sol. $10r > 6R$

$$\frac{r}{R} > \frac{6}{10}; \frac{r}{R} > 0.6 \text{ cannot oscillate \& if } \frac{r}{R} = 0.2$$

$$\Rightarrow T = 0.24 \pi.$$

4. **ABD**

Sol. Apply in impulse momentum equation.

$$\int_0^{0.1} (N - m_y) dt = mv_{yf} - mv_{y1}$$

$$\int_0^{0.1} N df = \frac{10 \times 10}{2} + 10 \times 10 + 10 \times 10 \times 0.1$$

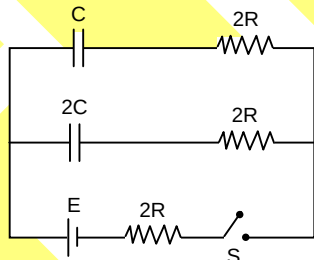
$$\Rightarrow \int_0^{0.1} N dt = 160$$

$$\langle N \rangle \Rightarrow 160 = \frac{\int N dt}{0.1}$$

$$\Rightarrow 1600$$

5. **AD**

Sol.



$$q = q_{\max} \left(1 - e^{-\frac{t}{CR_{\text{eq}}}} \right)$$

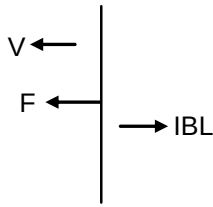
 $Q_{\max} \rightarrow$ Max. charge at $t = \infty$ R_{eq} = Net resistance about capacitor keeping battery as short circuit.6. **BCD**

Sol.

At any instant motional emf = BVl Charge on capacitor $\Rightarrow q = C(BVl)$

$$\text{Current } I \text{ in rod} = \frac{dq}{dt} + \frac{BVI}{R}$$

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when terminal velocity achieved $F = IBL$

$$F = \left(\frac{dq}{dt} + \frac{Bvl}{R} \right) BL$$

$$F = \left(\frac{CB|dv}{dt} + \frac{Bvl}{R} \right) BL$$

$$\frac{dv}{dt} = 0$$

$$F = \frac{B^2 v l^2}{R} ; \quad v_t = \frac{FR}{B^2 l^2}$$

7. **BD**

Sol. At time t , the distance traveled by the rod is

$$ED = vt$$

$$\tan \alpha = \frac{AD}{ED} \text{ or } AD = ED \tan \alpha$$

$$AD = vt \tan \alpha$$

$$\tan(90 - \alpha) = \frac{DC}{ED} \text{ or } DC = vt \cot \alpha$$

So, $AC = AD + DC = vt(\tan \alpha + \cot \alpha)$

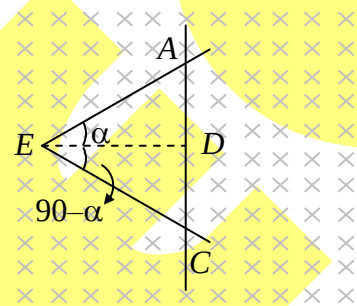
$$\text{Induced emf} = Bvl = Bv(AC)$$

$$= Bv \cdot vt(\tan \alpha + \cot \alpha)$$

$$\text{Induced emf} = Bv^2 t(\tan \alpha + \cot \alpha)$$

Hence, Induced emf $\propto t$ and v^2

Therefore, choices (B) and (D) are correct and choices (A) and (C) are wrong.



8. **ABC**

Sol. $\frac{x}{R} + \frac{x}{2} - 10 + \frac{q}{2} + x - 18 = 0$

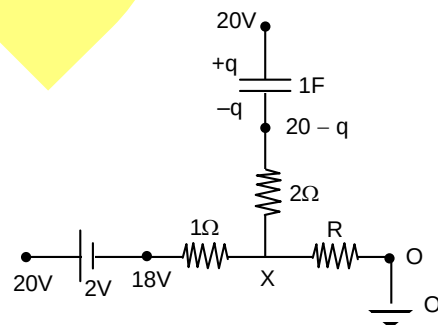
$$\frac{x-0}{R} + \frac{x-20+q}{2} + \frac{x-18}{1} = 0$$

$$x \left(\frac{3}{2} + \frac{1}{R} \right) + \frac{q}{2} = 28$$

$$x = \frac{28 - \frac{q}{2}}{\frac{3}{2} + \frac{1}{R}}$$

$$\text{Current in AP} = \frac{20 - q - (x)}{2} = 3A$$

$$\text{Current through R} = \frac{x}{R} = 7A \quad + R = 2\Omega.$$



9. **A**

10. **A**

11. **C**

Sol. $T - mg = \frac{mv_A^2}{R} ; \quad T + mg = \frac{mv_B^2}{R}$

12. **A**

Sol. (P) $\rightarrow$ Work done $= F \cos \theta = (F \cos \theta) \times s = 0.01 \times 1 \times 4 \times 2 \times 10^2 = 2J$

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(Q) $\rightarrow$ Work done = 0 as bob is not getting displaced relative to car

(R) $\rightarrow v = u + at = 0 + 2 \times 10 = 20$

(S) $\rightarrow$ Work done = 0 as bob is not getting displaced in vertical direction

(PART – B)

1. **2.25**

Sol. $\lambda_m = \frac{b}{T}$

$$\Rightarrow 500 = \frac{b}{T_A}$$

And $1500 = \frac{b}{T_B}$

$$\Rightarrow \frac{T_A}{T_B} = 3$$

$$\frac{U_A}{U_B} = \frac{\sigma(4\pi r_A^2)T_A^4}{\sigma(4\pi r_B^2)T_B^4} = \frac{9}{4} = 2.25$$

2. **3**

Sol. $\mu \cdot 2 \frac{n\lambda}{12u} = \frac{\lambda}{2}$

$$\Rightarrow n = 3$$

3. **1.20**

Sol. $\omega = \sqrt{\frac{k}{m_1 + m_2 + m_3}} = \sqrt{\frac{24}{1+2+3}} = 2$

$$0.4 \times g = \omega^2 A_{\max}, A_{\max} = 1.$$

4. **6**

Sol. Current through 30Ω is $\frac{12}{40}$ and apply loop rule then charge on the capacitor C_2 is $60 \mu C$.

5. **1.25**

Sol. $B \int_0^R \frac{\omega}{2\pi} Q \frac{2\pi r}{\pi R^2} dr \pi r^2 = \frac{T_0 D}{2}$

$$\Rightarrow \omega = \frac{T_0 D}{QR^2 B}$$

6. **6**

Sol. Consider five more identical surfaces, such that a Hexagonal prism is formed with point charge at the centre. Flux will be equally divided between these surfaces.

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SECTION – II : CHEMISTRY**(PART – A)**1. **ABCD**Sol. $\text{HgCl}_2 + \text{SnCl}_2 \longrightarrow \text{Hg}_2\text{Cl}_2 \downarrow \longrightarrow \text{Hg}$ 2. **AD**

Sol. Stable alkyl carbocation

3. **ABC**

Sol. Theoretical

4. **CD**

Sol. Conceptual

5. **AC**

Sol. Heisenberg contradict Bohr

6. **BCD**

Sol. Conceptual

7. **CD**Sol. $T_B = \frac{a}{Rb}$
 $T_C = \frac{8}{27} \frac{a}{Rb}$ 8. **ABCD**

Sol. Positive deviation

$$\left(\begin{matrix} A & \text{---} & A \\ B & \text{---} & B \end{matrix} \right) > (A \text{---} B)$$

9. **A**10. **D**

Sol. 9 & 10: 7 crystal system

11. **B**12. **D**

Sol. 11 & 12: General organic reaction

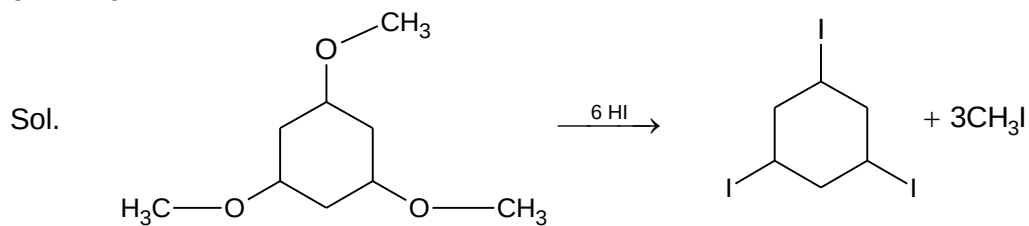
(PART – B)1. **7.83**Sol. $\text{pH} = \text{pH} = 7 + \frac{1}{2}(\text{pK}_a - \text{pK}_b)$ 2. **4**

Sol. Amphoteric

3. **4**Sol. $-\Delta T_f = iK_f m \Rightarrow i = 1, \alpha = 1$
 $\Rightarrow i + \alpha = 4$ 4. **7***TG ~ @bohring_bot*

Sol. $\text{Na}_2\text{S}_2\text{O}_3$, $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, Na_2SO_4

5. 6



6. 108.8

Sol. $\% \text{SO}_3 = 39.5$

$$\% \text{SO}_3 = (\% \text{ oleum} - 100) \times \frac{80}{18}$$

$$\% \text{oleum} = 108.88$$

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SECTION – III : MATHEMATICS**(PART – A)**

1. AB

Sol. $\frac{dy}{2x dx} + \frac{y}{x^2} = \frac{\sin^{-1} x}{2x^2}$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{2}{x}\right)y = \frac{\sin^{-1} x}{x}$$

I.F. $e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2$

Solution is given by $y \times x^2 = \int \frac{\sin^{-1} x}{x} \times x^2 dx$

$$\Rightarrow y \times x^2 = \frac{1}{2}(2x^2 - 1)\sin^{-1} x + \frac{1}{4}x\sqrt{1-x^2} + C$$

2. ABCD

Sol. $f(x) = \begin{vmatrix} 5 + \sin^2 x & \cos^2 x & 4 \sin 2x \\ \sin^2 x & 5 + \cos^2 x & 4 \sin 2x \\ \sin^2 x & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}$

Apply $C_1 \rightarrow C_1 + C_2 + C_3$

$$= (6 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 1 & 5 + \cos^2 x & 4 \sin 2x \\ 1 & \cos^2 x & 5 + 4 \sin 2x \end{vmatrix}$$

Apply $R_3 \rightarrow R_3 + R_1$

$$= (6 + 4 \sin 2x) \begin{vmatrix} 1 & \cos^2 x & 4 \sin 2x \\ 1 & 5 + \cos^2 x & 4 \sin 2x \\ 0 & 0 & 5 \end{vmatrix}$$

$$= (6 + 4 \sin 2x) 5 \times 5 = 25 \times (6 + 4 \sin 2x)$$

$$[50, 250]$$

3. AB

Sol. $A = \cos \theta \cdot \cos 2\theta \cdot \cos 3\theta \dots \cos(n-1)\theta$

$$B = \sin \theta \cdot \sin 2\theta \cdot \sin 3\theta \dots \sin(n-1)\theta \text{ where } \theta = \frac{2\pi}{2019}$$

$$n = 2019$$

$$2^{1009} \times A \times B = (2 \sin \theta \cdot \cos \theta)(2 \sin 2\theta \cdot \cos 2\theta)$$

$$\dots (2 \sin(n-1)\theta \cdot \cos(n-1)\theta)$$

$$= \sin 2\theta \cdot \sin 4\theta \sin 6\theta \dots \sin(2n-2)\theta$$

$$\Rightarrow 2^{1009} \times A \times B = B$$

$$A = \frac{1}{2^{1009}} = 2^{-1009}$$

$$\frac{p}{q} = \frac{1009}{1} \Rightarrow p + q = 1010$$

4. ABC

Sol. $\sin x + \sin y = 1$

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$$\Rightarrow \cos x + \cos y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\cos x}{\cos y}$$

$$\Rightarrow -\sin x - (\sin y) \left(\frac{dy}{dx} \right)^2 + (\cos y) \frac{d^2y}{dx^2} = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{\sin x + \sin y \left(-\frac{\cos x}{\cos y} \right)^2}{\cos y}$$

$$= \frac{\cos^2 y \cdot \sin x + \sin y \cdot \cos^2 x}{(\cos y)^3}$$

$$\text{Put } \sin x = t, \sin y = 1-t, \cos y = \sqrt{2t-t^2}$$

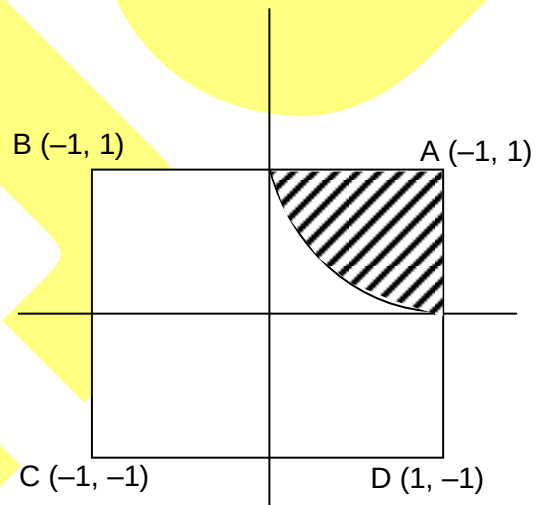
$$\lim_{x \rightarrow 0} x^\alpha \frac{d^2y}{dx^2} = \lim_{t \rightarrow 0} \frac{t^\alpha \left[(2t-t^2)t + (1-t)(1-t^2) \right]}{(\sqrt{2t-t^2})^3}$$

$$\text{exists} \Rightarrow \alpha \geq \frac{3}{2}$$

5. BC

Sol. Required area = 4 × area in quadrant 1

$$= 4 \times \left(1 - \frac{\pi}{4} \right) = 4 - \pi$$



6. AB

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{x^2 \frac{\sin \beta x}{\beta x} \times \beta x}{\alpha n + \frac{(\alpha x)^3}{3} + \dots - \left(x - \frac{x^3}{3} + \dots \right)} = 4$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \times \beta}{x(\alpha - 1) + \left(\frac{\alpha^3}{3} + \frac{1}{3!} \right) x^3 + \dots} = 4$$

$$\Rightarrow \alpha = 1 \text{ and } \frac{\beta}{\frac{1}{3} + \frac{1}{3!}} = 4$$

$$\Rightarrow \beta = 4 \times \left(\frac{1}{3} + \frac{1}{6} \right)$$

$$= 4 \times \frac{1}{2} = 2$$

7. ABC

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Sol. $A^{n+2} - B^{n+2} = (A+B)(A^{n+1} - B^{n+1}) - AB(A^n - B^n)$
 $\Rightarrow A^n - B^n = (A+B)(A^n - B^n) - AB(A^n - B^n)$
 $\Rightarrow I = A+B-AB \quad [\because A^n - B^n \text{ is invertible}]$
 $\Rightarrow (I-A)(I-B) = O.$
 As $A, B \neq I$, we get $I-A$ and $I-B$ are singular matrices.

8. CD

Sol. $I(a, b) = a^3 I_1 + 3a^2 b I_2 + 3ab^2 I_3 + b^3 I_4$ where

$$I_1 = \int_0^\pi \sin^3 x \, dx = 2 \int_0^{\pi/2} \sin^3 x \, dx = \frac{4}{3}$$

$$I_2 = \int_0^\pi \sin^2 x \cos x \, dx = 0$$

$$I_3 = \int_0^\pi \sin x \cos^2 x \, dx = -\frac{1}{3} \cos^3 x \Big|_0^\pi = \frac{2}{3}$$

$$I_4 = \int_0^\pi \cos^3 x \, dx = 0$$

$$\text{Thus, } I(a, b) = \frac{4}{3}a^3 + 2ab^2$$

$$I(a, 1) = \frac{4}{3}a^3 + 2a \leq \frac{4}{3} + 2 = \frac{10}{3} = I(1, 1)$$

$$\text{and } I(-1, b) = \frac{-4}{3} + 2b^2 \geq \frac{-10}{3} = I(-1, 1)$$

9. A

10. B

Sol. (9 to 10)

(I) Odd numbers:

One – digit number $\rightarrow 3$

Two – digit numbers (Choices 3, 5, 7) $\rightarrow 3 \times 2 = 6$

Three – digit number (Choices = 3, 5, 7) $3 \times 2 \times 2 = 12$

$$(n-1)(2n-1) = 3 + 6 + 12 = 21 \Rightarrow 2n^2 - 3n - 20 = 0$$

$\Rightarrow n = 4$, (Option S)

(II) Number of triangles $= {}^{10}C_3 - (10 + 10 \times 6)$
 $n = 50$ (option – S)

			12, 52	$\rightarrow 4$ choices
			32, 24	

(III)

$\underbrace{\hspace{10em}}_{3!}$
 $\{3!\}$ (Last two digits)

$$(n+2)(n-3) = 4 \times 3! = 24 \Rightarrow n^2 - n - 30 = 0 \Rightarrow n = 6$$

Option (R)

(IV) Assumption: Each question carries one mark for correct and zero for other cases.

$$x_1 + x_2 + x_3 + x_4 = 8 \quad (0 \leq x_1, x_2, x_3 \leq x_4 \leq 4).$$

Number of non-negative solutions = Coefficient of x^8 in

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$$\begin{aligned}
 (1+x+x^2+x)^4 &= \left(\frac{1-x^5}{1-x} \right)^4 = (1+x^{10}-2x^5)^2 (1-x)^{-4} \\
 &= (x^{20} + 4x^{10} - 4x^{15} - 4x^5 + 2x^{10} + 1)(1-x)^{-4} \\
 \text{Coefficient of } x^8 &= (-4)^{3+4-1} C_{4-1} + {}^{8+4-1} C_8 = {}^{11} C_8 - 4 {}^6 C_3 = 85 \\
 n &= 85 \cdot \left[\frac{n}{16} \right] = \left[\frac{85}{16} \right] = 5.
 \end{aligned}$$

11. B

12. C

Sol. (11 & 12)(I) Vector along the line is $2\hat{i} + \hat{j} - \hat{k}$ None of the vectors from (P) (Q) (R) (S) are perpendicular to $2\hat{i} + \hat{j} - \hat{k}$

(II) Vectors perpendicular to the planes are

$$\vec{n}_1 = \hat{i} + \hat{j} + \hat{k} \quad (\text{Plane } x + y + z - 3 = 0)$$

$$\vec{n}_2 = 2\hat{i} - \hat{j} + 3\hat{k} \quad (\text{Plane } 2x - y + 3z = 0)$$

Vector parallel to both planes is along $\vec{n}_1 \times \vec{n}_2$.

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 3 \end{vmatrix} = 4\hat{i} - \hat{j} - 3\hat{k}. \quad [\text{Option (Q)}]$$

$$(III) \quad \frac{x}{2} = \frac{y}{-3} = \frac{z}{-1} \quad (\text{line - 1})$$

$$\frac{x-3}{1} = \frac{y+1}{1} = \frac{z-1}{-2} \quad (\text{line - 2})$$

Vectors along the lines are $2\hat{i} - 3\hat{j} - \hat{k}$ and $\hat{i} + \hat{j} - 2\hat{k}$.

$$\text{Vector along the shortest distance : } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 7\hat{i} + 3\hat{j} + 5\hat{k} \quad (\text{Option P})$$

$$(IV) \quad \text{Vector normal to the plane is } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -2 \\ -1 & 2 & -5 \end{vmatrix} = -11\hat{i} + 7\hat{j} + 5\hat{k} \quad (\text{Option R})$$

(I) $\rightarrow$ none, (II) $\rightarrow$ (Q), (III) $\rightarrow$ (P) (IV) $\rightarrow$ (R)**(PART - B)**

1. 0.25

Sol. The given family of lines can be represented by the equation $x \cos \alpha + y \sin \alpha = 10$ (1)where α is an arbitrary constant. Differentiating we have

$$\cos \alpha + \sin \alpha y_1 = 0 \quad \dots\dots\dots(2)$$

Multiplying (2) by x and subtracting it from (1)

$$y \sin \alpha - x \sin \alpha \frac{dy}{dx} = 10$$

$$\Rightarrow (y - y_1) \sin \alpha = 10 \quad \dots\dots\dots(3)$$

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Multiplying (1) by y_1 and (2) by y and subtract

$$\Rightarrow xy_1 \cos \alpha - y \cos \alpha = 10y_1 \dots\dots\dots(4)$$

Square and add (3) and (4) we get, $(y - xy_1)^2 = 100(1 + y_1^2)$.

2. 0.17

Sol. Let E_i denote the event that the i th object goes to the i th place.

We have $P(E_i \cap E_j \cap E_k) = \frac{(n-3)!}{n!}$ for $i < j < k$

Since we can choose 3 places out of n in nC_3 ways, the probability of the required event is

$$p = {}^nC_3 \cdot \frac{(n-3)!}{n!} = \frac{n!}{3!(n-3)!} \cdot \frac{(n-3)!}{n!} = \frac{1}{6}.$$

3. 0.4

Sol. Putting $t = \cos \theta$, we get $\int_0^{\pi/2} \sin \theta \log \sqrt{1 - \cos^2 \theta} d\theta$

$$= \frac{1}{2} \int_0^{\pi/2} \sin \theta \log(1 - \cos^2 \theta) d\theta$$

$$= -\frac{1}{2} \int_1^0 \log(1 - t^2) dt$$

$$= \frac{1}{2} \int_0^1 [\log(1 - t) + \log(1 + t)] dt$$

$$= \frac{1}{2} \int_0^1 \log[1 - (1 - t)] dt + \frac{1}{2} \int_0^1 \log(1 + t) dt$$

$$= \frac{1}{2} \int_0^1 [\log t + \log(1 + t)] dt$$

$$= \frac{1}{2} \left(t \log t \Big|_0^1 - \int_0^1 dt + t \log(1 + t) \Big|_0^1 - \int_0^1 \frac{1+t-1}{1+t} dt \right)$$

$$= \frac{1}{2} (-1 + \log 2 - 1 + \log 2) = \log 2 - \log e$$

$$= \log \left(\frac{2}{e} \right).$$

4. 1.25

Sol. $p_1 = \frac{|12 - (3i - 5j + 8k) \cdot (2i + 2j - k)|}{|2i + 2j - k|}$

$$= \frac{|12 - (6 - 10 - 8)|}{\sqrt{4 + 4 + 1}} = \frac{24}{3} = 8$$

$$\text{Similarly, } p_2 = \frac{1}{3} |12 - (4 - 82 - 21)| = \frac{111}{3} = 37.$$

5. 2.25

Sol. Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

$$\Rightarrow A(a, 0, 0), B(0, b, 0), C(0, 0, c)$$

$$\Rightarrow (x, y, z) = \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3} \right)$$

$$k = x^{-2} + y^{-2} + z^{-2} = 9(a^{-2} + b^{-2} + c^{-2})$$

Since the distance of the plane from the origin is 1

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$$\left| \frac{-1}{\sqrt{\left(\frac{1}{a}\right)^2 + \left(\frac{1}{b}\right)^2 + \left(\frac{1}{c}\right)^2}} \right| = 1 \Rightarrow a^{-2} + b^{-2} + c^{-2} = 1$$

Hence $k = 9$.

6. 0

Sol. $\lim_{x \rightarrow 0} \left(1 + \frac{p(x)}{x^2} \right) = 2 \Rightarrow \lim_{x \rightarrow 0} \left(\frac{p(x)}{x^2} \right) = 1$

Let $p(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4$. The given limit is finite is possible only if $a_3 = a_4 = 0$ and is 1 if $a_2 = 1$. Thus $p(x) = a_0x^4 + a_1x^3 + x^2$ has extremum at $x = 1, 2$ if $p'(1) = 0, p'(2) = 0 \Rightarrow 4a_0 + 2a_1 = -2, 32a_0 + 12a_1 = -4$.

Solving we get, $a_0 = \frac{1}{4}, a_1 = -1$

$$\therefore p(x) = \frac{1}{4}x^4 - x^3 + x^2 \Rightarrow p(2) = \frac{1}{4}(16) - 8 + 4 = 0.$$

TG ~ @bohring_bot

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 1

QP Code: 100191

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 234

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (i) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **Part-B (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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SECTION-1 : PHYSICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- If the dimensions of length are expressed as $G^x c^y h^z$; where G , c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then

(A) $x = \frac{1}{2}, y = \frac{1}{2}$

(C) $y = \frac{1}{2}, z = \frac{3}{2}$

(B) $x = \frac{1}{2}, z = \frac{1}{2}$

(D) $y = -\frac{3}{2}, z = \frac{1}{2}$
- If the electric field E is plotted, with distance from q_1 along the line joining of two, then (E is positive along +ve x -axis) it looks as shown in figure. From the plot we can say that

(A) q_1 and q_2 both are negative

(B) $\left| \frac{q_1}{q_2} \right| = 4$

(C) q_1 is positive and q_2 is negative

(D) $\left| \frac{q_1}{q_2} \right| = \frac{1}{4}$
- One mole of an ideal monoatomic gas is taken from A to C along the path ABC. The temperature of the gas at A is T_0 . For the process ABC (where R is gas constant)

(A) Heat absorbed by the gas is $\frac{11}{12}RT_0$

(B) Heat absorbed by the gas is $\frac{11}{2}RT_0$

(C) Work done by the gas = RT_0

(D) Change in internal energy of gas is $\frac{9}{2}RT_0$
- The threshold wavelength for photoelectric mission from a material is $5200 \text{ \AA}$. Photoelectrons will be emitted when this material is illuminated with monochromatic radiation from a:

(A) 50 watt infrared lamp

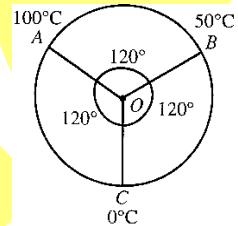
(C) 50 watt ultraviolet lamp

(B) 20 watt infrared lamp

(D) 20 watt ultraviolet lamp

Space For Rough Work

TG ~ @bohring_bot

5. A capacitor is charged to a potential of V_0 . It is connected with an inductor through a switch S. The switch is closed at time $t = 0$. Which of the following statements are correct?
- (A) The maximum current in the circuit is $V_0 \sqrt{\frac{C}{L}}$
- (B) Potential across capacitor becomes zero for the first time at time $t = \pi\sqrt{LC}$
- (C) Energy stored in the inductor at time $t = \frac{\pi}{2}\sqrt{LC}$ is $\frac{1}{4}CV_0^2$
- (D) Maximum energy stored in the inductor is $\frac{1}{2}CV_0^2$
6. A circular ring (centre O) of radius a , and of uniform cross section is made up of three different metallic rods AB, BC & CA joined together at the points A, B and C in pairs) of thermal conductivities α_1 , α_2 and α_3 , respectively (see diagram). The junction A, B and C are maintained at the temperatures 100°C , 50°C and 0°C , respectively. All the rods are of equal lengths and cross sections. Under steady state conditions, assume that no heat is lost from the sides of the rods. Let Q_1 , Q_2 and Q_3 be the rates of transmission of heat along the three rods AB, BC and CA. Then which of the following is/are correct?
- (A) $Q_1 = Q_2 = Q_3$ and all are transmitted in the clockwise sense
- (B) Q_1 and Q_2 flow in clockwise sense and Q_3 in the anticlockwise sense
- (C) $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$
- (D) $\frac{Q_1}{\alpha_1} + \frac{Q_2}{\alpha_2} = \frac{Q_3}{\alpha_3}$
- 
7. A 10 m long horizontal stainless steel wire AB of mass 1 kg whose end A is fixed, is connected to a massless string BC passing over a smooth pulley. String BC is connected to a container of mass 2 kg at end C. Water (density $= 1 \times 10^3 \text{ kg/m}^3$) is poured in the container at a constant rate of 2.25 litre/sec at $t = 0$. Also at $t = 0$, a pulse is generated at end A. Choose from the following the correct option(s).
- (A) Time taken by the pulse to reach point B is 0.612 sec
- (B) Time taken by the pulse to reach point B is 0.212 sec
- (C) Tension in the string at this moment is 33.77 N
- (D) Tension in the string at this moment is 24.77 N

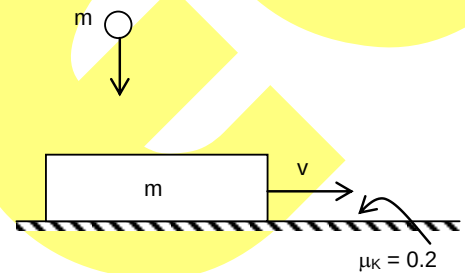
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(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. A dipole of dipole moment p is kept at the centre of a ring of radius R and charge q . The dipole moment has direction along the axis of the ring. The resultant force on the ring due to the dipole is
 (A) zero
 (B) $\frac{pq}{4\pi\epsilon_0 R^3}$
 (C) $\frac{pq}{2\pi\epsilon_0 R^3}$
 (D) $\frac{pq}{4\pi\epsilon_0 R^3}$ only if the charge is uniformly distributed on the ring.
9. A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is $\mu_k = 0.2$, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block just after the collision decreases by
 (A) $0.5\sqrt{2gh}$ (B) 0 (C) $0.1\sqrt{2gh}$ (D) $0.3\sqrt{2gh}$
10. Consider elastic collision of a particle of mass m moving with a velocity u with another particle of the same mass at rest. After the collision the projectile and the struck particle move in direction making angle θ_1 and θ_2 respectively with the initial direction of motion. The sum of the angle $\theta_1 + \theta_2$ is
 (A) 45° (B) 90° (C) 135° (D) 180°
11. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is $4/3$. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant when the ball is 12.8 m above the water surface the fish see the speed of ball as
 (A) 9 m/s (B) 12 m/s (C) 16 m/s (D) 21.33 m/s
12. In Bohr model of the hydrogen atom, let R , V and E represent the radius of the orbit, speed of the electron and the total energy of the electron respectively. Which of the following quantities are proportional to the quantum number n :
 (A) VR (B) RE (C) V (D) $\frac{R}{E}$



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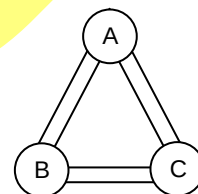
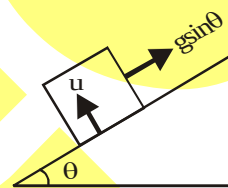
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13. A particle of mass m and charge q is projected in a region where an electric field is existing and given by $\vec{E} = E_0 \hat{i}$, with a velocity $v_0 \hat{j}$ from the origin at time $t = 0$, then choose the correct statement (assuming $m^2 v_0^2 = 2qE_0 m x_0$).
- (A) radius of curvature of the particle when its x -coordinate becomes x_0 is $2x_0$.
 (B) radius of curvature of the particle when its x -coordinate becomes x_0 is $4\sqrt{2} x_0$.
 (C) speed of the particle when its x -coordinate becomes x_0 is v_0 .
 (D) speed of the particle when its x -coordinate becomes x_0 is $2v_0$.

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

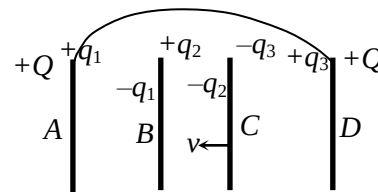
1. A cabin is moved up the inclined plane with constant acceleration $g \sin \theta$. A particle is projected with some velocity in a direction perpendicular to the inclined plane. If maximum height attained by particle perpendicular to inclined plane is same as range of particle with respect to the cabin parallel to plane then calculate values of $\cot \theta$.
2. Three metallic blocks A, B and C have masses m , m and $2m$ respectively. Specific heat of A, B and C are C , $2C$ and C respectively. Initial temperature of A, B and C are 10°C , 5°C and 5°C respectively. Now the blocks are connected by 3-identical rods as shown. Find the final temperature of block A on Celsius scale (Neglect any heat loss due to radiation).
3. In an AC circuit the phase difference between emf and current is $\pi/3$ then the power factor of the circuit is n . Find 'n'.
4. A thin uniform rod of mass ' m ', length ' ℓ ' rotates uniformly about a vertical axis ω , form a conical pendulum, the upper end of the rod is hinged and θ is angle between rod and vertical. If the change of angular momentum of the rod about hinge is $\frac{m\ell^2}{x} \omega^2 \sin 2\theta$ then the value of 'x' will be



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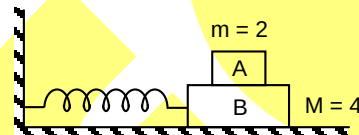
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5. In the diagram there are four conducting plates A, B, C and D placed parallel to each other at equal separation L . If plate C starts moving towards plate B with velocity v . The current (in mA) flowing in the wire connecting A and D is '2y'. Find 'y'. (assume all other plates to be fixed) (Given: $q_2 = 5\mu\text{C}$, $q_3 = 10\mu\text{C}$, $v = 10\text{ m/s}$, $L = 0.05\text{ m}$)



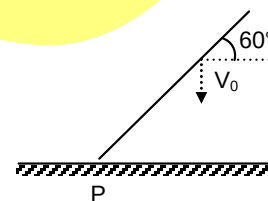
6. A mono-atomic gas undergo a process such that $P \propto \frac{1}{T^2}$. If final temperature is twice that of initial temperature of the gas in given process and molar heat capacity for this process is $\frac{nR}{2}$ then value of 'n' is

7. Block B of mass 4 kg resting on a frictionless horizontal surface and connected with spring of spring constant $K = 10$ as shown in figure.



Another block A of mass 2 kg is placed on the top of block B. Co-efficient of friction between the blocks is $\mu = 0.5$. Body B is displaced and released such that both bodies are moving together and perform simple harmonic motion. Maximum oscillation amplitude that permits body A and B to move together without slipping between the surfaces in contact of body A and B. (Consider $g = 10\text{ m/sec}^2$).

8. A rod of length $\ell = \frac{6}{7}$ meter forming an angle 60° with horizontal strikes a frictionless floor at point P with its velocity $v_0 = 2$ meter/second in downward direction and no angular velocity. Assume that collision at point P is perfectly elastic then angular velocity of the rod just after collision is



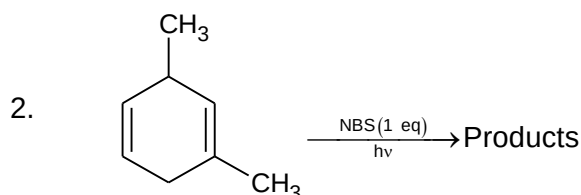
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SECTION-2 : CHEMISTRY**PART – A****(Multi Correct Choice Type)**

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Which order is/are correct for OCl_2 , NCl_3 and BF_3 molecules?
- (A) Tendency to form back π -bond: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$
(B) Polarity of covalent bond: $\text{B} - \text{F} > \text{O} - \text{Cl} > \text{N} - \text{Cl}$
(C) Dipole moment: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$
(D) Hydrolysis: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$

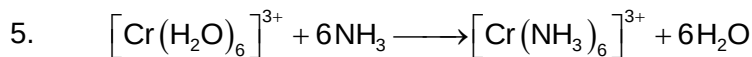


The characteristic(s) of the major product of above reaction is

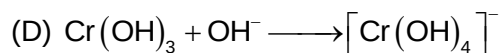
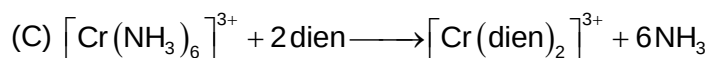
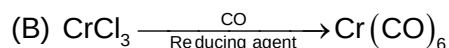
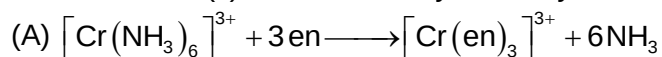
- (A) it is a conjugated cyclic diene.
(B) an α -keto acid is one of its oxidation product.
(C) it contains five allylic hydrogen atoms.
(D) it forms an ether when reacts with aq. KOH
3. Which of the following are colourless or pale colour complexes?
- (A) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Mn}(\text{CN})_6]^{4-}$
(C) $[\text{FeF}_6]^{3-}$ (D) $[\text{Fe}(\text{CN})_6]^{3-}$
4. Choose the incorrect statement(s) regarding the face centred cubic unit cell of a metal lattice?
- (A) The metal atoms are present at the lattice points as well as voids.
(B) The metal atoms touch each other along the face diagonal.
(C) The body diagonal contains a relative number of $\frac{5}{4}$ atoms.
(D) The interatomic attraction force is uniform through out the crystal.

Space For Rough Work

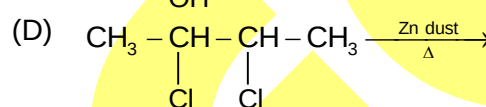
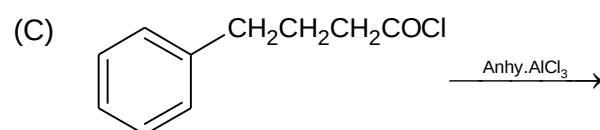
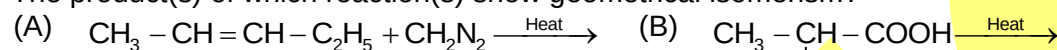
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Which reaction(s) is/are thermodynamically more favourable than the one given above



6. The product(s) of which reaction(s) show geometrical isomerism?



7. Which of the following solutions mixtures show identical colligative properties?

[Assume complete dissociation of salts]

(A) 0.2 M CaCl_2 and 0.3 M KCl

(B) 0.4 M $\text{C}_6\text{H}_{12}\text{O}$ and 0.2 M NaCl

(C) 0.1 M AlCl_3 and 0.1 M NH_2CONH_2

(D) 0.4 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ and 0.5 M $\text{K}_3[\text{Fe}(\text{CN})_6]$

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The incorrect statement regarding $\text{Fe}(\text{CO})_5$ is

(A) it has trigonal bipyramidal structure

(B) both sigma donation and pi-acceptance takes place by the CO ligands

(C) on heating the complex, the oxidation number of Fe decreases as CO is a reducing agent

(D) the CO bond length in the complex increases as compared to the free CO

9. Which is correct statement for an ideal solution?

(A) The mole fraction of solute should be less than that of the solvent.

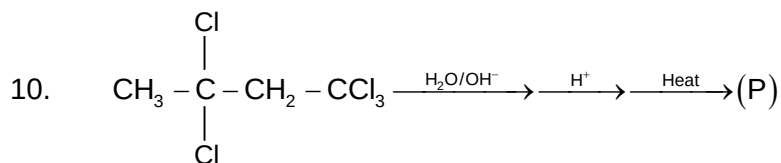
(B) The entropy of the system increases and that of the surrounding does not change

(C) $\Delta H_{\text{mix}} > 0$

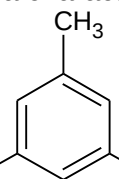
(D) $\Delta G_{\text{mix}} = 0$

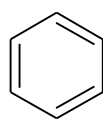
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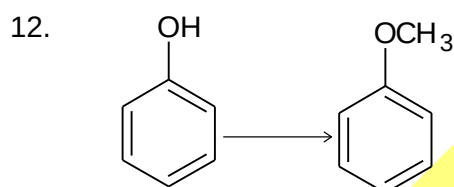


Which is a characteristic of (P)?

- (A)  is formed when (P) reacts with hot conc. H_2SO_4
 (B) in aqueous solution (P) remains in four different forms
 (C) reaction of (P) with CH_3MgBr , forms secondary alcohol

(D) when heated in Al-tube, (P) forms 

11. How many elements of the fourth period of periodic table, contains 10 electrons in their orbitals, which azimuthal quantum number is 2?
 (A) 10 (B) 5 (C) 6 (D) 8



Which reagent can't be used for above reaction?

- (A) $\text{CH}_2\text{N}_2/\Delta$ (B) KOH and CH_3Cl
 (C) CH_3Br and Na (D) $\text{Cl}_2/\text{H}_2\text{O}$ and CH_3OH
13. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 (\text{s}) \xrightarrow{\text{Heating}} \text{P} (\text{s}) + \text{Q} (\text{g}) + \text{H}_2\text{O}$
 Select the incorrect statement

- (A) the equivalent mass of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ in above reaction is $\frac{M}{6}$
 $M = \text{Molar mass of } (\text{NH}_4)_2\text{Cr}_2\text{O}_7 \text{ in g mol}^{-1} \text{ unit}$
 (B) $\text{Q} (\text{g})$ is NH_3
 (C) $\text{P} (\text{s})$ is a green solid
 (D) $\text{P} (\text{s})$ is dissolve in HCl

Space For Rough Work

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PART – B

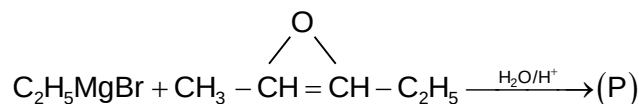
(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. One mole of an ideal gas at 0.1 atm and 300 K, was heated to 320 K at constant pressure, this results in expansion of the gas. What is the entropy change of this reversible process in $\text{JK}^{-1} \text{mol}^{-1}$ unit?

$$[\ell n(1.066) = 0.064, R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}, C_p = \frac{5}{2}R]$$

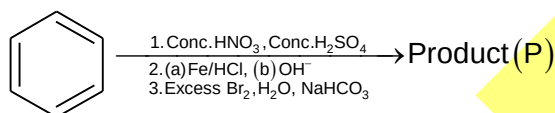
2.



If x = Number of chiral carbon atoms in organic product P
and y = Number of optical isomers possible for P

What is the value of $\frac{x+y}{4}$?

3.



If the molar mass of the product (P) is expressed as $(120x + 30) \text{ g mol}^{-1}$, what is the value of x ? [At mass of Br = 80 g mol^{-1}]

4. One litre of a buffer solution contains 0.5 mole of CH_3COONa and 0.3 mole of CH_3COOH . The pH of the solution is x . Now 0.1 mole of H^+ is added to the buffer solution, the pH changes to y . What is the value of $|x - y|$?

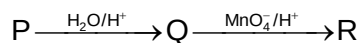
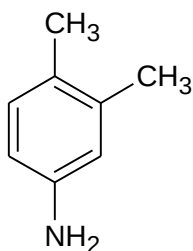
$$[\text{K}_a \text{ of } \text{CH}_3\text{COOH} = 10^{-5}][\log \frac{0.5}{0.3} = 0.22]$$

5. The octacarbonyl of Scandium does not involve back donation from metal to ligand. So the complex is weaker than that of other 3d-series metal. If the oxidation number of scandium in the carbonyl is $+x$, what is the value of x ?

Space For Rough Work

TG ~ @bohring_bot

6. Reaction of an organic compound(P) with Br_2 and KOH form an amine having the following structure.



If the sum of the number of oxygen atoms present in(P), (Q) and (R) is expressed as $10x$, what is the value of 'x'?

7. The specific conductance or conductivity of an electrolyte AB is $4 \times 10^{-3} \text{ ohm}^{-1}\text{cm}^{-1}$. What will be the degree of dissociation of the electrolyte in 0.1 M aqueous solution, if the limiting molar conductance Λ_m^0 is $100 \text{ ohm}^{-1} \text{ cm}^2\text{mol}^{-1}$
8. The molecular formula of an octahedral complex is $\text{CrN}_6\text{H}_{18}\text{Cl}_3$. it shows cis- and trans isomerism and ionise in aqueous solution. If one mole of the complex is dissolved in water. How much gram of AgNO_3 is needed to produce sufficient amount of AgCl precipitate?

Space For Rough Work

TG ~ @bohring_bot

SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The solution of $\frac{dy}{dx} = \sqrt{y-x}$ is given by
 (A) $x + C = 2\sqrt{y-x} + 2\log(\sqrt{y-x} - 1)$ (B) $x^2 + C = 2\sqrt{y-x} + \log(\sqrt{y-x} - 1)$
 (C) $x + C = (y-x)^2 + \log(y-x-1)$ (D) $\sqrt{y-x} - 1 = C'e^{x/2 - \sqrt{y-x}}$
2. If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is
 (A) $\frac{1}{6}(n-2)(n-3)(n-4)$ (B) ${}^{n-2}C_3$
 (C) ${}^{n-3}C_3 + {}^{n-3}C_2$ (D) none of these
3. A four digit number (numbered from 0000 to 9999) is said to be lucky if the sum of first two digits is equal to the sum of its last two digits. If a four digit number is picked up at random, then the probability that it is lucky is
 (A) > 0.065 (B) < 0.068
 (C) 0.066 (D) 0.067
4. If $\sum_{k=1}^n \left(\sum_{m=1}^k m^2 \right) = an^4 + bn^3 + cn^2 + dn + e$, then
 (A) $a = \frac{1}{12}$ (B) $b = \frac{1}{16}$
 (C) $d = \frac{1}{6}$ (D) $e = 0$
5. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x^2)f''(x) = f'(x)f'(x^2) \forall x \in \mathbb{R}$, given that $f(1) = 1$ and $f'''(1) = 8$, then
 (A) $f'(1) = 2$ (B) $f''(1) = 4$
 (C) $f'(1) = 4$ (D) $f''(1) = 2$

Space For Rough Work

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6. Which of the following are CORRECT?

(A) If A and B are symmetric matrices of order 3 then $|AB - BA| = 0$

(B) If A and B are square matrices of order 'n' such that $AB = I_n$, then $BA = I_n$.

(C) The matrix $A = \begin{bmatrix} 9017 & 5148 & 7220 \\ 1234 & 2117 & 4848 \\ 4320 & 3648 & 1179 \end{bmatrix}$ is non-invertible.

(D) If two square matrices A and B of same order commute, then all positive integral powers of A and B commute.

7. Let solutions of the equation $(z - 1 - 2i)^4 = 1$ be $\omega_1, \omega_2, \omega_3, \omega_4$ such that

$|\omega_1| > |\omega_2| > |\omega_3| > |\omega_4|$. If $\arg\left(\frac{z - \omega_i}{z - \omega_j}\right) = \pm \frac{\pi}{2}$ represents a circle, then for different values

of i, j where $i, j \in \{1, 2, 3, 4\}$, $i \neq j$ the number of different circles is λ , where λ lies in

(A) $\left[\frac{42\pi}{22}, \frac{49\pi}{22}\right]$ (B) $\left[\frac{35\pi}{22}, \frac{42\pi}{22}\right]$ (C) $\left[\frac{31\pi}{22}, \frac{37\pi}{22}\right]$ (D) $\left[\frac{29\pi}{22}, \frac{3\pi}{2}\right]$

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The number of values of k for which the equation $x^3 - 3x + k = 0$ has two distinct roots lying in the interval (0, 1) is

(A) three (B) two
(C) infinitely many (D) zero

9. Let $a, b, p, q \in \mathbb{Q}$ and suppose that $f(x) = x^2 + ax + b = 0$ and $g(x) = x^3 + px + q = 0$ have a common irrational root, then

(A) $f(x)$ divides $g(x)$ (B) $g(x) = x f(x)$
(C) $g(x) = (x - b - q)f(x)$ (D) none of these

10. The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible value of the slope of this chord, are

(A) $\{-1, 1\}$ (B) $\{-2, 2\}$ (C) $\left\{-2, \frac{1}{2}\right\}$ (D) $\left\{2, \frac{-1}{2}\right\}$

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11. Consider parabola $y^2 = 40x$, A circle $S = 0$ is drawn taking one of its focal chord as diameter. The extremities of the focal chord being P_1 and P_2 . A point 'P' is taken on the circumference of $S = 0$, then
- (A) $\angle P_1PP_2 = \frac{\pi}{4}$
 (B) $x = 10$ is a tangent to $S = 0$
 (C) Combined equation of the tangents at P_1 and P_2 has $(\text{coeff. } x^2) + (\text{coeff. } y^2) = -2$
 (D) Orthocenter of ΔP_1PP_2 lies on it
12. If $\cos^{-1}p + \cos^{-1}q + \cos^{-1}r = \pi$. Then value of $p^2 + q^2 + r^2 + 2pqr + 4$ is equal to
- (A) 1 (B) 0
 (C) 5 (D) 3
13. If $x_{n+1} = \sqrt{\frac{1}{2}(1+x_n)}$, then $\cos \left[\frac{\sqrt{1-x_0^2}}{x_1x_2x_3\ldots\text{to infinite}} \right] (-1 < x_0 < 1)$ is equal to
- (A) -1 (B) 1
 (C) x_0 (D) $\frac{1}{x_0}$

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. The integer n for which $\lim_{n \rightarrow \infty} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number, then $\frac{1}{n}$ is
2. The sum of the products of the ten numbers $\pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 taking two at a time is x , then $|x| =$
3. The circles having radii 1, 2, 3 touch each other externally then find the radius of the circle which cuts the three circles orthogonally.
4. If $x + \frac{1}{x} = 2\cos 20^\circ$, then the value of $\frac{1}{\left(x^3 + \frac{1}{x^3}\right)^{-1} + 1}$ is equal to

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5. Let $A = \{3, 4, 5, 6\}$ and $B = \{1, 2, 3, \dots, 10\}$ be two sets. Let a function f be defined from set A to set B , such that $f(i) - f(j) \geq 2 \forall i > j$, where $i, j \in A$. If the number of such functions is N , then the value of $\frac{N}{5}$ is
6. In a scalene triangle ABC , the value of $\Delta \left[\frac{\tan \frac{A}{2}}{(a-b)(a-c)} + \frac{\tan \frac{B}{2}}{(b-a)(b-c)} + \frac{\tan \frac{C}{2}}{(c-a)(c-b)} \right]$ is (all notations are standard)
7. Let $a_1, a_2, a_3, \dots, a_{4001}$ are in A.P such that $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{4000} a_{4001}} = 10$ and $a_2 + a_{4000} = 50$ then $\frac{10}{|a_1 - a_{4001}|}$ is equal to
8. If CF is perpendicular from the centre C of the ellipse $\frac{x^2}{49} + \frac{y^2}{25} = 1$ on the tangent at any point P , and G is the point where the normal at P meets the minor axis, then $(C.F. \cdot PG)^2$ is equal to $(40 + \lambda)^2$ then $\lambda =$

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Q.P. Code: 100191

Answers

SECTION-1 : PHYSICS**PART – A**

- | | | | |
|--------------|---------------|---------------|--------------|
| 1. BD | 2. AB | 3. BCD | 4. CD |
| 5. AD | 6. BCD | 7. AC | 8. B |
| 9. D | 10. B | 11. C | 12. A |
| 13. B | | | |

PART – B

- | | | | |
|---------------|-------------|---------------|-------------|
| 1. 8 | 2. 6 | 3. 0.5 | 4. 6 |
| 5. 0.5 | 6. 9 | 7. 3 | 8. 8 |

SECTION-2 : CHEMISTRY**PART – A**

- | | | | |
|--------------|---------------|---------------|--------------|
| 1. AB | 2. ABC | 3. AC | 4. AC |
| 5. AC | 6. ABD | 7. ABD | 8. C |
| 9. B | 10. A | 11. D | 12. D |
| 13. B | | | |

PART – B

- | | | |
|-----------------------------------|---------------|---------------|
| 1. 1.33 (Range 1.3 to 1.4) | 2. 1.5 | 3. 2.5 |
| 4. 0.22[Range 0.2 to 0.3] | 5. 3 | 6. 0.9 |
| 7. 0.4 | 8. 170 | |

SECTION-3 : MATHEMATICS**PART – A**

- | | | | |
|--------------|---------------|---------------|---------------|
| 1. AD | 2. ABC | 3. ABD | 4. ACD |
| 5. AB | 6. ABD | 7. BD | 8. D |
| 9. A | 10. A | 11. D | 12. C |
| 13. C | | | |

PART – B

- | | | | |
|----------------|--------------|----------------|---------------|
| 1. 0.33 | 2. 55 | 3. 1 | 4. 0.5 |
| 5. 7 | 6. 1 | 7. 0.33 | 8. 9 |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **BD**

Sol. Length $\propto G^x c^y h^z$

$$L = [M^{-1}L^3T^{-2}]^x [LT^{-1}]^y [ML^2T^{-1}]^z$$

By comparing the power of M, L and T in both sides we get
 $-x + z = 0, 3x + y + 2z = 1$ and $-2x - y - z = 0$

By solving above three equations we get

$$x = \frac{1}{2}, y = -\frac{3}{2}, z = \frac{1}{2}$$

2. **AB**

Sol. q_1 and q_2 both are negative and $\frac{kq_1}{\left(\frac{2a}{3}\right)^2} = \frac{kq_2}{\left(\frac{a}{3}\right)^2} \Rightarrow \frac{q_1}{q_2} = 4$.

3. **BCD**

Sol. $f = 3, n = 1 \quad P_0 V_0 = nRT_0$

$$H_{AB} = P_0(2V_0 - V_0) + \frac{3}{2}P_0(2V_0 - V_0) = \frac{5}{2}P_0V_0 = \frac{5RT_0}{2}$$

$$H_{BC} = \frac{fn}{2}R(T_C - T_B) = \frac{3}{2}(2P_0V_0 - P_0V_0) = 3P_0V_0 = 3RT_0$$

$$H_{AB-C} = \frac{5}{2}RT_0 + 3RT_0 = \frac{11}{2}RT_0$$

$$W_{AB-C} = P_0V_0 + 0 = RT_0$$

$$\Delta U_{A-B-C} = H_{ABC} - W_{ABC} = \frac{9}{2}RT_0$$

4. **CD**

Sol. Threshold frequency is independent of intensity.

5. **AD**

Sol. $\frac{1}{2}Li_{\max}^2 = \frac{1}{2}CV_0^2$

6. **BCD**

Sol. The flow of heat will always be in the direction of the temperature gradient from higher to lower temperature. Hence Q_1 in rod AB, Q_2 in rod BC will both be in clockwise sense while Q_3 in CA will be in anti-clockwise sense. Also, we have if L is the length of each rod and A its area of cross-section,

$$Q_1 = \frac{\alpha_1 A (100 - 50)}{L} = (50\alpha_1) \frac{A}{L}$$

$$Q_2 = \frac{\alpha_2 A (50 - 0)}{L} = (50\alpha_2) \frac{A}{L}$$

$$Q_3 = \frac{\alpha_3 A (100 - 0)}{L} = (100\alpha_3) \frac{A}{L}$$

Hence, $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$

Also,

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$$\frac{Q_1}{\alpha_1} + \frac{Q_2}{\alpha_2} = \left(50 \frac{A}{L}\right) + \left(50 \frac{A}{L}\right) = \frac{100A}{L} = \frac{Q_3}{\alpha_3}$$

7. **AC**

Sol. Mass per unit length of the wire = 0.1 kg/m.

At any instant, tension in the string is $(2 + 2.25t)$ g N

$$v = \sqrt{\frac{(2 + 2.25t)g}{0.1}}$$

$$\int_0^{10} dx = 10 \int_0^t \sqrt{2 + 2.25t} dt$$

$$t = 0.612 \text{ s}$$

$$T = (2 + 2.25 \times 0.612) \times 10 = 33.77 \text{ N}$$

8. **B**

Sol. The electric field due to the dipole on the circumference of the ring $E = \frac{p}{4\pi\epsilon_0 R^3}$ and it is directed normal to the plane of charged ring.

Hence force on the charged ring $F = qE = \frac{pq}{4\pi\epsilon_0 R^3}$

9. **D**

Sol. $\int N dt = \frac{3MV}{2}$; $\frac{\int \mu N dt}{m} = \Delta V$

10. **B**

Sol. Property of perfectly elastic collision.

11. **C**

Sol. $V = \frac{4}{3} \sqrt{2 \times 10 \times 7.2} = 16$

12. **A**

Sol. $Vr = \frac{nh}{2\pi m}$, $rV \propto n$; $r \propto n^2$

$$E \propto 1/n^2 \text{ and } V \propto 1/n$$

13. **B**

Sol. $v^2 = v_0^2 + 2 \left(\frac{qE_0}{m} \right) x_0$

$$v = \sqrt{2} v_0$$

$$a_n = \frac{qE_0}{m} \frac{v_0}{\sqrt{v_x^2 + v_0^2}}$$

$$R = \frac{v^2}{a_n} = \frac{[m^2 v_0^2 + 2qE_0 m x_0]^{\frac{3}{2}}}{qE_0 v_0 m^2} = 4\sqrt{2} x_0$$

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PART – B

1. 8

Sol. Maximum height = $\frac{u^2}{2g \cos \theta}$... (i)

Range = $\frac{1}{2} \times 2g \sin \theta \times \left(\frac{2u}{g \cos \theta} \right)^2$... (ii)

From (i) and (ii) $\cot \theta = 8$.

2. 6

Sol. $mc(10 - T) = m 2c(T - 5) + 2mC(T - 5)$
 $T = 6^\circ\text{C}$.

3. 0.5

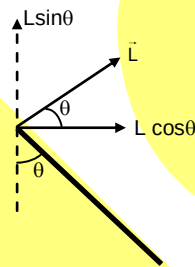
Sol. Power factor = $\cos \phi = \cos \pi/3 = 1/2$.

4. 6

Sol. $\left| \frac{d\vec{L}}{dt} \right| = L \cos \theta \omega$

$$= \frac{m\ell^2}{3} \omega \sin \theta \cos \theta \omega$$

$$= \frac{m\ell^2}{3} \omega^2 \sin \theta \cos \theta$$



5. 0.5

Sol. $\frac{2q_1 L}{2A \epsilon_0} + \frac{2q_2 x}{2A \epsilon_0} - \frac{2q_3 (2L - x)}{2A \epsilon_0} = 0$... (i)

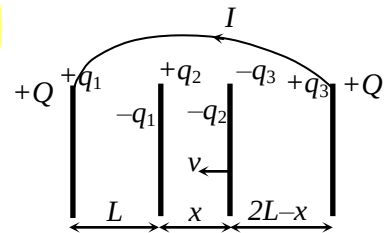
$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt}$... (ii)

From (i),

$$q_1 L + q_2 x - 2q_3 L + q_3 x = 0$$

$$\Rightarrow L \frac{dq_1}{dt} + x \frac{dq_2}{dt} + q_2 \frac{dx}{dt} - 2 \frac{dq_3}{dt} L + x \frac{dq_3}{dt} + q_3 \frac{dx}{dt} = 0$$

$$\Rightarrow I = \frac{V}{3L} (q_2 + q_3) = 1 \text{ mA}$$



6. 9

Sol. $P \propto \frac{1}{T^2} \Rightarrow PT^2 = \text{constant}$

$$P \left(\frac{PV}{nR} \right)^2 = \text{constant}$$

$$P^3 V^2 = \text{constant}$$

$$PV^{2/3} = \text{constant}$$

Molar heat capacity $C = C_V + \frac{R}{1-x}$

$$C = C_V + \frac{R}{1-\frac{2}{3}} = \frac{3R}{2} + 3R = \frac{9R}{2}$$

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7. 3

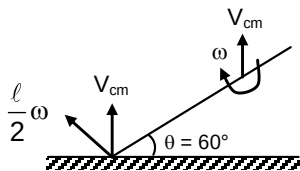
Sol. $m\omega^2 A \leq \mu mg$

$$A \leq \frac{\mu g}{\left[\frac{K}{(m+M)} \right]}$$

$$A \leq \frac{0.5 \times 10}{\left(\frac{10}{6} \right)} \leq 3$$

8. 8

Sol.



Impulse $I = m(V_{cm} + V_0) \dots(1)$

Angular Impulse $I \left(\frac{l}{2} \cos \theta \right) = \frac{m l^2}{12} \omega \dots(2)$

$\therefore e = 1$

$\Rightarrow V_0 = V_{cm} + \frac{l}{2} \omega \cos \theta \dots(3)$

From equation (1) and (2)

$$m(V_{cm} + 2) \frac{l}{2} \cos \theta = \frac{m l^2}{12} \omega$$

$$(V_{cm} + 2) \frac{3}{l} = \omega$$

$\Rightarrow V_{cm} = \frac{\omega l}{3} - 2 \dots(4)$

From equation (3) and (4)

$$2 = \frac{7\omega l}{12} - 2$$

$\Rightarrow \omega = 8.$

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SECTION-2 : CHEMISTRY

PART – A

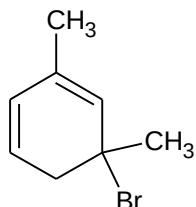
1. AB

Sol. Back bond is stronger in BF_3 as compared to OCl_2 due to energy difference between overlapping orbitals. NCl_3 does not form back π -bond due to less electronegativity difference between N and Cl.

2. ABC

Sol.

The major product is



3. AC

Sol. Due to $t_{2g}^3 e_g^2$ configuration of Mn^{2+} and Fe^{3+} in their respective weak field complexes.

4. AC

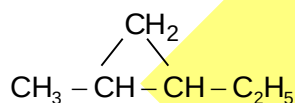
Sol. Close packing takes place along the face diagonal. No atom occupy the voids, as voids can hold smaller atoms or ions.

5. AC

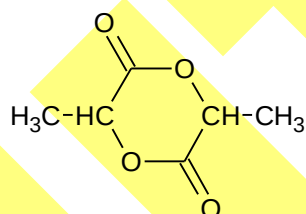
Sol. In (A) and (C) entropy of solution increases.

6. ABD

Sol. (A)



(B)

(C) $\text{CH}_3 - \text{CH} = \text{CHCH}_3$

7. ABD

Sol. Colligative properties depend on the number of solute particles.

0.2 MCaCl_2 has $0.2 \times 3 = 0.6$ mole particles.

0.3 MKCl has $0.3 \times 2 = 0.6$ mole particles

Similarly, B and D contain same number of particles

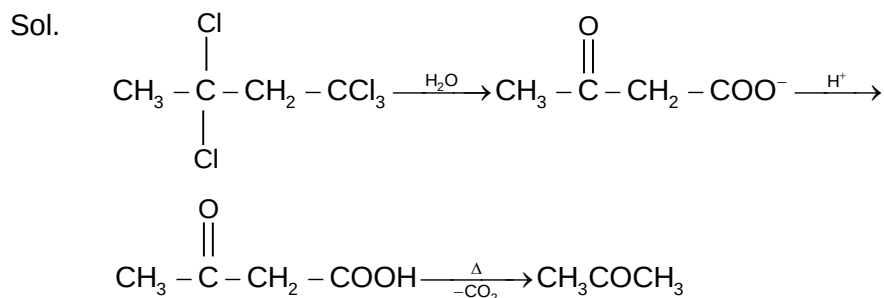
8. C

Sol. Oxidation number of iron remains zero even after heating.

9. B

10. A

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11. D

Sol. The elements are
Cu, Zn, Ga, Ge, As, Se, Br and Kr

12. D

Sol. Reagents in A,B,C convert phenol to anisole.

13. B

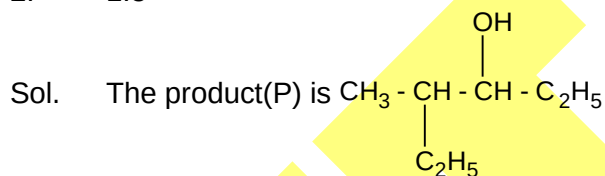
Sol. Q(g) is N₂

PART - B

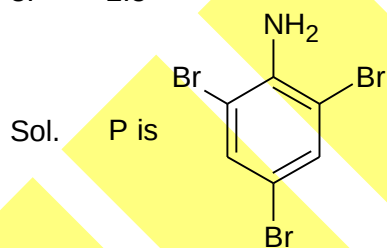
1. 1.33 (Range 1.3 to 1.4)

Sol. $\Delta S(\text{adiabatic}) = nC_p \ln \frac{T_2}{T_1}$

2. 1.5



3. 2.5



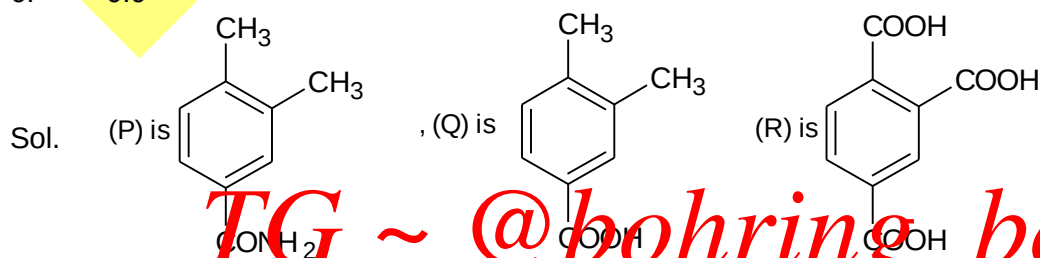
4. 0.22 [Range 0.2 to 0.3]

Sol. $\text{p}^{\text{H}} = \text{p}^{\text{K}_a} + \log \frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]}$

5. 3

Sol. Sc³⁺ contains no d-electron. So back donation does not take place.

6. 0.9



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7. 0.4

Sol. $K = 4 \times 10^{-3} \text{ ohm}^{-1} \text{ cm}^2$

$$\Lambda_m = K \times \frac{1000}{M}$$

$$= 4 \times 10^{-3} \times \frac{1000}{0.1} = 40 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{40}{100} = 0.4$$

8. 170

Sol. The complex is $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot 2\text{NH}_3$

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SECTION-3 : MATHEMATICS

PART – A

1. AD

Sol. Let $\sqrt{y-x} = t$

$$y-x=t^2$$

$$\frac{dy}{dx} = 2t \frac{dt}{dx} + 1$$

$$2t \frac{dt}{dx} + 1 = \sqrt{t}$$

$$\frac{2t}{\sqrt{t}-1} dt = dx \text{ which leads A and D as solutions}$$

2. ABC

Sol. Let x_0 be the number of objects to the left of the first object chosen, x_1 the number of objects between the first and the second, x_2 the number of objects between the second and the third and x_3 the number of objects to the right of the third object. We have

$$x_0, x_3 \geq 0, x_1, x_2 \geq 1 \text{ and } x_0 + x_1 + x_2 + x_3 = n-3 \quad (1)$$

Put $x_1 = y_1 + 1$ and $x_2 = y_2 + 1$. Then (1) reads as

$$x_0 + y_1 + y_2 + x_3 = n-5 \quad (2)$$

where $x_0, y_1, y_2, x_3 \geq 0$

The number of non-negative integral solution of (2) is ${}^{n-5+3}C_3 = {}^{n-2}C_3$

$$\text{We have } {}^{n-2}C_3 = \frac{1}{6}(n-2)(n-3)(n-4)$$

$$\text{Also, } {}^{n-3}C_3 + {}^{n-3}C_2 = {}^{n-2}C_3$$

3. ABD

Sol. The total number of ways of choosing the ticket is 1000. Let the four digits number on the ticket is 1000.

Let the four digits number on the ticket be $x_1x_2x_3x_4$.

Note that $0 \leq x_1 + x_2 \leq 18$ and $0 \leq x_3 + x_4 \leq 18$.

Also, the number of non-negative integral solutions of $x+y=m$ (with $0 \leq x, y \leq 9$) is $m+1$ if $0 \leq m \leq 9$ and is $19-m$ if $10 \leq m \leq 18$.

Thus, the number of favourable ways

$$= 1 \times 1 \times 1 + 2 \times 2 + \dots + 10 \times 10 + 9 \times 9 + 8 \times 8 + \dots + 1 \times 1$$

$$= 2 \left\{ \frac{9 \times 10 \times 19}{6} \right\} + 100 = 670$$

$$\therefore \text{Probability of required event} = \frac{670}{1000} = 0.67$$

Note that $0.067 > 0.065$ and $0.067 < 0.068$.

4. ACD

$$\text{Sol. } \sum_{k=1}^n \left(\sum_{m=1}^k m^2 \right) = an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \sum_{k=1}^n \left[\frac{k(k+1)(2k+1)}{6} \right] = an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \frac{1}{6} \left\{ 2 \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \sum_{k=1}^n k \right\} = an^4 + bn^3 + cn^2 + dn + e$$

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$$\Rightarrow \frac{1}{6} \left\{ \frac{k^2(k+1)^2}{2} + \frac{k(k+1)(2k+1)}{2} + \frac{k(k+1)}{2} \right\} :$$

$$an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \frac{n^4}{12} + \frac{n^3}{3} + \frac{5n^2}{12} + \frac{n}{6} = an^4 + bn^3 + cn^2 + dn + e$$

$$\text{Therefore, } a = \frac{1}{12}; b = \frac{1}{3}; c = \frac{5}{12}; d = \frac{1}{6}; e = 0$$

5. AB

Sol: Let $f'(1) = a$ and $f''(1) = b$, putting $x = 1$ in the given equation we get $b = a^2$.

Differentiating the equation and putting $x = 1$ gives

$$2ab + 8 = ab + 2ab \Rightarrow ab = 8 \Rightarrow a = 2, b = 4$$

6. ABD

Sol: $AB - BA$ is skew symmetric. Also in option C, $|A| \neq 0$. Rest are obvious.

7. BD

$$\text{Sol: } (z - (1 + 2i))^4 = 1 \Rightarrow z = 1^{1/4} + 1 + 2i$$

$$z_1 = 1 + 2i + 1 = 2 + 2i = \omega_2$$

$$z_2 = 1 + 2i - 1 = 2i = \omega_3$$

$$z_3 = 1 + 2i + i = 1 + 3i = \omega_1$$

$$z_4 = 1 + 2i - i = 1 + i = \omega_4$$

$\lambda = 5$ (circle on $\omega_1 - \omega_4$ and $\omega_3 - \omega_2$ will overlap)

8. D

Sol: Let there be a value of k for which $x^3 - 3x + k = 0$ has two distinct roots between 0 and 1.

Let a, b be two distinct roots of $x^3 - 3x + k = 0$ lying between 0 and 1 such that $a < b$. Let

$f(x) = x^3 - 3x + k$. Then $f(a) = f(b) = 0$. Since between any two roots of a polynomial $f(x)$

there exist at least one root of its derivative $f'(x)$. Therefore $f'(x) = 3x^2 - 3$ has at least one

root between a and b . But $f'(x) = 0$ has two roots equal to ± 1 which do not lie between a

and b . Hence $f(x) = 0$ has no real roots lying between 0 and 1 for any value of k .

9. A

Sol. Let $\alpha \in \mathbb{R} - \mathbb{Q}$ be a common root of $f(x) = 0$ and $g(x) = 0$.

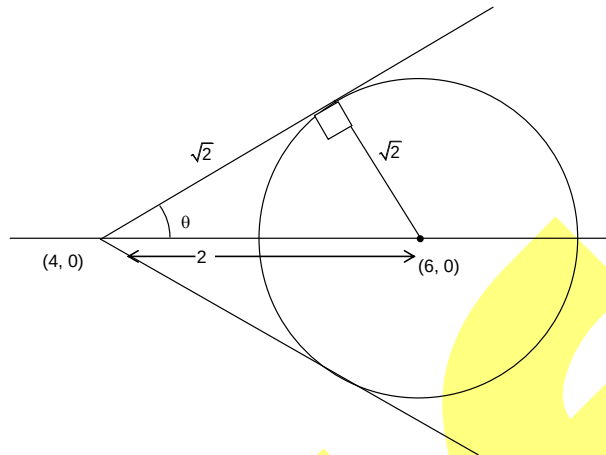
Then $\alpha^2 = -a\alpha - b$. Substituting this in $\alpha^3 + p\alpha + q = 0$, we get $(a^2 - b + p)\alpha + ab + q = 0$

As α is irrational and $a, b, p, q \in \mathbb{Q}$, $p = b - a^2$, $q = -ab$. This gives, $g(x) = (x - a)f(x)$.

10. A

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Sol. From diagram, $\theta = 45^\circ$
 $\Rightarrow \text{Slope} = \pm 1$,



11. D

Sol. $x = -10$ is the directrix of the given parabola. Hence the circle $S = 0$ will touch it. P_1P_2 is the diameter. So $\angle P_1PP_2 = 90^\circ$ and hence ΔP_1PP_2 is right angled and its orthocentre is point P. Tangents at extremities of focal chord are perpendicular. Hence in their combined equation (coefficient x^2) + (coefficient y^2) = 0.

12. C

Sol. Let $\cos^{-1}p = \alpha$, $\cos^{-1}q = \beta$, $\cos^{-1}r = \gamma$
 $\cos \alpha = p$
 $\cos \beta = q$
 $\cos \gamma = r$
 $\alpha + \beta + \gamma = \pi$
 $\Rightarrow \cos(\alpha + \beta) = \cos(\pi - \gamma)$
 $\Rightarrow \cos \alpha \cos \beta - \sin \alpha \sin \beta = -\cos \gamma$
 $\Rightarrow pq - \sqrt{1-p^2} \sqrt{1-q^2} = -r$
 $\Rightarrow pq + r = \sqrt{1-p^2} \sqrt{1-q^2}$
 $2pqr + p^2q^2 + r^2 = (1-p^2)(1-q^2)$
 $p^2 + q^2 + r^2 + 2pqr = 1$

13. C

Sol. Let $x_0 = \cos \theta$, then $x_1 = \sqrt{\frac{1}{2}(1 + \cos \theta)} = \cos \frac{\theta}{2}$, $x_2 = \cos \left(\frac{\theta}{2^2} \right)$, $x_3 = \cos \left(\frac{\theta}{2^3} \right)$ and so on.

$$\begin{aligned} \text{So that } & \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 x_3 \dots \text{to infinite}} \right] \\ &= \frac{\sin \theta}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \\ &= \frac{2^2 \sin \frac{\theta}{2^2} \cos \frac{\theta}{2^2}}{\cos \frac{\theta}{2^2} \cos \frac{\theta}{2^3} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \end{aligned}$$

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$$= \lim_{n \rightarrow \infty} \frac{2^n \sin \frac{\theta}{2^n}}{\cos \frac{\theta}{2^{n+1}}}$$

$$= \lim_{n \rightarrow \infty} \theta \left(\frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \right) \frac{1}{\cos \frac{\theta}{2^{n+1}}} = \theta$$

$$\text{So that } \cos \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 \dots \inf} \right] = \cos \theta = x_0$$

PART – B

1. 0.33

Sol. $\lim_{n \rightarrow \infty} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{2^2 \left(\frac{x}{2} \right)^2} \frac{e^x - \cos x}{x^{n-2}} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x - \cos x}{x^{n-2}}$$

($n \neq 1$ for then the limit = 0)

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x + \sin x}{(n-2)x^{n-3}}$$

So, if $n=3$, the limit is $\frac{1}{2(n-2)}$ which is finite.

If $n=4$, the limit is infinite.

Answer is 0.33.

2. 55

Sol. $(1-1+2-2+\dots+5-5)^2 =$
 $1^2 + 1^2 + 2^2 + 2^2 + \dots + 5^2 + 5^2 + 2S.$

3. 1

Sol. $\Delta = \sqrt{s(s-a)(s-b)(s-c)} = 6$

$$\therefore r = \frac{\Delta}{s} = 1$$

Answer is 1.

4. 0.5

Sol. Cubing on both sides apply
 $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$

5. 7

Sol. The problem corresponds to a selection of 4 non-consecutive integers out of first 10 natural numbers. Therefore $N = {}^7C_4$.

6. 1

Sol: Given expression = $\frac{(s-b)(s-c)}{(a-b)(a-c)} + \frac{(s-a)(s-c)}{(b-a)(b-c)} + \frac{(s-a)(s-b)}{(c-a)(c-b)} = 1$

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7. 0.33

Sol: $\frac{4000}{a_1 a_{4001}} = 10 \Rightarrow a_1 a_{4001} = 400$

also $a_1 + a_{4001} = 50$

$$\Rightarrow |a_1 - a_{4001}|^2 = 2500 - 1600$$

$$\Rightarrow |a_1 - a_{4001}| = 30$$

8. 9

Sol. Equation of the tangent at P $(7 \cos \theta, 5 \sin \theta)$ on the ellipse is $\frac{x}{7} \cos \theta + \frac{y}{5} \sin \theta = 1$

$$\begin{aligned} \text{then } (CF)^2 &= \frac{7^2 \times 5^2}{5^2 \cos^2 \theta + 7^2 \sin^2 \theta} \\ &= \frac{25 \times 49}{25 \cos^2 \theta + 49 \sin^2 \theta} \end{aligned}$$

Equation of the normal at P is $\frac{7x}{\cos \theta} - \frac{5y}{\sin \theta} = 7^2 - 5^2$

Coordinates of G are $\left(0, \frac{-24 \sin \theta}{5}\right)$

$$\begin{aligned} (PG)^2 &= (7 \cos \theta)^2 + \left(5 \sin \theta + \frac{24 \sin \theta}{5}\right)^2 \\ &= \frac{49}{25} (25 \cos^2 \theta + 49 \sin^2 \theta) \end{aligned}$$

So $(CF \cdot PG)^2 = (49)^2 = 2401$

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Pattern – 1

QP Code: 100191

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 234

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-B (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

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SECTION-1 : PHYSICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

- If the dimensions of length are expressed as $G^x c^y h^z$; where G , c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then

(A) $x = \frac{1}{2}, y = \frac{1}{2}$ (B) $x = \frac{1}{2}, z = \frac{1}{2}$

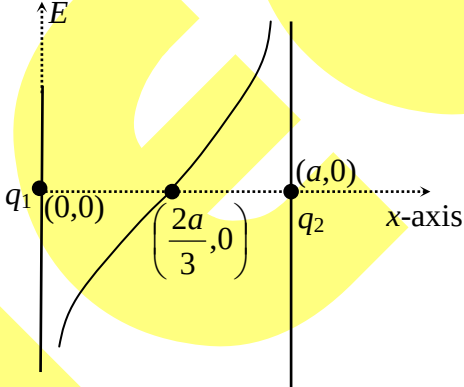
(C) $y = \frac{1}{2}, z = \frac{3}{2}$ (D) $y = -\frac{3}{2}, z = \frac{1}{2}$
- If the electric field E is plotted, with distance from q_1 along the line joining of two, then (E is positive along +ve x -axis) it looks as shown in figure. From the plot we can say that

(A) q_1 and q_2 both are negative

(B) $\left| \frac{q_1}{q_2} \right| = 4$

(C) q_1 is positive and q_2 is negative

(D) $\left| \frac{q_1}{q_2} \right| = \frac{1}{4}$

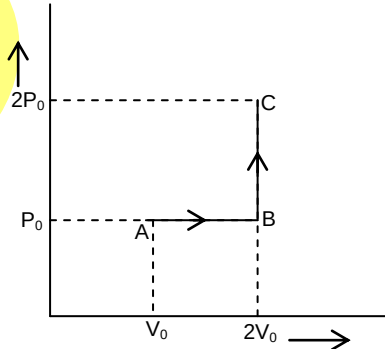

- One mole of an ideal monoatomic gas is taken from A to C along the path ABC. The temperature of the gas at A is T_0 . For the process ABC (where R is gas constant)

(A) Heat absorbed by the gas is $\frac{11}{12}RT_0$

(B) Heat absorbed by the gas is $\frac{11}{2}RT_0$

(C) Work done by the gas = RT_0

(D) Change in internal energy of gas is $\frac{9}{2}RT_0$


- The threshold wavelength for photoelectric mission from a material is $5200 \text{ \AA}$. Photoelectrons will be emitted when this material is illuminated with monochromatic radiation from a:

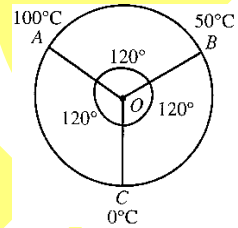
(A) 50 watt infrared lamp (B) 20 watt infrared lamp

(C) 50 watt ultraviolet lamp (D) 20 watt ultraviolet lamp

Space For Rough Work

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5. A capacitor is charged to a potential of V_0 . It is connected with an inductor through a switch S. The switch is closed at time $t = 0$. Which of the following statements are correct?
- (A) The maximum current in the circuit is $V_0 \sqrt{\frac{C}{L}}$
- (B) Potential across capacitor becomes zero for the first time at time $t = \pi\sqrt{LC}$
- (C) Energy stored in the inductor at time $t = \frac{\pi}{2}\sqrt{LC}$ is $\frac{1}{4}CV_0^2$
- (D) Maximum energy stored in the inductor is $\frac{1}{2}CV_0^2$
6. A circular ring (centre O) of radius a , and of uniform cross section is made up of three different metallic rods AB, BC & CA joined together at the points A, B and C in pairs) of thermal conductivities α_1 , α_2 and α_3 , respectively (see diagram). The junction A, B and C are maintained at the temperatures 100°C , 50°C and 0°C , respectively. All the rods are of equal lengths and cross sections. Under steady state conditions, assume that no heat is lost from the sides of the rods. Let Q_1 , Q_2 and Q_3 be the rates of transmission of heat along the three rods AB, BC and CA. Then which of the following is/are correct?
- (A) $Q_1 = Q_2 = Q_3$ and all are transmitted in the clockwise sense
- (B) Q_1 and Q_2 flow in clockwise sense and Q_3 in the anticlockwise sense
- (C) $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$
- (D) $\frac{Q_1}{\alpha_1} + \frac{Q_2}{\alpha_2} = \frac{Q_3}{\alpha_3}$
7. A 10 m long horizontal stainless steel wire AB of mass 1 kg whose end A is fixed, is connected to a massless string BC passing over a smooth pulley. String BC is connected to a container of mass 2 kg at end C. Water (density $= 1 \times 10^3 \text{ kg/m}^3$) is poured in the container at a constant rate of 2.25 litre/sec at $t = 0$. Also at $t = 0$, a pulse is generated at end A. Choose from the following the correct option(s).
- (A) Time taken by the pulse to reach point B is 0.612 sec
- (B) Time taken by the pulse to reach point B is 0.212 sec
- (C) Tension in the string at this moment is 33.77 N
- (D) Tension in the string at this moment is 24.77 N

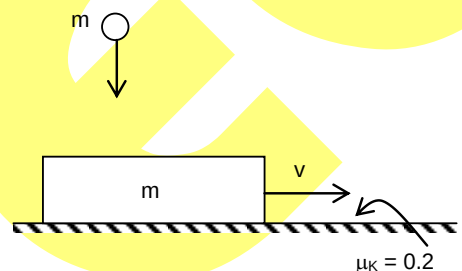


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(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. A dipole of dipole moment p is kept at the centre of a ring of radius R and charge q . The dipole moment has direction along the axis of the ring. The resultant force on the ring due to the dipole is
 (A) zero
 (B) $\frac{pq}{4\pi\epsilon_0 R^3}$
 (C) $\frac{pq}{2\pi\epsilon_0 R^3}$
 (D) $\frac{pq}{4\pi\epsilon_0 R^3}$ only if the charge is uniformly distributed on the ring.
9. A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is $\mu_k = 0.2$, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block just after the collision decreases by
 (A) $0.5\sqrt{2gh}$ (B) 0 (C) $0.1\sqrt{2gh}$ (D) $0.3\sqrt{2gh}$
- 
10. Consider elastic collision of a particle of mass m moving with a velocity u with another particle of the same mass at rest. After the collision the projectile and the struck particle move in direction making angle θ_1 and θ_2 respectively with the initial direction of motion. The sum of the angle $\theta_1 + \theta_2$ is
 (A) 45° (B) 90° (C) 135° (D) 180°
11. A ball is dropped from a height of 20 m above the surface of water in a lake. The refractive index of water is $4/3$. A fish inside the lake, in the line of fall of the ball, is looking at the ball. At an instant when the ball is 12.8 m above the water surface the fish see the speed of ball as
 (A) 9 m/s (B) 12 m/s (C) 16 m/s (D) 21.33 m/s
12. In Bohr model of the hydrogen atom, let R , V and E represent the radius of the orbit, speed of the electron and the total energy of the electron respectively. Which of the following quantities are proportional to the quantum number n :
 (A) VR (B) RE (C) V (D) $\frac{R}{E}$

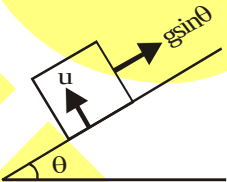
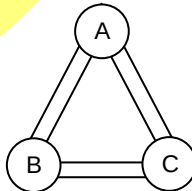
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13. A particle of mass m and charge q is projected in a region where an electric field is existing and given by $\vec{E} = E_0 \hat{i}$, with a velocity $v_0 \hat{j}$ from the origin at time $t = 0$, then choose the correct statement (assuming $m^2 v_0^2 = 2qE_0 m x_0$).
- (A) radius of curvature of the particle when its x -coordinate becomes x_0 is $2x_0$.
 (B) radius of curvature of the particle when its x -coordinate becomes x_0 is $4\sqrt{2} x_0$.
 (C) speed of the particle when its x -coordinate becomes x_0 is v_0 .
 (D) speed of the particle when its x -coordinate becomes x_0 is $2v_0$.

PART – B (Numerical based)

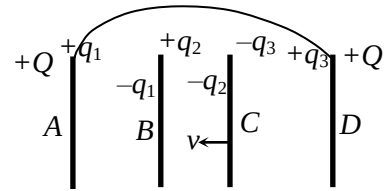
This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

- A cabin is moved up the inclined plane with constant acceleration $g \sin \theta$. A particle is projected with some velocity in a direction perpendicular to the inclined plane. If maximum height attained by particle perpendicular to inclined plane is same as range of particle with respect to the cabin parallel to plane then calculate values of $\cot \theta$. 
- Three metallic blocks A, B and C have masses m , m and $2m$ respectively. Specific heat of A, B and C are C , $2C$ and C respectively. Initial temperature of A, B and C are 10°C , 5°C and 5°C respectively. Now the blocks are connected by 3-identical rods as shown. Find the final temperature of block A on Celsius scale (Neglect any heat loss due to radiation). 
- In an AC circuit the phase difference between emf and current is $\pi/3$ then the power factor of the circuit is n . Find 'n'.
- A thin uniform rod of mass ' m ', length ' ℓ ' rotates uniformly about a vertical axis ω , form a conical pendulum, the upper end of the rod is hinged and θ is angle between rod and vertical. If the change of angular momentum of the rod about hinge is $\frac{m\ell^2}{x} \omega^2 \sin 2\theta$ then the value of 'x' will be

Space For Rough Work

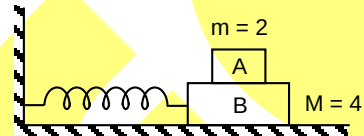
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5. In the diagram there are four conducting plates A, B, C and D placed parallel to each other at equal separation L . If plate C starts moving towards plate B with velocity v . The current (in mA) flowing in the wire connecting A and D is '2y'. Find 'y'. (assume all other plates to be fixed) (Given: $q_2 = 5\mu\text{C}$, $q_3 = 10\mu\text{C}$, $v = 10\text{ m/s}$, $L = 0.05\text{ m}$)



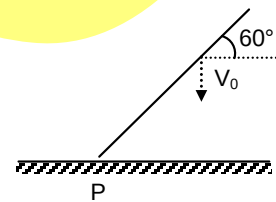
6. A mono-atomic gas undergo a process such that $P \propto \frac{1}{T^2}$. If final temperature is twice that of initial temperature of the gas in given process and molar heat capacity for this process is $\frac{nR}{2}$ then value of 'n' is

7. Block B of mass 4 kg resting on a frictionless horizontal surface and connected with spring of spring constant $K = 10$ as shown in figure.



Another block A of mass 2 kg is placed on the top of block B. Co-efficient of friction between the blocks is $\mu = 0.5$. Body B is displaced and released such that both bodies are moving together and perform simple harmonic motion. Maximum oscillation amplitude that permits body A and B to move together without slipping between the surfaces in contact of body A and B. (Consider $g = 10\text{ m/sec}^2$).

8. A rod of length $\ell = \frac{6}{7}$ meter forming an angle 60° with horizontal strikes a frictionless floor at point P with its velocity $v_0 = 2$ meter/second in downward direction and no angular velocity. Assume that collision at point P is perfectly elastic then angular velocity of the rod just after collision is



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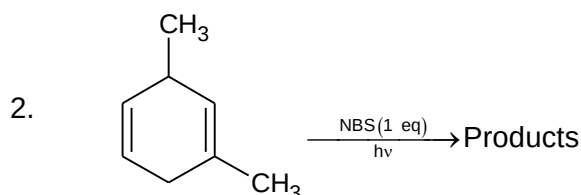
SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Which order is/are correct for OCl_2 , NCl_3 and BF_3 molecules?
 (A) Tendency to form back π -bond: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$
 (B) Polarity of covalent bond: $\text{B} - \text{F} > \text{O} - \text{Cl} > \text{N} - \text{Cl}$
 (C) Dipole moment: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$
 (D) Hydrolysis: $\text{BF}_3 > \text{OCl}_2 > \text{NCl}_3$

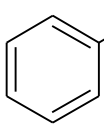


The characteristic(s) of the major product of above reaction is

- (A) it is a conjugated cyclic diene.
 (B) an α -keto acid is one of its oxidation product.
 (C) it contains five allylic hydrogen atoms.
 (D) it forms an ether when reacts with aq. KOH
3. Which of the following are colourless or pale colour complexes?
 (A) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Mn}(\text{CN})_6]^{4-}$
 (C) $[\text{FeF}_6]^{3-}$ (D) $[\text{Fe}(\text{CN})_6]^{3-}$
4. Choose the incorrect statement(s) regarding the face centred cubic unit cell of a metal lattice?
 (A) The metal atoms are present at the lattice points as well as voids.
 (B) The metal atoms touch each other along the face diagonal.
 (C) The body diagonal contains a relative number of $\frac{5}{4}$ atoms.
 (D) The interatomic attraction force is uniform through out the crystal.

Space For Rough Work

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5. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+} + 6\text{NH}_3 \longrightarrow [\text{Cr}(\text{NH}_3)_6]^{3+} + 6\text{H}_2\text{O}$
Which reaction(s) is/are thermodynamically more favourable than the one given above
(A) $[\text{Cr}(\text{NH}_3)_6]^{3+} + 3\text{en} \longrightarrow [\text{Cr}(\text{en})_3]^{3+} + 6\text{NH}_3$
(B) $\text{CrCl}_3 \xrightarrow[\text{Reducing agent}]{\text{CO}} \text{Cr}(\text{CO})_6$
(C) $[\text{Cr}(\text{NH}_3)_6]^{3+} + 2\text{dien} \longrightarrow [\text{Cr}(\text{dien})_2]^{3+} + 6\text{NH}_3$
(D) $\text{Cr}(\text{OH})_3 + \text{OH}^- \longrightarrow [\text{Cr}(\text{OH})_4]^-$
6. The product(s) of which reaction(s) show geometrical isomerism?
(A) $\text{CH}_3 - \text{CH} = \text{CH} - \text{C}_2\text{H}_5 + \text{CH}_2\text{N}_2 \xrightarrow{\text{Heat}}$ (B) $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{COOH} \xrightarrow{\text{Heat}}$
(C)  $\xrightarrow{\text{Anhy. AlCl}_3}$ (D) $\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3 \xrightarrow[\Delta]{\text{Zn dust}}$
7. Which of the following solutions mixtures show identical colligative properties?
[Assume complete dissociation of salts]
(A) 0.2 M CaCl_2 and 0.3 M KCl (B) 0.4 M $\text{C}_6\text{H}_{12}\text{O}$ and 0.2 M NaCl
(C) 0.1 M AlCl_3 and 0.1 M NH_2CONH_2 (D) 0.4 M $\text{K}_4[\text{Fe}(\text{CN})_6]$ and 0.5 M $\text{K}_3[\text{Fe}(\text{CN})_6]$

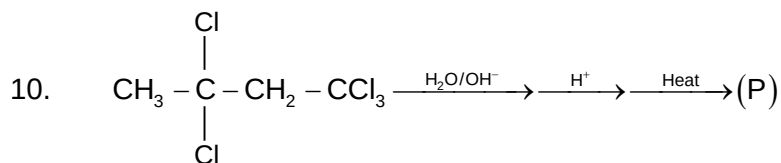
(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

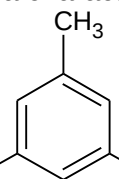
8. The incorrect statement regarding $\text{Fe}(\text{CO})_5$ is
(A) it has trigonal bipyramidal structure
(B) both sigma donation and pi-acceptance takes place by the CO ligands
(C) on heating the complex, the oxidation number of Fe decreases as CO is a reducing agent
(D) the CO bond length in the complex increases as compared to the free CO
9. Which is correct statement for an ideal solution?
(A) The mole fraction of solute should be less than that of the solvent.
(B) The entropy of the system increases and that of the surrounding does not change
(C) $\Delta H_{\text{mix}} > 0$
(D) $\Delta G_{\text{mix}} = 0$

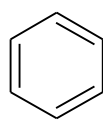
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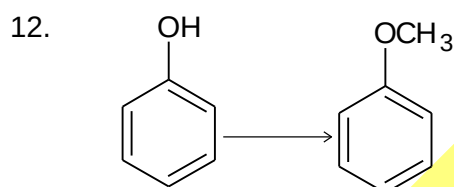


Which is a characteristic of (P)?

- (A)  is formed when (P) reacts with hot conc. H_2SO_4
 (B) in aqueous solution (P) remains in four different forms
 (C) reaction of (P) with CH_3MgBr , forms secondary alcohol

(D) when heated in Al-tube, (P) forms 

11. How many elements of the fourth period of periodic table, contains 10 electrons in their orbitals, which azimuthal quantum number is 2?
 (A) 10 (B) 5 (C) 6 (D) 8



Which reagent can't be used for above reaction?

- (A) $\text{CH}_2\text{N}_2/\Delta$ (B) KOH and CH_3Cl
 (C) CH_3Br and Na (D) $\text{Cl}_2/\text{H}_2\text{O}$ and CH_3OH
13. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 (\text{s}) \xrightarrow{\text{Heating}} \text{P} (\text{s}) + \text{Q} (\text{g}) + \text{H}_2\text{O}$
 Select the incorrect statement

- (A) the equivalent mass of $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ in above reaction is $\frac{M}{6}$
 $M = \text{Molar mass of } (\text{NH}_4)_2\text{Cr}_2\text{O}_7 \text{ in g mol}^{-1} \text{ unit}$
 (B) $\text{Q} (\text{g})$ is NH_3
 (C) $\text{P} (\text{s})$ is a green solid
 (D) $\text{P} (\text{s})$ is dissolve in HCl

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PART – B

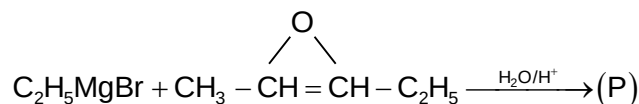
(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. One mole of an ideal gas at 0.1 atm and 300 K, was heated to 320 K at constant pressure, this results in expansion of the gas. What is the entropy change of this reversible process in $\text{JK}^{-1} \text{mol}^{-1}$ unit?

$$[\ell n(1.066) = 0.064, R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}, C_p = \frac{5}{2}R]$$

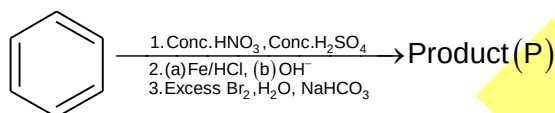
2.



If x = Number of chiral carbon atoms in organic product P
and y = Number of optical isomers possible for P

What is the value of $\frac{x+y}{4}$?

3.



If the molar mass of the product (P) is expressed as $(120x + 30) \text{ g mol}^{-1}$, what is the value of x ? [At mass of Br = 80 g mol^{-1}]

4. One litre of a buffer solution contains 0.5 mole of CH_3COONa and 0.3 mole of CH_3COOH . The pH of the solution is x . Now 0.1 mole of H^+ is added to the buffer solution, the pH changes to y . What is the value of $|x - y|$?

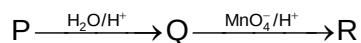
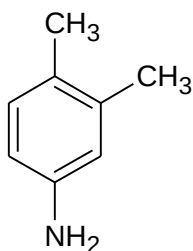
$$[\text{K}_a \text{ of } \text{CH}_3\text{COOH} = 10^{-5}][\log \frac{0.5}{0.3} = 0.22]$$

5. The octacarbonyl of Scandium does not involve back donation from metal to ligand. So the complex is weaker than that of other 3d-series metal. If the oxidation number of scandium in the carbonyl is $+x$, what is the value of x ?

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6. Reaction of an organic compound(P) with Br_2 and KOH form an amine having the following structure.



If the sum of the number of oxygen atoms present in(P), (Q) and (R) is expressed as $10x$, what is the value of 'x'?

7. The specific conductance or conductivity of an electrolyte AB is $4 \times 10^{-3} \text{ ohm}^{-1}\text{cm}^{-1}$. What will be the degree of dissociation of the electrolyte in 0.1 M aqueous solution, if the limiting molar conductance Λ_m^0 is $100 \text{ ohm}^{-1} \text{ cm}^2\text{mol}^{-1}$
8. The molecular formula of an octahedral complex is $\text{CrN}_6\text{H}_{18}\text{Cl}_3$. it shows cis- and trans isomerism and ionise in aqueous solution. If one mole of the complex is dissolved in water. How much gram of AgNO_3 is needed to produce sufficient amount of AgCl precipitate?

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The solution of $\frac{dy}{dx} = \sqrt{y-x}$ is given by
 (A) $x + C = 2\sqrt{y-x} + 2\log(\sqrt{y-x} - 1)$ (B) $x^2 + C = 2\sqrt{y-x} + \log(\sqrt{y-x} - 1)$
 (C) $x + C = (y-x)^2 + \log(y-x-1)$ (D) $\sqrt{y-x} - 1 = C'e^{x/2 - \sqrt{y-x}}$
2. If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is
 (A) $\frac{1}{6}(n-2)(n-3)(n-4)$ (B) ${}^{n-2}C_3$
 (C) ${}^{n-3}C_3 + {}^{n-3}C_2$ (D) none of these
3. A four digit number (numbered from 0000 to 9999) is said to be lucky if the sum of first two digits is equal to the sum of its last two digits. If a four digit number is picked up at random, then the probability that it is lucky is
 (A) > 0.065 (B) < 0.068
 (C) 0.066 (D) 0.067
4. If $\sum_{k=1}^n \left(\sum_{m=1}^k m^2 \right) = an^4 + bn^3 + cn^2 + dn + e$, then
 (A) $a = \frac{1}{12}$ (B) $b = \frac{1}{16}$
 (C) $d = \frac{1}{6}$ (D) $e = 0$
5. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x^2)f''(x) = f'(x)f'(x^2) \forall x \in \mathbb{R}$, given that $f(1) = 1$ and $f'''(1) = 8$, then
 (A) $f'(1) = 2$ (B) $f''(1) = 4$
 (C) $f'(1) = 4$ (D) $f''(1) = 2$

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6. Which of the following are CORRECT?

(A) If A and B are symmetric matrices of order 3 then $|AB - BA| = 0$

(B) If A and B are square matrices of order 'n' such that $AB = I_n$, then $BA = I_n$.

(C) The matrix $A = \begin{bmatrix} 9017 & 5148 & 7220 \\ 1234 & 2117 & 4848 \\ 4320 & 3648 & 1179 \end{bmatrix}$ is non-invertible.

(D) If two square matrices A and B of same order commute, then all positive integral powers of A and B commute.

7. Let solutions of the equation $(z - 1 - 2i)^4 = 1$ be $\omega_1, \omega_2, \omega_3, \omega_4$ such that

$|\omega_1| > |\omega_2| > |\omega_3| > |\omega_4|$. If $\arg\left(\frac{z - \omega_i}{z - \omega_j}\right) = \pm \frac{\pi}{2}$ represents a circle, then for different values

of i, j where $i, j \in \{1, 2, 3, 4\}$, $i \neq j$ the number of different circles is λ , where λ lies in

(A) $\left[\frac{42\pi}{22}, \frac{49\pi}{22}\right]$ (B) $\left[\frac{35\pi}{22}, \frac{42\pi}{22}\right]$ (C) $\left[\frac{31\pi}{22}, \frac{37\pi}{22}\right]$ (D) $\left[\frac{29\pi}{22}, \frac{3\pi}{2}\right]$

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The number of values of k for which the equation $x^3 - 3x + k = 0$ has two distinct roots lying in the interval (0, 1) is

(A) three (B) two
(C) infinitely many (D) zero

9. Let $a, b, p, q \in \mathbb{Q}$ and suppose that $f(x) = x^2 + ax + b = 0$ and $g(x) = x^3 + px + q = 0$ have a common irrational root, then

(A) $f(x)$ divides $g(x)$ (B) $g(x) = x f(x)$
(C) $g(x) = (x - b - q)f(x)$ (D) none of these

10. The focal chord to $y^2 = 16x$ is tangent to $(x - 6)^2 + y^2 = 2$, then the possible value of the slope of this chord, are

(A) $\{-1, 1\}$ (B) $\{-2, 2\}$ (C) $\left\{-2, \frac{1}{2}\right\}$ (D) $\left\{2, \frac{-1}{2}\right\}$

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11. Consider parabola $y^2 = 40x$, A circle $S = 0$ is drawn taking one of its focal chord as diameter. The extremities of the focal chord being P_1 and P_2 . A point 'P' is taken on the circumference of $S = 0$, then
- (A) $\angle P_1PP_2 = \frac{\pi}{4}$
 (B) $x = 10$ is a tangent to $S = 0$
 (C) Combined equation of the tangents at P_1 and P_2 has $(\text{coeff. } x^2) + (\text{coeff. } y^2) = -2$
 (D) Orthocenter of ΔP_1PP_2 lies on it
12. If $\cos^{-1}p + \cos^{-1}q + \cos^{-1}r = \pi$. Then value of $p^2 + q^2 + r^2 + 2pqr + 4$ is equal to
- (A) 1 (B) 0
 (C) 5 (D) 3
13. If $x_{n+1} = \sqrt{\frac{1}{2}(1+x_n)}$, then $\cos \left[\frac{\sqrt{1-x_0^2}}{x_1x_2x_3\ldots\text{to infinite}} \right] (-1 < x_0 < 1)$ is equal to
- (A) -1 (B) 1
 (C) x_0 (D) $\frac{1}{x_0}$

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. The integer n for which $\lim_{n \rightarrow \infty} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non – zero number, then $\frac{1}{n}$ is
2. The sum of the products of the ten numbers $\pm 1, \pm 2, \pm 3, \pm 4$, and ± 5 taking two at a time is x , then $|x| =$
3. The circles having radii 1, 2, 3 touch each other externally then find the radius of the circle which cuts the three circles orthogonally.
4. If $x + \frac{1}{x} = 2\cos 20^\circ$, then the value of $\frac{1}{\left(x^3 + \frac{1}{x^3}\right)^{-1} + 1}$ is equal to

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5. Let $A = \{3, 4, 5, 6\}$ and $B = \{1, 2, 3, \dots, 10\}$ be two sets. Let a function f be defined from set A to set B , such that $f(i) - f(j) \geq 2 \forall i > j$, where $i, j \in A$. If the number of such functions is N , then the value of $\frac{N}{5}$ is
6. In a scalene triangle ABC , the value of $\Delta \left[\frac{\tan \frac{A}{2}}{(a-b)(a-c)} + \frac{\tan \frac{B}{2}}{(b-a)(b-c)} + \frac{\tan \frac{C}{2}}{(c-a)(c-b)} \right]$ is (all notations are standard)
7. Let $a_1, a_2, a_3, \dots, a_{4001}$ are in A.P such that $\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{4000} a_{4001}} = 10$ and $a_2 + a_{4000} = 50$ then $\frac{10}{|a_1 - a_{4001}|}$ is equal to
8. If CF is perpendicular from the centre C of the ellipse $\frac{x^2}{49} + \frac{y^2}{25} = 1$ on the tangent at any point P , and G is the point where the normal at P meets the minor axis, then $(C.F. \cdot PG)^2$ is equal to $(40 + \lambda)^2$ then $\lambda =$

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Q.P. Code: 100191

Answers

SECTION-1 : PHYSICS**PART – A**

- | | | | |
|--------------|---------------|---------------|--------------|
| 1. BD | 2. AB | 3. BCD | 4. CD |
| 5. AD | 6. BCD | 7. AC | 8. B |
| 9. D | 10. B | 11. C | 12. A |
| 13. B | | | |

PART – B

- | | | | |
|---------------|-------------|---------------|-------------|
| 1. 8 | 2. 6 | 3. 0.5 | 4. 6 |
| 5. 0.5 | 6. 9 | 7. 3 | 8. 8 |

SECTION-2 : CHEMISTRY**PART – A**

- | | | | |
|--------------|---------------|---------------|--------------|
| 1. AB | 2. ABC | 3. AC | 4. AC |
| 5. AC | 6. ABD | 7. ABD | 8. C |
| 9. B | 10. A | 11. D | 12. D |
| 13. B | | | |

PART – B

- | | | |
|-----------------------------------|---------------|---------------|
| 1. 1.33 (Range 1.3 to 1.4) | 2. 1.5 | 3. 2.5 |
| 4. 0.22[Range 0.2 to 0.3] | 5. 3 | 6. 0.9 |
| 7. 0.4 | 8. 170 | |

SECTION-3 : MATHEMATICS**PART – A**

- | | | | |
|--------------|---------------|---------------|---------------|
| 1. AD | 2. ABC | 3. ABD | 4. ACD |
| 5. AB | 6. ABD | 7. BD | 8. D |
| 9. A | 10. A | 11. D | 12. C |
| 13. C | | | |

PART – B

- | | | | |
|----------------|--------------|----------------|---------------|
| 1. 0.33 | 2. 55 | 3. 1 | 4. 0.5 |
| 5. 7 | 6. 1 | 7. 0.33 | 8. 9 |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **BD**

Sol. Length $\propto G^x c^y h^z$

$$L = [M^{-1}L^3T^{-2}]^x [LT^{-1}]^y [ML^2T^{-1}]^z$$

By comparing the power of M, L and T in both sides we get
 $-x + z = 0, 3x + y + 2z = 1$ and $-2x - y - z = 0$

By solving above three equations we get

$$x = \frac{1}{2}, y = -\frac{3}{2}, z = \frac{1}{2}$$

2. **AB**

Sol. q_1 and q_2 both are negative and $\frac{kq_1}{\left(\frac{2a}{3}\right)^2} = \frac{kq_2}{\left(\frac{a}{3}\right)^2} \Rightarrow \frac{q_1}{q_2} = 4$.

3. **BCD**

Sol. $f = 3, n = 1 \quad P_0 V_0 = nRT_0$

$$H_{AB} = P_0(2V_0 - V_0) + \frac{3}{2}P_0(2V_0 - V_0) = \frac{5}{2}P_0V_0 = \frac{5RT_0}{2}$$

$$H_{BC} = \frac{fn}{2}R(T_c - T_B) = \frac{3}{2}(2P_0V_0 - P_0V_0) = 3P_0V_0 = 3RT_0$$

$$H_{AB-C} = \frac{5}{2}RT_0 + 3RT_0 = \frac{11}{2}RT_0$$

$$W_{AB-C} = P_0V_0 + 0 = RT_0$$

$$\Delta U_{A-B-C} = H_{ABC} - W_{ABC} = \frac{9}{2}RT_0$$

4. **CD**

Sol. Threshold frequency is independent of intensity.

5. **AD**

Sol. $\frac{1}{2}Li_{\max}^2 = \frac{1}{2}CV_0^2$

6. **BCD**

Sol. The flow of heat will always be in the direction of the temperature gradient from higher to lower temperature. Hence Q_1 in rod AB, Q_2 in rod BC will both be in clockwise sense while Q_3 in CA will be in anti-clockwise sense. Also, we have if L is the length of each rod and A its area of cross-section,

$$Q_1 = \frac{\alpha_1 A (100 - 50)}{L} = (50\alpha_1) \frac{A}{L}$$

$$Q_2 = \frac{\alpha_2 A (50 - 0)}{L} = (50\alpha_2) \frac{A}{L}$$

$$Q_3 = \frac{\alpha_3 A (100 - 0)}{L} = (100\alpha_3) \frac{A}{L}$$

Hence, $Q_1 : Q_2 : Q_3 :: \alpha_1 : \alpha_2 : 2\alpha_3$

Also,

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$$\frac{Q_1}{\alpha_1} + \frac{Q_2}{\alpha_2} = \left(50 \frac{A}{L}\right) + \left(50 \frac{A}{L}\right) = \frac{100A}{L} = \frac{Q_3}{\alpha_3}$$

7. **AC**

Sol. Mass per unit length of the wire = 0.1 kg/m.

At any instant, tension in the string is $(2 + 2.25t)$ g N

$$v = \sqrt{\frac{(2 + 2.25t)g}{0.1}}$$

$$\int_0^{10} dx = 10 \int_0^t \sqrt{2 + 2.25t} dt$$

$$t = 0.612 \text{ s}$$

$$T = (2 + 2.25 \times 0.612) \times 10 = 33.77 \text{ N}$$

8. **B**

Sol. The electric field due to the dipole on the circumference of the ring $E = \frac{p}{4\pi\epsilon_0 R^3}$ and it is directed normal to the plane of charged ring.

Hence force on the charged ring $F = qE = \frac{pq}{4\pi\epsilon_0 R^3}$

9. **D**

Sol. $\int N dt = \frac{3MV}{2}$; $\frac{\int \mu N dt}{m} = \Delta V$

10. **B**

Sol. Property of perfectly elastic collision.

11. **C**

Sol. $V = \frac{4}{3} \sqrt{2 \times 10 \times 7.2} = 16$

12. **A**

Sol. $Vr = \frac{nh}{2\pi m}$, $rV \propto n$; $r \propto n^2$

$$E \propto 1/n^2 \text{ and } V \propto 1/n$$

13. **B**

Sol. $v^2 = v_0^2 + 2\left(\frac{qE_0}{m}\right)x_0$

$$v = \sqrt{2}v_0$$

$$a_n = \frac{qE_0}{m} \frac{v_0}{\sqrt{v_x^2 + v_0^2}}$$

$$R = \frac{v^2}{a_n} = \frac{[m^2 v_0^2 + 2qE_0 m x_0]^{\frac{3}{2}}}{qE_0 v_0 m^2} = 4\sqrt{2} x_0$$

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PART – B1. **8**

Sol. Maximum height = $\frac{u^2}{2g \cos \theta}$... (i)

Range = $\frac{1}{2} \times 2g \sin \theta \times \left(\frac{2u}{g \cos \theta} \right)^2$... (ii)

From (i) and (ii) $\cot \theta = 8$.

2. **6**

Sol. $mc(10 - T) = m 2c(T - 5) + 2mC(T - 5)$
 $T = 6^\circ\text{C}$.

3. **0.5**

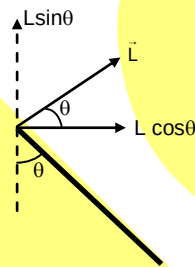
Sol. Power factor = $\cos \phi = \cos \pi/3 = 1/2$.

4. **6**

Sol. $\left| \frac{d\vec{L}}{dt} \right| = L \cos \theta \omega$

$$= \frac{m\ell^2}{3} \omega \sin \theta \cos \theta \omega$$

$$= \frac{m\ell^2}{3} \omega^2 \sin \theta \cos \theta$$

5. **0.5**

Sol. $\frac{2q_1 L}{2A \epsilon_0} + \frac{2q_2 x}{2A \epsilon_0} - \frac{2q_3 (2L - x)}{2A \epsilon_0} = 0$... (i)

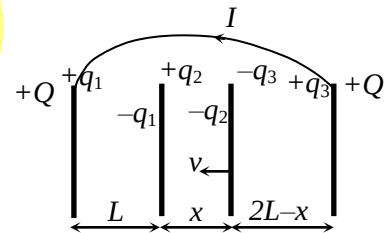
$I = \frac{dq_1}{dt} = \frac{-dq_3}{dt} = \frac{dq_2}{dt}$... (ii)

From (i),

$$q_1 L + q_2 x - 2q_3 L + q_3 x = 0$$

$$\Rightarrow L \frac{dq_1}{dt} + x \frac{dq_2}{dt} + q_2 \frac{dx}{dt} - 2 \frac{dq_3}{dt} L + x \frac{dq_3}{dt} + q_3 \frac{dx}{dt} = 0$$

$$\Rightarrow I = \frac{V}{3L} (q_2 + q_3) = 1 \text{ mA}$$

6. **9**

Sol. $P \propto \frac{1}{T^2} \Rightarrow PT^2 = \text{constant}$

$$P \left(\frac{PV}{nR} \right)^2 = \text{constant}$$

$$P^3 V^2 = \text{constant}$$

$$PV^{2/3} = \text{constant}$$

Molar heat capacity $C = C_V + \frac{R}{1-x}$

$$C = C_V + \frac{R}{1-\frac{2}{3}} = \frac{3R}{2} + 3R = \frac{9R}{2}$$

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7. **3**

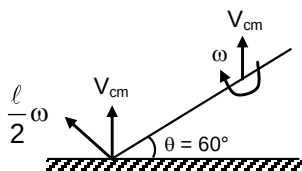
Sol. $m\omega^2 A \leq \mu mg$

$$A \leq \frac{\mu g}{\left[\frac{K}{(m+M)} \right]}$$

$$A \leq \frac{0.5 \times 10}{\left(\frac{10}{6} \right)} \leq 3$$

8. **8**

Sol.



Impulse $I = m(V_{cm} + V_0) \quad \dots(1)$

Angular Impulse $I \left(\frac{l}{2} \cos \theta \right) = \frac{m l^2}{12} \omega \quad \dots(2)$

$\therefore e = 1$

$\Rightarrow V_0 = V_{cm} + \frac{l}{2} \omega \cos \theta \quad \dots(3)$

From equation (1) and (2)

$$m(V_{cm} + 2) \frac{l}{2} \cos \theta = \frac{m l^2}{12} \omega$$

$$(V_{cm} + 2) \frac{3}{l} = \omega$$

$\Rightarrow V_{cm} = \frac{\omega l}{3} - 2 \quad \dots(4)$

From equation (3) and (4)

$$2 = \frac{7\omega l}{12} - 2$$

$\Rightarrow \omega = 8.$

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SECTION-2 : CHEMISTRY

PART – A

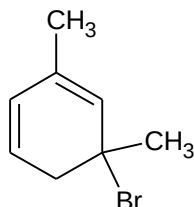
1. AB

Sol. Back bond is stronger in BF_3 as compared to OCl_2 due to energy difference between overlapping orbitals. NCl_3 does not form back π -bond due to less electronegativity difference between N and Cl.

2. ABC

Sol.

The major product is



3. AC

Sol. Due to $t_{2g}^3 e_g^2$ configuration of Mn^{2+} and Fe^{3+} in their respective weak field complexes.

4. AC

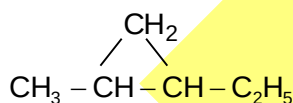
Sol. Close packing takes place along the face diagonal. No atom occupy the voids, as voids can hold smaller atoms or ions.

5. AC

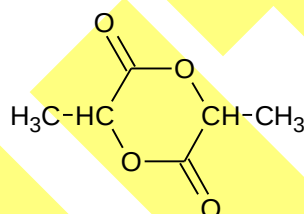
Sol. In (A) and (C) entropy of solution increases.

6. ABD

Sol. (A)



(B)

(C) $\text{CH}_3 - \text{CH} = \text{CHCH}_3$

7. ABD

Sol. Colligative properties depend on the number of solute particles.

0.2 M CaCl_2 has $0.2 \times 3 = 0.6$ mole particles.

0.3 M KCl has $0.3 \times 2 = 0.6$ mole particles

Similarly, B and D contain same number of particles

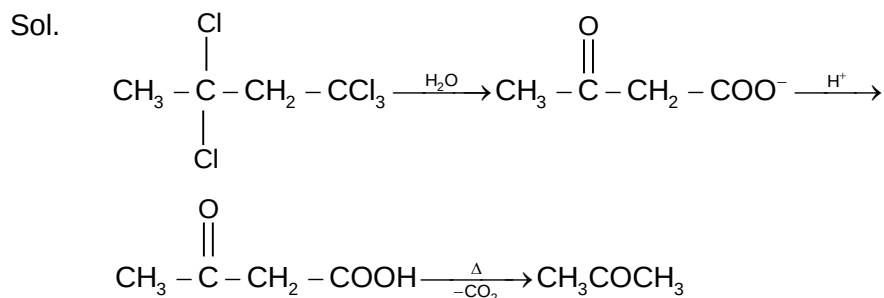
8. C

Sol. Oxidation number of iron remains zero even after heating.

9. B

10. A

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11. D

Sol. The elements are
Cu, Zn, Ga, Ge, As, Se, Br and Kr

12. D

Sol. Reagents in A,B,C convert phenol to anisole.

13. B

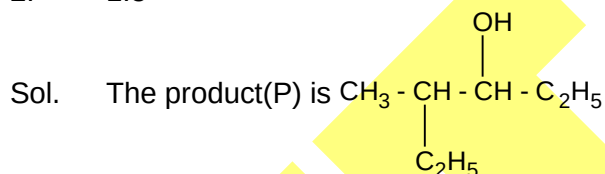
Sol. Q(g) is N₂

PART - B

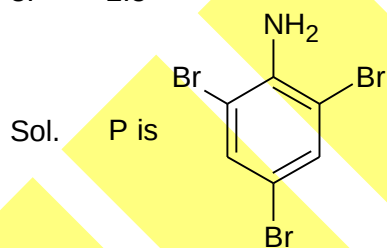
1. 1.33 (Range 1.3 to 1.4)

Sol. $\Delta S(\text{adiabatic}) = nC_p \ln \frac{T_2}{T_1}$

2. 1.5



3. 2.5



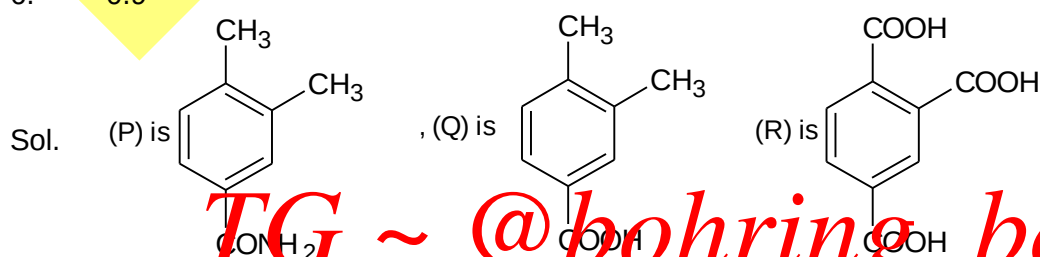
4. 0.22 [Range 0.2 to 0.3]

Sol. $\text{pH} = \text{p}K_a + \log \frac{[\text{CH}_3\text{COONa}]}{[\text{CH}_3\text{COOH}]}$

5. 3

Sol. Sc³⁺ contains no d-electron. So back donation does not take place.

6. 0.9



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7. 0.4

Sol. $K = 4 \times 10^{-3} \text{ ohm}^{-1} \text{ cm}^2$

$$\Lambda_m = K \times \frac{1000}{M}$$

$$= 4 \times 10^{-3} \times \frac{1000}{0.1} = 40 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{40}{100} = 0.4$$

8. 170

Sol. The complex is $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot 2\text{NH}_3$

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SECTION-3 : MATHEMATICS

PART – A

1. AD

Sol. Let $\sqrt{y-x} = t$

$$y-x=t^2$$

$$\frac{dy}{dx} = 2t \frac{dt}{dx} + 1$$

$$2t \frac{dt}{dx} + 1 = \sqrt{t}$$

$$\frac{2t}{\sqrt{t}-1} dt = dx \text{ which leads A and D as solutions}$$

2. ABC

Sol. Let x_0 be the number of objects to the left of the first object chosen, x_1 the number of objects between the first and the second, x_2 the number of objects between the second and the third and x_3 the number of objects to the right of the third object. We have

$$x_0, x_3 \geq 0, x_1, x_2 \geq 1 \text{ and } x_0 + x_1 + x_2 + x_3 = n-3 \quad (1)$$

Put $x_1 = y_1 + 1$ and $x_2 = y_2 + 1$. Then (1) reads as

$$x_0 + y_1 + y_2 + x_3 = n-5 \quad (2)$$

where $x_0, y_1, y_2, x_3 \geq 0$

The number of non-negative integral solution of (2) is ${}^{n-5+3}C_3 = {}^{n-2}C_3$

$$\text{We have } {}^{n-2}C_3 = \frac{1}{6}(n-2)(n-3)(n-4)$$

$$\text{Also, } {}^{n-3}C_3 + {}^{n-3}C_2 = {}^{n-2}C_3$$

3. ABD

Sol. The total number of ways of choosing the ticket is 1000. Let the four digits number on the ticket is 1000.

Let the four digits number on the ticket be $x_1x_2x_3x_4$.

Note that $0 \leq x_1 + x_2 \leq 18$ and $0 \leq x_3 + x_4 \leq 18$.

Also, the number of non-negative integral solutions of $x+y=m$ (with $0 \leq x, y \leq 9$) is $m+1$ if $0 \leq m \leq 9$ and is $19-m$ if $10 \leq m \leq 18$.

Thus, the number of favourable ways

$$= 1 \times 1 \times 1 + 2 \times 2 + \dots + 10 \times 10 + 9 \times 9 + 8 \times 8 + \dots + 1 \times 1$$

$$= 2 \left\{ \frac{9 \times 10 \times 19}{6} \right\} + 100 = 670$$

$$\therefore \text{Probability of required event} = \frac{670}{1000} = 0.67$$

Note that $0.067 > 0.065$ and $0.067 < 0.068$.

4. ACD

$$\text{Sol. } \sum_{k=1}^n \left(\sum_{m=1}^k m^2 \right) = an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \sum_{k=1}^n \left[\frac{k(k+1)(2k+1)}{6} \right] = an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \frac{1}{6} \left\{ 2 \sum_{k=1}^n k^3 + 3 \sum_{k=1}^n k^2 + \sum_{k=1}^n k \right\} = an^4 + bn^3 + cn^2 + dn + e$$

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$$\Rightarrow \frac{1}{6} \left\{ \frac{k^2(k+1)^2}{2} + \frac{k(k+1)(2k+1)}{2} + \frac{k(k+1)}{2} \right\} :$$

$$an^4 + bn^3 + cn^2 + dn + e$$

$$\Rightarrow \frac{n^4}{12} + \frac{n^3}{3} + \frac{5n^2}{12} + \frac{n}{6} = an^4 + bn^3 + cn^2 + dn + e$$

$$\text{Therefore, } a = \frac{1}{12}; b = \frac{1}{3}; c = \frac{5}{12}; d = \frac{1}{6}; e = 0$$

5. AB

Sol: Let $f'(1) = a$ and $f''(1) = b$, putting $x = 1$ in the given equation we get $b = a^2$.

Differentiating the equation and putting $x = 1$ gives

$$2ab + 8 = ab + 2ab \Rightarrow ab = 8 \Rightarrow a = 2, b = 4$$

6. ABD

Sol: $AB - BA$ is skew symmetric. Also in option C, $|A| \neq 0$. Rest are obvious.

7. BD

$$\text{Sol: } (z - (1 + 2i))^4 = 1 \Rightarrow z = 1^{1/4} + 1 + 2i$$

$$z_1 = 1 + 2i + 1 = 2 + 2i = \omega_2$$

$$z_2 = 1 + 2i - 1 = 2i = \omega_3$$

$$z_3 = 1 + 2i + i = 1 + 3i = \omega_1$$

$$z_4 = 1 + 2i - i = 1 + i = \omega_4$$

$\lambda = 5$ (circle on $\omega_1 - \omega_4$ and $\omega_3 - \omega_2$ will overlap)

8. D

Sol: Let there be a value of k for which $x^3 - 3x + k = 0$ has two distinct roots between 0 and 1.

Let a, b be two distinct roots of $x^3 - 3x + k = 0$ lying between 0 and 1 such that $a < b$. Let

$f(x) = x^3 - 3x + k$. Then $f(a) = f(b) = 0$. Since between any two roots of a polynomial $f(x)$

there exist at least one root of its derivative $f'(x)$. Therefore $f'(x) = 3x^2 - 3$ has at least one

root between a and b . But $f'(x) = 0$ has two roots equal to ± 1 which do not lie between a

and b . Hence $f(x) = 0$ has no real roots lying between 0 and 1 for any value of k .

9. A

Sol. Let $\alpha \in \mathbb{R} - \mathbb{Q}$ be a common root of $f(x) = 0$ and $g(x) = 0$.

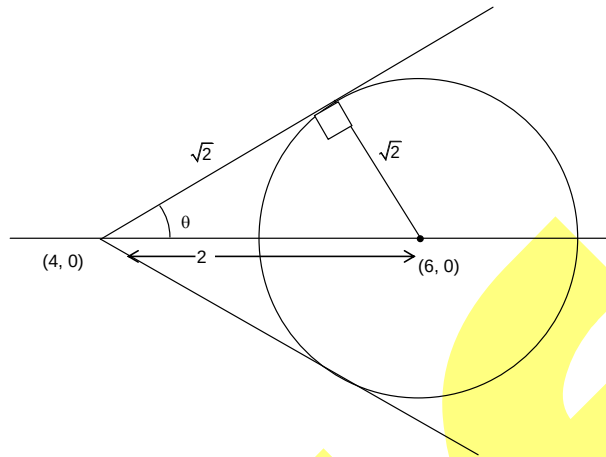
Then $\alpha^2 = -a\alpha - b$. Substituting this in $\alpha^3 + p\alpha + q = 0$, we get $(a^2 - b + p)\alpha + ab + q = 0$

As α is irrational and $a, b, p, q \in \mathbb{Q}$, $p = b - a^2$, $q = -ab$. This gives, $g(x) = (x - a)f(x)$.

10. A

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Sol. From diagram, $\theta = 45^\circ$
 $\Rightarrow \text{Slope} = \pm 1$,



11. D

Sol. $x = -10$ is the directrix of the given parabola. Hence the circle $S = 0$ will touch it. P_1P_2 is the diameter. So $\angle P_1PP_2 = 90^\circ$ and hence ΔP_1PP_2 is right angled and its orthocentre is point P. Tangents at extremities of focal chord are perpendicular. Hence in their combined equation (coefficient x^2) + (coefficient y^2) = 0.

12. C

Sol. Let $\cos^{-1}p = \alpha$, $\cos^{-1}q = \beta$, $\cos^{-1}r = \gamma$
 $\cos \alpha = p$
 $\cos \beta = q$
 $\cos \gamma = r$
 $\alpha + \beta + \gamma = \pi$
 $\Rightarrow \cos(\alpha + \beta) = \cos(\pi - \gamma)$
 $\Rightarrow \cos \alpha \cos \beta - \sin \alpha \sin \beta = -\cos \gamma$
 $\Rightarrow pq - \sqrt{1-p^2} \sqrt{1-q^2} = -r$
 $\Rightarrow pq + r = \sqrt{1-p^2} \sqrt{1-q^2}$
 $2pqr + p^2q^2 + r^2 = (1-p^2)(1-q^2)$
 $p^2 + q^2 + r^2 + 2pqr = 1$

13. C

Sol. Let $x_0 = \cos \theta$, then $x_1 = \sqrt{\frac{1}{2}(1 + \cos \theta)} = \cos \frac{\theta}{2}$, $x_2 = \cos \left(\frac{\theta}{2^2} \right)$, $x_3 = \cos \left(\frac{\theta}{2^3} \right)$ and so on.

$$\begin{aligned} \text{So that } & \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 x_3 \dots \text{to infinite}} \right] \\ &= \frac{\sin \theta}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \\ &= \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\cos \frac{\theta}{2} \cos \frac{\theta}{2^2} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \\ &= \frac{2^2 \sin \frac{\theta}{2^2} \cos \frac{\theta}{2^2}}{\cos \frac{\theta}{2^2} \cos \frac{\theta}{2^3} \dots \cos \frac{\theta}{2^n} \dots \text{infinite}} \end{aligned}$$

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$$= \lim_{n \rightarrow \infty} \frac{2^n \sin \frac{\theta}{2^n}}{\cos \frac{\theta}{2^{n+1}}}$$

$$= \lim_{n \rightarrow \infty} \theta \left(\frac{\sin \frac{\theta}{2^n}}{\frac{\theta}{2^n}} \right) \frac{1}{\cos \frac{\theta}{2^{n+1}}} = \theta$$

$$\text{So that } \cos \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 \dots \inf} \right] = \cos \theta = x_0$$

PART – B

1. 0.33

Sol. $\lim_{n \rightarrow \infty} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$

$$\text{Limit} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{2^2 \left(\frac{x}{2} \right)^2} \frac{e^x - \cos x}{x^{n-2}} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x - \cos x}{x^{n-2}}$$

($n \neq 1$ for then the limit = 0)

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{e^x + \sin x}{(n-2)x^{n-3}}$$

So, if $n=3$, the limit is $\frac{1}{2(n-2)}$ which is finite.

If $n=4$, the limit is infinite.

Answer is 0.33.

2. 55

Sol. $(1-1+2-2+\dots+5-5)^2 =$
 $1^2 + 1^2 + 2^2 + 2^2 + \dots + 5^2 + 5^2 + 2S.$

3. 1

Sol. $\Delta = \sqrt{s(s-a)(s-b)(s-c)} = 6$

$$\therefore r = \frac{\Delta}{s} = 1$$

Answer is 1.

4. 0.5

Sol. Cubing on both sides apply
 $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$

5. 7

Sol. The problem corresponds to a selection of 4 non-consecutive integers out of first 10 natural numbers. Therefore $N = {}^7C_4$.

6. 1

Sol: Given expression = $\frac{(s-b)(s-c)}{(a-b)(a-c)} + \frac{(s-a)(s-c)}{(b-a)(b-c)} + \frac{(s-a)(s-b)}{(c-a)(c-b)} = 1$

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7. 0.33

Sol: $\frac{4000}{a_1 a_{4001}} = 10 \Rightarrow a_1 a_{4001} = 400$

also $a_1 + a_{4001} = 50$

$$\Rightarrow |a_1 - a_{4001}|^2 = 2500 - 1600$$

$$\Rightarrow |a_1 - a_{4001}| = 30$$

8. 9

Sol. Equation of the tangent at P $(7 \cos \theta, 5 \sin \theta)$ on the ellipse is $\frac{x}{7} \cos \theta + \frac{y}{5} \sin \theta = 1$

$$\begin{aligned} \text{then } (CF)^2 &= \frac{7^2 \times 5^2}{5^2 \cos^2 \theta + 7^2 \sin^2 \theta} \\ &= \frac{25 \times 49}{25 \cos^2 \theta + 49 \sin^2 \theta} \end{aligned}$$

Equation of the normal at P is $\frac{7x}{\cos \theta} - \frac{5y}{\sin \theta} = 7^2 - 5^2$

Coordinates of G are $\left(0, \frac{-24 \sin \theta}{5}\right)$

$$\begin{aligned} (PG)^2 &= (7 \cos \theta)^2 + \left(5 \sin \theta + \frac{24 \sin \theta}{5}\right)^2 \\ &= \frac{49}{25} (25 \cos^2 \theta + 49 \sin^2 \theta) \end{aligned}$$

So $(CF \cdot PG)^2 = (49)^2 = 2401$

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Pattern – 2

QP Code: 100192

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 195

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+4 marks** for the correct answer and **– 2 marks** for wrong answer.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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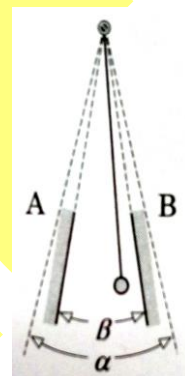
SECTION-1 : PHYSICS

PART – A

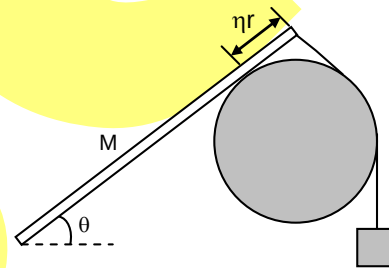
(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. A simple pendulum initially oscillating simple harmonically with angular amplitude α and period T_0 is symmetrically confined between two rigid fixed planes A and B making angle $\beta < \alpha$ with each other as shown in the figure.
 - (A) If collisions at both the walls are elastic, period is $T_0(1 - \beta/\alpha)$.
 - (B) If collisions at both the walls are inelastic, period is T_0 .
 - (C) If collision at one wall is elastic and at the other is inelastic, the period is T_0 .
 - (D) If collision at one wall is elastic and at the other is inelastic, the period is less than T_0 .



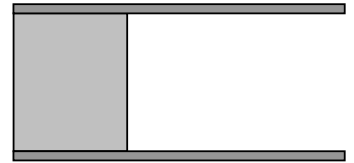
2. A load is suspended from one end of a uniform rod of mass $m = 10 \text{ kg}$ and length ℓ with the help of a light inextensible thin cord and the arrangement is placed on a fixed horizontal frictionless cylinder of radius r as shown in the figure. In equilibrium, the rod stays at angle $\theta = 30^\circ$ with the horizontal and distance between point of contact of the rod with the cylinder and top end of the rod is $\eta = 1/\sqrt{2}$ times the radius of the cylinder. Then
 - (A) ratio of length of the rod to the radius of the cylinder = $\frac{4 + \sqrt{6}}{\sqrt{3}}$
 - (B) ratio of length of the rod to the radius of the cylinder = $\frac{4 + \sqrt{3}}{\sqrt{6}}$
 - (C) mass of the load = 15 kg
 - (D) mass of the load = 10 kg



Space For Rough Work

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3. A parallel plate capacitor of capacitance C_0 is charged to a voltage V and then the battery is disconnected. A dielectric covering one-third area of each plate is now inserted as shown in the figure. If charges on the capacitor plates get redistributed such that the portions covered with dielectric and not covered with the dielectric share equal amounts of charge, which of the following statements is/are true?



- (A) Dielectric constant of the dielectric is 2.0.
 (B) Charge appearing due to polarization on the surface of the dielectric is $0.25 C_0 V$.
 (C) Force of electrostatic interaction between portions of the plates covered with dielectric is equal to that between uncovered portions.
 (D) Force of electrostatic interaction between the plates after insertion of the dielectric becomes $9/8$ times of its value before insertion of the dielectric.

4. Four waves of the equation given by

$$y_1 = 5A \sin (wt - Kx + \pi/2)$$

$$y_2 = 2A \sin (wt - Kx + 3\pi/2)$$

$$y_3 = 6A \sin (wt - Kx)$$

$$\text{and } y_4 = 2A \sin (wt - Kx + \pi)$$

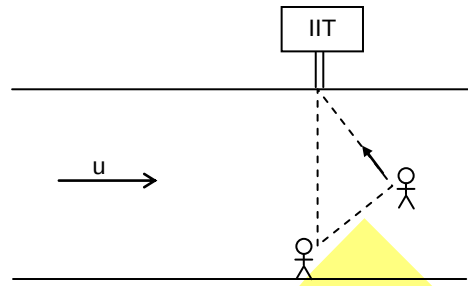
- (A) The resulted wave will have amplitude $5A$.
 (B) It will make a phase difference of 37° with y_3 .
 (C) Its equation will be $y = 5A \sin (wt + 37^\circ)$.
 (D) Its equation will be $y = 5A \sin (wt - Kx + 37^\circ)$.

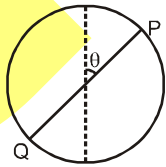
5. During an experiment, an ideal gas is found to obey a condition $\frac{P^2}{\rho} = \text{constant}$ [ρ = density of the gas]. The gas is initially at temperature T , pressure P and density ρ . The gas expands such that density changes to $\rho/2$
 (A) The pressure of the gas changes to $\sqrt{2}P$.
 (B) The temperature of the gas changes to $\sqrt{2}T$.
 (C) The graph of the above process on the P - T diagram is parabola.
 (D) The graph of the above process on the P - T diagram is hyperbola.

Space For Rough Work

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6. There is a board on which 'IIT' is written. It is situated on the opposite side of a river while IIT-JEE aspirant swim the river such that he/she always approach the IIT board [i.e. he/she always try to swim towards the board]. If speed of river and speed of aspirants w.r.t. river are same and width of river is d .



- (A) drift suffered by the aspirant will be $\frac{d}{4}$
 (B) drift suffered by the aspirant will be $\frac{d}{2}$
 (C) If IIT board is taken away from the bank by distance ' d ' in perpendicular direction of river flow then drift suffered will be $\frac{4d}{3}$
 (D) If IIT board is taken away from the bank by distance ' d ' in perpendicular direction of river flow then drift suffered will be $\frac{3d}{4}$
7. A bead is sliding down a smooth, massless rod PQ attached on the circumference of a vertical circle of radius R . The bead starts from rest from P:
- 
- (A) Time taken by bead to go from P to Q is $2\sqrt{\frac{R \cos \theta}{g}}$
 (B) Time taken by bead to go from P to Q is $2\sqrt{\frac{R}{g \cos \theta}}$
 (C) The velocity of bead when it reaches centre of circle is $\sqrt{2gR \cos \theta}$
 (D) The velocity of bead when it reaches centre of circle is $2\sqrt{gR \cos \theta}$

(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The pair(s) of physical quantities that do not have the same dimensions, is
 (A) Reynolds number and coefficient of friction
 (B) Latent heat and gravitational potential
 (C) Curie and frequency of a light wave
 (D) Planck's constant and torque

Space For Rough Work

TG ~ @bohring_bot

9. In a LR decay circuit, the current at $t = 0$ is 10 A. After 5 s it reduces to 9 A then incorrect statement is
- (A) The time constant is $\frac{5}{\ln \frac{10}{9}}$ (B) The time constant is $5 \ln \frac{10}{9}$
- (C) At $t = 10$ s, the current will be 8.1 A (D) At $t = 15$ s, the current will be 7.3 A
10. A ball of mass m projected horizontally from the top of a tower with kinetic energy K flies in the air for a time interval τ before it hits the horizontal ground. If force of air drag is proportional to the speed, and the proportionality constant is k_0 , find horizontal range of the ball.
- (A) $\frac{\sqrt{Km}}{k_0} \left\{ 1 - e^{-\frac{k_0 \tau}{m}} \right\}$ (B) $\frac{\sqrt{2Km}}{k_0} \left\{ 1 - e^{-\frac{k_0 \tau}{m}} \right\}$
- (C) $\frac{\sqrt{Km}}{2k_0} \left\{ 1 - e^{-\frac{k_0 \tau}{m}} \right\}$ (D) $\frac{\sqrt{4Km}}{k_0} \left\{ 1 - e^{-\frac{k_0 \tau}{m}} \right\}$
11. Find suitable expression for the moment of inertia of a uniform ring of mass m and radius r about an axis that passes through the centre of the ring and makes an angle θ with the plane of the ring.
- (A) $\frac{mr^2}{2}$ (B) $\frac{mr^2}{2} (\sin^2 \theta)$
- (C) $mr^2(1 + \sin^2 \theta)$ (D) $\frac{mr^2}{2} (1 + \sin^2 \theta)$
12. If electric field in a region is radially outward with magnitude $E = Ar$. The charge contained in a sphere of radii r_0 centered at origin is
- (A) $\frac{4\pi\epsilon_0 A}{r_0^3}$ (B) $\frac{A}{4\pi\epsilon_0 r_0^3}$
- (C) $4\pi\epsilon_0 r_0^3 A$ (D) $2\pi\epsilon_0 r_0^3 A$
13. A ball is dropped from a height h above a massive platform that is moving upward with a constant speed u . If the ball collides with the platform elastically, find height above its initial position the ball will rebound. Acceleration of free fall is g .
- (A) $\frac{2u\sqrt{u^2 + gh}}{g}$ (B) $\frac{u\sqrt{u^2 + 2gh}}{g}$
- (C) $\frac{2u\sqrt{u^2 + 2gh}}{g}$ (D) $\frac{u\sqrt{u^2 + 2gh}}{2g}$

Space For Rough Work

TG ~ @bohring_bot

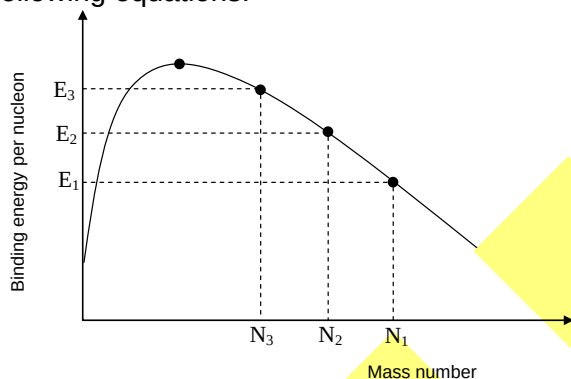
14. A person walking at the rate of 3km/hour, the rain appears to fall vertically when he increase his to speed 6 km/hr it appears to meet him at angle of 45° with vertical. The speed of rain is
- (A) $3\sqrt{2}$ km / hr (B) $\frac{3}{\sqrt{2}}$ km / hr
- (C) $6\sqrt{2}$ km / hr (D) $2\sqrt{3}$ km / hr

(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Consider the nuclear fission reaction $W \rightarrow X + Y$. Using the graph given, answer the following equations:



15. What is the Q value of the reaction?
- (A) $E_1N_1 - (E_2N_2 + E_3N_3)$ (B) $(E_2N_2 + E_3N_3 - E_1N_1)$
- (C) $E_2N_2 + E_1N_1 + E_3N_3 - E_3N_3$ (D) $E_1N_1 + E_3N_3 - E_2N_2$
16. If M_W is the mass of W, M_X is mass of X and M_Y is mass of Y nucleus, choose correct statement.
- (A) $\frac{M_W}{N_1} > \frac{M_Y}{N_2} > \frac{M_X}{N_3}$ (B) $\frac{M_W}{N_1} < \frac{M_Y}{N_2} < \frac{M_X}{N_3}$
- (C) $\frac{M_W}{N_1} < \frac{M_Y}{N_2} > \frac{M_X}{N_3}$ (D) $\frac{M_W}{N_1} > \frac{M_Y}{N_2} > \frac{M_X}{N_3}$

Space For Rough Work

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Paragraph for Question no. 17 to 18

A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest. Its length is parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane. There is sufficient friction present at the edge so that a very small displacement causes the cylinder to roll off the edge without slipping

17. The angle θ through which the cylinder rotates before it leaves contact with the edge is
(A) $\tan^{-1} \frac{4}{7}$ (B) $\cos^{-1} \frac{7}{4}$ (C) $\cos^{-1} \frac{4}{7}$ (D) $\tan^{-1} \frac{7}{4}$
18. The speed of the centre of mass of the cylinder before leaving contact with the edge is
(A) $\sqrt{\frac{gR}{7}}$ (B) $\sqrt{\frac{4gR}{7}}$ (C) $\sqrt{\frac{9gR}{7}}$ (D) $\sqrt{\frac{3gR}{7}}$

Space For Rough Work

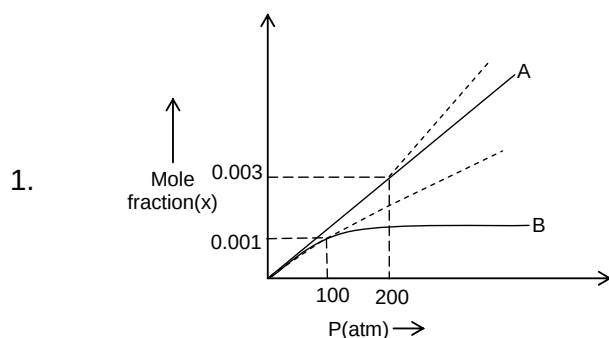
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SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

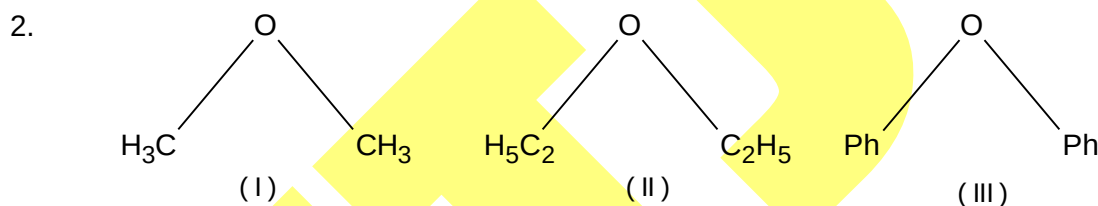
This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.



The solubility of two gases A and B in water at 50°C is plotted above. The variation of mole fraction and partial pressure of the gases is given in above graph.

Choose the correct statements

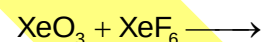
- (A) Gas 'A' follows Henry's law upto 200 atm
- (B) The Henry's law constant for gas B is 10^{-5} atm^{-1}
- (C) Gas 'A' is more soluble than gas 'B'
- (D) Gas 'B' does not follow ideal gas behaviour after 100 atm



Choose correct statements

- (A) the dipole moment of (I) is higher than that of (II)
- (B) (II) is stable in basic medium whereas (III) is stable in acid as well as basic medium
- (C) (I) is more reactive than (III) towards Cl_2 in presence of sunlight
- (D) (II) is more reactive than (III) towards electrophilic substitution reaction

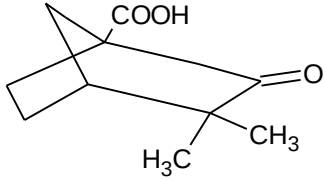
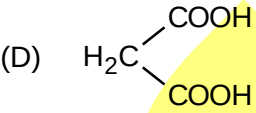
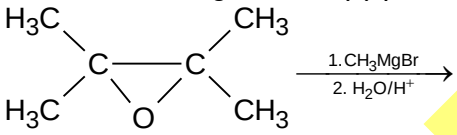
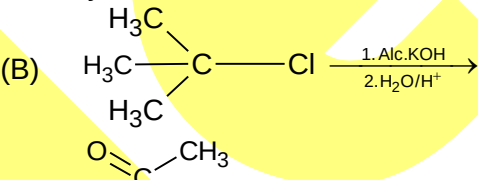
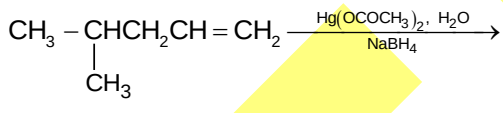
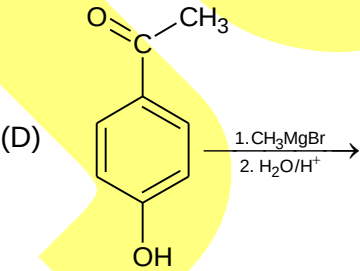
3. Which of the following oxo fluoride(s) is/are formed by the following reaction?



- (A) XeO_2F_2
- (B) XeOF_4
- (C) XeOF_2
- (D) XeO_3F_2

Space For Rough Work

TG ~ @bohring_bot

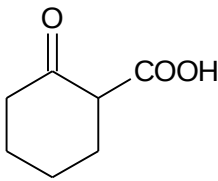
4. An aqueous solution of 50 mL of a weak dibasic acid H_2A of concentration 0.02 M is titrated with 0.1 M NaOH solution $K_{a1} = 10^{-2}$, $K_{a2} = 10^{-5}$
Choose the correct statement [Assume $1 - \alpha = 1$]
(A) the pH of the solution before titration is approx 2
(B) the pH of solution at first equivalence point is 3.5
(C) the volume of NaOH solution consumed in complete titration is 180 mL
(D) the degree of dissociation of HA^- is greater than that of H_2A
5. Which of the following compound(s) undergo(es) decarboxylation on heating?
(A) $(CH_3)_2C(OH)COOH$
(B) $HOOC-CH_2-C(=O)-CH_2-COOH$
(C) 
(D) 
6. Which of the following reaction(s) produce(s) tertiary alcohol?
(A) 
(B) 
(C) 
(D) 
7. Which of the following statement(s) is/are correct for zinc blend structure of ZnS ?
(A) The Zn^{2+} ions occupy half of the tetrahedral voids of the unit cell.
(B) Pure ZnS fluoresces when light falls on it.
(C) If the sulphide ions of ZnS are replaced by, X^{4-} ions, two cationic vacancy will be created
(D) It is more stable than the wurtzite structure of zinc sulphide.

Space For Rough Work

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(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. Reaction of an aqueous solution of $\text{Hg}(\text{NO}_3)_2$ with an aqueous solution of KNO_2 forms an eight-coordinate complex which is obtained as a yellow crystal. The correct statement for the complex is
 (A) it contains Hg – Hg bond (B) it contains eight NO_3^- ligands
 (C) the oxidation state of Hg is +8 (D) it contains NO_2^- as a bidentate ligand
9. $\text{Ag(s)} \mid \text{AgCl(s), KCl(0.2 M)} \parallel \text{KBr(0.001 M), AgBr(s)} \mid \text{Ag(s)}$
 K_{sp} of $\text{AgCl} = 2.8 \times 10^{-10}$
 K_{sp} of $\text{AgBr} = 3.3 \times 10^{-13}$
 What is the emf of the cell?
 (A) -0.41 V (B) -0.037 V
 (C) 0.037 V (D) +0.41 V
10. Which of the following substance undergoes dehydration on heating?
 (A) $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ (B) $\text{CH}_3\text{CH}=\text{NOH}$
 (C)  (D) $\text{CH}_3\text{CH}_2\text{OH}$
11. Which of the following substance is not needed in order to elucidate the open chain structure of glucose?
 (A) $\text{Br}_2/\text{H}_2\text{O}$ (B) $(\text{CH}_3\text{CO})_2\text{O}/\text{catalyst}$
 (C) Conc. $\text{HI}/\text{red 'P'}$ (D) $\text{LiAlH}_4/\text{ether}$
12. $\text{XY}_2(\text{s}) \rightleftharpoons \text{X}(\text{g}) + 2\text{Y}(\text{g})$
 Above reaction takes place by taking only XY_2 in container. The reaction attains equilibrium at 1.2 atm.
 Choose the correct statement
 (A) if the reaction quotient Q_P is 0.12 atm^3 , then the forward reaction will take place
 (B) if the partial pressure of $\text{X}(\text{g})$ is doubled then the partial pressure of $\text{Y}(\text{g})$ will change to $\frac{p_y}{2\sqrt{2}}$
 (C) if the reaction takes place in a larger container at constant temperature then the equilibrium constant K_p will decrease
 (D) addition of XY_2 favours forward reaction

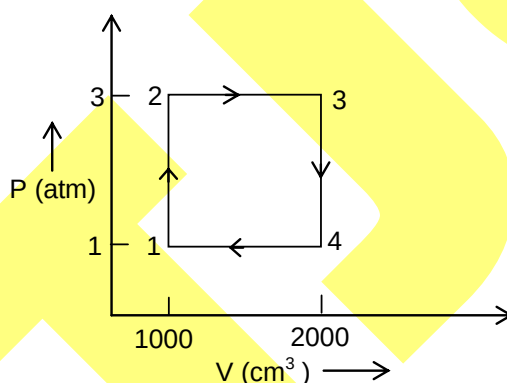
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13. $\text{Co}_2(\text{CO})_8 \xrightarrow{\text{NO}} (\text{X}) + \text{CO}$
 In above reaction 'X' may be
 (A) $\text{Co}(\text{NO})_2(\text{CO})_2$ (B) $\text{Co}(\text{NO})(\text{CO})_3$
 (C) $\text{Co}(\text{NO})_3(\text{CO})_2$ (D) $\text{Co}(\text{NO})(\text{CO})_4$
14. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} \xrightarrow[\text{Heat}]{\text{Anhy. AlCl}_3} \text{Product}$
 The product and type of above reaction is
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$, dimerisation
 (B) $\text{CH}_3\underset{\text{Cl}}{\text{CH}}\text{CH}_3$, isomerisation
 (C) $\text{CH}_3\text{CH}_2\underset{\text{Cl}}{\overset{\text{Cl}}{\text{CH}}}$, nucleophilic substitution
 (D) $\text{CH}_3\text{CH}_2\text{Cl}$, CH_3Cl , pyrolysis

(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

0.1 mole of an ideal gas having $C_{V, M} = \frac{3R}{2}$

(independent of temperature) undergoes a reversible cyclic process $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ as shown above. Each step is accompanied with either constant P or constant V.

Answer the following questions on the basis of above write up.

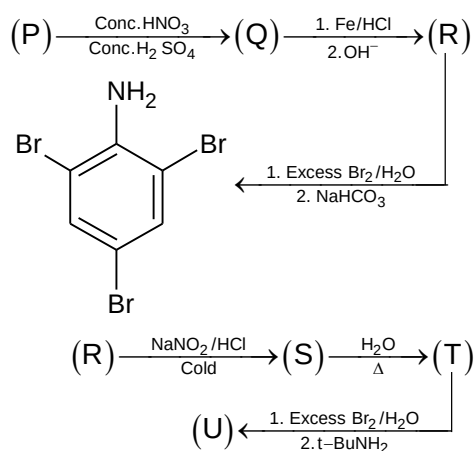
15. How much work is done for the complete cycle in Joule unit?
 (A) -281 (B) -408 (C) -203 (D) -513

Space For Rough Work

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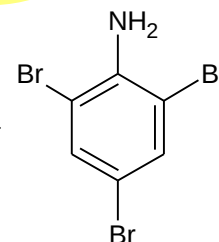
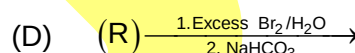
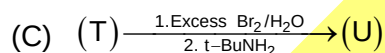
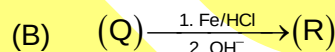
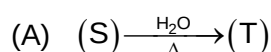
16. What is the sum of the heat absorbed in the steps $1 \rightarrow 2$ and $2 \rightarrow 3$ ($q_{1 \rightarrow 2} + q_{2 \rightarrow 3}$) in joule unit?
 (A) 1218 (B) 1120 (C) 1065 (D) 975

Paragraph for Question no. 17 to 18

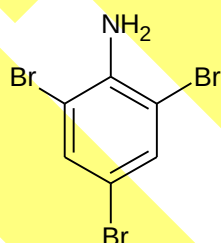
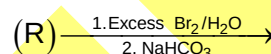


Answer the following question on the basis of above write up

17. Which of the following reaction is accompanied with formation of aromatic side products?



18.



Which of the following change(s) is/are observed if NaHCO_3 is not used in the reaction?

- (I) Rate of reaction will decrease
 (II) Meta orientation products are formed
 (III) Nucleophilic substitution will take place
 (IV) Dehydrogenation reaction will take place

The correct answer(s) is/are

- (A) II, III (B) I, II (C) I, II, III (D) III, IV

Space For Rough Work

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. The sum of two non-negative numbers is $2a$. If P is the probability that the product of these numbers is not less than m times their greatest possible product, then

(A) $P = \frac{2}{3}, m = \frac{4}{5}$ (B) $P = \frac{1}{3}, m = \frac{3}{5}$
 (C) $P = \frac{1}{2}, m = \frac{3}{4}$ (D) $P = \frac{3}{4}, m = \frac{7}{16}$

2. If $\theta \in \left[-\frac{\pi}{9}, -\frac{\pi}{36}\right]$, such that $f(\theta) = \tan\left(\theta + \frac{5\pi}{18}\right) + \cos\left(\theta + \frac{5\pi}{18}\right) - \tan\left(\theta + \frac{7\pi}{9}\right)$ has maximum and minimum values as a and b respectively, then

(A) $a = \frac{11}{\sqrt{3}}$ (B) $b = \frac{-3 + 2\sqrt{2}}{2}$
 (C) $b = \frac{2\sqrt{2} + 1}{\sqrt{2}}$ (D) $a = \frac{11}{2\sqrt{3}}$

3. If $a_0 = \sqrt{2} - \sqrt{3} + \sqrt{6}$ and $a_n = \tan\left(\frac{2^{n-3}\pi}{3}\right) + 2$ then the value of

$M = \sum_{n=0}^3 \tan\left(\frac{2^{n-3}\pi}{3}\right) \cdot \sec\left(\frac{2^{n-2}\pi}{3}\right)$ is equal to

(A) $a_4 - a_0$ (B) $a_3 - a_1$
 (C) $a_1 - a_0$ (D) $a_1 - a_0 - 2$

Space For Rough Work

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4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{[x]}{1+n^2} + \frac{3[x]}{2+n^2} + \frac{5[x]}{3+n^2} + \dots + \frac{(2n-1)[x]}{n+n^2} \right) & x \neq \frac{\pi}{2} \\ 1 & x = \frac{\pi}{2} \end{cases}, \text{ where } [.] \text{ denotes GIF, then}$$

which of the following statement(s) is (are) correct?

- (A) $f(x)$ is injective but not surjective
- (B) $f(x)$ is non – differentiable at $x = \frac{\pi}{2}$
- (C) $f(x)$ is discontinuous at all integers and continuous at $x = \frac{\pi}{2}$
- (D) $f(x)$ is unbounded function.
5. If $x \in \mathbb{R}$, then the roots of the equation $x^4 + 4x^3 - 8x^2 + k = 0$ with respect to the values of k , ($k \in [0, 2014]$), are:
- (A) for $k \in [0, 3]$: 4 real and distinct roots
- (B) for $k \in (3, 128)$: two real and two imaginary roots
- (C) for $k \in (128, 583)$: no real root
- (D) for $k \in (0, 3)$: four real and distinct roots
6. If $f(x) = \int_0^x \frac{\sin t}{t} dt$, $x > 0$ then
- (A) $f(x)$ has a local maxima at $x = n\pi$ ($n = 2k$, $k \in \mathbb{I}^+$)
- (B) $f(x)$ has a local minima at $x = n\pi$ ($n = 2k$, $k \in \mathbb{I}^+$)
- (C) $f(x)$ has neither maxima nor minima at $x = n\pi$ ($n \in \mathbb{I}^+$)
- (D) $f(x)$ has local maxima at $x = n\pi$ ($n = 2k - 1$, $k \in \mathbb{I}^+$)
7. Consider the planes $3x - 6y + 2z + 5 = 0$ and $4x - 12y + 3z = 3$. The plane $67x - 162y + 47z + 44 = 0$ bisects the angle between the given planes which
- (A) contains the origin
- (B) is acute
- (C) is obtuse
- (D) none of these

Space For Rough Work

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(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The maximum value of n for which the quadratic equation $a_{n+1}x^2 - 2x\sqrt{\sum_{k=1}^{n+1} a_k^2} + \sum_{k=1}^n a_k = 0$ has real roots for every choice of real numbers $a_1, a_2, \dots, a_{n+1}$ is
 (A) 2 (B) 3
 (C) 4 (D) 5
9. Let $f : \mathbb{R} \rightarrow \mathbb{R} - \{3\}$ be a function which satisfies $f(x+10) = \frac{f(x)-5}{f(x)-3}$, then $\int_{10}^{20} f(x) dx$ is equal to
 (A) $\int_{100}^{110} f(x) dx$ (B) $\int_{110}^{120} f(x) dx$
 (C) $\int_{120}^{130} f(x) dx$ (D) $\int_{130}^{140} f(x) dx$
10. C the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The tangents at any point P on this hyperbola meets the straight lines $bx - ay = 0$ and $bx + ay = 0$ in the points Q and R respectively. Then CQ . CQ . CR =
 (A) $a^2 + b^2$ (B) $a^2 - b^2$
 (C) $\frac{1}{a^2} + \frac{1}{b^2}$ (D) $\frac{1}{a^2} - \frac{1}{b^2}$
11. The ratio of the areas of two regions in which the curve $C_1 \equiv 4x^2 + \pi^2 y^2 = 4\pi^2$ is divided by the curve $C_2 \equiv y = -\left(\operatorname{sgn}\left(x - \frac{\pi}{2}\right)\right)\cos x$ is equal to (where sgn denotes signum function)
 (A) $\frac{\pi^2 - 2}{\pi^2 + 2}$ (B) $\frac{\pi^2 + 4}{\pi^2 - 2\sqrt{2}}$
 (C) $\frac{\pi^2 + 3\sqrt{2}}{\pi^2 + 6}$ (D) $\frac{\pi^2 - \sqrt{2}}{\pi^2 + 1}$

Space For Rough Work

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12. The plane $\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1$ intersects x-axis, y-axis and z-axis at points A, B and C respectively. The distance between origin and orthocentre of $\triangle ABC$ is equal to
 (A) $\frac{1}{\sqrt{7}}$ units (B) $\frac{6}{\sqrt{7}}$ units
 (C) $\frac{6}{7}$ units (D) none of these
13. If the equation $4x^4 - ax^3 + bx^2 - cx + 6 = 0$, $a, b, c \in \mathbb{R}^+$ has four distinct real roots say $x_1 < x_2 < x_3 < x_4$ such that $\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4} = 8$, then $\frac{x_4}{x_1}$ is equal to
 (A) 2 (B) 3
 (C) 4 (D) 6
14. The perpendicular bisectors $x + y + 2 = 0$ and $x - y - 1 = 0$ of sides AB and AC of a triangle ABC intersect them at $(-1, -1)$ and $(2, 1)$ respectively. If the midpoint of side BC is P, then the distance of P from the orthocentre of triangle ABC is
 (A) $\sqrt{85}$ (B) $\sqrt{41}$
 (C) $\sqrt{13}$ (D) none of these

(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Consider the equation $x^2 - [x^2] = (x - [x])^2$ where $[.]$ denotes Greatest integer function. For $n \in \mathbb{N}$, let $f(n)$ denote the number of roots of the above equation in the interval $1 \leq x \leq n$. Answer the following questions:

15. The number of solutions of the equation $f(n) = n$ is
 (A) 0 (B) 1
 (C) 2 (D) none of these

Space For Rough Work

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16. The value of $\lim_{n \rightarrow \infty} \left(\sqrt{f(n)} - \left[\sqrt{f(n)} \right] \right)$ (where $[.]$ denotes GIF) is
- (A) 1 (B) $\frac{1}{4}$
 (C) $\frac{1}{2}$ (D) none of these

Paragraph for Question no. 17 to 18

Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be unit vectors such that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$ and $\vec{d} \cdot \vec{a} = \vec{d} \cdot \vec{b} = \vec{d} \cdot \vec{c} = \cos \alpha$.

17. If $\theta = \frac{\pi}{3}$ then α is equal to
- (A) $\frac{\pi}{4}$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 (C) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (D) $\frac{\pi}{2}$
18. If $\theta = \cos^{-1}\left(\frac{1}{4}\right)$ then α is equal to
- (A) $\frac{\pi}{4}$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
 (C) $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$ (D) $\frac{\pi}{2}$

Space For Rough Work

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Q.P. Code: 100192**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|-------|-------|--------|--------|
| 1. BC | 2. AC | 3. ABD | 4. ABD |
| 5. BD | 6. BD | 7. BC | 8. D |
| 9. B | 10. B | 11. D | 12. C |
| 13. C | 14. A | 15. B | 16. A |
| 17. C | 18. B | | |

SECTION-1 : CHEMISTRY**PART – A**

- | | | | |
|---------|--------|-------|-------|
| 1. ABCD | 2. ABC | 3. AB | 4. AB |
| 5. ABD | 6. AB | 7. AD | 8. D |
| 9. B | 10. A | 11. D | 12. A |
| 13. B | 14. B | 15. C | 16. C |
| 17. A | 18. B | | |

SECTION-1 : MATHEMATICS**PART – A**

- | | | | |
|--------|-------|-------|-------|
| 1. CD | 2. CD | 3. AD | 4. CD |
| 5. BCD | 6. BD | 7. AB | 8. C |
| 9. D | 10. A | 11. A | 12. C |
| 13. C | 14. C | 15. B | 16. C |
| 17. C | 18. A | | |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **BC**

Sol. If collisions at both the walls are inelastic, period is T_0 .

If collision at one wall is elastic and at the other is inelastic, the period is T_0 .

2. **AC**

Sol. $\frac{\ell}{r} = \frac{4 + \sqrt{6}}{\sqrt{3}}$

$$\text{Mass of load} = \frac{m(1 + \eta^2)}{(1 - \eta^2)} \sin \theta = 15 \text{ kg.}$$

3. **ABD**

Sol. Dielectric constant = $\frac{\ell - \frac{\ell}{3}}{\frac{\ell}{3}} = 2$.

Charge appearing due to polarization on the surface of the dielectric is $0.25 C_0 V$.

4. **ABD**

Sol. Resultant amplitude = $\sqrt{(5A - 2A)^2 + (6A - 2A)^2} = 5A$

$$\tan \phi = \frac{3A}{4A} ; \phi = 37^\circ$$

$$\text{Resultant wave} = 5A \sin (\omega t - kx + 37^\circ)$$

5. **BD**

Sol. $PT = \text{constant}$

$$P_i = P \sqrt{\frac{\rho}{2}} = \frac{P}{\sqrt{2}}$$

$$T_i = T \sqrt{\frac{\rho}{2}} = T\sqrt{2}$$

6. **BD**

Sol. Drift = $\frac{d}{2}$, when board at bank of river.

Say drift = x , when board taken away from bank by a distance 'd'.

$$2d - x = \sqrt{d^2 + x^2}$$

$$4d^2 + x^2 - 4dx = d^2 + x^2$$

$$4dx = 3d^2$$

$$x = \frac{3d}{4}$$

7. **BC**

Sol.

$$PQ = 2R$$

$$a = g \cos \theta$$

$$u^2 = 2(g \cos \theta) (2R)$$

$$u = 2\sqrt{gR \cos \theta}$$

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$$t = \frac{u}{a} = \frac{2\sqrt{gR\cos\theta}}{g\cos\theta} = 2\sqrt{\frac{R}{g\cos\theta}}$$

In reaching upto centre.

$$u^2 = 2(g\cos\theta) \cdot R = \sqrt{2gR\cos\theta}.$$

8. **D**

Sol. Reynolds number and coefficient of friction are dimensionless.

Latent heat and gravitational potential both have dimension $[L^2T^{-2}]$

Curie and frequency of a light wave both have dimension $[T^{-1}]$. But dimensions of Planck's constant is $[ML^2T^{-1}]$ and torque is $[ML^2T^{-2}]$.

9. **B**

Sol. $I = I_0(e^{-Rt/L})$

10. **B**

Sol. $x = \frac{\sqrt{2Km}}{k_0} \left\{ 1 - e^{\frac{-k_0 x}{m}} \right\}$

At $t = \tau$, $x = R$

$$R = \frac{\sqrt{2Km}}{k_0} \left\{ 1 - e^{\frac{-k_0 R}{m}} \right\}$$

11. **D**

Sol. $I = \frac{mr^2}{2} + \frac{mr^2 \sin^2 \theta}{2}$

$$= \frac{mr^2}{2} (1 + \sin^2 \theta)$$

12. **C**

Sol. $\oint \vec{E} \cdot d\vec{s} = \frac{q_{en}}{\epsilon_0}$

$$A r_0 \cdot 4\pi r_0^2 = \frac{q}{\epsilon_0}$$

$$q = 4\pi\epsilon_0 A r_0^3$$

13. **C**

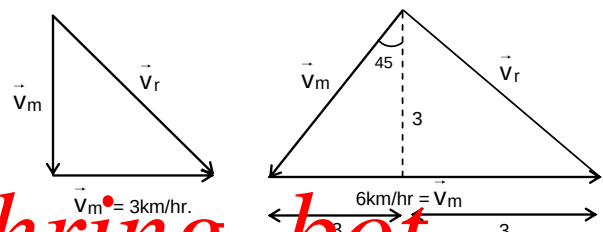
Sol. $H = \frac{(u + \sqrt{u^2 + 2gh})^2}{2g} - h + \frac{\sqrt{u^2 + 2gh} - u}{g}$

$$= \frac{2u\sqrt{u^2 + 2gh}}{g}$$

14. **A**

Sol. $\vec{v}_r$ should be same in both the situations.

Thus, $|\vec{v}_r| = 3\sqrt{2} \text{ km/hr.}$



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15. **B**Sol. W correspond to N_1 , X and Y correspond to N_2 and N_3 and Q value = $E_2N_2 + E_3N_3 - E_1N_1$ 16. **A**Sol. Binding energy per nucleon for N_1 is least hence mass of nucleus of W is greatest.17. **C**Sol. $\frac{3}{4}MV^2 = mgr(1 - \cos \theta)$ and $Mg \cos \theta = \frac{MV^2}{r} \Rightarrow \cos \theta = \frac{4}{7}$ 18. **B**Sol. $V = \sqrt{gR \cos \theta} = \sqrt{\frac{4}{7}gR}$

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SECTION-2 : CHEMISTRY

PART – A

1. ABCD

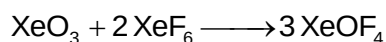
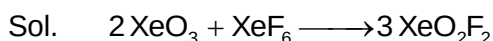
Sol. Henry's law is followed by ideal gas for which mole fraction is directly proportional to partial pressure.

$$x = K_H P$$

$$\therefore k_h = \frac{x}{p} = \frac{10^{-3}}{100} = 10^{-5} \text{ atm}^{-1}$$

2. ABC

3. AB



4. AB

Sol. Before titration, the pH of acid is calculates as

$$[\text{H}^+] = C\alpha = 0.02 \times 0.5 = 0.01$$

$$\therefore \text{pH} = 2$$

At first equivalence point the solution contains HA^-

$$\text{So pH} = \frac{1}{2}(\text{p}^{K_{a1}} + \text{p}^{K_{a2}}) = \frac{1}{2}(2 + 5) = 3.5$$

5. ABD

Sol. (A) forms aldehyde on decarboxylation reaction (C) does not undergo decarboxylation as six-membered ring is not formed.

6. AB

Sol. In (C) 3° -alcohol is not formed
(D) CH_4 is formed

7. AD

Sol. Zn^{2+} ions are present in half of the tetrahedral voids which are present on body diagonals.
Pure ZnS does not fluoresce it does so in presence of impurities like Ag^+ and Cd^{2+}
Anionic vacancy will be created not cationic vacancy.

8. D

Sol. The complex is:
 $\text{K}_2[\text{Hg}(\text{NO}_2)_4] \cdot \text{KNO}_3$

9. B

Sol.
$$E = E^0 - \frac{0.0591}{1} \log \frac{[\text{Ag}^+]_{\text{LHS}}}{[\text{Ag}^+]_{\text{RHS}}}$$

$$= 0 - 0.0591 \log \frac{(2.8 \times 10^{-10}) / 0.2}{(3.3 \times 10^{-13}) / 10^{-3}} = -0.03 \text{ V}$$

10. A

Sol. α -hydroxy acids undergoes dehydration on heating to form a cyclic compound.

11. D

Sol. $\text{Br}_2/\text{H}_2\text{O}$ confirms CHO group
 $(\text{CH}_2\text{CO})_n$ confirms OH group
Conc. H_2SO_4 confirm six membered carbon chain

12. A

Sol. $P_{\text{equ}^m} = 1.2 \text{ atm}$

$$\therefore p_x + p_y = 1.2 \text{ atm}$$

$$\text{or, } p_x + 2p_x = 1.2 \text{ atm}$$

$$\therefore p_x = 0.4 \text{ atm}$$

$$K_p = p_x \times p_y^2 = 0.4 \times (0.8)^2 = 0.256$$

$Q_p < K_p$, so forward reaction takes place

$$K_p = p_x \times p_y^2 = 2p_x \times p_y'^2$$

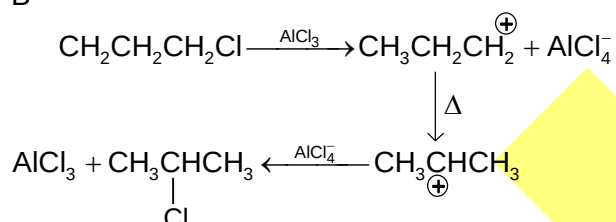
$$p_y'^2 = \frac{p_y^2}{2} \Rightarrow p_y' = \sqrt{\frac{p_y^2}{2}} = \frac{p_y}{\sqrt{2}}$$

13. B

Sol. One NO can substituted three CO groups.

14. B

Sol.



15. C

Sol.

$$\begin{aligned} W_T &= W_{1 \rightarrow 2} + W_{2 \rightarrow 3} + W_{3 \rightarrow 4} + W_{4 \rightarrow 1} \\ &= 0 + \left[-\int_2^3 P dv \right] + 0 + \left[-\int_4^1 P dv \right] \\ &= 0 + (-304 \text{ J}) + 0 + 101 \text{ J} = -203 \text{ J} \end{aligned}$$

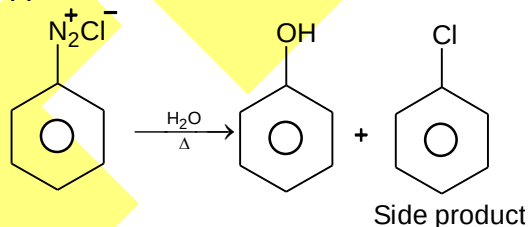
16. C

Sol.

$$\begin{aligned} q_{1 \rightarrow 2} &= \int_1^2 C_v dT = nC_{v,m} \int_1^2 (T_2 - T_1) = 304 \text{ J} \\ q_{2 \rightarrow 3} &= \int_2^3 C_p dT = nC_{p,m} \int_2^3 (T_3 - T_2) = 761 \text{ J} \\ \therefore q_T &= 304 + 761 = 1065 \text{ J} \end{aligned}$$

17. A

Sol.



18. B

Sol.

HBr will be formed during reaction. It is consumed by NaHCO_3 . Unless it is consumed, 'H⁺' will change $-\text{NH}_2$ group to $-\text{NH}_3^+$ which decreases the reaction rate and direct meta substitution.

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SECTION-3 : MATHEMATICS

PART – A

1. CD

Sol: Let numbers be x and y and let $z = xy$

$$z_{\max} = a^2$$

$$\text{Now, } P(xy \geq ma^2) = P(x(2a-x) \geq ma^2) \Rightarrow x^2 - 2ax + ma^2 \leq 0$$

$$\text{For } m = \frac{3}{4}, P = \frac{1}{2}, \text{ and for } m = \frac{7}{16}, P = \frac{3}{4}.$$

2. CD

$$\text{Sol: Let } A = \theta + \frac{5\pi}{18} \Rightarrow A \in \left[\frac{\pi}{6}, \frac{\pi}{4} \right]$$

$$f(A) = \tan A + \cos A + \cot A \Rightarrow a = \frac{11}{2\sqrt{3}}, b = \frac{2\sqrt{2}+1}{\sqrt{2}}$$

3. AD

$$\text{Sol: } M = \sum_{n=0}^3 \tan\left(\frac{2^{n-3}\pi}{3}\right) \cdot \sec\left(\frac{2^{n-2}\pi}{3}\right)$$

$$= \frac{\sin\left(\frac{\pi}{12} - \frac{\pi}{24}\right)}{\cos\frac{\pi}{24} \cdot \cos\frac{\pi}{24}} + \dots + \frac{\sin\left(\frac{2\pi}{3} - \frac{\pi}{3}\right)}{\cos\frac{\pi}{3} \cdot \cos\frac{2\pi}{3}}$$

$$= \tan\frac{2\pi}{3} - \tan\frac{\pi}{24}$$

$$= -\sqrt{3} - \sqrt{2} + \sqrt{3} + 2 - \sqrt{6}$$

$$M = 2 - \sqrt{2} - \sqrt{6}$$

$$a_0 = \sqrt{2} - \sqrt{3} - \sqrt{6}$$

$$a_1 = \tan\frac{\pi}{12} + 2 = 4 - \sqrt{3}$$

$$a_4 = \tan\left(\frac{2\pi}{3}\right) + 2 = 2 - \sqrt{3}$$

$$\text{then } a_1 - a_0 - 2 = 2 - \sqrt{2} - \sqrt{6} \text{ and } a_4 - a_0 = 2 - \sqrt{2} - \sqrt{6}$$

4. CD

$$\text{Sol: } f(x) = \begin{cases} \lim_{n \rightarrow \infty} \left(\frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+1^2} \right) [x], & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases}$$

$$\therefore \frac{1}{n+n^2} + \frac{3}{n+n^2} + \dots + \frac{(2n-1)}{(n+n^2)} < \frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{(n+n^2)} < \frac{1}{1+n^2} + \frac{3}{1+n^2} + \dots + \frac{(2n-1)}{1+n^2}$$

$$\Rightarrow \frac{n^2}{n+n^2} < \frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+n^2} < \frac{n^2}{1+n^2}$$

$$\therefore \lim_{n \rightarrow \infty} \frac{n^2}{n+n^2} = 1 \text{ and } \lim_{n \rightarrow \infty} \frac{n^2}{1+n^2} = 1$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{1}{1+n^2} + \frac{3}{2+n^2} + \dots + \frac{(2n-1)}{n+n^2} \right) = 1$$

$$\therefore f(x) = \begin{cases} [x], & x \neq \frac{\pi}{2} \\ 1, & x = \frac{\pi}{2} \end{cases} \Rightarrow f(x) = [x] \forall x \in \mathbb{R}.$$

Hence range of $f(x)$ is set of all integers.
 $\Rightarrow f(x)$ is unbounded.

Now, at $x = \frac{\pi}{2}$

Clearly, L.H.L = R.H.L = 1 = $f\left(\frac{\pi}{2}\right)$

But $f(x)$ is discontinuous at all integers as $[x]$ is discontinuous at integers.

5. BCD

Sol:

$$x^4 + 4x^3 - 8x^2 = -k$$

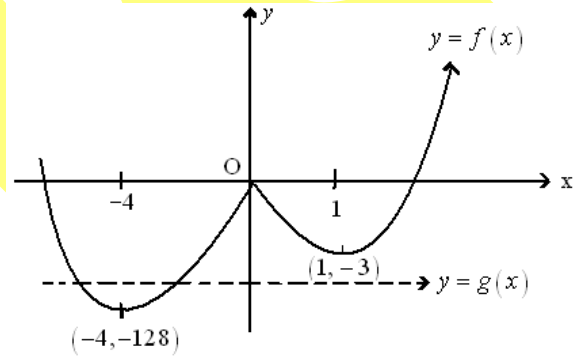
$$\Rightarrow f(x) = -k$$

Where

$$f(x) = x^4 + 4x^3 - 8x^2 = x^2(x^2 + 4x - 8)$$

Let $g(x) = -k$

From the graph the following cases arise :



- When $-3 \leq -k \leq 0, \Rightarrow 0 \leq k \leq 3$
 In this case, $y = -k$ intersect at four points.
- When $-4 \leq -k < -3, \Rightarrow 3 < k \leq 4$
 In this case, $y = -k$ intersect at two points, the given equation has two real roots.
- When $k < 0, \Rightarrow -k > 0$
 In this case, there are two points of intersection. So, the equation has two real roots.

6. BD

Sol. $f'(x) = \frac{\sin x}{x}$

For $f'(x) = 0, \frac{\sin x}{x} = 0 \Rightarrow x = n\pi (n \in \mathbb{I}, n \neq 0)$

$$f''(x) = \frac{x \cos x - \sin x}{x^2}$$

$$f''(n\pi) = \frac{\cos n\pi}{n\pi} < 0 \text{ if } n = 2k - 1 \text{ and } > 0 \text{ if } n = 2k, k \in \mathbb{I}^+$$

Hence, $f(x)$ has local maxima at $x = n\pi$, where $n = 2k - 1$ and local minima at $x = n\pi, n = 2k$, where $k \in \mathbb{I}^+$.

7. AB

Sol: $3x - 6y + 2z + 5 = 0$

$$-4x + 12y - 3z + 3 = 0$$

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Bisectors are $\frac{3x - 6y + 2z + 5}{\sqrt{9 + 36 + 4}} = \pm \frac{-4x + 12y - 3z + 3}{\sqrt{16 + 144 + 9}}$

The plane which bisects the angle between the planes that contains the origin is

$$13(3x - 6y + 2z + 5) = 7(-4x + 12y - 3z + 3)$$

$$67x - 162y + 47z + 44 = 0$$

Further, $3x(-4) + (-6)(12) + 2 \times (-3) < 0$

Hence, the origin lies in the acute angle.

8. C

Sol: Clearly $D_{\min} = 4a_{n+1}^2 \left[1 - \frac{n}{4} \right]$

For real roots, $D_{\min} \geq 0 \Rightarrow n \leq 4$.

9. D

Sol: $f(x)$ is periodic with period 40.

10. A

Sol: P is $(a \sec \theta, b \tan \theta)$

Tangent at P is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$

It meets $bx - ay = 0$ i.e. $\frac{x}{a} = \frac{y}{b}$ in Q

$\therefore$ Q is $\left(\frac{a}{\sec \theta - \tan \theta}, \frac{-b}{\sec \theta - \tan \theta} \right)$

It meets $bx + ay = 0$ i.e. $\frac{x}{a} = -\frac{y}{b}$ in R.

$\therefore$ R is $\left(\frac{a}{\sec \theta + \tan \theta}, \frac{-b}{\sec \theta + \tan \theta} \right)$

$$\therefore CQ \cdot CR = \frac{\sqrt{a^2 + b^2}}{(\sec \theta - \tan \theta)} \cdot \frac{\sqrt{a^2 + b^2}}{(\sec \theta + \tan \theta)}$$

$$= a^2 + b^2, (\because \sec^2 \theta - \tan^2 \theta = 1).$$

11. A

Sol: The areas of the regions are respectively $\pi^2 - 2$ and $\pi^2 + 2$.

12. C

Sol: Orthocentre is the foot of perpendicular from origin to the plane.

13. C

Sol: All roots are positive. Applying $AM \geq GM$ on $\frac{1}{x_1}, \frac{2}{x_2}, \frac{3}{x_3}, \frac{4}{x_4}$ we get $x_1 = \frac{1}{2}$ and $x_4 = 2$.

14. C

Sol: Triangle is right angled. Distance AP is equal to distance between $(-1, -1)$ and $(2, 1)$ which is equal to $\sqrt{13}$.

15. B

16. C

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Sol: (15 & 16)

It is obvious that $x = n$ is a required root. Let $1 \leq \alpha < n$ be another root of the given equation.

Let $m = [\alpha]$ and $t = \{\alpha\}$. Then the equation becomes: $2mt - [2mt + t^2] = 0$.

Therefore $2mt$ is a non-negative integer, i.e. $t = 0, \frac{1}{2m}, \frac{2}{2m}, \dots, \frac{2m-1}{2m}$ for

$m = 1, 2, 3, \dots, n-1$. Hence the number of required roots is

$$f(n) = 2[1 + 2 + 3 + \dots + (n-1)] + 1 = n^2 - n + 1$$

17. **C**18. **A****Sol 17 & 18:**

Obviously, $\vec{d} \parallel (\vec{a} + \vec{b} + \vec{c})$ and since $\vec{d}$ is a unit vector $\therefore \vec{d} = \frac{\vec{a} + \vec{b} + \vec{c}}{|\vec{a} + \vec{b} + \vec{c}|}$

$$\text{Now, } |\vec{a} + \vec{b} + \vec{c}| = \sqrt{3 + 6\cos\theta}$$

$$\text{And } \vec{d} \cdot \vec{a} = \frac{\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}}{\sqrt{3 + 6\cos\theta}}$$

$$\Rightarrow \frac{1 + 2\cos\theta}{\sqrt{3 + 6\cos\theta}} = \cos\alpha$$

$$\Rightarrow \cos\alpha = \frac{1}{\sqrt{3}} \sqrt{1 + 2\cos\theta}$$

$$\therefore \theta = \frac{\pi}{3} \Rightarrow \alpha = \cos^{-1} \frac{\sqrt{2}}{\sqrt{3}} \text{ and } \theta = \cos^{-1} \frac{1}{4} \Rightarrow \alpha = \frac{\pi}{4}$$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 3

QP CODE: 100202

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 204

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & B** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are **not allowed**.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-04)** – Contains **Four (04)** multiple choice questions which have ONLY ONE CORRECT answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **PART-A (05–12)** contains Eight (8) Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (iii) **Part-B (1 – 8)** contains Eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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SECTION – I : PHYSICS

PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

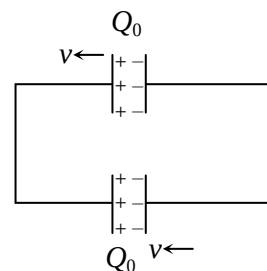
- A ball is projected from a point in one of the two smooth parallel vertical walls against the other in a plane perpendicular to both. After being reflected at each wall impinge again on the second at a point in the same horizontal plane as it started from. If “e” is the coefficient of restitution, “a” is the distance between the walls and “R” is the free range on horizontal surface, then which of the following is correct?

(A) $Re^3 = a(1 + e + e^2)$ (B) $Re = a(1 + e^2 + e^3)$
 (C) $Re^2 = a(1 + e + e^2)$ (D) $R = a(1 + e + e^2)$
- A hemisphere rests in equilibrium on a rough ground and against an equally rough wall, if the equilibrium is limiting, then the inclination θ of the base to the horizontal is

(A) $\theta = \sin^{-1} \left[\frac{3}{8} \left\{ \frac{\mu + \mu^2}{1 + \mu^2} \right\} \right]$ (B) $\theta = \sin^{-1} \left[\frac{8}{3} \left\{ \frac{\mu + \mu^2}{1 + \mu^2} \right\} \right]$
 (C) $\theta = \cos^{-1} \left[\frac{8}{3} \left\{ \frac{\mu + \mu^2}{1 + \mu^2} \right\} \right]$ (D) $\theta = \cos^{-1} \left[\frac{3}{8} \left\{ \frac{\mu + \mu^2}{1 + \mu^2} \right\} \right]$
- A cubical container of side ‘a’ and wall thickness x ($x \ll a$) is suspended in air and filled n moles of diatomic gas (adiabatic exponent γ) in a room where room temperature is T_0 . If at time $t = 0$ gas temperature is T_1 ($T_1 > T_0$), the temperature of gas ‘T’ at time ‘t’ is

(A) $T = T_0 + T_1 e^{-\frac{6ka^2(\gamma-1)t}{nRx}}$ (B) $T = T_0 - (T_1 - T_0) e^{-\frac{6ka^2(\gamma-1)t}{nRx}}$
 (C) $T = T_0 + (T_1 + T_0) e^{-\frac{6ka^2(\gamma+1)t}{nRx}}$ (D) $T = T_0 + (T_1 - T_0) e^{-\frac{6ka^2(\gamma-1)t}{nRx}}$
- Two identical capacitor connected as shown and having initial charge Q_0 each. Separation between plates of capacitor is d. Suddenly the left plate of upper capacitor and right plate of lower capacitor start moving with speed v towards left while other plate of capacitor remains fixed. The current in the circuit is

(A) $\frac{Q_0 v}{2d}$ (B) $\frac{Q_0 v}{d}$ (C) $\frac{Q_0 v t}{2d}$ (D) zero



Space For Rough Work

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PART – A (Maximum Marks: 32)

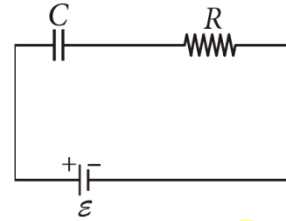
This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A strip of wood of mass M and length ℓ is placed on a smooth horizontal surface. An insect of mass m starts at one end on the strip and walks to the other end in time t , moving with a constant speed.
- (A) the speed of the insect as seen from the ground is $< \frac{\ell}{t}$.
- (B) the speed of the strip as seen from the ground is $\frac{\ell}{t} \left(\frac{M}{M+m} \right)$
- (C) the speed of the strip as seen from the ground is $\frac{\ell}{t} \left(\frac{m}{M+m} \right)$
- (D) the total kinetic energy of the system is $\frac{1}{2}(m+M) \left(\frac{\ell}{t} \right)^2$
6. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50)$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$ then
- (A) the work function of A is 2.25 eV (B) the work function of B is 4.20 eV
- (C) $T_A = 2.00$ eV (D) $T_B = 2.75$ eV
7. A tube of length l and radius R carries a steady flow of fluid whose density is ρ and viscosity is η . The fluid flow velocity depends on the distance r from the axis of the tube as $v = v_0 \left(1 - \frac{r^2}{R^2} \right)$. Choose the correct option(s) from the following.
- (A) the volume of the fluid flowing across the section of the tube per unit time is $\frac{1}{2} \pi^2 v_0 R^2$
- (B) the kinetic energy of the fluid within the tube's volume is $\frac{1}{6} \pi \ell R^2 \rho v_0^2$
- (C) the friction force exerted on the tube the fluid is $\frac{4\pi\eta\ell}{v_0}$
- (D) the pressure difference at the ends of the tube is $\frac{4\eta\ell v_0}{R^2}$

Space For Rough Work

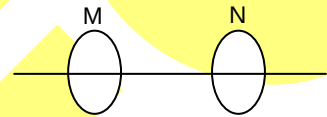
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8. In the following R – C circuit, the capacitor is in the steady state. The initial separation of the capacitor plates is x_0 . If at $t = 0$, the separation between the plates starts changing so that a constant current flows through R, find the velocity of the moving plates as a function of time. The plate area is A.



- (A) $\left(\frac{I/\epsilon_0 A}{IR - \xi}\right) \left[\left(\frac{I/\epsilon_0 A}{\xi - IR}\right)^2 t + \frac{1}{x_0}\right]^{-2}$
- (B) $\left(\frac{I/\epsilon_0 A}{IR - \xi}\right) \left[\left(\frac{I/\epsilon_0 A}{\xi - IR}\right) t + \frac{1}{x_0}\right]^{-2}$
- (C) $\left(\frac{I/\epsilon_0 A}{IR - \xi}\right) \left[\left(\frac{I/\epsilon_0 A}{\xi - IR}\right) t + \frac{1}{x_0}\right]^2$
- (D) $\left(\frac{I/\epsilon_0 A}{IR - \xi}\right) \left[\left(\frac{I/\epsilon_0 A}{\xi - IR}\right)^{-1} t + \frac{1}{x_0}\right]^2$

9. Two identical circular coils M and N are arranged coaxially as shown in the figure. Separation between the coils is large as compared to their radii. The arrangement is viewed from left along the common axis. The sign convention adopted is that currents are taken to be positive when they appear to flow in clockwise direction. Then



- (A) if M carries a constant positive current and is moved towards N, a positive current is induced in N
- (B) if M carries a constant positive current and N is moved towards M, a negative current is induced in N
- (C) if a positive current in M is switched off, a positive current is momentarily induced in N
- (D) if both coils carry positive currents, they will attract each other
10. An external magnetic field is decreased to zero due to which a current is induced in a circular wire loop of radius r and resistance R placed in the field. This current will not become zero.
- (A) at the instant when external magnetic field stops changing ($t = 0$), the current in the loop as a function of time for $t > 0$ is given by $i_0 e^{-2Rt/\mu_2 \pi F}$
- (B) at the instant when B stops changing ($t = 0$), the current in the loop as a function of time $t > 0$ is given by $\frac{\mu_0 IR}{2r}$
- (C) the time in which current in loop decreases to $10^{-3} I_0$ (from $t = 0$) for $R = 100\Omega$ and $r = 5\text{ cm}$ is given by $\frac{3\pi^2 \ln 10}{10^{10}}$
- (D) the time in which current in loop decreases to $10^3 I_0$ (from $t = 0$) for $R = 100\Omega$ and $r = 5\text{ cm}$ is given by $\frac{3\pi^2}{10^6} \text{ s}$

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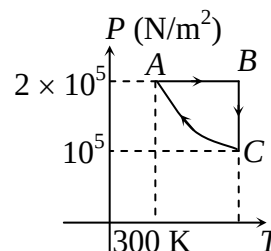
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11. In the series L – C – R circuit, the voltage across resistance, capacitance and inductance are 30V each at frequency $f = f_0$.
- (A) If the inductor is short-circuited, the voltage across the capacitor will be $30\sqrt{2}$ V.
- (B) If the capacitor is short-circuited, the voltage drop across the inductor will be $\frac{30}{\sqrt{2}}$ V.
- (C) If the frequency is changed to $2f_0$, the ratio of reactance of the inductor to that of the capacitor is 4 : 1.
- (D) If the frequency is changed to $2f_0$, the ratio of the reactance of the inductor to that of the capacitor is 1 : 4.
12. A wave disturbance in a medium is described by
 $y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x)$,
 where 'x' and 'y' are in metre and 't' in seconds.
- (A) A node occurs at $x = 0.15$ m
- (B) An antinode occurs at $x = 0.3$ m
- (C) The speed of the component wave is 5.0 m/s
- (D) The wavelength is 0.2 m.

PART – B (Maximum Marks: 24)**(Numerical Type)**

This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. A machine is blowing spherical soap bubbles of different radii filled with helium gas. It is found that if the bubbles have a radius smaller than 1 cm, then they sink to the floor in still air. Larger bubbles float in the air. Assume that the thickness of the soap film in all bubbles is uniform and equal. Assume that the density of soap solution is same as that of water ($=1000 \text{ kg m}^{-3}$). The density of helium inside the bubbles and air are 0.18 kg m^{-3} and 1.23 kg m^{-3} , respectively. If d is the thickness (in μm) of the soap film of the bubbles which are just floating, find the value of $2d$.
2. Find the mass ratio M_D/M_H of deuterium and hydrogen (in near integer value) if their $H\alpha$ lines have wavelengths of $6561.01 \text{ \AA}$ and $6562 \text{ \AA}$ respectively.
3. Two moles of a monatomic ideal gas is taken through a cyclic process shown on pressure(P) - temperature(T) diagram in figure. Process CA is represented as $PT = \text{Constant}$. If efficiency of given cyclic process is $1 - \frac{3x}{12\ln 2 + 15}$, then find the value of 'x'.

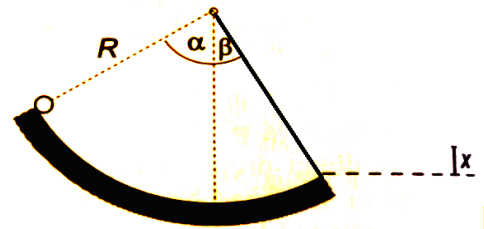


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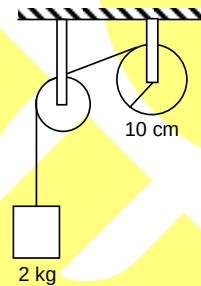
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4. A solid sphere of radius $R = 0.2$ m and of negligible mass is swimming in the water tank of base surface area $A = 0.5$ m². The depth of the water in the tank is $h = 1$ m. Determine the work (in joules) needed to push the sphere down the bottom of tank. (Assume that no water flows out of the tank. The density of water is $\rho = 1000$ kg/m³ and $g = 9.81$ m/s²).

5. A small wheel, initially at rest, rolls down a ramp in the shape of a quarter circle without slipping. The radius of the circle is $R = 1$ m and $\alpha = 60^\circ$, $\beta = 30^\circ$. Find the height 'x' (in cm) reached by the wheel after leaving the track.



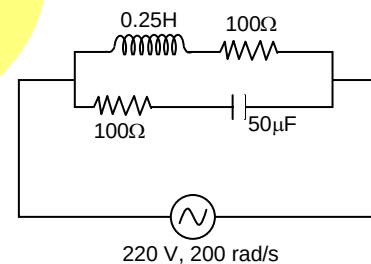
6. A string is wrapped on a wheel of moment of inertia 0.20 kg m² and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. Find the acceleration of the block. (in m/s²)



7. A battery is made by joining identical cells in m rows each row having n cells. The current flowing in an external resistance R is i_1 . Now if the number of rows ' m ' and number of cells in each row ' n ' are interchanged, the current in external resistance is i_2 . If $\frac{m}{2} = 2$ and $\frac{R}{r} = 4$

then the value of $\frac{i_2}{i_1}$ is

8. In the given Ac circuit, what is the power developed in watts?



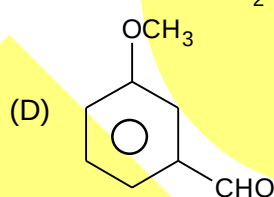
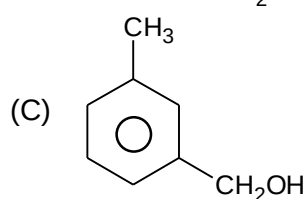
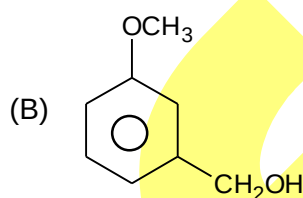
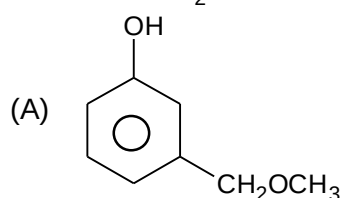
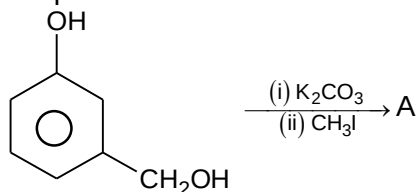
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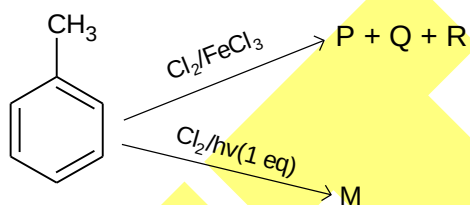
SECTION – II : CHEMISTRY**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. The product A is



- 2.



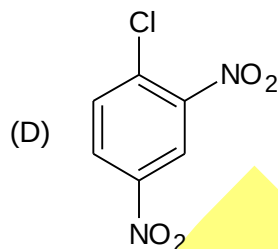
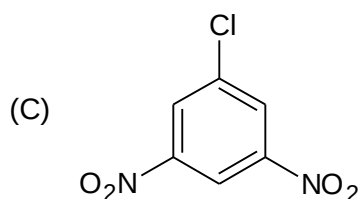
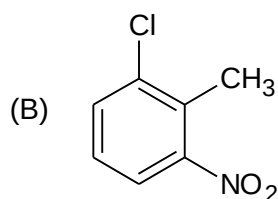
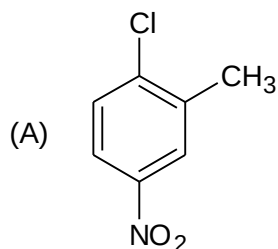
Which product can easily undergo nucleophilic substitution reaction (S_N1 and S_N2 with equal ease).

- (A) P (B) Q
(C) R (D) M

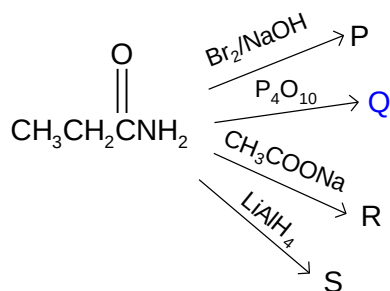
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3. Which is most reactive towards aromatic nucleophilic substitution reaction towards NaOH?



4.



Choose correct statement

- (A) (P) and (S) are chain isomers (B) hydrolysis of (Q) produces carboxylic acid
(C) (R) is obtained by the fastest reaction (D) (S) contains four carbon atom

PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

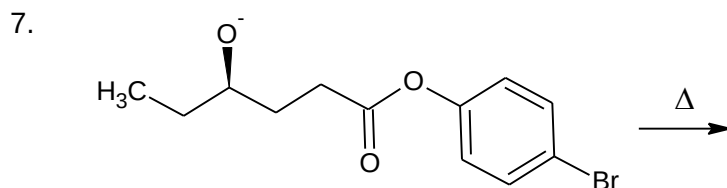
5. Choose incorrect statements:

- (A) AlCl_3 undergoes hydrolysis to form $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$
(B) Amalgamated Al reacts with cold water to liberate H_2
(C) The ash formed on burning magnesium in air fumes on wetting and exposing to HCl
(D) B^{3+} ion can exist in aqueous solution in the hydrated form

Space For Rough Work

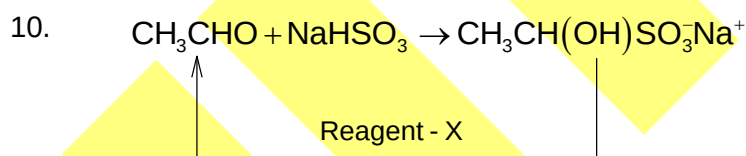
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6. Given $\Delta H_f^\circ (\text{C}_2\text{H}_6, \text{g}) = -85 \text{ kJ/mole}$,
 $\Delta H_f^\circ (\text{C}_3\text{H}_8, \text{g}) = -104 \text{ kJ/mole}$,
 $\Delta H_{\text{sub}}^\circ (\text{C}, \text{s}) = 718 \text{ kJ/mole}$ & B.E. of $(\text{H}-\text{H}) = 436 \text{ kJ/mole}$.
 Then, in kJ/mole, the
 (A) C – C bond enthalpy is 218 (B) C – H bond enthalpy is 414
 (C) C – C bond enthalpy is 345 (D) C – H bond enthalpy is 448



Which of the statements regarding the product formed in the above reaction is/are correct?

- (A) If Br is replaced with NO_2 rate of reaction increases
 (B) Inversion of configuration takes place.
 (C) Nucleophilic acyl substitution takes place.
 (D) Configuration at chiral carbon does not change
8. Which of the following compounds are more reactive than $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ towards attack of nucleophile?
 (A) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$ (B) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CN}$
 (C) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ (D) $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
9. Compound/s which contain $3\text{c} - 2\text{e}^-$ bond.
 (A) $(\text{BeH}_2)_n$ (B) $\text{Al}_2(\text{CH}_3)_6$
 (C) B_2H_6 (D) Al_2Cl_6



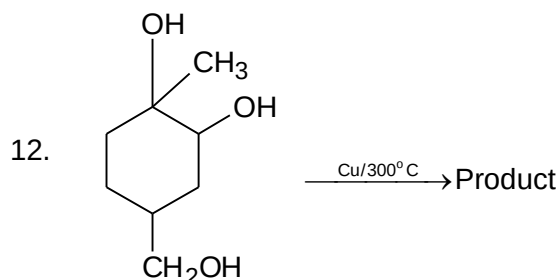
In the above change, the reagent (X) can be

- (A) CCl_4 (B) HCl
 (C) NaOH (D) DMF

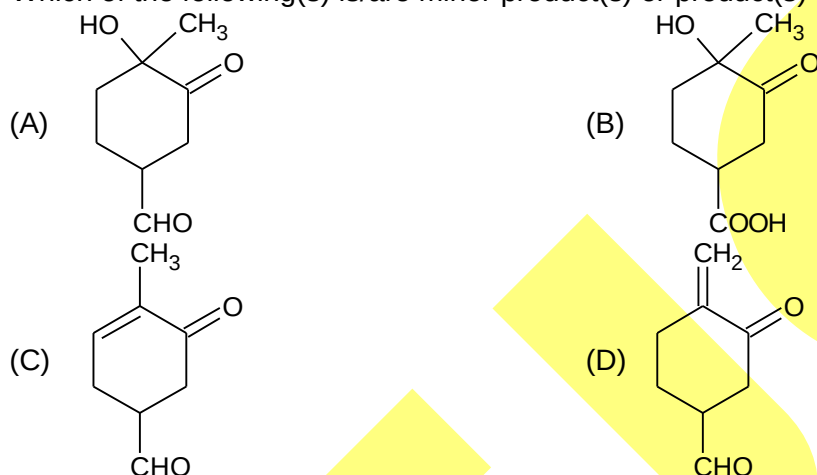
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11. Acetone can form an aromatic compound when treated with conc. H_2SO_4 . Which of the following reaction(s) do(es) not take place?
 (A) oxidation (B) dehydration
 (C) sulphonation (D) dehydrogenation



Which of the following(s) is/are minor product(s) or product(s) that are not formed at all?



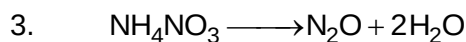
PART – B (Maximum Marks: 24) (Numerical Type)

This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

- 4 moles of a liquid A (V.P = 100 mm of Hg) was mixed with 6 moles of B (V.P = 400 mm of Hg). If the mole fraction of A in the vapour above the solution is expressed as $\frac{x}{y}$, what is the value of $(x + y)$?
- How many moles of O_2 gas can be produced by passing eight Faraday of electricity into sufficient water?

Space For Rough Work

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What is the n-factor of NH_4NO_3 in above reaction?

4. An organic compound(X) upon ozonolysis reaction, produces 2-heptanone and formaldehyde. Reaction of (X) with HCl forms major product(Y) and minor product (Z). Reaction of (Y) with sodium metal in presence of dry ether(Wurtz reaction) forms the only product(P).

If x = the number of quaternary carbon atoms present in (P)

y = the number of monochloro products formed by (P) excluding stereoisomers

z = number of allylic hydrogen atoms present in (X)?

What is the value of $\left(\frac{x+y+z}{10}\right)$?

5. 11.7 g of NaCl solid is added to 176.4 g of H_2O . The degree of dissociation of NaCl in the solution is 0.8. If the relative lowering of vapour pressure of water in the solution is expressed as $x \times 10^{-2}$, what is the value of x?
6. In a mixed metallic oxide, the oxide ion forms the hexagonal close packing(hcp) lattice. The M^{2+} ions occupy $\frac{1}{2}$ of the octahedral voids and the N^{3+} ions occupy $\frac{1}{6}$ th of the tetrahedral voids. If the formula of the solid is written as $\text{M}_x\text{N}_y\text{O}_z$, what is the value of $\left(\frac{x+y+z}{10}\right)$?
7. A 0.5 M aqueous solution of $\text{Al}_2(\text{SO}_4)_3$ is isotonic to a 0.2 M aqueous solution of glucose. If the solute undergoes completely ionization, then the ratio of their osmotic pressure is expressed as x : y. What is the value of $\frac{x}{y}$ in decimal form?
8. 5.35 g of a salt of a strong acid and weak base is dissolved in water to make one litre solution. The pH of the solution is found to be 5. The crystal of the salt contains NaCl type unit cell. What is the mass of one mole of unit cell of the salt in gram unit?
[K_b of the weak base = 10^{-5}]

Space For Rough Work

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SECTION – III : MATHEMATICS**PART – A (Maximum Marks: 12)**

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- A varying parabola 'S' of latus rectum '4a' touches $y^2 = 4ax$, the axis of 'S' being parallel to x – axis. The locus of vertex of 'S' is

(A) $x + y + a = 0$ (B) $y^2 = 16ax$
 (C) $x + y - a = 0$ (D) $y^2 = 8ax$
- On the open interval $(-c, c)$, where c is a positive real number, $y(x)$ is an infinitely differentiable solution of the differential equation $\frac{dy}{dx} = y^2 - 1 + \cos x$, with the initial condition $y(0) = 0$. Then which one of the following is correct?

(A) $y(x)$ has a local maximum at the origin
 (B) $y(x)$ has a local minimum at the origin.
 (C) $y(x)$ is strictly increasing on the open interval $(-\delta, \delta)$ for some positive real number δ
 (D) $y(x)$ is strictly decreasing on the open interval $(-\delta, \delta)$ for some positive real number δ .
- Let $f : (0, \infty) \rightarrow (0, \infty)$, $f(1) = \sqrt{2}$ be a twice differentiable function such that $f'\left(\frac{1}{x}\right) = \frac{1}{f(x)}$ for all x in given domain, then $f(3) =$

(A) $\sqrt{2}$ (B) $\sqrt{10}$
 (C) $3\sqrt{3}$ (D) $\sqrt{3}$
- The values that 'a' can take so that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = x^3 + x^2 + a \cos x$ is injective

(A) $a \in \left(\frac{1}{3}, \infty\right)$ (B) $a \in \left(-\infty, -\frac{1}{3}\right)$
 (C) $a = 2$ (D) No value of a

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PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Equation of ellipse E is $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Suppose C be a circle concentric with E intersecting the ellipse in four points. Let A be a point on E and B be a point on C such that line joining A and B is common tangent of ellipse E and circle C. If ℓ be maximum value of length AB and m be its corresponding slope, then
 (A) $\ell = 2$ (B) $m = \pm\sqrt{2}$
 (C) $\ell = 6$ (D) $m = \frac{\pm 1}{\sqrt{2}}$
6. Let $S = \{w_1, w_2, \dots\}$ be a sample space associated to a random experiment. Let $P(w_n) = \frac{P(w_{n-1})}{2}, n \geq 2$. Let $A = \{2k + 3\ell : k, \ell \in \mathbb{N}\}$ and $B = \{w_n : n \in A\}$. Then
 (A) $P(w_n) = \frac{1}{2^{n-1}}$ (B) $P(B) = \frac{3}{32}$
 (C) $P(w_n) = \frac{1}{2^n}$ (D) $P(B) = \frac{3}{64}$
7. If $|2z + 3| = |3z + 4|$, then
 (A) $|z|_{\min} = 2$ (B) $|z|_{\max} = \frac{7}{5}$
 (C) $|z|_{\max} = 2$ (D) $|z|_{\min} = 1$
8. Let $f : X \rightarrow Y$ be a function. If $P \subset X$ and $Q \subset Y$, define $f(P) = \{f(x) : x \in P\}$ and $f^{-1}(Q) = \{x : f(x) \in Q\}$. Then the true statement(s) is/are:
 (A) $f(f^{-1}(Q)) = Q$ (B) $f^{-1}(Q) \cup f^{-1}(R) = f^{-1}(Q \cup R)$
 (C) $f^{-1}(f(P)) = P$ (D) $f^{-1}(Q) \cap f^{-1}(R) = f^{-1}(Q \cap R)$
9. Consider function $f : A \rightarrow \bar{B}$ and $g : B \rightarrow C$ ($A, B, C \subseteq \mathbb{R}$) such that $(g \circ f)^{-1}$ exists then
 (A) f is ONTO and g is ONE – ONE (B) f is ONE – ONE
 (C) g is ONTO (D) f and g are both ONTO

Space For Rough Work

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10. The function $f(x)$ is differentiable, continuous and $f(x) \neq 0$ for all x on the interval $[4, 8]$.

Also $f(4) = \frac{1}{4}$, $f(8) = \frac{1}{2}$ and $\int_4^8 \frac{(f'(x))^2}{(f(x))^4} dx = 1$. Find $f(6)$.

- (A) $f(6) = \frac{1}{9}$ (B) $f(6) = \frac{1}{3}$
 (C) $f(5) = \frac{2}{7}$ (D) $f(5) = 3$

11. A relation R on the set of complex numbers is defined by $z_1 R z_2$ if and only if $\frac{z_1 - z_2}{z_1 + z_2}$ is

real. Then

- (A) R is reflexive relation (B) R is symmetric relation
 (C) R is transitive relation (D) R is equivalence relation

12. Consider 4 persons A, B, C, D

- (A) If A, B, C, D each have four houses then number of ways in which they can enter these houses so that none of them enter their own house is **9**
 (B) If A has two houses whereas B, C, D have one house each then number of ways in which they can enter these houses such that no one enters their own house is **44**
 (C) If A, B, C, D each have four houses then number of ways in which they can enter these houses so that none of them enter their own house is **16**
 (D) If A has two houses whereas B, C, D have one house each then number of ways in which they can enter these houses such that no one enters their own house is **42**

PART – B (Maximum Marks: 24)

(Numerical Type)

This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. If number of arrangements of letters of the word MISSISSIPPI in which the first S precedes the first I equals α , then $\frac{\alpha}{10}$ equals
2. For all real numbers x , $f(x)$ is an increasing function that is differentiable and satisfies the following conditions
- (i) $f(1) = 1$, $\int_1^2 f(x) dx = \frac{5}{4}$
- (ii) Let $g(x)$ be inverse of $f(x)$, then for all $x \geq 1$, $g(2x) = 2f(x)$.
- If $\int_1^8 x f'(x) dx = \frac{p}{q}$; $\text{HCF}(p, q) = 1$, then $p + q =$

Space For Rough Work

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3. The value of $\left[\frac{\sum_{n=1}^{99} \sqrt{10 + \sqrt{n}}}{\sum_{n=1}^{99} \sqrt{10 - \sqrt{n}}} \right]$ is equal to (here $[.]$ represents greatest integer function) is equal to
4. If the length of the shortest distance between any two opposite edges of the tetrahedron formed by the planes whose equations are $y + z = 0$, $z + x = 0$, $x + y = 0$ and $x + y + z = \sqrt{3}$ is λ , then the value of 2λ is equal to
5. f is differentiable function such that $f(f(x)) = x$ where $x \in [0, 1]$. Also, $f(0) = 1$. Then the value of $2024 \int_0^1 (x - f(x))^{2023} dx$ is
6. Let position vector of points A, B and C of triangle $\triangle ABC$ respectively be $\hat{i} + \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$. Let l_1, l_2 and l_3 be the lengths of perpendiculars drawn from the orthocenter 'O' on the sides AB, BC and CA, then $(l_1 + l_2 + l_3)$ equals.
7. Let $f(x) = \frac{1}{2}x \sin x - (1 - \cos x)$. The smallest positive integer k such that $\lim_{x \rightarrow 0} \frac{f(x)}{x^k} \neq 0$ is
8. Consider the real valued function $h: \{0, 1, 2, \dots, 100\} \rightarrow \mathbb{R}$ such that $h(0) = 5$, $h(100) = 20$ and satisfying $h(i) = \frac{1}{2}(h(i+1) + h(i-1))$ for every $i = 1, 2, \dots, 99$. Then the value of $h(1)$ is

Space For Rough Work

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Answers**SECTION – I : PHYSICS****PART – A**

- | | | | |
|--------|--------|--------|----------|
| 1. C | 2. B | 3. D | 4. B |
| 5. AC | 6. ABC | 7. BCD | 8. B |
| 9. BCD | 10. AC | 11. BC | 12. ABCD |

PART – B

- | | | | |
|---------|---------|---------|-----------|
| 1. 7 | 2. 2 | 3. 7 | 4. 274.00 |
| 5. 4.58 | 6. 0.89 | 7. 1.50 | 8. 629.20 |

SECTION – II : CHEMISTRY**PART – A**

- | | | | |
|--------|--------|---------|---------|
| 1. B | 2. D | 3. D | 4. B |
| 5. AD | 6. BC | 7. ACD | 8. ABD |
| 9. ABC | 10. BC | 11. ACD | 12. ABD |

PART – B

- | | | | |
|--------|--------|---------|--------|
| 1. 8 | 2. 2 | 3. 4 | 4. 1.4 |
| 5. 3.6 | 6. 1.1 | 7. 12.5 | 8. 214 |

SECTION – III : MATHEMATICS**PART – A**

- | | | | |
|-------|--------|----------|--------|
| 1. D | 2. D | 3. B | 4. C |
| 5. AD | 6. CD | 7. BD | 8. BD |
| 9. BC | 10. BC | 11. ABCD | 12. AD |

PART – B

- | | | | |
|--------|---------|---------|---------|
| 1. 924 | 2. 143 | 3. 2 | 4. 2.82 |
| 5. 1 | 6. 1.22 | 7. 4.00 | 8. 5.15 |

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Answers & Solutions

SECTION – I : PHYSICS

PART – A

1. **C**

Sol. $\frac{a}{u_x} + \frac{q}{eu_x} + \frac{a}{e^2 u_x} = \frac{2u_y}{g} \quad \dots(1)$

Horizontal range:

$$R = \frac{2u_x u_y}{g} \quad \dots(2)$$

2. **B**

Sol. $N_1 = \mu N_2 \quad \dots(1)$

$$W = \mu N_1 + N_2 \quad \dots(2)$$

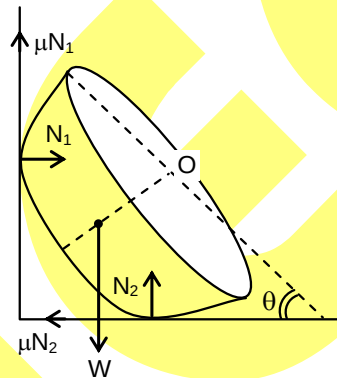
By (1) and (2)

$$N_1 = \frac{\mu W}{1 + \mu^2} \quad \dots(2)$$

Balancing torque about 'O'

$$\mu N_1 \times R + \mu N_2 \times R = W \times \frac{3R}{8} \sin \theta$$

$$\sin \theta = \frac{8}{3} \left[\frac{\mu + \mu^2}{1 + \mu^2} \right]$$

3. **D**

Sol. $\frac{dQ}{dt} = \frac{K(6a^2)(T - T_0)}{x} = n \left(\frac{R}{\gamma - 1} \right) \frac{dT}{dt}$

$$\int_{T_1}^T \frac{dT}{T - T_0} = \frac{6Ka^2(\gamma - 1)}{nRx} \int_0^t dt$$

$$T = T_0 + (T_1 - T_0) e^{\frac{6Ka^2(\gamma - 1)t}{nRx}}$$

4. **B**

Sol. $\frac{q_1}{C_1} = \frac{q_2}{C_2} ; q_1 + q_2 = 2Q_0$

$$C_1 = \frac{\epsilon_0 A}{d_0 + vt} ; C_2 = \frac{\epsilon_0 A}{d_0 - vt}$$

$$\frac{q_1}{q_2} = \frac{d_0 - vt}{d_0 + vt}$$

$$q_2 \left(\frac{d_0 - vt}{d_0 + vt} \right) + q_2 = 2Q_0$$

$$q_2 \left[\frac{2d_0}{d_0 + vt} \right] = 2Q_0$$

$$q_2 = \frac{2Q_0}{2d_0} (d_0 + vt)$$

$$I = \frac{dq_2}{dt} = \frac{Q_0 v}{d_0}$$

5. **AC**

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Sol. Velocity of insect w.r.t. strip = $\frac{\ell}{t}$

Let strip moves with speed v
Initial momentum was 0

$$\Rightarrow 0 = m\left(\frac{\ell}{t} + v\right) + Mv$$

$$v = -\frac{m\ell/t}{M+m} \text{ i.e. } \frac{m\ell/t}{M+m} \text{ toward left}$$

$$\therefore \text{Velocity of insect w.r.t. ground} = \frac{\ell}{t} - \frac{m\ell/t}{M+m} < \frac{\ell}{t}.$$

6. **ABC**

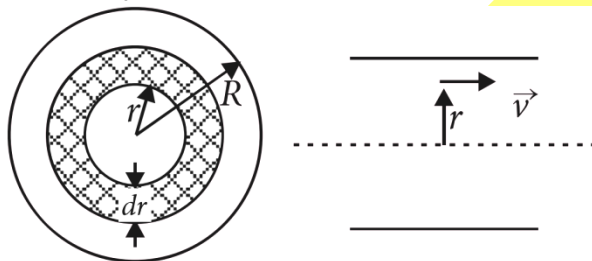
Sol. $KE = h\nu - \phi$

7. **BCD**

Sol. Let dV be the volume flowing per second through the cylindrical shell of thickness dr then,

$$dV = (2\pi r dr)v_0 \left(1 - \frac{r^2}{R^2}\right) = 2\pi v_0 \left(r - \frac{r^3}{R^2}\right) dr \text{ and the total volume,}$$

$$V = 2\pi v_0 \int_0^R \left(r - \frac{r^3}{R^2}\right) dr = 2\pi v_0 \frac{R^2}{4} = \frac{\pi R^2 v_0}{2}$$



Let, dK be the kinetic energy, within the above cylindrical shell. Then

$$dK = \frac{1}{2} (dm)v^2 = \frac{1}{2} (2\pi r \ell \rho) v^2 dr$$

$$= \frac{1}{2} (2\pi \ell \rho) r v_0^2 \left(1 - \frac{r^2}{R^2}\right) dr$$

$$= \pi \ell \rho v_0^2 \left[r - \frac{2r^3}{R^2} + \frac{r^5}{R^4} \right] dr$$

Hence, kinetic energy of the fluid,

$$K = \pi \ell \rho v_0^2 \int_0^R \left(r - \frac{2r^3}{R^2} + \frac{r^5}{R^4} \right) dr = \frac{\ell \pi R^2 \rho v_0^2}{6}$$

Hence frictional force is the shearing force on the tube, exerted by the fluid, which equals $-\eta S \frac{dv}{dr}$, where S is the surface area of tube.

$$\text{Given, } v = v_0 \left(1 - \frac{r^2}{R^2}\right) \text{ So } \frac{dv}{dr} = -2v_0 \frac{r}{R^2}$$

$$\text{and at } r = R, \frac{dv}{dr} = -\frac{2v_0}{R}$$

Then, viscous force is given by,

$$\begin{aligned} dF &= -\eta (2\pi R l) \left(\frac{dv}{dr} \right)_{r=R} \\ &= -2\pi R \eta l \left(-\frac{2v_0}{R} \right) = 4\pi \eta l v_0 \end{aligned}$$

Taking a cylindrical shell of thickness dr and radius r Viscous force, $F = -\eta (2\pi r l) \frac{dv}{dr}$,

Let, ΔP be the pressure difference, the net force on the element $= \Delta P \pi r^2 + 2\pi \eta l r \frac{dv}{dr}$

But, since the flow is steady, $F_{\text{net}} = 0$

$$\text{or } \Delta P = \frac{-2\pi \eta l r \frac{dv}{dr}}{\pi r^2} = \frac{2\pi \eta l r \left(2v_0 \frac{r}{R^2} \right)}{\pi r^2} = \frac{4\eta l v_0}{R^2}$$

8. **B**

Sol. Let q be the instantaneous charge on the capacitor when a steady current I flows through the circuit. Applying KVL on the circuit, we have

$$\xi = \frac{q}{C} + IR \text{ or } \xi = \frac{qx}{\epsilon_0 A} + IR \left(\because C = \frac{\epsilon_0 A}{x} \right) \quad \dots (i)$$

Differentiating eqn. (i) with respect to time, we get

$$0 = \frac{q}{\epsilon_0 A} \left(\frac{dx}{dt} \right) + \frac{Ix}{\epsilon_0 A} + 0 \quad (\because q = It)$$

$$\text{or } q = - \frac{Ix}{(dx/dt)} \left(\text{where } \frac{dx}{dt} = \text{velocity} \right) \quad \dots (ii)$$

From equation (ii) substituting the value of q in equation (i). we have

$$\xi = -I \frac{x^2}{\epsilon_0 A (dx/dt)} + IR$$

$$\text{or } \frac{dx}{dt} = v = \left(\frac{-I/\epsilon_0 A}{\xi - IR} \right) x^2 \quad \dots (iii)$$

$$\text{or } - \frac{dx}{x^2} = \left(\frac{I/\epsilon_0 A}{\xi - IR} \right) dt \quad \dots (iv)$$

Integrating the above expression w.r.t. time, we get

$$\frac{1}{x} - \frac{1}{x_0} = \left(\frac{I/\epsilon_0 A}{\xi - IR} \right) t \quad \dots (v)$$

From eqn (iii) and (v), we get

$$v = \left(\frac{I/\epsilon_0 A}{IR - \xi} \right) \left[\left(\frac{I/\epsilon_0 A}{\xi - IR} \right) t + \frac{1}{x_0} \right]^{-2}$$

9. **BCD**

Sol. Apply Faraday law for direction of induced current.

10. **AC**

Sol. Flux linked with loop due to its own magnetic field,

$$\phi = \frac{\mu_0 I}{2r} (\pi r^2) = \frac{\mu_0 \pi r I}{2}$$

$$\text{emf induced} = \frac{d\phi}{dt} = \epsilon = - \frac{\mu_0 \pi r}{2} \frac{dI}{dt}$$

$$I = \frac{\epsilon}{R} = - \frac{\mu_0 \pi r}{2R} \frac{dI}{dt}$$

$$\int_{I_0}^1 \frac{dI}{I} = - \int_0^t \frac{2R}{\mu_0 \pi r} .dt; I = I_0 e^{-2Rt/\mu_0 \pi r}$$

$$\text{Now, } 10^{-3} I_0 = I_0 e^{-\frac{2Rt}{\mu_0 \pi r}} = I_0 e^{-2Rt/\mu_0 \pi r}$$

$$\text{which gives } t = \frac{3\pi^2 \ln 10}{10^{10}} \text{ s.}$$

11. **BC**

$$V_L = V_C = \frac{E}{2}$$

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$$\Rightarrow x_L = x_C = R$$

when inductor is short circuited

$$Z = \sqrt{R^2 + x_C^2} = \sqrt{2} R$$

$$\therefore I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$$

$$\therefore V_L = i x_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here f_0 is the resonance frequency as $v_L = v_C$

$$\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$$

Given $f = 2f_0$

$$\Rightarrow \omega = 2\omega_0$$

$$\therefore \frac{x_L}{x_C} = 4$$

$\therefore$ (C) is also correct.

12. ABCD

Sol. $y = 2a \cos(\omega t + \phi) \cos(Kx)$ comparison with given equation gives

$$K = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = \frac{1}{5} \text{ m}$$

$$\omega = 2\pi f = 50\pi \Rightarrow f = 25 \text{ Hz}$$

$$\therefore v = 5 \text{ ms}^{-1}$$

at $x = 0.15 \text{ m}$

$$\cos(10\pi \times 0.15) = \cos(1.5\pi)$$

$= -1$ for all t

at $x = 0.3$

$$\cos(10\pi \times 0.3) = \cos 3\pi = -1 \text{ for all } t$$

PART - B

1. 7

Sol. For equilibrium of bubble.

$$4\pi R^2 d \rho_{\text{sol}} g + \frac{4}{3} \pi R^3 \rho_{\text{He}} g = \frac{4}{3} \pi R^3 \rho_{\text{air}} g$$

$d \rightarrow$ Thickness of soap solution.

$$\therefore d = 3.5 \mu\text{m}.$$

2. 2

Sol. Use formula in terms of reduced mass

$$\frac{1}{\lambda} = \frac{R_\infty z^2}{\left(1 + \frac{m}{M}\right)} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\therefore \frac{\lambda_D}{\lambda_H} = \frac{1 + \frac{m}{M_D}}{1 + \frac{m}{M_H}}$$

$$\frac{M_D}{M_H} \approx 2.0$$

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3. 7

Sol. For process AB $T_A = 300 \text{ K}$, $T_B = 600 \text{ K}$

$$W = nR\Delta T = nR(T_B - T_A) = 300 nR = 600R.$$

$$Q = n C_p \Delta T = 2 \times \frac{5}{2} R (300) = 1500R.$$

For process BC

$$W = nRT \ln \frac{V_f}{V_c} = nRT \ln \frac{p_i}{p_f} = nRT \ln 2 = 1200R \ln 2$$

$$Q = W = 1200R \ln 2$$

For process CA

$$W = \int P dV = \int_{600}^{300} \frac{K}{T} \frac{2nRT}{K} dT.$$

$$= -2nR(300) = -1200R.$$

$$Q = nC_v \Delta T + W$$

$$= 2 \times \frac{3}{2} R(-300) - 1200R.$$

$$= -900R - 1200R = -2100R$$

$$\eta = \frac{600R + 1200R \ln 2 - 1200R}{1500R + 1200R \ln 2}$$

$$= 1 - \frac{21}{12 \ln 2 + 15} \Rightarrow x = 7.$$

4. 274.00

$$\text{Sol. } W = \frac{4}{3} \pi R^3 \rho g \left[h - R + \frac{x}{2} \right] = \frac{4}{3} \pi R^3 \rho g \left[h - R + \frac{\frac{4}{3} \pi R^3}{2A} \right]$$

5. 4.58

$$\text{Sol. } h = (R - r) [\cos 30^\circ - \cos 60^\circ]$$

$$mgh = \frac{1}{2} mv^2 + \frac{1}{2} mr^2 \omega^2$$

$$x = \frac{v^2 \sin^2 30^\circ}{2g}$$

6. 0.89

$$\text{Sol. } 2g - T = 2a \quad \dots(i)$$

$$TR = I\alpha \quad \dots(ii)$$

$$a = R\alpha \quad \dots(iii)$$

$$\text{From (ii) and (iii) } T = \frac{Ia}{R^2}$$

$$\therefore 2g = a \left(2 + \frac{I}{R^2} \right)$$

$$\Rightarrow a = \frac{2g}{\left(2 + \frac{I}{R^2} \right)} = \frac{2 \times 9.8}{2 + \frac{0.2}{0.01}} = \frac{9.8}{11} \approx 0.89 \text{ m/s}^2$$

7. 1.50

$$\text{Sol. } i_1 = \frac{E}{\frac{r}{m} + \frac{R}{n}} \quad \& \quad i_2 = \frac{E}{\frac{r}{n} + \frac{R}{m}}$$

$$\frac{i_2}{i_1} = \frac{\frac{r}{m} + \frac{R}{n}}{\frac{r}{n} + \frac{R}{m}} = \frac{1 + \frac{m}{n} \cdot \frac{R}{r}}{\frac{m}{n} + \frac{R}{r}} = \frac{3}{2} = 1.5$$

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8. **629.20**Sol. $X_L = 50 \, \Omega$, $X_C = 100 \, \Omega$

$$Z_1 = 100 - 50i ; Z_2 = 100 + 100i \quad (i = \sqrt{-1})$$

$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{100}{17} (13 + i)$$

$$|Z| = \frac{100}{17} \sqrt{170} ; \cos \theta = \frac{13}{\sqrt{170}}$$

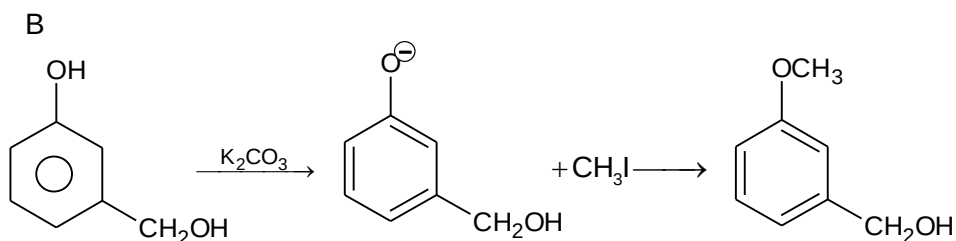
$$P = \frac{V^2}{|Z|} \cdot \cos \phi = 629.20$$

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SECTION – II : CHEMISTRY

PART – A

1. Sol.



2. Sol.

D
P, Q and R are formed by EAS reaction M is formed by free radical chlorination.

3. Sol.

D
Electron withdrawing groups favour nucleophilic substitution when they are ortho and para to leaving group.

4. Sol.

B
P = $CH_3CH_2NH_2$, Q = CH_3CH_2CN , R = Not formed, S = $CH_3CH_2CH_2NH_2$.

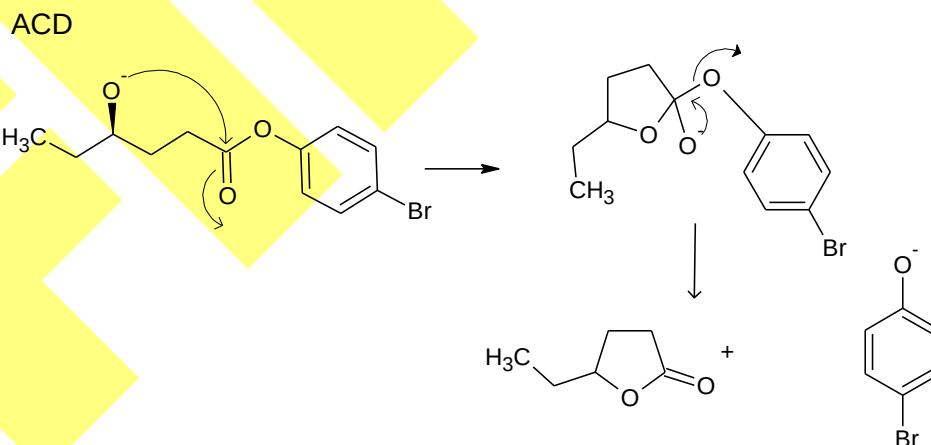
5. Sol.

AD
 $AlCl_3 + 3H_2O \xrightarrow{\text{Hydrolysis}} Al(OH)_3 + 3HCl$
 B^{3+} cannot exist in aqueous media

6. Sol.

BC
 $2C_{(s)} + 3H_{2(g)} \rightarrow C_2H_{6(g)}$
 $\Delta H_f^0 = [2 \times \Delta H_{sub}^0 + 3 \times B.E.(H-H)] - [B.E.(C-C) + 6 \times B.E.(C-H)]$
 $\Rightarrow -85 = [(2 \times 718) + (3 \times 436)] - (x + 6y)$
 $\therefore x + 6y = 2829 \quad \dots\dots 1$
Similarly for $C_3H_8(g)$
 $2x + 8y = 4002 \quad \dots\dots 2$
Solving (1) & (2), $x = 345$
 $y = 414$

7. Sol.



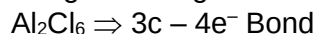
8. Sol.

ABD
 $R-C(=O)-NH_2$ has least +ve charge density among all of the given compound.

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9. ABC

Sol. Bridge bonding

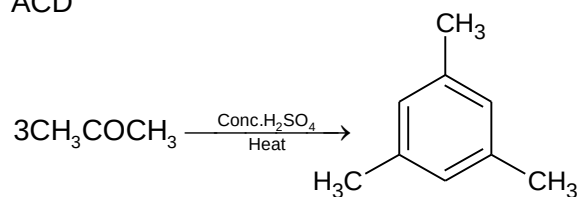


10. BC

Sol. Original carbonyl compound can be generated from product by weak acid or base treatment.

11. ACD

Sol.



12. ABD

Sol. Dehydrogenation reaction takes place.

PART - B

1. 8

$$\text{Sol. } P_T = p_A^0 x_A + p_B^0 x_B = 100 \times \left(\frac{4}{6+4} \right) + 400 \times \left(\frac{6}{6+4} \right) = 280 \text{ mm of Hg}$$

$$p_A = y_A P_T$$

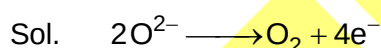
$$\text{or, } p_A^0 x_A = y_A P_T \text{ or, } 100 \times 0.4 = y_A \times 280$$

$$\text{or, } y_A = \frac{40}{280} = \frac{1}{7}$$

$$\therefore \frac{x}{y} = \frac{1}{7}$$

$$\text{so, } (x + y) = 8$$

2. 2

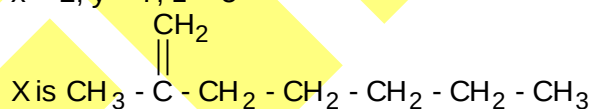
One mole of O_2 produced by passing 4F electricity. $\therefore$ Two moles of O_2 will be produced by passing 8F electricity.

3. 4

Sol. The oxidation number of N changes from -3 to +1 and +5 to +1.

4. 1.4

$$\text{Sol. } x = 2, y = 7, z = 5$$



5. 3.6

$$\text{Sol. } \frac{p^0 - p}{p^0} = iX_{\text{solute}} = (1 + \alpha) \left(\frac{0.2}{10} \right)$$

$$= (1 + 0.8) (0.02) = 0.036$$

$$\therefore x \times 10^{-2} = 0.036 = 3.6 \times 10^{-2}$$

$$\therefore x = 3.6$$

6. 1.1

Sol. Formula of solid $\text{M}_3\text{N}_2\text{O}_6$

$$\therefore \frac{x + y + z}{10} = \frac{3 + 2 + 6}{10} = 1.1$$

~@bohring_bot

7. 12.5

Sol. π of $\text{Al}_2(\text{SO}_4)_3 = \pi$ of glucose

or, $iC_1RT = C_2RT$

or, $iC_1 = C_2$

or, $5 \times 0.5 = 0.2$

Ratio = $\frac{5 \times 0.5}{0.2} = 12.5$

8. 214

Sol. $\text{pH} = \frac{1}{2} [\text{p}^{K_w} - \text{p}^{K_b} - \log C] = \frac{1}{2} [14 - 5 - \log C] = 5$

or $\log C = -1$ or $C = 10^{-1} = 0.1 \text{ M}$

Mass of 0.1 mole of salt = 5.35 g

Mass of one mole of salt = 53.5 g

NaCl unit cell contains 4 units of NaCl

The unit cell of the salt contains 4 units of the salt

Mass of one mole of unit cell = $53.5 \times 4 = 214$

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SECTION – III : MATHEMATICS

PART – A

1. D

Sol. Let the vertex be (h, k) .So, 'S' can be assumed as $(y - k)^2 = -4a(x - h)$ Solving with $y^2 = 4ax$

$$y^2 - 2ky + k^2 = -y^2 + 4ah$$

$$\Rightarrow 2y^2 - 2ky + k^2 - 4ah = 0$$

Disc. = 0 ($\because$ the parabolas touch each other)

$$4k^2 - 8(k^2 - 4ah) = 0$$

$$-k^2 + 8ah = 0$$

$$\Rightarrow y^2 = 8ax$$

2. D

Sol. $y(0) = y'(0) = y''(0) = 0$

$$y'''(0) = -1$$

 $\Rightarrow y'''(x)$ is negative in the neighbourhood of $x = 0$ $\Rightarrow y''(x)$ is a decreasing function in the neighbourhood of $x = 0$. $\Rightarrow$ For $x > 0$

$$y''(x) < y''(0)$$

$$\Rightarrow y''(x) < 0$$

For $x < 0$

$$y''(x) > y''(0)$$

$$y''(x) > 0$$

Clearly, the concavity of curve changes from positive to negative across $x = 0$ So $x = 0$ is point of inflection

So the answer should therefore be (D).

Also, by n -th derivative test if $f^n(c) \neq 0$ (n is odd) then $x = c$ is neither point of maxima nor minima.

3. B

Sol. $f'\left(\frac{1}{x}\right)f(x) = 1$ Replace x by $\frac{1}{x}$, we get $f'(x)f\left(\frac{1}{x}\right) = 1$

$$\text{Differentiating } f''(x)f\left(\frac{1}{x}\right) + f'(x)f'\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) = 0$$

$$\frac{f''(x)}{f'(x)} - \frac{f'(x)}{f(x)} \cdot \frac{1}{x^2} = 0$$

$$\Rightarrow \frac{f''(x)f(x)}{(f'(x))^2} f(x) dx = \int \frac{1}{x^2} dx$$

$$-\frac{1}{f'(x)} f(x) - \int \left(-\frac{1}{f'(x)}\right) f'(x) dx = -\frac{1}{x} + c$$

$$-\frac{f(x)}{f'(x)} + x = -\frac{1}{x} + c$$

$$\text{Put } x = 1 \Rightarrow -\frac{f(1)}{f'(1)} + 1 = -1 + c \Rightarrow c = 2 - \frac{\sqrt{2}}{1} = 2 - 2 = 0$$

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$$\Rightarrow \frac{f(x)}{f'(x)} = x + \frac{1}{x}$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{x}{x^2+1} \Rightarrow \ln(f(x)) = \frac{1}{2} \ln(1+x^2) + c'$$

$$\ln \sqrt{2} = \frac{1}{2} \ln 2 + c' \Rightarrow c' = 0$$

$$\Rightarrow f(x) = \sqrt{1+x^2} \Rightarrow f(3) = \sqrt{10}$$

4. C

Sol. $f'(x) = 3x^2 + 2x - a \sin x$; $f''(x) = 6x + 2 - a \cos x$

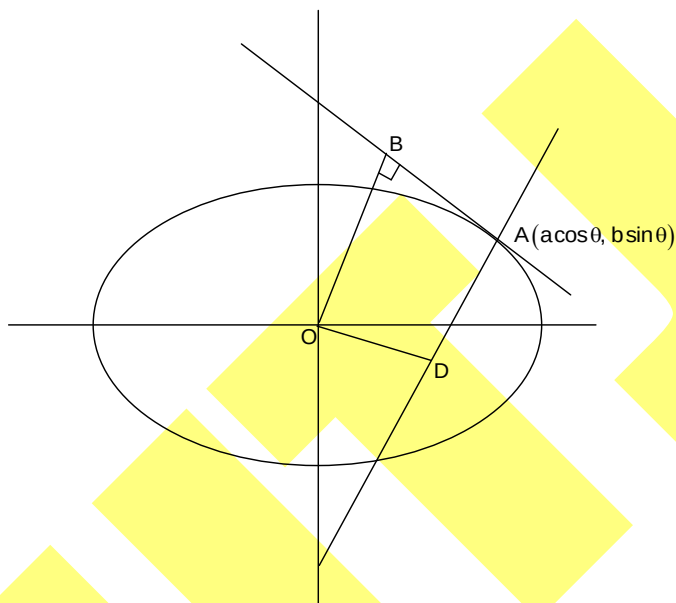
Clearly $f'(0) = 0$

So, for f to be injective $f'(x)$ must attain minima at $x = 0$

$$\Rightarrow f''(0) = 0 \Rightarrow 6(0) + 2 - a = 0 \Rightarrow a = 2$$

5. AD

Sol.



AD is normal to ellipse at A $\Rightarrow$ Equation of AD is $a \sec \theta - b y \operatorname{cosec} \theta = a^2 - b^2$

$$\Rightarrow AB = OD = \frac{a^2 - b^2}{\sqrt{a^2 \sec^2 \theta + b^2 \operatorname{cosec}^2 \theta}} = \frac{a^2 - b^2}{\sqrt{(a+b)^2 + (a \tan \theta - b \cot \theta)^2}} \leq \frac{a^2 - b^2}{a+b}$$

$$\Rightarrow AB \leq a - b \Rightarrow AB \leq 4 - 2 \Rightarrow AB \leq 2 \Rightarrow \ell = 2$$

$$\ell = 2 \text{ happens when } a \tan \theta - b \cot \theta = 0 \Rightarrow \tan^2 \theta = \frac{b}{a} = \frac{1}{2} \Rightarrow \tan \theta = \frac{1}{\sqrt{2}}$$

$$\text{Slope of tangent at A} = -\frac{\cot \theta}{2} = -\frac{1}{\sqrt{2}}$$

6. CD

Sol. Since P is a probability distribution, we have $\sum_{n=1}^{\infty} P(w_n) = 1$

$$P(w_n) = \frac{1}{2}(w_{n-1}) = \frac{1}{2^2} P(w_{n-2}) = \dots = \frac{1}{2^{n-1}} P(w_1)$$

$$\sum_{n=1}^{\infty} P(w_n) = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} P(w_1) = 1 \Rightarrow P(w_1) \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 1 \Rightarrow P(w_1)(2) = 1$$

$$\text{So, } P(w_1) = \frac{1}{2}$$

Clearly, the set A contains all natural numbers greater than 4 (excluding 6)

$$\text{So, } P(B) = 1 - \{P(w_1) + P(w_2) + P(w_3) + P(w_4) + P(w_6)\}$$

$$= 1 - \left\{ \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{64} \right\} = \frac{3}{64}$$

7. BD

$$\text{Sol. } |2z + 3|^2 = |3z + 4|^2 \Rightarrow 4|z|^2 + 9 + 6(z + \bar{z}) = 9|z|^2 + 16 + 12(z + \bar{z})$$

$$\Rightarrow 5|z|^2 + 7 = -6(z + \bar{z}) = -12\text{Re}(z) \leq 12|z|$$

$$\Rightarrow 5|z|^2 - 12|z| + 7 \leq 0 \Rightarrow (5|z| - 7)(|z| - 1) \leq 0 \Rightarrow 1 \leq |z| \leq \frac{7}{5}$$

8. BD

Sol.

$$(B) \text{ If } x \in f^{-1}(Q) \cup f^{-1}(R) \Rightarrow x \in f^{-1}(Q) \text{ OR } x \in f^{-1}(R) \Rightarrow f(x) \in Q \text{ OR } f(x) \in R$$

$$\Rightarrow f(x) \in Q \cup R \Rightarrow x \in f^{-1}(Q \cup R)$$

$$\text{If } x \in f^{-1}(Q \cup R) \Rightarrow f(x) \in Q \text{ or } f(x) \in R \Rightarrow x \in f^{-1}(Q) \text{ or } x \in f^{-1}(R) \Rightarrow x \in (f^{-1}(Q) \cup f^{-1}(R))$$

$$\Rightarrow f^{-1}(Q) \cup f^{-1}(R) = f^{-1}(Q \cup R)$$

$$(D) \text{ If } x \in f^{-1}(Q) \cap f^{-1}(R) \text{ then } x \in f^{-1}(Q) \text{ and } x \in f^{-1}(R) \Rightarrow f(x) \in Q \text{ and } f(x) \in R.$$

$$\Rightarrow f(x) \in Q \cap R \Rightarrow x \in f^{-1}(Q \cap R)$$

$$\text{If } x \in f^{-1}(Q \cap R) \text{ then } f(x) \in Q \cap R \Rightarrow f(x) \in Q \text{ and } f(x) \in R \Rightarrow x \in f^{-1}(Q) \text{ and } x \in f^{-1}(R)$$

$$\Rightarrow x \in (f^{-1}(Q) \cap f^{-1}(R))$$

$$\Rightarrow f^{-1}(Q) \cap f^{-1}(R) = f^{-1}(Q \cap R)$$

$$(A) \text{ Consider } f: \{1, 2, 3\} \rightarrow \{1, 2, 3\}$$

$$\text{where } f(1) = f(2) = f(3) = 1$$

$$\text{If } y \in (f^{-1}(Q))$$

$$\text{the } f^{-1}(y) \in f^{-1}(Q)$$

not possible for $y = 2, 3$

$$\text{If } y \in Q$$

then $f^{-1}(y)$ is not possible since f is not surjective

(C) Same counter example as in A can be used

9. BC

Sol. If $(\text{gof})^{-1}$ exists then gof must be one – one and onto

gof is one – one $\Rightarrow f$ is one – one

gof is onto $\Rightarrow g$ is onto

Consequently, since $(\text{gof})^{-1}$ exists, then gof is one – one and onto, hence f is one – one and g is onto

Claim 1: gof is one – one $\Rightarrow f$ is one – one

Proof Let a_1 and a_2 be two elements of the set A.

$$f(a_1) = f(a_2) \Rightarrow \text{gof}(a_1) = \text{gof}(a_2) \text{ since gof is one – one we get that } a_1 = a_2$$

So, we have proved that $f(a_1) = f(a_2)$ implies f is one – one

Claim 2 : gof is ONTO $\Rightarrow g$ is ONTO

Proof Since gof is onto it follows that for any $c \in C$ there exists $a \in A$ such that $\text{gof}(a) = c$

Consequently, for any $c \in C$ there exists $b = f(a) \in B$ such that $g(b) = \text{gof}(a) = c$

Hence g is ONTO

10. BC

Sol. Let $g(x) = \frac{f'(x)}{(f(x))^2} \Rightarrow \int g(x) dx = -\frac{1}{f(x)} + c$

$$\Rightarrow \int_4^8 g(x) dx = -\frac{1}{f(8)} + \frac{1}{f(4)} = -2 + 4 = 2$$

Also, $\int_4^8 (g(x))^2 dx = 1$

$$\Rightarrow \int_4^8 \left(g(x) - \frac{1}{2} \right)^2 dx = 0$$

$$\Rightarrow g(x) = \frac{1}{2}$$

$$\Rightarrow \frac{f'(x)}{(f(x))^2} = \frac{1}{2}$$

$$\Rightarrow -\frac{1}{f(x)} = \frac{x}{2} + d$$

$$\Rightarrow d = -6 \text{ using } f(4) = \frac{1}{4}$$

$$\Rightarrow f(x) = \frac{2}{12-x}$$

$$\Rightarrow f(6) = \frac{1}{3}$$

11. ABCD

Sol. $z_1 R z_1 \Rightarrow \frac{z_1 - z_2}{z_1 + z_2} = 0 = \text{a real number} \Rightarrow R \text{ is reflexive}$

If $z_1 R z_2$ holds i.e. $\frac{z_1 - z_2}{z_1 + z_2}$ is real then clearly $\frac{z_2 - z_1}{z_2 + z_1} = -\left(\frac{z_1 - z_2}{z_1 + z_2}\right)$ is also real

$\Rightarrow R$ is symmetric

If $\frac{z_1 - z_2}{z_1 + z_2}$ is real i.e. $\frac{z_1 - z_2}{z_1 + z_2} = \frac{\bar{z}_1 - \bar{z}_2}{\bar{z}_1 + \bar{z}_2}$

$$z_1 \bar{z}_1 + z_1 \bar{z}_2 - \bar{z}_1 z_2 - z_2 \bar{z}_2 = z_1 \bar{z}_1 - z_1 \bar{z}_2 + \bar{z}_1 z_2 - z_2 \bar{z}_2$$

$$z_1 \bar{z} = \bar{z}_1 z_2$$

$$\Rightarrow \frac{z_1}{z_1} = \frac{z_2}{z_2}$$

So, if $\frac{z_2 - z_3}{z_2 + z_3}$ is real, in same manner we have $\frac{z_2}{z_2} = \frac{z_3}{z_3}$

$$\Rightarrow \frac{z_1}{z_1} = \frac{z_2}{z_2} = \frac{z_3}{z_3} \Rightarrow R \text{ is transitive.}$$

12. AD

Sol. Derangement of 4 persons in 4 houses $= D_4 = 9$ ways. Hence option 1 is correct.

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If A has two houses. Then let there be a hypothetical person E. So that A, B, C, D, E have one house each now.

Among the total derangements ($D_5 = 44$). A is equally likely to go to B, C, D, E houses. So

A does not go to E's house in $\frac{1}{4}D_5$ ways.

$$\therefore \text{Number of ways} = D_5 - \frac{1}{4}D_5 = 44 - \frac{1}{4}(44) = 33 \text{ ways}$$

But it is OK, for E to go to A's house, it means one of A's houses is occupied.

So, number of derangements in this case $= D_4 = 9$

$\therefore$ Required number of ways $= 33 + 9 = 42$ ways

PART - B

1. 924

Sol. Consider the first S, I, P

* S * I * P *

The second P can only be placed in the last gap and hence we have * S * I * P *

The three I's can be placed only in the last three gaps.

Choosing the places for I's is put 3 identical object into 3 boxes i.e.

$${}^{3+3-1}C_{3-1} = {}^5C_2 = 10 \text{ ways } (x_1 + x_2 + x_3 = 3)$$

We have so far placed 7 letters. We have to put 3S's in the gaps created by these. However, for placing the 3 - S's, we cannot choose the gap before the first S and hence we have 7 - gaps to choose from

This can be done in ${}^{7+3-1}C_{3-1} = {}^9C_2 = 36$ ways.

Only, the letter M remains to be placed. M can go in any of these 11 - gaps.

$\Rightarrow$ Required number of ways $= 10 \times 36 \times 11 = 3960$

2. 143

Sol. $f(2) = g(2), f(4) = g(4), f(8) = g(8)$

$$\int_1^8 x f'(x) dx = x f(x) \Big|_1^8 - \int_1^8 f(x) dx = 8f(8) - f(1) = 63 - \int_1^8 f(x) dx$$

$$\int_1^2 f(x) dx = \int_1^2 \frac{g(2x)}{2} dx = \int_2^4 \frac{g(t)}{4} dt \quad (t = 2x)$$

$$\int_1^2 f(x) dx = \frac{5}{4} \Rightarrow \frac{5}{4} = \frac{1}{4} \int_2^4 g(t) dt \Rightarrow \int_2^4 g(x) dx = 5$$

$$\int_2^4 f(x) dx + \int_2^4 g(x) dx = 16 - 4 = 12 \Rightarrow \int_2^4 f(x) dx = 7$$

$$\int_2^4 \frac{g(2x)}{2} dx = 7 \Rightarrow \int_2^4 \frac{g(t)}{4} dt = 7 \Rightarrow \int_4^8 g(x) dx = 28$$

$$\Rightarrow \int_4^8 f(x) dx + \int_4^8 g(x) dx = 64 - 16 = 48$$

$$\Rightarrow \int_4^8 f(x) dx = 48 - 28 = 20$$

$$\Rightarrow \int_1^8 x f'(x) dx = 63 - \left[\int_1^2 f(x) dx + \int_2^4 f(x) dx + \int_4^8 f(x) dx \right]$$

$$= 63 - \left[\frac{5}{4} + 7 + 20 \right] = \frac{139}{4} = \frac{p}{q}$$

$$\Rightarrow p + q = 143$$

3. 2

Sol. $A = \sqrt{10 + \sqrt{1}} + \sqrt{10 + \sqrt{2}} + \dots + \sqrt{10 + \sqrt{99}}$

$$B = \sqrt{10 - \sqrt{1}} + \sqrt{10 - \sqrt{2}} + \dots + \sqrt{10 - \sqrt{99}}$$

$$\frac{A}{B} = 1 + \frac{A - B}{B}$$

$$A - B = \sum_{i=1}^{99} (\sqrt{10 + \sqrt{i}} - \sqrt{10 - \sqrt{i}})$$

Consider, for each i

$$(\sqrt{10 + \sqrt{i}} - \sqrt{10 - \sqrt{i}})^2 = 10 + \sqrt{i} + 10 - \sqrt{i} - 2\sqrt{10^2 - i}$$

$$= 20 - 2\sqrt{100 - i}$$

$$\therefore \sqrt{10 + \sqrt{i}} - \sqrt{10 - \sqrt{i}} = \sqrt{20 - 2\sqrt{100 - i}} = \sqrt{2}\sqrt{10 - \sqrt{100 - i}}$$

$$A - B = \sum_{i=1}^{99} (\sqrt{10 + \sqrt{i}} - \sqrt{10 - \sqrt{i}}) = \sqrt{2} \sum_{i=1}^{99} \sqrt{10 - \sqrt{100 - i}} = \sqrt{2} \sum_{i=1}^{99} \sqrt{10 - \sqrt{i}} = B\sqrt{2}$$

$$\frac{A - B}{B} = \sqrt{2}$$

4. 2.82

Sol. The equation of two edges can be taken to be

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{z-0}{-1} \quad \text{and} \quad \frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-\sqrt{3}}{0}$$

The shortest distance between them is $\sqrt{2}$.

5. 1

Sol. $x = f(t) \Rightarrow dx = f'(t) dt$

$$I = \int_1^0 (f(t) - t)^{2023} f'(t) dt$$

$$= - \int_0^1 (t - f(t))^{2023} f'(t) dt$$

$$= - \int_0^1 (t - f(t))^{2023} (-1 + f'(t) + 1) dt$$

$$= + \int_0^1 (t - f(t))^{2023} (1 - f'(t)) dt - I$$

$$2I = \int_0^1 (t - f(t))^{2023} (1 - f'(t)) dt = \left. \frac{(t - f(t))^{2024}}{2024} \right|_0^1 = \frac{2}{2024}$$

6. 1.22

Sol. Clearly, triangle formed by the given points $\hat{i} + \hat{j} + 2\hat{k}$, $\hat{i} + 2\hat{j} + \hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$ is equilateral as $AB = BC = AC = \sqrt{2}$.

$\therefore$ Distance of orthocentre 'O' from the sides is equal to inradius of the triangle.

IG ~ @bohring_bot

$$\therefore l_1 = l_2 = l_3 = \text{inradius} = r = \frac{\Delta}{s} = \frac{\frac{\sqrt{3}}{4}(\sqrt{2})^2}{\frac{4}{2}(\sqrt{2})} = \frac{1}{\sqrt{6}}$$

$$\Rightarrow (l_1 + l_2 + l_3) = \frac{3}{\sqrt{6}} = \frac{\sqrt{6}}{2}$$

7. 4.00

Sol. $\lim_{x \rightarrow 0} \frac{f(x)}{x^k} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x \sin x - 1 + \cos x}{x^k}$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \sin x + \frac{1}{2}x \cos x - \sin x}{kx^{k-1}}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \cos x + \frac{1}{2} \cos x - \frac{1}{2}x \sin x - \cos x}{k(k-1)x^{k-2}}$$

$$= \lim_{x \rightarrow 0} \frac{-\frac{1}{2} \sin x}{k(k-1)x^{k-3}}$$

If $k-3=1$ i.e. $k=4$, $\lim_{x \rightarrow 0} \frac{f(x)}{x^k} = \frac{-\frac{1}{2}}{4 \times 3} = \frac{-1}{24}$

8. 5.15

Sol. Clearly $h(i)$ is an A.P.

$$h(100) = h(0) + 100d \Rightarrow 100d = 20 - 5 = 15 \Rightarrow d = 0.15$$

$$\Rightarrow h(1) = 5.15$$

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FIITJEE RBT-7 for (JEE-Advanced)

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 4

QP CODE: 100203

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

- Attempt ALL the questions. Answers have to be marked on the OMR sheets.
- This question paper contains **Three Sections**.
- Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
- Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
- Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
- Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

- Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
- On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
- OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Part.

- (i) **PART-A (01-08)** contains (8) Multiple Choice Questions which have **One or More Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (09-12)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **-1 marks** for wrong options chosen.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

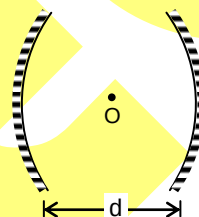
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SECTION – I : PHYSICS**(PART – A)****(One or More Than One Options Correct Type)**

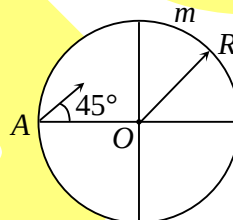
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. The potential energy ϕ in joule of a particle of mass 1 kg moving in the $x - y$ plane, obeys the law $\phi = 3x + 4y$, where (x, y) are the coordinate of the particle in metre. If the particle is at rest at $(6, 4)$ at time $t = 0$, then (only conservative force are acting on particle).
 (A) the particle has constant acceleration
 (B) the work done by the conservative forces from position of rest of the particle to the instant of the particle crossing x axis is 25 J
 (C) the speed of the a particle when it crosses the y axis is 10 ms^{-1}
 (D) the coordinates of the particle at time $t = 4\text{s}$ are $(-18, -28)$

2. The distance d between two identical concave mirrors of radius of curvature R . So, as to coincide the image I on the object O itself, which is placed at the mid point can be
 (A) R
 (B) $2R$
 (C) $3R$
 (D) $R/2$



3. A ring of mass m and radius R is placed on a frictionless horizontal surface. A particle of mass m is projected from point A with velocity v at an angle of 45° with AO as shown. The correct statements are



- (A) The particle reaches the same point A on the ring after time $\frac{4R\sqrt{2}}{v}$.
 (B) Magnitude of impulse transformed during first collision is $\frac{mv}{\sqrt{2}}$.
 (C) Magnitude of impulse transformed during second collision is $\frac{mv}{\sqrt{2}}$.
 (D) Particle reaches diametrically opposite point on the ring in time $\frac{2R}{v}$.

Space For Rough Work

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4. A point moves with deceleration along the circle of radius R so that at any time its tangential and normal acceleration are equal in magnitude. At the initial moment $t = 0$, the velocity of the point is v_0 . The velocity of the point will be

(A) $v = \frac{v_0}{1 + \frac{v_0 t}{R}}$ at t second

(B) $v = v_0 e^{-s/R}$ where s is path travelled on circle

(C) $v = v_0 e^{-sR}$ where s is path travelled on circle

(D) $v = \frac{v_0}{1 - \frac{v_0 t}{R}}$ at t second

5. A horizontal disc rotates freely about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, they rotate with a common angular speed.

(A) some friction exists between the disc and the ring.

(B) the angular momentum of the disc plus ring is conserved.

(C) the final common angular speed is $\frac{2}{3}$ rd of the initial angular velocity of the disc.

(D) $\frac{2}{3}$ rd of the initial kinetic energy changes to heat.

6. If the potential difference of Coolidge tube producing X-ray is increased, then choose the correct option(s).

(A) the interval between $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ increases

(B) the interval between $\lambda_{k\alpha}$ and λ_0 increases

(C) the interval between $\lambda_{k\beta}$ and λ_0 increases

(D) λ_0 does not change

Here λ_0 is cutoff wavelength and $\lambda_{k\alpha}$ and $\lambda_{k\beta}$ are wavelength of k_α and k_β characteristic X-rays.

7. A coil of radius r ($r \ll R$) and mass m carries current i_2 . Planes of two coils are mutually perpendicular and both the coils are free to rotate about common diameter.

Choose the correct statement/statements.

(A) The maximum Kinetic energy of smaller coil when both the coils are released, is

$$K_{\max} = \frac{\mu_0 \pi r^2 i_1 i_2 MR}{2(MR^2 + mr^2)}$$

(B) Total energy of both the coils remains constant.

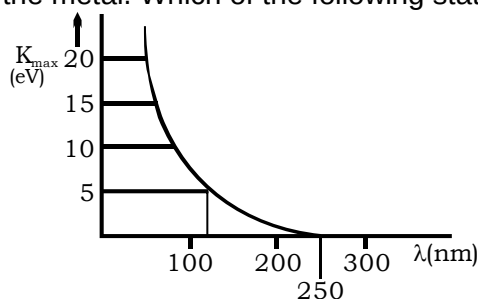
(C) Total angular momentum of both the coils remains constant.

(D) The potential energy of the system will be minimum when both the coils are in same plane and current in them are in same direction.

Space For Rough Work

TG ~ @bohring_bot

8. In a photoelectric effect experiment, the maximum kinetic energy of the ejected photoelectrons is measured for various wavelengths of the incident light. Figure shows a graph of this maximum kinetic energy $K_{\max}$ as a function of the wavelength λ of the light falling on the surface of the metal. Which of the following statements is/are correct?

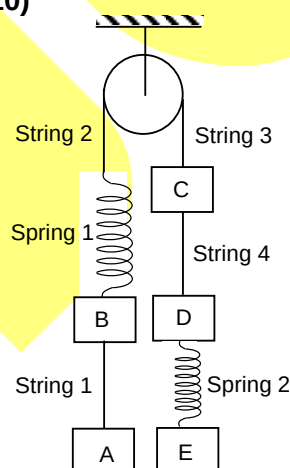


- (A) Threshold frequency for the metal is 1.2×10^{15} Hz
 (B) Work function of the metal is 4.968 eV.
 (C) Maximum kinetic energy of photoelectrons corresponding to light of wavelength 100 nm is nearly 7.4 eV.
 (D) Photoelectric effect takes place with red light.

This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

The system shown below is initially in equilibrium
 $m_A = m_B = 3\text{kg}$
 $m_C = m_D = m_E = 2\text{kg}$
 Take $g = 10 \text{ m/s}^2$



List-I gives the four strings while List-II the value of the tension in the strings.

List - I		List - II	
(I)	String 1	(P)	10 N
(II)	String 2	(Q)	20 N
(III)	String 3	(R)	30 N
(IV)	String 4	(S)	40 N
		(T)	60 N
		(U)	0 N

9. If the spring 1 is cut, match the correct for the tension in the strings just after the cutting.
 (A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ P
 (B) I $\rightarrow$ U, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ P
 (C) I $\rightarrow$ U, II $\rightarrow$ U, III $\rightarrow$ U, IV $\rightarrow$ P
 (D) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ S, IV $\rightarrow$ P

Space For Rough Work

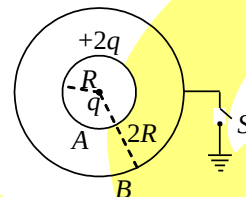
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10. If the spring 2 is cut, match the correct for the tension in the strings just after the cutting.
 (A) I $\rightarrow$ R, II $\rightarrow$ T, III $\rightarrow$ T, IV $\rightarrow$ R (B) I $\rightarrow$ R, II $\rightarrow$ U, III $\rightarrow$ U, IV $\rightarrow$ R
 (C) I $\rightarrow$ U, II $\rightarrow$ R, III $\rightarrow$ R, IV $\rightarrow$ P (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ Q, IV $\rightarrow$ R

List Match Set (11-12)

Match the List-I with the List-II for question no. 11 and 12.

Two concentric conducting shells A and B have radii R and $2R$. A charge q is placed at the centre of the shells and a charge $2q$ is given to shell A.



List – I		List – II	
(I)	Charge on inner surface of shell A	(P)	$-3q$
(II)	Charge on inner surface of shell B	(Q)	Zero
(III)	Charge on outer surface of shell A	(R)	$-q$
(IV)	Charge on outer surface of shell B	(S)	$3q$
		(T)	$2q$
		(U)	$-2q$

11. When switch is open
 (A) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ S (B) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ Q
 (C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ T (D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U
12. When switch is closed
 (A) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ Q (B) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ Q
 (C) I $\rightarrow$ T, II $\rightarrow$ U, III $\rightarrow$ R, IV $\rightarrow$ P (D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

(PART – B)

(Numerical Type)

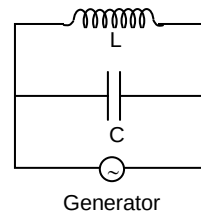
Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. A sphere of radius 0.1 m and mass $8\pi\text{ kg}$ is attached to the lower end of a steel wire of length 5 m and diameter 10^{-3} m . The wire is suspended from 5.22 m high ceiling of a room. When the sphere is made to swing as a simple pendulum, it just grazes the floor at its lowest point. Calculate the velocity of the sphere at the lowest position (in m/s). Young's modulus of steel is $1.994 \times 10^{11}\text{ N/m}^2$.

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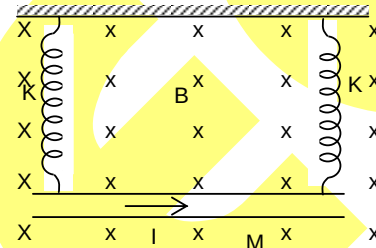
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2. For the circuit shown in the figure, the current through the inductor is 0.6 A, while the current through the capacitor is 0.4 A. Find the current (in A) drawn from the generator

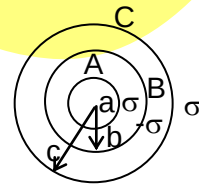


3. An ice cube of mass 0.1 kg at 0°C is placed in an isolated container which is at 227°C . The specific heat s of the container varies with temperature T according to the empirical relation $s = A + BT$, where $A = 100 \text{ cal kg}^{-1} \text{K}^{-1}$ and $B = 2 \times 10^{-2} \text{ cal kg}^{-1} \text{K}^{-2}$. The final temperature of the container is 27°C , the mass of the container is $x \times 10^{-1} \text{ kg}$. What is the value of x ? (Latent heat of fusion of water $= 8 \times 10^4 \text{ cal kg}^{-1}$, Specific heat of water $= 10^3 \text{ cal kg}^{-1} \text{K}^{-1}$).

4. A metal rod of mass 10g and length 25 cm is suspended on two springs as shown in figure. The springs are extended by 4 cm. When a 20 ampere current passes through the rod it rises by 1 cm. The magnetic field is $x \times 10^{-2} \text{ T}$ ($g = 10 \text{ m/s}^2$). Find the value of $2x$.



5. The adjacent figure shows charged spherical shells A, B and C having charge densities σ , $-\sigma$, σ and radii a , b , c respectively. If $V_A = V_C$ then c is equal to (in m) (assuming $a = 0.10 \text{ m}$, $b = 0.20 \text{ m}$)



6. In YDSE experiment if the screen is shifted by a distance of 0.5 m away from the slit, 3rd maxima is shifted by $3 \times 10^{-4} \text{ m}$. The slit width is $2 \times 10^{-3} \text{ m}$. If the wavelength (in 100 nm) used in the experiment is 'k', find the value of 'k'.

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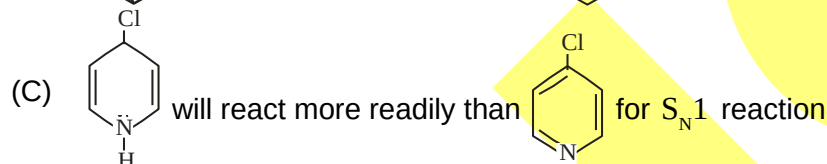
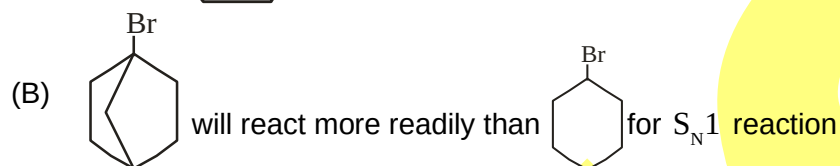
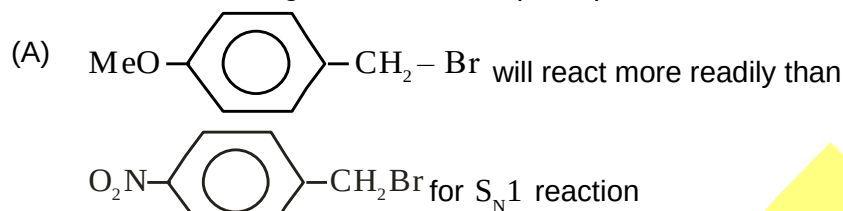
SECTION – II : CHEMISTRY

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. Consider the following statements and pick up the correct statements:



(D) Generally $\text{S}_\text{N}1$ reaction occurs in polar protic solvent

2. Which of the following statement(s) is/are correct for PCl_5 ?

- (A) Phosphorus undergoes sp^3d hybridization.
 (B) Two sets of P–Cl bond length are observed in the molecule
 (C) It's solid contains $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$ ions
 (D) The maximum value of $\angle \text{ClPCl}$ angle is 120° .

3. The standard electrode potential of a metal-metal ion ($\text{Ag}|\text{Ag}^+$) and metal-sparingly soluble salt anion ($\text{Ag}|\text{AgCl}|\text{Cl}^-$) are not related as

(A) $E^\circ_{\text{Ag}^+|\text{Ag}} = E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}} + \frac{RT}{F} \ln K_{\text{sp}}$

(B) $E^\circ_{\text{Ag}^+|\text{Ag}} = E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}} + \frac{RT}{F} \ln K_{\text{sp}}$

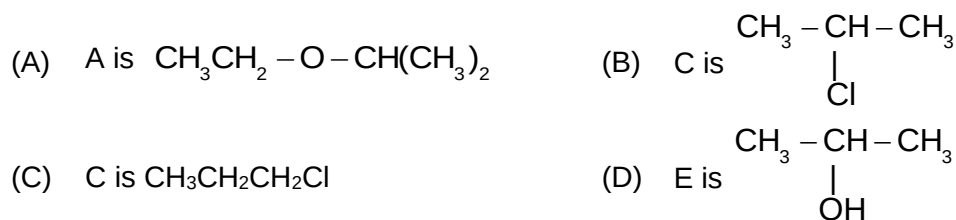
(C) $E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}} = E^\circ_{\text{Ag}^+|\text{Ag}} - \frac{RT}{F} \ln \frac{[\text{Cl}^-]}{K_{\text{sp}}}$

(D) $E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}} = E^\circ_{\text{Ag}^+|\text{Ag}} - \frac{RT}{F} \ln \frac{K_{\text{sp}}}{[\text{Cl}^-]}$

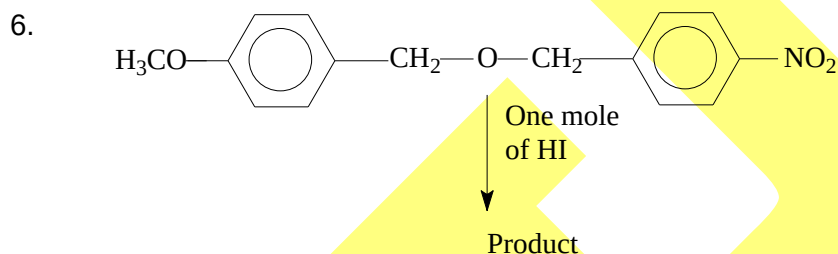
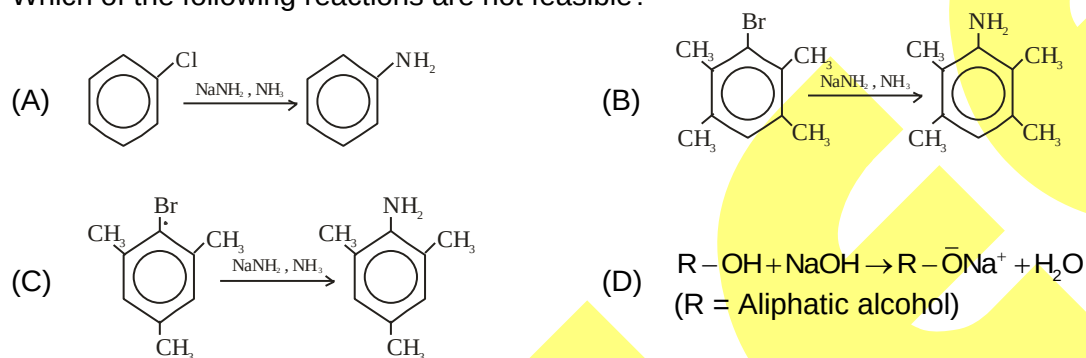
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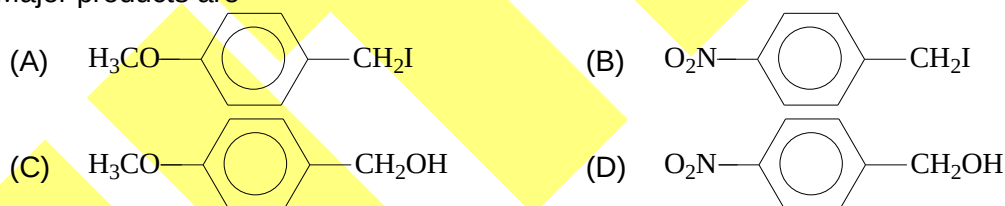
4. A ($C_5H_{12}O$) ether, on reaction with PCl_5 form alkyl chloride B and C. B and C both on reaction with aqueous KOH form alcohol D and E. Both D and E give iodoform test. Identify correct answers.



5. Which of the following reactions are not feasible?



Major products are

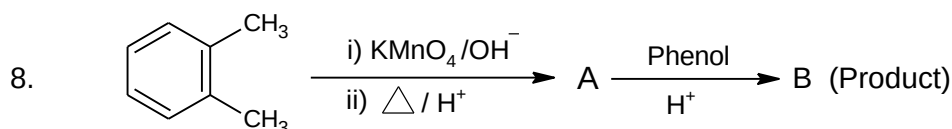


7. For a non-ideal solution with a negative deviation



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Which of the following is correct regarding product 'B'?

- (A) 'B' is colourless in acidic medium
 (B) 'B' is pink colour in basic medium
 (C) 'B' has Benzenoid form in Basic medium
 (D) 'B' exists in Ionic form in Basic medium

This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Match the reversible reactions mentioned in List – I with the effect of Le – Chatelier principles on them mentioned in List - II and answer the question.

	List – I		List - II
(I)	$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g}); \Delta H = +ve$	(P)	The concentration of NO_2 increases by increasing temperature.
(II)	$2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}); \Delta H = +ve$	(Q)	The concentration of NO_2 increases by decreasing temperature
(III)	$\text{N}_2\text{O}_3(\text{g}) \rightleftharpoons \text{NO}_2(\text{g}) + \text{NO}(\text{g}); \Delta H = +ve$	(R)	The concentration of NO_2 increases by increasing the volume of reaction vessel.
(IV)	$2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g}); \Delta H = -ve$	(S)	The concentration of NO_2 increases by increasing pressure.
		(T)	Inert gas addition does not have any result at constant pressure or at constant volume.

9. The correct matching according to List-I & List-II is
 (A) I $\rightarrow$ PQ (B) II $\rightarrow$ T
 (C) III $\rightarrow$ P (D) IV $\rightarrow$ Q
10. The correct matching according to List-I & List-II is
 (A) I $\rightarrow$ R (B) II $\rightarrow$ Q
 (C) III $\rightarrow$ T (D) IV $\rightarrow$ S

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List Match Set (11-12)

Match the compound(s) mentioned in column-I with their properties mentioned in column-II.

List – I		List– II	
(I)		(P)	Shows optical isomerism before hydrogenation with H_2/Ni
(II)		(Q)	Shows optical isomerism after hydrogenation with H_2/Ni
(III)		(R)	Shows optical isomerism before dehydration
(IV)		(S)	Shows optical isomerism after dehydration
		(T)	Shows optical isomerism before addition of HBr
		(U)	Shows optical isomerism after addition of HBr

11. The correct matching according to above list is
 (A) I $\rightarrow$ Q (B) II $\rightarrow$ U
 (C) III $\rightarrow$ R (D) IV $\rightarrow$ T
12. The correct matching according to above list is
 (A) I $\rightarrow$ P (B) IV $\rightarrow$ S
 (C) III $\rightarrow$ U (D) IV $\rightarrow$ R

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(PART – B)**(Numerical Type)**

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30).

- How many of them are correctly matched for purification process(refining of metal)?
 (a) Liquation – Sn
 (b) Distillation – Zn
 (c) Parke's process – Pb
 (d) Zone refining – Ge
 (e) Cupellation – Cu
 (f) Mond's process – Ni
 (g) Electrolytic refining – Ag
 (h) Baeyer's process – Al
 (i) Parting process – Au
 (j) Poling process – Cu
- A dipositive metal ion is oxidized to it's tripositive ion by acidified permangante solution ($\text{MnO}_4^- / \text{H}^+$). A solution containing 134.04 g of dipositive ion, required 1.2 L of 0.4 M acidified solution of MnO_4^- for complete oxidation. What is the ionic mass of the ion?

- Which of the following is/are correctly matched?

	Drugs used for treatment	Class of drug
(a)	Ranitidine	Antacid
(b)	Histamine	Antacid
(c)	Seldane	Anti histamines
(d)	Seretonin	Tranquilizers
(e)	Aspirin	Analgesic & antipyretic
(f)	Heroin	Analgesic(Narcotic)
(g)	Pencillin	Antibiotics(Bactericidal)
(h)	1% Phenol	Antiseptic
(i)	Norethindrone	Antifertility

- A radiation travels with an uniform velocity of $2116.52 \times 10^4 \text{ m s}^{-1}$. What is the wavelength of the radiation in nm unit if it's energy is $3.3 \times 10^{-20} \text{ Joule}$?
- $$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$$

A container contains 6 moles of PCl_5 and 4 moles of PCl_3 . On heating, PCl_5 decomposes to PCl_3 and Cl_2 according to above reaction. The total number of gaseous moles present at equilibrium is 14. What will be the volume of the reaction container in liter unit, if the equilibrium constant, K_c is $\frac{1}{35.45} \text{ mol L}^{-1}$.
- What is the maximum limit of NO_3^- (Nitrate ion) in drinking water prescribed, otherwise excess nitrate in drinking water can cause disease Methemoglobinemia(Blue Baby Syndrome)
 [Give answer in ppm or mg dm^{-3}]

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SECTION – III : MATHEMATICS

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. Let a and b be two complex numbers such that $|a + b| = 20$ and $|a^2 + b^2| = 16$ then
 (A) maximum value of $|a^3 + b^3|$ is 4480 (B) minimum value of $|a^3 + b^3|$ is 2022
 (C) minimum value of $|a^3 + b^3|$ is 3520 (D) maximum value of $|a^3 + b^3|$ is 4044

2. If $y = \alpha x^2 + \alpha x + \frac{1}{24}$ and $x = \alpha y^2 + \alpha y + \frac{1}{24}$ are tangent to each other, then
 (A) $\alpha = \frac{2}{3}$ (B) $\alpha = \frac{3}{2}$
 (C) $\frac{13 - \sqrt{601}}{12}$ (D) $\frac{13 + \sqrt{601}}{12}$

3. A parabola touches x – axis at $A(4, 0)$ and y – axis at $B(0, 3)$
 (A) Equations of directrix is $4x + 3y = 0$ (B) Co – ordinates of focus are $\left(\frac{36}{25}, \frac{48}{25}\right)$
 (C) Equation of directrix is $3x + 4y = 0$ (D) Co – ordinates of focus are $\left(\frac{48}{25}, \frac{36}{25}\right)$

4. If $8 \sin x = \frac{\sqrt{3}}{\cos x} + \frac{1}{\sin x}$ then $x =$ _____
 (A) $n\pi \pm \frac{\pi}{4}$ (B) $n\pi \pm \frac{\pi}{6}$
 (C) $n\pi + \frac{\pi}{6}$ (D) $\frac{n\pi}{2} - \frac{\pi}{12}$

5. Let a_n be the number of non-empty subsets of $S = \{1, 2, 3, 4, \dots, n-1, n\}$ such that there are no two consecutive numbers in the same set, then
 (A) $a_n = a_{n-1} + a_{n-2}$ (B) $a_n = a_{n-1} + a_{n-2} + 1$
 (C) $a_7 = 33$ (D) $a_7 = 21$

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6. Suppose **100** students appeared for IIT Genius Test. The test carries a maximum of **100** marks and there is no negative marking. The arithmetic mean of their scores is **40** with standard deviation **20**. Let p be the number of students who have scored marks equal to or more than **80**. Then the possible value(s) of p can be
 (A) 10 (B) 20
 (C) 22 (D) 18
7. If $\triangle ABC$, let G be the centroid, and let I be the centre of the inscribed circle. If IG be parallel to AB and $\tan \frac{B}{2} = \frac{1}{3}$. Then
 (A) $\triangle ABC$ is obtuse angled (B) $\triangle ABC$ is right angled but not isosceles
 (C) $3c = 4b$ (D) $5b = 3a$
8. Let $a_1, a_2, a_3, \dots, a_n$ be first 'n' terms of a G.P. with first term 'a' and common ratio 'r' then
 (A) $\frac{1}{a_1^2 - a_2^2} + \frac{1}{a_2^2 - a_3^2} + \dots + \frac{1}{a_{n-1}^2 - a_n^2} = \frac{(1-r^{2n})}{a^2 r^{2n-4} (1-r^2)^2}$
 (B) $\frac{1}{a_1^2 - a_2^2} + \frac{1}{a_2^2 - a_3^2} + \dots + \frac{1}{a_{n-1}^2 - a_n^2} = \frac{(1-r^{2n-2})}{a^2 r^{2n-4} (1-r^2)^2}$
 (C) $\frac{1}{a_1^m + a_2^m} + \frac{1}{a_2^m + a_3^m} + \dots + \frac{1}{a_{n-1}^m + a_n^m} = \frac{(r^{mn-m} - 1)}{a^m (1+r^m)(r^{mn-m} - r^{mn-2m})}$
 (D) $\frac{1}{a_1^m + a_2^m} + \frac{1}{a_2^m + a_3^m} + \dots + \frac{1}{a_{n-1}^m + a_n^m} = \frac{(r^{mn-m} - 1)}{a^m (1-r^m)(r^{mn-m} - r^{mn-2m})}$

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Three players A, B and C alternatively throw a die in that order, the first player to throw a being deemed the winner. A's die is fair whereas B and C throw dice with probabilities p_1 and p_2 respectively, or throwing a 6.

Column-I		Column-II	
(I)	If $p_1 = \frac{1}{5}, p_2 = \frac{1}{4}$ and probability that A wins the game is $\frac{1}{\lambda_1}$, then λ_1 is divisor of	(P)	6
(II)	If $p_1 = \frac{1}{5}, p_2 = \frac{1}{4}$ and probability that C wins the game is $\frac{1}{\lambda_2}$, then λ_2 is divisor of	(Q)	8
(III)	If $P(A \text{ wins}) = P(B \text{ wins})$ and $\frac{1}{p_1} = \lambda_3$, then λ_3 is divisor of	(R)	12
(IV)	If game is equiprobable to all the three players and $\frac{1}{p_1} = \lambda_4$, then λ_4 is divisor of	(S)	15

9. Which of the following is the correct combination?
 (A) I $\rightarrow$ Q (B) II $\rightarrow$ PRS
 (C) III $\rightarrow$ R (D) IV $\rightarrow$ P
10. Which of the following is the correct combination?
 (A) I $\rightarrow$ PQS (B) II $\rightarrow$ Q
 (C) III $\rightarrow$ S (D) IV $\rightarrow$ P

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List Match Set (11-12)

Matching Type Questions

Column-I		Column-II	
(I)	If number of dissimilar terms in the expansion of $(x + 2y + 3z)^n$ ($n \in \mathbb{N}$) is $am^2 + bn + c$, then	(P)	$a + b + c = 3$
(II)	If number of dissimilar terms in the expansion of $(x + y + z)^{2n+1} - (x + y - z)^{2n+1}$ ($n \in \mathbb{N}$) is $an^2 + bn + c$, then	(Q)	$a + b + c = 4$
(III)	If number of dissimilar terms in the expansion $(x - y + z)^n + (x + y - z)^n$ ($n \in$ is even natural number) is $an^2 + bn + c$, then	(R)	$a + b = 2c$
(IV)	If number of dissimilar terms in the expansion of $\left(\frac{x^2 + 1 + x^4}{x^2}\right)^{\sum n}$ ($n \in \mathbb{N}$) is $an^2 + bn + c$, then	(S)	$b + c = 8a$

11. Which of the following is the correct combination?
 (A) I $\rightarrow$ RS (B) I $\rightarrow$ PRS
 (C) II $\rightarrow$ P (D) IV $\rightarrow$ PR
12. Which of the following is the correct combination?
 (A) I $\rightarrow$ QS (B) I $\rightarrow$ R
 (C) II $\rightarrow$ Q (D) III $\rightarrow$ Q

(PART – B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimal Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. Let A and B be two square matrices having real elements satisfying $A^2 + B^2 = 2AB$ then value of $\det(AB - BA) =$
2. If $p(t) = \frac{t(t-1)\dots(t-2023)}{2023!}$, then the value of $\int_0^1 \left(\frac{1}{t+1} + \frac{1}{t+2} + \dots + \frac{1}{t+2024} \right) p(-t-1) dt = k^2$ then $k - 2000$ equals

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3. If $\cot k^\circ = \tan 78^\circ + \tan 72^\circ + \tan 60^\circ$ then $k =$ _____
4. 6 given points on a circle are joined by line segments of which no three are concurrent. The number of triangles inside the circle equals _____
5. The number of continuous functions $f: \left[0, \frac{3}{2}\right] \rightarrow (0, \infty)$ satisfy the equation $4 \int_0^{3/2} f(x) dx + 125 \int_0^{3/2} \frac{dx}{\sqrt{f(x) + x^2}} = 108$ is _____.
6. If $S = \frac{1}{1.2.3} + \frac{1}{5.6.7} + \frac{1}{9.10.11} + \dots$ (value of $\ln 2 \approx 0.693$) then value of $S =$ _____

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Q. P Code: 100203
Answers
SECTION – I : PHYSICS

(PART – A)

- | | | | |
|---------|-------|---------|--------|
| 1. ABCD | 2. AB | 3. ABC | 4. AB |
| 5. ABD | 6. BC | 7. ABCD | 8. ABC |
| 9. C | 10. A | 11. A | 12. B |

(PART – B)

- | | | | |
|---------|---------|---------|------|
| 1. 8.74 | 2. 0.20 | 3. 4.95 | 4. 3 |
| 5. 0.30 | 6. 4 | | |

SECTION – II : CHEMISTRY

(PART – A)

- | | | | |
|--------|--------|--------|--------|
| 1. ACD | 2. ABD | 3. ACD | 4. ABD |
| 5. BC | 6. AD | 7. ABD | 8. ABD |
| 9. C | 10. D | 11. B | 12. A |

(PART – B)

- | | | |
|------------------------------|-----------------------------|------|
| 1. 8 | 2. 55.85 (Range 55 to 56) | 3. 7 |
| 4. 423.30 (Range 421 to 425) | 5. 567.2 (range 566 to 568) | |
| 6. 50 (Range 49 to 51) | | |

SECTION – III : MATHEMATICS

(PART – A)

- | | | | |
|-------|---------|--------|-------|
| 1. AC | 2. ABCD | 3. AB | 4. CD |
| 5. BC | 6. ABD | 7. BCD | 8. BC |
| 9. B | 10. C | 11. D | 12. C |

(PART – B)

- | | | | |
|------|---------|------|--------|
| 1. 0 | 2. 24 | 3. 6 | 4. 111 |
| 5. 1 | 6. 0.17 | | |

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Answers & Solutions

SECTION – I : PHYSICS

(PART – A)

1. **ABCD**

Sol. $\vec{F} = 3\hat{i} - 4\hat{j}$, $W_{\text{con.f}} = U_i - U_f = 34 - 9 = 25$ & $\frac{1}{2}mv^2 - 16 = 34$

2. **AB**

Sol. $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

3. **ABC**

Sol. Assume x-axis along OA and y-axis perpendicular to it in the plane of ring.
After 1st collision

Velocity of particle = $\frac{v}{\sqrt{2}}$ along x-axis

And velocity of ring = $\frac{v}{\sqrt{2}}$ along y-axis

After second collision velocity of ring = $\frac{v}{\sqrt{2}}\hat{i} + \frac{v}{\sqrt{2}}\hat{j}$

Velocity of particle = 0

Time to return at point A = $4 \times \frac{R\sqrt{2}}{v}$

4. **AB**

Sol. $-\frac{dv}{dt} = \frac{v^2}{R}$ & $-v \frac{dv}{ds} = \frac{v^2}{R}$

5. **ABD**

Sol. By conservation of angular momentum,

$$I_1\omega_1 = (I_1 + I_2)\omega_f$$

$$\omega_f = \omega_1 / 3$$

$$E_i = \frac{1}{2}I_1\omega_1^2$$

$$E_f = \frac{1}{2}(I_1 + I_2)\omega_f^2$$

$$\text{Ratio of the heat produced to initial kinetic energy} = \frac{E_i - E_f}{E_i} = \frac{2}{3}$$

6. **BC**

Sol. As $\lambda_0 = \frac{hc}{eV}$ and as V increases. λ_0 decreases but characteristic wavelengths do not change so interval between $\lambda_{K\alpha}$ & λ_0 increases and the same for the interval between $\lambda_{K\beta}$ & λ_0

7. **ABCD**

Sol. $MB(\cos 0^\circ - \cos 90^\circ) = \frac{1}{2}I_1\omega_1^2 + \frac{1}{2}I_2\omega_2^2 \quad \dots(1)$

Also, $I_1\omega_1 = I_2\omega_2 \quad \dots(2)$

By (1) and (2), we get

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$$K_{\max} = \frac{\mu_0 \pi r^2 i_1 i_2 M R}{2(MR^2 + mr^2)}$$

8. **ABC**

Sol. From graph,

$$\lambda_{th} = 250 \text{ nm}$$

$$f_{th} = \frac{c}{\lambda_{th}} = \frac{3 \times 10^8 \text{ m/s}}{250 \times 10^{-9} \text{ m}} = 1.2 \times 10^{15} \text{ Hz}$$

$$\text{Work function } \phi = \frac{\lambda f_{th}}{e} (eV)$$

$$= 4.968 \text{ eV}$$

$$K.E_{\max} = h\nu - h\nu_{th}(\phi)$$

$$= 7.4 \text{ eV}$$

9. **C**

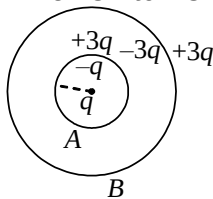
Sol. Just after cutting force of cut spring will be zero whereas the force of other spring will be unchanged.

10. **A**

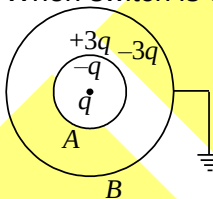
Sol. Just after cutting force of cut spring will be zero whereas the force of other spring will be unchanged.

11. **A**

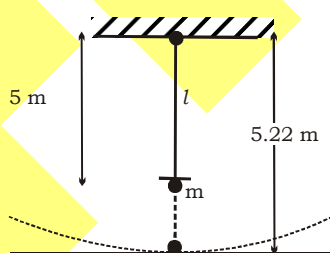
Sol. When switch is open

12. **B**

Sol. When switch is closed

**(PART - B)**1. **8.74**

Sol.



$$T = mg + \frac{mv^2}{R}$$

$$v = \sqrt{\left(\frac{T}{m} - g\right)R}$$

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$$\frac{mv^2}{R} = T - mg$$

$$mv^2 = TR - mgR$$

$$v^2 = \frac{TR}{m} - gR$$

$$v^2 = \left(\frac{T}{m} - g \right) R$$

$$v = \sqrt{(24.925 - 10)5.12}$$

$$= 8.74 \text{ m/s}$$

$$\frac{T}{m} = \frac{yA\Delta L}{mL}, g = 10$$

$$R = 5.22 - 0.1$$

$$(R = 5.12)$$

$$\frac{T}{m} = \frac{1.994 \times 10^{11}}{1000 \cdot 8A} \times \frac{A \times 10^{-6}}{4} \times \frac{0.02}{5100}$$

$$= \frac{1994}{84} \times \frac{2}{20}$$

$$\frac{T}{m} = \frac{1994}{80}$$

2. **0.20**

Sol. I_L and I_C will be in opposite phases.

$$I_{\text{net}} = I_L - I_C$$

$$= 0.6 - 0.4 = 0.2 \text{ A}$$

3. **4.95**

Sol. Heat lost by contains = $msdt$.

$$fdQ = \int_{500}^{300} m(A + Bt)dt$$

$$-Q = m \left[AT + \frac{BT^2}{2} \right]_{500}^{300}$$

$$+Q = m[+21600] \text{ cal}$$

$$\text{Heat gained by ice at } 0^\circ \quad Q_1 = mL$$

$$Q_1 = 8000 \text{ cal}$$

$$\text{Heat gained by water to raise temp.} = ms\Delta T$$

$$Q_2 = 0.1 \times 1000 \times 27$$

$$= 2700$$

$$\text{Total heat gained} = Q_1 + Q_2$$

$$= 8000 + 2700$$

$$= 10700 \text{ cal}$$

$$\text{Heat lost} = \text{Heat gained}$$

$$m21600 = 10700$$

$$m = 0.495 \text{ kg} = 4.95 \times 10^{-1} \text{ kg}$$

4. **3**

Sol. Initially the rod will be in equilibrium if

$$2T_0 = Mg \text{ with } T_0 = kx_0 \quad \dots(i)$$

when the current I is passed through the rod, it will experience a force

$F = BIL$ vertically up,

In equilibriums

$$2T + BIL = Mg \text{ with } T = kx \quad \dots(ii)$$

from (i) & (ii)

$$\frac{T}{T_0} = \frac{Mg - BIL}{Mg} \text{ i.e. } \frac{x}{x_0} = 1 - \frac{BIL}{Mg}$$

$$\text{or, } B = \frac{Mg(x_0 - x)}{ILx_0}$$

Putting the values we get $B = 1.5 \times 10^{-2} \text{ T}$

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5. **0.30**Sol. $V_A = V_C$
 $c = a + b$ 6. **4**Sol. Here 3rd maxima is shifted by 3×10^{-4} m. It indicates fringe width increases by 1×10^{-4} m.

$$\text{Hence } \beta = \frac{\lambda(D+0.5)}{d} = \frac{\lambda D}{d} + 1 \times 10^{-4}$$

$$\text{or } \frac{0.5\lambda}{d} = 1 \times 10^{-4} \text{ m} \quad \text{or } \lambda = \frac{2 \times 10^{-3} \times 1 \times 10^{-4}}{0.5} = 4 \times 10^{-7} \text{ m} = 400 \text{ nm.}$$

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SECTION – II : CHEMISTRY**(PART – A)**

1. ACD

Sol. S_N1 Rate $\propto$ Steric $\propto$ Stability of C^+

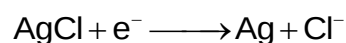
$$S_N2 \text{ Rate} \propto \frac{1}{\text{Steric}}$$

2. ABD

Sol. Basic hybridization.

3. ACD

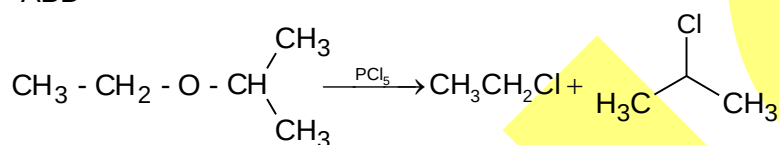
$$\text{Sol. } E_{\text{AgCl/Ag}}^0 = E_{\text{AgCl/Ag}}^0 + \frac{RT}{F} \ln [Cl^-]$$



$$K_{sp} = [Ag^+][Cl^-]$$

4. ABD

Sol.



5. BC

Sol. Benzyne mechanism, $K_{eq} < 1$

6. AD

Sol. S_N1 mechanism

7. ABD

Sol. Conceptual

8. ABD

Sol. A = Phthalic anhydride
B = Phenol/orthalic acid

9. C

Sol. $\Delta H = +ve$, $T \uparrow = K_{eq} \uparrow$

10. D

Sol. $P \uparrow = \text{Conc}^n \uparrow$

11. B

Sol. Chiral centre generated, without symmetry.

12. A

Sol. $\left. \begin{array}{l} \text{POS X} \\ \text{COS X} \end{array} \right\} \text{Optically active}$ **(PART – B)**

1. 8

Sol. Refining of metals

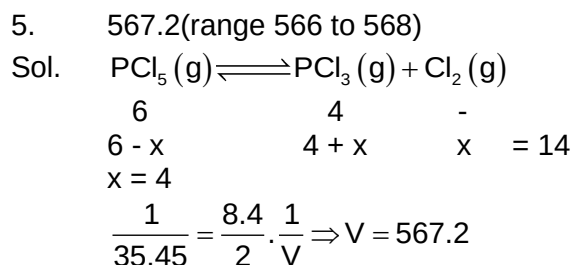
2. 55.85 (Range 55 to 56)

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Sol. $\frac{134.04}{M} \times 1 = 1.2 \times 0.4 \times 5$

3. 7
Sol. Theoretical

4. 423.30 (Range 421 to 425)
Sol. Speed = λv



6. 50 (Range 49 to 51)
Sol. Theoretical

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SECTION – III : MATHEMATICS**(PART – A)**

1. AC

Sol. $|a + b| = 20 \Rightarrow a + b = 20e^{i0}$

Assume $a + b = 20$ (Coordinate axes can be related so that $\theta = 0$)

$$|a^2 + b^2| = 16 \Rightarrow a^2 + b^2 = 16e^{i\alpha}$$

$$|a^3 + b^3| = |(a + b)^3 - 3ab(a + b)|$$

$$= \left| (a + b)^3 - \frac{3}{2} \{ (a + b)^2 - (a^2 + b^2) \} (a + b) \right|$$

$$= \left| \frac{3}{2} (a^2 + b^2)(a + b) - \frac{1}{2} (a + b)^3 \right|$$

$$= |480e^{i\alpha} - 4000|$$

$$= 160 |3e^{i\alpha} - 25|$$

$$= 160 \sqrt{634 - 150 \cos \alpha}$$

$$= 4480 (\cos \alpha = -1) \text{ and } 3520 (\cos \alpha = 1)$$

2. ABCD

Sol. Let (x, y) be point common to both parabolas at point of tangency.

$$y - x = \alpha(x^2 - y^2) + \alpha(x - y)$$

$$\Rightarrow (x - y) \{ \alpha(x + y) + \alpha + 1 \} = 0$$

$$\text{Either } x = y \text{ OR } \alpha(x + y) + \alpha + 1 = 0$$

$$\text{Solving } y = x \text{ with } y = \alpha x^2 + \alpha x + \frac{1}{24}$$

$$x = \alpha x^2 + \alpha x + \frac{1}{24} \Rightarrow \alpha x^2 + (\alpha - 1)x + \frac{1}{24} = 0$$

Since the curves touch $\Rightarrow$ above quadratic should have one root only

$$\Rightarrow \text{Disc} = 0 \Rightarrow (\alpha - 1)^2 - 4\alpha \left(\frac{1}{24} \right) = 0 \Rightarrow 6\alpha^2 - 13\alpha + 6 = 0$$

$$\Rightarrow \alpha = \frac{2}{3}, \frac{3}{2}$$

$$\text{Solving } \alpha(x + y) + \alpha + 1 = 0 \text{ and } y = \alpha x^2 + \alpha x + \frac{1}{24}$$

$$\alpha x^2 + (\alpha + 1)x + \frac{\alpha + 1}{\alpha} + \frac{1}{24} = 0$$

$$\Rightarrow \text{Disc} = 0 \Rightarrow (\alpha + 1)^2 - 4\alpha \left(\frac{25}{24} + \frac{1}{\alpha} \right) = 0 \Rightarrow \alpha = \frac{13 \pm \sqrt{601}}{12}$$

3. AB

Sol. By default, x – axis and y – axis (both tangents) are perpendicular

 $\Rightarrow$ Origin lies on the directrix

Also, line joining point of intersection of tangents at A and B and mid – point of chord AB is parallel to axis or perpendicular to directrix.

$$\Rightarrow \text{midpoint of AB} = \left(2, \frac{3}{2}\right) \equiv M$$

$$\Rightarrow \text{slope of OM} = \frac{\frac{3}{2} - 0}{2 - 0} = \frac{3}{4} \Rightarrow \text{slope of directrix} = -\frac{4}{3}$$

$$\Rightarrow \text{Equation of directrix : } y = -\frac{4}{3}x \text{ i.e. } 4x + 3y = 0$$

Line segment joining point of contact (0, 3) and (0, 0) subtend a right angle at focus (h, k)

$$\text{Slope of AB (focal chord)} = -\frac{3}{4}$$

$$\text{Slope of OS} = \frac{4}{3} = \frac{k-0}{h-0} \Rightarrow k = \frac{4}{3}h$$

Since tangents at A and B are perpendicular AB $\left(\frac{x}{4} + \frac{y}{3} = 1\right)$ is the focal chord so

$$\frac{h}{4} + \frac{k}{3} = 1$$

$$\text{Solving for h and k, we get } h = \frac{36}{25}, k = \frac{48}{25}$$

4. CD

Sol. $8\sin^2 x \cos x = \sqrt{3} \sin x + \cos x$

$$4\sin 2x \sin x = \sqrt{3} \sin x + \cos x$$

$$2(\cos x - \cos 3x) = \sqrt{3} \sin x + \cos x$$

$$2\cos 3x = \cos x - \sqrt{3} \sin x$$

$$\cos 3x = \cos x \cdot \frac{1}{2} - \sin x \cdot \frac{\sqrt{3}}{2}$$

$$\cos 3x = \cos \left(x + \frac{\pi}{3}\right)$$

$$3x = 2n\pi \pm \left(x + \frac{\pi}{3}\right)$$

$$\Rightarrow x = n\pi + \frac{\pi}{6} \text{ or } x = \frac{n\pi}{2} - \frac{\pi}{12}$$

5. BC

Sol. a_n = Number of sets containing the element 'n' + Number of sets that do not contain 'n'

Number of subsets containing 'n' = set containing 'n' itself + number of subsets to be

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$$= 1 + a_{n-2}$$

Number of subsets not containing 'n' = subset to be chosen from $\{1, 2, 3, 4, \dots, n-2, n-1\}$

$$= a_{n-1}$$

$$\text{So, } a_n = (1 + a_{n-2}) + a_{n-1}$$

6. ABD

Sol. Let $\bar{x} = 40$ be the mean of the scores and let $x_1, x_2, \dots, x_p$ be the scores higher than or equal to 80. Then

$$(\bar{x} - x_{p+1}) + (\bar{x} - x_{p+2}) + \dots + (\bar{x} - x_{100}) = (x_1 - \bar{x}) + \dots + (x_p - \bar{x}) \geq (80 - 40)p = 40p$$

$$100 \cdot 20^2 = (x_1 - \bar{x})^2 + \dots + (x_p - \bar{x})^2 + (x_{p+1} - \bar{x})^2 + \dots + (x_{100} - \bar{x})^2$$

By RMS $\geq$ AM inequality $\sqrt{\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n}} \geq \frac{a_1 + a_2 + \dots + a_n}{n}$

We have

$$\sqrt{\frac{(\bar{x} - x_{p+1})^2 + (\bar{x} - x_{p+2})^2 + \dots + (\bar{x} - x_{100})^2}{100 - p}} \geq \frac{(\bar{x} - x_{p+1}) + (\bar{x} - x_{p+2}) + \dots + (\bar{x} - x_{100})}{100 - p} \geq \frac{(40 - 0)p}{100 - p}$$

$$\text{So } (x_{p+1} - \bar{x})^2 + (x_{p+2} - \bar{x})^2 + \dots + (x_{100} - \bar{x})^2 \geq \frac{(40p)^2}{100 - p}$$

Similarly,

$$\sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_p - \bar{x})^2}{p}} \geq \frac{(x_1 - \bar{x}) + (x_2 - \bar{x}) + \dots + (x_p - \bar{x})}{p} \geq \frac{(80 - 40)p}{p} = 40$$

$$(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_p - \bar{x})^2 \geq (40)^2 p$$

$$100 \cdot 20^2 = (x_1 - \bar{x})^2 + \dots + (x_p - \bar{x})^2 + (x_{p+1} - \bar{x})^2 + \dots + (x_{100} - \bar{x})^2 \geq (40)^2 p + \frac{(40p)^2}{100 - p}$$

On simplifying we get

$$100 - p \geq 4p$$

$$p \leq 20$$

7. BCD

Sol. Let AB lie along the x – axis.

Let $A \equiv (0,0)$

$$\tan \frac{B}{2} = \frac{1}{3} \Rightarrow \tan B = \frac{3}{4}$$

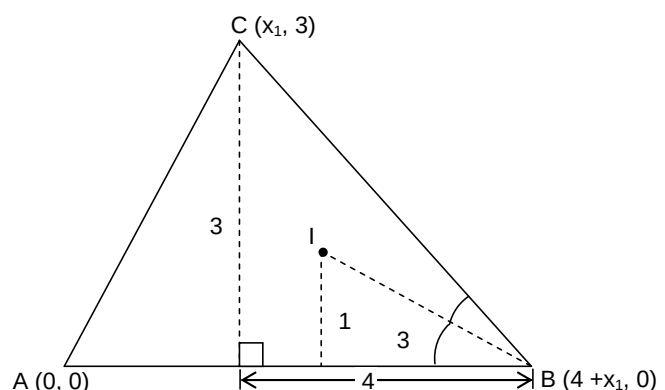
Assume, $B \equiv (4 + x_1, 0)$, so that

tanB can be equal to $\frac{3}{4}$

and $C \equiv (x_1, 3)$

$$\text{Centroid} \equiv \left(\frac{4 + 2x_1}{3}, 1 \right)$$

Since $IG \parallel AB$, we have inradius, $r = 1$



$$r = \frac{\Delta}{S}$$

$$\Rightarrow \Delta =$$

$r = \frac{\Delta}{S}$
 $\Rightarrow \Delta = S$

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$$\Rightarrow \frac{1}{2}(3)(4+x_1)$$

$$= \frac{(4+x_1)+5+\sqrt{9+x_1^2}}{2}$$

$$\Rightarrow 3x_1^2 + 12x_1 = 0$$

$$\Rightarrow x_1 = 0 \text{ } (-4 \text{ is rejected otherwise}$$

area of triangle will become zero) .

So the foot of perpendicular from C on AB coincides with A

$$\Rightarrow \angle CAB = 90^\circ$$

$$\Rightarrow \sin B = \frac{b}{a} = \frac{3}{5}; \tan B = \frac{3}{4} = \frac{b}{c}$$

8. BC

$$\text{Sol. (B) } \frac{1}{a^2(1-r^2)} \left[1 + \frac{1}{r^2} + \frac{1}{r^4} + \dots + \frac{1}{r^{2(n-2)}} \right] = \frac{1 - \frac{1}{r^{2(n-1)}}}{\left(1 - \frac{1}{r^2}\right) a^2 (1-r^2)}$$

$$= \frac{(1-r^{2n})}{a^2 (1-r^2)^2 r^{2n-4}}$$

$$\text{(C) } \frac{1}{a^m(1+r^m)} \left[1 + \frac{1}{r^m} + \frac{1}{r^{2m}} + \dots + \frac{1}{r^{(n-2)m}} \right] = \frac{\left(1 - \frac{1}{r^{(n-1)m}}\right)}{\left(1 - \frac{1}{r^m}\right) a^m (1+r^m)}$$

$$= \frac{(1-r^{mn-m})}{(1-r^m)(r^{mn-m} - r^{mn-2m})}$$

9. B

10. C

Sol.

(I) Let $q_1 = 1 - p_1$ and $q_2 = 1 - p_2$

$\therefore$ A can win the game at the 1st, 4th, 7th... trials.

$$\therefore P(\text{A wins}) = \frac{1}{6} + \frac{5}{6}(q_1)(q_2)\left(\frac{1}{6}\right) + \left(\frac{5}{6}\right)^2$$

$$(q_1)^2 (q_2)^2 \left(\frac{1}{6}\right) + \dots$$

$$= \frac{\frac{1}{6}}{1 - \frac{5}{6}q_1q_2} = \frac{1}{6 - 5q_1q_2} = \frac{1}{6 - 5\left(\frac{4}{3}\right)\left(\frac{3}{4}\right)} = \frac{1}{\lambda_1} [\text{given}]$$

$$\therefore \lambda_1 = 3$$

$$\text{(II) } P(\text{C wins}) = \frac{5}{6} \cdot q_1 \cdot p_2 + \left(\frac{5}{6}\right)^2 \cdot q_1^2 \cdot q_2 \cdot p_2 + \dots$$

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$$\begin{aligned}
 &= \frac{\frac{5}{6}q_1 \cdot p_2}{1 - \frac{5}{6} \cdot q_1 q_2} = \frac{5q_1 p_2}{6 - 5q_1 q_2} = \frac{5 \times \frac{4}{5} \times \frac{1}{4}}{6 - 5 \times \frac{4}{5} \times \frac{3}{4}} = \frac{1}{3} \\
 &= \frac{1}{\lambda_2} \quad \text{[given]} \\
 \therefore \lambda_2 &= 3
 \end{aligned}$$

$$\begin{aligned}
 \text{(III)} \quad \therefore P(A \text{ wins}) &= P(B \text{ wins}) \\
 \Rightarrow \frac{1}{6 - 5q_1 q_2} &= \frac{5p_1}{6 - 5q_1 q_2} \\
 \therefore p_1 &= \frac{1}{5} = \frac{1}{\lambda_3} \quad \text{[given]} \\
 \therefore \lambda_3 &= 5
 \end{aligned}$$

$$\begin{aligned}
 \text{(IV)} \quad P(Q \text{ wins}) &= P(B \text{ wins}) = P(C \text{ wins}) \\
 \Rightarrow \frac{1}{6 - 5q_1 q_2} &= \frac{5p_1}{6 - 5q_1 q_2} = \frac{5q_1 p_2}{6 - 5q_1 q_2} \\
 \Rightarrow 1 &= 5p_1 = 5q_1 p_2 \\
 \Rightarrow p_1 &= \frac{1}{5}, \frac{1}{p_2} = 5q_1 = 5 \left(1 - \frac{1}{5}\right) = 4 = \lambda_4 \quad \text{[given]} \\
 \therefore \lambda_4 &= 4
 \end{aligned}$$

11. D

12. C

(I) Number of dissimilar terms in the expansion of $(x + 2y + 3z)^n$ ($n \in \mathbb{N}$)

$$= {}^{n+3-1}C_{3-1} = {}^{n+2}C_2 = \frac{1}{2}n^2 + \frac{3}{2}n + 1$$

$$\therefore a = \frac{1}{2}, b = \frac{3}{2}, c = 1$$

$$\text{Hence, } a + b + c = \frac{1}{2} + \frac{3}{2} + 1 = 3 \text{ and } a + b = \frac{1}{2} + \frac{3}{2} = 2 = 2c$$

$$\begin{aligned}
 \text{(II)} \quad \text{We have } (x + y + z)^{2n+1} &= \{(x + y) + z\}^{2n+1} \\
 &= (x + y)^{2n+1} + {}^{2n+1}C_1 (x + y)^{2n} z + {}^{2n+1}C_2 (x + y)^{2n-1} z^2 + {}^{2n+1}C_3 (x + y)^{2n-2} z^3 + \dots \\
 &\quad {}^{2n+1}C_{2n} (x + y) z^{2n} + {}^{2n+1}C_{2n+1} z^{2n+1} \text{ and } (x + y - z)^{2n+1} = \{(x + y) - z\}^{2n+1} \\
 &= (x + y)^{2n+1} - {}^{2n+1}C_1 (x + y)^{2n} z + {}^{2n+1}C_2 (x + y)^{2n-1} z^2 \\
 &\quad - {}^{2n+1}C_3 (x + y)^{2n-2} z^3 + \dots + {}^{2n+1}C_{2n} (x + y) z^{2n} - {}^{2n+1}C_{2n+1} z^{2n+1} \\
 \therefore (x + y + z)^{2n+1} - (x + y - z)^{2n+1} &= 2 \left\{ {}^{2n+1}C_1 (x + y)^{2n} z + {}^{2n+1}C_3 (x + y)^{2n-2} z^3 + \dots + z^{2n+1} \right\}
 \end{aligned}$$

$$\therefore \text{The number of dissimilar terms in the expansion of } (x + y + z)^{2n+1} - (x + y - z)^{2n+1}$$

$$= (2n+1) + (2n-1) + (2n-3) + \dots + 5 + 3 + 1$$

$$= \frac{(n+1)}{2} (n+1+1) = (n+1)^2$$

$$= n^2 + 2n + 1$$

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$$\therefore a=1, b=2, c=1$$

$$\text{Hence, } a+b+c=1+2+1=4$$

(III) We have, $(x-y+z)^n = \{x-(y-z)\}^n$
 $= x^n - {}^nC_1 x^{n-1}(y-z) + {}^nC_2 x^{n-2}(y-z)^2 + {}^nC_3 x^{n-3}(y-z)^3$
 $+ \dots - {}^nC_{n-1} x(y-z)^{n-1} + {}^nC_n (y-z)^n$ and $(x+y-z)^n = (x+y-z)^n$
 $= x^n + {}^nC_1 x^{n-1}(y-z) + {}^nC_2 x^{n-2}(y-z)^2 + {}^nC_3 x^{n-3}(y-z)^3 + \dots + {}^nC_{n-1} x(y-z)^{n-1} + (y-z)^n$
 $\therefore (x+y+z)^n + (x+y-z)^n$
 $= 2 \left[x^n + {}^nC_2 x^{n-2}(y-z)^2 + {}^nC_4 x^{n-4}(y-z)^4 + \dots + (y-z)^n \right]$
 $\therefore$ The number of dissimilar terms in the expansion of
 $(x-y+z)^n + (x+y-z)^n = 1+3+5+\dots+(n+1)$

$$\frac{(n+2)}{2} (1+n+1) = \frac{(n+2)^2}{4} = \frac{1}{4} (n^2 + n + 1)$$

$$\therefore a = \frac{1}{4}, b=1, c=1$$

$$\text{Hence, } b+c=1+1=2=8a$$

(IV) $\therefore \left(\frac{x^2+1+x^4}{x^2} \right)^{\sum n}$
 $= \frac{a_0 + a_2 x^2 + a_4 x^4 + \dots + a_{2n(n+1)} x^{2n(n+1)}}{x^{n(n+1)}}$
 $\therefore$ Number of terms $= \frac{1}{2} \cdot 2n(n+1) + 1 = n^2 + n + 1$

$$\therefore a=1, b=1, c=1$$

$$\text{Hence, } a+b+c=1+1+1=3 \text{ and } a+b=1+1=2=2c$$

(PART - B)

1. 0

Sol. Assume $\det(AB-BA) \neq 0$

$$\text{Since } A^2 + B^2 = 2AB \text{ we have } A^2 + B^2 - AB - BA = 2AB - AB - BA$$

$$\Rightarrow (A-B)^2 = AB-BA$$

$$\Rightarrow \det(A-B)^2 = \det(AB-BA) \neq 0$$

$$\Rightarrow \det(A-B) \neq 0 \Rightarrow A-B \text{ is invertible.}$$

$$\text{Also, if } A^2 + B^2 = 2AB$$

$$\text{then } A^2 - AB = AB - B^2$$

$$A(A-B) = (A-B)B$$

$$\Rightarrow A(A-B)(A-B)^{-1} = (A-B)B(A-B)^{-1}$$

$$\Rightarrow A = (A-B)B(A-B)^{-1}$$

Subtracting B from both sides.

$$\Rightarrow A-B = (A-B)B(A-B)^{-1} - B$$

$$\Rightarrow (A-B)^{-1}(A-B) = (A-B)^{-1}(A-B)B(A-B)^{-1} - (A-B)^{-1}B$$

$$\Rightarrow I = B(A-B)^{-1} - (A-B)^{-1}B$$

$$\text{trace}(I) = \text{trace} \{ B(A-B)^{-1} - (A-B)^{-1}B \}$$

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$$\Rightarrow \text{trace}(I) = \text{trace}\{B(A-B)^{-1}\} - \text{trace}\{(A-B)^{-1}B\}$$

$$\Rightarrow \text{trace}(I) = 0 \quad (\because \text{trace}(AB) = \text{trace}(BA))$$

which is a contradiction.

$$\text{Hence, } \det(AB - BA) = 0$$

2. 24

Sol.
$$p(-t-1) = \frac{(-t-1)(-t-1-1)(-t-1-2)\dots(-t-1-2023)}{2023!}$$

$$= \frac{(t+1)(t+2)(t+3)\dots(t+2024)}{2023!}$$

$$\int_0^1 \left\{ \frac{1}{t+1} + \frac{1}{t+2} + \dots + \frac{1}{t+2024} \right\} \frac{(t+1)(t+2)(t+3)\dots(t+2024)}{2023!} dt$$

$$= \frac{1}{2023!} \left| (t+1)(t+2)(t+3)\dots(t+2024) \right|_0^1$$

$$= \frac{1}{2023!} (2025! - 2024!)$$

$$= \frac{(2024)!}{2023} \{2025 - 1\}$$

$$= (2024)^2$$

$$\Rightarrow k = 2024$$

3. 6

Sol.
$$\tan 78^\circ + \tan 72^\circ + \tan 60^\circ$$

$$= \frac{1}{2 \cos 72^\circ \cos 75^\circ} + \cot 30^\circ$$

$$= \frac{2 \cos 36^\circ}{\cos 78^\circ} + 2 \cos 30^\circ \quad 2 \cos 72^\circ = 2 \sin 18^\circ = \frac{\sqrt{5}-1}{2}$$

$$= \frac{2 \cos 36^\circ + 2 \cos 30^\circ \cos 78^\circ}{\cos 78^\circ} \quad = \frac{2}{\sqrt{5}+1}$$

$$= \frac{2 \cos 36^\circ + \cos 108^\circ + \cos 48^\circ}{\cos 78^\circ} \quad = \frac{1}{2 \cos 36^\circ}$$

$$= \frac{2 \cos 36^\circ - \cos 72^\circ + \cos 48^\circ}{\cos 78^\circ} \quad \cos 36^\circ - \cos 72^\circ = \frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} = \frac{1}{2}$$

$$= \frac{1 + \cos 72^\circ + \cos 48^\circ}{\cos 78^\circ} = \frac{1 + 2 \cos 60^\circ \cos 12^\circ}{\sin 12^\circ} = \frac{1 + \cos 12^\circ}{\sin 12^\circ} = \cot 6^\circ$$

$$\Rightarrow k = 6$$

4. 111

Sol. In general, number of triangles is ${}^nC_3 + 4({}^nC_4) + 5({}^nC_5) + {}^nC_6$

So, putting $n = 6$, number of triangles is 111

5. 1

Sol.
$$4 \int_0^{\pi/2} (f(x) - x^2) dx + 125 \int_0^{3/2} \frac{dx}{\sqrt{f(x) + x^2}} = \frac{225}{2}$$

TG ~ @bohring_bot

Let $g(x) = f(x) + x^2$

$$4 \int_0^{3/2} g(x) dx + 125 \int_0^{3/2} \frac{1}{2\sqrt{g(x)}} dx + 125 \int_0^{3/2} \frac{1}{2\sqrt{g(x)}} dx$$

$$\geq 3 \int_0^{3/2} \left(4g(x) \cdot \frac{125}{2\sqrt{g(x)}} \cdot \frac{125}{2\sqrt{g(x)}} \right)^{1/3} dx$$

$$= \frac{225}{2}$$

$$\text{Equality occurs if } 4g(x) = \frac{125}{2\sqrt{g(x)}} \Rightarrow (g(x))^{\frac{3}{2}} = \left(\frac{5}{2}\right)^3$$

$$\Rightarrow g(x) = \frac{25}{4}$$

6. 0.17

Sol. $S = \sum_{i=0}^{\infty} \int_0^1 (x^{4i} - 2x^{4i+1} + x^{4i+2}) dx = \frac{1}{2} \int_0^1 \left(\frac{1}{1-x^4} - \frac{2x}{1-x^4} + \frac{x^2}{1-x^4} \right) dx = \frac{1}{2} \int_0^1 \frac{(1-x)^2}{1-x^4} dx]$

$$\text{So, } S = \frac{1}{2} \int_0^1 \left(\frac{1}{1+x} - \frac{x}{1+x^2} \right) dx = \frac{\ln 2}{4}$$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 1

QP Code: 100204

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 234

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **Part-B (01-08)** contains eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

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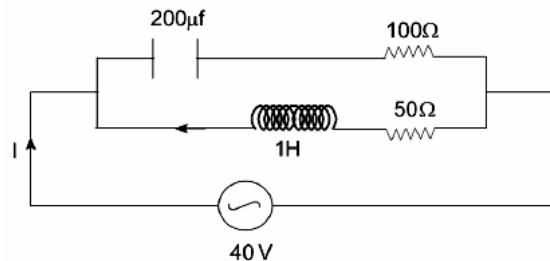
SECTION-1 : PHYSICS

PART – A

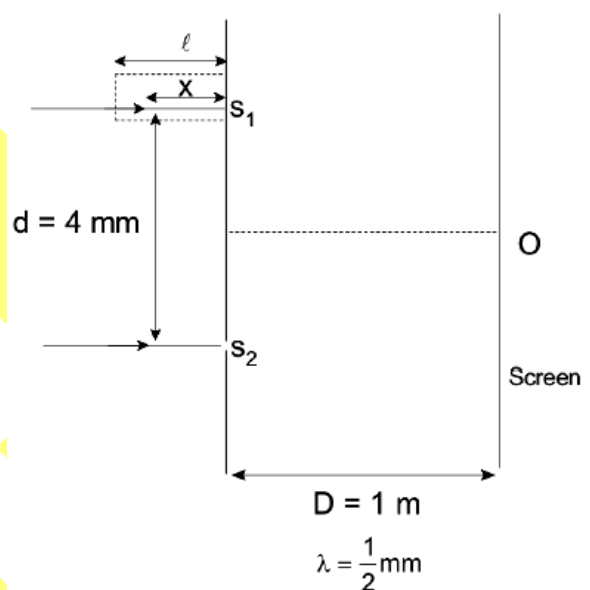
(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. In the given circuit, the AC source has $\omega = 50 \text{ rad/s}$ considering the inductor and capacitor to be ideal, the correct choice(s) is/are
- (A) The voltage across 100Ω resistor = 20.
 (B) The voltage across 50Ω resistor = 20.
 (C) The current through the circuit is $I = \frac{\sqrt{5}}{\sqrt{2}} \text{ A}$
 (D) the current through the circuit is $I = 1.2 \text{ A}$



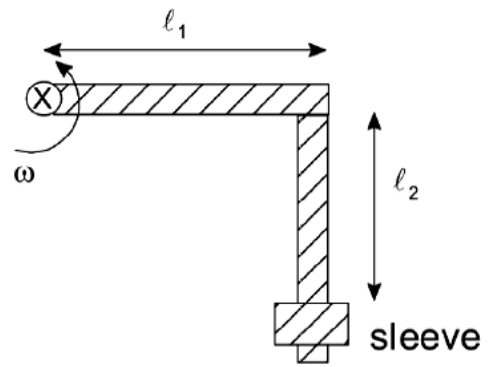
2. In YDSE, $D = 1 \text{ m}$, $d = 4 \text{ mm}$ and $\lambda = \frac{1}{2} \text{ mm}$ in the figure shown, a parallel beam of light incident on the plane of the slits of YDSE. Light incident on the slit S_1 , passes through a medium of variable refractive index $\mu = (2x + 1)$ (where x is the distance from the plane of slits in mm) up to a distance $\ell = 1 \text{ mm}$ before falling on S_1 . Rest of the space filled with air. Then choose the correct option(s):
- (A) Position of central bright fringe is at a distance $y = \frac{1}{\sqrt{3}} \text{ m}$ above the central line.
 (B) Position of central bright fringe is at a distance $y = \frac{1}{2} \text{ m}$ above the central line.
 (C) Number of maxima obtaining on the screening is 7.
 (D) Number of maxima obtaining on the screening is 8.



Space For Rough Work

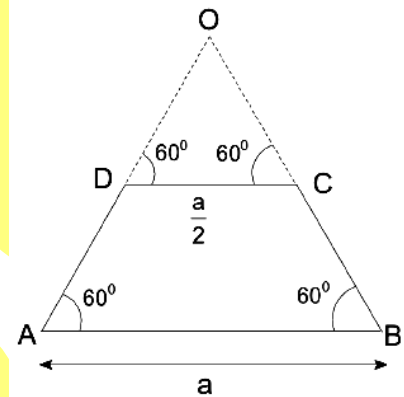
TG ~ @bohring_bot

3. A rough L – shaped rod is located in a horizontal plane and a sleeve of mass m is inserted in the rod. The rod is rotated with a constant angular velocity ω in the horizontal plane. The lengths ℓ_1 and ℓ_2 are shown in figure. The normal reaction and frictional force acting on the sleeve when it starts slipping are (μ : coefficient of friction between rod and sleeve).



- (A) $N = m\omega^2\ell_1$
 (B) $\omega^4 = \frac{\mu^2 g^2}{\ell_2^2 - \mu^2 \ell_1^2}$
 (C) $N = m\sqrt{g^2 + \omega^4 \ell_1^2}$
 (D) $f = \mu N$

4. Consider a uniformly charged sheet ABCD, which is a part of an equilateral triangular sheet of a side a as shown in the figure. Choose the correct options regarding the electric field E at point O due to this sheet.



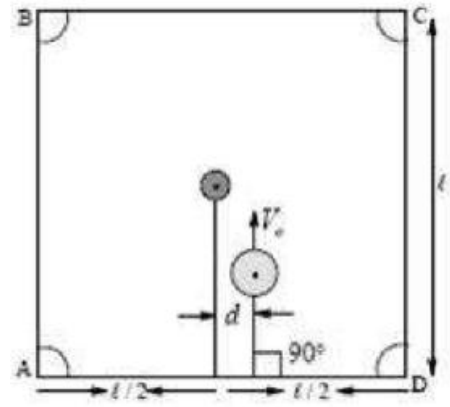
- (A) Magnitude of $\vec{E}$, increases with the increase in 'a' (Keeping charge density same)
 (B) Magnitude of E , decreases with the increase in 'a' (Keeping total charge same)
 (C) If charge density is σ and $a = 1$ m, magnitude of E is equal to $\frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$
 (D) If charge density is σ and $a = 2$ m, magnitude of E is equal to $\frac{7\sigma}{22\epsilon_0} \ln \sqrt{2}$

5. Which of the following statements is/are correct for mechanical standing wave on a stretched wire?
- (A) Elastic potential energy of a small element at antinode is constant and minimum.
 (B) Elastic potential energy of a small element at node is constant and maximum.
 (C) Elastic potential energy of a small element at node is constant and maximum.
 (D) Total kinetic energy between two consecutive nodes become maximum twice in one-time period.

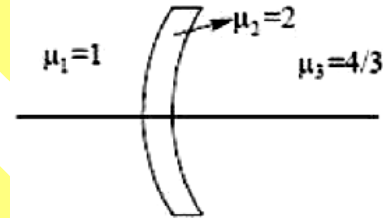
Space For Rough Work

TG ~ @bohring_bot

6. The queen is put at the centre of a perfectly smooth carom board (square with side ℓ). The striker strikes the queen with a speed V_e as shown in the figure. Radius of the queen is $\sqrt{10}$ cm and that of the striker is $2\sqrt{10}$ cm. Coefficient of restitution for the collision between the queen and the striker is $\frac{1}{2}$ and that for the collision between the queen and the walls of the board is 1. (Assume $\ell \gg \text{radius of queen}$)



- (A) The value of 'd' for which the queen gets in the hole A is 3 cm.
 (B) The value of 'd' for which the queen gets in the hole A is 2 cm.
 (C) The value of 'd' depends on the coefficient of restitution between the queen and the striker.
 (D) The value of 'd' independent on the coefficient of restitution between the queen and the striker.
7. A thin lens of same radius of curvature 20 cm is having two different medium on its two sides extending upto infinity as shown in the figure. Then
 (A) it may behave as a converging lens of focal length 60 cm.
 (B) it may behave as a diverging lens of focal length 60 cm.
 (C) it may behave as a converging lens of focal length 80 cm.
 (D) it may behave as a diverging lens of focal length 80 cm.



Space For Rough Work

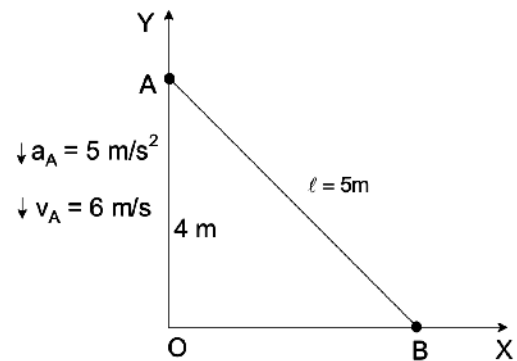
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(Single Correct Choice Type)

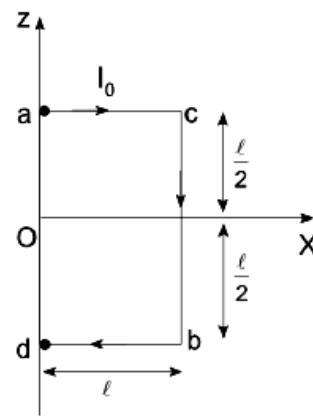
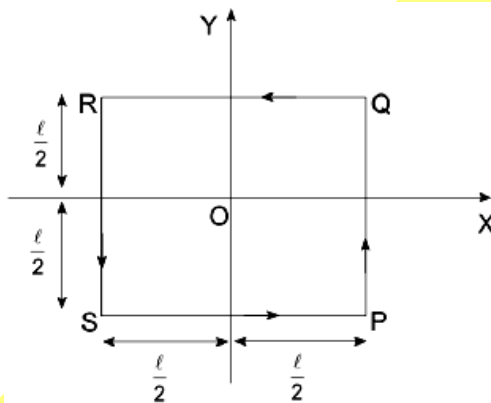
This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

8. Point A on the rod AB has an acceleration of 5 m/s^2 and a velocity of 6 m/s at an instant as shown in the figure. The acceleration of the end B at the same moment is:

- (A) $-\frac{80}{3} \hat{i} \text{ m/s}^2$
 (B) $\frac{80}{3} \hat{i} \text{ m/s}^2$
 (C) $-\frac{40}{3} \hat{i} \text{ m/s}^2$
 (D) $-\frac{75}{6} \hat{i} \text{ m/s}^2$



9. Current I_0 is flowing through a bent wire $a \rightarrow b \rightarrow c \rightarrow d$ in $z-x$ plane as shown in figure, then $\oint \vec{B} \cdot d\vec{\ell}$ over the loop PQRS lying in the $X-Y$ plane as shown in figure, due to the bent wire abcd



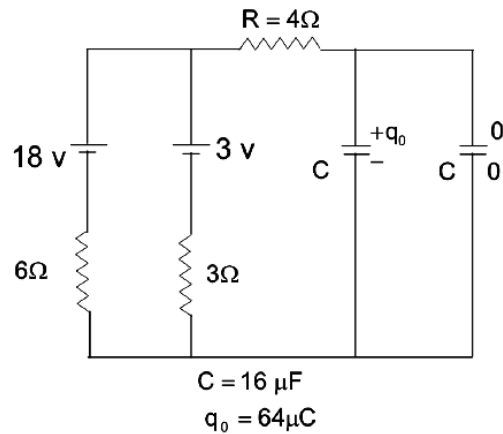
- (A) $\frac{\mu_0 I_0}{8}$ (B) $\frac{\mu_0 I_0}{7}$ (C) $\frac{\mu_0 I_0}{5}$ (D) $\frac{\mu_0 I_0}{3}$

Space For Rough Work

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10. The circuit is completed at $t = 0$, then heat loss in $R = 4\Omega$ resistance, till the capacitor gets fully charged:

(A) $64 \mu\text{J}$
 (B) $384 \mu\text{J}$
 (C) $320 \mu\text{J}$
 (D) $576 \mu\text{J}$



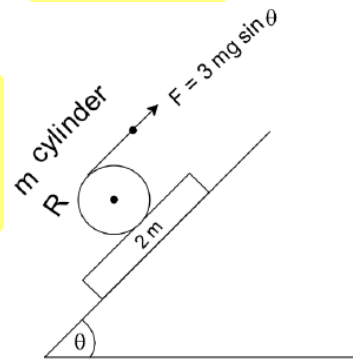
11. A simple harmonic plane progressive wave is travelling along the +ve x-axis having amplitude A and $\omega = \pi \text{ rad/s}$. At $t = 1 \text{ sec}$, particle at $x = 2 \text{ m}$ and $x = 3 \text{ m}$ are moving in downward and upward direction respectively at the position $y = +\frac{A}{2}$. The time when

maximum power transfer will take place at $x = \frac{11}{4} \text{ m}$?

(A) $\frac{5}{4} \text{ sec}$ (B) $\frac{5}{3} \text{ sec}$ (C) $\frac{7}{4} \text{ sec}$ (D) $\frac{7}{5} \text{ sec}$

12. There is sufficient friction between long plank of mass $2m$ and cylinder of mass m and radius R to prevent slipping. Whole system is placed on the smooth inclined surface. Now, force $F = 3mg \sin \theta$ is applied on the ideal string wrapped over the cylinder as shown in figure. Choose correct option:

(A) If the plank moves down a distance of $S \text{ m}$, work done by the applied force F on the string is $5mg \sin \theta S$.
 (B) If the plank moves down a distance of $S \text{ m}$, work done by the applied force F on the string is $6mg \sin \theta S$.
 (C) If the plank moves down a distance of $S \text{ m}$, work done by the applied force F on the string is $12mg \sin \theta S$.
 (D) If the plank moves down a distance of $S \text{ m}$, work done by the applied force F on the string is $15mg \sin \theta S$.



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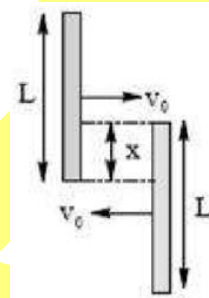
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13. A vertical thermally insulated cylinder of volume V contains n moles of an ideal monoatomic gas under a weightless piston. A load of weight W is placed on the piston as a result of which the piston is displaced. If the initial temperature of the gas is 300 K , area of piston is A and atmospheric pressure P_0 . (take $W = P_0 A$). Determine the value of final temperature of the gas.
- (A) 375 K (B) 425 K
(C) 475 K (D) 515 K

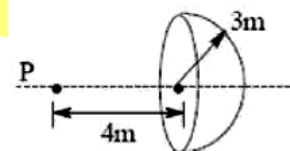
PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. Two identical rigid rods, each of mass 1 kg & length $\sqrt{3}\text{ m}$, are moving opposite to each other without rotation on a smooth horizontal table as shown. For what maximum value of ' x ' (after rounding off), the direction of motion of each rod would not change after collision, irrespective of the type of collision. Assume no sticking.

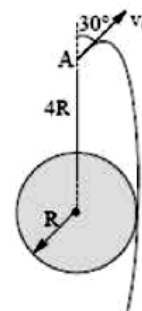


2. A point source of power ' P ' is placed on the axis of a thin hemispherical shell as shown. The shell has a radius of 3 m and behaves like a perfect black body. If the steady state temperature of shell is T . Find T^4 (after rounding off). (Take $P = 2400\pi \sigma W$, σ = Stefan's constant)



3. A particle is projected from point 'A', that is at a distance $4R$ from the centre of the Earth, with speed V_0 as shown. If the particle passes grazing the earth's surface, then it is found that $V_0 = 100 \times N$. Find the value of N (after rounding off).

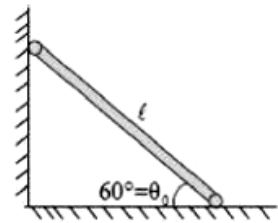
[Take $\frac{GM}{R} = 6.4 \times 10^7 \text{ m}^2 / \text{s}^2$, $\sqrt{2} = 1.414$]



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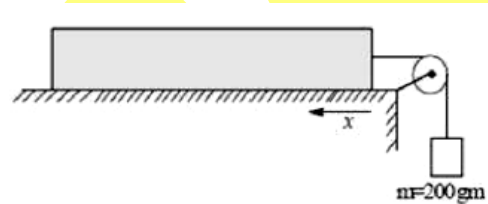
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4. A ladder of mass m and length ℓ stands against a frictionless wall with its feet on a frictionless floor. If it is let go at an initial angle $\theta_0 = 60^\circ$ then the angle ' θ ' at which the ladder loses contact with the wall is given as $\sin^{-1}(1/\sqrt{N})$, find 'N'.

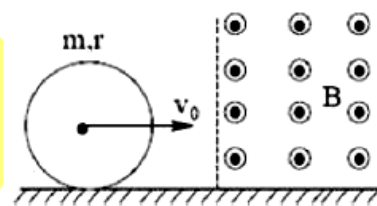


5. An electric charge distribution produces an electric field $\vec{E} = C(1 - e^{-\alpha r}) \frac{\hat{r}}{r^2}$ where $C = \frac{1}{4\pi\epsilon_0}$ and α are constant. If the net charge within the radius $r = \frac{1}{\alpha}$ is $(1 - e^{-N})$, then find the value of 'N'?

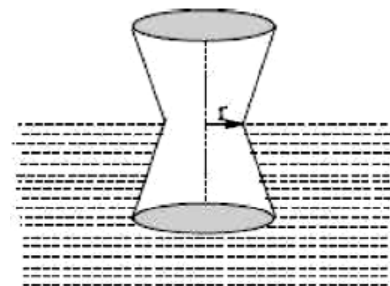
6. Uniform rope of mass = 5 kg and length 1 meter is lying on a rough horizontal surface. Coefficient of static friction varies from right end of the rope as $\mu = \mu_0 x$ where $\mu_0 = 0.5$ per meter. A block of mass 200 gm is hanging from an ideal string which passes over an ideal pulley as shown in the figure. The minimum value of x (in cm) for which tension at some cross-section of rope becomes zero is $10 \times N$. Find the value of 'N'.



7. A ring of mass m and radius r is made of an insulating material carries uniformly distributed charge. Initially it rests on a frictionless horizontal table top with its plane vertical. The charge on the ring, such that it starts rolling on entering completely into the region of the magnetic field, is $\frac{\sqrt{N}mv_0}{rB}$, then the value of 'N'.



8. A solid object of mass $\frac{22}{7}$ kg is in the shape of a drum, its half submerged in water of density 1000 kg/m^3 with dimensions as shown in the figure. Find the time period (in seconds) of small vertical oscillations of the drum. [Take $r = \frac{22}{7} \text{ cm}$]



Space For Rough Work

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SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

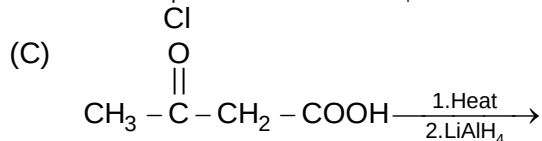
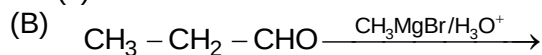
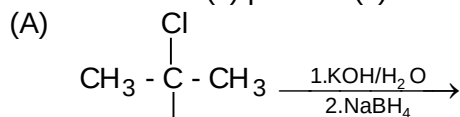
- The correct statement(s) regarding the wave function $\psi_{4, 1, -1}$ is/are
 (A) it represent the 4p-subshell
 (B) orbital associated with the wave function has one nodal plane and two radial nodes
 (C) the energy of this orbital is higher than that of 3s orbital
 (D) arsenic(Z = 33) contains one electron in this orbital
- The correct statement(s) regarding a complex ion $[\text{MnCl}_4]^{2-}$ is/are that
 (A) according to crystal field theory, the electronic configuration of the 3d-orbital of Mn^{2+} in the complex is $e^2 t_2^3$.
 (B) it is a square planar complex
 (C) the Cl^- ions acts as weak field ligands
 (D) Mn^{2+} undergoes sp^3 -hybridization in the complex
- The solubility product constant(K_{sp}) of $\text{Be}(\text{OH})_2$ is 5×10^{-10} at 298K. Choose correct statement(s)
 (A) the concentration of it's saturated solution is $5 \times 10^{-4} \text{ mol L}^{-1}$
 (B) it's solubility increases in a buffer of $\text{pH} > 11$ and decreases in a buffer of $\text{pH} < 11$
 (C) addition of NH_4Cl increase it's solubility
 (D) addition of NaCl does not affect solubility
- The product(s) of the given reaction

$$\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{CH} - \text{CH}_3 \xrightarrow{\text{NBS}(\text{excess})/h\nu} \text{Products}$$
 is/are
 (A) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} = \text{CH} - \text{CH}_2\text{Br}$
 (B) $\text{CH}_2 = \underset{\text{CH}_3}{\text{C}} - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$
 (C) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}}(\text{Br})(\text{Br}) - \text{CH} - \text{CH}_3$
 (D) $\text{BrCH}_2 - \underset{\text{CH}_3}{\text{C}} = \text{CH} - \text{CH}_3$

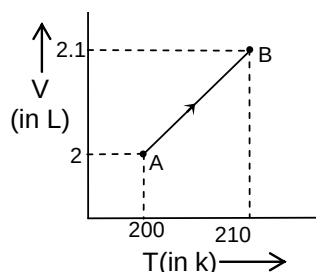
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TG ~ @bohring_bot

5. Which reaction(s) produce(s) secondary alcohol(s)?



6.



One mole of an ideal gas undergoes above thermodynamic process. C_p of the gas is $\frac{5R}{2}$.

[$R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$]

Choose correct statement(s) $\left[\ln \frac{21}{20} = 0.048 \right]$

- (A) The irreversible work done along path $A \rightarrow B$ is -83 J mol^{-1} .
 (B) The system absorbs 207.5 J mol^{-1} heat from the surrounding.
 (C) The entropy [$\Delta S(\text{system})$] change along the path $A \rightarrow B$ in reversible way is $0.996 \text{ J K}^{-1} \text{ mol}^{-1}$.
 (D) The internal energy of the system will increase to 124.5 J mol^{-1} .

7. The correct hydrolysis order of the given compound(s) is/are

- (A) $\text{SiCl}_4 > \text{SiF}_4$ (B) $\text{NCl}_3 > \text{CCl}_4$
 (C) $\text{BI}_3 > \text{BF}_3$ (D) $\text{BBr}_3 > \text{BCl}_3$

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

8. BeCl_2 undergoes polymerization and BCl_3 undergo dimerisation because

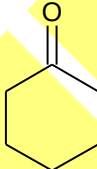
- (A) BeCl_2 is more electron deficient than BCl_3
 (B) BeCl_2 is linear and BCl_3 is angular
 (C) Electronegativity difference between Be and Cl is higher than that between B and Cl
 (D) BeCl_2 is less covalent than BCl_3 molecule

Space For Rough Work

TG ~ @bohring_bot

9. $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$ $\xrightarrow{1, 4\text{-addition of HCl}}$ (P)
 $\xrightarrow{1, 2\text{-addition of HCl}}$ (Q) Major product

Which statement is correct regarding properties of P and Q?

- (A) Reactivity with aq.KOH ($\text{S}_{\text{N}}1$ path) $\text{P} > \text{Q}$
 (B) Number of stereoisomers: $\text{P} = \text{Q}$
 (C) Rate of reaction with alc.KOH: $\text{P} > \text{Q}$
 (D) Heat of hydrogenation: $\text{P} > \text{Q}$
10. Which solubility order is incorrect?
 (A) $\text{LiI} > \text{NaI} > \text{KI} > \text{RbI}$
 (B) $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4$
 (C) $\text{Be}(\text{OH})_2 > \text{Mg}(\text{OH})_2 > \text{Ca}(\text{OH})_2 > \text{Sr}(\text{OH})_2$
 (D) $\text{LiF} < \text{NaF} < \text{KF} < \text{RbF}$
11. Which forms the most stable complex with carbon monoxide?
 (A) Fe
 (B) Fe^{2+}
 (C) Fe^{3+}
 (D) $\text{Fe}^{+8/3}$
12. Gases having which of the following value of 'a' and 'b' (van der Waal's constant) can be liquefied easily?
 [a in $\text{atm L}^2 \text{mol}^{-2}$, b in L mol^{-1}]
 (A) $a = 0.04$, $b = 0.03$
 (B) $a = 0.08$, $b = 0.4$
 (C) $a = 0.06$, $b = 0.03$
 (D) $a = 0.004$, $b = 0.04$
13. Two moles of  forms a single product when they react with
 (A) LiAlH_4
 (B) Zn-Hg/Conc.HCl
 (C) NaOH
 (D) $\text{C}_6\text{H}_5\text{NH} - \text{NH}_2$

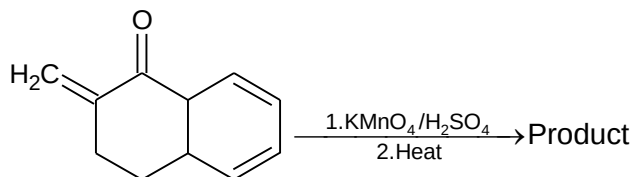
Space For Rough Work

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PART – B
(Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1.

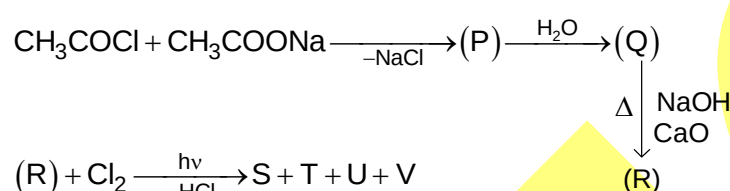


How many maximum number of CO_2 molecule(s) is/are formed in the above reaction?

2. A container contains 200 mL of 0.5 M NH_4NO_3 solution. 200 mL of 0.4 M NaOH solution was added to it. What is the pH of the resulting solution?

$[K_b \text{ of } \text{NH}_4\text{OH} = 10^{-5}] \quad [\log 0.25 = -0.6]$

3.



Order of molar mass = $S < T < U < V$

Let x = The number of chlorine atoms present in 'T'.

y = Number of hydrogen atoms present in 'U'

What is the value of $\left(\frac{x+y}{10}\right)$?

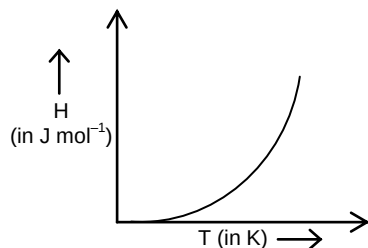
4. The crystal field stabilization energy of $[\text{Fe}(\text{CN})_6]^{3-}$ is $-x\Delta_o$ and that of $[\text{Fe}(\text{CN})_6]^{4-}$ is $-y\Delta_o$. What is the value of $(x + y)$?

5. In a first order reaction $A(g) \longrightarrow \text{Product}$. What is the value of $\frac{[A_t]}{[A_0]}$ after two half-lives.

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6.

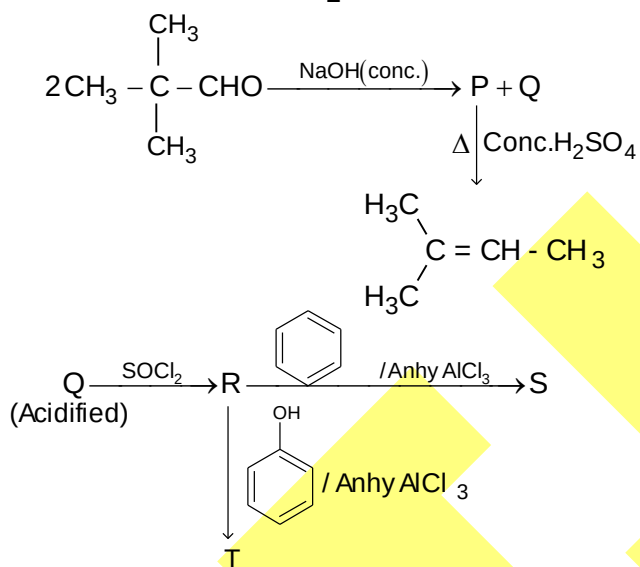


If the slope of the above curve is $20.75 \text{ J K}^{-1} \text{ mol}^{-1}$, what is the value of C_v in $\text{J K}^{-1} \text{ mol}^{-1}$ unit? $[R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}]$

7.

A tetrahedral chloro complex of cobalt contains same number of electrons in the 'e' and t_2 orbitals according to crystal field theory. If the oxidation number of cobalt in the complex is $+x$, what is the value of $\frac{x}{2}$ is

8.



If the molar mass difference between the organic product S and T is X, what is $\frac{X}{10}$?

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. Consider the equation $z^6 + 6z + 20 = 0$, Then which of following can be true?
 (A) The number of the roots in the first quadrant can be 1 or 2
 (B) The number of the roots in the second quadrant can be 1 or 2
 (C) The number of the roots in the third quadrant can be 1 or 2
 (D) The number of the roots in the fourth quadrant can be 1 or 2

2. A function f from integers to integers is defined as

$$f(x) = \begin{cases} n+3 & n \in \text{odd} \\ n/2 & n \in \text{even} \end{cases}$$

suppose $k \in \text{odd}$ and $f(f(f(k))) = 27$ then

- (A) k can be 45
 (B) sum of the digit of k is 6
 (C) sum of the digit of k is 9
 (D) k can be 105

3. Let α and $f(\alpha)$ be the eccentricity of the ellipse $\frac{x^2}{3b^2 - 2a^2} + \frac{y^2}{2b^2 - a^2} = 1$ ($3b^2 > 2a^2$) and

$$\frac{x^2}{2b^2 - a^2} + \frac{y^2}{b^2} = 1 \quad (2b^2 > a^2) \text{ then}$$

(A) $f(\alpha) = \frac{\alpha}{\sqrt{1-\alpha^2}}, b \in \mathbb{R} - \{0\}$

(B) $\int_0^{1/2} ffff(\alpha) d\alpha = \frac{1}{4}$

(C) $\int e^{\alpha} (f(\alpha) - f''(\alpha)) d\alpha = e^{\alpha} \left[\frac{\alpha}{\sqrt{1-\alpha^2}} - \frac{1}{(1-\alpha^2)^{3/2}} \right] + C$

(D) $f(\alpha) = \frac{\alpha}{2\sqrt{1-\alpha^2}}, b \in \mathbb{R} - \{0\}$

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4. If $f(x) = (a^2 + a + 1)x^2 + bx + c$ and $f(x)$ is symmetrical about the line $x = 1$, then
 (A) $f(1 - \sqrt{3}) < f(\sqrt{3}) < f(2 + \sqrt{3}) < f(4)$ (B) $f(\sqrt{3}) < f(1 - \sqrt{3}) < f(2 + \sqrt{3}) < f(4)$
 (C) $f(2 + \sqrt{3}) > f(4) > f(\sqrt{3}) > f(1 - \sqrt{3})$ (D) $f(4) > f(2 + \sqrt{3}) > f(-\sqrt{2}) > f(\sqrt{2})$
5. If $\int_0^1 (4x^3 - f(x))f(x)dx = \frac{4}{7}$, then which of the following is/are true
 (A) $f(x)$ has a point of inflexion
 (B) $\left|f(x) - \frac{1}{4}\right|$ has exactly one point of non-differentiability
 (C) area bounded by $y = f(x)$, x -axis, $x = 1$ and $x = 2$ is $\frac{15}{2}$
 (D) The function $f(x)$ is concave upward for $\forall x \in \mathbb{R}$
6. In triangle ABC, $a=4$ and $b=c=2\sqrt{2}$. A point P moves with the triangle such that the square of its distance from BC is half the rectangle contained by its distances from the other two sides. If D be the centre of P, then
 (A) locus of P is an ellipse with eccentricity $\sqrt{\frac{2}{3}}$
 (B) locus of P is a hyperbola with eccentricity $\sqrt{\frac{3}{2}}$
 (C) area of the quadrilateral ABCD = $\frac{16}{3}$ sq. units
 (D) area of the quadrilateral ABCD = $\frac{32}{3}$ sq. units
7. If $f: \mathbb{R} \rightarrow [2, 4]$ be a periodic function such that the equation $f(x) = g(x)$ has a unique solution, if $g(x) = 1 - \cos \pi x$ then period of $f(x)$ can be
 (A) e (B) $\sqrt{3}$
 (C) π (D) 2020

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(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. A boy have 32 cards out of which 10 cards, each of blue, green and red colours, have denominations as $\{2^1, 2^2, \dots, 2^{10}\}$ and one black and one white each has value unity. The number of ways in which the boy can get a sum of 2012 if he can choose any number of cards, is
 (A) $(1001)^2$ (B) $(1002)^2$
 (C) $(1007)^2$ (D) $(1111)^2$
9. The minimum value of $(\sin \alpha - \cot \beta)^2 + (\tan \beta - \cos \alpha)^2$ for all admissible real Values of α and β is
 (A) $3 - 2\sqrt{2}$ (B) 0
 (C) $3 + 2\sqrt{2}$ (D) $2 - 3\sqrt{3}$
10. On the circle $|z - i\sqrt{3}| = 1$ points z_1, z_2, z_3 are so located that $3z_1 = 2z_2 + 2z_3 - i\sqrt{3}$. Then which is true ?
 (A) $|z_1 - z_2| = \frac{1}{\sqrt{2}}$
 (B) $|z_2 - z_3| = \frac{\sqrt{7}}{4}$
 (C) The value of $\frac{z_2 - i\sqrt{3}}{z_3 - i\sqrt{3}} = \frac{1 \pm i\sqrt{63}}{4}$
 (D) $|z_2 - z_3| = \frac{\sqrt{7}}{2}$
11. If $f(x)$ be a positive, continuous and differentiable on the interval (a, b) . If $\lim_{x \rightarrow a^+} f(x) = 1$ and $\lim_{x \rightarrow b^-} f(x) = 3^{1/4}$. Also $f'(x) + \frac{1}{f(x)}$, then
 (A) $b - a \geq \frac{\pi}{24}$ (B) $b - a \leq \frac{\pi}{24}$
 (C) $b - a = \frac{\pi}{12}$ (D) none of these

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12. If P be a point inside and equilateral triangle ABC such that $PA=3, PB=4$ and $PC=5$, then the side length of the equilateral triangle ABC is
 (A) $\sqrt{25 - 12\sqrt{3}}$ (B) 13
 (C) $\sqrt{25 + 12\sqrt{3}}$ (D) 17
13. If $f(x)$ is continuous function and satisfying $f(x) + f(1+x) = |2^x - 1| + |x - 1|$ in $0 \leq x \leq 2$, then the value of $\int_0^2 f(x) dx$ is equal to
 (A) $\frac{1}{\ln 2} - \frac{1}{2}$ (B) $\frac{1}{\ln 2} - 1$
 (C) $\frac{2}{\ln 2} - 1$ (D) none of these

PART – B (Numerical based)

This section contains **8 Numerical based questions**, the answer of which maybe positive or negative numbers or decimals to **two decimal places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)

1. Let f be a continuous function and $f(x) = f(2x)$, $x \in \mathbb{R}$. If $f(x) = 3$ then $\int_{-1}^{+1} f(f(f(x))) dx$ is equal to
2. The number of ordered triplet (x, y, z) such that $\text{LCM}(x, y) = 3375$, $\text{LCM}(y, z) = 1125$, $\text{LCM}(z, x) = 3375$ is equal to
3. Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, then find the value of determinant $((A - I)^3 - 4A)$. [Note I is an identity matrix of order 3].

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4. A biased coin shows head with probability of $\frac{3}{4}$ and tail with a probability of $\frac{1}{4}$. Let 'A' be the event that three or more heads occurs in four tosses and 'B' be the event that three heads do not occur in first three tosses. If $P\left(\frac{A}{B}\right) = \frac{m}{n}$ where $m, n \in \mathbb{N}$ then find the least value of $(m + n - 225)$.
5. Let foci of the conic represented by the equation $ax^2 + 2(a+2)xy + ay^2 + 2fy = 0$, where $a < -1$ and $f \neq 0$ be $F_1(\alpha, -3)$ and $F_2(\beta, -5)$. If the feet of perpendicular from F_1 and F_2 upon x-axis be M & N and eccentricity of the conic given by the equation $\sqrt{10}e^2 - 7e + \sqrt{10} = 0$ then the square of the radius of director circle of the conic is
6. If $[\sin x] + [\cos x] + 2 = 0$, then number of integral values in the range of $f(x) = \sin x - \cos x + 3$, corresponding to the solution set of the given equation is (where $[.]$ denotes the greatest integer function)
7. $f(n) \sum_{r=1}^n \left[r^2 \left({}^nC_r - {}^nC_{r-1} \right) + (2r+1) \left({}^nC_r \right) \right]$, then $f(30)$ is
8. If p is a positive integer and 'f' be a function defined for positive numbers and attains only positive values such that $f(xf(y)) = x^p y^4$, then $p =$

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Answers

SECTION-1 : PHYSICS**PART – A**

- | | | | |
|--------|-------|--------|-------|
| 1. ABC | 2. AD | 3. BCD | 4. BC |
| 5. AD | 6. AD | 7. AC | 8. A |
| 9. D | 10. B | 11. B | 12. D |
| 13. A | | | |

PART – B

- | | | | |
|---------|---------|----------|---------|
| 1. 0.73 | 2. 8.89 | 3. 56.56 | 4. 3.00 |
| 5. 1.00 | 6. 4.00 | 7. 2.00 | 8. 2.00 |

SECTION-2 : CHEMISTRY**PART – A**

- | | | | |
|--------|---------|---------|---------|
| 1. BCD | 2. ACD | 3. ACD | 4. ABCD |
| 5. ABC | 6. ABCD | 7. ABCD | 8. A |
| 9. C | 10. C | 11. A | 12. C |
| 13. C | | | |

PART – B

- | | | | |
|---------|----------|--------|--------|
| 1. 4 | 2. 9.6 | 3. 0.3 | 4. 4.4 |
| 5. 0.25 | 6. 12.45 | 7. 1.5 | 8. 4.4 |

SECTION-3 : MATHEMATICS**PART – A**

- | | | | |
|---------|-------|--------|-------|
| 1. ABCD | 2. BD | 3. ABC | 4. BD |
| 5. ABC | 6. AC | 7. ABC | 8. C |
| 9. A | 10. A | 11. B | 12. C |
| 13. A | | | |

PART – B

- | | | | |
|-------|-------|--------|------|
| 1. 6 | 2. 50 | 3. 8 | 4. 4 |
| 5. 40 | 6. 1 | 7. 960 | 8. 2 |

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Answers & Solutions

SECTION-1 : PHYSICS PART – A

1. **ABC**

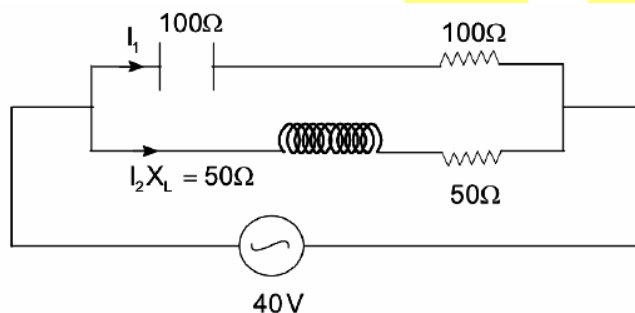
Sol. $I_1 = \frac{20}{Z_1} \times 2$

$$= \frac{2}{5\sqrt{2}} \text{ A at } 45^\circ \text{ leading}$$

$$I_2 = \frac{20}{Z_2} = \frac{\sqrt{2}}{5} \text{ A at } 45^\circ \text{ lagging}$$

$$I_{1\text{RMS}} = \frac{1}{5} \text{ A}$$

$$I_{2\text{RMS}} = \frac{2}{5} \text{ A}$$

2. **AD**

Sol. For central fringe

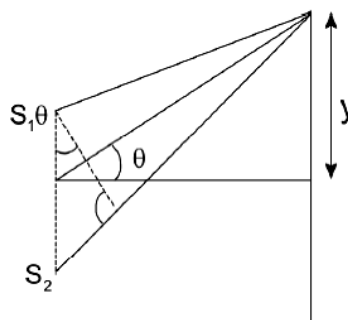
$$d \sin \theta = 2 \text{ mm}$$

$$4 \text{ mm} \sin \theta = 2 \text{ mm}$$

$$\theta = 30^\circ$$

$$\tan 30^\circ = \frac{y}{2}$$

$$y = \frac{1}{\sqrt{3}} \text{ m}$$

For bright fringe $(d \sin \theta \pm 2) = n\lambda$ Where n is 0, 1, 2, 3,3. **BCD**

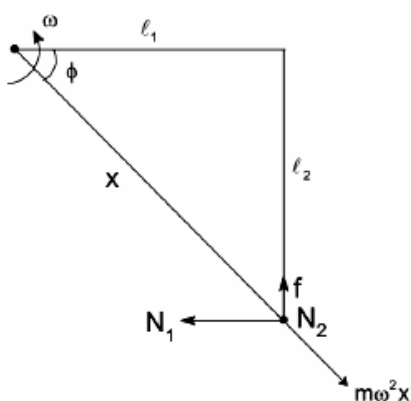
Sol. $N_1^2 + N_2^2 = N^2$

$$N_2 = mg$$

$$m\omega^2 \times \sin \phi = f$$

$$f = m\omega^2 \ell_2$$

$$N_1 = m\omega^2 \ell_1$$

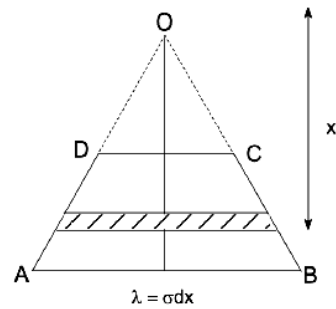
4. **BC**

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Sol.
$$dE = 2 \left(\frac{1}{4\pi\epsilon_0} \right) \frac{\sigma dx}{x} \left(\frac{1}{2} \right)$$

$$dE = \frac{\sigma}{4\pi\epsilon_0} \frac{dx}{x}$$

$$E = \frac{7\sigma}{44\epsilon_0} \ln \sqrt{2}$$



5. **AD**

Sol. Elastic potential energy $\propto \left(\frac{\partial y}{\partial x} \right)^2$, kinetic energy $\propto \left(\frac{\partial y}{\partial t} \right)^2$. So for antinode elastic potential energy is constant & minimum. $\frac{\partial y}{\partial x}$ always changes for all other points.

6. **AD**

Sol. Let's assume, time taken by the queen to get into the hole is t

$$t = \frac{\ell}{2V \sin \theta} \text{ \& also } t = \frac{\ell}{2V \cos \theta} + \frac{\ell}{eV \cos \theta}$$

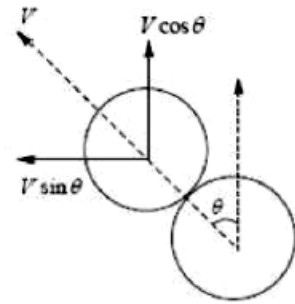
Therefore

$$\frac{\ell}{2V \sin \theta} = \frac{\ell}{2V \cos \theta} + \frac{\ell}{eV \cos \theta}$$

$$\therefore \tan \theta = \frac{1}{3}$$

$$\therefore \frac{d}{\sqrt{(r_1 + r_2)^2 - d^2}} = \frac{1}{3}$$

$$d^2 = 9 \Rightarrow d = 3 \text{ cm.}$$



7. **AC**

Sol. When parallel light falls from left side

$$\frac{\mu_3}{f} = \frac{(\mu_3 - \mu_2)}{R} + \frac{(\mu_2 - \mu_1)}{R}$$

$$\frac{4/3}{f_1} = \frac{(4/3 - 2)}{R} + \frac{(2 - 1)}{R} \Rightarrow f_1 = 80 \text{ cm}$$

When parallel light falls from right side

$$\frac{1}{f_2} = \frac{(1 - 2)}{-R} + \frac{(2 - 4/3)}{-R} \Rightarrow f_2 = 60 \text{ cm}$$

8. **A**

Sol.
$$\vec{a}_B = \vec{a}_A + (\vec{\alpha} \times \vec{r}_{B,A}) - \omega^2 \vec{r}_{B,A}$$

$$= -5\hat{j} + 4\alpha\hat{i} + 3\alpha\hat{j} - (12\hat{i} - 16\hat{j})$$

$\vec{a}_B$ along y-axis should be zero.

$$\Rightarrow 11 + 3\alpha = 0$$

$$\Rightarrow \alpha = -\frac{4}{3} \text{ rad/s}^2$$

$$\vec{a}_B = (4\alpha - 12)\hat{i} = \frac{80}{3} \hat{i} \text{ m/s}^2$$

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9. **D**

Sol. $y = \frac{\ell}{2} \tan \theta$

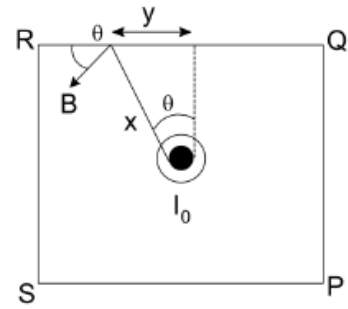
$$dy = \frac{\ell}{2} \sec^2 \theta d\theta$$

$$8 \int_0^{\frac{\pi}{4}} B dy \cos \theta + \oint_{a \rightarrow b \rightarrow c \rightarrow d} \vec{B} \cdot d\vec{\ell} = \mu_0 I_0$$

$$\int_0^{\frac{\pi}{4}} \frac{8\mu_0 I_0}{4\pi x} \cdot \frac{2\frac{\ell}{2}}{x^2 + \frac{\ell^2}{4}} \cdot \frac{\ell}{2} \sec^2 \theta \cos \theta + \oint_{a \rightarrow b \rightarrow c \rightarrow d} \vec{B} \cdot d\vec{\ell} = \mu_0 I_0$$

$$\frac{2}{3} \mu_0 I_0 + \oint_{a \rightarrow b \rightarrow c \rightarrow d} \vec{B} \cdot d\vec{\ell} = \mu_0 I_0$$

$$\oint_{a \rightarrow b \rightarrow c \rightarrow d} \vec{B} \cdot d\vec{\ell} = \mu_0 I_0 - \frac{2}{3} \mu_0 I_0 = \frac{\mu_0 I_0}{3}$$

10. **B**

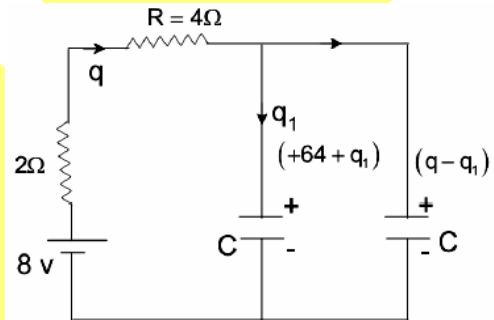
Sol. Let $R + 2 = R_1$

$$q(t) = \frac{3CE}{2} \left(1 - e^{-\frac{t}{2R_1 C}} \right)$$

$$I = \frac{dq}{dt} = \frac{3E}{4R_1} e^{-\frac{t}{2R_1 C}}$$

$$\text{Heat loss in } R_1 = \int I^2 R_1 dt = 576 \mu\text{J}$$

$$\text{So, heat loss in } R = 4\Omega \text{ is } = 576 \times \frac{2}{3} = 384 \mu\text{J}.$$

11. **B**

Sol. Maximum power transfer will take place when v_p is maximum.

12. **D**

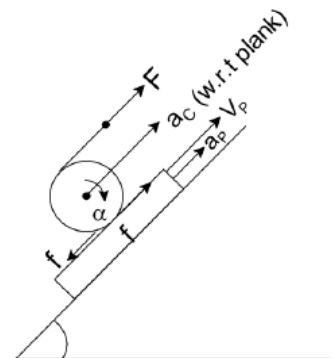
Sol. $a_c = R\alpha$ (for no slipping)

$$V_c = R\omega \quad (V_c \text{ is the speed of cylinder w.r.t plank})$$

$$a_c + 3a_p = 0$$

$$V_c + 3V_p = 0$$

$$S_c + 3S_p = 0$$

13. **A**

Sol. Let T_1 and T_2 be the initial and final temperature, then

$$P_0 V = nRT_1 \quad \dots(i)$$

$$\left(P_0 + \frac{W}{A} \right) (V - Ah) = nRT_2 \quad \dots(ii)$$

Where h is the displacement of piston.

Also, work done on gas $-\Delta U$

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$$Wh = nC_v \Delta T = \frac{3}{2} nR(T_2 - T_1) \quad \dots(iii)$$

From equation (i), (ii) and (iii)

$$\frac{2}{3} Wh = \frac{WV}{A} - Wh - P_0 Ab$$

$$\text{or, } Ah = \frac{WV}{P_0 A + \frac{5}{3} W}$$

$$T_2 - \left(P_0 + \frac{W}{A} \right) \left(V - \frac{W_v}{P_0 A + \frac{5}{3} W} \right)$$

PART – B

1. **0.73**

Sol. Impulse due to 'N' would act at the centre of the colliding parts.

Applying impulse momentum theorem

$$-\int N dt = M(v - v_0) \quad \dots(1)$$

& Applying angular impulse, angular momentum theorem

$$\int N \left(\frac{L-x}{2} \right) dt = \frac{ML^2}{12} \omega \quad \dots(2)$$

$$\text{Also } e = \frac{\omega \left(\frac{L-x}{2} \right) - v}{v_0} \quad \dots(3)$$

From (1), (2) & (3)

$$v \left[1 + 3 \left(\frac{L-x}{2} \right)^2 \right] - v_0 \left[3 \frac{(L-x)^2}{L^2} - e \right]$$

$\therefore$ for $v > 0$

$$x < L \left[1 - \sqrt{\frac{e}{3}} \right]$$

For inequality to hold true for all 'e'

$$x < L \left[1 - \frac{1}{\sqrt{3}} \right]$$

$$\text{or } x < (\sqrt{3} - 1)L, \therefore x_{\max} = 0.732L$$

2. **8.89**

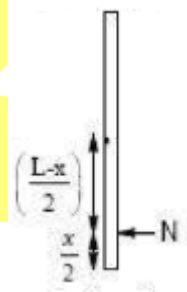
Sol. Power absorbed by shell

$$\Rightarrow \frac{P \times 2\pi(1 - \cos 37^\circ)}{4\pi}$$

$$\Rightarrow \frac{P}{10}$$

For equilibrium

$$\Rightarrow \frac{P}{10} = 3\pi R^2 \sigma T^4$$



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$$\therefore T^4 = \frac{P}{30\pi R^2 \sigma} = \frac{2400 \sigma \pi}{30\pi \times 9 \times \sigma} \Rightarrow 8.89.$$

3. **56.56**

Sol. Applying conservation of angular momentum w.r.t. the centre of earth,

$$4R \times mv_0 \sin 30 = mvR, v = 2v_0$$

Applying energy conservation

$$\frac{1}{2}m(2v_0)^2 - \frac{GMm}{R} = -\frac{GMm}{4R} + \frac{1}{2}mv^2$$

$$\therefore x = 40\sqrt{2} \Rightarrow 56.56$$

4. **3.00**

Sol. x coordinate of centre of mass of the rod

$$x = \frac{\ell}{2} \sin \theta$$

$$v_x = \frac{\ell}{2} \cos \theta \omega \text{ and } a_x = \frac{\ell}{2} (\cos \theta \alpha - \sin \theta \omega^2)$$

$$\text{When } N_1 = 0 \Rightarrow \cos \theta \alpha = \sin \theta \omega^2 \quad \dots(1)$$

Using COME:

$$mg \frac{\ell}{2} (\cos \beta - \cos \theta) = \frac{1}{2}m \frac{\ell^2}{4} \omega^2 + \frac{1}{2}m \frac{\ell^2}{12} \omega^2$$

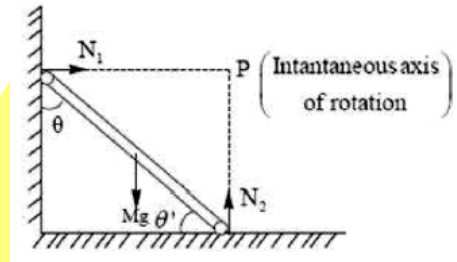
$$\left[v = \frac{\ell}{2} \omega \text{ from IAOR} \right]$$

$$\omega^2 = \frac{3g}{\ell} (\cos \beta - \cos \theta) \quad \dots(2)$$

$$\alpha = \frac{3g}{2\ell} \sin \theta \quad \dots(3)$$

Substituting the value of ω^2 and α in equation (1) gives

$$\Rightarrow \cos \theta = \frac{2}{3} \cos \beta \Rightarrow \cos \theta = \frac{1}{\sqrt{3}} \Rightarrow \sin \theta = \frac{1}{\sqrt{3}} \text{ So } N = 3.$$

5. **1.00**

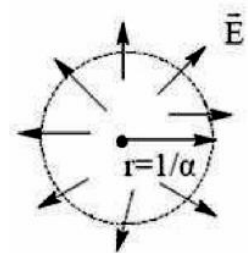
Sol. As the electric field is radial, by applying gauss law, we can write

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0}$$

$$\text{For } r = \frac{1}{\alpha}, \vec{E} = C(1 - e^{-\alpha x / \alpha}) \frac{\hat{r}}{(1/\alpha)^2}$$

$$\therefore \oint \vec{E} \cdot d\vec{s} = C(1 - e^{-1}) \alpha^2 \times 4\pi (1/\alpha)^2$$

$$\Rightarrow \frac{Q}{\epsilon_0} = 4\pi C(1 - e^{-1}) \Rightarrow Q = (1 - e^{-1}) \Rightarrow \therefore N = 1$$

6. **4.00**

$$\text{Sol. } \int_0^{x_{\min}} 0.5x \frac{mg}{\ell} dx = 2$$

$$\Rightarrow x_{\min} = 40 \text{ cm.}$$

7. **2.00**

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Sol. $mr^2\beta = \int_0^\theta \frac{2q}{2\pi} d\alpha vBr \cos \alpha$

$$\beta = \frac{dw}{dt} = \frac{qvB \sin \theta}{\pi mr} \quad \dots(1)$$

$$-ma = \int_0^\theta 2 \frac{q}{2\pi} d\alpha r\omega B \cos \alpha$$

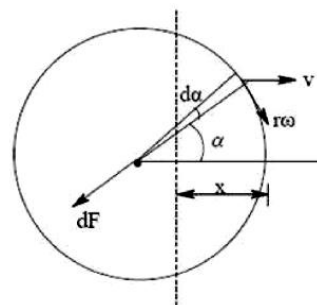
$$a = \frac{dv}{dt} = \frac{qr\omega B \sin \theta}{\pi m} \quad \dots(2)$$

$$= \frac{d\omega}{dv} = \frac{v}{r^2\omega} \Rightarrow r^2 \int_0^\omega \omega d\omega = - \int_{v_0}^v v dv$$

$$\Rightarrow v = \frac{v_0}{\sqrt{2}}$$

$$\frac{v dv}{dx} = \frac{qr\omega B \sin \theta}{\pi m} ; \quad \int_{v_0}^{v_0/\sqrt{2}} \frac{v dv}{\sqrt{v_0^2 - v^2}} = \frac{qrB}{m\pi} \int_0^\pi \sin^2 \theta d\theta$$

$$\Rightarrow q = \frac{\sqrt{2} mv_0}{Br}$$



$$x = r \cos \theta$$

$$dx = r \sin \theta d\theta$$

8. **2.00**

Sol. For slight displacement $(\pi r^2 x) \rho g = ma$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{\pi r^2 \rho g}}$$

$$\Rightarrow T = 2 \text{ sec.}$$

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SECTION-2 : CHEMISTRY

PART – A

1. BCD

Sol. $\psi_{4,1,-1}$ is an orbital of 4p-subshell. It may be $4p_x$ or $4p_y$ or $4p_z$.

2. ACD

Sol. $[\text{MnCl}_4]^{2-}$ is a tetrahedral complex.

3. ACD

Sol. $K_{sp} = 4s^3 = 5 \times 10^{-10}$

$$\therefore s = 5 \times 10^{-4}$$

$$\therefore \text{Conc. of saturated solution} = 5 \times 10^{-4} \text{ M}$$

$$\text{pOH of saturated solution} = -\log[\text{OH}^-]$$

$$= -\log[2 \times 5 \times 10^{-4}] = -\log 10^{-3} = 3$$

$$\therefore \text{pH} = 14 - 3 = 11$$

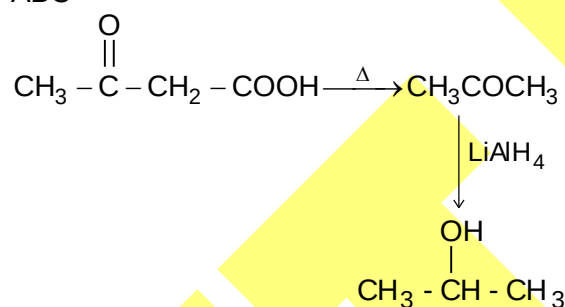
$$\therefore \text{Solubility decreases in a solution of pH} > 11 \text{ and increases in a solution of pH} < 11.$$

4. ABCD

Sol. With excess of NBS, addition as well as allylic substitution reactions take place.

5. ABC

Sol.

 $\text{CH}_3\text{COOC}_2\text{H}_5$ forms primary alcohols.

6. ABCD

Sol. It is an isobaric process (pressure remains constant)

$$W_{\text{irr}} = -P(V_2 - V_1)$$

$$q = nC_P\Delta T$$

$$\Delta S = nC_P \ln \frac{T_2}{T_1}$$

$$\Delta U = nC_V\Delta T$$

7. ABCD

8. A

Sol. BeCl_2 and BCl_3 remove electron deficiency by forming polymers and dimers respectively.

9. C

Sol. P will show geometrical isomerism (cis and trans) and Q will show optical isomerism (R and S).

10. C

Sol. For group-1 iodides and fluorides, mismatch of ionic radius favours solubility.

11. A

Sol. Metals in lower oxidation state forms stable complexes with CO.

12. C

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Sol. $T_C = \frac{8a}{27Rb}$

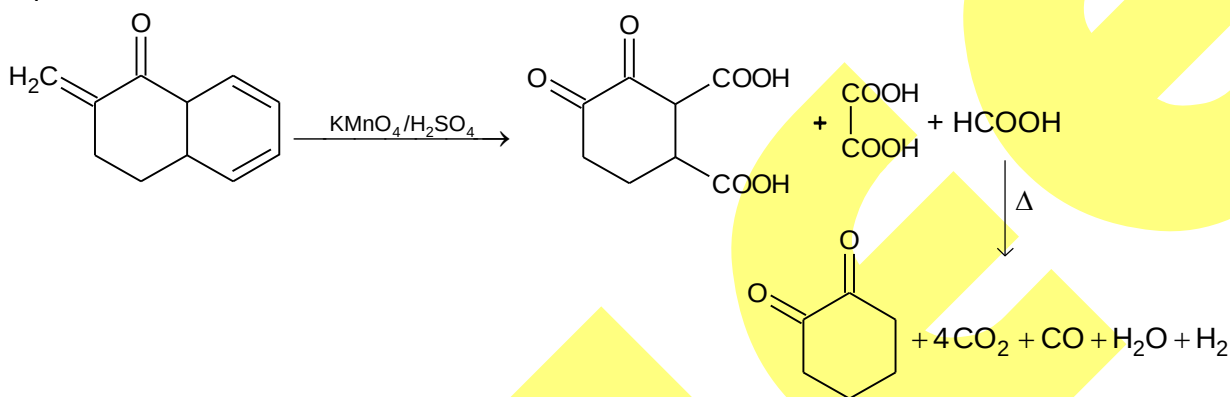
$\therefore$ Liquification $\propto \frac{a}{b}$ ratio.

13. C

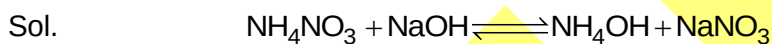
Sol. Aldol condensation will take place.

PART – B

1. 4
Sol.



2. 9.6



Before reac ⁿ M_{eq}	200×0.5 = 100	200×0.4 = 80	0	0
After reac ⁿ M_{eq}	$100 - 80$ = 20	$80 - 80$ = 0	80	80

$$\therefore \text{pOH} = \text{p}K_b + \log \frac{[\text{NH}_4\text{NO}_3]}{[\text{NH}_4\text{OH}]}$$

$$= 5 + \log \frac{20}{80} = 5 - 0.6 = 4.4$$

$$\text{pH} = 14 - 4.4 = 9.6$$

3. 0.3

Sol. $\text{R} = \text{CH}_4$, $x = 2$, $y = 1$

4. 4.4

Sol. $\text{Fe}^{3+} (t_{2g}^5 e_g^0)$: CFSE = $5 \times -0.4\Delta_0 = -2\Delta_0$

$\text{Fe}^{2+} (t_{2g}^6 e_g^0)$: CFSE = $6 \times -0.4\Delta_0 = -2.4\Delta_0$

$$\therefore x + y = 2 + 2.4 = 4.4$$

5. 0.25

Sol. $[A_t] = [A_0](\frac{1}{2})^n$

$$\text{or, } \frac{[A_t]}{[A_0]} = \left(\frac{1}{2}\right)^2 = \frac{1}{4} = 0.25$$

6. 12.45

Sol. Slope = $C_P = 20.75 \text{ JK}^{-1} \text{ mol}^{-1}$.

$$C_P - C_V = R$$

$$\therefore C_V = C_P - R = 20.75 - 8.3 = 12.45$$

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7. 1.5

Sol. The oxidation number of cobalt in the complex is +3.

8. 4.4

Sol. Cannizzaro Reaction takes place

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SECTION-3 : MATHEMATICS

PART – A

1. ABCD

Sol. Consider a function $f(x) = x^6 + 6x + 20$ and then use monotonicity to plot the graph and nature of the function

2. BD

Sol. $\therefore k \in \text{odd (even)}$

$$f(k) = k + 3$$

$$f(f(k)) = \frac{k+3}{2}$$

$$\text{if } \frac{k+3}{2} \in \text{odd} \Rightarrow 27 = \frac{k+3}{2} + 3 \Rightarrow k = 45 \text{ not possible}$$

$$\text{Now let } \frac{k+3}{2} \in \text{even}$$

$$\therefore 27 = f(f(f(k))) = f\left(\frac{k+3}{2}\right) = \frac{k+3}{4}$$

$$\therefore k = 105$$

$$\text{verifying } f(f(f(105))) = f(f(108)) = f(54) = 27$$

3. ABC

Sol: eliminate a and b by using the formulae for eccentricity

4. BD

Sol $f(x)$ is vertical upward parabola & is sym. about its axis $x=1$, so points situated at greater distance from axis have greater outputs

5. ABC

Sol Make integrand perfect square

6. AC

Sol. $PM=k$

$$\text{Equation of AB} \equiv -x + y = 2$$

$$\text{Equation of AC} \equiv x + y = 2$$

According to question

$$\left(\frac{2-h-k}{\sqrt{2}}\right)\left(\frac{2+h+k}{\sqrt{2}}\right) = 2k^2$$

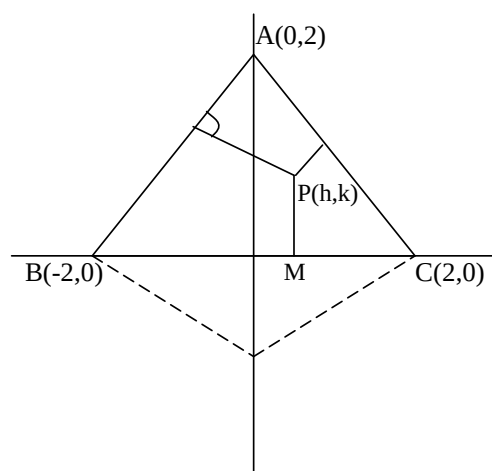
$$\Rightarrow h^2 + 3k^2 + 4k = 4$$

$$\Rightarrow h^2 + 3\left(k^2 + \frac{4}{3}k + \frac{4}{9}\right) = 4 + \frac{4}{3}$$

$$\Rightarrow h^2 + 3\left(k + \frac{2}{3}\right)^2 = \frac{16}{3}$$

$$\Rightarrow \frac{h^2}{16/3} + \frac{\left(k + \frac{2}{3}\right)^2}{16/9} = 1$$

$$\Rightarrow \text{ellipse with } e = \sqrt{\frac{2}{3}} \text{ and } D \equiv \left(0, -\frac{2}{3}\right)$$



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7. ABC

Sol. Period of $f(x)$ must be rational

8. C

Sol. Coefficient of x^{2012} in $(1+x)^2 \left((1+x^2) \dots (1+x^{2^{10}}) \right)^3$

$$\Rightarrow \text{Coefficient of } x^{2012} \text{ in } \frac{(1-x^{2^{11}})^3}{(1+x)(1-x)^3}$$

Coefficient of x^{2012} in $(1+x^2+x^4+\dots)(1+2x+3x^2+\dots)$

$$\Rightarrow 1 \cdot (2013) + 1 \cdot (2011) + 1 \cdot (2009) + \dots + 1$$

$$\Rightarrow 2k+1=2013 \Rightarrow k=1006$$

We know that $1+3+5+\dots+(2k+1)=(k+1)^2$

9. A

Sol. Minimise the distance btw a circle and rectangular hyp which is along the common normal

10. A

Sol. The mid – point of z_2 & z_3 divides the line joining the points z_1 & $i\sqrt{3}$ in the ratio 1:3 internally.

11. B

Sol. Since $f'(x) \geq f^3(x) + \frac{1}{f(x)}$

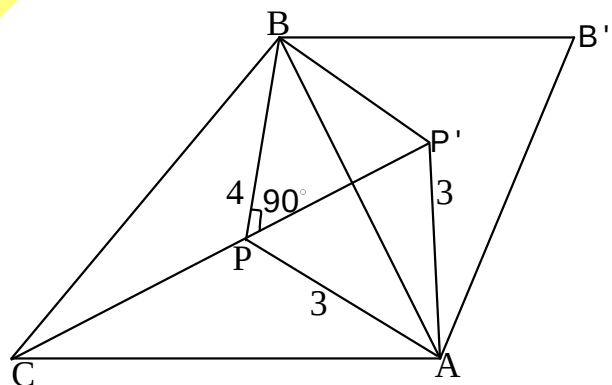
$$f(x)f'(x) \geq f^4(x) + 1 \Rightarrow \frac{f(x)f'(x)}{1+f^4(x)} \geq 1$$

On integrating w.r.t.x from $x=a$ to b

$$\frac{1}{2} \left(\tan^{-1} f^2(x) \right)_a^b \geq b-a$$

$$\text{Or } b-a \leq \frac{1}{2} \left[\lim_{x \rightarrow b^-} \tan^{-1}(f^2(x)) - \lim_{x \rightarrow a^+} \tan^{-1}(f^2(x)) \right] \text{ or } b-a \leq \frac{\pi}{24}$$

12. C

Sol. Rotate the triangle in clockwise direction through an angle 60° . Let the points A,B,C and P will be A',B', B and P' respectively after the rotation.We have $PA=P'A=3$ And $\angle PAP' = 60^\circ \Rightarrow PP' = 3$ Also $CP=BP'=5$ So $\triangle BPP'$ is right angle triangle which $\angle BPP' = 90^\circ$.

Now apply cosine rule in triangle BPA because

 $\angle BPA = 90^\circ + 60^\circ = 150^\circ$, $PA=3$ and $BP=4$, we can get AB

13. A

Sol. Use $\int_0^2 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx = \int_0^1 f(x) dx + \int_0^1 f(x+1) dx$

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PART – B

1. 6

Sol. $f(2x) = f(x) = f(x/2) = f(x/2^2) = f\left(\frac{x}{2^n}\right)$

f is continuous, when $n \rightarrow \infty$, $f(2x) = f(0)$

$\Rightarrow f(x)$ is constant

2. 50

Sol. $1125 = 5^3 \cdot 3^2, 3375 = 5^3 \cdot 3^3$

Clearly 3^3 is a factor of x and 3^2 is a factor of atleast one of y and z . This can be done in 5 ways. Also 5^3 is a factor of atleast two of the numbers x, y, z which can be done in $({}^3C_2 \times 4 - 2) = 10$

3. 8

Sol. We have $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 0 & 2 \\ 2 & -\lambda & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - \lambda - 3 = 0$$

$$\text{So, } A^3 - 3A^2 - A = 3I_3$$

$$\text{Now } |(A - I)^3 - 4A|$$

$$= |A^3 - 3A^2 + 3A - I - 4A|$$

$$= |A^3 - 3A^2 - A - I| = |2I_3| = 8$$

4. 4

Sol. $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{{}^3C_1 \cdot \frac{1}{4} \left(\frac{3}{4}\right)^2 \cdot \frac{3}{4}}{1 - \left[\left(\frac{3}{4}\right)^4 \cdot \frac{1}{4} + \left(\frac{3}{4}\right)^4\right]}$

$$= \frac{81}{148}$$

$$(m+n)_{\text{least}} = 81 + 148 = 229$$

5. 40

Sol. Given conic is an ellipse and x - axis being the tangent to it.

6. 1

Sol. $[\sin x] + [\cos x] + 2 = 0$ is possible only when

$$[\sin x] = -1 \text{ and } [\cos x] = -1$$

$$\text{Or } -1 \leq \sin x < 0 \text{ and } -1 \leq \cos x < 0 \Rightarrow x \in \left(\pi, \frac{3\pi}{2}\right),$$

$$\text{Now } f(x) = \sin x - \cos x + 3 = \sqrt{2} \sin\left(x - \frac{\pi}{4}\right) + 3$$

$$\text{For } x \in \left(\pi, \frac{3\pi}{2}\right), \sin\left(x - \frac{\pi}{4}\right) \in \left(-\frac{1}{\sqrt{2}}, -\frac{1}{2}\right)$$

$$\Rightarrow y = f(x) = \sqrt{2} \sin\left(x - \frac{\pi}{4}\right) + 3, \text{ then } f(x) \in (2, 4)$$

7. 960

$$\begin{aligned} \text{Sol. } f(n) \sum_{r=1}^n \left[r^2 \binom{n}{r} - \binom{n}{r-1} \right] + (2r)^n \binom{n}{r} + \binom{n}{r} &= \sum_{r=1}^n \left[(r^2 + 2r + 1)^n \binom{n}{r} - r^2 \binom{n}{r-1} \right] \\ &= \sum_{r=1}^n \left[(r+1)^2 \binom{n}{r} - r^2 \binom{n}{r-1} \right] = \sum_{r=1}^n [V_{r+1} - V_r] \\ &= V_2 - V_1 + V_3 - V_2 + \dots + V_{n+1} - V_n \\ V_{n+1} - V_1 &= (n+1)^2 \binom{n}{n+1} - 1 = n^2 + 2n \end{aligned}$$

8. 2

$$\text{Sol. } f(x(y)) = x^p y^4, \text{ put } x = \frac{1}{f(y)}$$

$$f(1) = \left(\frac{1}{f(y)} \right)^p y^4 = \frac{y^4}{(f(y))^p} \text{ for } y=1, f(1) = \frac{1}{(f(1))^p} \Rightarrow f(1) = 1$$

$$\text{So } f(y) = y^{4/p} \quad \dots(1)$$

$$\text{Hence } f(xy^{4/p}) = x^p y^4$$

$$\text{Put } y = z^{p/4}$$

$$f(xz) = x^p z^p \Rightarrow f(x) = x^p \quad \dots(2)$$

$$\text{From (1) and (2) } \frac{4}{p} = p \Rightarrow p = 2$$

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- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For Only One Part.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-14)** – Contains seven (07) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-A (15-18)** - This section contains Two paragraphs. Based on each paragraph, there are Two multiple choice questions. Each question has only one correct answer and carries **+4 marks** for the correct answer and **– 2 marks** for wrong answer.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

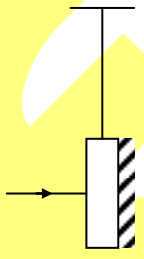
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SECTION-1 : PHYSICS

PART – A

(Multi Correct Choice Type)

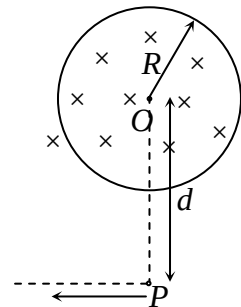
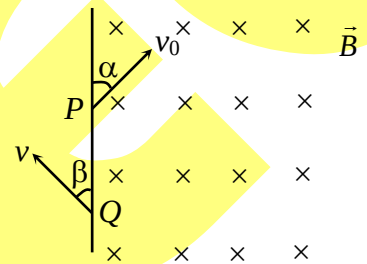
This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. A small mirror is suspended by a thread as shown in figure. A short pulse of monochromatic light rays is incident normally on the mirror and gets reflected. Which of the following statements are correct?
 (A) mirror will start oscillating
 (B) wavelength of reflected rays will be greater than that of incident rays
 (C) wavelength of reflected rays may be less than that of incident rays
 (D) mirror will be at rest after some time
 
2. A particle of mass m is moving in a field where the potential energy is given by $U(x) = U_0(1 - \cos ax)$, where U_0 and a are constants and x is the displacement from mean position. Then (for small oscillations)
 (A) the time period is $T = 2\pi\sqrt{\frac{m}{aU_0}}$ (B) the speed of particle is maximum at $x = 0$
 (C) the amplitude of oscillations is $\frac{\pi}{a}$ (D) the time period is $T = 2\pi\sqrt{\frac{m}{a^2U_0}}$
3. Two different coils have self inductances $L_1 = 8 \text{ mH}$ and $L_2 = 2 \text{ mH}$. The current in one coil is increased at a constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At this time the current, the induced voltage and the energy stored in the first coil are i_1 , V_1 and U_1 respectively. Corresponding values for the second coil at the same instant are i_2 , V_2 and U_2 respectively. Then
 (A) $\frac{i_1}{i_2} = \frac{1}{4}$ (B) $\frac{i_1}{i_2} = 4$ (C) $\frac{U_2}{U_1} = 4$ (D) $\frac{V_2}{V_1} = \frac{1}{4}$

Space For Rough Work

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4. The magnetic field perpendicular to the plane of conducting ring of radius r changes at the rate $\frac{dB}{dt} = \alpha$. Then
 (A) Emf induced in the ring is $\pi r^2 \alpha$
 (B) Emf induced in the ring is $2\pi r \alpha$
 (C) the potential difference between diametrically opposite points on the ring is half of induced emf
 (D) all points on the ring are at same potential
5. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.70 eV is $T_B = (T_A - 1.50)$ eV. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$ then
 (A) the work function of A is 2.25 eV (B) the work function of B is 4.20 eV
 (C) $T_A = 2.00$ eV (D) $T_B = 2.75$ eV
6. A particle of charge $-q$ and mass m enters a uniform magnetic field $\vec{B}$ (perpendicular to paper inwards) at P with a velocity v_0 at an angle α and leaves the field at Q with velocity v at angle β as shown in the figure. Then
 (A) $\alpha = \beta$
 (B) $v = v_0$
 (C) $PQ = \frac{2mv_0 \sin \alpha}{Bq}$
 (D) particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$
7. In a cylindrical region of radius R , there exists a time varying magnetic field B such that $\frac{dB}{dt} = k (> 0)$. A charged particle having charge q is placed at the point P at a distance $d (> R)$ from its centre O . Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.
- (A) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 2R$
 (B) Work done by external agent is $\frac{q\pi R^2}{8} k$ if $d = 4R$
 (C) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 4R$
 (D) Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 6R$



Space For Rough Work

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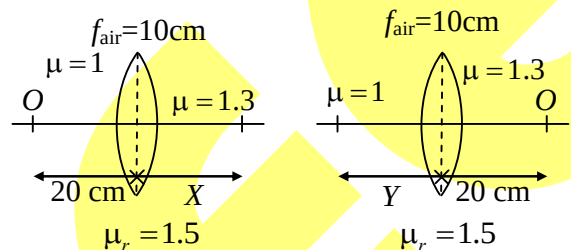
(Single Correct Choice Type)

This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. A diatomic molecule having atoms of masses m_1 and m_2 has its potential energy function about the equilibrium position r_0 as given by $U(r) = -A + B(r - r_0)^2$ where A and B are constants. When the atom vibrate at high temperature condition, the square of angular frequency of vibration will be

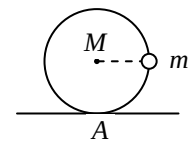
(A) $\frac{2B}{m_1}$ (B) $\frac{2B}{m_2}$ (C) $\frac{2B(m_1 + m_2)}{m_1 m_2}$ (D) $\frac{B(m_1 + m_2)}{2m_1 m_2}$

9. An equiconvex lens made up of a material of refractive index 1.5 has focal length of 10 cm when placed in air as shown in the figure. One side of the medium is replaced by another medium of refractive index 1.3. If X and Y are the image distances when the object is placed at a distance of 20 cm from optical centre in the medium with refractive index 1 and 1.3 respectively, then



- (A) $X > 1.3Y$ (B) $X < 1.3Y$
(C) $X = 1.3Y$ (D) cannot be determined

10. A uniform body of mass M of radius R has a small mass m attached at edge as shown in the figure. The system is placed on a perfectly rough horizontal surface such that mass m is at the same horizontal level as the centre of body. It is assumed that there is no slipping at point A. If I_A is the moment of the inertia of combined system about point of contact A then the normal reaction at point A just after the system is released from rest is ($M = 6$ kg, $m = 2$ kg, $I_A = 4$ kg m², $R = 1$ m, $g = 10$ m/s²)

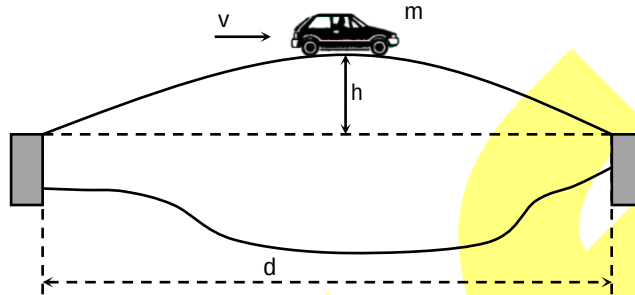


- (A) 60 N (B) 80 N (C) 75 N (D) 70 N
11. A circuit consists of a capacitor and a resistor having resistance $R = 220 \Omega$ connected in series. When an alternating e.m.f. of peak voltage $V_0 = 220 \sqrt{2}$ V is applied to the circuit, the peak current in steady state is observed to be $I_0 = 1$ A. The phase difference between the current and the voltage is
- (A) 30° (B) 45° (C) 60° (D) 90°

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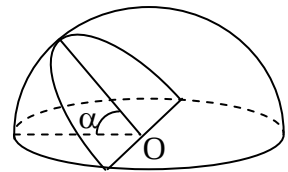
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12. A stone of mass m tied to one end of thread of length ℓ . The diameter of thread is 'd' and it is suspended vertically. The stone is now rotated in horizontal plane and makes an angle ' θ ' with vertical. The increase in length of wire is (young's modulus of wire is y)
- (A) $\frac{4mg\ell}{\pi d^2 y \cos \theta}$ (B) $\frac{4mg\ell}{\pi d^2 y \sin \theta}$
 (C) $\frac{4mg\ell}{\pi d^2 y}$ (D) $\frac{4mg\ell}{\pi d^2 y \sec \theta}$
13. There is a parabolic-shaped bridge across a river of width 100 m. The highest point of the bridge is 5 m above the level of the banks. A car of mass 1000 kg is crossing the bridge at a constant speed of 20 ms^{-1} .



Using the notation indicated in the figure, find the force exerted on the bridge by the car when it is at the highest point of the bridge (Ignore air resistance and take g as 10 ms^{-2} .)

- (A) 6.4 KN (B) 7.2 KN
 (C) 8.4 KN (D) 9.2 KN
14. The strength of the electric field produced by charges uniformly distributed over the surface of a hemisphere at its centre O is E_0 . A part of the surface is isolated from this hemisphere by two planes passing through the same diameter and forming an angle α with each other. Determine the electric field strength E produced at the same point O by the charges located on the isolated surface (on the "mericarp").



- (A) $\frac{E_0}{2} \sin \frac{\alpha}{2}$ (B) $\frac{2E_0}{3} \sin \frac{\alpha}{2}$ (C) $E_0 \sin \frac{\alpha}{2}$ (D) $2E_0 \sin \frac{\alpha}{2}$

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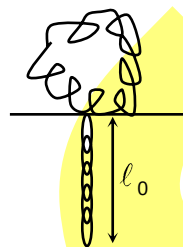
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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

A rope of mass m , length ' ℓ ' is lying on a table with ℓ_0 length lying outside a hole on the table on which it is kept. It is kept at $t = 0$. Answer the following questions:



15. Speed of the free end of the chain when it is at a distance of ' y ' from the table is?

(A) $v(y) = \sqrt{\frac{2g}{3} \left(y - \frac{\ell_0^3}{y^2} \right)}$

(B) $v(y) = \sqrt{\frac{g}{3} \left(y - \frac{2\ell_0^3}{y^2} \right)}$

(C) $v(y) = \sqrt{2g \left(y - \frac{\ell_0^3}{y^2} \right)}$

(D) $v(y) = \sqrt{g \left(y - \frac{2\ell_0^3}{y^2} \right)}$

16. If $\ell_0 = 0$, the time in which the entire rope will slip off the table is?

(A) $\sqrt{4\ell/g}$

(B) $\sqrt{12\ell/g}$

(C) $\sqrt{\frac{6\ell}{g}}$

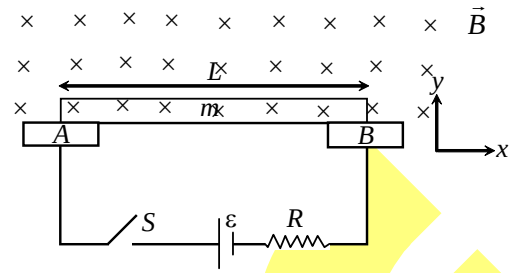
(D) $\sqrt{\frac{2\ell}{g}}$

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Paragraph for Question no. 17 to 18

In a vertical plane, a metal rod of length L and mass m is placed over two conducting platforms A and B . A region of magnetic field $\vec{B} = -B_0 \hat{k}$ starting from line joining A and B and lying over it, is switched on. Now, at $t = 0$ switch S is closed such that the charge q is passes through the rod in time dt due to which magnetic field exert an impulsive force which causes the rod to jump with certain velocity and since the gravitational field is present the rod comes back to AB after some time and collides inelastically and the process is repeated.



17. The maximum height reached by the rod is
- (A) $\frac{2q^2 L^2 B_0^2}{m^2 g}$ (B) $\frac{q^2 L^2 B_0^2}{2m^2 g}$
- (C) $\frac{q^2 L^2 B_0^2}{m^2 g}$ (D) $\frac{q^2 L^2 B_0^2}{4m^2 g}$
18. The maximum emf induced in the rod is
- (A) $\frac{B_0^2 L^2 q}{m}$ (B) $\frac{B_0^2 L^2 q}{2m}$
- (C) $\frac{2B_0^2 L^2 q}{m}$ (D) $\frac{4B_0^2 L^2 q}{m}$

Space For Rough Work

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SECTION-2 : CHEMISTRY

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

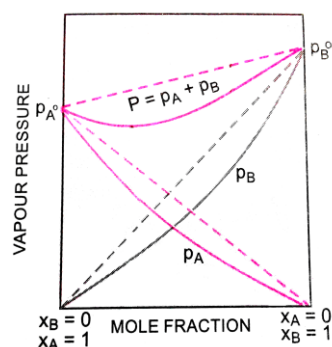
1. Which one of the following reactions of Xenon compounds is/are feasible ?

- (A) $\text{XeO}_3 + 6\text{HF} \longrightarrow \text{XeF}_6 + 3\text{H}_2\text{O}$
 (B) $3\text{XeF}_4 + 6\text{H}_2\text{O} \longrightarrow 2\text{Xe} + \text{XeO}_3 + 12\text{HF} + 1.5\text{O}_2$
 (C) $2\text{XeF}_2 + 2\text{H}_2\text{O} \longrightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
 (D) $\text{XeF}_6 + \text{RbF} \longrightarrow \text{Rb}[\text{XeF}_7]$

2. Which of the following reactions form racemic mixture?

- (A) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow{\text{Br}_2/\text{CCl}_4}$ (B) $\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{NaOH}]{\text{HCN}}$
 (C) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2 \xrightarrow[\text{CCl}_4]{\text{HCl}}$ (D) $\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{NH}_2\text{OH}/\text{H}^+]{\text{NH}_2\text{OH}}$

3.



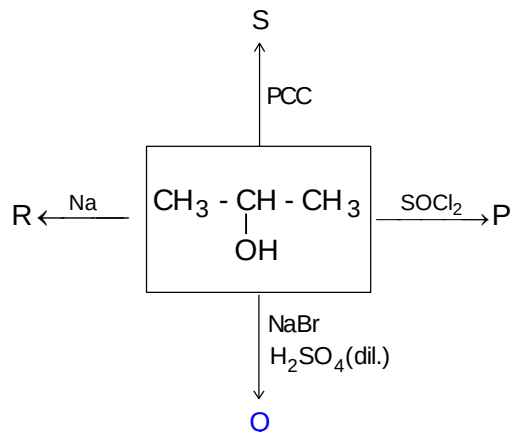
Which of the following statement(s) is/are correct for a solution of liquid A and B as given in the above figure?

- (A) Enthalpy change of mixing of A with B (ΔH_{mix}) < 0
 (B) Volume of mixing (ΔV_{mix}) < 0
 (C) $p < p_A^0$
 (D) Entropy of mixing $\Delta S_{\text{mixing}} < 0$

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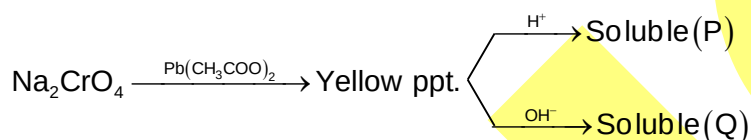
4.



P, Q, R and S are the organic products of above reactions. Choose correct statements.

- (A) 'P' is less reactive than 'Q' towards dehydrohalogenation reaction in presence of alcoholic KOH through E_1 path.
 (B) Formation of 'R' is accompanied with emission of H_2 gas.
 (C) Reaction of 'S' with $\text{CH}_3\text{MgBr}/\text{H}_3\text{O}^+$ forms a tertiary alcohol.
 (D) 'R' is more soluble in water than 'P'.

5.



The anions present in the soluble compounds of (P) and (Q) in the above reaction is/are

- (A) $[\text{Pb}(\text{OH})_4]^{2-}$ (B) $\text{Cr}_2\text{O}_7^{2-}$
 (C) $[\text{Pb}(\text{CrO}_4)_2]^{2-}$ (D) PbO_4^{2-}

6.

FeTiO_3 contains Fe^{2+} , Ti^{4+} and O^{2-} ions. It's crystal contains hcp unit cells in which O^{2-} form the hcp array.

Choose the correct statements

- (A) Fe^{2+} ions occupy $\frac{1}{3}$ rd of octahedral voids
 (B) Ti^{4+} ions occupy $\frac{1}{3}$ rd of octahedral voids
 (C) It's colour is due to only Fe^{2+} ions
 (D) It is a paramagnetic substance

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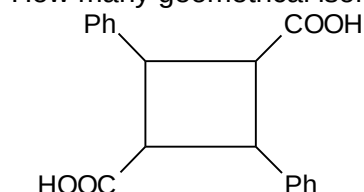
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7. The value of $\Delta H_{\text{transition}}$ of C (graphite) $\longrightarrow$ C (diamond) is 1.9 kJ/mol at 25°C entropy of graphite is higher than entropy of diamond. This implies that :
- (A) C (diamond) is more thermodynamically stable than C (graphite) at 25°C
 (B) C (graphite) is more thermodynamically stable than C (diamond) at 25°C
 (C) diamond will provide more heat on complete combustion at 25°C
 (D) $\Delta G_{\text{transition}}$ of C (diamond) $\longrightarrow$ C (graphite) is -ve

(Single Correct Choice Type)

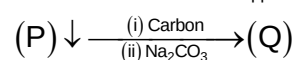
This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. How many geometrical isomers are possible for the following compound?



- (A) 3
(C) 5
(B) 4
(D) 6

9. $\text{SO}_4^{2-} \xrightarrow{\text{BaCl}_2} (\text{P}) \downarrow$
 White ppt.



Compound (Q) containing sulphur is

- (A) Na_2SO_4
(C) Na_2SO_3
(B) Na_2S
(D) CS_2

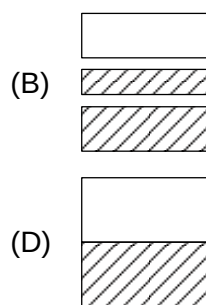
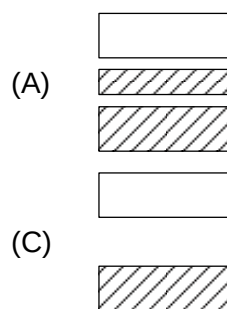
10. A solution of 'P' and 'Q' with 30 mole percent of 'P' is in equilibrium with its vapour which contains 60 mole percent of 'P'. The solution and the vapour behave ideally. If the ratio of vapour pressure of pure 'P' to pure 'Q' is expressed as $x : y$, the value of $x + y$ is
- (A) 3.5
(C) 18
(B) 9
(D) 7

11. A mixture of NaOH and Na_2CO_3 requires 100 mL of 3 M HCl for complete reaction upto phenolphthalein indicator. How many gram of NaOH is present in the mixture which contains 10.6 g of Na_2CO_3 ?
- (A) 8 g
(C) 4 g
(B) 12 g
(D) 12 mg

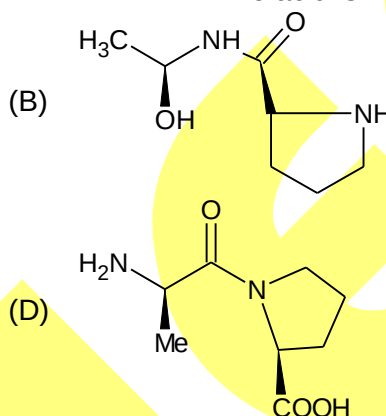
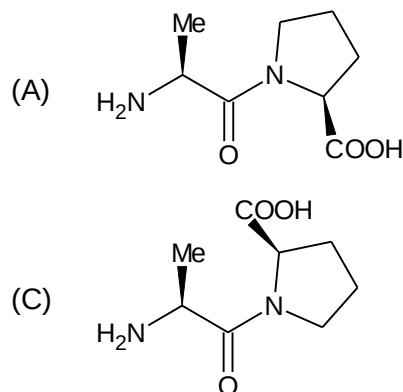
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12. The band structure of n-type semiconductor is



13. The structure of dipeptide "Ala-Pro" derived from Natural amino acid is



14. In the isoelectronic series VO_4^{3-} , CrO_4^{2-} & MnO_4^- , all members have CT transition. The incorrect statement is
- (A) CT transition attributed to excitation of electrons from ligand to metal
- (B) MnO_4^- exhibits charge transfer at shortest wavelength among three
- (C) CT transition lapporte & spin allowed transition
- (D) Charge on metal nucleus increases in order : $\text{VO}_4^{3-} < \text{CrO}_4^{2-} < \text{MnO}_4^-$

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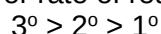
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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

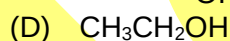
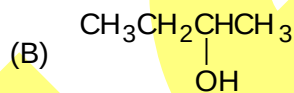
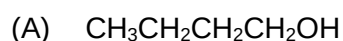
Lucas test is used to distinguish the different types of alcohols i.e., the primary, secondary and tertiary alcohols. Lucas reagent is anhydrous ZnCl_2 and conc. HCl . The alcohols are converted to alkyl chlorides in this test. The alcohols are distinguished through the rate of their reactions with HCl . Anhydrous ZnCl_2 is used as a catalyst. It removes the alcoholic OH to form a carbocation. The speed or rate of reaction of the different alcohols is



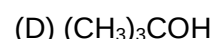
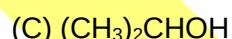
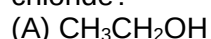
Formation of alkyl chlorides by the alcohols is observed due to formation of white turbidity.

Answer the following questions on the basis of above write up.

15. Which alcohol can form the white turbidity instantaneously?



16. Which alcohol mostly require the catalyst anhy. ZnCl_2 to form the corresponding alkyl chloride?

**Paragraph for Question no. 17 to 18**

Silicates are the oxo-anions of silicon, which contains $\text{Si} - \text{O}$ sigma bonds. The anions are associated with metal cations. So, they are called ionic crystals. The silicate anions are very simple like ortho silicates SiO_4^{4-} and complex like cyclic or sheet silicates. The SiO_4^{4-} has no sharing oxygen atoms. The higher silicate contain shared oxygen atoms. For example $\text{Si}_2\text{O}_7^{6-}$ is called pyrosilicate in which one oxygen atom is shared. $(\text{Si}_3\text{O}_9)^{6-}$ is called cyclic silicate in which three oxygen atoms are shared.

Answer the following questions on the basis of above write up.

17. How many oxygen atom(s) is/are shared in the silicate anion $(\text{Si}_6\text{O}_{15})^{6-}$?

(A) 3

(B) 6

(C) 9

(D) 8

18. How many O^- ions per tetrahedra are present in pyrosilicate?

(A) 2

(B) 3

(C) 1

(D) 4

Space For Rough Work

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SECTION-3 : MATHEMATICS

PART – A

(Multi Correct Choice Type)

This section contains 7 **multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

1. If $I_1 \equiv (1, -2)$ and $I_2 \equiv (-3, 2)$ are the ex-centers of $\triangle ABC$ and ex-centre I_3 lies on the circle $x^2 + y^2 + 3x + y - 6 = 0$, then
 - (A) locus of in-centre of $\triangle ABC$ is given by $x^2 + y^2 + x - y - 5 = 0$
 - (B) locus of in-centre of $\triangle ABC$ is given by $x^2 + y^2 + x - y + 5 = 0$
 - (C) locus of locus lies on the line $y = -x$
 - (D) origin lies inside the locus
2. Let N be the number of 7-digit numbers the sum of whose digits is even.
 - (A) The number of positive divisors of N is 126
 - (B) The number of positive divisors of N of the form $4k+1$, $k \geq 0$ is 14
 - (C) The number of positive divisors of N of the form $4k+2$, $k \geq 1$ is 9
 - (D) The number of positive divisors of N of the form $4k+2$, $k \geq 0$ is 12
3. Consider $f(x) = a_5x^5 + a_4x^4 + a_3x^3 + 3x^2 + 2x + 1 = 0$, $a_5, a_4, a_3 \in \mathbb{R} - \{0\}$. Then
 - (A) $f(x)=0$ will have atleast one non-real root
 - (B) $f(x)=0$ will have atleast one real root
 - (C) for $a_5 = 2021$ and $a_3 = 2019$, $f(x)=0$ has atleast one positive root
 - (D) for $a_5 = 1$, $a_4 = 9$ and $a_3 = 16$, the above equation has atleast one negative root
4. If A and B are different matrices satisfying $A^3 = B^3$, $A^2B = B^2A$ then which of the following is/are incorrect.
 - (A) $|A^2 + B^2| = 0$
 - (B) $|A - B| = 0$
 - (C) $|A^2 + B^2|$ and $|A - B|$ must be zero
 - (D) At least $|A^2 + B^2|$ or $|A - B|$ must be zero

Space For Rough Work

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5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then
- (A) $g'(2) = \frac{1}{3}$ (B) $h'(1) = 666$
 (C) $h(0) = 16$ (D) $h(g(3)) = 38$
6. If lines $x + y - 3 = 0$ and $y = x + 1$ are two tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then angle between the pair of tangents drawn from point $P(4, -\sqrt{5})$ to the curve $\frac{x^2}{3a^2 + b^2} + \frac{y^2}{a^2 + 3b^2} = 1$ can be
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{6}$
 (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$
7. Given that f is a real valued differentiable function such that $f(x) \cdot f'(x) \leq 0$, for all real x , there exists 'a' such that $f'(x) \neq 0 \forall x \in [a, \infty)$ then it follows that:
- (A) $f^2(x)$ is non-decreasing function
 (B) $f^2(x)$ is non-increasing function
 (C) $f(x)$ has no point of local maxima or minima
 (D) $f(x) = 0$ does not have any real root

(Single Correct Choice Type)

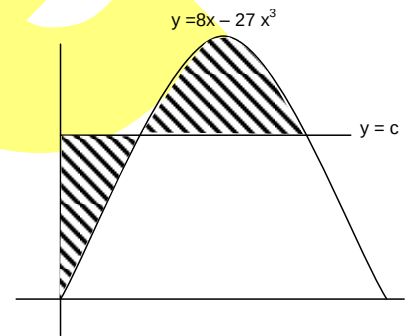
This section contains **7 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. If $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are three points on the parabola $y^2 = 4ax$ the normals at which meet in a point, then $\frac{x_1 - x_2}{y_3} + \frac{x_2 - x_3}{y_1} + \frac{x_3 - x_1}{y_2}$ is equal to
- (A) 4 (B) 8
 (C) 0 (D) 1
9. Let $P(x)$ be a polynomial with integer coefficients. It is known that $P(x)$ takes the value 2015 for four distinct integers. Then the number of integral values of x for which $P(x)$ equals 2022
- (A) 1 (B) 7
 (C) 2011 (D) 0

Space For Rough Work

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10. If α is a root of the equation $x^2 - 3x + 1 = 0$, then $\frac{2\alpha^5 - 5\alpha^4 + 2\alpha^3 - 8\alpha^2}{\alpha^2 + 1}$ is
 (A) -1 (B) -3
 (C) $\frac{2}{3}$ (D) 1
11. If the points of intersection of the curves $x^2 - y^2 = a^2$ and $y = x^2$ lie on a unique circle, and S is the set of all such values of a , then S contains
 (A) $(-1, 1)$ (B) $(0, 1)$
 (C) $(-1, 0)$ (D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
12. A fair coin is repeatedly tossed. If the probability that the first time heads is tossed, twice in a row, is on the 9th and 10th toss, is
 (A) $\frac{17}{512}$ (B) $\frac{15}{512}$
 (C) $\frac{11}{512}$ (D) $\frac{1}{2}$
13. The value of c such that areas of shaded region are equal.
 (A) $\frac{32}{27}$ (B) $\frac{1}{7}$
 (C) $\frac{16}{9}$ (D) none of these
14. If p, q are two real numbers satisfies the relations $2p^2 - 3p - 1 = 0$ and $q^2 + 3q - 2 = 0$ and $pq \neq 1$, then the value of $\frac{1 + 2p + pq}{q}$ is
 (A) $-\frac{3}{4}$ (B) $\frac{(12 + \sqrt{17})(3 + \sqrt{17})}{24}$
 (C) 1 (D) $\frac{1}{2}$



Space For Rough Work

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(Paragraph Type)

This section contains **2 paragraphs**. Based upon the paragraphs **2 multiple choice questions** have to be answered. Each of these questions has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Paragraph for Question no. 15 to 16

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R . C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R . Both the circles C_1 and C_2 pass through the focus of the parabola.

15. Area of the ΔPQR equals

- (A) $\frac{1}{2}$ (B) 1
(C) 2 (D) $\frac{1}{4}$

16. Radius of the circle C_2 is

- (A) $5\sqrt{5}$ (B) $5\sqrt{10}$
(C) $10\sqrt{2}$ (D) $\sqrt{210}$

Paragraph for Question no. 17 to 18

If $f(x) = (x - \alpha)^n g(x)$, then $f(\alpha) = f'(\alpha) = f''(\alpha) = \dots = f^{n-1}(\alpha) = 0$ where $f(x)$ and $g(x)$ are polynomials for a polynomial $f(x)$ with rational coefficients, answer the following questions

17. If $f(x)$ touches x -axis at only one point, then the point of touching is
(A) always a rational number (B) may or may not be a rational number
(C) never a rational number (D) none of these
18. $f(\alpha) = f'(\alpha) = f''(\alpha) = 0, f(\beta) = f'(\beta) = f''(\beta) = 0$ and $f(x)$ is a polynomial of degree 6, then
(A) atleast three roots of $f'(x) = 0$ are always real
(B) atleast two roots of $f'(x) = 0$ are always real
(C) exactly two roots of $f'(x) = 0$ are real
(D) none of these

Space For Rough Work

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Q.P. Code: 100205**Answers****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|--------|---------|--------|-------|
| 1. AB | 2. BCD | 3. ACD | 4. AB |
| 5. ABC | 6. ABCD | 7. ACD | 8. C |
| 9. B | 10. D | 11. B | 12. A |
| 13. C | 14. C | 15. A | 16. B |
| 17. B | 18. A | | |

SECTION-1 : CHEMISTRY**PART – A**

- | | | | |
|--------|---------|--------|---------|
| 1. BCD | 2. ABC | 3. AB | 4. ABCD |
| 5. AB | 6. ABCD | 7. BCD | 8. C |
| 9. B | 10. B | 11. A | 12. A |
| 13. A | 14. B | 15. C | 16. A |
| 17. C | 18. B | | |

SECTION-1 : MATHEMATICS**PART – A**

- | | | | |
|---------|---------|--------|--------|
| 1. ACD | 2. ABCD | 3. ABD | 4. ABC |
| 5. ABCD | 6. AB | 7. BCD | 8. C |
| 9. D | 10. A | 11. D | 12. A |
| 13. A | 14. D | 15. A | 16. B |
| 17. A | 18. B | | |

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Answers & Solutions

SECTION-1 : PHYSICS

PART – A

1. **AB**

2. **BCD**

Sol. $F = -\frac{\partial U}{\partial x}$

3. **ACD**

Sol. $U = \frac{1}{2} Li^2$

4. **AB**

Sol. $-\int \vec{E} \cdot d\vec{l} = -\frac{dB}{dt}$

5. **ABC**

Sol. $h\nu = \phi + KE$

6. **ABCD**

Sol. $R = \frac{mv}{qB}$

7. **ACD**

Sol. $\int \vec{E} \cdot d\vec{l} = A \frac{dB}{dt}$

$$E 2\pi \sqrt{x^2 + d^2} = \pi R^2 k$$

$$E = \frac{\pi R^2 k}{2\sqrt{x^2 + d^2}}$$

$$W_{\text{ext}} = \int_0^\infty q\vec{E} \cdot d\vec{x} = \frac{q\pi R^2}{4} k$$

8. **C**

Sol. $F = -\frac{\partial U}{\partial r}$

9. **B**

Sol. $\frac{\mu_2}{v'} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}, \frac{\mu_3}{v} - \frac{\mu_2}{v'} = \frac{\mu_3 - \mu_2}{-R} \Rightarrow \frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{2\mu_2 - \mu_1 - \mu_3}{R}$

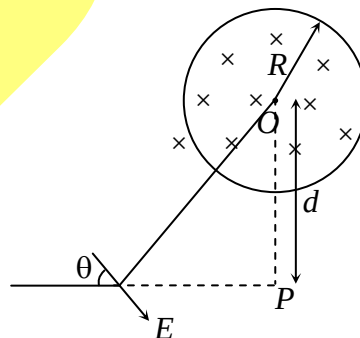
for Ist case: $\mu_3 = 1.3, \mu_2 = 1.5, \mu_1 = 1, u = -20, v = X$

we get $\frac{1.3}{X} + \frac{1}{20} = \frac{0.7}{R} \dots (i)$

for IInd case: $\mu_3 = 1, \mu_2 = 1.5, \mu_1 = 1.3, u = -20, v = Y$

$\frac{1}{Y} + \frac{1.3}{20} = \frac{0.7}{R} \dots (ii)$

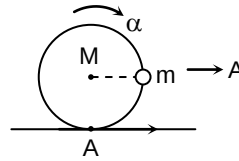
From (i) and (ii), $1.3Y > X$



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10. **D**

Sol. $(A_{CM})_x = \frac{mA + MA}{m + M} = A$
 $(A_{CM})_y = \frac{M \times 0 + mR\alpha}{m + M} = \frac{mR\alpha}{m + M}$
 $f = (M + m)A$
 $(M + m)g - N = (M + m)(A_{CM})_y$
 $mgR = I_A \alpha$
 $A = R\alpha$
 $\therefore N = 70 \text{ N}$



... (i)

... (ii)

... (iii)

... (iv)

11. **B**12. **A**

Sol. $y = \frac{T\ell}{A\Delta\ell}, \Delta L = \frac{T\ell}{Ay} = \frac{mg\ell}{Ay \cos \theta}$

13. **C**

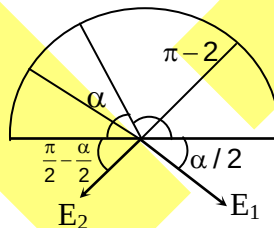
Sol. $mg - N = \frac{mv^2}{R}$

'R' can be measured by $\frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{d^2y/dx^2}$

14. **C**

Sol. $E_1 \cos \frac{\alpha}{2} = E_2 \sin \frac{\alpha}{2} \quad \dots (i)$

$E_1 \sin \frac{\alpha}{2} + E_2 \cos \frac{\alpha}{2} = E_0 \quad \dots (ii)$

Solving we get E_1 15. **A**

Sol. $V_{cm} = \frac{yV}{\ell}$

$\lambda yg - \lambda v^2 = \lambda \ell \cdot \frac{d}{dt}(V_{cm})$

$\frac{dy}{dt} = v.$

16. **B**17. **B**

Sol. $H_{\max} = \frac{v^2}{2g} = \frac{q^2 L^2 B_0^2}{2m^2 g}$

18. **A**

Sol. $E_{\max} = B_0 L v = \frac{B_0^2 L^2 q}{m} \quad (\because v_{\max} = v)$

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SECTION-2 : CHEMISTRY

PART – A

1. BCD

Sol. All fluorides of 'Xe' undergo hydrolysis. Perxenate ion is formed by XeF_6 .

2. ABC

Sol. $\text{CH}_3\text{CH}_2\underset{\text{Br}}{\text{CH}}\text{CH}_2\text{Br}$, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CN}$ and $\text{CH}_3\text{CH}_2\underset{\text{Cl}}{\text{CH}}\text{CH}_3$

contains chiral carbon atoms. So they form a pair of enantiomers.

3. AB

Sol. Negative deviation takes place.

4. ABCD

Sol. $\text{P} = \text{CH}_3\underset{\text{Cl}}{\text{CH}}\text{CH}_3$, $\text{Q} = \text{CH}_3 - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$, $\text{R} = \text{CH}_3 - \underset{\text{O}^-\text{Na}^+}{\text{CH}} - \text{CH}_3$, $\text{S} = \text{CH}_3 - \overset{\text{O}}{\underset{\text{||}}{\text{C}}} - \text{CH}_3$

contains chiral carbon atoms. So they form a pair of enantiomers.

5. AB

Sol. $\text{P} = \text{Cr}_2\text{O}_7^{2-}$ and $\text{Q} = \text{Pb}(\text{OH})_4^{2-}$

6. ABCD

Sol. Formula(empirical) = FeTeO_3 As there are six oxide ions per hcp unit cell. The molecular formula will be $\text{Fe}_2\text{Ti}_2\text{O}_6$.

7. BCD

Sol. Graphite is thermodynamically more stable than diamond.

8. C

Sol. Five isomers are possible.

9. B

Sol. $\text{P} = \text{BaSO}_4$ and $\text{Q} = \text{Na}_2\text{S}$

10. B

Sol. $\frac{X_P p_P^0}{X_Q p_Q^0} = \frac{y_P P_T}{y_Q P_T}$
 $\frac{0.3 p_P^0}{0.7 p_Q^0} = \frac{0.6}{0.4}$

$$\frac{p_P^0}{p_Q^0} = \frac{0.6}{0.4} \times \frac{0.7}{0.3} = \frac{7}{2} = \frac{x}{y}$$

$$\therefore x + y = 9$$

11. A

Sol. M_{eq} of $\text{HCl} = M_{\text{eq}}$ of $\text{NaOH} + M_{\text{eq}}$ of Na_2CO_3 ($n = 1$)

12. A

Sol. The conduction band contains more electrons than that of p-type semiconductor.

13. A

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Sol. The amino acids are alanine and proline.

14. B

Sol. MnO_4^- exhibit charge transfer band at longer wavelength than that of the other three.

15. C

Sol. Reactivity of alcohols towards Lucas test is $3^\circ > 2^\circ > 1^\circ > \text{CH}_3$

16. A

Sol. Primary alcohols can't remove the OH group as water easily by HCl alone. So they form a co-ordinated complex with ZnCl_2 which facilitates the removal of OH group and form the carbocation, which then forms the alkyl chloride.

17. C

Sol. $(\text{Si}_6\text{O}_{15})^{6-}$ shares nine oxygen atoms.

18. B

Sol. In pyrosilicate $\text{Si}_2\text{O}_7^{6-}$ contain one sharing oxygen atom. So other three atoms oxygen per tetrahedra carries 6-charge. Therefore per tetrahedra three O^- ions are present.

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SECTION-3 : MATHEMATICS

PART – A

1. ACD

Sol. Clearly $x^2 + y^2 + 3x + y - 6 = 0$ is circumcircle of $\Delta l_1 l_2 l_3$ and in-centre of ΔABC is orthocenter of $l_1 l_2 l_3$ let (h, k)

As mirror image of orthocenter about any side of triangle lies on the circumcircle. Hence required locus is $x^2 + y^2 + x - y - 5 = 0$

2. ABCD

Sol. The seven digit numbers are 1000000, 1000001, 9999999
The sums of their digits are odd, even, odd, even,etc.

$$\therefore N = \frac{9 \times 10^6}{2} = 2^5 3^2 5^6$$

Required no. number of divisors of N is $1.2.7 = 14$

3. ABD

Sol. If x_1, x_2, x_3, x_4, x_5 be the five roots, then $\sum \frac{1}{x_i^2} = -2$

So, the equation has atleast one complex root and since it is an odd degree polynomial equations with coefficients real always have atleast one real root

4. ABC

Sol. $A^3 - A^2 B = B^3 - B^2 A$
 $A^2(A - B) + B^2(A - B)$
 $(A^2 + B^2)(A - B) = 0$
 $\therefore$ Either $A^2 + B^2 = 0$ or $A - B = 0$

5. ABCD

Sol. $g'(2) = \frac{1}{f'(0)} = \frac{1}{3}$

$$h(g(g(x))) = x$$

$$h(y) = f(f(y))$$

$$h'(y) = f'(f(y)) \cdot f'(y)$$

$$h(0) = 16,$$

$$\text{let } g(g(x)) = y$$

$$g(x) = f(y)$$

$$x = f(f(y))$$

$$h'(1) = 666$$

6. AB

Sol. Tangents are perpendicular $\Rightarrow$ their point of intersection $(1, 2)$ lies on director circle $x^2 + y^2 = a^2 + b^2 \Rightarrow a^2 + b^2 = 5$

Equation of director circle of ellipse $x^2 + y^2 = 4(a^2 + b^2) = 20$

Clearly point $(4, -\sqrt{5})$ lies outside the director circle

$\Rightarrow$ Angle is acute

7. BCD

Sol. Case possible $f(x) > 0$ and $f(x)$ decreasing or $f(x) < 0$, $f(x)$ increasing and $f(x)$ and $f'(x)$ have to change sign simultaneously to maintain the condition which is not possible also $(f^2(x))' = 2f(x) \cdot f'(x) \leq 0$

8. C

Sol. $\frac{x_1 - x_2}{y_3} + \frac{x_2 - x_3}{y_1} + \frac{x_3 - x_1}{y_2}$

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$$= \frac{a(t_1^2 - t_2^2)}{2at_3} + \frac{a(t_2^2 - t_3^2)}{2at_1} + \frac{a(t_3^2 - t_1^2)}{2at_2}$$

$$= \frac{(t_1 - t_2)(-t_3)}{2t_3} + \frac{(t_2 - t_3)(-t_1)}{2t_1} + \frac{(t_3 - t_1)(-t_2)}{2t_2} = 0$$

9. D

Sol. $P(x) - 2015 = q(x)(x - x_1)(x - x_2)(x - x_3)(x - x_4)$

Let $P(\alpha) = 2022$

$$\Rightarrow P(\alpha) - 2015 = q(\alpha)(\alpha - x_1)(\alpha - x_2)(\alpha - x_3)(\alpha - x_4)$$

$$\Rightarrow 7 = q(\alpha)(\alpha - x_1)(\alpha - x_2)(\alpha - x_3)(\alpha - x_4)$$

Impossible since 7 is prime

10. A

Sol. α is a root of $x^2 - 3x + 1 = 0$

$$\Rightarrow \alpha^2 - 3\alpha + 1 = 0, \text{ yields}$$

$$\alpha^2 + 1 = 3\alpha (\alpha \neq 0)$$

$$\text{Now } \frac{2\alpha^5 - 5\alpha^4 + 2\alpha^3 - 8\alpha^2}{\alpha^2 + 1} = \frac{(\alpha^2 - 3\alpha + 1)(2\alpha^3 + \alpha^2 + 3\alpha) - 3\alpha}{\alpha^2 + 1}$$

$$= \frac{-3\alpha}{\alpha^2 + 1} = \frac{-3\alpha}{3\alpha} = -1$$

11. D

Sol. The points of intersection lies on $(x^2 - y^2 - a^2) + \lambda(x^2 - y) = 0$

It represents a circle if $\lambda = -2$

$$\therefore \text{equation of circle is } x^2 + (y - 1)^2 = 1 - a^2$$

$$\Rightarrow 1 - a^2 > 0 \Rightarrow a \in (-1, 1)$$

But both curves will intersect in real points if $y^2 - y + a^2 = 0$ for some real y

$$\text{i.e. } a \in \left(-\frac{1}{2}, \frac{1}{2}\right).$$

12. A

Sol. All 8 tails | T | T | T | T | T | T | T | T H H $\Rightarrow 1$ way

Exactly 7 tails | T | T | T | T | T | T | T H H $\Rightarrow {}^7C_1 = 7$ ways

7 gaps and one to be selected for a head

Exactly 6 tails | T | T | T | T | T | T H H $\Rightarrow {}^6C_2 = 15$ ways

6 gaps and 2 to be selected for heads

Exactly 5 tails | T | T | T | T | T H H $\Rightarrow {}^5C_3 = 10$ ways

5 gaps and 3 to be selected for heads

Exactly 4 tails | T | T | T | T H H $\Rightarrow {}^4C_4 = 1$ ways

Total ways = 34

$$\therefore \text{Required probability} = \frac{34}{1024} \cdot \frac{17}{512} = \frac{p}{512} \Rightarrow p = 17.$$

13. A

Sol. $\int_0^a (c - (8x - 27x^3)) dx = \int_a^b ((8x - 27x^3) - c) dx$

$$O = 4b^2 - 27b^4 - bc$$

$$O = 4b^2 - \frac{27}{4} - b(8b - 27b^3)$$

$$b^4 \left(\frac{81}{4} b^2 - 4 \right) = 0$$

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$$b > 0 \quad b^2 = \frac{4^2}{81} \quad b = \frac{4}{81}$$

$$C = 8b - 27b^3$$

$$= \frac{32}{27}$$

14. D

Sol. We can write the 2nd equation in form $2\left(\frac{1}{q}\right)^2 - 3\left(\frac{1}{q}\right) - 1 = 0$

then it is found that p and $\frac{1}{q}$ are both roots of equation $2x^2 - 3x - 1 = 0$, ($pq \neq 1$) from viete

$$p + \frac{1}{q} = \frac{3}{2} \text{ and } \frac{p}{q} = \frac{-1}{2}$$

$$\text{so, } \frac{1+2p+pq}{q} = \frac{1}{q} + p + \frac{2p}{q} = \frac{3}{2} - \frac{2}{2} = \frac{3}{2} - 1 = \frac{1}{2}$$

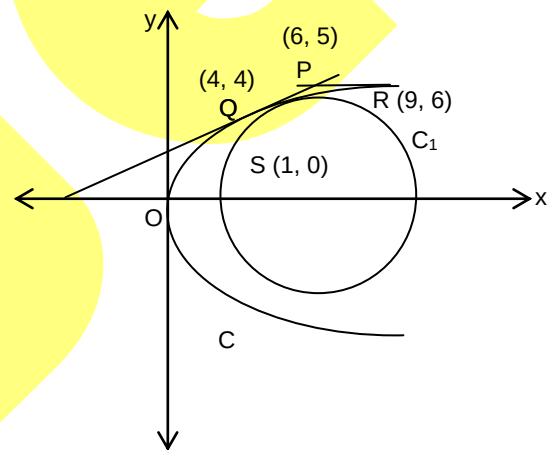
15. A

16. B

Sol. (15 to 16)

Equation of tangent of slope m to $y^2 = 4x$ is

$$y = mx + \frac{1}{m} \quad \dots(1)$$



(i) As (1) passes through $P(6, 5)$, so $5 = 6m = \frac{1}{m}$

$$\Rightarrow 6m^2 - m + 1 = 0 \Rightarrow m = \frac{1}{2} \text{ or } m = \frac{1}{3}$$

Points of contact are $\left(\frac{1}{m_1^2}, \frac{2}{m_1}\right)$ and $\left(\frac{1}{m_2^2}, \frac{2}{m_2}\right)$

Hence $P(4, 4)$ and $Q(9, 6)$

$$\text{Area of } \triangle PQR = \frac{1}{2} \begin{vmatrix} 6 & 5 & 1 \\ 4 & 4 & 1 \\ 9 & 6 & 1 \end{vmatrix} = \frac{1}{2} \Rightarrow (A)$$

(ii) $y = \frac{1}{2}x + 2 \Rightarrow x - 2y + 4 = 0 \quad \dots(2)$

$$\text{And } y = \frac{1}{3}x + 3 \Rightarrow x - 3y + 9 = 0$$

Now equation of circle C_2 touching $x - 3y + 9 = 0$ at $(9, 6)$, is

$$(x - 9)^2 + (y - 6)^2 + \lambda(x - 3y + 9) = 0$$

As above circle passes through $(1, 0)$, so

$$64 + 36 + 10\lambda = 0 \Rightarrow \lambda = -10$$

$$\text{Circle } C_2: x^2 + y^2 - 28x + 18y - 127 = 0 \quad \dots(3)$$

Radius of C_2 is

$$r_2^2 = 196 + 81 - 27 = 277 - 27 = 250 \Rightarrow r_2 = 5\sqrt{10} \Rightarrow (B)$$

17. A

Sol. If $f(x)$ touches x -axis at only one irrational point, then $f(x) = (x - \alpha)^2 g(x)$, where α is irrational

$\Rightarrow$ coefficient of $f(x)$ can't be rational

$\Rightarrow$ for $f(x)$ with rational coefficient, then point of touching is rational

18. B

Sol. $f(x) = (x - \alpha)^3 (x - \beta)^3$

$$\Rightarrow f'(x) = f(x - \alpha)^2 (x - \beta)^2 (2x - (\alpha + \beta))$$

$\Rightarrow f''(x)$ has roots α, β and a root between $\left(\alpha, \frac{\alpha + \beta}{2}\right)$ and $\left(\frac{\alpha + \beta}{2}, \beta\right)$

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PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 3

QP CODE: 100215

PAPER - 1

Time Allotted: 3 Hours

Maximum Marks: 204

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. All the section can be filled in **PART-A & B** of OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-04)** – Contains **Four (04)** multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (ii) **PART-A (05–12)** contains Eight (8) Multiple Choice Questions which have **One or More Than One Correct** answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: –2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–2 marks**, as a wrong option is also darkened.
- (iii) **Part-B (1 – 8)** contains Eight (08) Numerical based questions, the answer of which maybe positive or negative numbers or decimals **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

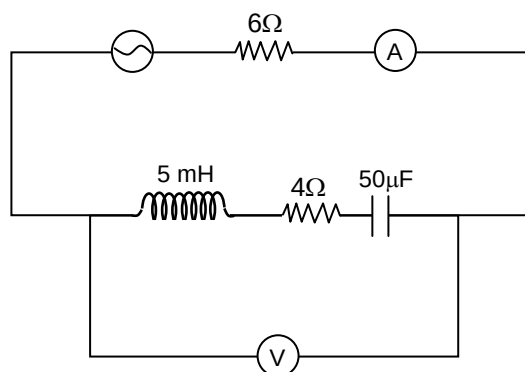
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SECTION – I : PHYSICS

PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A particle moves along the y-axis of a coordinate system, with a force component $F_y = (2 \text{ N/m}^3)y^3$ acting on it. As the particle moves from the origin to $y = 3 \text{ m}$, how much work is done on it by the force?
 (A) 0 J (B) 40.5 J
 (C) -40.5 J (D) 162 J
2. In the circuit shown in given below figure, the AC source gives a voltage $V = 20 \cos (2000 t)$. Neglecting source resistance, the voltmeter and ammeter reading will be: (approximately)



- (A) 0 V, 0.47 A (B) 1.68 V, 0.47 A
 (C) 0 V, 1.4 A (D) 5.6 V, 1.4 A
3. Two coils X and Y are placed in a circuit such that a current change of 3 A in coil X causes the change in magnetic flux by 1.2 Wb in coil Y. The value of mutual inductance of the coil is:
 (A) 0.2 H (B) 0.4 H
 (C) 0.6 H (D) 3.6 H
 4. The temperature gradient in a rod of 0.5 m length is 80°C/m . If the temperature of hotter end of the rod is 30°C , then the temperature of the cooler end is:
 (A) 40°C (B) -10°C
 (C) 10°C (D) 0°C

Space For Rough Work

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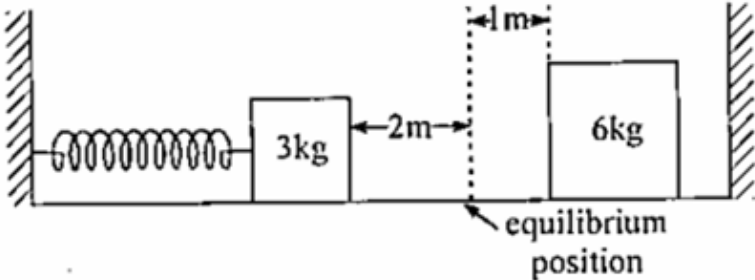
PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. Two particles A and B, each carrying a charge Q , are held fixed with a separation d between them. A particle C having a mass m and charge $-q$, is kept at the middle point of line AB. It is displaced through a distance x perpendicular to AB. Assume $x \ll d$. Then choose the correct option(s).
- (A) Force experienced by C is proportional to x
 (B) Force experienced by C is proportional to d
- (C) Particle C may execute SHM with time period $\left(\frac{m\pi^3 \epsilon_0 d^3}{qQ} \right)^{1/2}$
- (D) Particle C may execute SHM with time period $\left(\frac{m\pi^2 \epsilon_0 d^2}{qQ} \right)^{1/2}$
6. The equivalent resistance of a group of resistances is R . If another resistance is connected in parallel to the group, its new equivalent becomes R_1 and if it is connected in series to the group, its new equivalent becomes R_2 we have:
- (A) $R_1 > R$ (B) $R_1 < R$
 (C) $R_2 > R$ (D) $R_2 < R$
7. Radiations of monochromatic waves of wavelength 400 nm are made incident on the surface of metals Zn, Fe and Ni of work functions 3.4 eV, 4.8 eV and 5.9 eV respectively. Which of the following is (are) correct?
- (A) Maximum KE associated with photoelectrons from the surface of any metal is 0.3 eV.
 (B) No photoelectron are emitted from the surface of Ni.
 (C) If the wavelength of source of radiation is doubled then KE of photoelectrons is also doubled.
 (D) Photoelectrons will be emitted from the surface of all the three metals if the wavelength of incident radiations < 200 nm.
8. A diverging lens of focal length f_1 is placed in front of and coaxially with a concave mirror of focal length f_2 . Their separation is d . A parallel beam of light incident on the lens returns as a parallel beam from the arrangement.
- (A) The beam diameters of the incident and reflected beam must be the same.
 (B) $d = 2 |f_2| - |f_1|$
 (C) $d = 2 |f_1| - |f_2|$
 (D) If the entire arrangement is immersed in water, the conditions will remain unaltered.
-

Space For Rough Work

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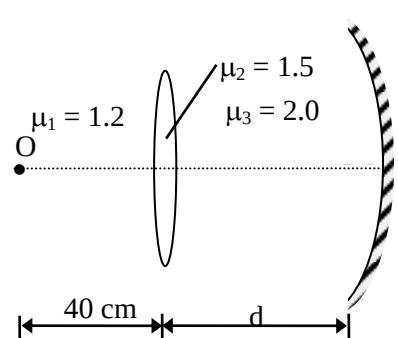
9. A positive charge is passing through an electromagnetic field in which $\vec{E}$ & $\vec{B}$ are directed towards y-axis and z-axis respectively. If a charge particle passes through the region undeviated, then its velocity may be represented by (here a, b & c are constant)
- (A) $\vec{v} = \frac{E}{B} \hat{i} + a \hat{j}$ (B) $\vec{v} = \frac{E}{B} \hat{i} + b \hat{k}$
 (C) $\vec{v} = \frac{E}{B} \hat{i} + c \hat{j}$ (D) $\vec{v} = \frac{E}{B} \hat{i}$
10. An ideal gas undergoes a process such that $P \propto \frac{1}{T}$. The molar heat capacity of this process is 33.24 J/mol K. For one mole, choose correct options. ($R = 8.31$)
- (A) The work done by the gas is $2R\Delta T$ (B) Degree of freedom of the gas is 4.
 (C) Degree of freedom of the gas is 3. (D) $\gamma = \left(\frac{C_P}{C_V} \right)$
11. Two blocks of masses 3 kg and 6 kg rest on a horizontal smooth surface. The 3 kg block is attached to a spring with a force constant $k = 900 \text{ Nm}^{-1}$ which is compressed 2 m from the equilibrium position as shown in given below figure. The 6 kg mass is at rest at 1 m from mean position. 3 kg mass strikes the 6 kg mass and the two stick together:
- 
- (A) Velocity of the combined masses immediately after the collision is 10 ms^{-1}
 (B) Velocity of the combined masses immediately after the collision is 5 ms^{-1}
 (C) Amplitude of the resulting oscillation is $\sqrt{2} \text{ m}$.
 (D) Amplitude of the resulting oscillation is $\sqrt{5/2} \text{ m}$.
12. If x , v and a denote displacement, velocity and acceleration of a particle executing simple harmonic motion of time period T , then, which of the following does not change with time?
- (A) $aT + 2\pi v$ (B) $\frac{aT}{v}$
 (C) $a^2T^2 + 4\pi^2v^2$ (D) $\frac{aT}{x}$

Space For Rough Work

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PART – B (Maximum Marks: 24)**(Numerical Type)**

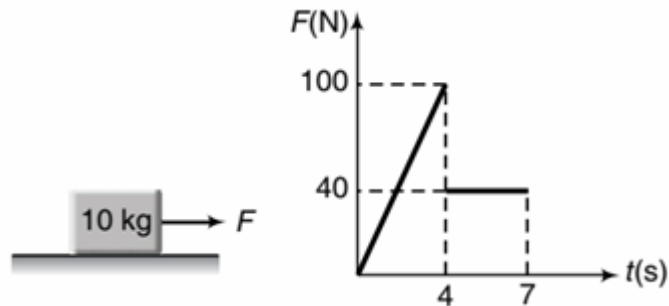
This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. A gas containing hydrogen like ions with atomic number Z , emits photons in transition $n + 2 \rightarrow n$, where $n = Z$. These photons fall on a metallic plate and eject electrons having minimum de-Broglie wavelength λ of $5000 \text{ \AA}$. Find the value of ' Z ' (nearest integer value) if the work function of metal is 4.2 eV .
2. Potential energy of a particle moving along x -axis is given by $U = \frac{x^3}{3} - \frac{9x^2}{2} + 20x$. Find out position (value of x) of stable equilibrium state.
3. The figure shows an arrangement of an equi-convex lens and a concave mirror. A point object O is placed on the principal axis at a distance 40 cm from the lens such that the final image is also formed at the position of the object. If the radius of curvature of the concave mirror is 80 cm . The distance d comes out to be $10n \text{ cm}$. Find n . The focal length of the lens in air is 20 cm .
 
4. A ring of mass 3.15 kg is rolling without slipping with linear velocity 1 m/sec on a smooth horizontal surface. A rod of same mass is fitted along its one diameter. Find total kinetic energy of the system (in J).
5. A source of sound is travelling towards a stationary observer. The frequency of sound heard by the observer is three times of the original frequency. The velocity of sound $v = 332.25 \text{ m/sec}$.
6. A light inextensible thread is wound round a solid cylindrical reel of mass $m = 1.5 \text{ kg}$ and radius $a = 10 \text{ cm}$. The end of the string is held fixed and the reel is allowed to fall so that the thread unwinds. If the axis of the reel remains horizontal and tension is 20 N , value of ' n ' is ($g = 10 \text{ m/sec}^2$)

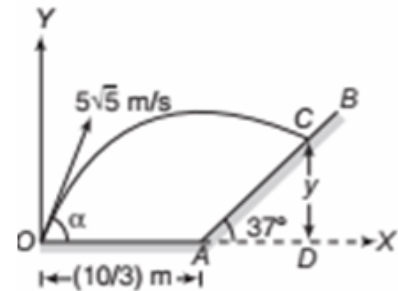
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7. The 10 kg block is resting on the horizontal surface when the force F is applied to it for 7s. The variation of F with time is shown. Calculate the maximum velocity reached by the block during which the block is in motion. The coefficients of static and kinetic friction are both 0.50. ($g = 9.8 \text{ m/s}^2$)



8. A particle is projected from point O on the ground with velocity $u = 5\sqrt{5} \text{ m/s}$ at angle $\alpha = \tan^{-1}(0.5)$. It strikes at a point C on a fixed smooth plane AB having inclination of 37° with horizontal as shown in figure. If the particle does not rebound, calculate maximum height from the ground to which the particle rises. ($g = 10 \text{ m/s}^2$).



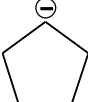
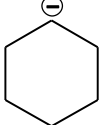
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SECTION – II : CHEMISTRY

PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

- Under which condition, the internal energy of a substance may be zero?
 (A) -273°C , 0.01 atm (B) 273°C , 1 atm
 (C) 298 K, 10^{-4} atm (D) None of these
- The strongest base out of the following is
 (A)  (B) 
 (C)  (D) 
- At constant volume, one mole of an ideal monoatomic gas was heated from 1400 to 1420 K.
 $\left[C_v = \frac{3R}{2}, R = 8.3 \text{ JK}^{-1} \text{ mol}^{-1} \right]$
 Choose the incorrect statement
 (A) ΔU or $\Delta E = 249 \text{ J mol}^{-1}$ (B) $Q = 249 \text{ J mol}^{-1}$
 (C) Change in kinetic energy = 249 J mol^{-1} (D) Entropy change = $249 \text{ J K}^{-1} \text{ mol}^{-1}$
- The product of which reaction can form a conjugated diene when it is treated with alcoholic KOH?
 (A) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH} \xrightarrow{\text{Cl}_2/\text{CCl}_4}$ (B) $\text{CH}_3\text{CH}=\text{CHCH}_3 \xrightarrow{\text{HCl}}$
 (C) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 \xrightarrow{\text{Cl}_2/\text{CCl}_4}$ (D) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 \xrightarrow{\text{HCl}}$

PART – A (Maximum Marks: 32)

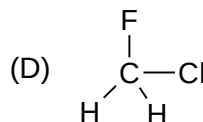
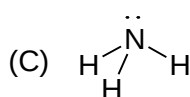
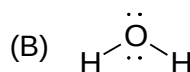
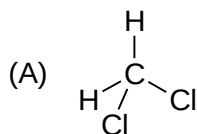
This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

- Which property(s) of NH_3 and NF_3 is/are given in correct order?
 (A) Dipole moment: $\text{NH}_3 > \text{NF}_3$
 (B) Bond angle: $\text{NH}_3 > \text{NF}_3$
 (C) Lewis basicity: $\text{NH}_3 > \text{NF}_3$
 (D) Number of hydrogen bonds formed with H_2O : $\text{NH}_3 > \text{NF}_3$

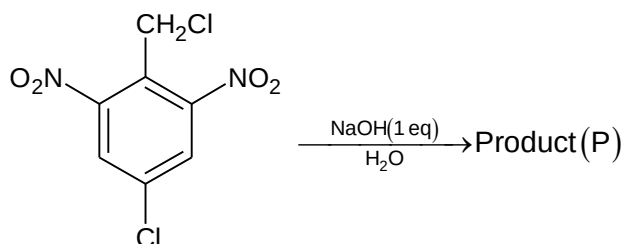
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6. Crystal field theory is not applied on complex(es) like
 (A) $[\text{Ni}(\text{CO})_4]$ (B) $[\text{Fe}(\text{CO})_5]$
 (C) $[\text{Zn}(\text{NH}_3)_4]\text{CO}_3$ (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
7. The boiling point of water in which aqueous solution(s) is/are higher than 100°C ?
 (A) $0.2 \text{ M Al}_2(\text{SO}_4)_3$ (B) $1 \text{ M CH}_3\text{OH}$ in water
 (C) 0.1 M NaCl (D) 1 M NaCONH_2
8. The first ionization enthalpies of two elements P and Q respectively are 8.1 and 10.2 eV. The electron gain enthalpies of P and Q are respectively -3.1 and -6.6 eV respectively. The percentage ionic character of the covalent bond P – Q is given as
 $\% \text{ ionic character} = 16(\text{END}) + 3.5(\text{END})^2$
 Where END is the electronegativity difference in **Mulliken scale** between P and Q.
 Choose correct statements
 (A) The electronegativity of P is 1.05
 (B) Q is more electronegative than P
 (C) percentage covalent character of P – Q bond is 80.5
 (D) P – Q bond is purely covalent with 100% covalent character
9. In which of the following molecule(s) the bond containing H and the central atom contains more than 25% s-orbital character?



10.



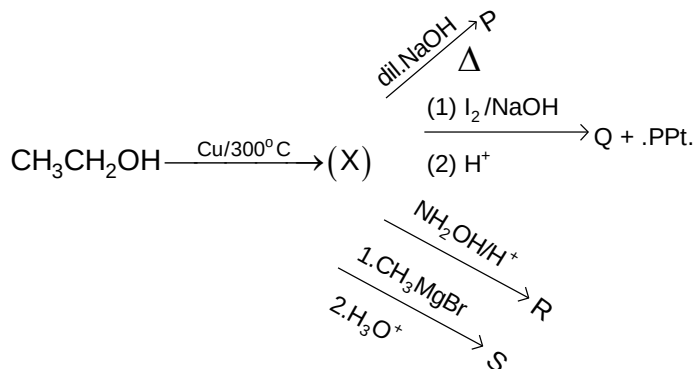
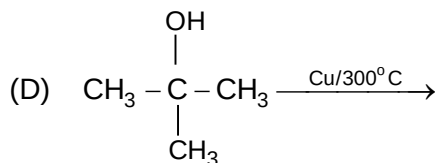
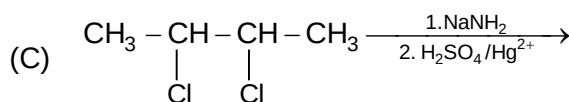
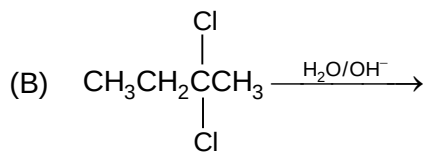
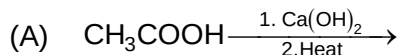
Correct statement(s) regarding product(P) is

- (A) it is an alcohol
 (B) it is a phenol
 (C) the longer C – Cl bond breaks in the reaction
 (D) the NO_2 groups favours substitution of benzylic chloride

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11. Which of the following reactions produce(s) a ketone?



Correct statement(s) regarding products of above reaction(s) is/are

- (A) P, Q and R contain pi-bonds (B) P and R exhibit geometrical isomerism
(C) R and S show optical isomerism (D) The ppt. is yellow colour

PART – B (Maximum Marks: 24)

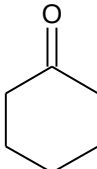
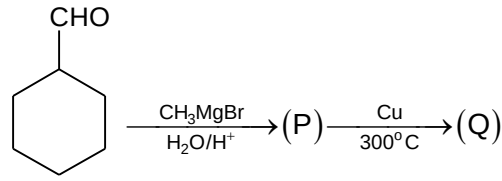
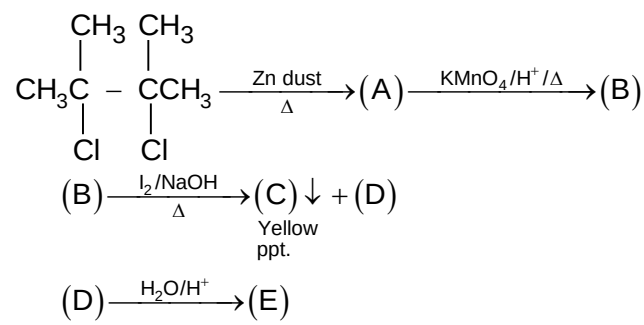
(Numerical Type)

This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

1. A dicarboxylic acid(P) forms a white precipitate when reacts with aqueous $\text{Ca}(\text{OH})_2$ solution. If (P) is the simplest dicarboxylic acid. How many moles of gases are evolved when one mole of (P) is heated at 85°C .

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2. Two moles of  reacts with dilute NaOH to produce one mole of product(P). Heating of (P) forms (Q). Reaction of (Q) with Zn-Hg/Conc.HCl(excess) forms product(R).
 If x = Number of C – C sigma bonds present in (R)
 y = Number of C = C bonds present in (R)
 z = Number of 3° -carbon atoms present in (R), then the value of $(x + y + z)$ is
3. A vessel contains equal number of moles of CH_4 and SO_2 gases at a particular temperature. If the ratio of rates of effusion of CH_4 to SO_2 through an orifice made on the vessel, is expressed as $x : y$, the value of $(x + y)$ is:
4. The $\text{p}^{\text{K}}_{\text{a}}$ of the α -COOH, α -NH₂ and side chain COOH groups of an α -amino acid are respectively 2.1, 9.82 and 3.86. What is the pl of the amino acid?
5. 
- If the number of hydrogen atom(s) is/are present in a molecule of (Q) is X . What is $\frac{X}{10}$?
6. 
- If the number of oxygen atom(s) are present in (E) is y , what is the value of $\frac{y}{10}$?
7. The depression in freezing point of an aqueous solution of NaCl is 18.6. How many moles of NaCl is present in 200 g of solvent(water)?
 $[K_f \text{ of water} = 1.86 \text{ K kg mol}^{-1}]$
 [Assume complete dissociation of NaCl]
8. A metal crystallized in b.c.c unit cell. What will be the diameter of the metal atoms in pm unit if the edge-length of the unit cell is $\sqrt{3}$ pm?

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SECTION – III : MATHEMATICS

PART – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. If $f(x) = x^4 \tan(x^3) - x \ln(1 + x^2)$, then the value of $\frac{d^4(f(x))}{dx^4}$ at $x = 0$ is
 (A) 0 (B) 6
 (C) 12 (D) 24
2. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, then incorrect statement is
 (A) $\vec{b}$ is in the plane of $\vec{a}$ and $\vec{c}$ (B) $\vec{b}$ is parallel to $\vec{a} \times \vec{c}$
 (C) angle between $\vec{a}$ and $\vec{c}$ is 30° (D) angle between $\vec{a}$ and $\vec{c}$ is 150°
3. Let a, b, c are distinct real numbers, not equal to one. If $ax + y + z = 0$, $x + by + z = 0$ and $x + y + cz = 0$ has a non-trivial solution, then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to:
 (A) -1 (B) 1
 (C) zero (D) none of these
4. The volume enclosed by the planes $|3x - 4| + |2y - 3| + |z + 4| = 3$ is
 (A) 4 (B) 6
 (C) 8 (D) None of these

PART – A (Maximum Marks: 32)

This section contains **EIGHT (08)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. The solution of the differential equation $\frac{d^2y}{dx^2} = \sin 3x + e^x + x^2$ when $y_1(0) = 1$ and $y(0) = 0$ is
 (A) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$ (B) $\frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x$
 (C) $\frac{-\cos 3x}{3} + e^x + \frac{x^4}{12} + \frac{1}{3}x + 1$ (D) none of these

Space For Rough Work

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6. If A and B are two events such that $P(A) = 3/4$ and $P(B) = 5/8$, then
 (A) $P(A \cup B) \geq 3/4$ (B) $P(A' \cap B) \leq 1/4$
 (C) $3/8 \leq P(A \cap B) \leq 5/8$ (D) $3/8 \leq P(A \cap B) \leq 5/8$
7. If $f(x) = 2x - \sin x$ and $g(x) = \sqrt[3]{x}$, then
 (A) range of gof is $\mathbb{R}$ (B) gof is one – one
 (C) both f and g are one – one (D) both f and g are onto
8. $\lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x})$ equals
 (A) $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$ (B) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$
 (C) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$ (D) $\lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x^2 + 2x}}}$
9. The solution/s of the equation $9\cos^{12} x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x - 2\cos 2x$ is/are
 (A) $x = n\pi + \frac{\pi}{2}, n \in \mathbb{I}$ (B) $x = n\pi \pm \cos^{-1} \sqrt[4]{\frac{2}{3}}, n \in \mathbb{I}$
 (C) $x = n\pi \pm \cos^{-1} \sqrt{\frac{2}{3}}, n \in \mathbb{I}$ (D) $x = n\pi, n \in \mathbb{I}$
10. If a straight line through a point $P(\alpha, 2), \alpha \neq 0$, meets the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at A and D and x – and y – axes at B and C respectively such that $PA \times PD = PB \times PC$, then ' α ' can lie in the interval
 (A) $[7, \infty)$ (B) $(-12, -8)$
 (C) $(-5, 0)$ (D) $(10, \infty)$
11. Tangent drawn at point $P(1, 3)$ of a parabola intersects its tangent at vertex at $M(-1, 5)$ and cuts the axis of parabola at T. If $R(-5, 5)$ is a point on SP; where S is focus of the parabola, then
 (A) slope of axis is -3
 (B) radius of circumcircle of ΔSMP is $\sqrt{\frac{5}{2}}$ units
 (C) $(ST)^2 - (SM)^2 = (PM)^2$
 (D) tangent cuts the axis of parabola at $T(-3, 7)$

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12. Suppose that $f(x)$ is a differentiable invertible function, $f(x) \neq 0$ and $h(x) = \int_1^x f(t)dt$. Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$,
 Let $G(x) = x^2g(x) - xh(g(x)) \quad \forall x \in \mathbb{R}$. Which of the following are correct:
 (A) $G'(1) = 2$ (B) $G'(1) = 3$
 (C) $G''(1) = 2$ (D) $G''(1) = 3$

PART – B (Maximum Marks: 24)

(Numerical Type)

This section contains **Eight (08)** Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **TWO** decimal places (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

- Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$ and $x = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$. If $P = AXB$, $Q = BX^T A$ and $\text{Tr}((PQ)^{10}) = a^{10} + b^{10}$, where $a < b$, then find the value of $b - 2a$ _____
 (Tr (M) denotes trace of Matrix M and M^T denotes transpose of matrix M)
- Let $\beta = \int_0^1 \frac{2+3x+4x^2}{2\sqrt{x^2+x+1}} dx$, then find the value of β^2 .
- Suppose that the circle $x^2 + y^2 - 4ax - 2ay + 4a^2 = 0$ ($a > 0$) touches the lines $y = mx$ at A and B such that $AB = 3$. Then the value of $\frac{4a}{\sqrt{5}}$ is _____
- If $|z - 3| = \text{Re } z$, $|w - 3| = \text{Re } w$ and $\arg(z - w) = \frac{\pi}{4}$, then find the value of $\text{Im}(z + w)$.
- The smallest value of 'a' for which the inequality $(a-1)x^2 - (a+1)x + (a-1) \geq 0$ holds for all $x \geq 2$ is k then the value of $\frac{3k}{14}$ is _____
- If roots of $x^3 + ax^2 + bx - \frac{1}{9} = 0$ are in A.P. the $\frac{121}{16}(2a^3 - 9ab + 1)$ is _____.
- If $\sum_{k=0}^{100} \left(\frac{k}{k+1} \right)^{100} C_k = \frac{a(2^{100}) + b}{c}$ where $a, b, c \in \mathbb{N}$, then find the least value of $\frac{(a+b+c)}{670}$.
- A straight line cuts the x-axis at point $A(1,0)$, and y-axis at point B, such that $\angle OAB = \alpha \left(\alpha > \frac{\pi}{4} \right)$. C is middle point of AB and B' is a mirror image of point B with respect to line OC and C' is a mirror image of point C with respect to line BB', then the ratio of the areas of triangles ABB' and BB'C' is equal to _____

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Q. P. Code: 100215

Answers**SECTION – I : PHYSICS****PART – A**

- | | | | |
|-------|---------|--------|--------|
| 1. B | 2. D | 3. B | 4. B |
| 5. AC | 6. BC | 7. BD | 8. AB |
| 9. BD | 10. ABD | 11. AC | 12. CD |

PART – B

- | | | | |
|-----------|---------|--------|---------|
| 1. 2 | 2. 5 | 3. 3 | 4. 5.25 |
| 5. 221.50 | 6. 0.25 | 7. 5.2 | 8. 4.45 |

SECTION – II : CHEMISTRY**PART – A**

- | | | | |
|---------|---------|---------|---------|
| 1. D | 2. D | 3. D | 4. C |
| 5. ABCD | 6. AB | 7. ACD | 8. BC |
| 9. AD | 10. ACD | 11. ABC | 12. ABD |

PART – B

- | | | | |
|--------|--------|------|---------|
| 1. 2 | 2. 15 | 3. 3 | 4. 2.98 |
| 5. 1.4 | 6. 0.2 | 7. 1 | 8. 1.5 |

SECTION – III : MATHEMATICS**PART – A**

- | | | | |
|-------|---------|----------|--------|
| 1. A | 2. B | 3. B | 4. B |
| 5. A | 6. ABCD | 7. ABCD | 8. BC |
| 9. AB | 10. ABD | 11. ABCD | 12. AD |

PART – B

- | | | | |
|---------|----------|---------|---------|
| 1. 3 | 2. 3 | 3. 3 | 4. 6 |
| 5. 0.50 | 6. 30.25 | 7. 0.30 | 8. 2.00 |

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Answers & Solutions

SECTION – I : PHYSICS

PART – A

1. **B**

Sol. $W = \int_0^3 F dy = 2 \int_0^3 y^3 dy = \frac{1}{2} [y^4]_0^3 = \frac{81}{2} = 40.5$

2. **D**

Sol. Circuit impedance for series RLC circuit is given as

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$R = 10\Omega, X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega.$$

$$X_C = \frac{1}{\omega C} = \frac{10^6}{2000 \times 5}$$

As $X_L = X_C$ so circuit is in resonance and current in circuit is given as

$$i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2A$$

$$\Rightarrow i_{rms} = \frac{2}{\sqrt{2}} = 1.41A$$

Voltmeter reading is equal to the voltage across the resistance as at resonance phasor voltage of capacitor and inductor gets cancelled out

$$V_{rms} = i_{rms} R = 4 \times 1.41 = 5.64 V$$

3. **B**

Sol. Flux change in coil Y is related to current change in coil X as

$$\phi = Mi$$

$$\Rightarrow d\phi = M di$$

$$\Rightarrow M = \frac{d\phi}{di}$$

$$\Rightarrow M = \frac{1.2}{3} = 0.4 H$$

4. **B**

Sol. Temperature gradient

$$\frac{dT}{dx} = 80^\circ C/m$$

$$\Rightarrow \int_{30}^T dT = -80 \int_0^{0.5} dx$$

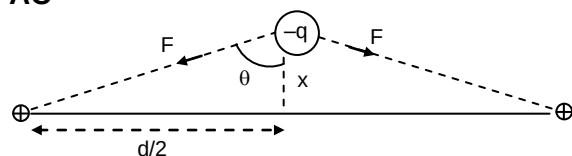
$$\Rightarrow T - 30 = -80(0.5)$$

$$\Rightarrow T - 30 = -40$$

$$\Rightarrow T = -10^\circ C$$

5. **AC**

Sol.



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$$F_{\text{net}} = 2F \cos \theta = 2 \frac{k Q q x}{\left(\frac{d^2}{4} + x^2\right)^{\frac{3}{2}}}$$

If $x \ll d$

$$\text{Acceleration } \bar{a} = -\frac{2 k Q q}{m \left(\frac{d}{2}\right)^3} \bar{x}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{md^3}{16 Q q}} = \left(\frac{m\pi^3 \epsilon_0 d^3}{q Q}\right)^{\frac{1}{2}}$$

6. BC

Sol. Suppose resistance connected is r .

$$R_2 = R + r > R$$

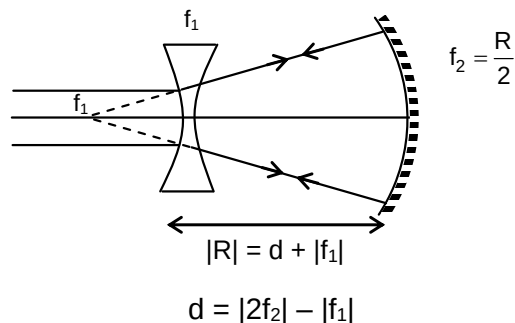
7. BD

$$\text{Sol. } \Delta E = \frac{hc}{\lambda} = \frac{12375}{4000} = 3.09375 \approx 3.1 \text{ eV.}$$

For photoelectric effect, ΔE should be greater than work function.

8. AB

Sol.



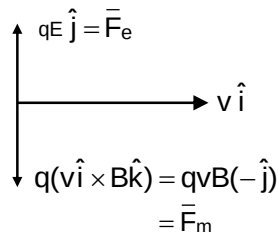
9. BD

$$\text{Sol. } q\bar{E} + q(\bar{v} \times \bar{B}) = 0$$

For, $\bar{E}$, $\bar{v}$, $\bar{B}$ mutually perpendicular.

$$qE = -qvB$$

$$v = \frac{E}{B}$$



$$q\bar{E} \hat{j} + q[(v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \times B(\hat{k})] = 0$$

$$\Rightarrow q(E - v_x B) \hat{j} + v_y B q \hat{i} = 0$$

$$\Rightarrow v_x = \frac{E}{B} \text{ and } v_y = 0$$

$$\therefore \bar{v} = \frac{E}{B} \hat{i} + b \hat{k}$$

10. ABD

Sol. Given $PT = \text{constant}$

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and $PV = nRT$

$\Rightarrow P^2V = \text{constant}$

$\Rightarrow PV^{1/2} = k$

From first law of thermodynamics

$$\Delta Q = \Delta U + \Delta W$$

$\Rightarrow C = C_V + \left(\frac{R}{1-\gamma} \right)$

$$C = C_V + 2R$$

$$33.24 = \frac{R}{\gamma-1} + 2R$$

$\Rightarrow \gamma = 1.5$

and $\gamma = 1 + \frac{2}{f}$

$\Rightarrow f = 4.$

11. **AC**

Sol. Just before collision

$$v = \omega \sqrt{A^2 - x^2}$$

$$v = \sqrt{\frac{k}{m}} \sqrt{A^2 - x^2} = \sqrt{\frac{900}{3}} [\sqrt{(4-1)}] = 30$$

COM

$$3 \times 30 = 9 v'$$

$\Rightarrow v = 10$ (velocity just after collision)

When 3 kg mass is released the amplitude of its oscillation is 2 m and at a distance 1 m from the equilibrium position we can find the speed of it using the relation $v = [(k/m)(A^2 - x^2)]^{1/2}$ then by conservation of momentum we can find the resulting speed of the combined mass and the new amplitude using the above relation which gives options (A) and (C) are correct.

12. **CD**

Sol. $a = -\omega^2 x, v = \omega \sqrt{A^2 - x^2}$

$$a^2 T^2 + 4\pi^2 v^2 = \omega^4 x^2 T^2 + 4\pi^2 \omega^2 (A^2 - x^2)$$

$$= \frac{4\pi^2}{T^2} \omega^2 x^2 T^2 + 4\pi^2 \omega^2 (A^2 - x^2)$$

$$= 4\pi^2 \omega^2 A^2 = \text{constant}$$

$$\frac{aT}{x} = -\frac{\omega^2 x T}{x} = -\omega^2 T = \text{constant.}$$

PART - B

1. **2**

Sol. $\Delta E = 4.2 + \frac{12375}{5000} = 6.675$

$$\therefore \Delta E = 13.6 Z^2 \left(\frac{1}{n^2} - \frac{1}{(n+2)^2} \right) = 13.6 \left[1 - \frac{1}{\left(1 + \frac{2}{Z} \right)^2} \right]$$

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$$\Rightarrow \frac{1}{2} \approx \left[1 - \frac{1}{\left(1 + \frac{2}{Z}\right)^2} \right] \Rightarrow \frac{1}{\left(1 + \frac{2}{Z}\right)^2} = \frac{1}{2} \Rightarrow 1 + \frac{2}{Z} = \sqrt{2}$$

$$\Rightarrow \frac{2}{Z} = 0.41 \Rightarrow Z = \frac{2}{0.41} \approx 5.$$

2. **5**

Sol. At stable equilibrium, U is minimum.

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} > 0$$

$$\Rightarrow \frac{1}{dx} \left(\frac{x^3}{3} - \frac{9x^2}{2} + 20x \right) = 0.$$

$$\Rightarrow x^2 - 9x + 20 = 0. \Rightarrow (x - 5)(x - 4) = 0.$$

$x = 5$ and $x = 4$ are points of equilibrium.

And U minimum when $\frac{d^2U}{dx^2} > 0$. i.e. at $x = 5$.

3. **3**

Sol. $\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$

$$\frac{2}{-(80-d)} - \frac{1.2}{-40} = \frac{1.5-1.2}{20} + \frac{2-1.5}{-20}$$

$$\Rightarrow \frac{2}{-(80-d)} + \frac{1.2}{40} = \frac{0.3}{20} + \left(\frac{-0.5}{20} \right)$$

$$\frac{1.2}{40} + \frac{0.2}{20} = \frac{2}{(80-d)}$$

$$\frac{1.6}{40} = \frac{2}{(80-d)}$$

$$80 - d = 50$$

$$d = 30$$

$$\Rightarrow n = 3.$$

4. **5.25**

Sol. $k = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}(mr^2)\left(\frac{v}{r}\right)^2 + \frac{1}{2}\left\{\frac{m(2r)^2}{12}\right\}\left(\frac{v}{r}\right)^2 + \frac{1}{2}mv_{rod}^2$

$$= \frac{1}{2}mv_{cm}^2 + \frac{1}{2}mv_{cm}^2 + \frac{1}{2}mv_{cm}^2 \times \frac{1}{3} + \frac{1}{2}mv_{cm}^2$$

$$= \frac{1}{2} \times 3.15 \times 1 \left[1 + 1 + 1 + \frac{1}{3} \right] = \frac{3.15}{2} \times \frac{10}{3} = 5.25$$

5. **221.50**

Sol. We use

$$n' = n \left(\frac{v}{v - v_s} \right)$$

$$\Rightarrow \frac{n'}{n} = \frac{v}{v - v_s} \quad \text{TG} \sim @bohring_bot$$

$$\Rightarrow \frac{v}{v - v_s} = 3$$

$$\Rightarrow v_s = \frac{2v}{3}.$$

6. **0.25**

Sol. Let T be the tension in the thread and f , the linear acceleration of the reel as it falls.

For the downward translation,

$$(mg - T) = mf \quad \dots(1)$$

For the rotational motion of the reel, angular acceleration is

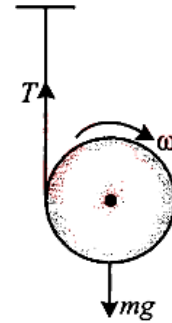
$$Ta = \frac{ma^2}{2} \alpha$$

$$T = \frac{ma\alpha}{2} = \frac{mf}{2} \quad \dots(2)$$

From equation (1) and (2)

$$\Rightarrow 3T = mg$$

$$\therefore T = \frac{mg}{3} = 1.5 \times \frac{10}{3} = 5 \text{ N}.$$

7. **5.2**

Sol. Block will start moving at, $F = \mu mg$

$$\text{or } 25t = (0.5)(10)(9.8) = 49 \text{ N}$$

$$\therefore t = 1.96 \text{ s}$$

Velocity is maximum at the end of 4 second.

$$\therefore \frac{dv}{dt} = \frac{25t - 49}{10} = 2.5t - 4.9$$

$$\therefore \int_0^{v_{\max}} dv = \int_{1.96}^4 (2.5t - 4.9) dt$$

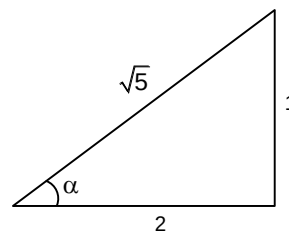
$$\therefore v_{\max} = 5.2 \text{ m/s}$$

8. **4.45**

Sol. Let, (x, y) be the coordinates of point C

$$x = OD = OA + AD$$

$$\therefore x = \frac{10}{3} + y \cot 37^\circ = \frac{10 + 4y}{3} \quad \dots(i)$$



As point C lies on the trajectory of a parabola, we have

$$y = x \tan \alpha - \frac{gx^2}{2u^2} (1 + \tan^2 \alpha) \quad \dots(ii)$$

$$\text{Given that, } \tan \alpha = 0.5 = \frac{1}{2}$$

Solving equations (i) and (ii), we get $x = 5 \text{ m}$ and $y = 1.25 \text{ m}$.

Hence, the coordinates of point C are $(5 \text{ m}, 1.25 \text{ m})$.

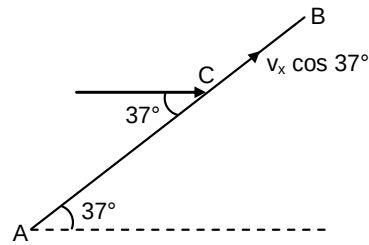
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Let v_y be the vertical component of velocity of the particle just before collision at C.

Using $v_y = u_y + a_y t$, we have

$$v_y = u \sin \alpha - g (x/u \cos \alpha) \quad (\because t = x/u \cos \alpha)$$

$$= \frac{5\sqrt{5}}{\sqrt{5}} - \frac{10 \times 5}{(5\sqrt{5} \times 2 / \sqrt{5})} = 0$$



Thus, at C, the particle has only horizontal component of velocity

$$v_x = u \cos \alpha = 5\sqrt{5} \times \left(\frac{2}{\sqrt{5}}\right) = 10 \text{ m/s}$$

Given, that the particle does not rebound after collision. So, the normal component of velocity (normal to the plane AB) becomes zero. Now, the particle slides up the plane

due to tangential component $v_x \cos 37^\circ = (10) \left(\frac{4}{5}\right) = 8 \text{ m/s}$

Let h be the further height raised by the particle. Then

$$mgh = \frac{1}{2} m (8)^2 \quad \text{or} \quad h = 3.2 \text{ m}$$

Height of the particle from the ground = $y + h$

$$\therefore H = 1.25 + 3.2 = 4.45 \text{ m}$$

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SECTION – II : CHEMISTRY

PART – A

1. D
Sol. The internal energy never becomes zero. Because even at 0 K, the vibration motions of the ideal solids exist.
2. D
Sol. Larger the bond angle of the cycloalkanes, stronger is their basic strength. Because the p-orbital character of carbon atom increases, if the bond angle increases. So due to less s-orbital character, the carbon atom can't strongly attract the negative charge or electron cloud.
3. D
Sol. $\Delta V = 0, \therefore W = 0$
So, $\Delta U = q = nC_v\Delta T = 1 \times \frac{3R}{2} \times (1420 - 1400) = 249 \text{ J mol}^{-1}$
 $\Delta K.E = \frac{3}{2} nR(\Delta T) = \frac{3}{2} \times 1 \times 8.3 \times 20 = 249 \text{ J mol}^{-1}$
 $\Delta S = nC_v \ln \frac{T_2}{T_1} = 0.176 \text{ J K}^{-1} \text{ mol}^{-1}$
4. C
Sol.
$$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 \xrightarrow{\text{Cl}_2/\text{CCl}_4} \text{CH}_3\text{CH}_2\underset{\text{Cl}}{\text{CH}}\underset{\text{Cl}}{\text{CH}}\text{CH}_3 \xrightarrow{\text{Alc.KOH}} \text{CH}_3\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$$
5. ABCD
Sol. In NH_3 , bond pair dipole-moment and lone-pair dipole moment are along same direction. In NF_3 , they are in opposite direction.
6. AB
Sol. CFT based on electrostatic interaction between metal ions and ligands. In the given metal carbonyls, neither metal nor ligand carry charges. Though CO carries partial positive charge, but the metal carries no charge, hence no electrostatic attraction takes place between them.
7. ACD
Sol. The boiling point of water increases by adding non-volatile solutes.
8. BC
Sol. E.N of P = $\frac{8.1+3.1}{5.6} = 2$
E.N of Q = $\frac{10.2+6.6}{5.6} = 3$
% ionic character = $16(3 - 2) + 3.5(3 - 2)^2 = 19.5$
% covalent character = $100 - 19.5 = 80.5$
9. AD
Sol. $\cos\theta = \frac{S}{S-1}$
Where θ = bond angle containing hydrogen and central atom
S = % s-orbital character.
In H_2O , $\theta = 104.5^\circ$

$$\cos 104.5^\circ = -0.25 = \frac{S}{S-1}$$

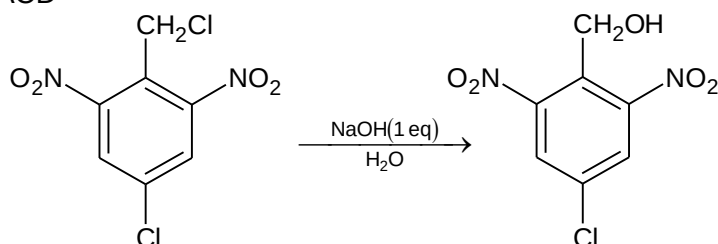
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On solving $S = 0.2$, hence the O – H bond contain 20% s-orbital character and 80% p-orbital character.

$\therefore$ (B) is not correct similarly (C) will not be correct.

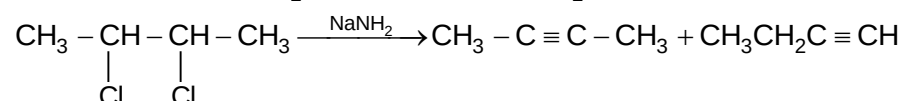
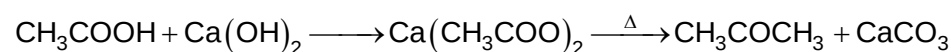
10. ACD

Sol.



11. ABC

Sol.



Both product will give $\text{CH}_3\text{COCH}_2\text{CH}_3$ upon hydrolysis with $\text{H}_2\text{SO}_4/\text{Hg}^{2+}$.

12. ABD

Sol.

X = CH_3CHO

P = $\text{CH}_3 - \text{CH} = \text{CH} - \text{CHO}$

Q = HCOOH , Ppt. = CHI_3

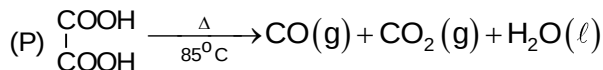
R = $\text{CH}_3 - \text{CH} = \text{NOH}$

S = CH_3COCH_3

PART – B

1. 2

Sol.



2. 15

Sol.

$x = 13, y = 0, z = 2$

3. 3

Sol.

When the moles are same

$$\frac{r_{\text{CH}_4}}{r_{\text{SO}_2}} = \sqrt{\frac{M_{\text{SO}_2}}{M_{\text{CH}_4}}} = \sqrt{\frac{64}{16}} = \sqrt{4} = 2$$

$$\therefore x : y = 2 : 1 \Rightarrow x + y = 3$$

4. 2.98

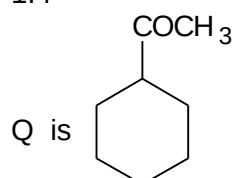
Sol.

$$\text{pI} = \frac{1}{2}(\text{p}^{\text{Ka}_1} + \text{p}^{\text{Ka}_2})$$

$$= \frac{1}{2}(2.1 + 3.86) = 2.98$$

5. 1.4

Sol.



6. 0.2

Sol.

(A) is $(\text{CH}_3)_2\text{C} = \text{C}(\text{CH}_3)_2$

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(B) is CH_3COCH_3

(C) = CHI_3

(D) = CH_3COONa

(E) = CH_3COOH

7. 1

Sol. $\Delta T_f = iK_f m = iK_f n \times \frac{1000}{W}$

or, $18.6 = 2 \times 1.86 \times n \times \frac{1000}{200}$

$\therefore n = 1$

8. 1.5

Sol. $\sqrt{3}a = 4r = 2d$

or, $(\sqrt{3})(\sqrt{3}) = 2d$

or, $d = 3/2 = 1.5 \text{ pm}$

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SECTION – III : MATHEMATICS

PART – A

1. A

Sol. As, $f(x) = x^4 \tan(x^3) - x \ln(1+x^2)$ is odd, $\frac{d^4 f(x)}{dx^4} = 0$ at $x = 0$.

2. B

Sol. On expanding we get $(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})(\vec{c}) = -\vec{b}$

$$\Rightarrow \vec{c} - (\vec{a} \cdot \vec{c})\vec{a} = \vec{b}$$

$\Rightarrow$ (a) is correct and (b) is ruled out.

$$\Rightarrow |\vec{c} - (\vec{a} \cdot \vec{c})\vec{a}|^2 = |\vec{b}|^2$$

If θ is the angle between $\vec{a}$ and $\vec{c}$ then

$$4 + (2\cos\theta)^2 - 2(2\cos\theta)^2 = 1$$

$$\Rightarrow \cos\theta = \pm \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ; 150^\circ$$

3. B

Sol. Since the system has non-trivial solution, therefore $\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$

Applying $R_1 \rightarrow R_1 - R_2, R_2 \rightarrow R_2 - R_3$, we get $\Delta = \begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$

$$\Rightarrow (a-1)(b-1)c - (1-c) + (1-b)(1-c) = 0$$

$$\Rightarrow (1-a)(1-b)c + (1-c) + (1-b)(1-c) = 0$$

Dividing throughout by $(1-a)(1-b)(1-c)$, we get $\frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$

$$\Rightarrow -1 + \frac{1}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$$

$$\Rightarrow \frac{1}{1-c} + \frac{1}{1-a} + \frac{1}{1-b} = 1.$$

4. B

Sol. The volume is invariant under translation.

If $X = x - \frac{4}{3}, Y = y - \frac{3}{2}, Z = z + 4$, then the planes

$$\text{become } 3|X| + 2|Y| + |Z| = 3$$

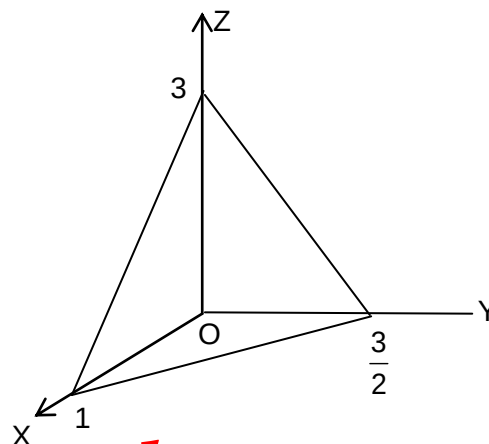
$$\text{The plane } 3X + 2Y + Z = \pm 3, -3X + 2Y + Z = \pm 3.$$

$3X + 2Y - Z = \pm 3, 3X + 2Y - Z = \pm 3$ form an octahedron.

The plane $3X + 2Y + Z = 3$ forms a tetrahedron with coordinate planes in the positive octant of

$$\text{volume } \frac{1}{6} \cdot 1 \cdot \frac{3}{2} \cdot 3 = \frac{3}{4}$$

$$\text{Total volume} = 8 \times \frac{3}{4} = 6$$



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5. A

Sol. Integrating the given differential equation, we have $\frac{dy}{dx} = \frac{-\cos 3x}{3} + e^x + \frac{x^3}{3} + C_1$ but $y_1(0) = 1$
 so $1 = \left(\frac{-1}{3}\right) + 1 + C_1 \Rightarrow C_1 = \frac{1}{3}$.

Again integrating, we get $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x + C_2$

but $y(0) = 0$ so $0 = 0 + 1 + C_2 \Rightarrow C_2 = -1$. Thus $y = \frac{-\sin 3x}{9} + e^x + \frac{x^4}{12} + \frac{1}{3}x - 1$

6. ABCD

Sol. $A \subseteq A \cup B$

$$\Rightarrow P(A) \leq P(A \cup B) \Rightarrow P(A \cup B) \geq \frac{3}{4}$$

Also,

$$\begin{aligned} P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &\geq P(A) + P(B) - 1 \\ &= \frac{3}{4} + \frac{5}{8} - 1 = \frac{3}{8} \end{aligned}$$

Now,

$$A \cap B \subseteq B$$

$$\Rightarrow P(A \cap B) \leq P(B) = \frac{5}{8}$$

$$\therefore \frac{3}{8} \leq P(A \cap B) = \frac{5}{8}$$

and

$$P(A \cap B') = P(A) - P(A \cap B)$$

$$\Rightarrow \frac{3}{4} - \frac{5}{8} \leq P(A \cap B') \leq \frac{3}{4} - \frac{3}{8}$$

$$\Rightarrow \frac{1}{8} \leq P(A \cap B') \leq \frac{3}{8}$$

$$\therefore P(A \cap B) = P(B) = P(A' \cap B) \quad [\text{Using Equation (1)}]$$

$$\Rightarrow \frac{3}{8} \leq P(B) - P(A' \cap B) \leq \frac{5}{8}$$

$$\Rightarrow 0 \leq P(A' \cap B) \leq \frac{1}{4}$$

7. ABCD

Sol. $(\text{gof})(x) = (2x - \sin x)^{1/3}$ Range of $(\text{gof})(x)$ is $\mathbb{R}$.

8. BC

Sol. $\lim_{x \rightarrow \infty} \sqrt{x}(\sqrt{x+1} - \sqrt{x})$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\sqrt{x+1} + \sqrt{x}} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} = \frac{1}{2}$$

$$\text{Option (A): } \lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2} = \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots\right) - x}{x^2} = -\frac{1}{2}$$

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$$\text{Option (B): } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{4 \times \left(\frac{x}{2}\right)^2} = \frac{1}{2}$$

$$\text{Option (C): } \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \lim_{x \rightarrow 0} \frac{(1+x)^{1/2} - (1)^{1/2}}{(1+x) - 1} = \frac{1}{2}$$

$$\text{Option (D): } \lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x^2 + 2x}}} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1 + \sqrt{1 + \frac{2}{x}}}} = \frac{1}{\sqrt{2}}$$

9. AB

$$\text{Sol. } 9\cos^{12} x + \cos^2 2x + 1 - 6\cos^6 x \cos 2x - 6\cos^6 x + 2\cos 2x = 0$$

$$\Rightarrow (3\cos^6 x - 1 - \cos 2x)^2 = 0$$

$$\Rightarrow \cos^2 x (3\cos^4 x - 2) = 0$$

$$\Rightarrow \cos x = 0$$

$$x = n\pi + \frac{\pi}{2}, n \in \mathbb{I}$$

$$\text{or } \cos^4 x = \frac{2}{3}$$

$$\cos x = \pm \sqrt[4]{\frac{2}{3}}$$

$$x = m\pi \pm \cos^{-1} \sqrt[4]{\frac{2}{3}}, m \in \mathbb{I}$$

10. ABD

$$\text{Sol. Let any point on the line through P be } x = \alpha + r \cos \theta, y = 2 + r \sin \theta.$$

$$\text{Put this in } \frac{x^2}{9} + \frac{y^2}{4} = 1 \text{ and } xy = 0.$$

$$PA \times PD = PB \times PC$$

$$\Rightarrow 2\alpha \sin 2\theta + 5 \cos 2\theta = 13$$

$$\Rightarrow 13 \leq \sqrt{4\alpha^2 + 25}$$

$$\Rightarrow |\alpha| \geq 6$$

11. ABCD

$$\text{Sol. Equation of tangent at P is } x + y = 4$$

Clearly mirror image of R(-5,5) lies on line PQ.

Now mirror image R' or R

$$\Rightarrow \frac{\alpha' + 5}{1} = \frac{\beta' - 5}{1} = \frac{-2(-5 + 5 - 4)}{2} = 4$$

$$\Rightarrow (\alpha', \beta') = (-1, 9)$$

Let PM cuts the axis at T; As M is midpoint of PT

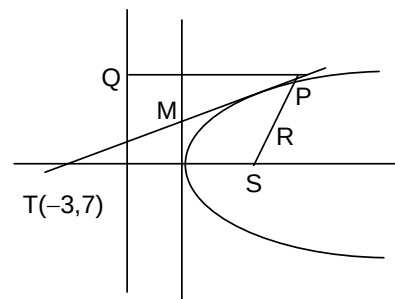
$$\Rightarrow T \text{ is } (-3, 7)$$

$$\text{We know that } SP = ST \text{ and } \angle SMP = \frac{\pi}{2}$$

$$\text{Equation SP } \Rightarrow y - 3 = -\frac{1}{3}(x - 1)$$

$$\Rightarrow x + 3y - 10 = 0$$

$$\text{Let S } \equiv (10 - 3\beta, \beta)$$



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$$\text{Again TS} \parallel \text{PQ} \Rightarrow \frac{\beta - 7}{13 - 3\beta} = \frac{9 - 3}{-1 - 1} = -3$$

$\therefore$ focus is $(-2, 4)$

12. AD

Sol. $h'(x) = f(x) \Rightarrow h''(x) = f'(x)$
 $h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = g(1)$
 $f(g(x)) = x$
 $f'(g(x)) \cdot g'(x) = 1$
 $f'(g(1)) \cdot g'(1) = 1 \Rightarrow g'(1) = 1$
 $G(x) = x^2 \cdot g(x) - x \cdot h(g(x))$
 $G'(x) = x^2 \cdot g'(x) + 2x \cdot g(x) - h(g(x)) - x \cdot h'(g(x)) \cdot g'(x)$
 $= 2x \cdot g(x) - h(g(x))$
 $G''(x) = 2 \cdot g(x) + x \cdot g'(x)$
 $G'(1) = 2g(1) - h(g(1)) = 2g(1) - h(1) = 2$
 $G''(1) = 2 \cdot g(1) + g'(1) = 3$

PART - B

1. 3

Sol. $XX^T = I$; Find PQ then $(PQ)^{10}$

2. 3

Sol. $\beta = \int_0^1 \frac{2x + 3x^2 + 4x^3}{2\sqrt{x^4 + x^3 + x^2}} dx$
Put $(x^2 + x^3 + x^4) = t \Rightarrow (2x + 3x^2 + 4x^3) dx = dt$
 $\beta = \int_0^3 \frac{dt}{2\sqrt{t}}$
 $\beta = \left[\sqrt{t} \right]_0^3 = \sqrt{3} \Rightarrow \beta^2 = 3$

3. 3

Sol. $y = mx$ touches circle
 $\Rightarrow D = 0$ for $(m^2 + 1)x^2 - 2a(m + 2)x + 4a^2 = 0$
 $\Rightarrow 4a^2(m + 2)^2 = 4(m^2 + 1)(4a^2)$
 $\Rightarrow m = 0, \frac{4}{3}$
Let $A : (x_1, m_1 x_1), B : (x_2, m_2 x_2)$
 $AB^2 = 9 \Rightarrow (x_1 - x_2)^2 + (m_1 x_1 - m_2 x_2)^2 = 9$ where
 $x_1 = \frac{(m_1 + 2)}{m_1^2 + 1} a, x_2 = \frac{(m_2 + 2)}{m_2^2 + 1} a, m_1 = 0, m_2 = \frac{4}{3}$
get $a = \frac{3\sqrt{5}}{4}$

4. 6

Sol. Let $z = x_1 + iy_1, w = x_2 + iy_2$
 $\therefore |z - 3| = \text{Re } z \Rightarrow (x_1 - 3)^2 + y_1^2 = x_1^2 \dots (i)$
 $|w - 3| = \text{Re } w \Rightarrow (x_2 - 3)^2 + y_2^2 = x_2^2 \dots (ii)$
 $\arg(z - w) = \frac{\pi}{4} \Rightarrow \frac{y_1 - y_2}{x_1 - x_2} = \tan \frac{\pi}{4} = 1 \dots (iii)$

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By solving (i), (ii) and (iii) we get

$$\Rightarrow \frac{y_1 - y_2}{x_1 - x_2} = \frac{6}{y_1 + y_2}$$

From (iii) $y_1 + y_2 = 6$

5. 0.50

Sol. $a(x^2 - x + 1) - (x^2 + x + 1) \geq 0$

$$\Rightarrow a \geq \frac{x^2 + x + 1}{x^2 - x + 1} \text{ or } a \geq \frac{(x^2 - x + 1) + 2x}{x^2 - x + 1}$$

$$\Rightarrow a \geq 1 + \frac{2x}{x^2 - x + 1} \text{ or } a \geq 1 + \frac{2}{\left(x + \frac{1}{x}\right) - 1}$$

Now, for $x \geq 2$, the minimum value of $x + \frac{1}{x}$ is $\frac{5}{2}$.

$$\Rightarrow a \geq \frac{7}{3}$$

6. 30.25

Sol. Let $\alpha - d, \alpha, \alpha + d$ are roots

$$\therefore 3\alpha = -a \quad (1)$$

$$3\alpha^2 - d^2 = b \quad (2)$$

$$\alpha(\alpha^2 - d^2) = \frac{1}{9} \quad (3)$$

$$\text{From (1) and (2), } d^2 = \frac{a^2}{3} - b$$

Put d^2 in (3) to get

$$2a^3 - 9ab + 1 = 4$$

7. 0.30

Sol. Let $S = \sum_{k=0}^{100} \left(\frac{k}{k+1}\right)^{100} C_k$

$$= \sum_{k=0}^{100} \frac{((k+1)-1)}{(k+1)}^{100} C_k = \left(\sum_{k=0}^{100} C_k\right) - \sum_{k=0}^{100} \frac{C_k}{k+1}$$

$$= 2^{100} - \frac{1}{101} \sum_{k=0}^{100} \left(\frac{101}{k+1}\right)^{100} C_k$$

$$= 2^{100} - \frac{1}{101} \sum_{k=0}^{100} C_{k+1}$$

$$2^{100} - \left(\frac{2^{101} - 1}{101}\right) = \frac{(101)2^{100} - 2^{101} + 1}{101} = \frac{99(2^{100}) + 1}{101}$$

$$= \frac{a(2^{100}) + b}{c} \quad (\text{Given})$$

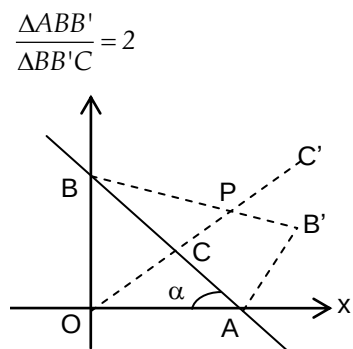
So, $a = 99, b = 1, c = 101$

Hence, $(a + b + c)_{\text{least}} = 99 + 1 + 101 = 201$.

8. 2.00

Sol. $A \equiv (1, 0), B \equiv (0, \tan \alpha), C \equiv \left(\frac{1}{2}, \frac{\tan \alpha}{2}\right), B' = \left(\frac{2 \tan^2 \alpha}{1 + \tan^2 \alpha}, \frac{\tan^3 \alpha - \tan \alpha}{1 + \tan^2 \alpha}\right)$

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FIITJEE RBT-9 for (JEE-Advanced)

PHYSICS, CHEMISTRY & MATHEMATICS

Pattern – 4

QP CODE: 100216

PAPER - 2

Time Allotted: 3 Hours

Maximum Marks: 186

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Part.

- (i) **PART-A (01-08)** contains (8) Multiple Choice Questions which have One or More Correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct options(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened.
Negative Marks: -2 In all other cases.
For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **-2 marks**, as a wrong option is also darkened.
- (ii) **Part-A (09-12)** – This section contains Two (02) List-Match Sets, each List-Match set has Two (02) Multiple Choice Questions. Each List-Match set has two lists: List-I and List-II. FOUR options are given in each Multiple Choice Question based On List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question. Each question carries **+3 Marks** for correct combination chosen and **-1 marks** for wrong options chosen.
- (iii) **Part-B (01-06)** contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCHES – Two Yr CRP2123(AII)

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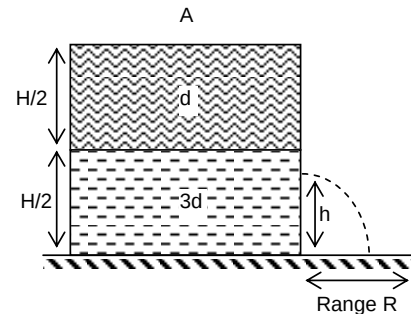
SECTION – I : PHYSICS

(PART – A)

(One or More Than One Options Correct Type)

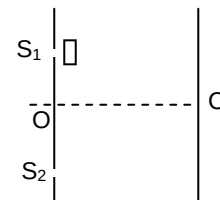
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE** or **MORE THAN ONE** is correct.

1. A container of large uniform cross sectional area A resting on a horizontal surface holds two immiscible non viscous and incompressible liquids of density d and $3d$, each of height $H/2$. The lower density liquid is open to the atmosphere having pressure P_0 . A tiny hole of area a ($a \ll A$) is punched to the vertical side of lower container at a height h ($0 < h < H/2$) for which range is maximum.



- (A) $h = H/3$ (B) Range $R = \frac{2H}{3}$
(C) Range $R = \frac{3H}{2}$ (D) Velocity of efflux $v = \sqrt{\frac{2}{3}gH}$

2. A YDSE is performed in a medium of refractive index μ_1 . In front of one of the slits say S_1 as shown, thin glass slab of refractive index $\mu_2 (< \mu_1)$ is kept. If initially the central maxima was formed on the central line OC then



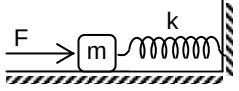
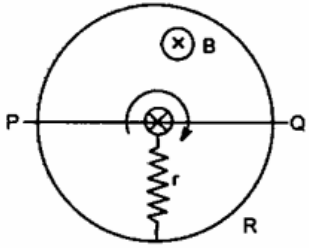
- (A) central maxima will shift upwards
(B) central maxima will shift downwards
(C) the waves reaching on the screen at C from S_1 will lead the waves reaching from S_2
(D) the waves reaching C from S_1 will lag from the waves reaching from S_2

3. Two projectile are thrown at the same time from two different points. The projectile thrown from the origin has initial velocity $3\hat{i} + 3\hat{j}$ with respect to earth. The projectile has initial velocity $a\hat{i} + b\hat{j}$ with respect to earth thrown from the point (10, 5). ($\hat{i}$ is a unit vector along horizontal, $\hat{j}$ along vertical). If the projectile collides after two second, then the

- (A) value of a is -2 (B) value of a is $\frac{1}{2}$
(C) value of b is $\frac{1}{2}$ (D) value of b is -2

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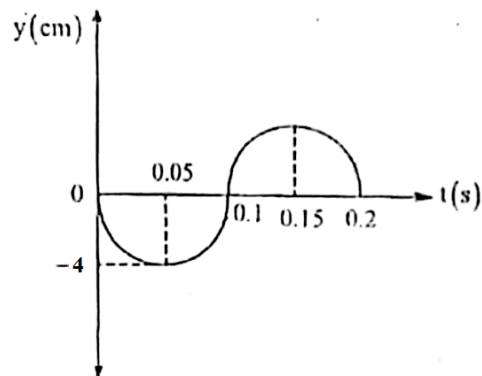
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4. A constant force F is applied on a spring block system as shown in figure. The mass of the block is m and spring constant is k . The block is placed over a smooth surface. Initially the spring was unstretched. Choose the correct alternative (s)
- 
- (A) the block will execute SHM (B) amplitude of oscillation is $F/2k$
- (C) time period of oscillation is $2\pi\sqrt{\frac{m}{k}}$ (D) the maximum speed of block is $\sqrt{\frac{2Fx - kx^2}{m}}$
(here $x = F/k$)
5. A particle is taken from point A to point B under the influence of a force field. Now it is taken back from B to A and it is observed that the work done in taking the particle from A to B is not equal to the work done in taking it from B to A. If W_{nc} and W_c is the work done by non conservative forces and conservative forces present in the system respectively, ΔU is the change in potential energy, Δk is the change in kinetic energy, then
- (A) $W_{nc} - \Delta U = \Delta k$ (B) $W_c = -\Delta U$
- (C) $W_{nc} + W_c = \Delta k$ (D) $W_{nc} - \Delta U = -\Delta k$
6. A particle of charge q and mass m moves rectilinearly under the action of an electric field $E = \alpha - \beta x$. Here, α and β are positive constants and x is the distance from the point where the particle was initially at rest. Then,
- (A) the motion of the particle is oscillatory.
- (B) the amplitude of the particle is $\left(\frac{\alpha}{\beta}\right)$
- (C) the mean position of the particle is at $x = \left(\frac{\alpha}{\beta}\right)$
- (D) the maximum acceleration of the particle is $\frac{q\alpha}{m}$
7. In figure, R is a fixed conducting ring of negligible resistance and radius a . PQ is a uniform rod of resistance r . It is hinged at the centre of the ring and rotated about this point in clockwise direction with a uniform angular velocity ω . There is a uniform magnetic field of strength B pointing inward and r is a stationary resistance. Then:
- 
- (A) current through r is zero.
- (B) current through r is $\frac{2B\omega a^2}{5r}$
- (C) direction of current in external resistance r is from centre to circumference.
- (D) direction of current in external resistance r is from circumference to center.

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8. For a certain transverse standing wave on a long string, an antinode is formed at $x = 0$ and next to it, a node is formed at $x = 0.10$ meter, the displacement $y(t)$ of the string particle at $x = 0$ is shown in figure.



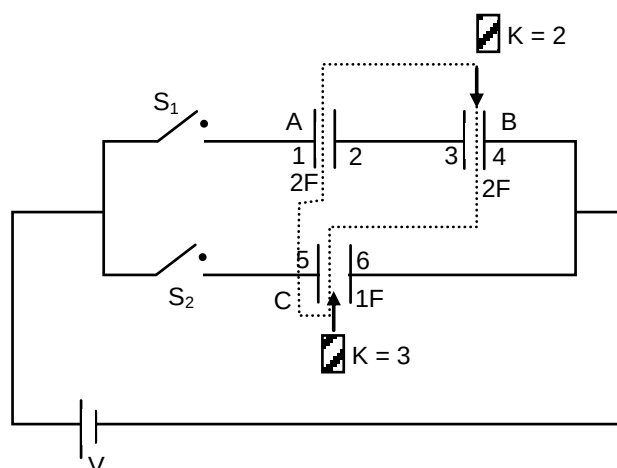
- (A) Transverse displacement of the particle at $x = 0.05$ m and $t = 0.05$ sec is $-2\sqrt{2}$ cm
 (B) Transverse displacement of the particle at $x = 0.04$ m and $t = 0.025$ sec is $-2\sqrt{2}$ cm
 (C) Speed of travelling wave that interfere to produce this standing wave is 2 m/s.
 (D) The transverse velocity of the string particle at $x = \frac{1}{15}$ m and $t = 0.1$ sec is 20π cm/sec.

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This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)



Change in quantities between initial and final situation is given in list-I and its numerical value in list-II.

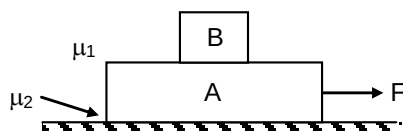
List-I		List-II	
(I)	Change in amount of + charge on capacitor	(P)	0
(II)	Change in energy	(Q)	$\frac{1}{3}$
(III)	Change in potential difference	(R)	$\frac{1}{36}$
(IV)	Final charge inside closed surface inclosing plate 2, 3 and 5 of different capacitor.]	(S)	$\frac{1}{6}$
		(T)	$\frac{1}{2}$

9. S_1 is closed and S_2 open, and $V = 1$ volt. In initial situation capacity are given. When capacitors are fully charged then dielectric of dielectric constant shown in figure is inserted in capacitor B to fill space between plates completely for final situation. Then for capacitor B. Choose correct option
- (A) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P (B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
 (C) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ P (D) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R
10. Now switch S_2 is also closed and $V = 1/6$ volt. In initial situation, capacitor C is completely charged then in final situation, di electric is inserted completely as shown in figure. For capacitor C.
- (A) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ S (B) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ R
 (C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ T (D) I $\rightarrow$ R, II $\rightarrow$ S, III $\rightarrow$ T, IV $\rightarrow$ P

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List Match Set (11-12)



List-I		List-II	
(I)	Acceleration of A or B	(P)	$\frac{4}{3}$
(II)	Normal reaction between blocks	(Q)	$\frac{2}{3}$
(III)	Frictional force between blocks f_1	(R)	20
(IV)	Frictional force on surface f_2	(S)	0
		(T)	1

11. $F = 4$ Newton, $m_A = 4$, $m_B = 2$, $\mu_1 = 0.1$ and $\mu_2 = 0$, frictional force between blocks is f_1 and between surface and block A is f_2 ($g = 10 \text{ m/sec}^2$)
- (A) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ S (B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R
 (C) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ P (D) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ S
12. $F = 4$ Newton, $m_A = 4$, $m_B = 2$, $\mu_1 = 0$ and $\mu_2 = \frac{1}{45}$
- (A) I $\rightarrow$ QS, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P (B) I $\rightarrow$ QS, II $\rightarrow$ RS, III $\rightarrow$ S, IV $\rightarrow$ T
 (C) I $\rightarrow$ QR, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ Q (D) I $\rightarrow$ QS, II $\rightarrow$ S, III $\rightarrow$ P, IV $\rightarrow$ R

(PART – B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

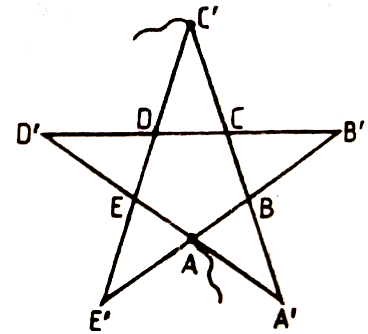
1. Toluene liquid of volume 300 cm^3 at 0°C is contained in a beaker. Another quantity of toluene of volume 110 cm^3 at 100°C is in another beaker. (the combined volume is 410 cm^3). $\Delta V \text{ cm}^3$ is change in volume of the mixture of the toluene liquids when they are mixed together. Given the coefficient of volume expansion $\gamma = 0.001/\text{C}$ and all forms of heat losses can be ignored. Value of ΔV is

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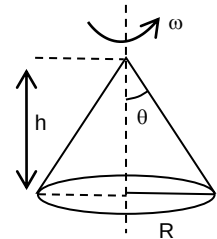
2. Two tuning forks with natural frequencies 340 Hz each move relative to a stationary observer. One fork moves away from the observer while the other moves towards him at the same speed. The observer hears beats of frequency 3 Hz. Find the speed of the tuning fork. Velocity of tuning fork is small in comparison to velocity of sound. (velocity of sound in air is 340 m/s).

3. A five-pointed regular star has been formed from a uniform wire. Calculate the equivalent resistance between points A and C'. Take $\sin 18^\circ = 1/3$ and the resistance of the sections $AA' = A'B = BB' = \dots = r = \frac{1}{2}\Omega$.



4. Three concentric thin spherical shells are of radii a , b and c ($a < b < c$). The first and third are connected by a fine wire through a small hole in the second. The second is connected to earth. Find the capacity of the condenser formed in μF . ($a : b : c = 1 : 2 : 3$, $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$, $a = 90 \text{ m}$)

5. 'Q' charge is uniformly distributed over the curve surface of a right circular cone of semi-vertical angle θ and height h . The cone is uniformly rotated about its axis at angular velocity ω . Magnetic moment associated with the cone is found to be $nQ\omega h^2 \tan^2 \theta$ where value of 'n' is



6. In an experiment of photoelectric effect light of wavelength 400 nm is incident on a metal plate at the rate of 5W. the potential of the collector plate is made sufficiently positive with respect to emitter so that the current reaches the saturation value. Assuming that on the average one out of every 10^6 photons is able to eject a photoelectron, find the photocurrent in the circuit. (in micro ampere)

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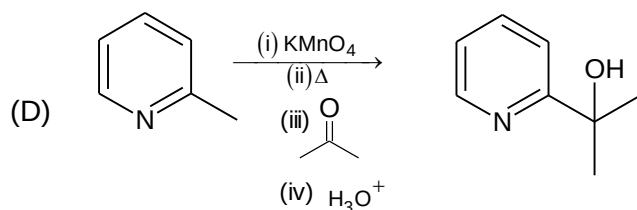
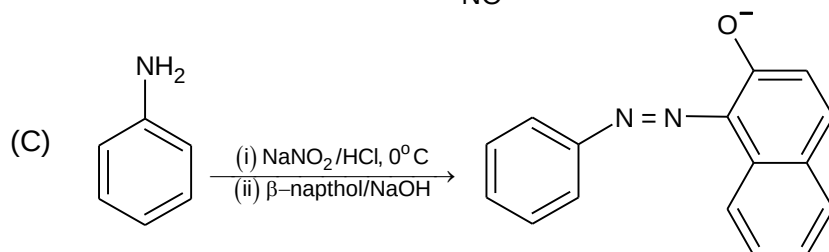
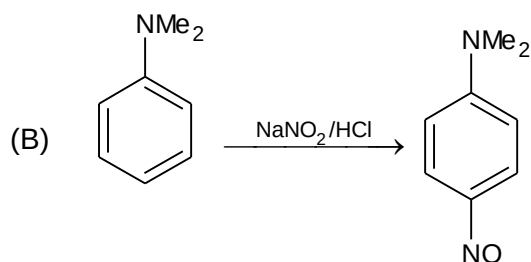
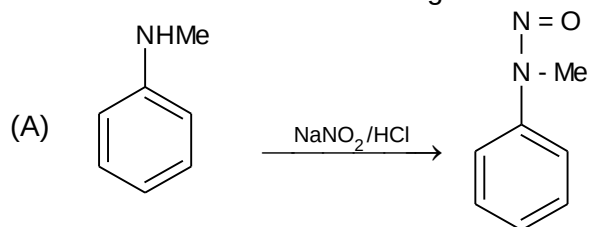
SECTION – II : CHEMISTRY

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

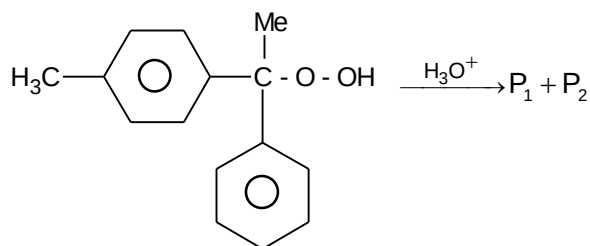
- Which of the following pairs produce same gas?
 (A) $(\text{Ca}_3\text{N}_2 + \text{dil. HCl})$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
 (B) $[(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \text{ on heating}]$ and $(\text{NH}_4\text{NO}_2 \text{ on heating})$
 (C) $(\text{NH}_4\text{NO}_3 \text{ on heating})$ and $[\text{Hg}(\text{NO}_3)_2 \text{ on heating}]$
 (D) $(\text{NH}_4\text{Cl on heating})$ and $[\text{NaNO}_3 + \text{Zn} + \text{NaOH}_{(\text{aq.})} \text{ on heating}]$
- A current is passed through 500 mL of an aqueous solution of CaI_2 . After sometime, it is observed that 50 millimoles of I_2 have been formed. Which of the following statements is(are) correct?
 (A) The number of faradays of charge passed through the solution is 0.10 F.
 (B) The volume of dry H_2 at STP that has been formed during electrolysis is 1120 mL.
 (C) The pH of the solution is nearly 0.7.
 (D) The mass of calcium produced is 2.0 g.
- Products of which of the following reactions is/are correctly represented.



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4.



$\text{P}_2 + \text{FeCl}_3 \rightarrow \text{Violet Colour}$

$\text{P}_1 + \text{NaOI} \rightarrow \text{P}_3 \downarrow + \text{P}_4$
yellow

The correct statement/s for the above sequence is/are

- (A) P_3 on reaction with Ag gives acetylene
 (B) P_4 on reaction with sodalime gives toluene
 (C) P_4 on reaction with sodalime gives benzene
 (D) P_1 on reaction with LiAlH_4 gives a compound which gives blue colour in Victor Meyer test

5.

During the production of iron and steel.

(A) The oxide ore is primarily reduced to iron by solid coke according to the reaction.



(B) The oxide ore is reduced by the carbon monoxide according to the reaction



(C) Major silica impurities are removed as calcium silicate slag by addition of a fluxing agent lime stone.

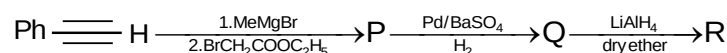
(D) The converter slag containing phosphorus is used as a fertilizer.

6.

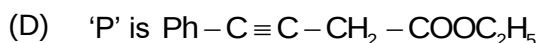
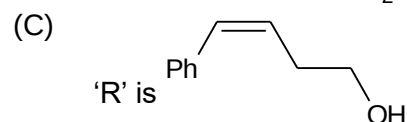
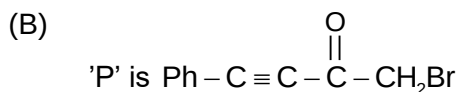
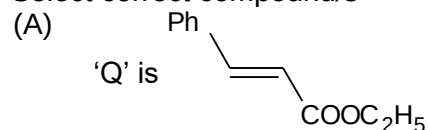
Select the correct statement(s)

- (A) the value of rate constant cannot exceed the value of Arrhenius factor
 (B) molecularity of multiple step reaction can be obtained from mechanism
 (C) half life of a fourth order reaction is linearly dependent on initial concentration of reactant
 (D) reactions with order ≥ 1 cannot get complete infinite time interval

7.



Select correct compound/s



Space For Rough Work

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8. Which of the following reactions will have α -hydroxy acid as a product?

- (A) $\text{CH}_3\text{-CHO} \xrightarrow[\text{(ii) H}_2\text{O/H}^+/\Delta]{\text{(i) NaCN/HCl}}$
- (B) $\text{CH}_3\text{-CHO} + \text{Br-CH}_2\text{-COOC}_2\text{H}_5 \xrightarrow[\text{(iii) HOH/H}^+/\Delta]{\text{(i) Zn, (ii) NH}_4\text{Cl/HOH}}$
- (C) $\begin{array}{c} \text{X} \\ | \\ \text{CH}_2\text{-CH}_2\text{-COOH} \end{array} \xrightarrow{\text{NaOH/HOH}}$
- (D) $\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{(iii) NaOH}]{\text{(i) NaCN followed by hydrolysis, (ii) Red P/Br}_2}$

This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-10)

Match the lists and answer the following question.

List – I		List – II	
Alcohols subjected to Lucas reagent		Observation	
(I)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$	(P)	White turbidity after boiling only
(II)	PhCH_2OH	(Q)	Turbidity instantaneously
(III)	$\begin{array}{c} \text{H}_3\text{C} \\ \diagdown \\ \text{C}-\text{OH} \\ \diagup \\ \text{H}_3\text{C} \end{array}$	(R)	Appearance of turbidity after 5-10 minutes of addition of reactants
(IV)	$\text{H}_3\text{C}-\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$	(S)	No visible change observed
		(T)	Red colour is obtained

9. Which combination is correct according to the above list?

- (A) I $\rightarrow$ R (B) II $\rightarrow$ Q
(C) III $\rightarrow$ S (D) IV $\rightarrow$ T

Space For Rough Work

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Match the lists and answer the following question.

List – I		List – II	
Alcohols subjected to Lucas reagent		Observation	
(I)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$	(P)	White turbidity after boiling only
(II)	PhCH ₂ OH	(Q)	Turbidity instantaneously
(III)	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{H}_3\text{C}-\text{C}-\text{OH} \\ \\ \text{H}_3\text{C} \end{array}$	(R)	Appearance of turbidity after 5-10 minutes of addition of reactants
(IV)	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$	(S)	No visible change observed
		(T)	Red colour is obtained

10. Which combination is correct according to the above list?

(A) I → R

(B) II → S

(C) III → Q

(D) IV → P

List Match Set (11-12)

Match the Lists and answer the following question.

List– I		List – II	
(I)	Neso Silicate	(P)	$(\text{SiO}_3)_n^{2n-}$
(II)	Soro Silicate	(Q)	$(\text{Si}_2\text{O}_5)_n^{2n-}$
(III)	Cyclic Silicate	(R)	$(\text{Si}_2\text{O}_7)^{6-}$
(IV)	Phyllo Silicate	(S)	None of the O atom is shared
		(T)	All oxygen atoms are shared

11. Which combination is correct according to the above list?

(A) I → Q

(B) II → S

(C) III → P

(D) IV → P

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Match the Lists and answer the following question.

List – I		List – II	
(I)	Neso Silicate	(P)	$(\text{SiO}_3)_n^{2n-}$
(II)	Soro Silicate	(Q)	$(\text{Si}_2\text{O}_5)_n^{2n-}$
(III)	Cyclic Silicate	(R)	$(\text{Si}_2\text{O}_7)^{6-}$
(IV)	Phyllo Silicate	(S)	None of the O atom is shared
		(T)	All oxygen atoms are shared

12. Which combination is correct according to the above list?
- (A) I $\rightarrow$ S (B) II $\rightarrow$ P
(C) III $\rightarrow$ Q (D) IV $\rightarrow$ T

(PART – B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

- 3.64 g of silver salt of an organic dibasic acid yields, on strong heating, 2.16 g of silver. If the weight percentage of carbon in acid 8 times the weight percentage of hydrogen and one-half the weight percentage of oxygen, if the molecular formula of the acid is written as $\text{C}_x\text{H}_y\text{O}_z$. Then find $x + y + z = ?$
[Atomic weight of Ag = 108]
- How many of the following statements are incorrect?
 - Spontaneous adsorption of gases on solid surface is an endothermic process as entropy decreases during adsorption.
 - Formation of micelles takes place when temperature is below Kraft Temperature (T_k) and concentration is above critical micelle concentration (CMC).
 - A colloid of $\text{Fe}(\text{OH})_3$ is prepared by adding a little excess [required to completely precipitate Fe^{3+} ions as $\text{Fe}(\text{OH})_3$] of NaOH in FeCl_3 solution the particles of this sol will move towards anode during electrophoresis.
 - According to Hardy-Schulze rule for the coagulation of As_2S_3 sol flocculation value of Fe^{3+} ion will be more than Ba^{2+} or Na^+ .
 - In Micelle formation of ionic soap, entropy decreases
- How many of following is/are paramagnetic in nature?
 $[\text{NiCl}_4]^{2-}$, $[\text{Ni}(\text{CO})_4]$, $[\text{PtCl}_4]^{2-}$, $[\text{Cu}(\text{NH}_3)_4]^+$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $\text{K}_2[\text{NiF}_6]$, $[\text{Ni}(\text{NH}_3)_6]^{2+}$, $[\text{Co}(\text{CN})_6]^{3-}$, $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cr}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{NH}_3)_6]^{2+}$, $[\text{PtCl}_6]^{2-}$

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4. Na_2HPO_4 and NaH_2PO_4 on heating at high temperature produce a chain sodium pentaphosphate quantitatively of ideal molar ratio of Na_2HPO_4 to NaH_2PO_4 is $x : y$ & total charge on pentaphosphate anion is z then calculate value of $\frac{x}{y} \times z$.
5. 10 g Neon initially at a pressure of 506.625 KPa and temperature of 473 K expand adiabatically to a pressure of 202.65 KPa. Calculate entropy change (in JK^{-1}) of universe of expansion is a free expansion.
6. The value of rate constant for the gas phase reaction $2\text{NO}_2 + \text{F}_2 \longrightarrow 2\text{NO}_2\text{F}$ is $38 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 300 K. the order of reaction is _____?
Write "1000" if the order cannot be predicated with given information
-

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SECTION – III : MATHEMATICS

(PART – A)

(One or More Than One Options Correct Type)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE or MORE THAN ONE is correct**.

1. If $f(x) = \begin{vmatrix} \sec^2 x & 1 & 1 \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cot^2 x \end{vmatrix}$ then

(A) $\int_{-\pi/4}^{\pi/4} f(x) dx = \frac{1}{16}(3\pi + 8)$

(C) Maximum value of $f(x)$ is 1

(B) $f'\left(\frac{\pi}{2}\right) = 0$

(D) Minimum value of $f(x)$ is 0.

2. If $A(\alpha, \beta) = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & e^\beta \end{bmatrix}$, then

(A) $A(\alpha, \beta)' = A(-\alpha, \beta)$

(C) $\operatorname{Adj}(A(\alpha, \beta)) = e^\beta A(-\alpha, -\beta)$

(B) $A(\alpha, \beta)^{-1} = A(-\alpha, -\beta)$

(D) $A(\alpha, \beta)' = A(\alpha, -\beta)$

3. If n objects are arranged in a row, then the number of ways of selecting three of these objects so that no two of them are next to each other is

(A) $\frac{1}{6}(n-2)(n-3)(n-4)$

(C) ${}^{n-3}C_3 + {}^{n-3}C_2$

(B) ${}^{n-2}C_3$

(D) None of these

4. Let $\alpha(n) = 1 - \frac{1}{2} + \frac{1}{3} - \dots + (-1)^{n-1} \frac{1}{n}$

(A) $\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} = \alpha(2n)$

(C) $\alpha(2n) \geq 0.5 \forall n$

(B) $\alpha(2n) < 1 \forall n$

(D) $0.5 < \alpha(n) < 1 \forall n$

5. If $\int_0^\alpha \frac{dx}{1 - \cos \alpha \cos x} = \frac{A}{\sin \alpha} + B (a \neq 0)$. Then possible values of A and B are

(A) $A = \frac{\pi}{2}, B = 0$

(C) $A = \frac{\pi}{6}, B = \frac{\pi}{\sin \alpha}$

(B) $A = \frac{\pi}{4}, B = \frac{\pi}{4 \sin \alpha}$

(D) $A = \pi, B = \frac{\pi}{\sin \alpha}$

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6. The solution of $\frac{dy}{dx} = \sqrt{y-x}$ is given by
 (A) $x + C = 2\sqrt{y-x} + 2\log(\sqrt{y-x} - 1)$ (B) $x^2 + C = 2\sqrt{y-x} + \log(\sqrt{y-x} - 1)$
 (C) $x + C = (y-x)^2 + \log(y-x-1)$ (D) $\sqrt{y-x} - 1 = C' e^{x/2 - \sqrt{y-x}}$
7. If sines of the angle A and B of a triangle ABC satisfy the equation $c^2x^2 - c(a+b)x + ab = 0$, then the triangle
 (A) is acute angled (B) is right angled
 (C) is obtuse angled (D) satisfies $\sin A + \cos A = \frac{(a+b)}{c}$
8. In the expansion of $\left(x^2 + 1 + \frac{1}{x^2}\right)^n$, $n \in \mathbb{N}$,
 (A) number of terms is $2n+1$ (B) coefficient of constant term is $2^n - 1$
 (C) coefficient of x^{2n-2} is n (D) coefficient of x^2 is n

This section contains **2 List-Match Sets**, each List-Match set has **2 Multiple Choice Questions**. Each List-Match set has two lists: List-I and List-II. Four options are given in each Multiple Choice Question based On List-I and List-II and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

List Match Set (09-12)

9. Match the following

List – I		List – II	
(I)	The radical axis of two circles	(P)	Subtends a right angle at a point of intersection.
(II)	The common tangent to two intersecting circles of equal radii	(Q)	Is perpendicular to the line joining the centres.
(III)	The common chord of two intersecting circles	(R)	Is parallel to the line joining the centres.
(IV)	The line joining the centres of two circles intersecting orthogonally	(S)	Is bisected by the line joining the centres.
		(T)	Is parallel to common normals

Which is correct option?

- (A) I $\rightarrow$ Q (B) II $\rightarrow$ P
 (C) III $\rightarrow$ R (D) IV $\rightarrow$ S

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10. Match the following

List - I		List - II	
(I)	$f(x) = x \operatorname{sgn}(x-1)$	(P)	$\lim_{x \rightarrow 1} f(x)$ does not exist
(II)	$f(x) = \frac{\sin\left(\sin\left(\tan\left(\frac{x^2}{2}\right)\right)\right)}{\log \cos 3x}$	(Q)	$\lim_{x \rightarrow 0} f(x)$ doesn't exist
(III)	$f(x) = \frac{\sqrt[3]{1+\tan^{-1} 3x} - \sqrt[3]{1-\sin^{-1} 3x}}{\sqrt{1-\sin^{-1} 2x} - \sqrt{1+\tan^{-1} 2x}}$	(R)	$\lim_{x \rightarrow 0} f(x) = \frac{-1}{9}$
(IV)	$f(x) = \frac{e^{1/x} - 1}{e^{1/x} + 1}$	(S)	$\lim_{x \rightarrow 0} f(x) = -1$
		(T)	$\lim_{x \rightarrow 0} f(x) = 2$

Which is correct option?

(A) I $\rightarrow$ S(B) II $\rightarrow$ R(C) III $\rightarrow$ Q(D) IV $\rightarrow$ P11. Match the sequence $a_1, a_2, a_3, \dots$ whose n th term is given on the left with properties of the sequence on the right.

List - I		List - II	
(I)	$a_n = \int_0^{\pi/2} \frac{1 - \cos 2nx}{1 - \cos 2x} dx$	(P)	$a_1, a_2, a_3, \dots$ are in A.P.
(II)	$I_n = \int_0^{\pi/4} \tan^n x dx$ and $a_n = \frac{1}{I_{n+1} + I_{n+3}}$	(Q)	$a_1, a_2, a_3, \dots$ are in G.P.
(III)	$I_n = \int_0^{\pi/4} \frac{\sin^2 nx}{\sin x} dx \quad \forall n \geq 0$ and $a_n = I_n - I_{n-1} \quad \forall n \geq 1$	(R)	$a_1, a_2, a_3, \dots$ are in H.P.
(IV)	$b_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x} dx, \quad a_n = b_n - b_{n-1} \quad \forall n \geq 1$	(S)	$a_1, a_2, a_3, \dots$ is constant sequence.
		(T)	$a_{n+1} = a_n + a_n^2$

Which is correct option?

(A) I $\rightarrow$ Q(B) II $\rightarrow$ T(C) III $\rightarrow$ R(D) IV $\rightarrow$ S

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12. Lines $y^2 - xy = 0$ make Intercepts on curve given on R.H.S of Matrix.

List – I		List – II	
(I)	2	(P)	$\frac{x^2}{3} + y^2 = 1$
(II)	4	(Q)	$x^2 + \frac{y^2}{3} = 1$
(III)	$\sqrt{6}$	(R)	$x^2 + y^2 = 4$
(IV)	$2\sqrt{3}$	(S)	$y^2 = 2\sqrt{2}x$
		(T)	$x^2 - y^2 = 2$

Which is correct option?

(A) I $\rightarrow$ Q

(B) II $\rightarrow$ P

(C) III $\rightarrow$ S

(D) IV $\rightarrow$ R

(PART – B)

(Numerical Type)

Part-B (01-06) contains six (06) Numerical based questions, the answer of which maybe positive or negative numbers or decimals to **Two decimals Places** (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30).

- If $f : [0,1] \rightarrow \mathbb{R}$ is a continuous function satisfying $\int_0^1 f(x) dx = \frac{1}{3} + \int_0^1 (f(x^2))^2 dx$, then the reciprocal of $\frac{1}{8} f\left(\frac{1}{4}\right)$ is _____
- Any chord of conic $x^2 + y^2 + xy = 1$ passing through origin is bisected at point (p, q), then $\frac{(p+q+1)}{8}$ equals _____
- Compute the square of the value of the expression $\frac{4 + \sec 20^\circ}{\operatorname{cosec} 20^\circ}$
- Let $f(x) = x - \frac{1}{x}$. Then find the number of different solutions to the equation $f(f(f(x))) = 1$.
- If the total number of selections that can be made from n different books each having 5 copies taking at least 2 copies of each book is 256, then n =
- The positive integral value of n such that $1.2^1 + 2.2^2 + 3.2^3 + 4.2^4 + \dots + n.2^n = 2 + 2^{n+5}$, is

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Q. P Code:
Answers
SECTION – I : PHYSICS

(PART – A)

- | | | | |
|--------|---------|-------|--------|
| 1. ABD | 2. BC | 3. AC | 4. ACD |
| 5. ABC | 6. ABCD | 7. BD | 8. ACD |
| 9. A | 10. C | 11. D | 12. A |

(PART – B)

- | | | | |
|---------|------------------------------|---------|---------|
| 1. 0.75 | 2. 1.50 | 3. 0.50 | 4. 0.11 |
| 5. 0.25 | 6. 1.6 (Range: 1.60 to 1.62) | | |

SECTION – II : CHEMISTRY

(PART – A)

- | | | | |
|--------|-------|---------|--------|
| 1. BD | 2. AB | 3. ABCD | 4. ACD |
| 5. BCD | 6. AD | 7. CD | 8. AD |
| 9. B | 10. D | 11. C | 12. A |

(PART – B)

- | | | |
|-------------------------------|---|------|
| 1. 16 | 2. 4 | 3. 6 |
| 4. -4.66 (Range -4.6 to -4.7) | 5. 3.81 JK ⁻¹ (Range 3.7 to 3.9) | |
| 6. 2 | | |

SECTION – III : MATHEMATICS

(PART – A)

- | | | | |
|---------|--------|--------|--------|
| 1. ABCD | 2. ABC | 3. ABC | 4. ABC |
| 5. AB | 6. AD | 7. BD | 8. AC |
| 9. A | 10. A | 11. C | 12. A |

(PART – B)

- | | | | |
|---------|----------|---------|---------|
| 1. 0.25 | 2. 0.125 | 3. 3.00 | 4. 1.00 |
| 5. 4.00 | 6. 17.00 | | |

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Answers & Solutions

SECTION – I : PHYSICS

(PART – A)

1. **ABD**

Sol. Bernoulli's Theorem for an orifice at depth 'x' in liquid '3d'.

$$P_0 + \left(\rho g \frac{H}{2} + 3\rho g x \right) = P_0 + \frac{1}{2} \times 3d \times v^2 \quad \dots(1)$$

$$\frac{H}{2} - x = \frac{1}{2} g t^2 \quad \dots(2)$$

$$R = Vt \quad \dots(3)$$

Solve for R and apply maxima/minima.

2. **BC**

Sol. Introduction of sheet decrease optical path length.

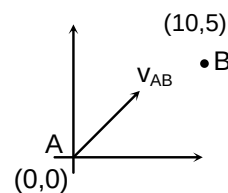
3. **AC**

$$\vec{v}_{AB} = (3-a)\hat{i} + (3-b)\hat{j}$$

$$\vec{a}_{AB} = \vec{0}$$

$$(3-a) \times 2 = 10 \text{ and } (3-b) \times 2 = 5$$

$$a = -2 \text{ and } b = \frac{1}{2}$$

4. **ACD**

$$\text{Sol. } Fx - \frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$\Rightarrow \sqrt{\frac{2Fx - kx^2}{m}} = v$$

$$\text{Where } x = \frac{F}{k}$$

5. **ABC**Sol. Using work energy theorem $W_c + W_{nc} = \Delta K$ Where $W_k = -\Delta U$.6. **ABCD**

$$\text{Sol. } a = \frac{F}{m} = \frac{qE}{m} = \frac{q}{m}(\alpha - \beta x) \quad \dots(1)$$

$$a = 0 \text{ at } x = \frac{\alpha}{\beta}$$

$$\text{i.e., force on the particle is zero at } x = \frac{\alpha}{\beta}$$

$$\text{So, mean position of the particle is at } x = \frac{\alpha}{\beta}$$

$$\text{Equation (1) can be written as } v \frac{dv}{dx} = \frac{q}{m}(\alpha - \beta x)$$

$$\therefore \int_0^v v dv = \frac{q}{m} \int_0^x (\alpha - \beta x) dx$$

$$\therefore v = \sqrt{\frac{q}{m} \left(\alpha x - \frac{\beta x^2}{2} \right)} \quad @bohrring_bot$$

$$v = 0 \text{ at } x = 0 \text{ and } x = \frac{2\alpha}{\beta}$$

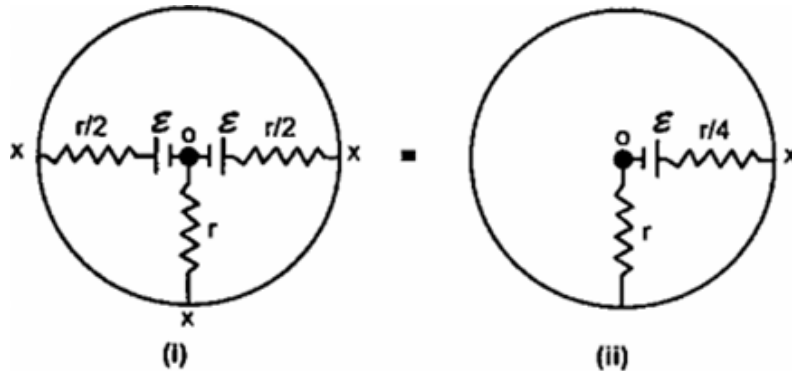
So, the particle oscillates between $x = 0$ and $x = \frac{2\alpha}{\beta}$

with mean position at $x = \frac{\alpha}{\beta}$.

Maximum acceleration of the particle is at extreme positions (at $x = 0$ or $x = 2\alpha/\beta$) and $a_{\max} = q \alpha/m$ [from equation (1)].

7. **BD**

Sol. Equivalent circuit:



$$\text{Induced e.m.f. } \varepsilon = \left(\frac{B\omega a^2}{2} \right) \quad (\because \text{radius} = a)$$

By nodal equation:

$$4 \left(\frac{x - \varepsilon}{r} \right) + \left(\frac{x - 0}{r} \right) = 0$$

$$5x = 4\varepsilon$$

$$\Rightarrow x = \frac{4\varepsilon}{5} = \frac{2B\omega a^2}{5}$$

$$\text{and } I = \frac{x}{r} = \frac{2B\omega a^2}{5r}$$

Also direction of current in (i) will be toward negative terminal, i.e., from rim to origin alternating, by equivalent of cells figure.

$$I = \frac{\varepsilon}{r + \frac{r}{4}} = \frac{4\varepsilon}{5r} = \frac{2B\omega a^2}{5r}$$

8. **ACD**

Sol. $y(x, t) = -\cos kx \sin \omega t$

$$\begin{aligned} Y(0.05, 0.05) &= -4 \cos \left(\frac{2\pi}{0.4} 0.05 \right) \sin \left(\frac{2\pi}{0.2} 0.05 \right) \\ &= -4 \cos \left(\frac{\pi}{4} \right) \sin \left(\frac{\pi}{2} \right) = -2\sqrt{2} \text{ cm} \end{aligned}$$

$$v = f\lambda = \frac{0.4}{0.2} = 2 \text{ m/s}$$

$$\frac{\partial y}{\partial t} = -A\omega \cos(kx) \cos(\omega t)$$

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$$= \sim 4 \left(\frac{2\pi}{0.2} \right) \cos \left(\frac{2\pi}{0.4} \frac{1}{15} \right) \cos \left(\frac{2\pi}{0.2} 0.1 \right)$$

$$\frac{\partial y}{\partial t} = 20\pi \text{ cm/sec} = \text{particle velocity.}$$

9. **A**

Sol. $C_{eq} = 1 \text{ F}$ of capacitors A and B before inserting dielectric. Final C_{eq} of A and B = $\frac{4}{3} \text{ F}$.

$$\therefore \Delta Q = (\Delta C)V = \left(\frac{4}{3} - 1 \right) \times 1 = \left(\frac{1}{3} \right)$$

$$\Delta U = \left| \frac{\left(\frac{4}{3} \right)^2}{2 \times 4} - \frac{(1)}{2 \times 2} \right| = \left| \frac{2}{9} - \frac{1}{4} \right| = \frac{1}{36}$$

$$\Delta V = \left| \frac{q_2}{C_2} - \frac{q_1}{C_1} \right| = \left| \frac{\left(\frac{4}{3} \right)}{4} - \frac{1}{2} \right| = \left| \frac{1}{3} - \frac{1}{2} \right| = \frac{1}{6}$$

10. **C**Sol. $C_{\text{initial}} = 1 \text{ F}$ $C_{\text{final}} = 3 \text{ F}$

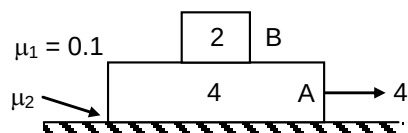
$$\Delta Q = (C_f - C_i)V = 2 \times \frac{1}{6} = \frac{1}{3}$$

$$\Delta U = \frac{1}{2} C_f V^2 - \frac{1}{2} C_i V^2 = \frac{1}{2} \times 2 \times \frac{1}{36} = \frac{1}{36}$$

$$Q_{\text{final}} = C_{\text{final}} V = 3 \times \frac{1}{6} = \frac{1}{2}$$

11. **D**

Sol.



If both block have together, their common acceleration

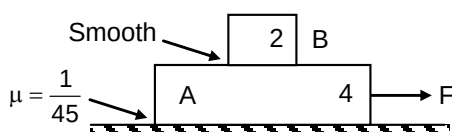
$$a = \frac{4}{6} = \left(\frac{2}{3} \right)$$

Maximum possible acceleration of block B = $\mu g = 1$ which is greater than common acceleration. Therefore both block will move together.

$$f_1 = ma = 2 \times \frac{2}{3} = \frac{4}{3}$$

12. **A**

Sol.



$$a = \frac{F - f}{4} = \frac{4 - \left(\frac{1}{45} \right) 60}{4} = \frac{2}{3}$$

B will be at rest

$$f = \mu(2 + 6)g = \frac{60}{45} = \frac{4}{3}$$

(PART – B)1. **0.75**Sol. $m_1 = \rho_0 \times 300$

$$m_2 = \rho_{100} \times 100$$

Suppose final temp = T

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$$m_1 S(T - 0) = m_2 S(100 - T)$$

$$\rho_0 \times 300 T = \rho_{100} \times 110(100 - T)$$

$$300 \rho_0 T = \frac{\rho_0 \times 110}{(1 + \gamma \times 100)} \times (100 - T)$$

$$300 T = \frac{110}{1.1} \times (100 - T)$$

$$3T = 100 - T$$

$$\Rightarrow T = 25^\circ$$

$$V_{\text{final}} = 300(1 + 0.001 \times 25) + 110(1 - 0.001 \times 75) \\ = 410 + 7.5 - 8.25 = 409.25 \text{ cm}^3$$

$$\text{Change} = 0.75 \text{ cm}^3.$$

2. **1.50**

Sol. Given: $f_1 - f_2 = 3$

$$\text{or} \quad \left(\frac{v}{v - v_s} \right) f - \left(\frac{v}{v + v_s} \right) f = 3$$

$$\text{or} \quad \left[\frac{1}{\left(1 - \frac{v_s}{v} \right)} - \frac{1}{\left(1 + \frac{v_s}{v} \right)} \right] f = 3$$

$$\text{or} \quad \left[\left(1 - \frac{v_s}{v} \right)^{-1} - \left(1 + \frac{v_s}{v} \right)^{-1} \right] f = 3$$

$$\text{or} \quad \left[\left(1 + \frac{v_s}{v} \right) - \left(1 - \frac{v_s}{v} \right) \right] f = 3$$

$$\text{or} \quad \frac{2v_s f}{v} = 3$$

$$\text{or} \quad \text{Speed of tuning fork } v_s = \frac{3v}{2f}$$

Substituting the values, we get

$$v_s = \frac{(3)(340)}{(2)(340)} = 1.5 \text{ m/s}$$

3. **0.50**

Sol. Each of angle A' , B' , C' , ... is 36° .

$$\therefore \text{Resistance of AB} = 2 \times r \sin 18^\circ = 2r \times \frac{1}{3} = \frac{2}{3} r$$

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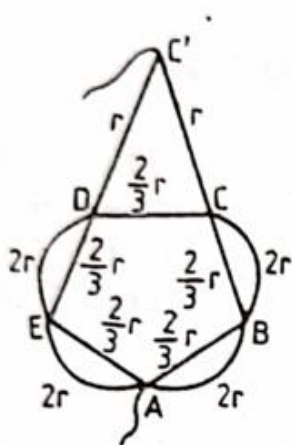


Fig. (a)

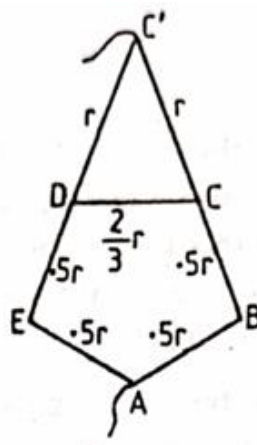


Fig. (b)

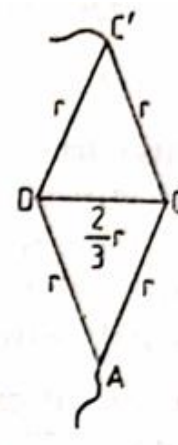


Fig. (c)

AA' and A'B are in series and so they sum up to $2r$. Similarly BB' and B'C, DD' and D'E, and EE' and E'A sum up to $2r$. This is shown in figure (a). In each side $2r$ and $\frac{2r}{3}$ in parallel sum up to $0.5r$, this is shown in figure b. The next reduction is shown in figure (c). D and C are equipotential points and hence DC' and CC' are in parallel and they sum up to $\frac{r}{2}$; AD and AC are in parallel and they sum up to $\frac{r}{2}$. So the required resistance is r .

4. 0.11

Sol.

If a charge q is given to the outermost sphere C, some of the charge will go to the outer surface of sphere A. Some will go to the inner surface of C and the rest will go to the outer surface of C. The charge q_1 on sphere A will induce a charge $-q_1$ on the inner surface of sphere B. Charge q_2 on the inner surface of sphere C will induce a charge $-q_2$ on the outer surface of sphere B.

In this way three capacitors are formed. One between the outer surface of A and inner surface of B. Second between the outer surface of B and inner surface of C and third by the outer surface of C.

Their capacities are

$$C_1 = 4\pi\epsilon_0 \frac{ab}{b-a};$$

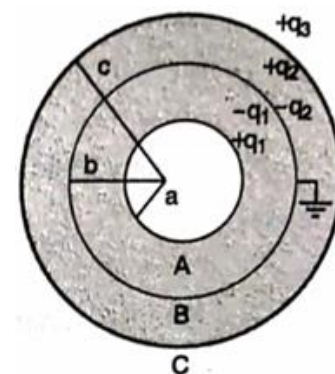
$$C_2 = 4\pi\epsilon_0 \frac{bc}{c-b} \quad \text{and} \quad C_3 = 4\pi\epsilon_0 c$$

Hence the combined capacity is

$$C = C_1 + C_2 + C_3$$

$$C = 4\pi\epsilon_0 \left[\frac{ab}{b-a} + \frac{bc}{c-b} + c \right] - 4\pi\epsilon_0 \left[\frac{ab}{b-a} + \frac{c^2}{c-b} \right]$$

5. 0.25



TG ~ @bohring_bot

Sol.

$$\text{Charge on ring} = q' = \frac{Q}{\pi RL} 2\pi x d\ell$$

$$\text{Magnetic moment due to ring} = \frac{q'\omega}{2\pi} \pi x^2$$

$$= \left(\frac{Q 2x d\ell}{RL} \right) \frac{\omega x^2}{2}$$

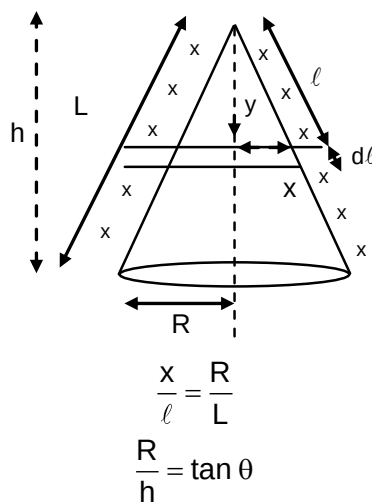
$$= \frac{Q \omega d\ell}{RL} \left(\frac{R}{L} \ell \right)^3$$

$$= \frac{Q \omega R^2}{L^4} \ell^3 d\ell$$

$$\text{Total magnetic moment} = \frac{Q \omega R^2}{L^4} \int_0^L \ell^3 d\ell = \frac{Q \omega R^2}{4}$$

$$= \frac{Q \omega h^2 \tan^2 \theta}{4}$$

$$= 0.25 Q \omega h^2 \tan^2 \theta$$

6. **1.6 (Range: 1.60 to 1.62)**

Sol. $E = \frac{12375}{4000} = 3.1 \text{ eV}$

Number of photoelectrons emitted per second,

$$n = \left(\frac{1}{10^6} \right) \left(\frac{5}{3.1 \times 1.6 \times 10^{-19}} \right)$$

$$= 1.0 \times 10^{13} \text{ per second}$$

$$\therefore i = ne$$

$$= 1.0 \times 10^{13} \times 1.6 \times 10^{-19}$$

$$= 1.6 \times 10^{-6} \text{ A}$$

$$= 1.6 \mu\text{A}$$

TG ~ @bohring_bot

SECTION – II : CHEMISTRY

(PART – A)

1. BD

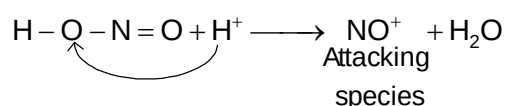
Sol. The gases formed in the options are

- (A) NH_3 and N_2
 (B) N_2 and N_2
 (C) N_2O and O_2
 (D) NH_3 and NH_3

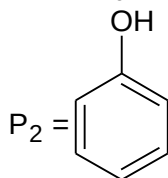
2. AB

Sol. $2\text{I}^- \longrightarrow \text{I}_2 + 2\text{e}^-$ $\therefore$ One mole I_2 needs 2F of electricity.

3. ABCD

Sol. $\text{NaNO}_2 + \text{HCl} \longrightarrow \text{NaCl} + \text{HNO}_2$ 

4. ACD

Sol. $\text{P}_1 = \text{CH}_3\text{COCH}_3$  $\text{P}_4 = \text{CH}_3\text{COONa}$

5. BCD

Sol. The correct reaction in (A) is $\text{Fe}_2\text{O}_3 + \text{C} \longrightarrow 2\text{FeO} + \text{CO}$

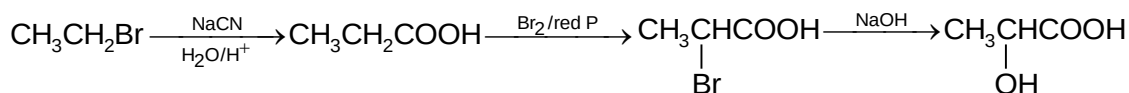
6. AD

Sol. All zero order reactions complete in two half lives.

7. CD

Sol. Q is $\text{Ph}-\text{CH}=\text{CHCH}_2\text{COOC}_2\text{H}_5$ (Cis-)

8. AD

Sol. $\text{CH}_3\text{CHO} \xrightarrow[\text{(ii) H}_2\text{O/H}^+/\Delta]{\text{(i) NaCN/HCl}} \text{CH}_3\text{CH}(\text{OH})\text{COOH}$ 

9. B

Sol. PhCH_2OH forms benzyl carbocation Ph^+CH_2 which is more stable and gives instantaneous result.

10. D

Sol. 1° -alcohols are less reactive than 2° - and 3° -alcohol.

11. C

Sol. Cyclic silicates contain $(\text{SiO}_3)_n^{2n-}$ units.

12. A

Sol. Meso silicates are not polymer. So they do not share any oxygen atom.

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(PART – B)

1. 16

Sol. $x = 4, y = 6, z = 6$

2. 4

Sol. Statements a, c, d & e are incorrect.

3. 6

Sol. There are six paramagnetic complexes.

4. -4.66(Range-4.6 to -4.7)

Sol. $2\text{Na}_2\text{HPO}_4 + 3\text{NaH}_2\text{PO}_4 \longrightarrow \text{Na}_7\text{P}_5\text{O}_{16} + 4\text{H}_2\text{O}$ $Z = -7, x = 2, y = 3$ 5. 3.81 JK^{-1} (Range 3.7 to 3.9)Sol. $nC_v(T_2 - T_1) = -Pdv$

6. 2

Sol. Since the unit of rate constant is $\text{dm}^3 \text{ mol}^{-1} \text{ s}^{-1} = \text{L mol}^{-1} \text{ s}^{-1}$
The reaction is second order.*TG ~ @bohring_bot*

SECTION – III : MATHEMATICS

(PART – A)

1. ABCD

Sol. Applying $C_2 \rightarrow C_1 - \cos^2 x C_1$, we get $f(x) = \begin{vmatrix} \sec^2 x & 0 & 1 \\ \cos^2 x & \cos^2 x - \cos^4 x & \operatorname{cosec}^2 x \\ 1 & 0 & \cot^2 x \end{vmatrix}$

$$= (\cos^2 x - \cos^4 x)(\sec^2 x \cot^2 x - 1) \text{ [expanding along } C_2]$$

$$= \cos^2 x \sin^2 x (\operatorname{cosec}^2 x - 1)$$

$$= \cos^2 x \sin^2 x \cot^2 x = \cos^4 x$$

$$= \frac{1}{4}(3 + 4 \cot 2x + \cos 4x)$$

Thus $\int_{-\pi/4}^{\pi/4} f(x) dx$

$$= \frac{1}{4} \left(3x + 2 \sin 2x + \frac{1}{4} \sin 4x \right) \Big|_0^{\pi/4} = \frac{1}{16} (3\pi + 8)$$

It can be easily checked that each of (b), (c) and (d) also holds.

2. ABC

Sol. We have $A(\alpha, \beta)' = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & e^\beta \end{bmatrix}$

$$= \begin{bmatrix} \cos(-\alpha) & \sin(-\alpha) & 0 \\ -\sin(-\alpha) & \cos(-\alpha) & 0 \\ 0 & 0 & e^\beta \end{bmatrix} = A(-\alpha, \beta)$$

Also, $A(\alpha, \beta)A(-\alpha, -\beta)$

$$= \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & e^\beta \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & e^{-\beta} \end{bmatrix}$$

$$= I$$

$$\Rightarrow A(\alpha, \beta)^{-1} = A(-\alpha, -\beta)$$

Next, $\operatorname{Adj}(\alpha, \beta) = |A(\alpha, \beta)| A(\alpha, \beta)^{-1}$

$$= e^\beta A(-\alpha, -\beta)$$

3. ABC

Sol. Let x_0 be the number of objects to the left of the first object chosen, x_1 the number of objects between the first and the second, x_2 the number of objects between the second and the third and x_3 the number of objects to the right of the third object. We have

$$x_0, x_3 \geq 0, x_1, x_2 \geq 1 \text{ and } x_0 + x_1 + x_2 + x_3 = n - 1 \quad (1)$$

Put $x_1 = y_1 + 1$ and $x_2 = y_2 + 1$. Then (1) reads as $x_0 + y_1 + y_2 + x_3 = n - 5 \quad (2)$

where $x_0, y_1, y_2, x_3 \geq 0$

The number of non-negative integral solution of (2) is ${}^{n-5+3}C_3 = {}^{n-2}C_3$

We have ${}^{n-2}C_3 = \frac{1}{6}(n-2)(n-3)(n-4)$.

Also, ${}^{n-3}C_3 + {}^{n-3}C_2 = {}^{n-2}C_3$ TG @bohring_bot

4. ABC

Sol.
$$\frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$$

$$= \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{2n}\right) - \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$= \left(1 + \frac{1}{3} + \dots + \frac{1}{2n-1}\right)$$

$$+ \frac{1}{2} \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + \dots + \frac{1}{n}\right)$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \dots + \frac{1}{2n-1} - \frac{1}{2n}$$

$$\alpha(2n) < \underbrace{\frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}}_{n \text{ times}} = 1 \text{ and } \alpha(2n) \geq \frac{1}{2n} \geq \frac{1}{2} \forall n$$

(D) is not correct as $\alpha(1) = 1$

5. AB

Sol.
$$I = \int_0^\alpha \frac{dx}{1 - \cos \alpha \cos x}$$

$$= \int_0^\alpha \frac{dx}{\left(\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2}\right) - \cos \alpha \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}\right)}$$

$$= \int_0^\alpha \frac{dx}{(1 - \cos \alpha) \cos^2 \frac{x}{2} + (1 + \cos \alpha) \sin^2 \frac{x}{2}}$$

$$= \int_0^\alpha \frac{dx}{2 \sin^2 \frac{\alpha}{2} \cos^2 \frac{x}{2} + 2 \cos^2 \frac{\alpha}{2} \sin^2 \frac{x}{2}}$$

$$= \frac{1}{2} \int_0^\alpha \frac{\left(\sec^2 \frac{\alpha}{2}\right) \sec^2 \frac{x}{2}}{\tan^2 \frac{\alpha}{2} + \tan^2 \frac{x}{2}} dx$$

Put $\tan \frac{x}{2} = t$, we have

$$I = \int_0^{\tan \frac{\alpha}{2}} \frac{\sec^2 \frac{\alpha}{2} dt}{t^2 + \tan^2 \frac{\alpha}{2}}$$

$$= \sec^2 \frac{\alpha}{2} \cot \frac{\alpha}{2} \left[\tan^{-1} \left(\frac{t}{\tan \frac{\alpha}{2}} \right) \right]_0^{\tan \frac{\alpha}{2}}$$

$$= \frac{2}{\sin \alpha} \cdot \frac{\pi}{4} = \frac{\pi}{2 \sin \alpha}$$

Thus $\frac{A}{\sin \alpha} + B = \frac{\pi}{2 \sin \alpha}$. $A = \frac{\pi}{2}$, $B = 0$ and $A = \frac{\pi}{4}$, $B = \frac{\pi}{4 \sin \alpha}$ satisfy the last equation.

6. AD

Sol. Let $\sqrt{y-x} = t$ *TG ~ @bohring_bot*

$$y - x = t^2$$

$$\frac{dy}{dx} = 2t \frac{dt}{dx} + 1$$

$$2t \frac{dt}{dx} + 1 = \sqrt{t}$$

$$\frac{2t}{\sqrt{t}-1} dt = dx \text{ which leads A and D as solutions}$$

7. BD

$$\text{Sol. } \sin A + \sin B = \frac{a+b}{c} = \frac{\sin A + \sin B}{\sin C} \Rightarrow \sin C = 1$$

$$\Rightarrow C = 90^\circ$$

$$A + B = 90^\circ \Rightarrow \cos A = \sin B$$

8. AC

$$\text{Sol. } \left(x^2 + 1 + \frac{1}{x^2} \right)^n$$

$$= {}^nC_0 + {}^nC_1 \left(x^2 + \frac{1}{x^2} \right) + {}^nC_2 \left(x^2 + \frac{1}{x^2} \right)^2 + \dots + {}^nC_n \left(x^2 + \frac{1}{x^2} \right)^n$$

This contains each of the term $x^0, x^2, x^4, \dots, x^{2n}, x^{-2}, x^{-4}, \dots, x^{-2n}$

Coefficient of constant term

$$= {}^nC_0 + ({}^nC_2)(2) + ({}^nC_4)({}^4C_2) + ({}^nC_6)({}^6C_3) + \dots$$

$$\neq 2^n - 1$$

Coefficient of x^{2n-2} in ${}^nC_{n-1} = n$

$$\text{Coefficient of } x^2 \text{ is } {}^nC_1 + ({}^nC_3)({}^3C_1) + ({}^nC_5)({}^5C_2) + \dots > n$$

9. A

$$\text{Sol. Let the equations of the circles be } x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0 \text{ and } x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0.$$

$$(I) \text{ Equation of the radical axis is } 2(g_1 - g_2)x + 2(f_1 - f_2)y + c_1 - c_2 = 0$$

$$\text{Slope of the radical axis} = \frac{g_1 - g_2}{f_1 - f_2}$$

Slope of the line joining the centres $= \frac{f_1 - f_2}{g_1 - g_2}$ So, the radical axis is perpendicular to the line joining the centre

(II) Common tangent to the intersecting circles of equal radii is at the same distance from the centres of the two circles and hence is parallel to the line joining the centres.

(III) Since the line joining the centres of the circles to the mid – point of the common chord is perpendicular to the chord, it bisects the chord.

(IV) The line joining the centre of one to a point of intersection of tangent to the other circle. So by definition of orthogonality, they are perpendicular.

10. A

$$\text{Sol. For function } f \text{ in (a) } f(x) = \begin{cases} x, & x > 1 \\ 0, & x = 1 \\ -x, & x < 1 \end{cases} \text{ so } \lim_{x \rightarrow 1} f(x) \text{ does not exist.}$$

TG ~ @bohring_bot

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\sin\left(\sin\left(\tan\left(\frac{x^2}{2}\right)\right)\right)}{\log \cos 3x} \\
&= \lim_{x \rightarrow 0} \frac{\sin\left(\sin\left(\tan\left(\frac{x^2}{2}\right)\right)\right)}{\sin\left(\frac{\tan x^2}{2}\right)} \times \frac{\sin\left(\sin\left(\tan\left(\frac{x^2}{2}\right)\right)\right)}{\sin\left(\frac{\tan x^2}{2}\right)} \times \frac{\sin\left(\tan\frac{x^2}{2}\right)}{\tan\frac{x^2}{2}} \times \frac{\tan\left(\frac{x^2}{2}\right)}{\frac{x^2}{2}} \times \frac{\frac{x^2}{2}}{\log \cos 3x} \\
&= \frac{1}{2} \lim_{x \rightarrow 0} \frac{2x}{-3 \tan 3x} = -\frac{1}{9}
\end{aligned}$$

$$\begin{aligned}
& \lim_{x \rightarrow 0} \frac{\sqrt[3]{1 + \tan^{-1} 3x} - \sqrt[3]{1 + \sin^{-1} 3x}}{\sqrt{1 - \sin^{-1} 2x} - \sqrt{1 + \tan^{-1} 2x}} \\
&= \lim_{x \rightarrow 0} \frac{\left(1 + \frac{1}{3} \tan^{-1} 3x + \dots\right) - \left(1 - \frac{1}{3} \sin^{-1} 3x + \dots\right)}{\left(1 - \frac{1}{2} \sin^{-1} 2x - \dots\right) - \left(1 + \frac{1}{2} \tan^{-1} 2x + \dots\right)} \\
&= \lim_{x \rightarrow 0} \frac{\frac{1}{3} \frac{\tan^{-1} 3x}{x} + \frac{1}{3} \frac{\sin^{-1} 3x}{x} + \frac{1}{x} (\text{highest power of } \sin^{-1} 3x \text{ and } \tan^{-1} 3x)}{-\frac{1}{2} \frac{\sin^{-1} 2x}{x} - \frac{1}{2} \frac{\tan^{-1} 2x}{x} + \frac{1}{x} (\text{highest power of } \sin^{-1} x \text{ and } \tan^{-1} 2x)} \\
&= \frac{1+1}{-1-1} = -1 \\
& \lim_{x \rightarrow 0} \frac{e^{1/x} - 1}{e^{1/x} + 1} = -1 \text{ and } \lim_{x \rightarrow 0^+} \frac{e^{1/x} - 1}{e^{1/x} + 1} = 1
\end{aligned}$$

11. C

Sol.

$$(I) \quad a_1 = \int_0^{\pi/2} dx = \frac{\pi}{2}, a_2 = \pi$$

Also, $a_{n+2} + a_n - 2a_{n+1}$

$$= \int_0^{\pi/2} \frac{2 \cos(2n+2)x - \cos(2n+4)x - \cos(2nx)}{1 - \cos 2x} dx$$

$$= \int_0^{\pi/2} \frac{2 \cos(2n+2)x - 2 \cos(2n+2)x \cos 2x}{1 - \cos 2x} dx$$

$$= \int_0^{\pi/2} 2 \cos(2n+2)x dx$$

$$= \frac{2}{2n+2} \sin(2n+2)x \Big|_0^{\pi/2} = 0$$

 $\therefore a_1, a_2, a_3, \dots$ are in A.P.

$$(II) \quad I_{n+1} + I_{n+3}$$

$$= \int_0^{\pi/4} \tan^{n+1} x \sec^2 x dx$$

$$= \frac{1}{n+2} \tan^{n+2} x \Big|_0^{\pi/4} = \frac{1}{n+2}$$

$$\therefore a_n = n+2$$

 $\Rightarrow a_1, a_2, \dots$ are in A.P.

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$$\begin{aligned}
 \text{(III)} \quad I_n - I_{n-1} &= \int_0^{\pi/2} \frac{\sin^2(nx) - \sin^2(n-1)x}{\sin x} dx \\
 &= \int_0^{\pi/2} \frac{\sin(2n-1)x \sin x}{\sin x} dx \\
 &= -\frac{1}{2n-1} \cos(2n-1)x \Big|_0^{\pi/2} = \frac{1}{2n-1} \\
 &\Rightarrow a_1, a_2, a_3, \dots \text{ are in H.P.}
 \end{aligned}$$

$$\begin{aligned}
 \text{(IV)} \quad b_n &= \int_0^{\pi/2} \frac{1 - \cos 2nx}{1 - \cos 2x} dx = \frac{n\pi}{2} [\text{see (a)}] \\
 a_n &= \frac{\pi}{2} \forall n.
 \end{aligned}$$

$\therefore a_1, a_2, a_3, \dots$ are in A.P., G.P. and H.P.

Also $a_1, a_2, a_3, \dots$ is a constant sequence.

12. A

Sol.

(I) The given lines are $y = x$ and $y = 0$. The line $y = x$ meets $\frac{x^2}{3} + y^2 = 1$ at the points

$$x = \pm \frac{\sqrt{3}}{2}, y = \pm \frac{\sqrt{3}}{2}. \text{ So the required intercept is } \sqrt{2 \left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right)^2} = \sqrt{6}, \text{ intercept on } y = 0 \text{ is } 2\sqrt{3}$$

(II) Required intercepts are $\sqrt{6}$ and 2.

(III) $y = x$ and $y = 0$ are diameters of the circle and hence the required intercepts is equal to the diameter of the circle which is 4.

(IV) $y = x$ meets the parabola $y^2 = 2\sqrt{2}x$ at points where $x = 0, 2\sqrt{2}$.

So that the points of intersection of the line and the parabola are $(0, 0)$ and $(2\sqrt{2}, 2\sqrt{2})$ and the required intercept is 4. Intercept on $y = 0$ is 0.

For T, $x = \pm\sqrt{2}$, and $0 = 2$ which is not possible hence no intercept or intercept $= 2\sqrt{2}$

(PART – B)

1. 0.25

Sol. Given $\int_0^1 f(x) dx = \frac{1}{3} + \int_0^1 (f(x^2))^2 dx$

Consider, $I_1 = \int_0^1 (f(x^2))^2 dx$

Put $x^2 = t \Rightarrow 2x dx = dt$

$$I_1 = \frac{1}{2} \int_0^1 \frac{(f(t))^2}{\sqrt{t}} dt$$

$\therefore$ Equation (1) becomes

$$\int_0^1 \frac{t}{2\sqrt{t}} dt + \int_0^1 \frac{(f(t))^2 - 2\sqrt{t}f(t)}{2\sqrt{t}} dt \Rightarrow \int_0^1 (f(t) - \sqrt{t})^2 dt = 0$$

$$\therefore f(t) = \sqrt{t}$$

TG ~ @bohring_bot

So, $f\left(\frac{1}{4}\right) = \frac{1}{2}$

So, reciprocal of $f\left(\frac{1}{4}\right) = 2$

2. **0.125**

Sol. Given curve is ellipse having centre (0, 0). All chords of ellipse through origin are bisected at origin
 $\Rightarrow p = q = 0$

3. 3.00

Sol. We have $\frac{4 + \sec 20^\circ}{\operatorname{cosec} 20^\circ} = \frac{\sin 20^\circ}{\cos 20^\circ} (4 \cos 20^\circ + 1)$

$$= \frac{2 \sin 40^\circ + \sin 20^\circ}{\cos 20^\circ}$$

$$= \frac{\sin 40^\circ + (\sin 40^\circ + \sin 20^\circ)}{\cos 20^\circ}$$

$$= \frac{\sin 40^\circ + 2 \sin 30^\circ \cos 10^\circ}{\cos 20^\circ}$$

$$= \frac{2 \sin 60^\circ \cos 20^\circ}{\cos 20^\circ} = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

Hence square of the value of expression = 3

4. 1.00

Sol. Note that for the x interval $(-\infty, 0)$, $f(x)$ is strictly increasing and has range $(-\infty, \infty)$. For the x interval $(0, \infty)$, again, $f(x)$ is strictly increasing and has range $(-\infty, \infty)$.

Hence for all a, there is exactly one positive and one negative solution to $f(x) = a$.

Thus, there are 2^n solutions to $f^n(x) = a$, and the answer is $2^3 = 8$

5. 4.00

Sol. 2 or 3 or 4 or 5 copies of each book out of 5 identical copies can be selected in 4 ways. Since number of books is n, therefore, total number of selections = 4^n

Given $4^n = 256 \Rightarrow n = 4$

6. 17.00

Sol. Let $S_n = 1.2^1 + 2.2^2 + 3.2^3 + 4.2^4 + \dots + n.2^n$ (1)

$\Rightarrow 2.S_n = 1.2^2 + 2.2^3 + \dots + (n-1)2^n + n.2^{n+1}$ (2)

$\therefore$ On Subtracting, we get

$\Rightarrow -S_n = (2^1 + 2^2 + 2^3 + \dots + 2^n) - n.2^{n+1}$

$$= 2(2^n - 1) - n.2^{n+1}$$

$\therefore S_n = n.2^{n+1} - 2(2^n - 1) = n.2^{n+1} - 2^{n+1} + 2 = 2 + (n-1)2^{n+1}$

But $2 + (n-1)2^{n+1} = 2 + 2^{n+5}$ (given)

$\Rightarrow n-1 = 2^4$

$\therefore n = 17$

IG ~ @bohring_bot